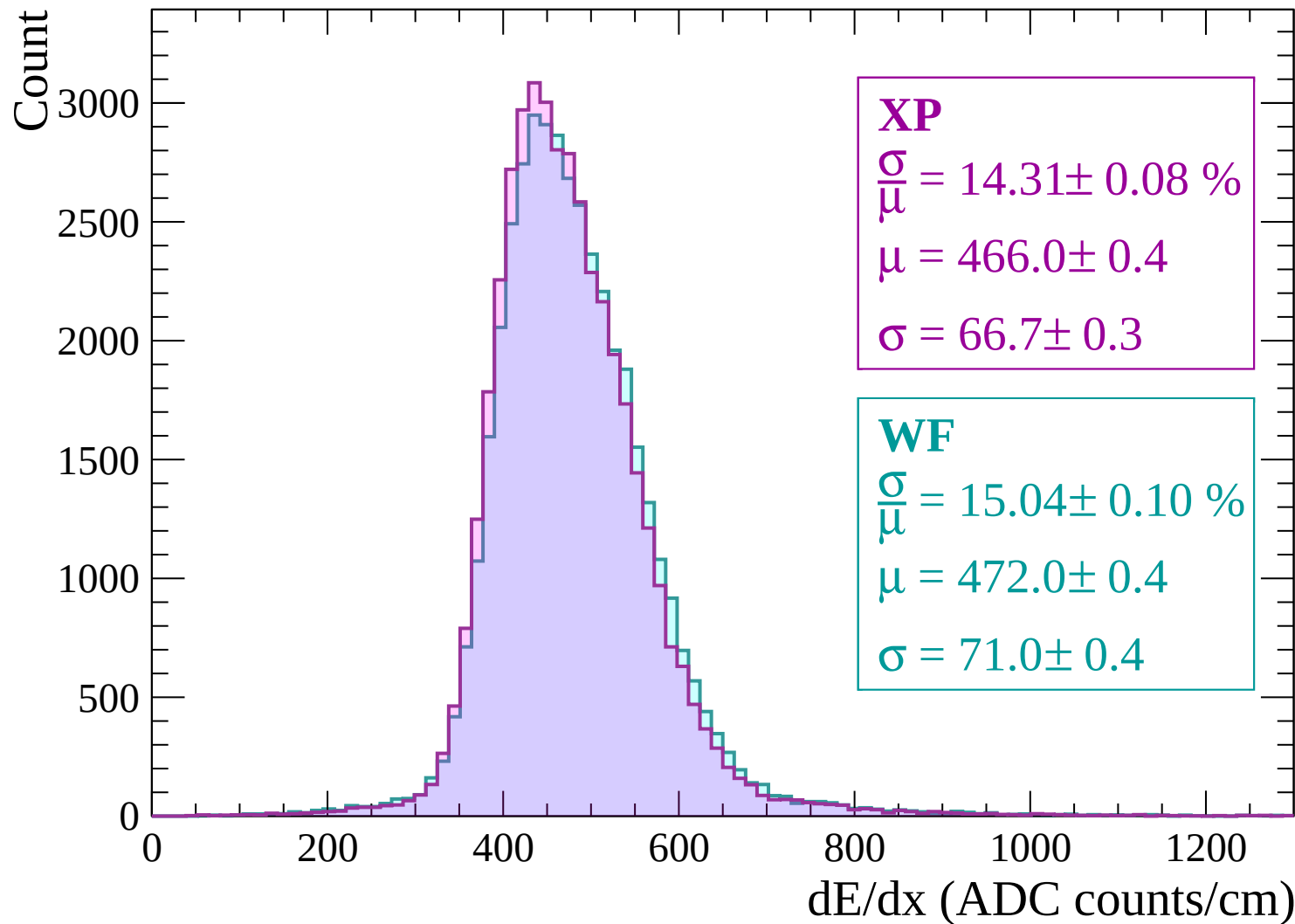
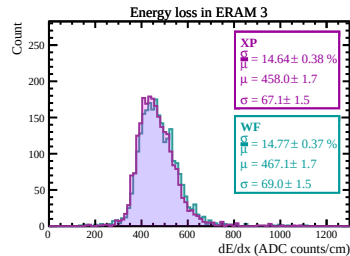
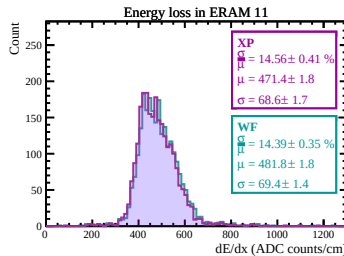
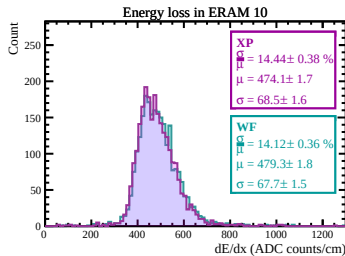
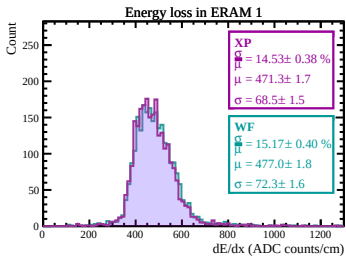
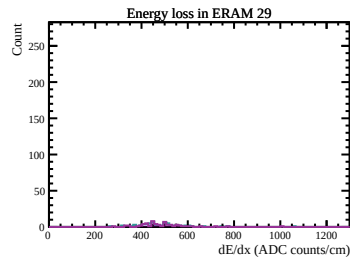
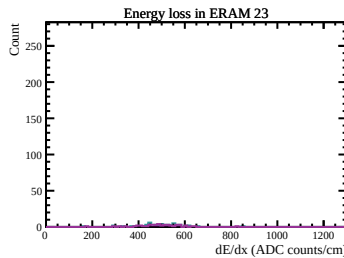
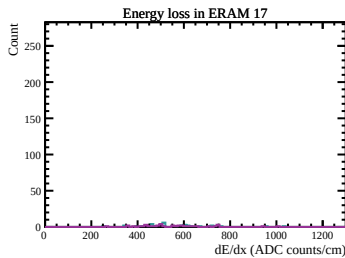
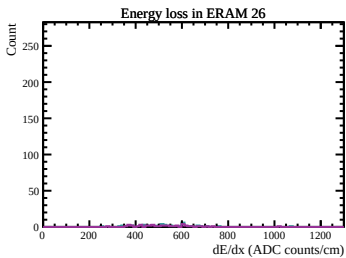
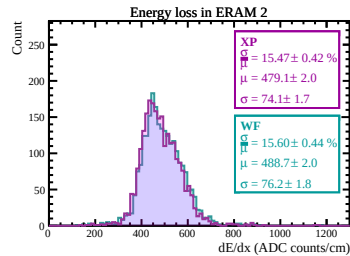
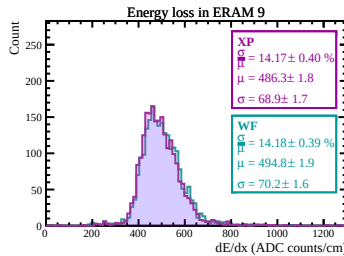
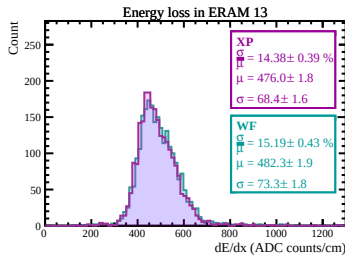
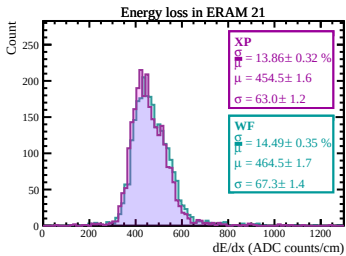
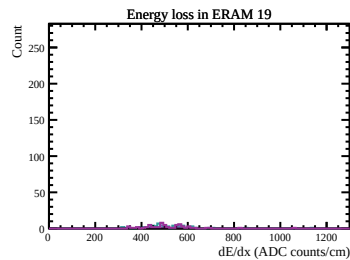
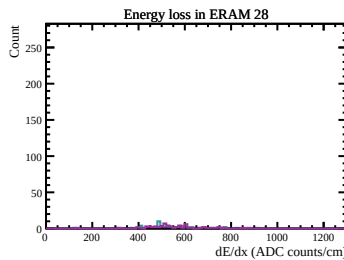
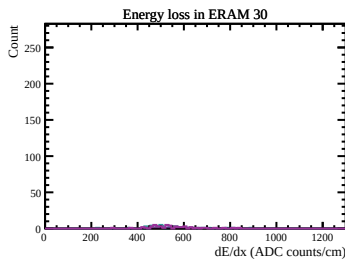
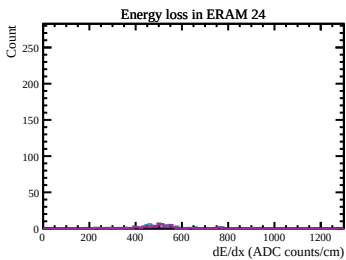
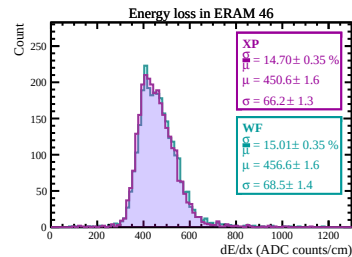
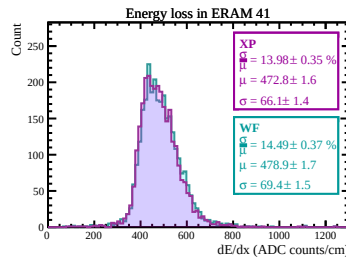
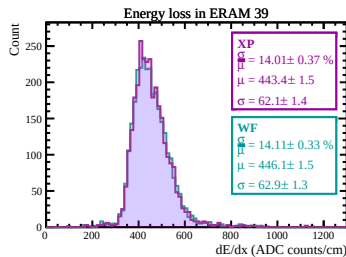
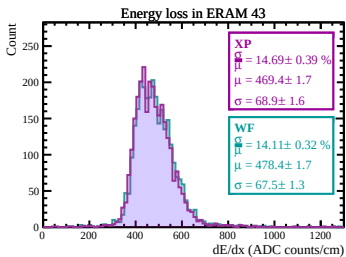
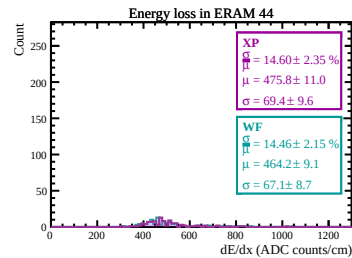
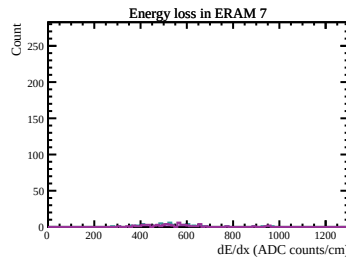
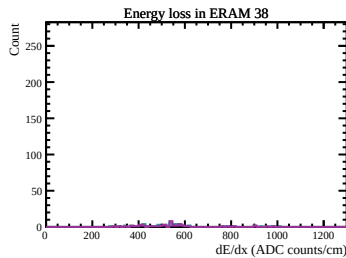
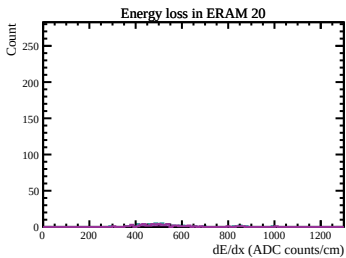
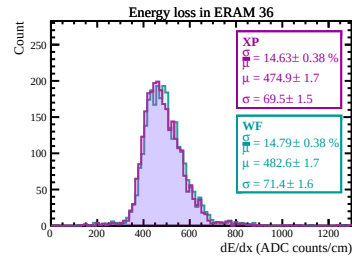
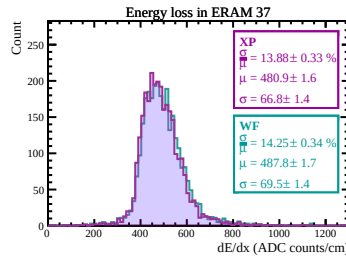
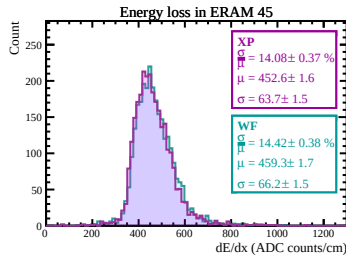
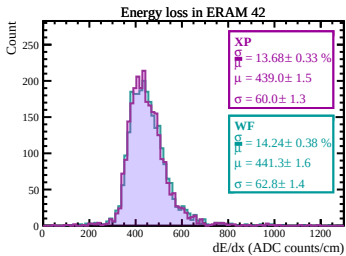
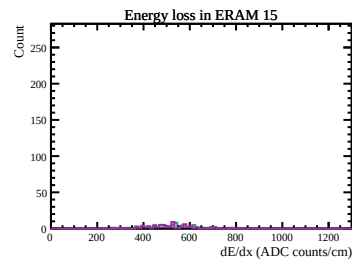
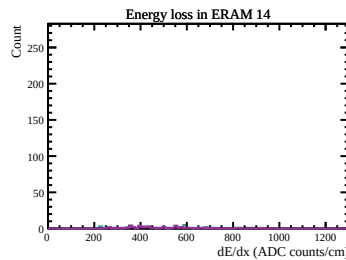
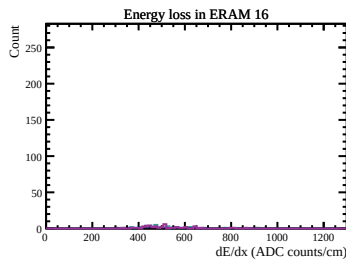
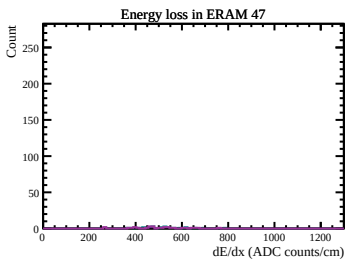
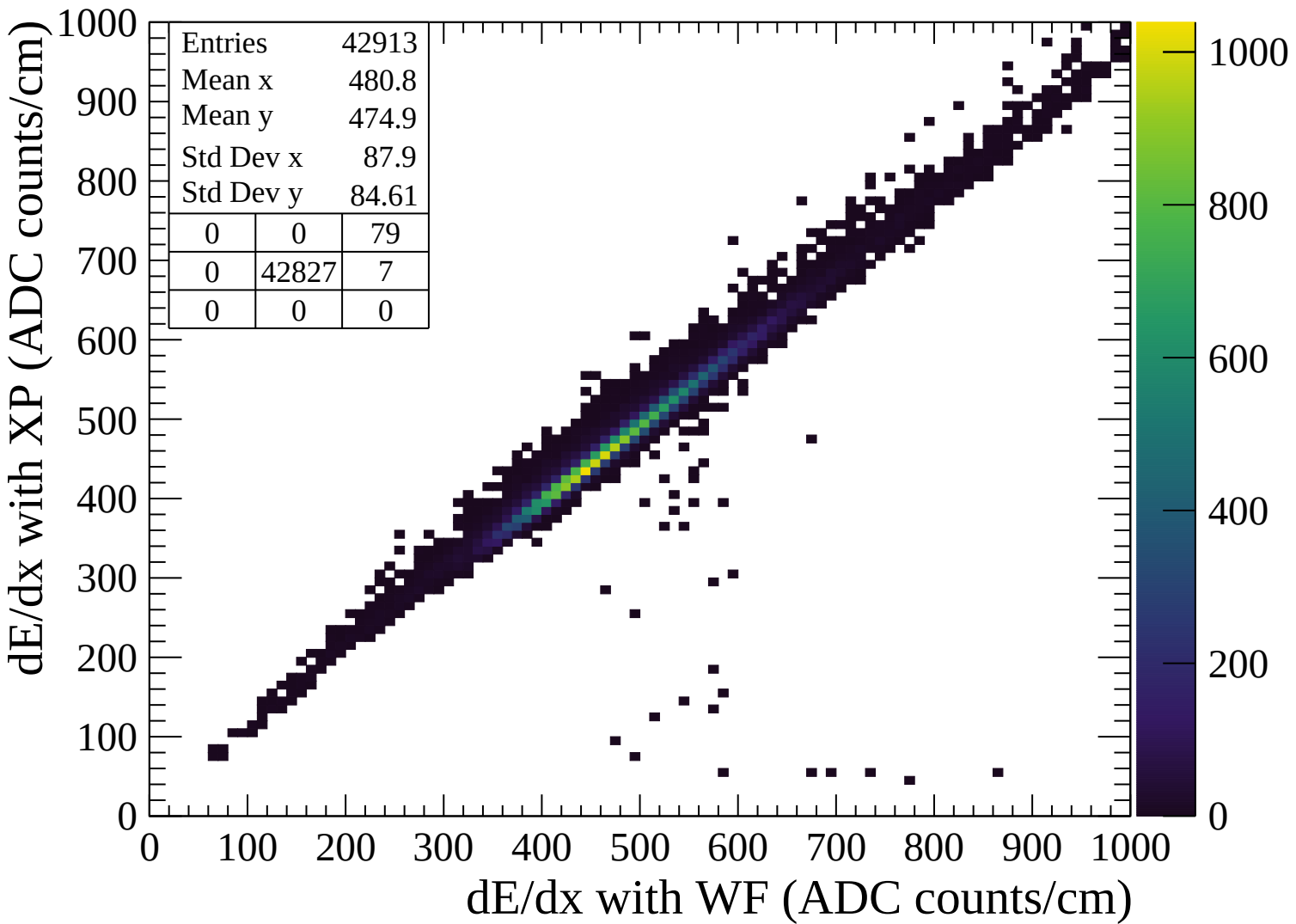
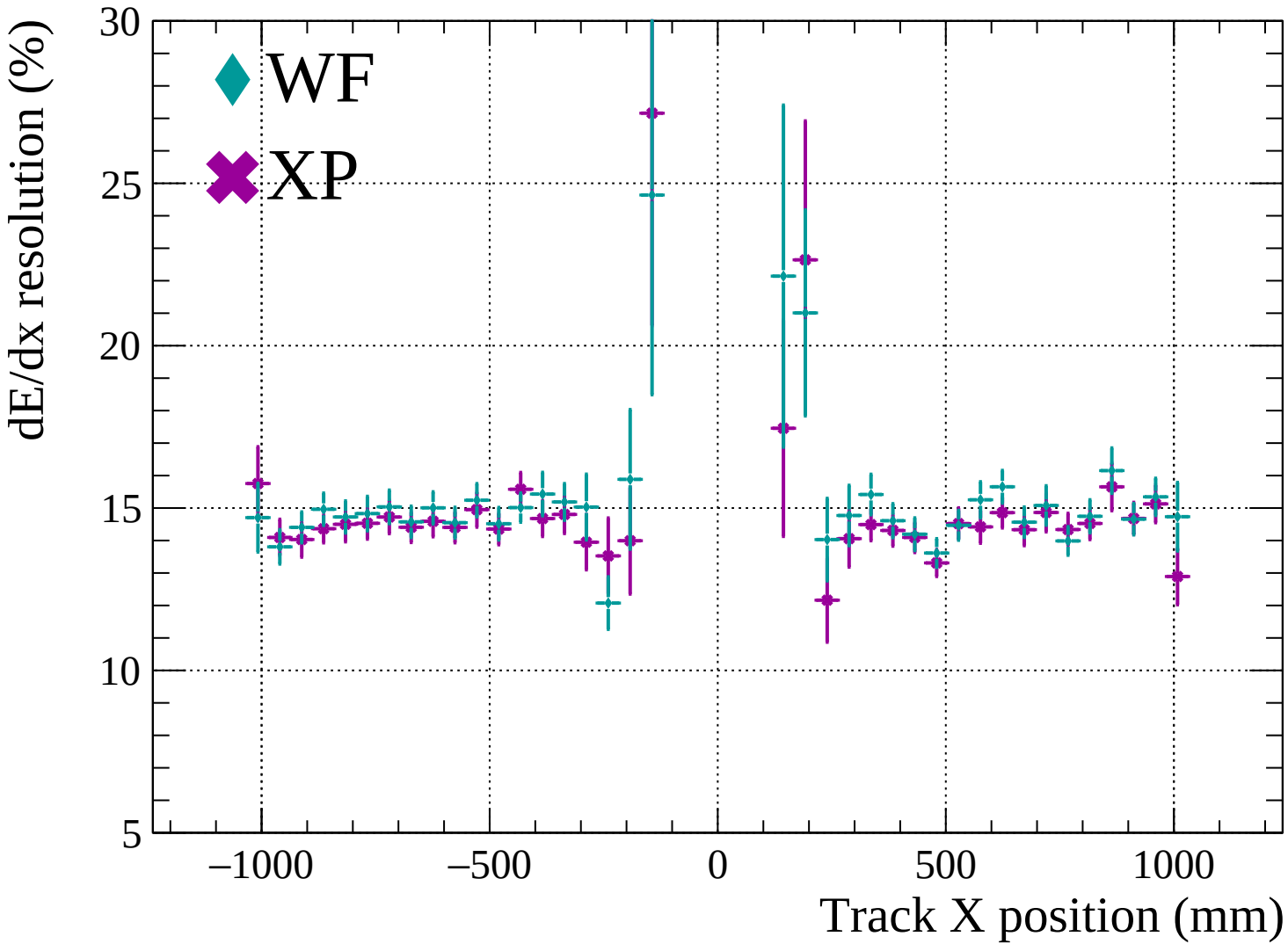


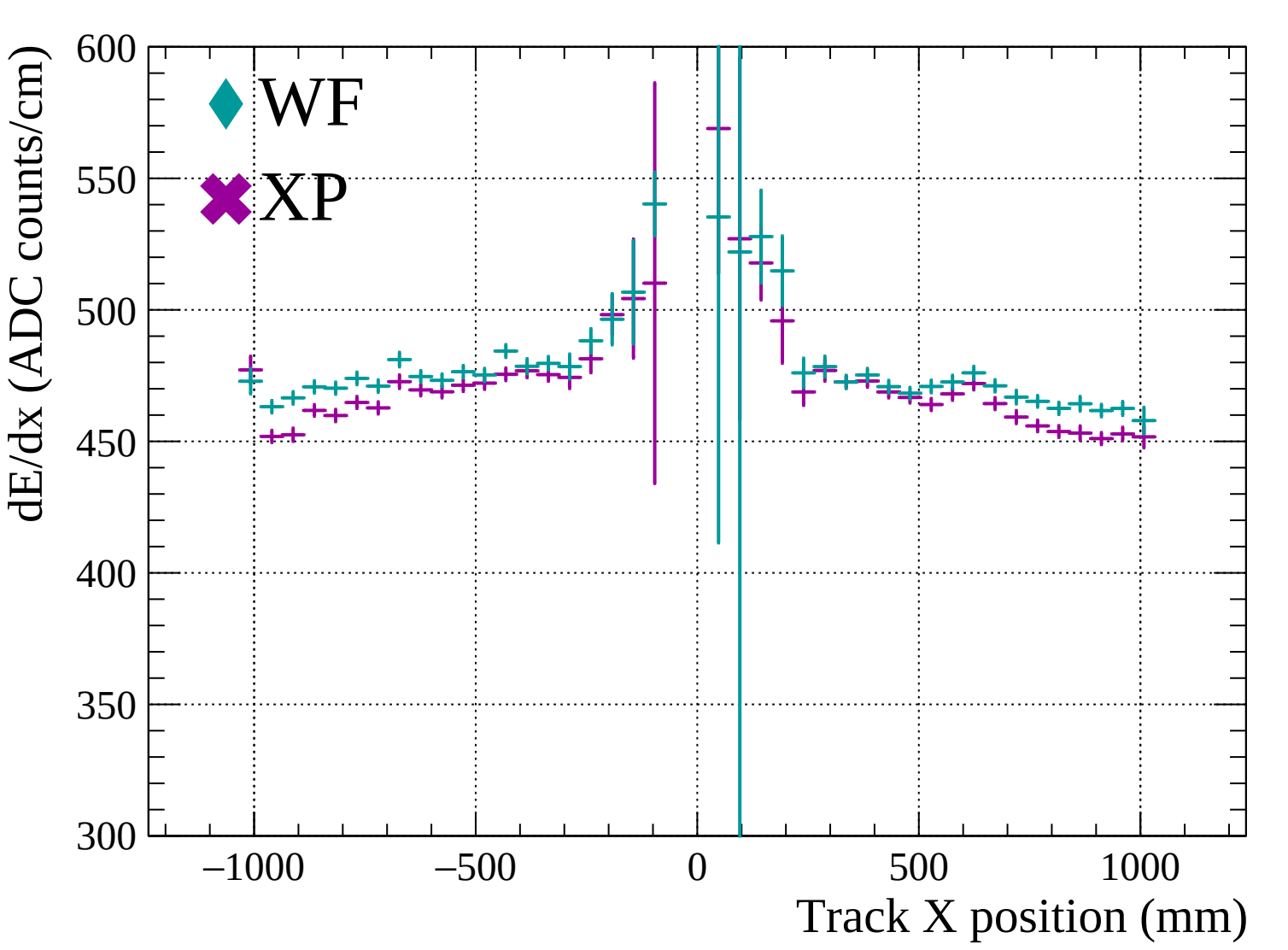


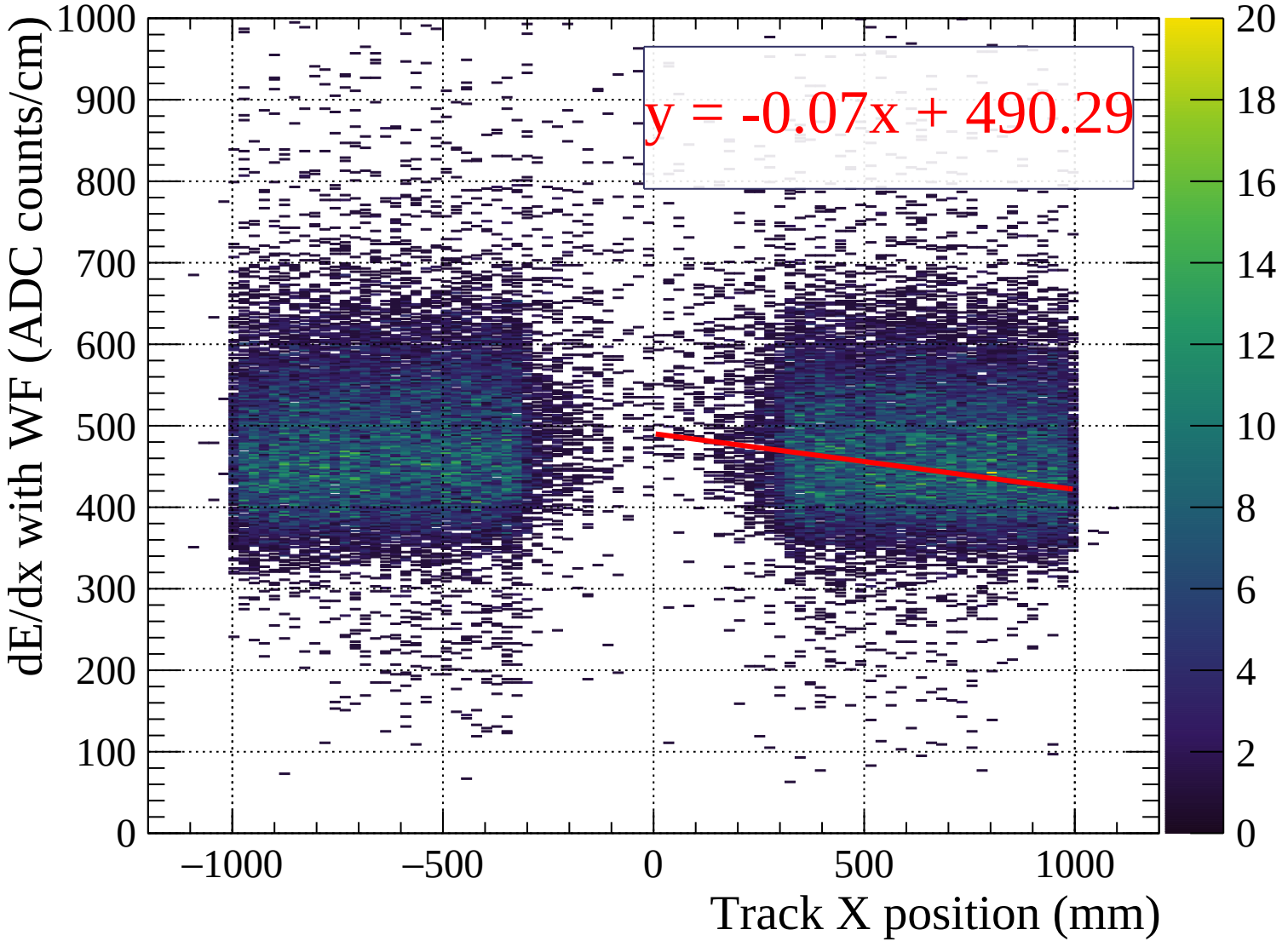


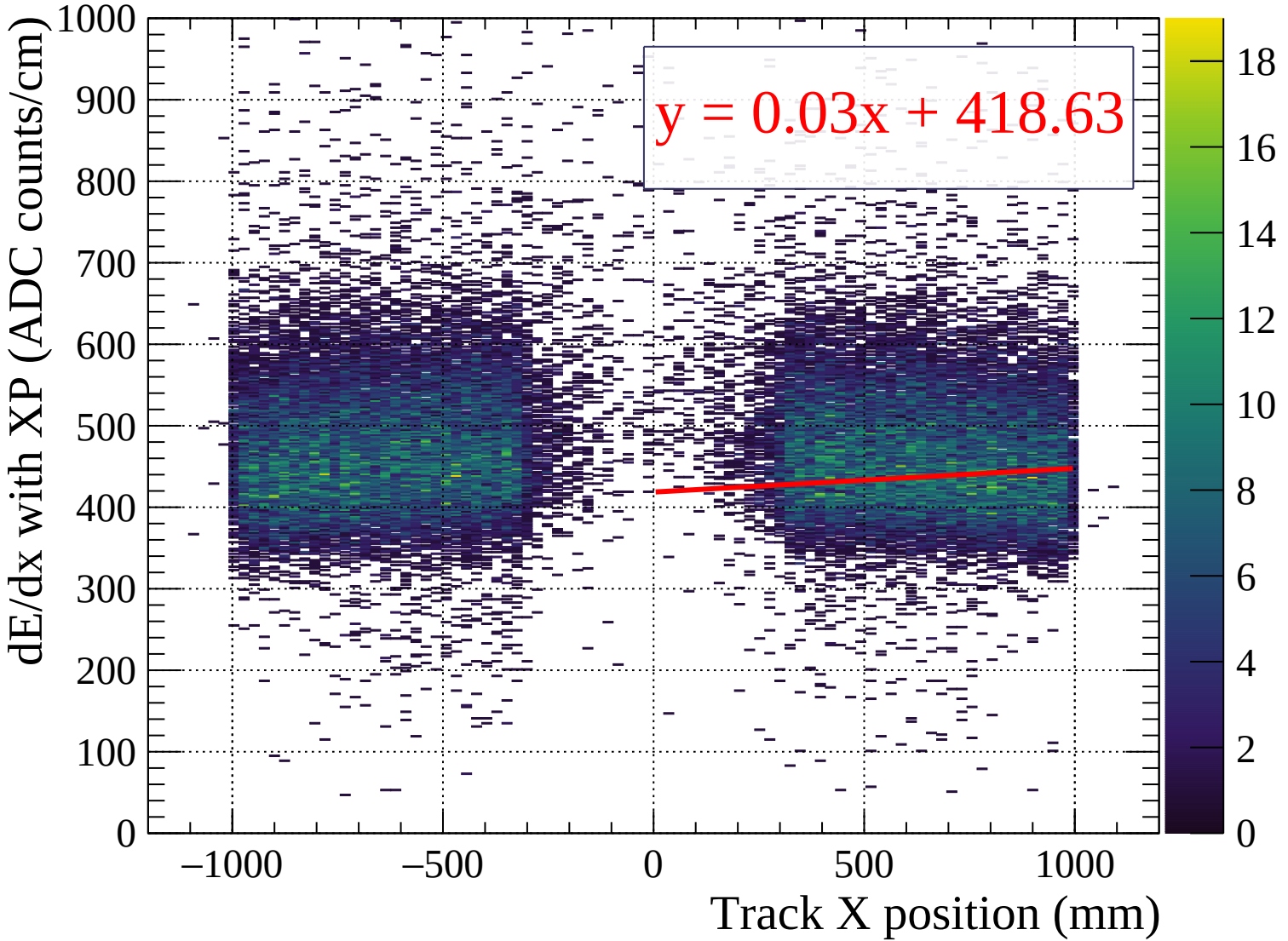


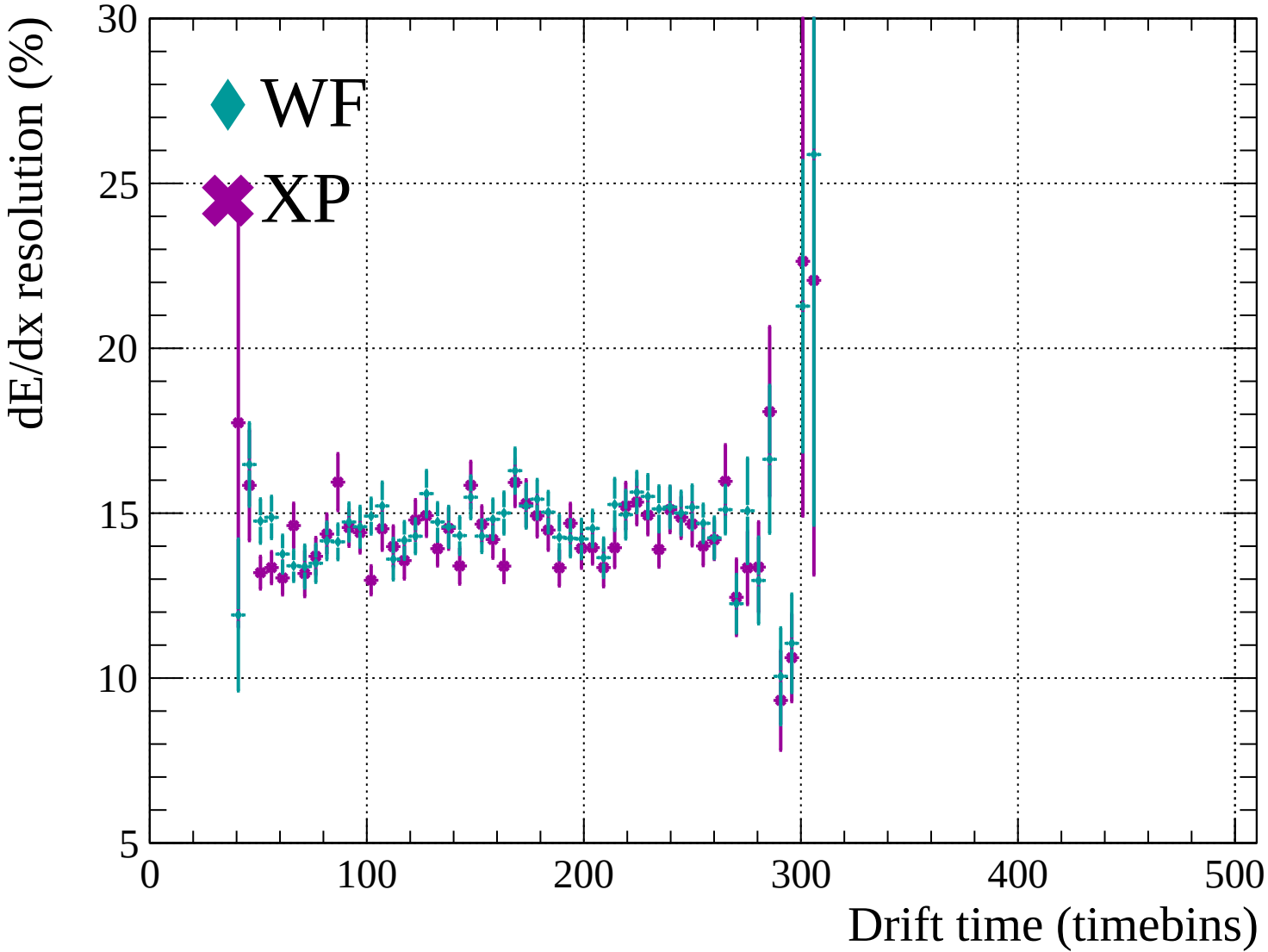


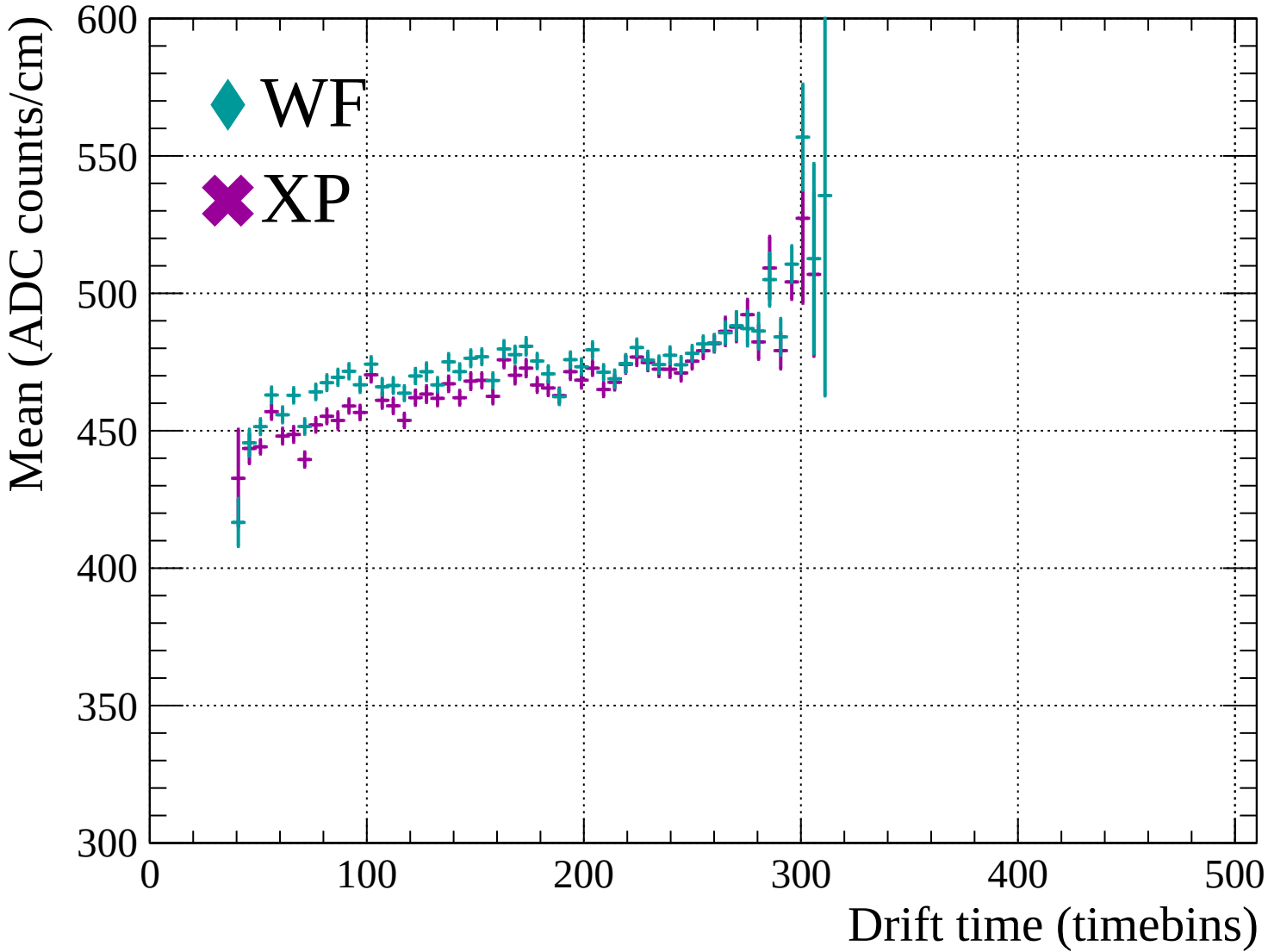


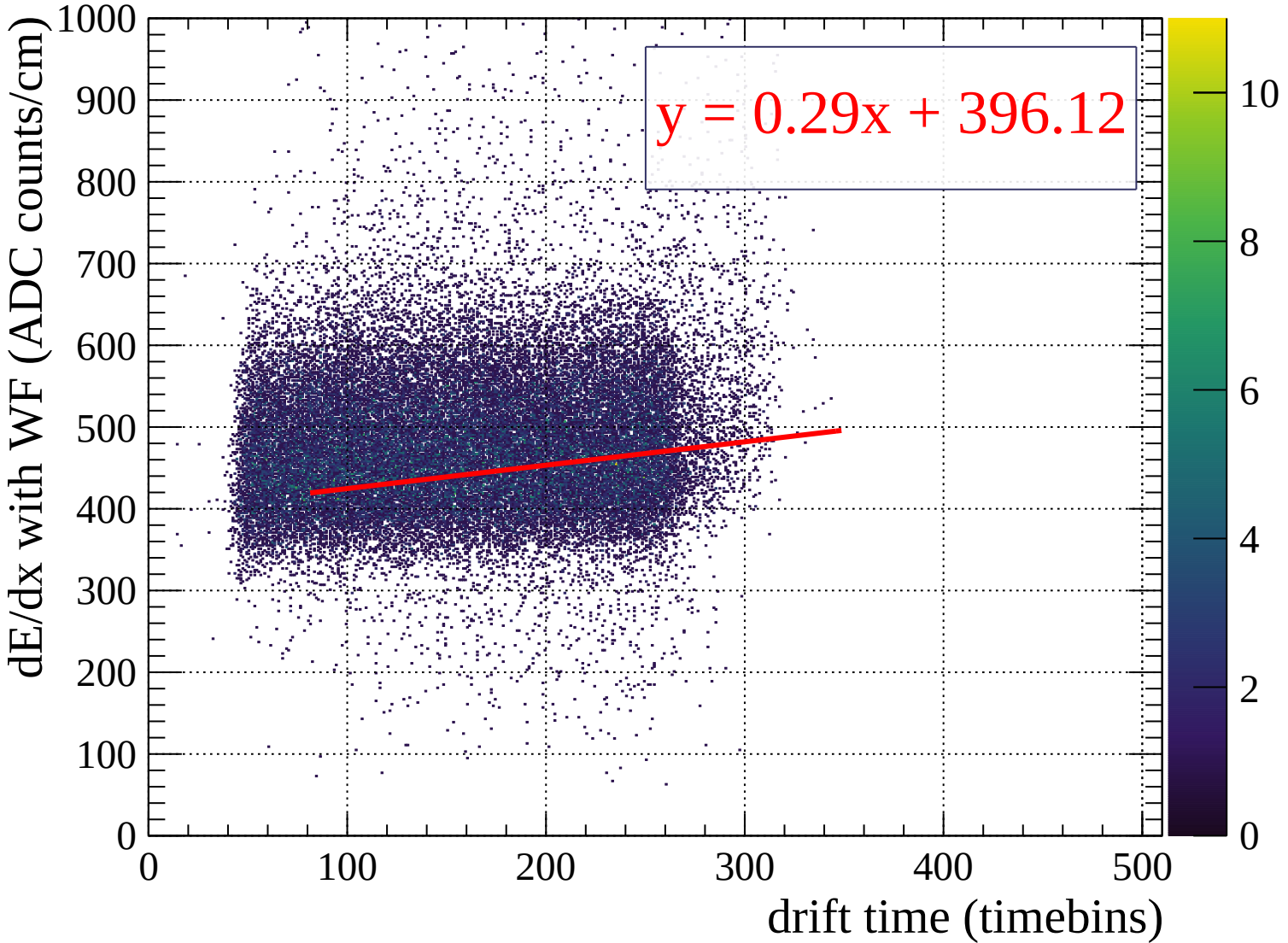


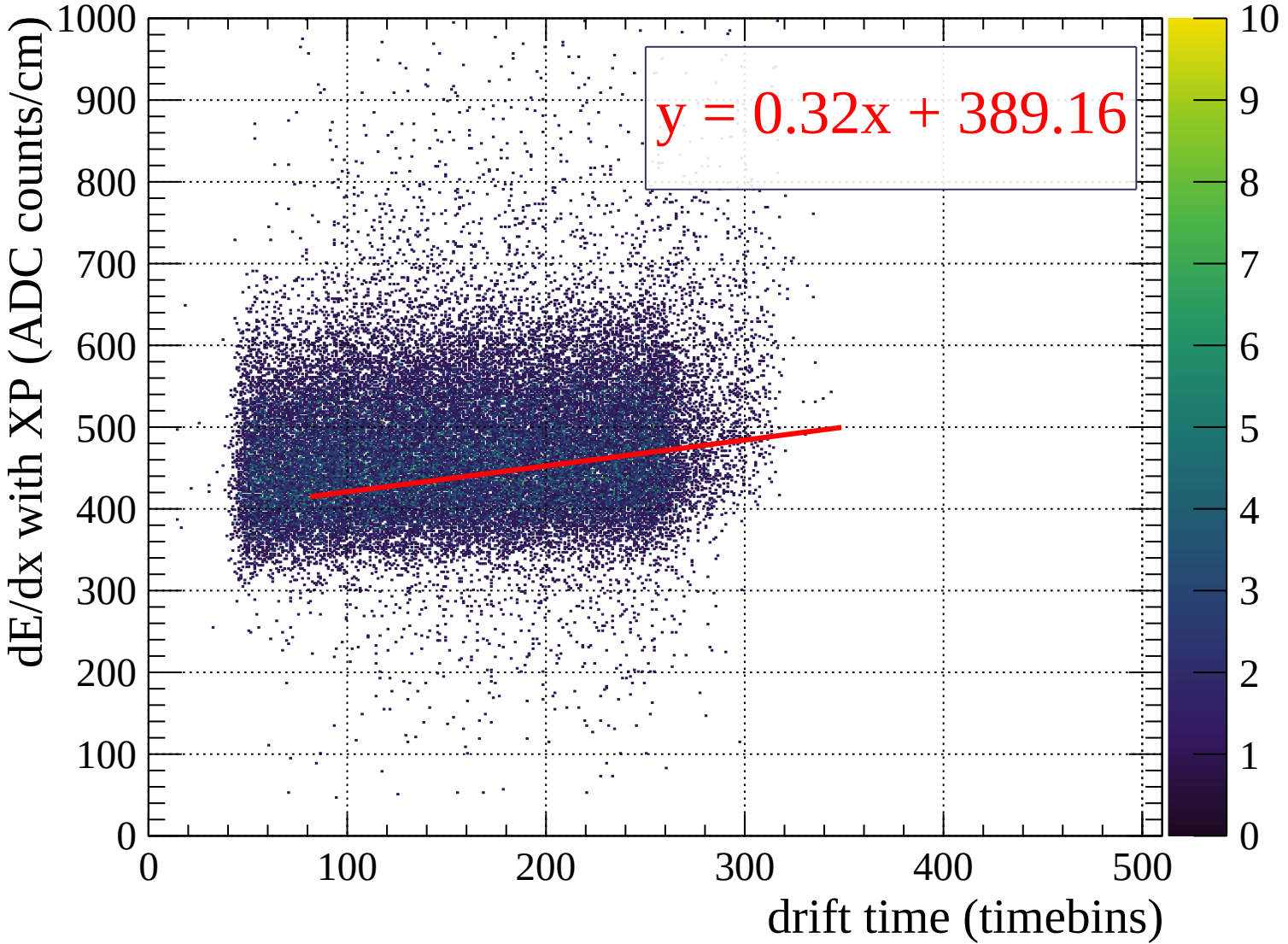


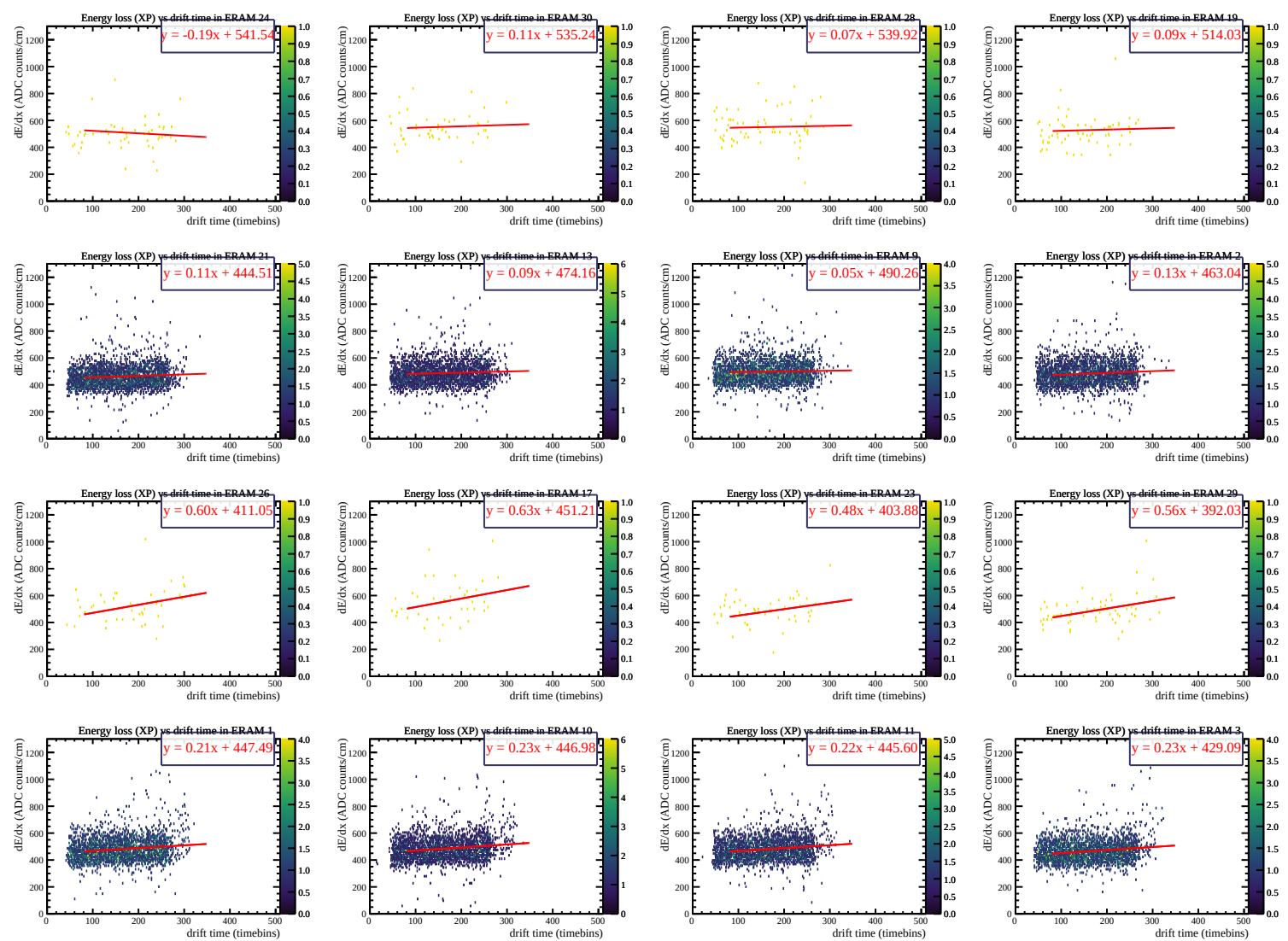


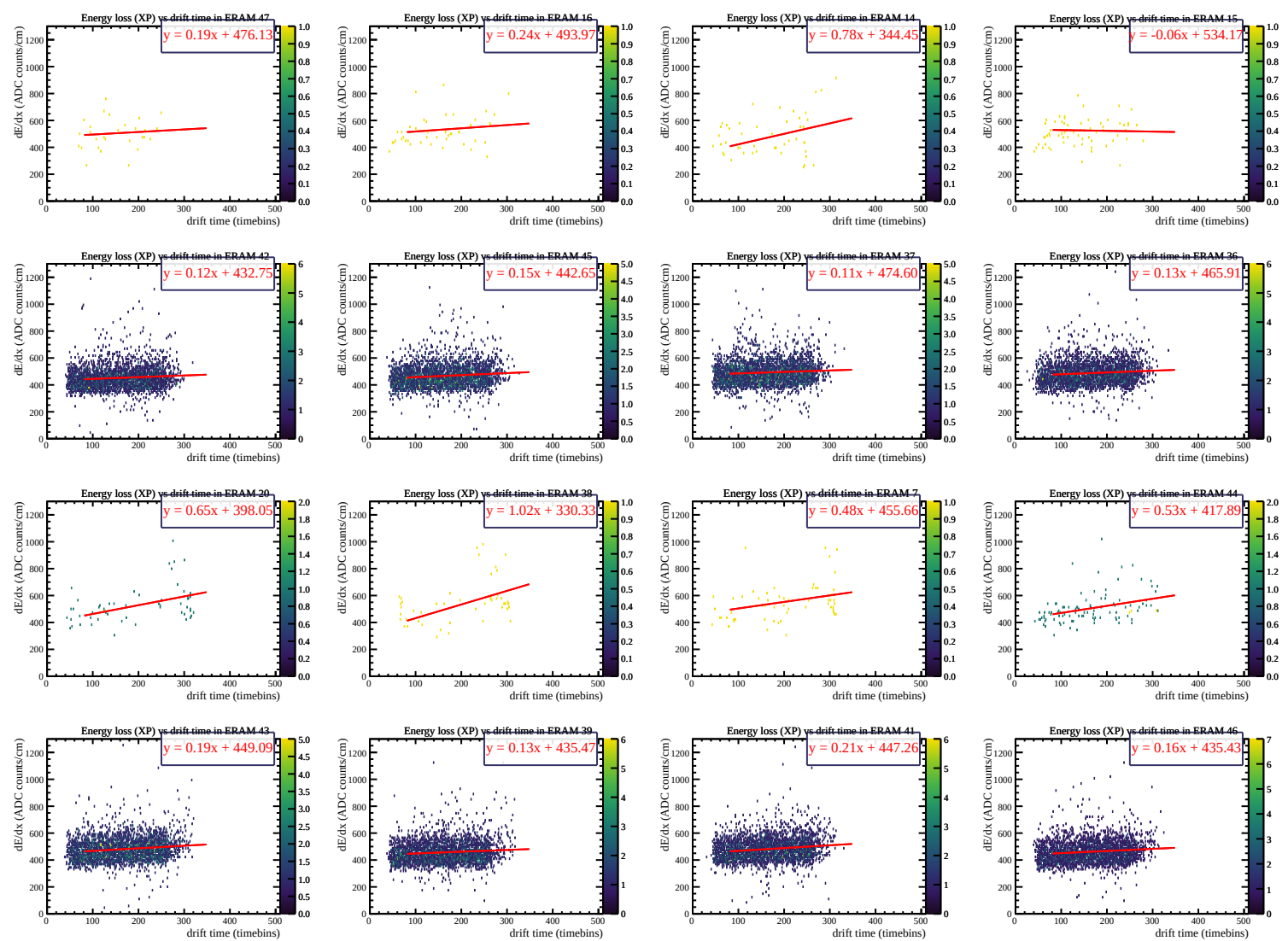


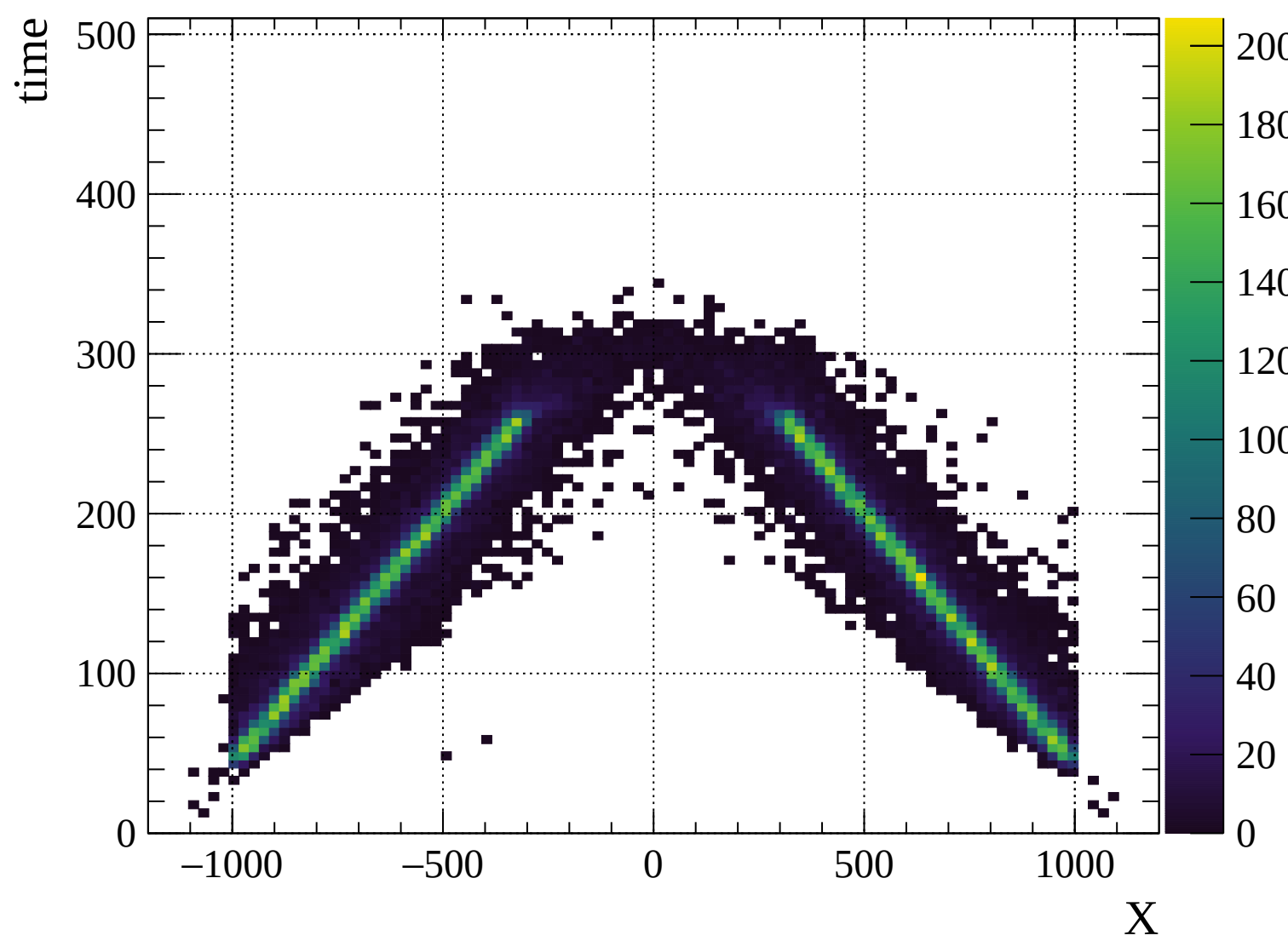


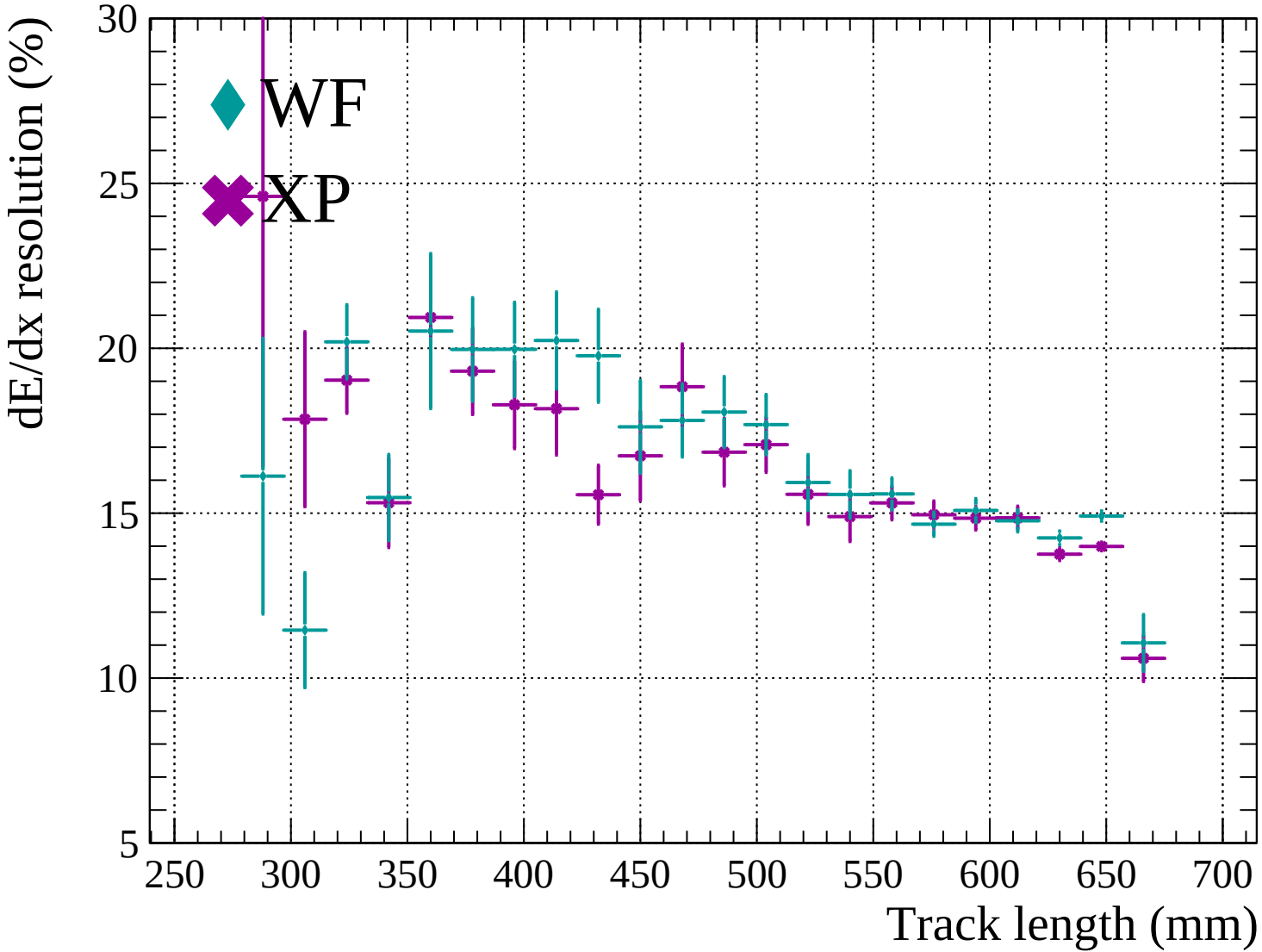


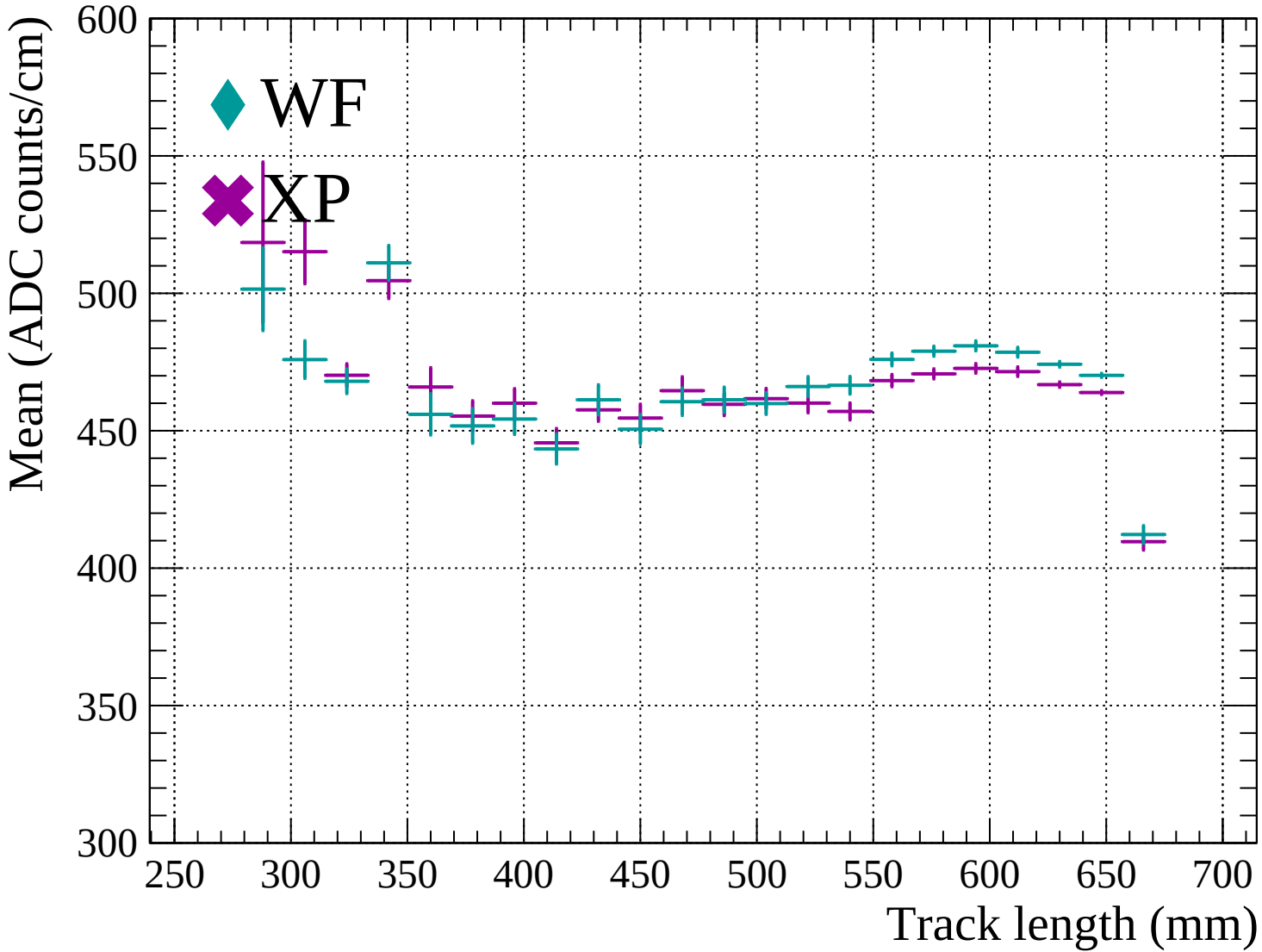


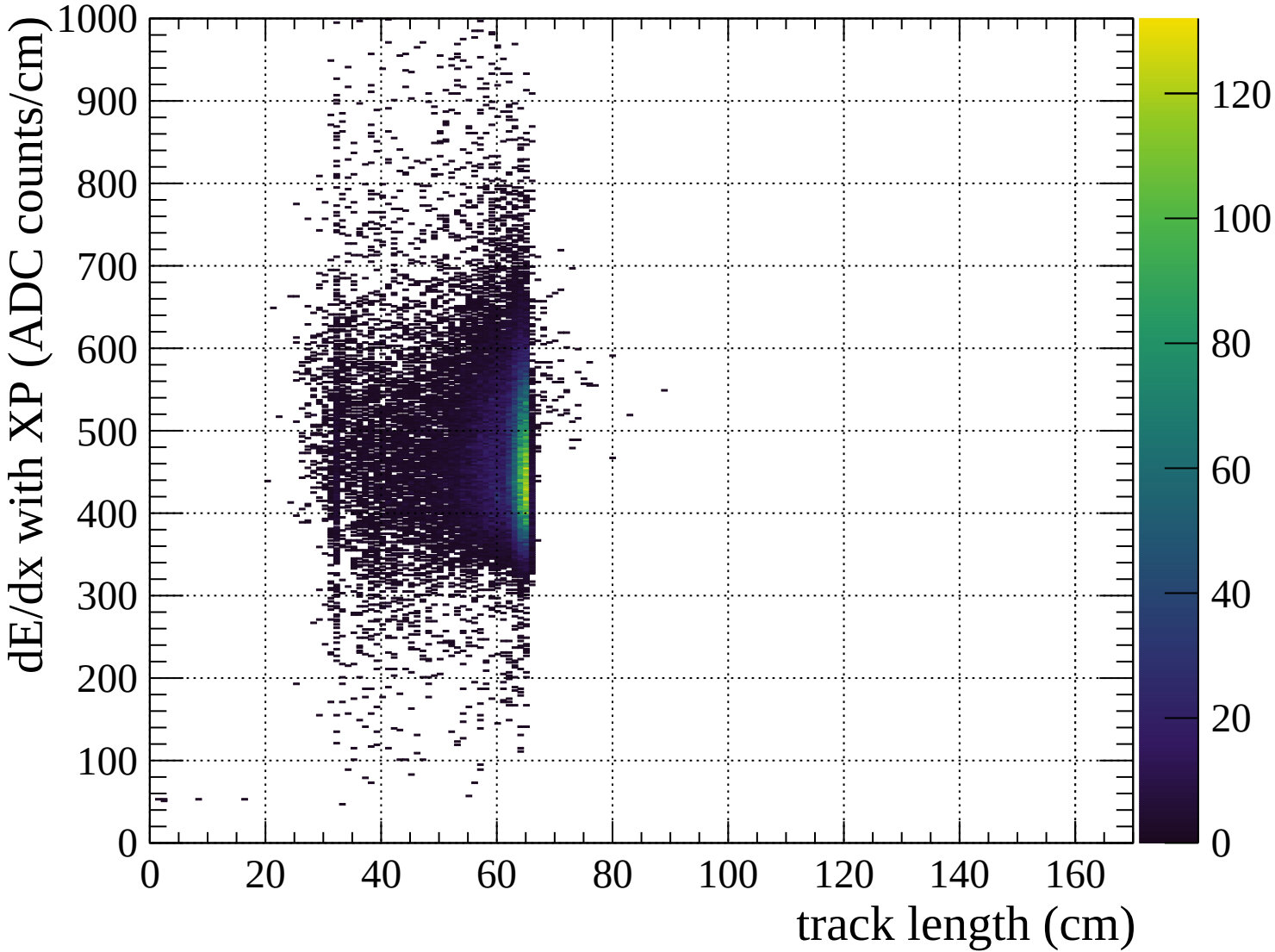


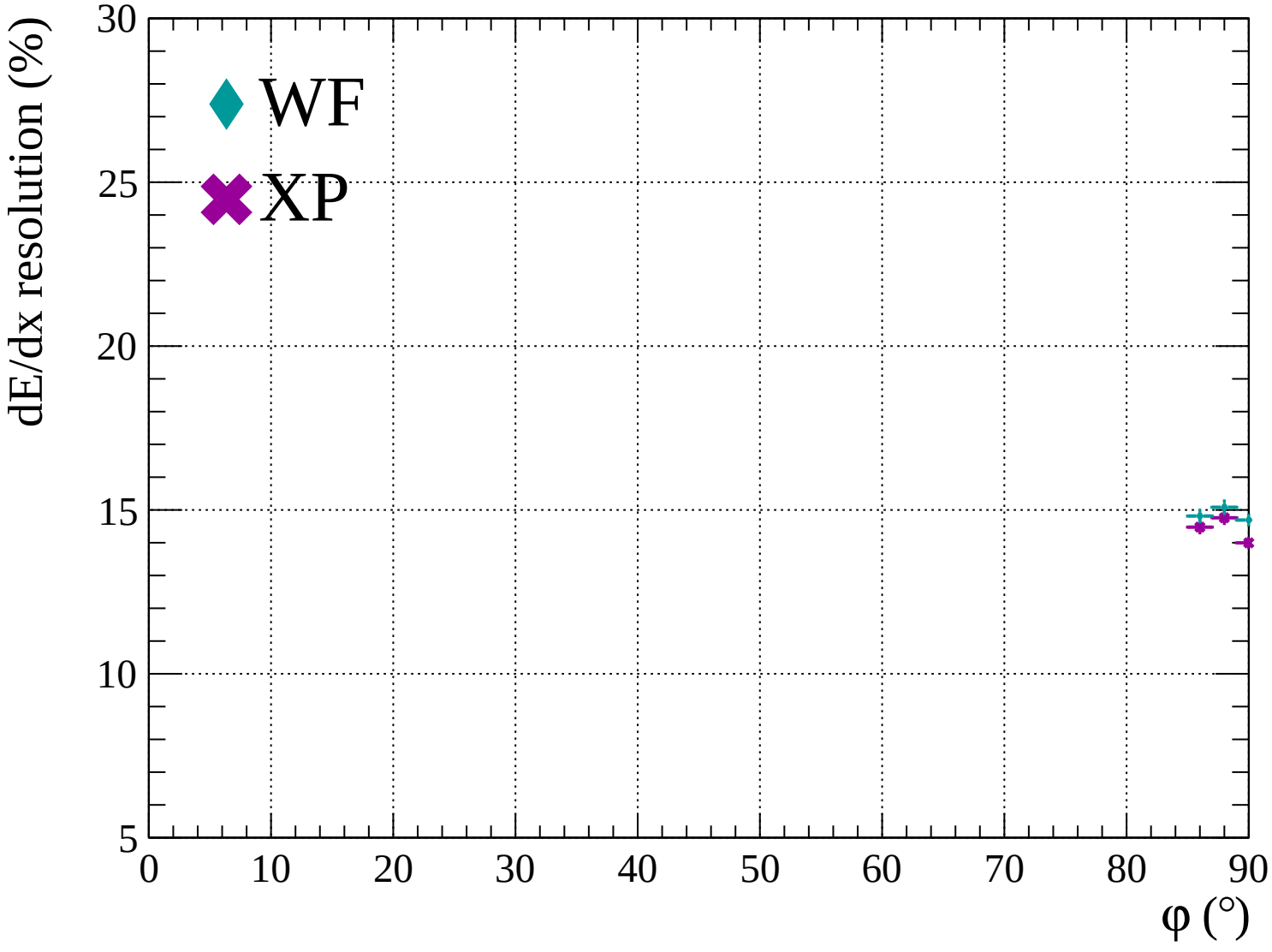


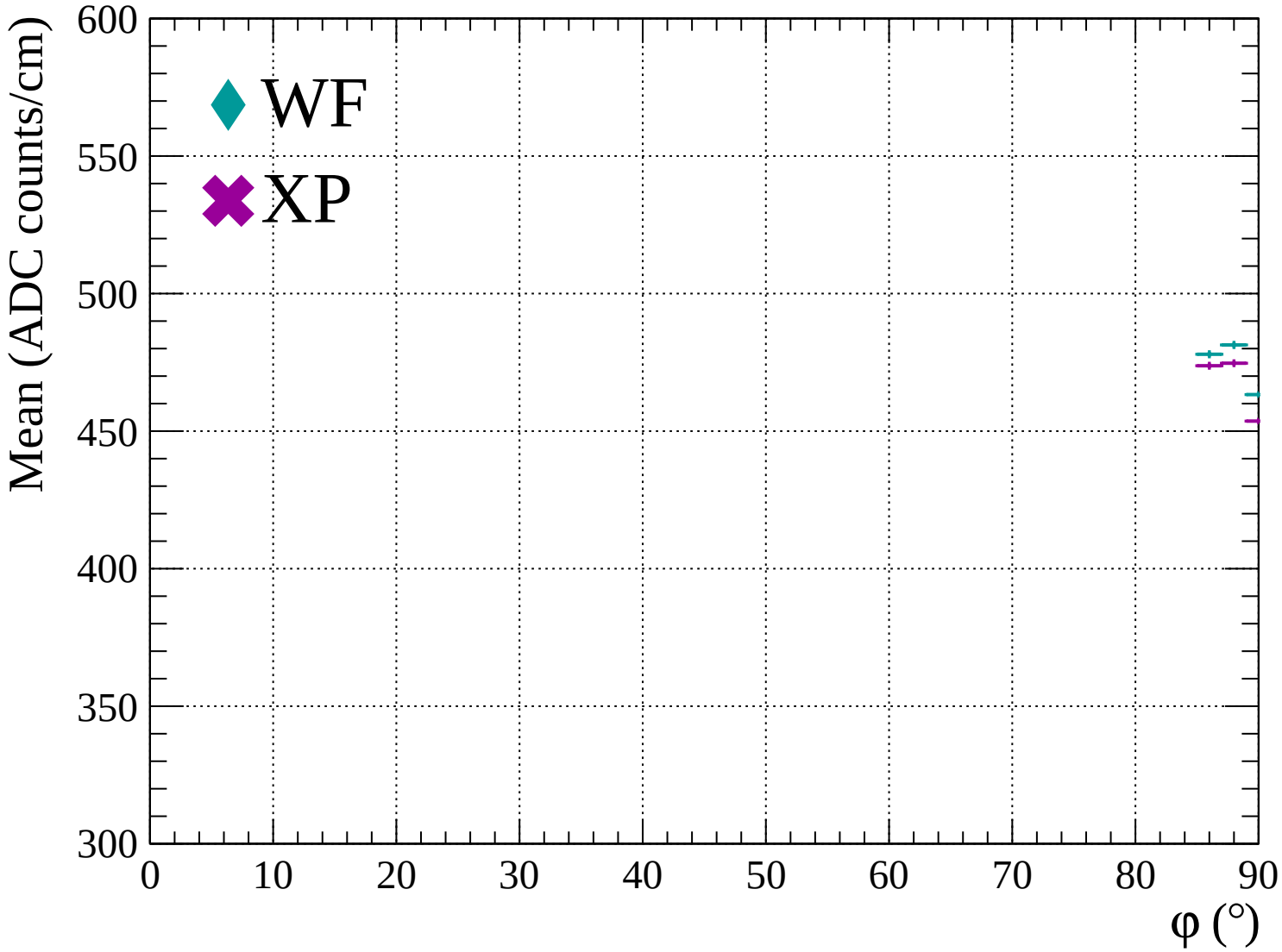


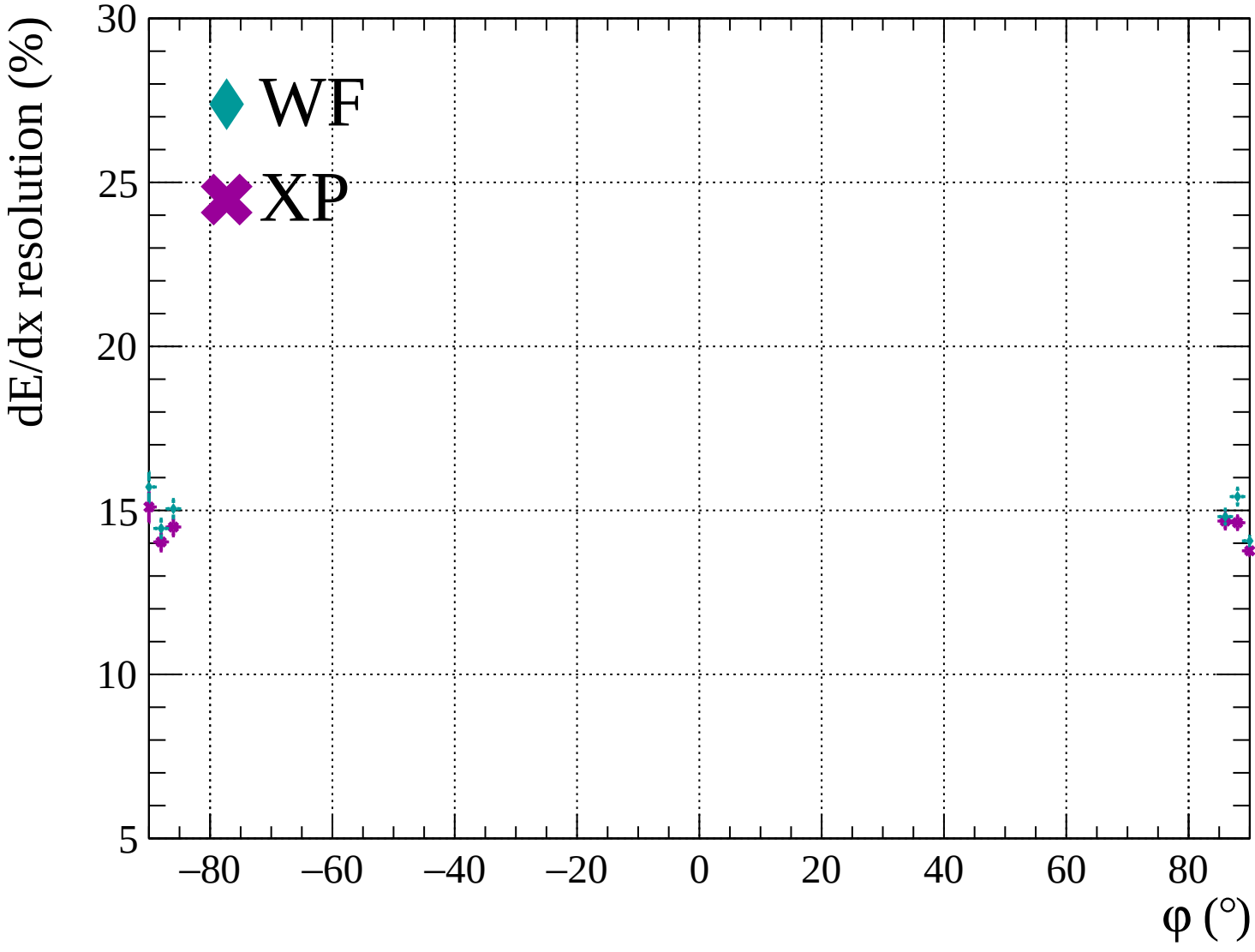


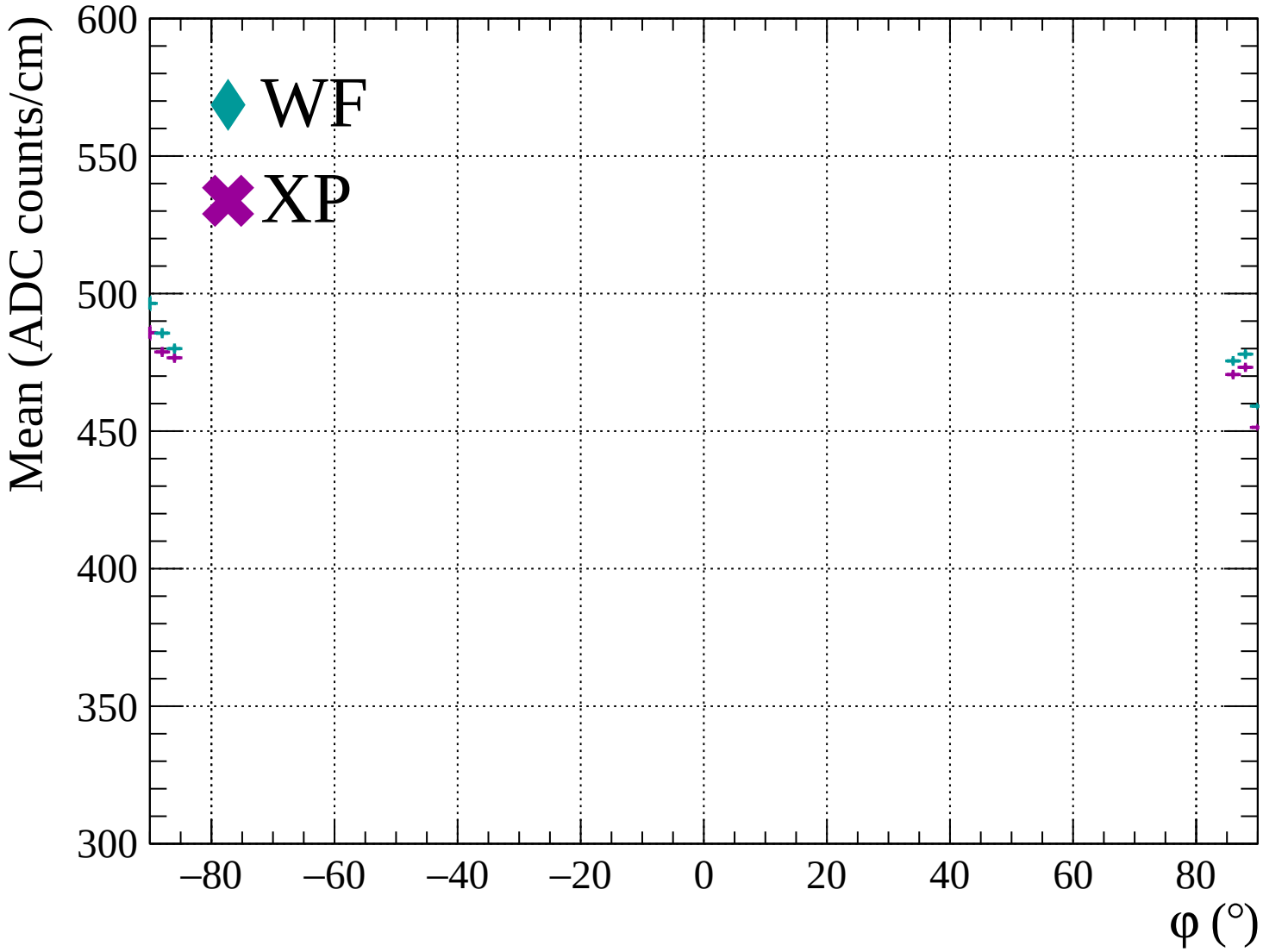


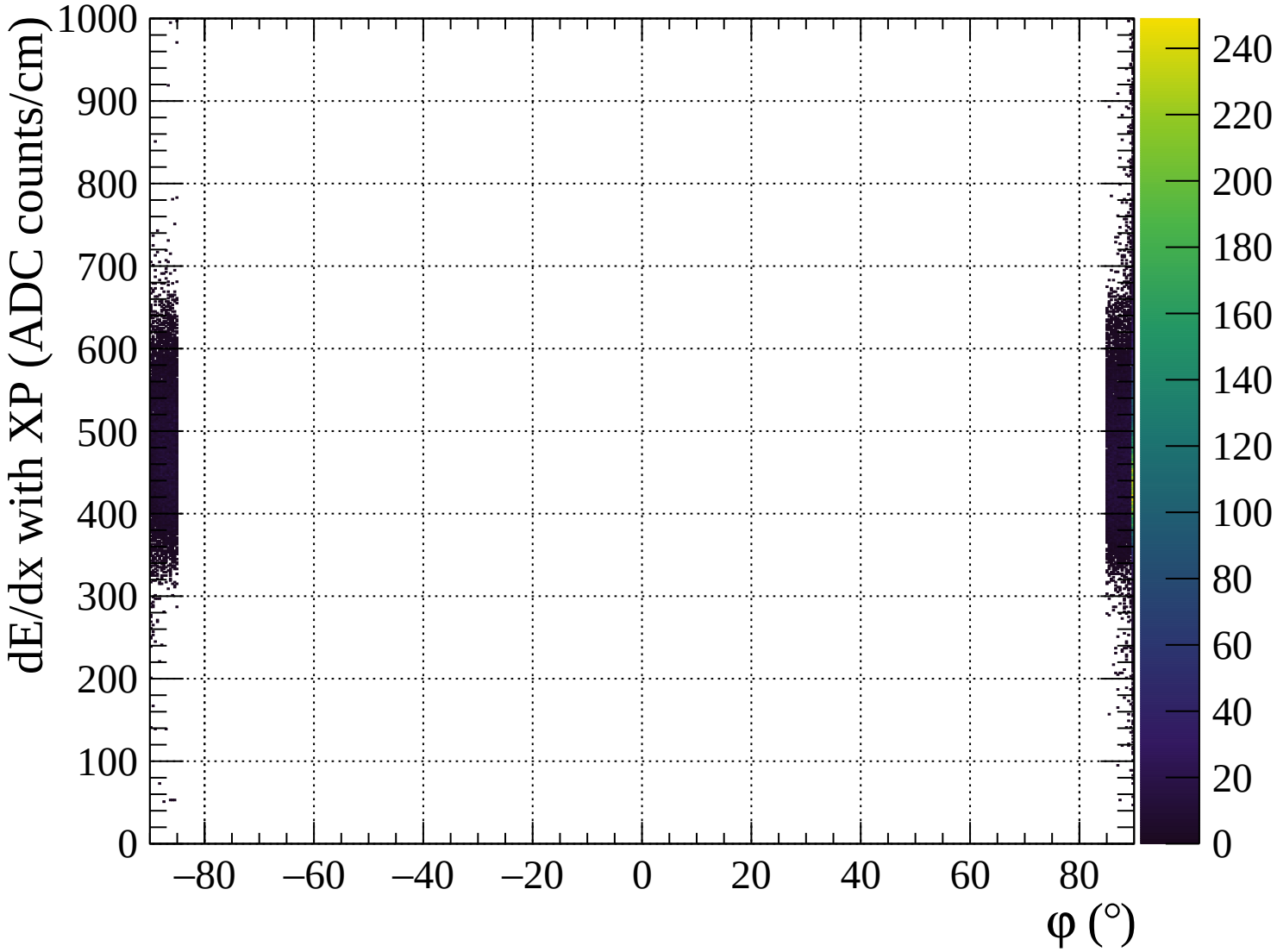


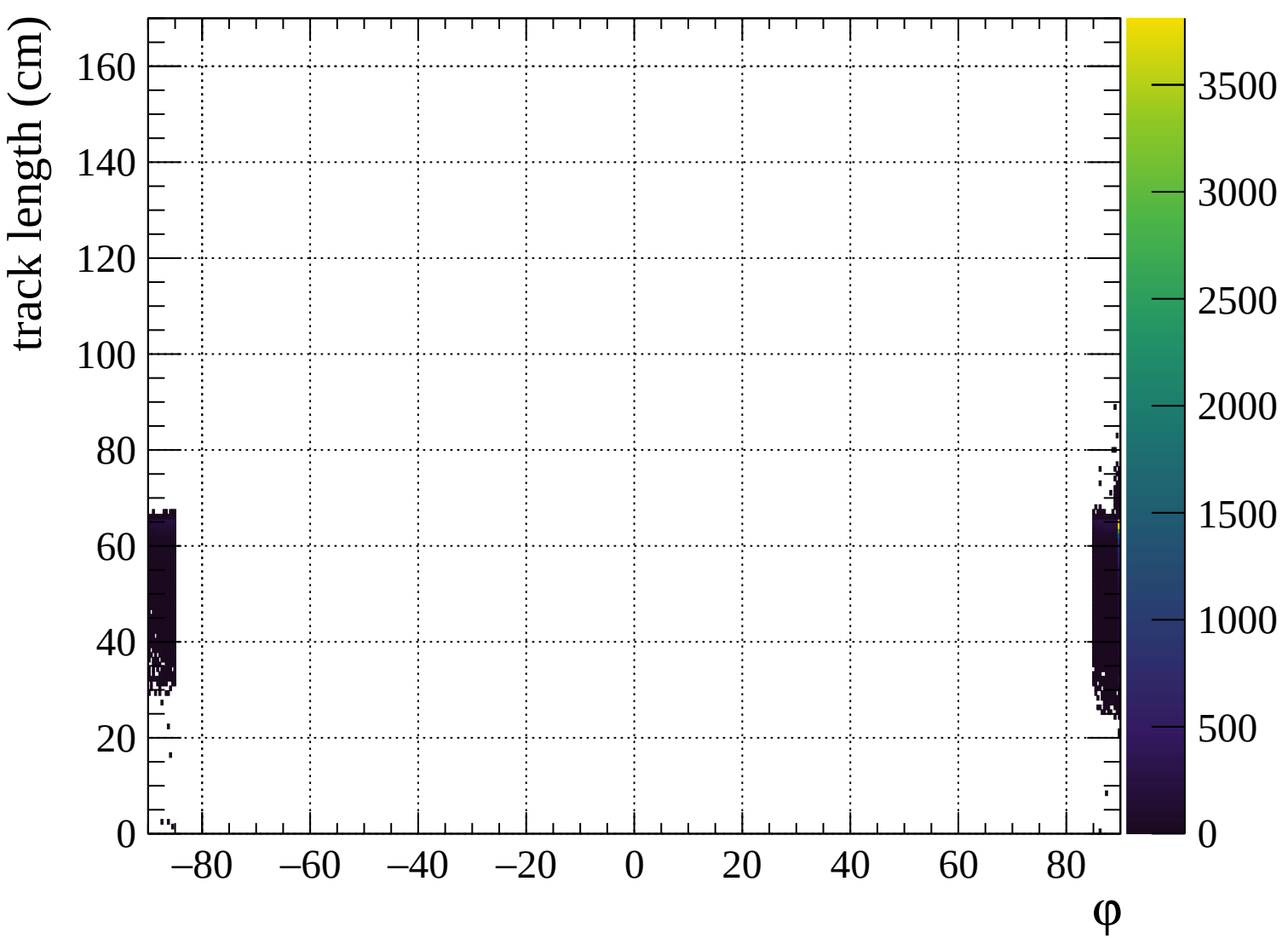


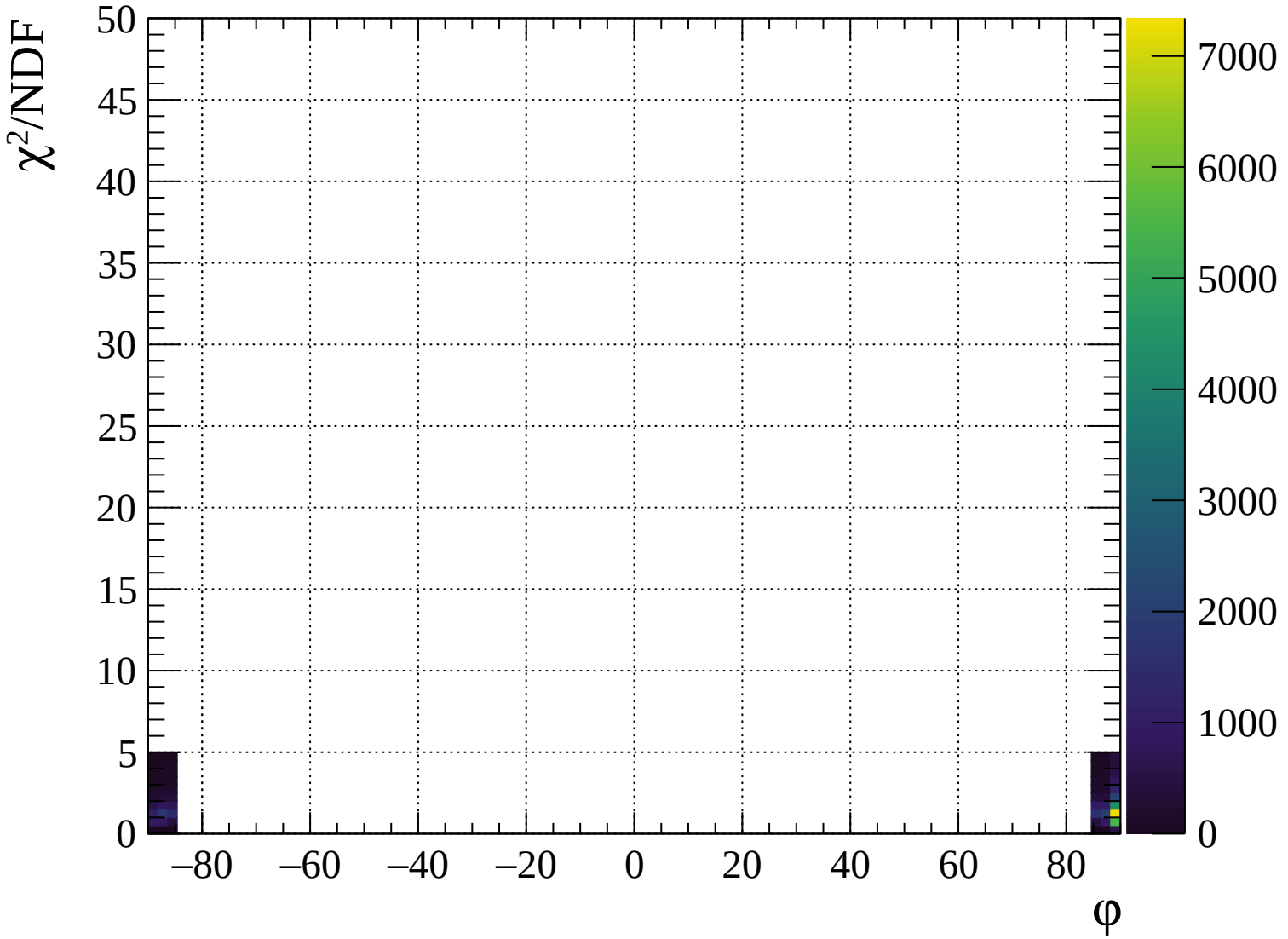


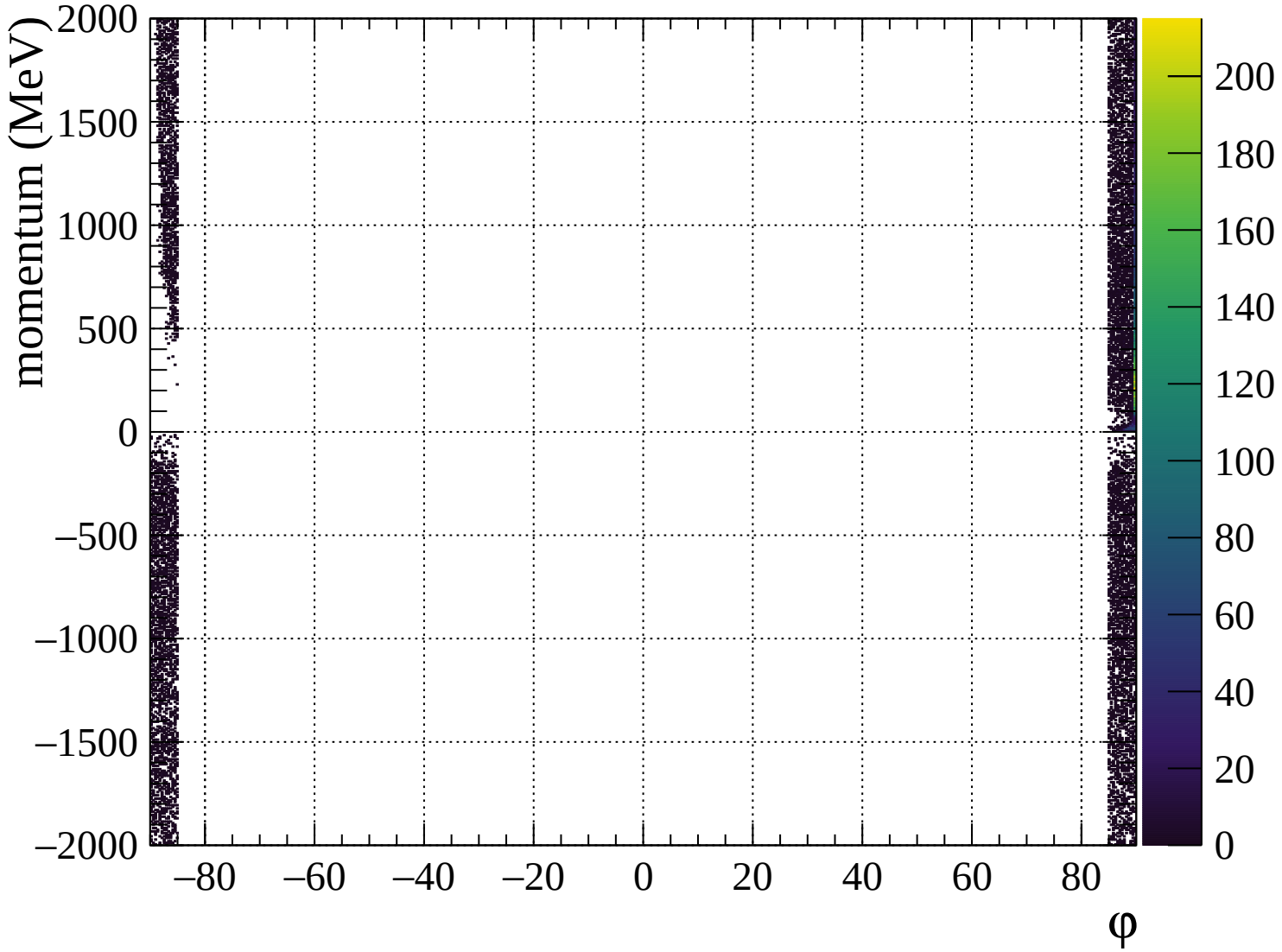


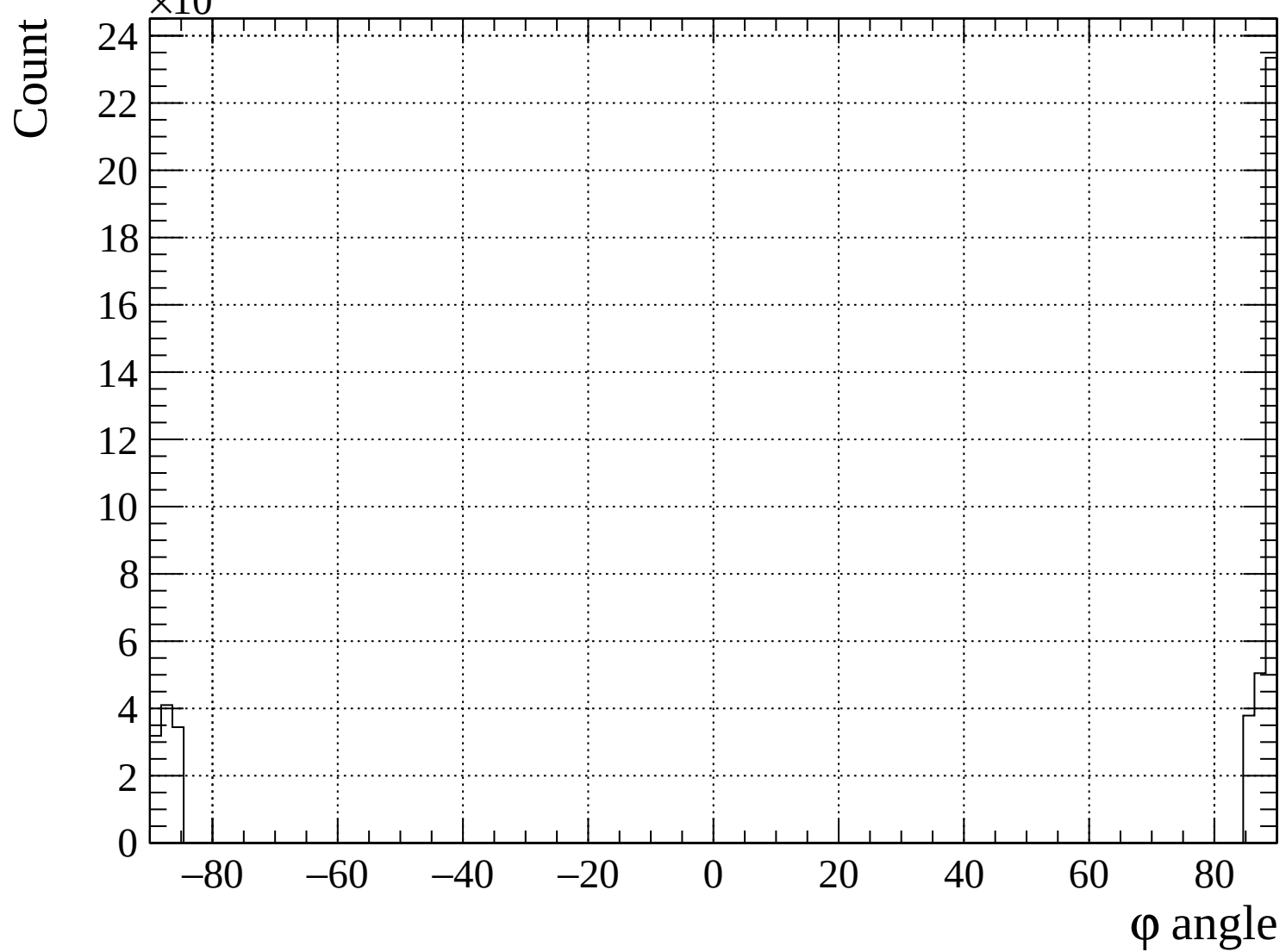


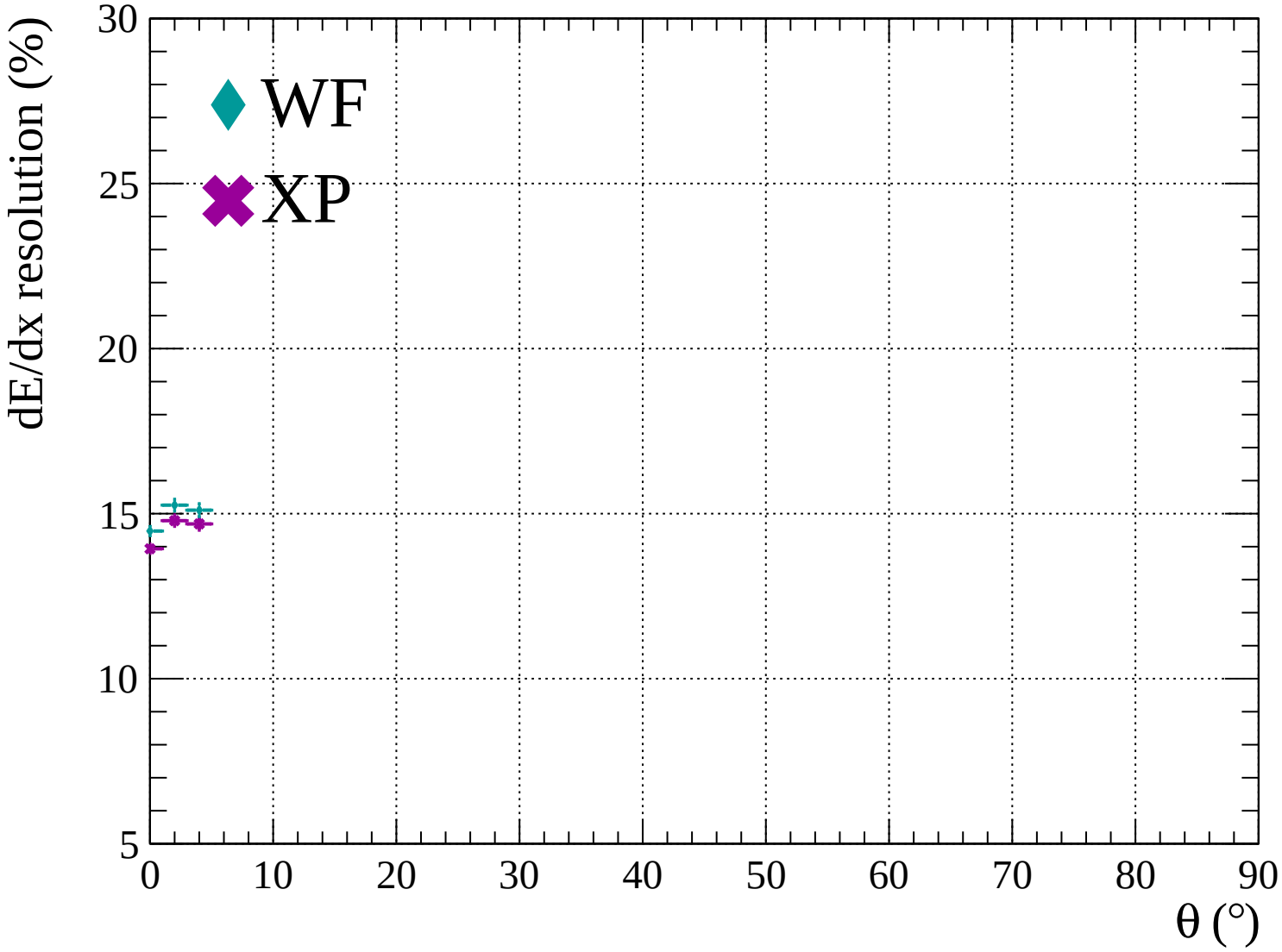


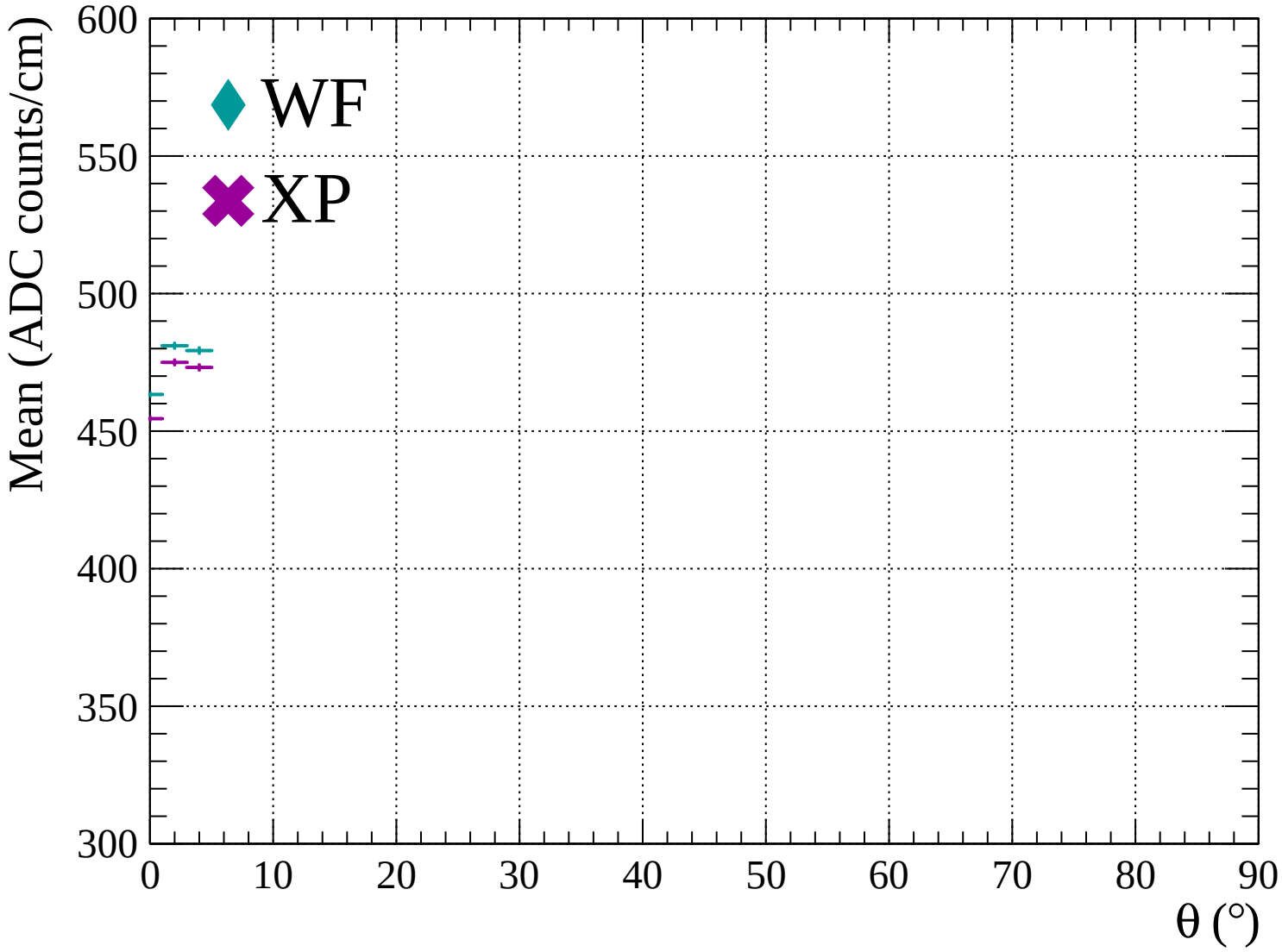


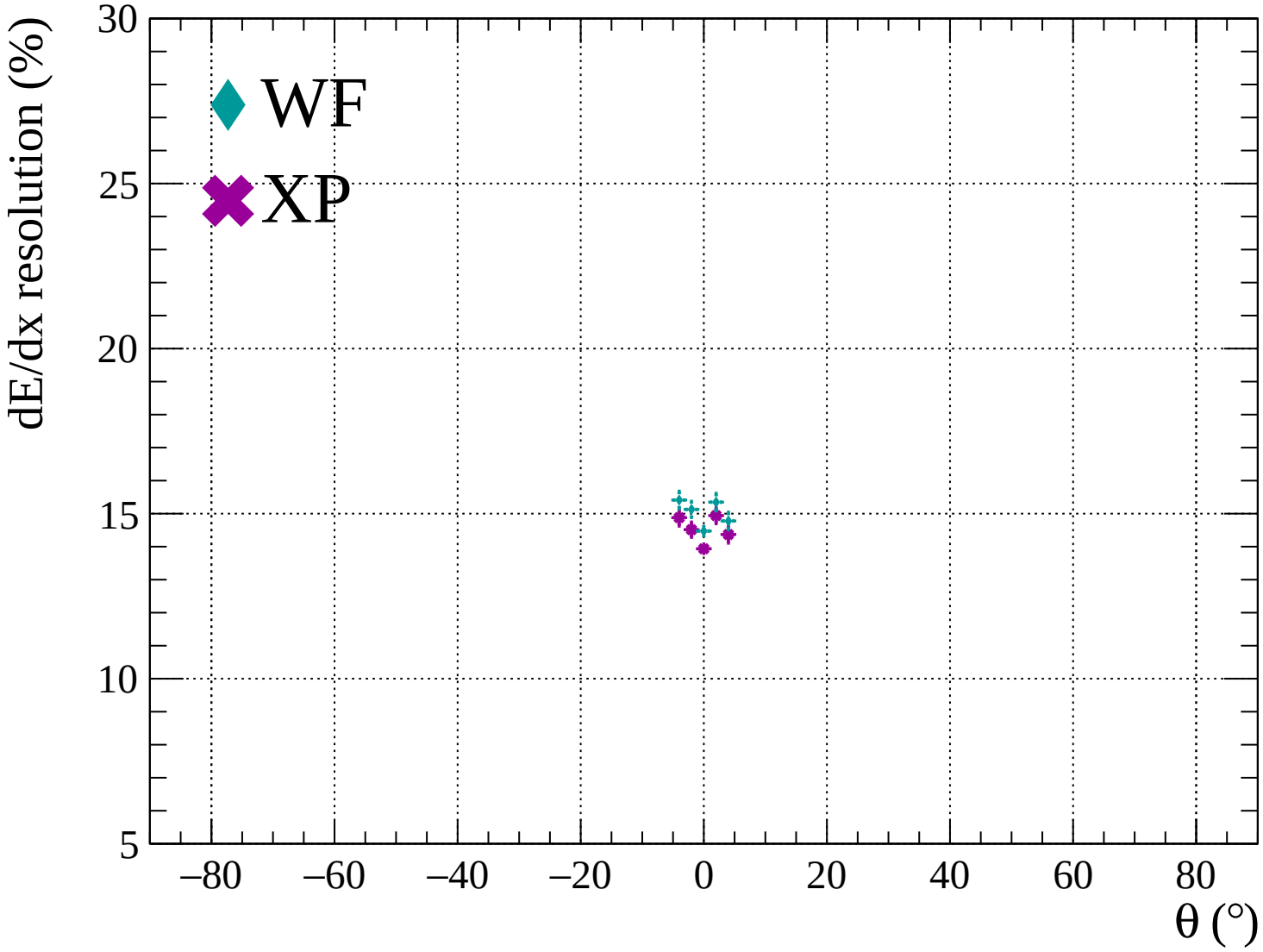


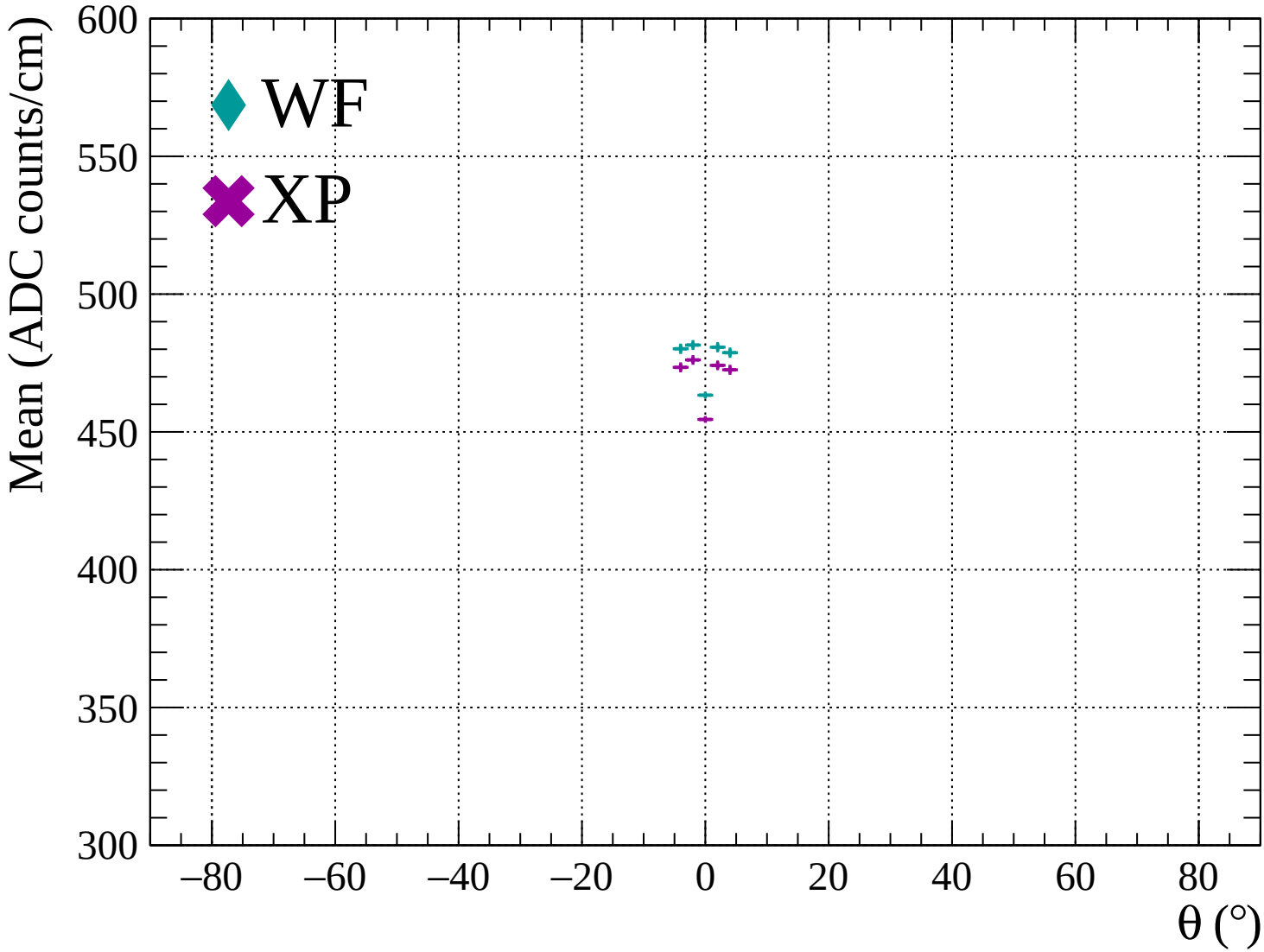


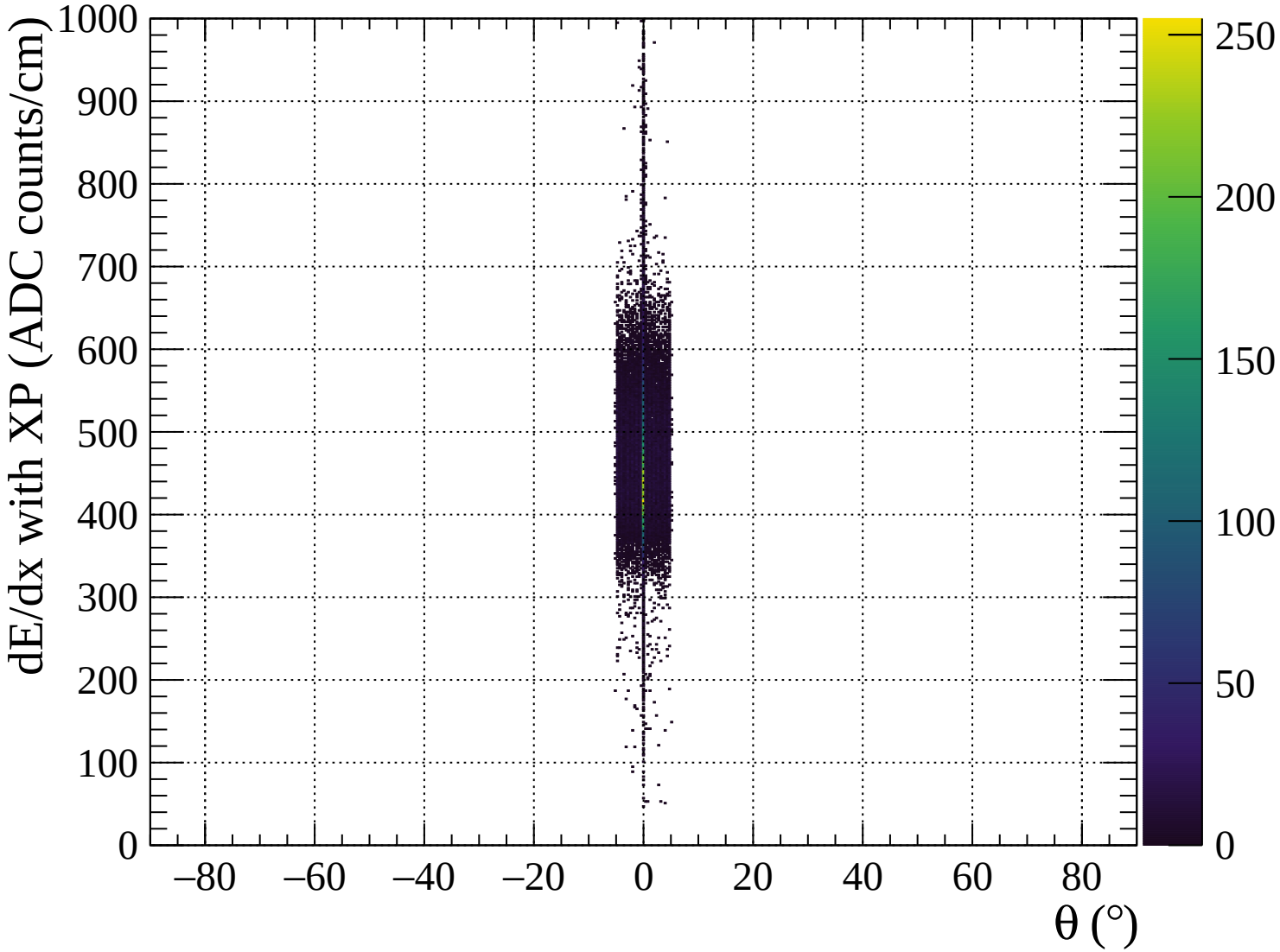


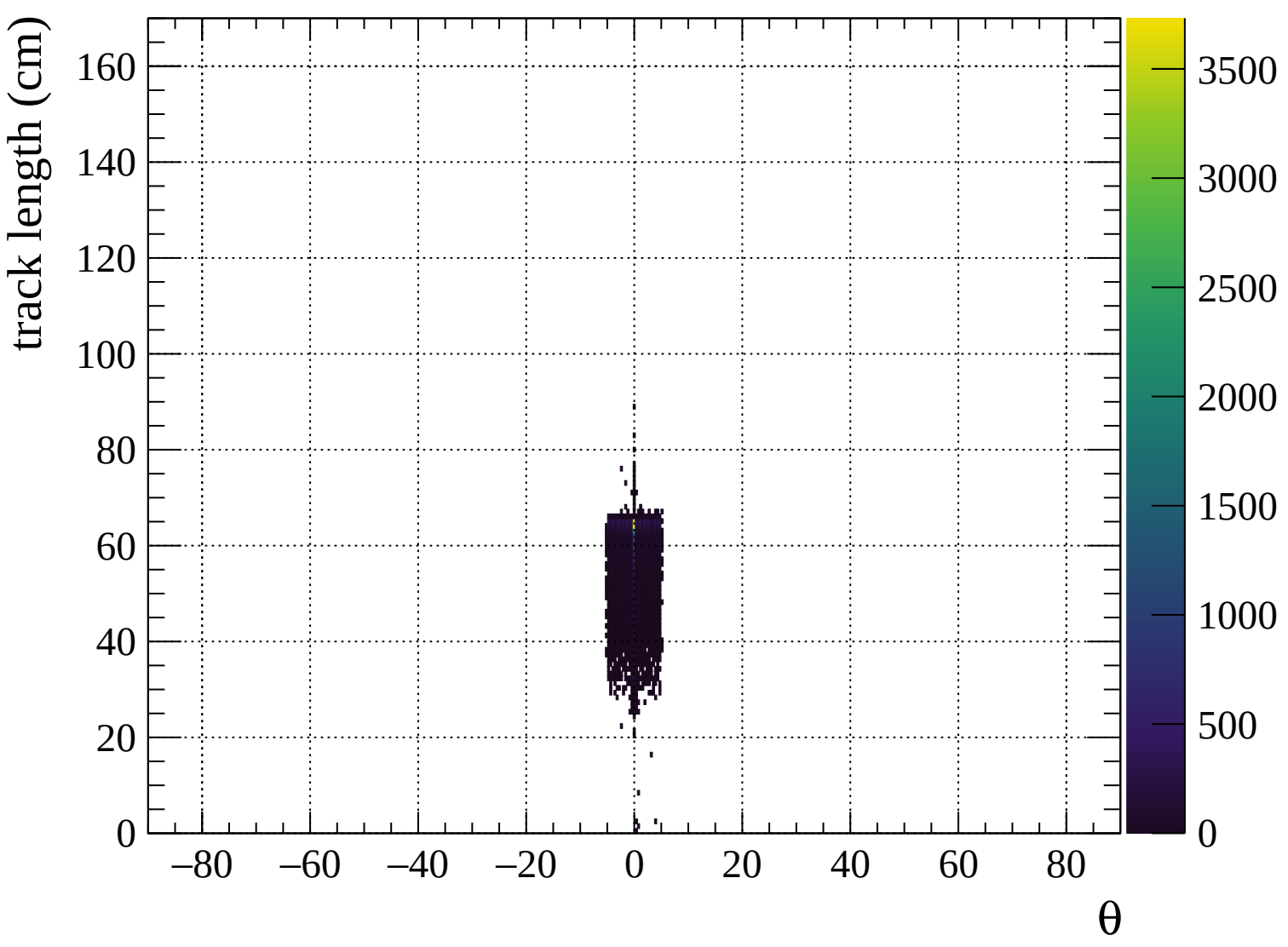


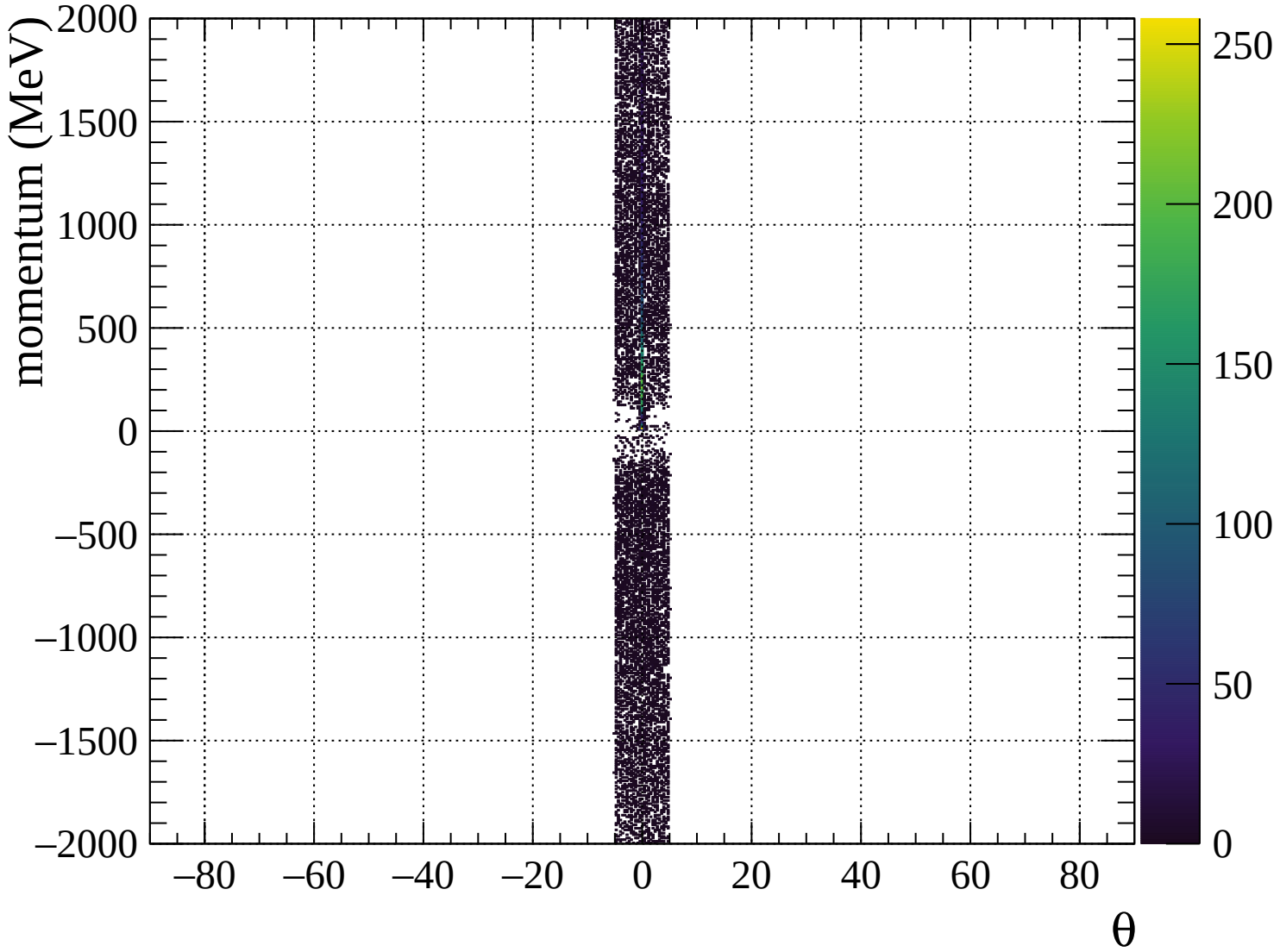


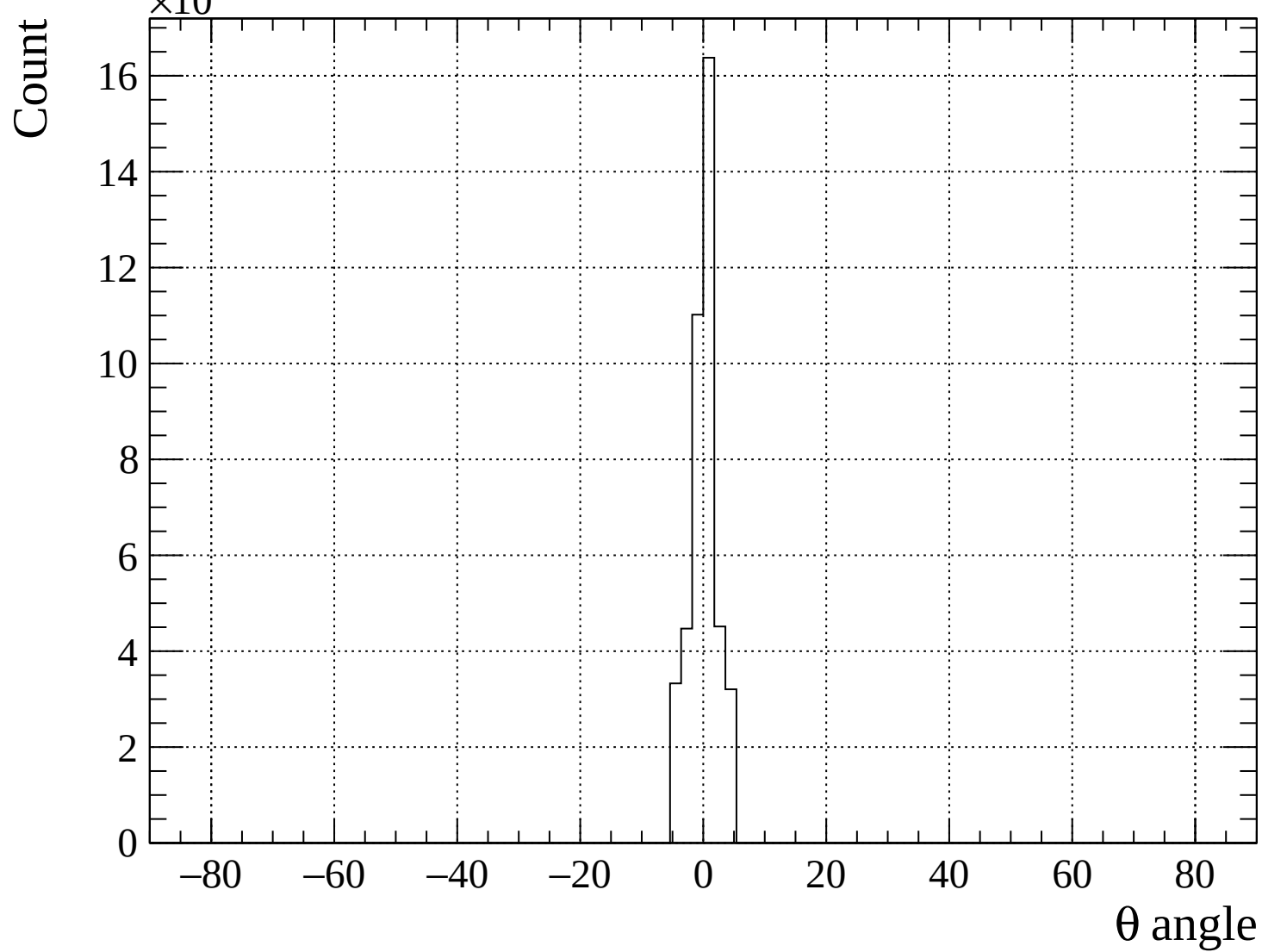


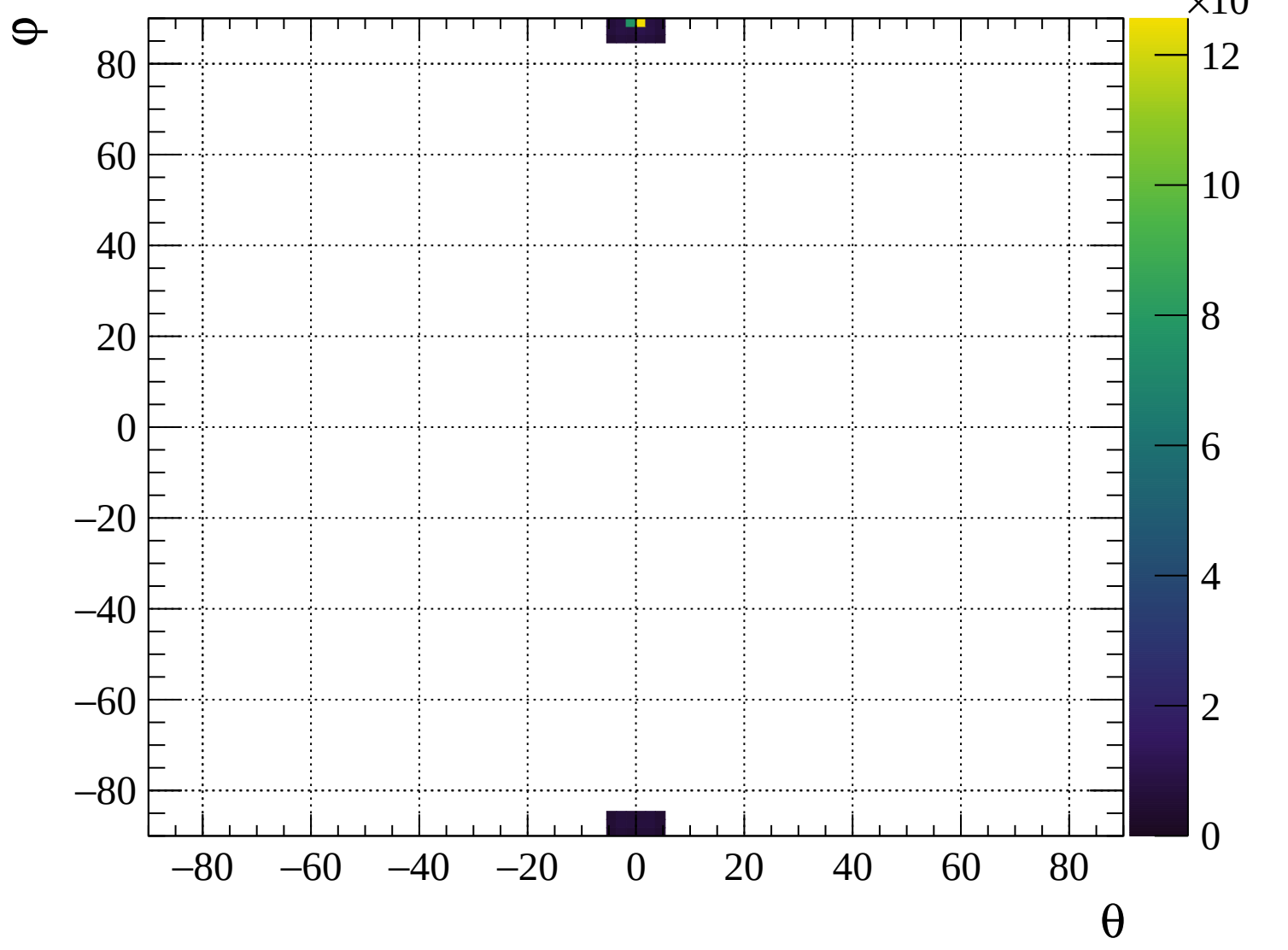


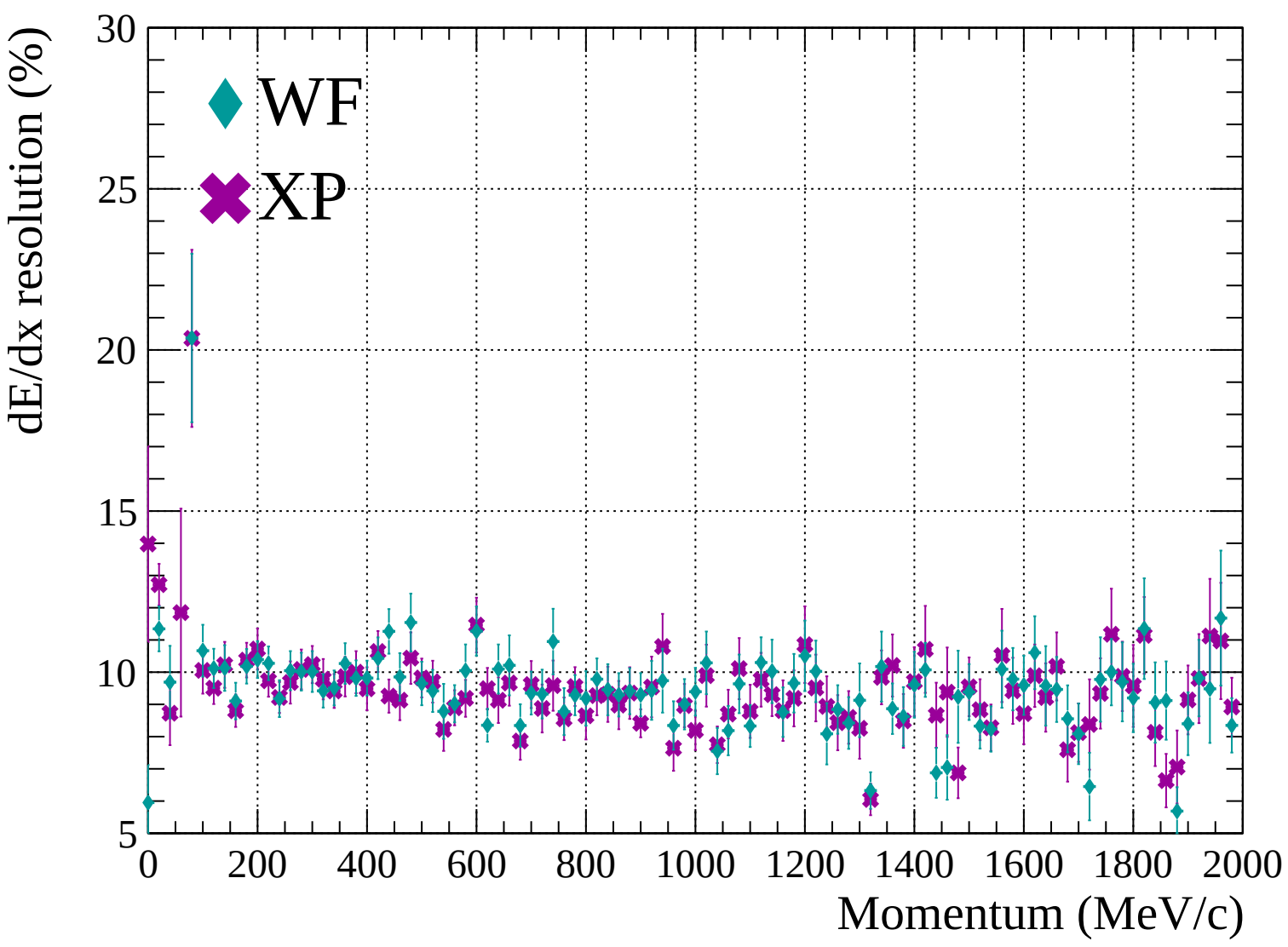


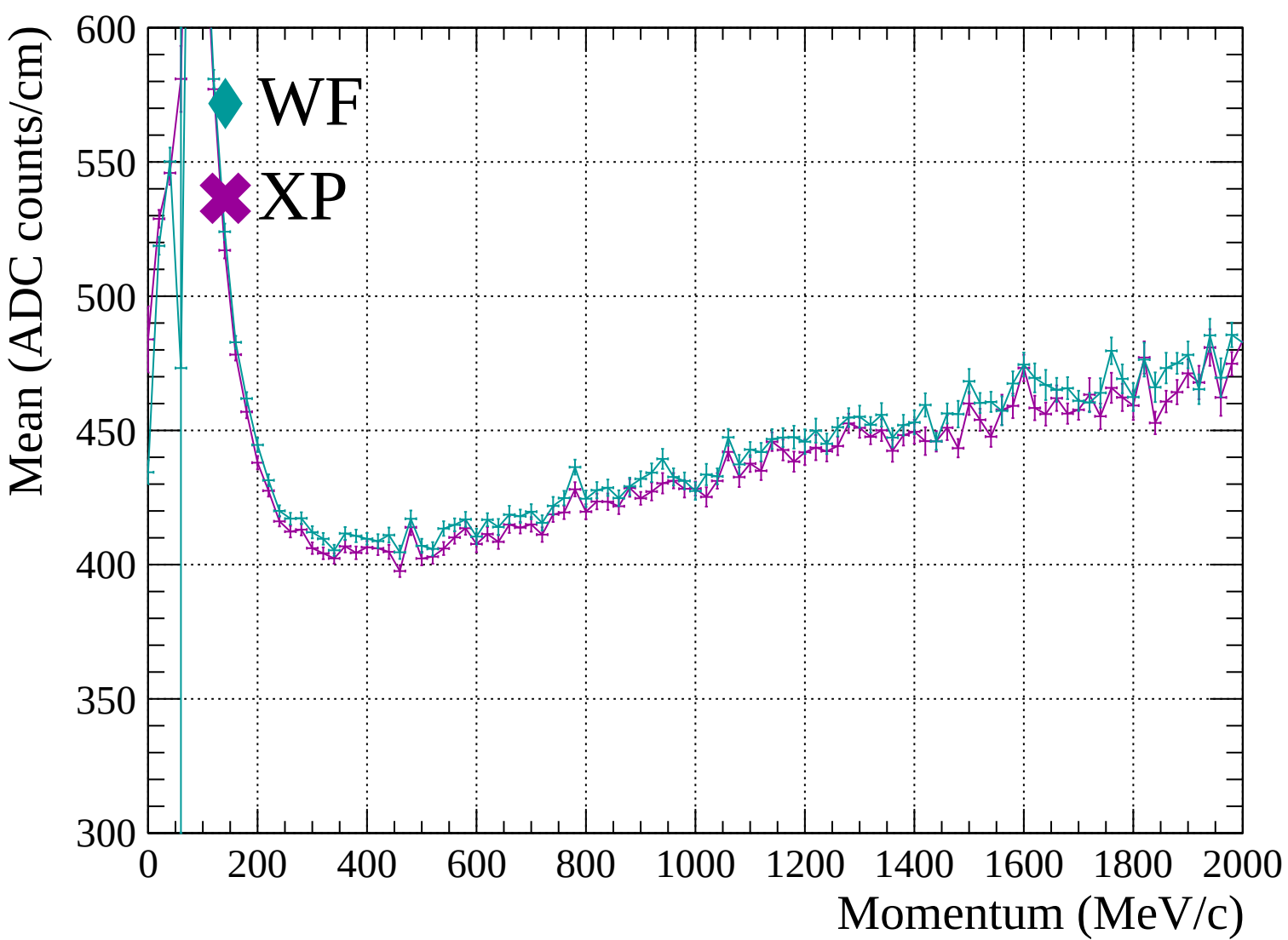


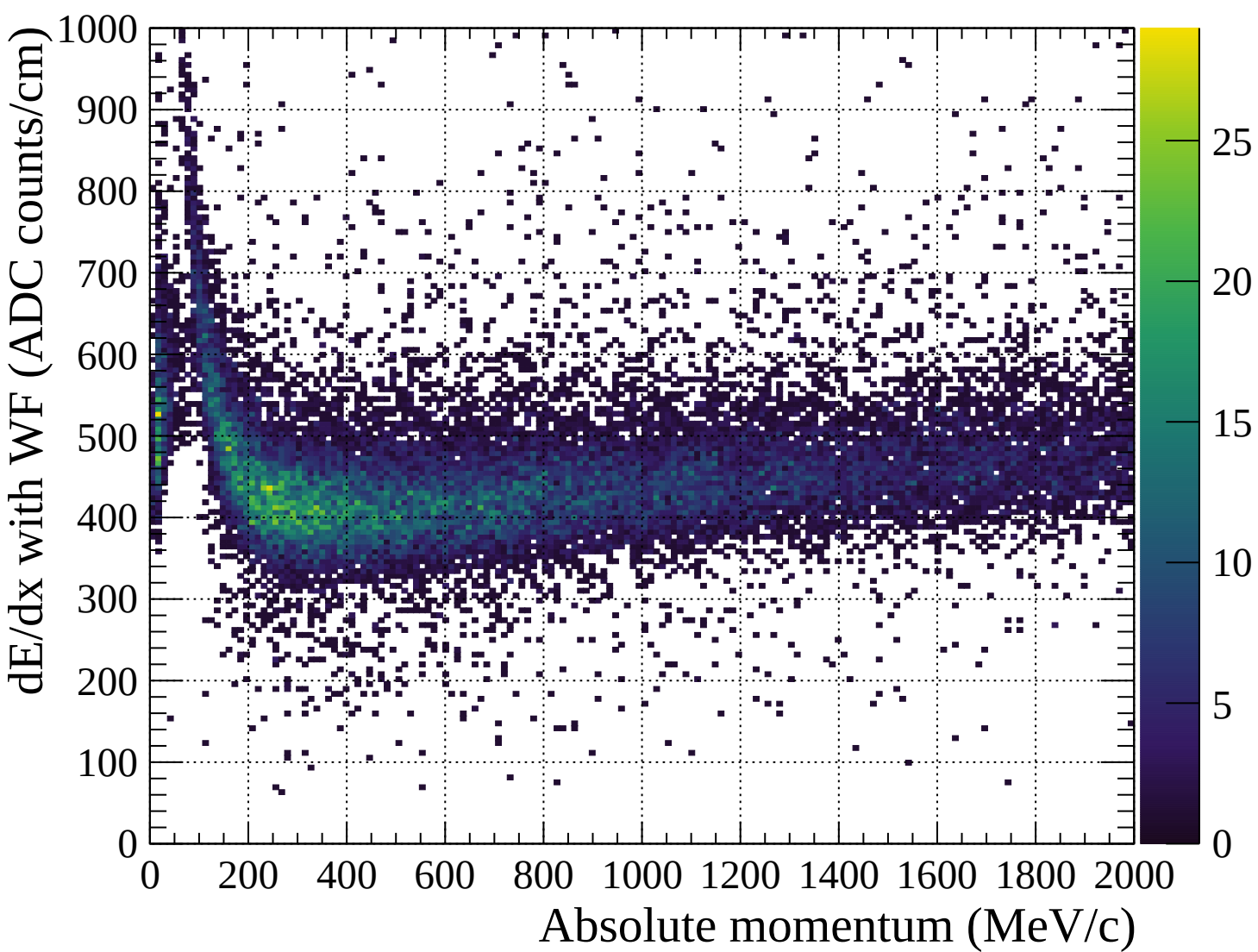


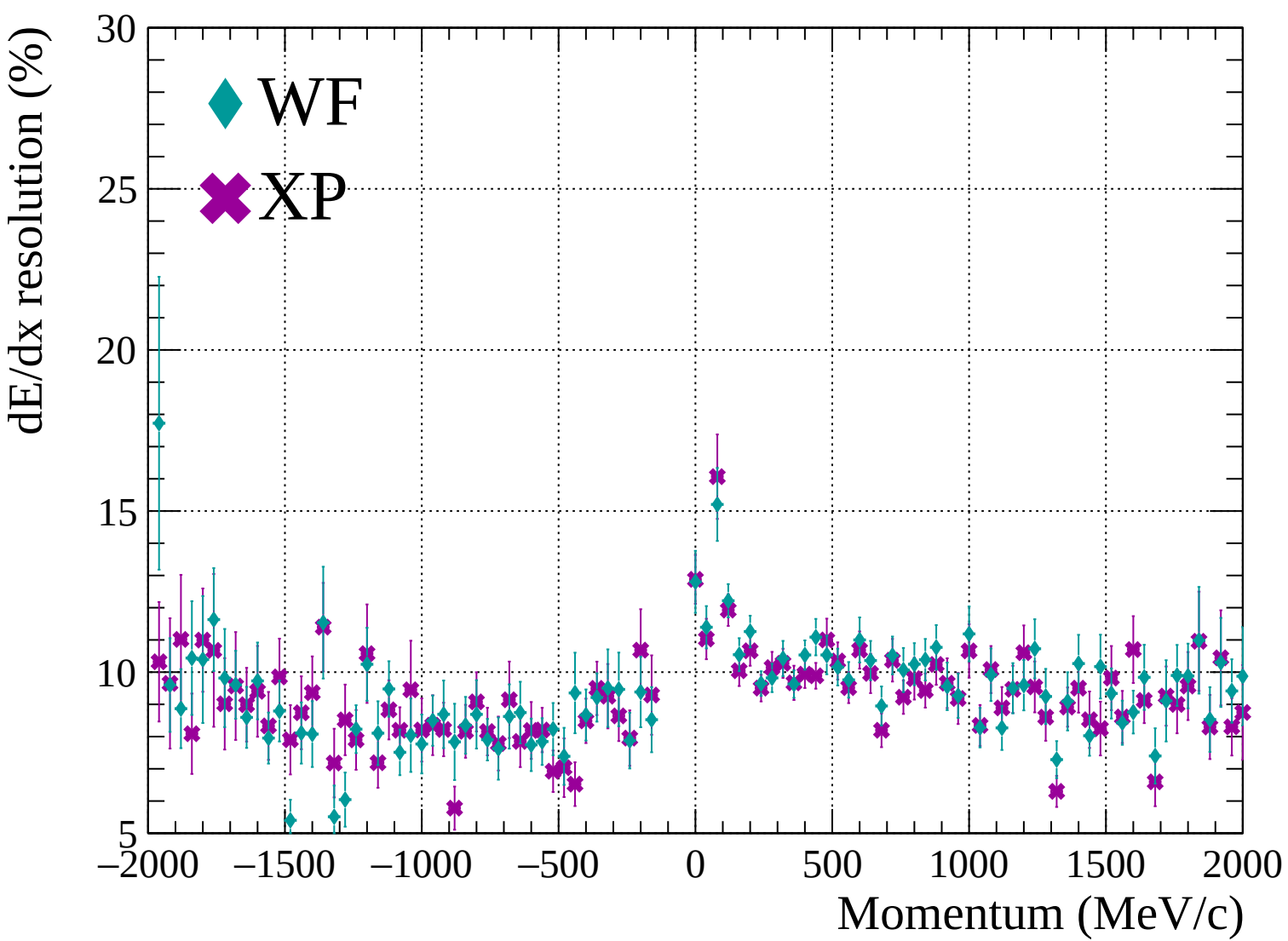


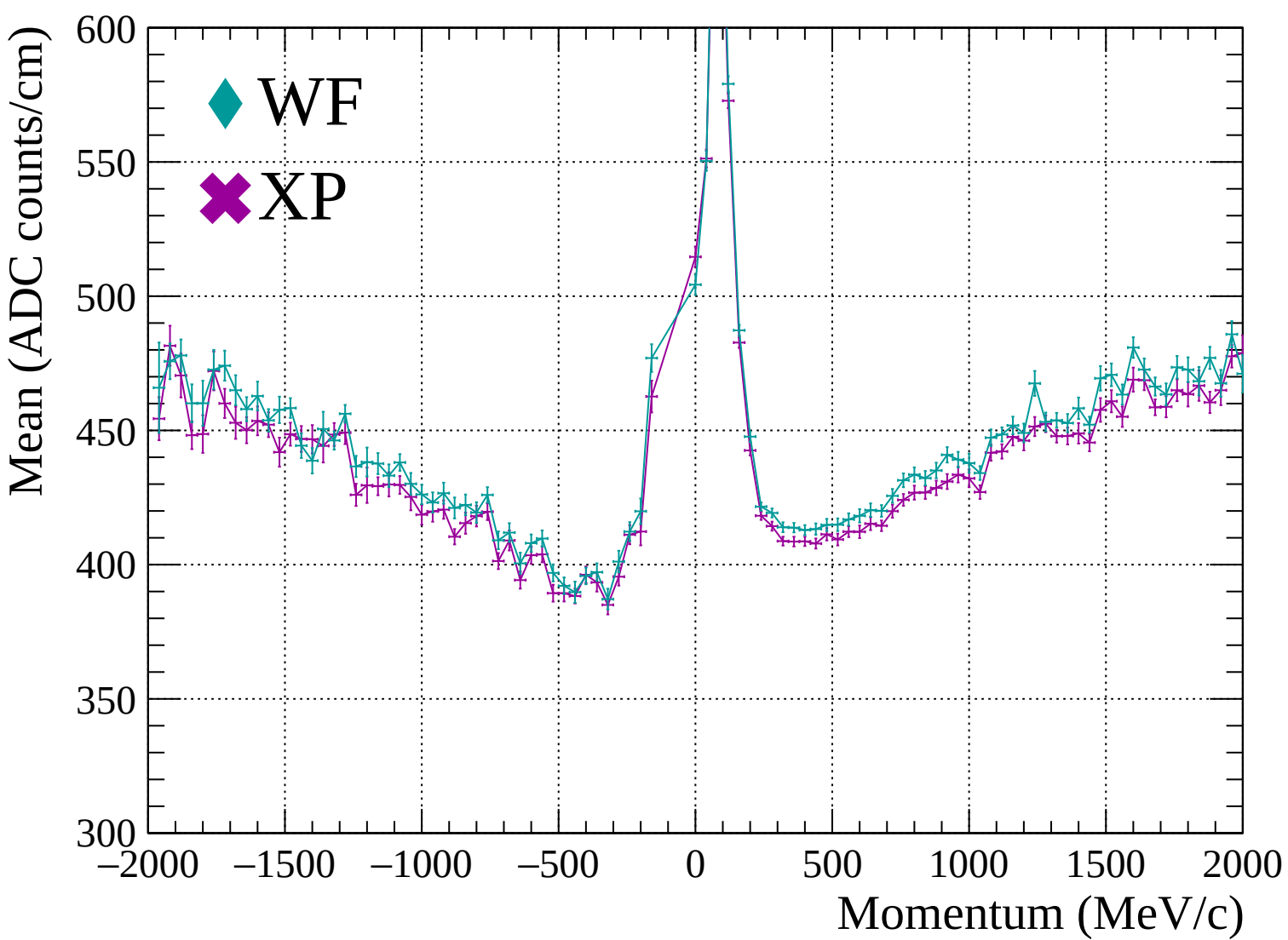


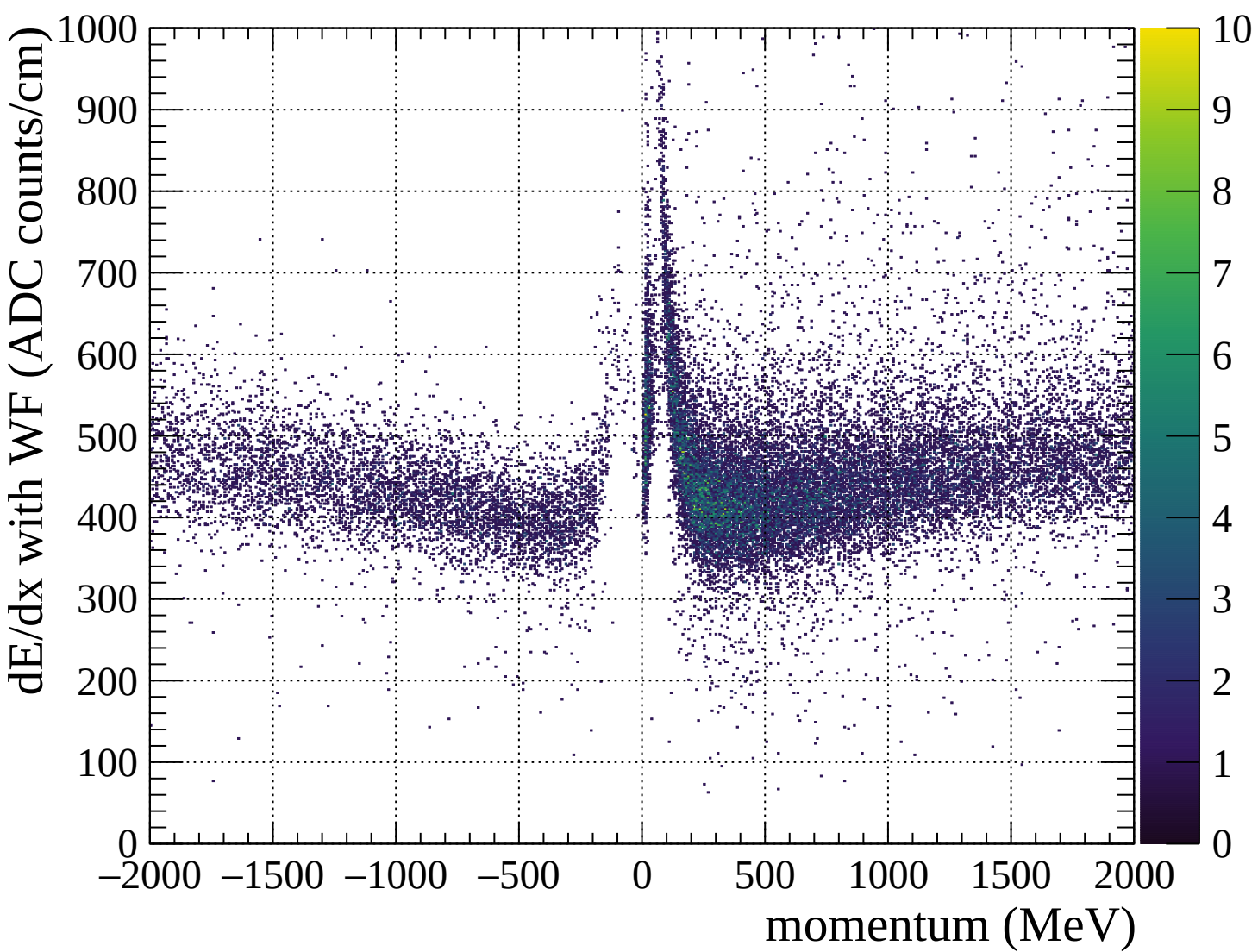


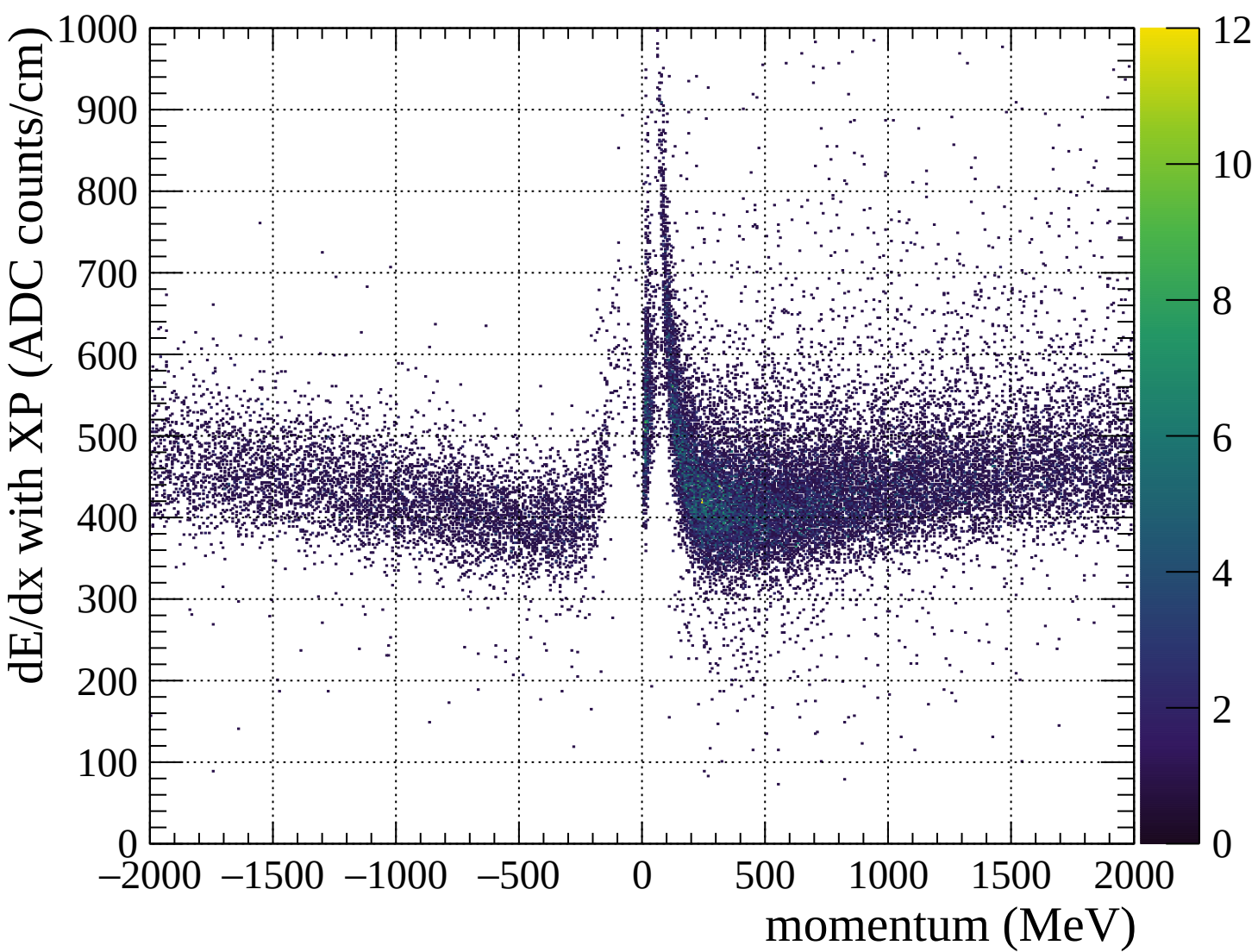


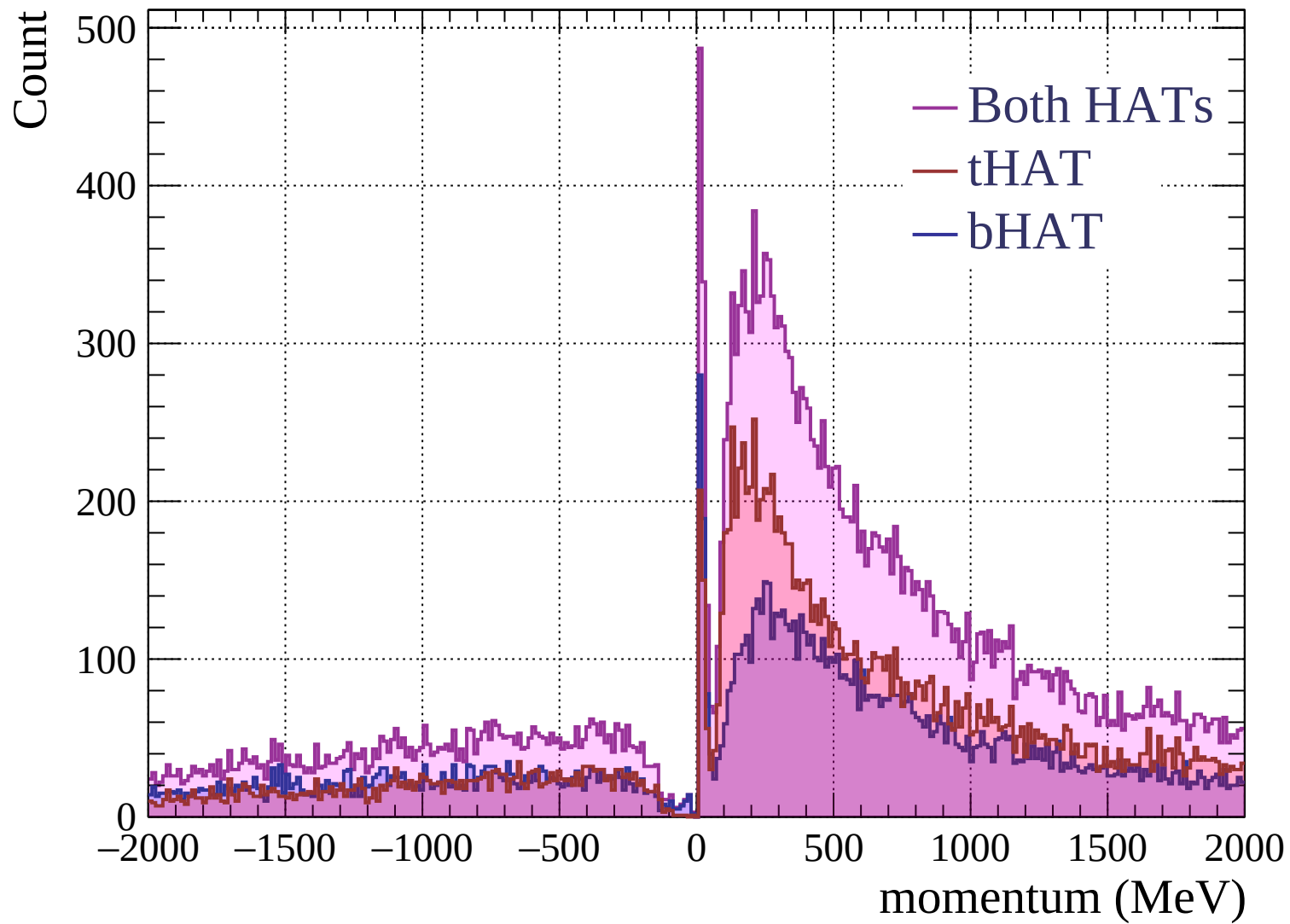


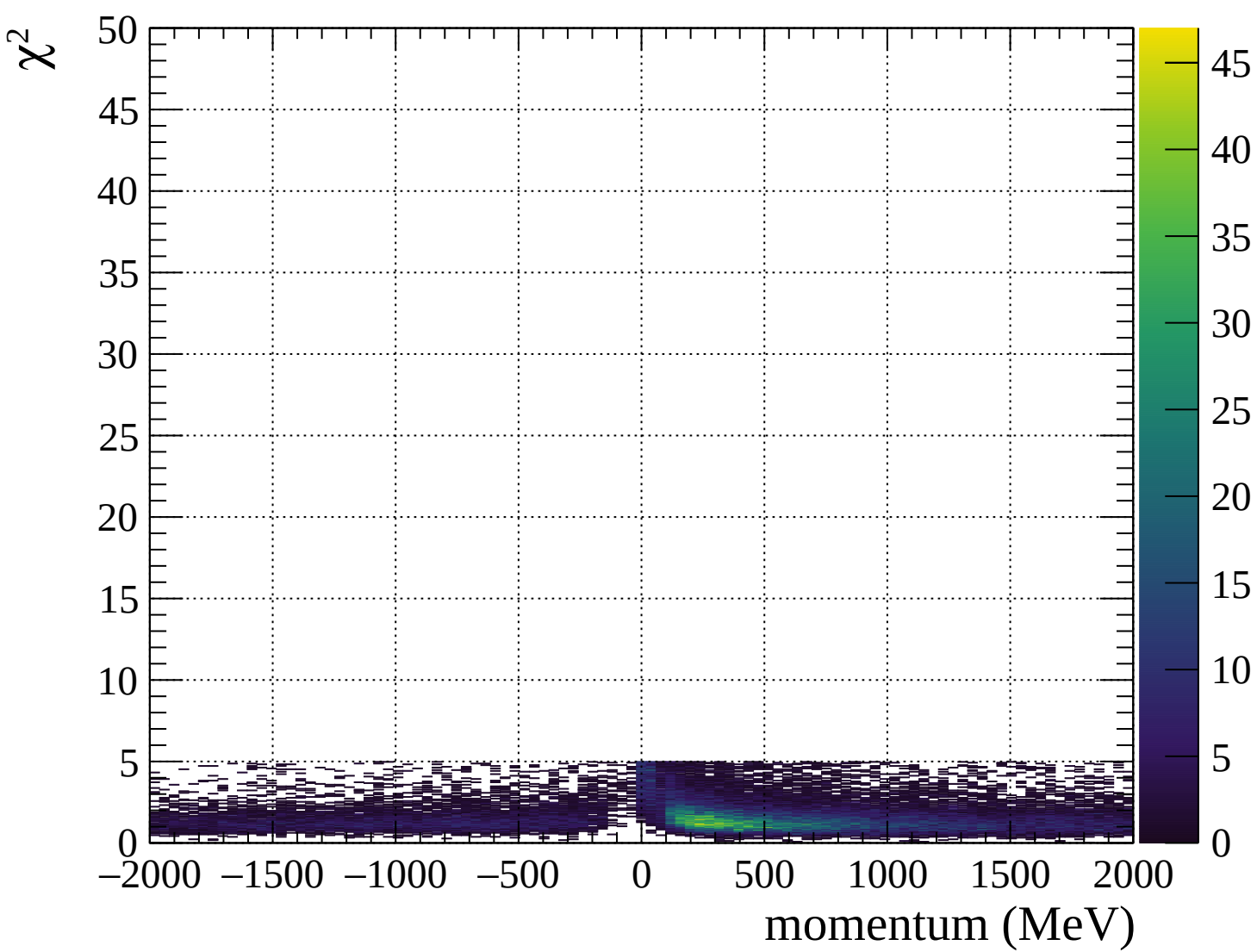




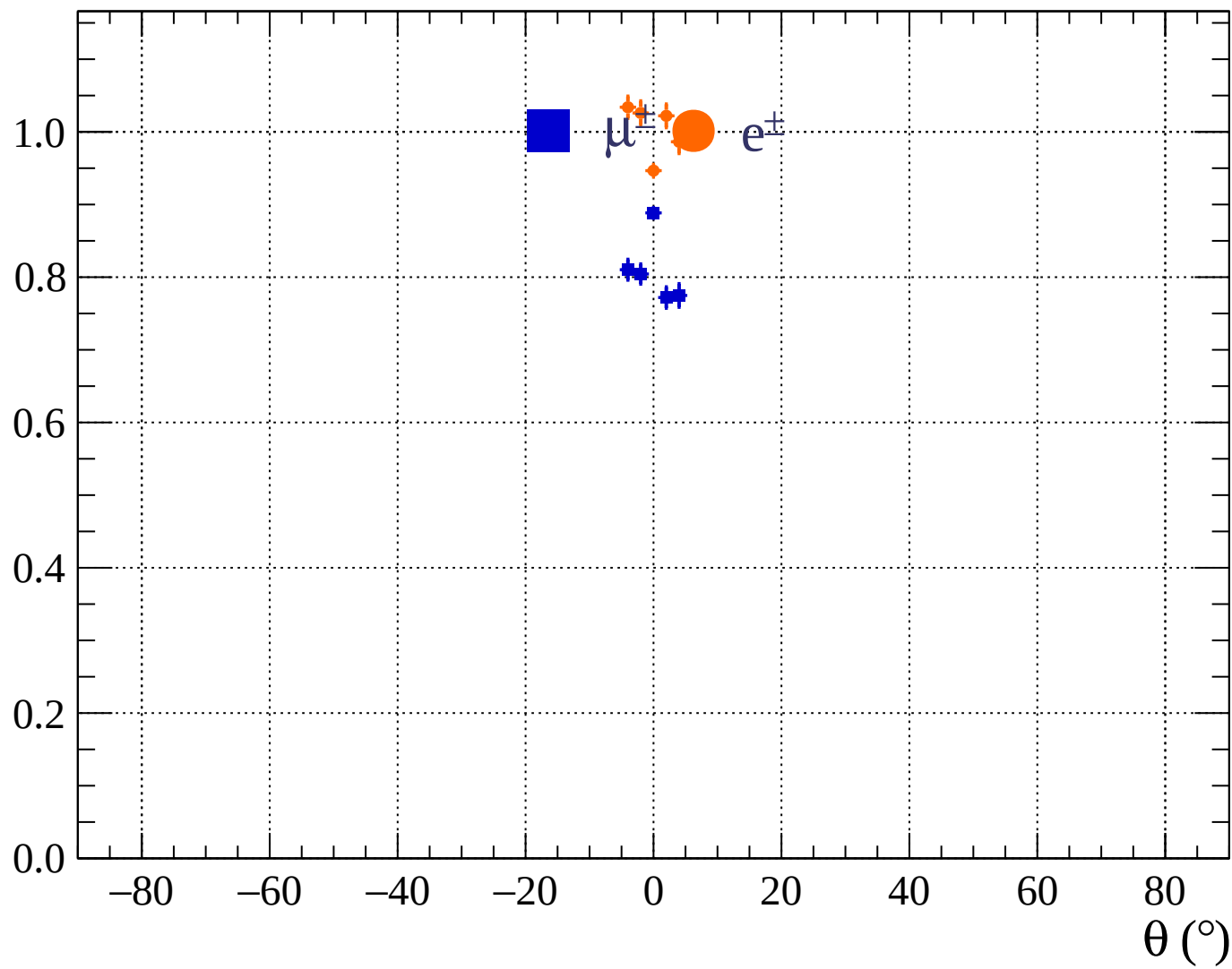


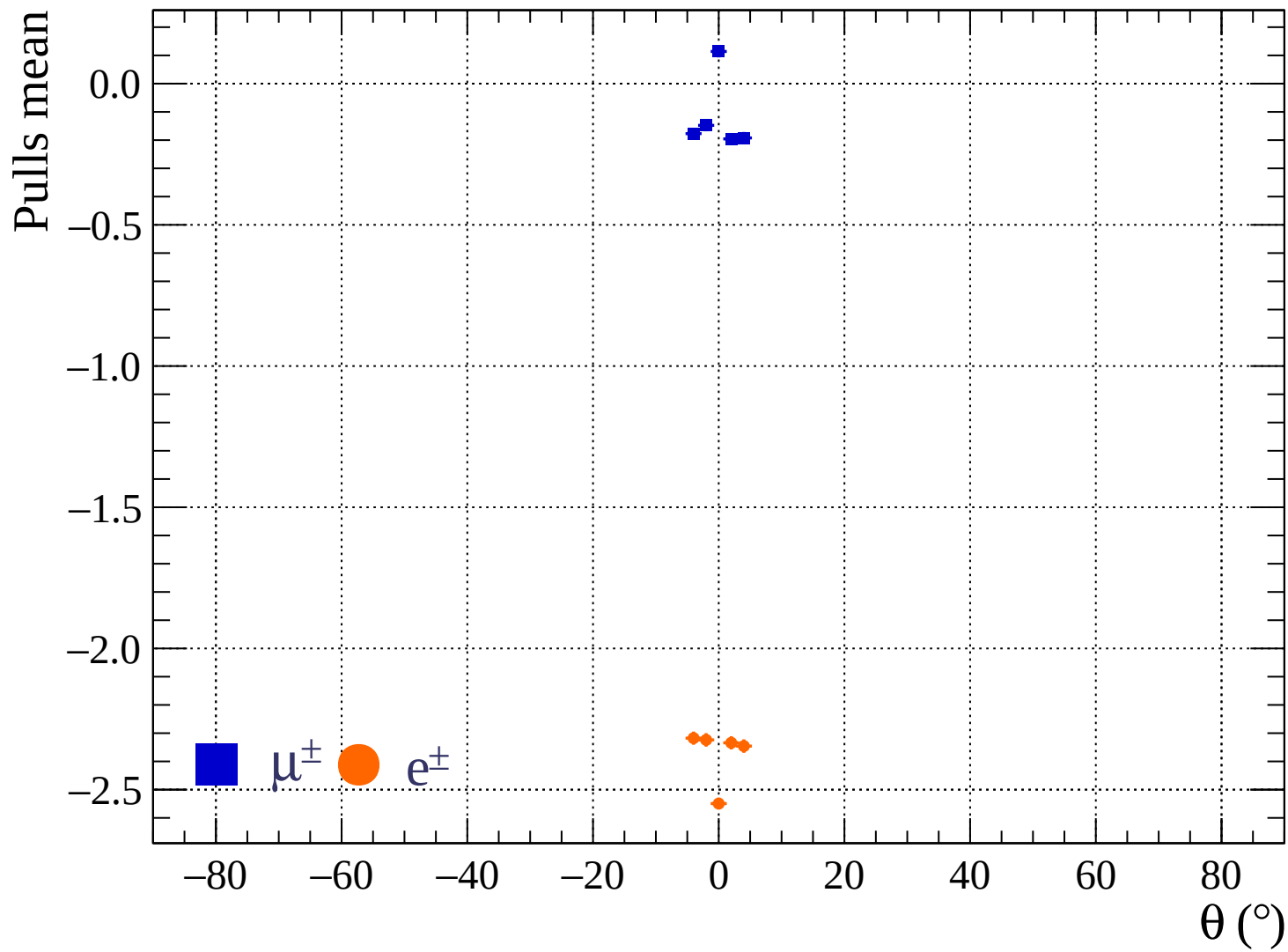




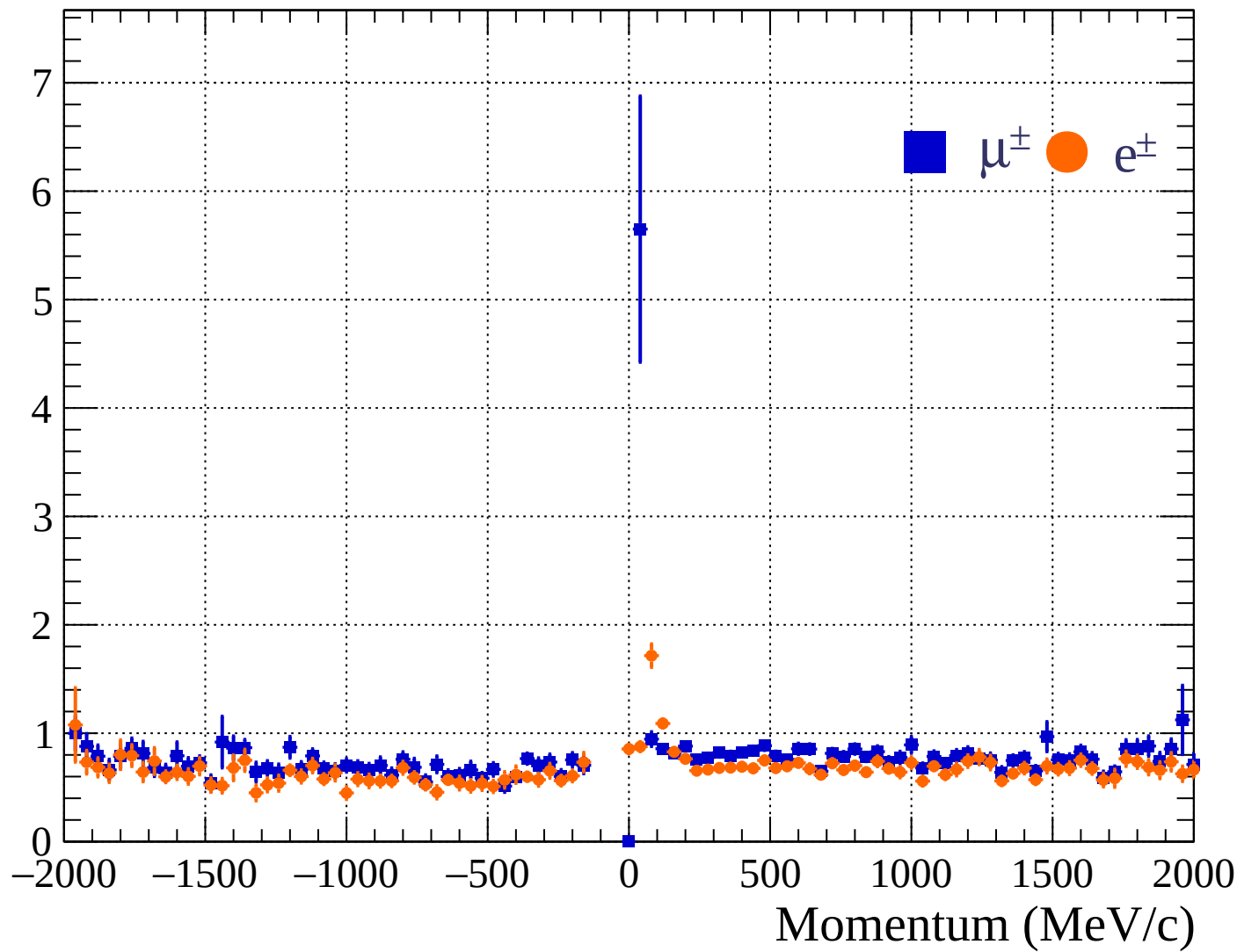


Pulls standard deviation

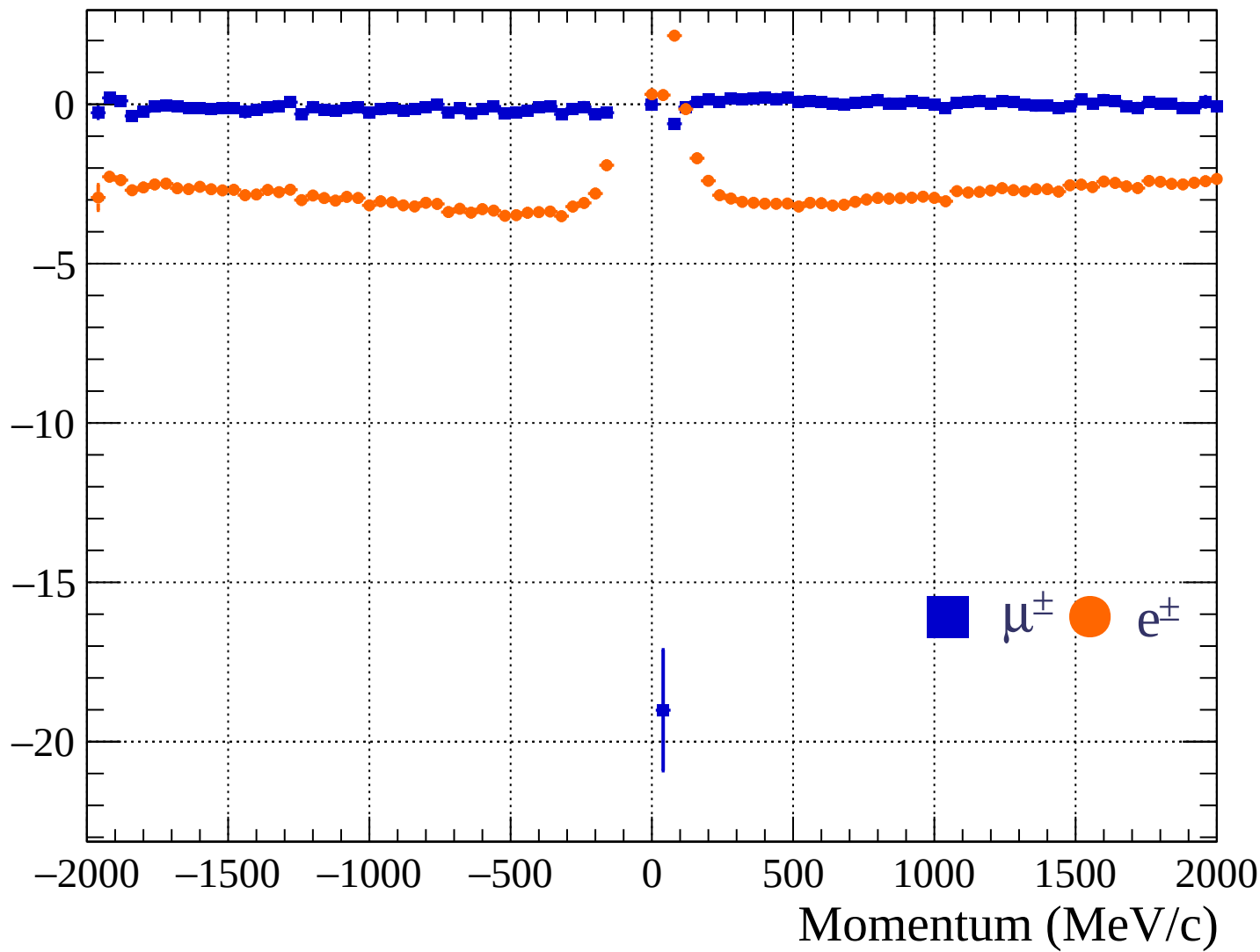


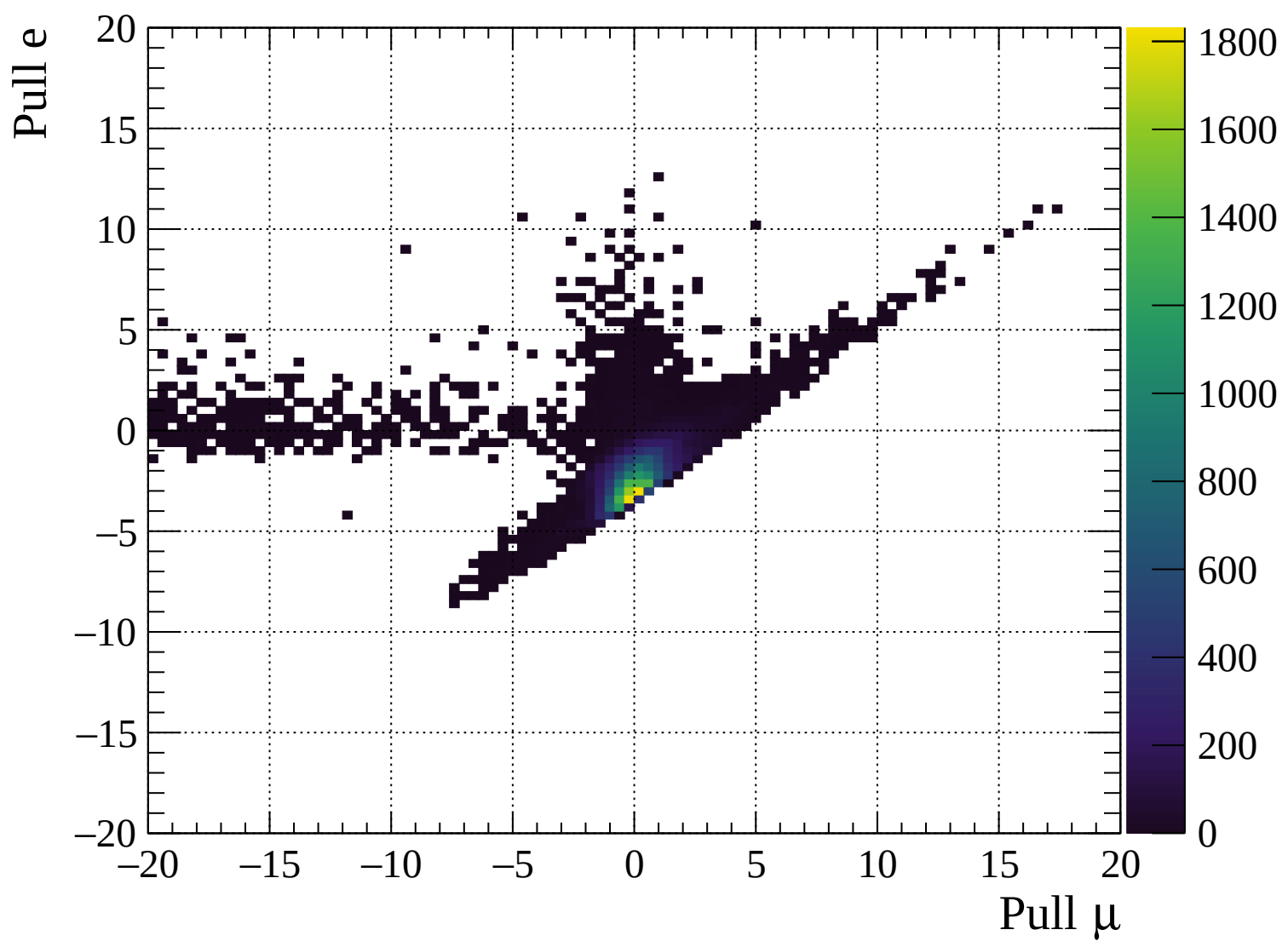


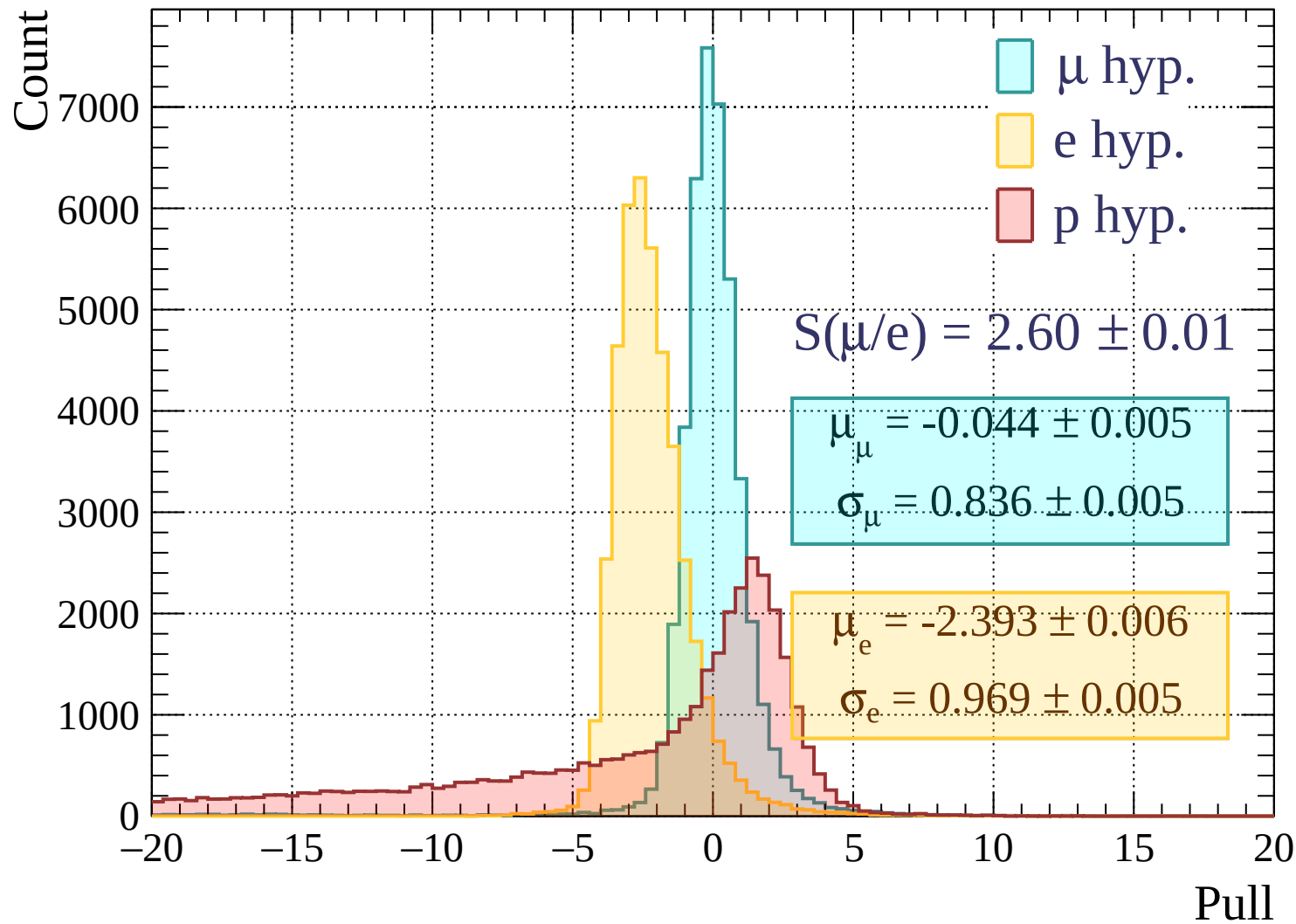
Pulls standard deviation

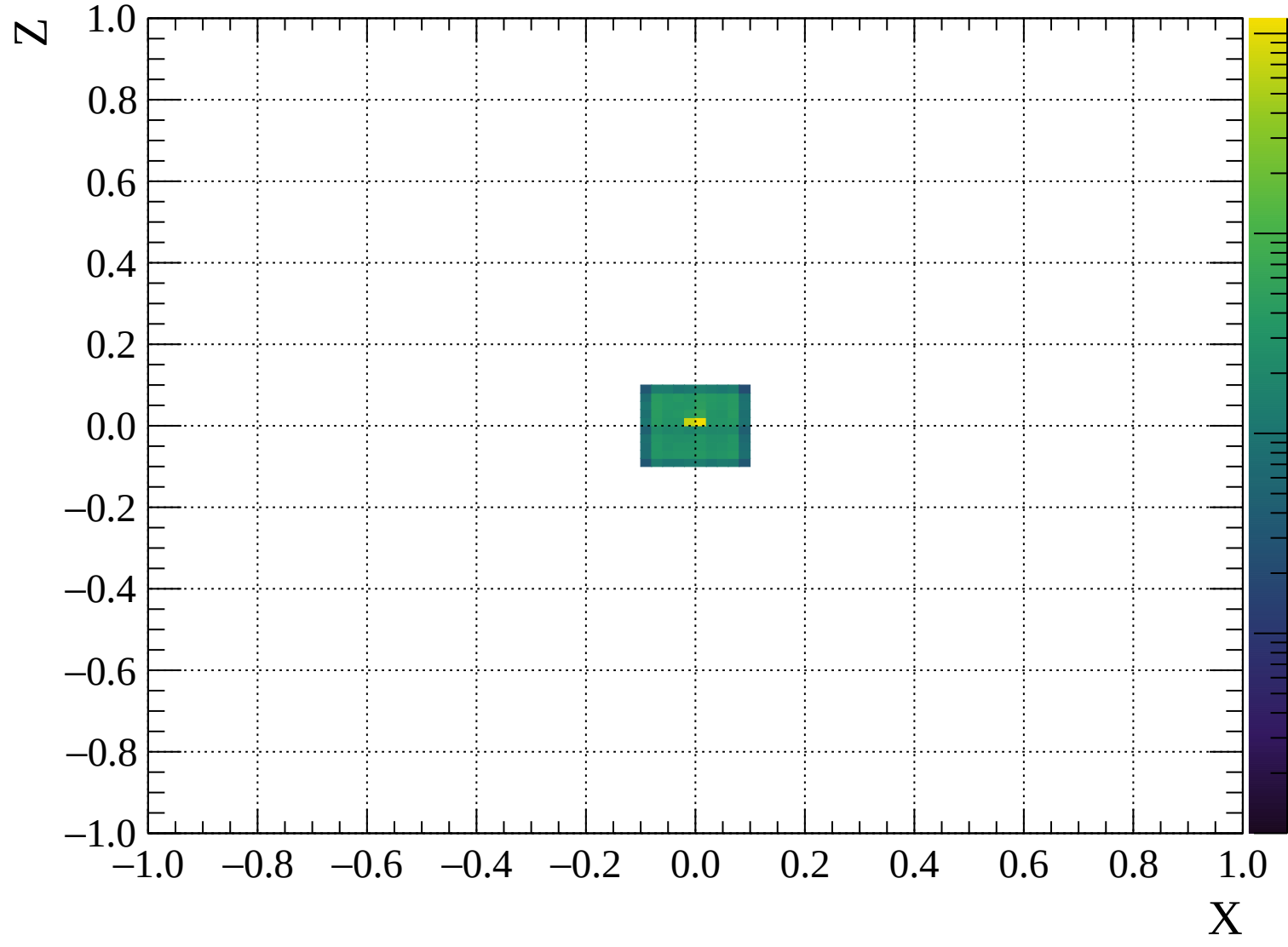


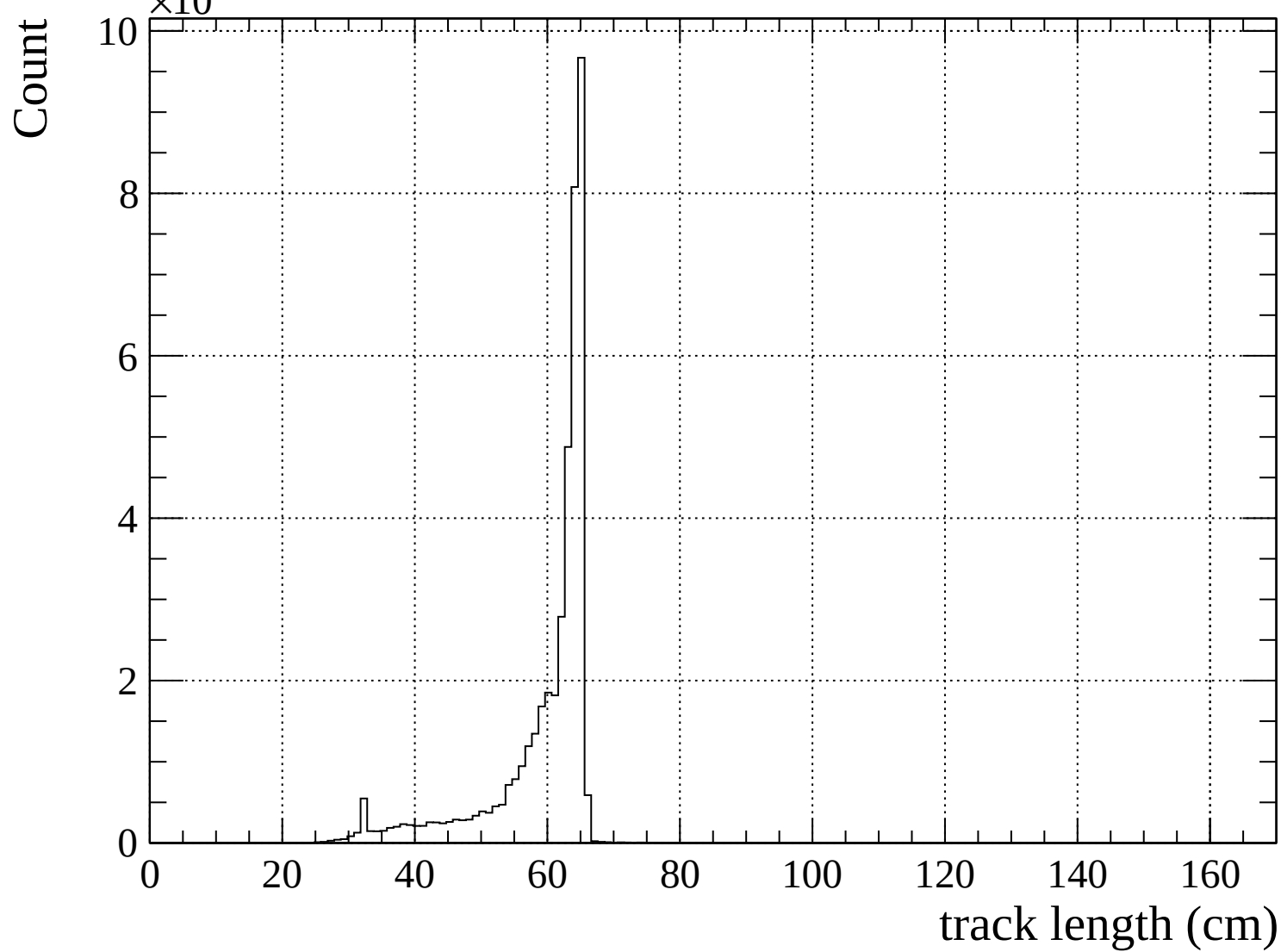
Pulls mean

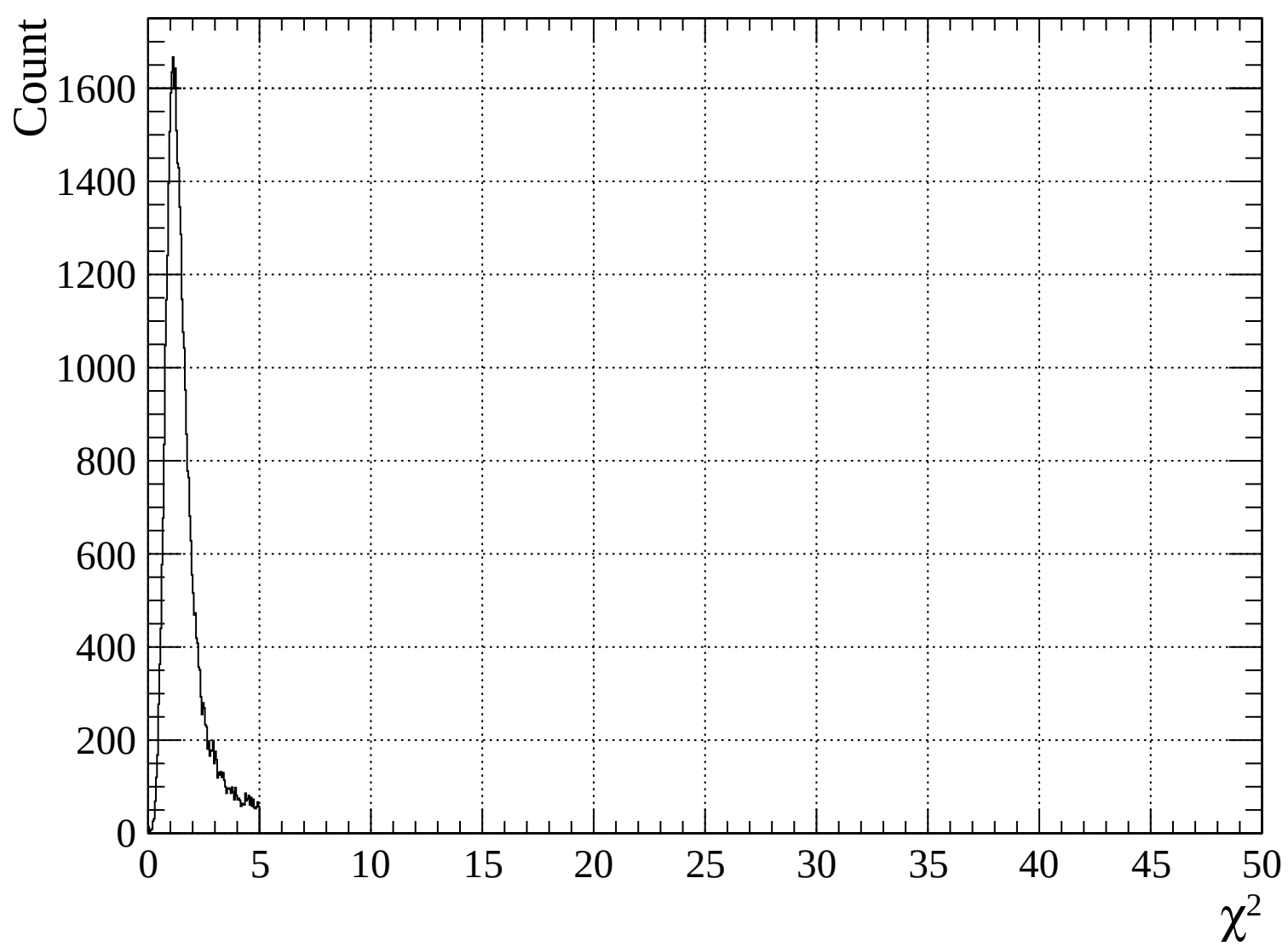


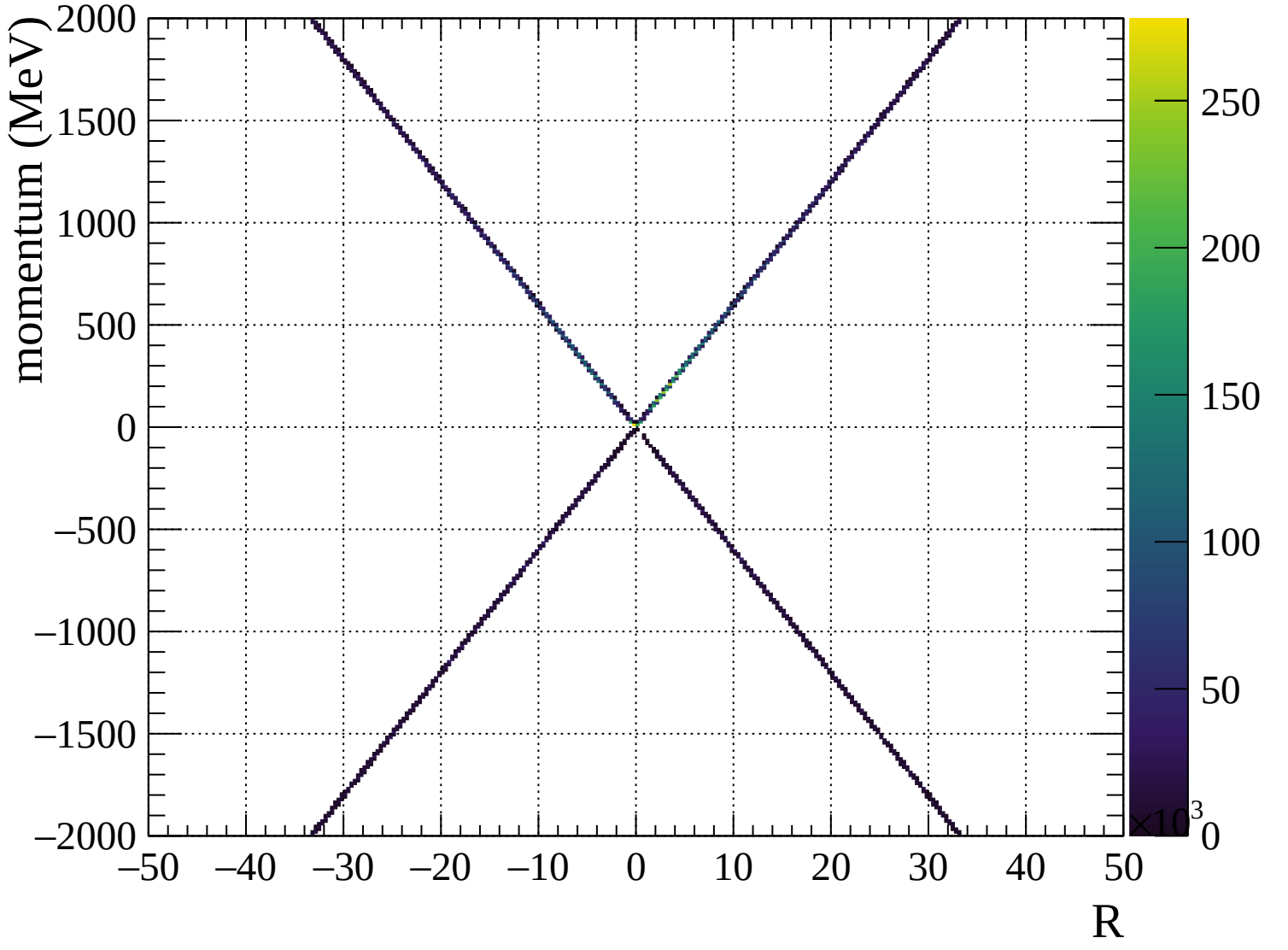


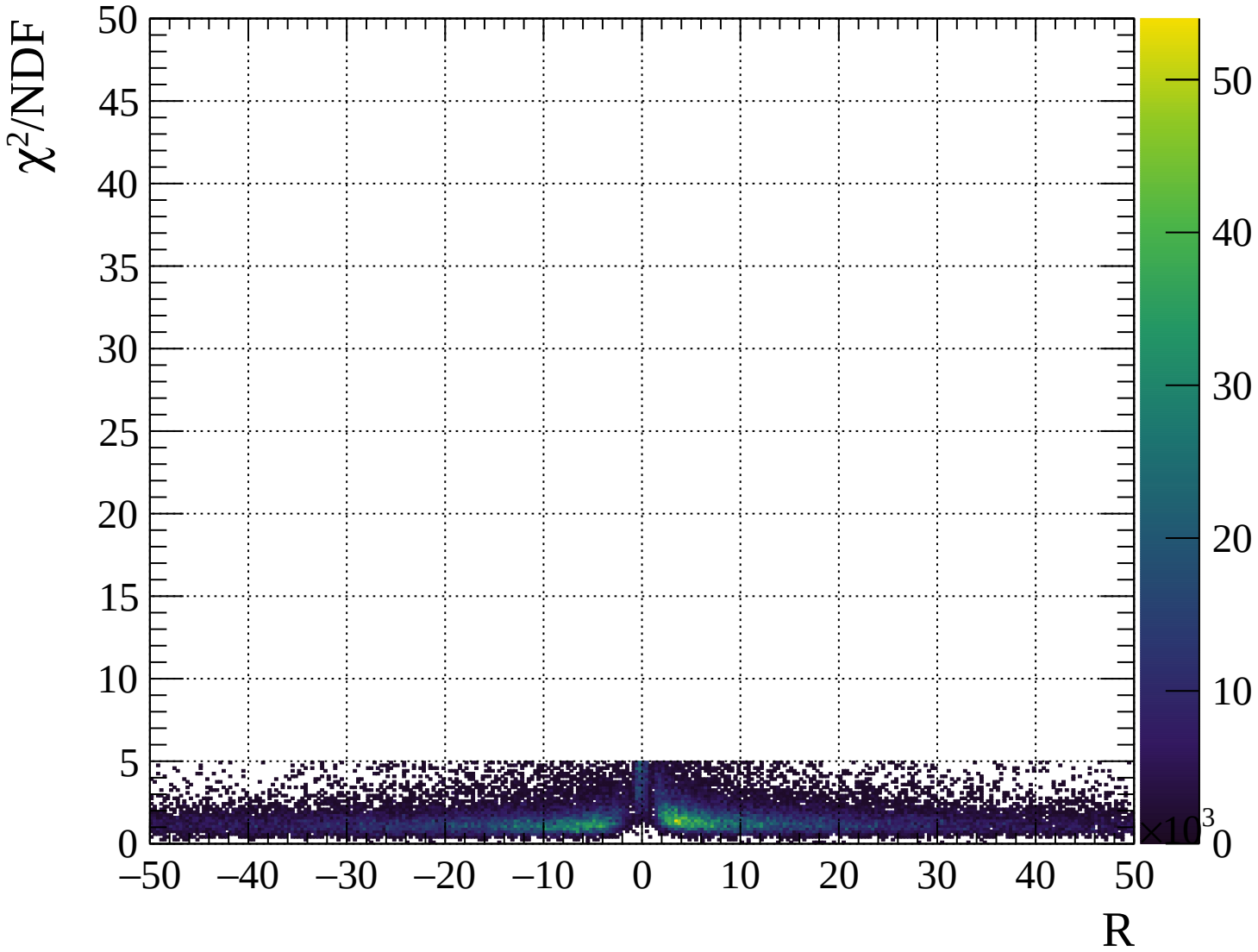






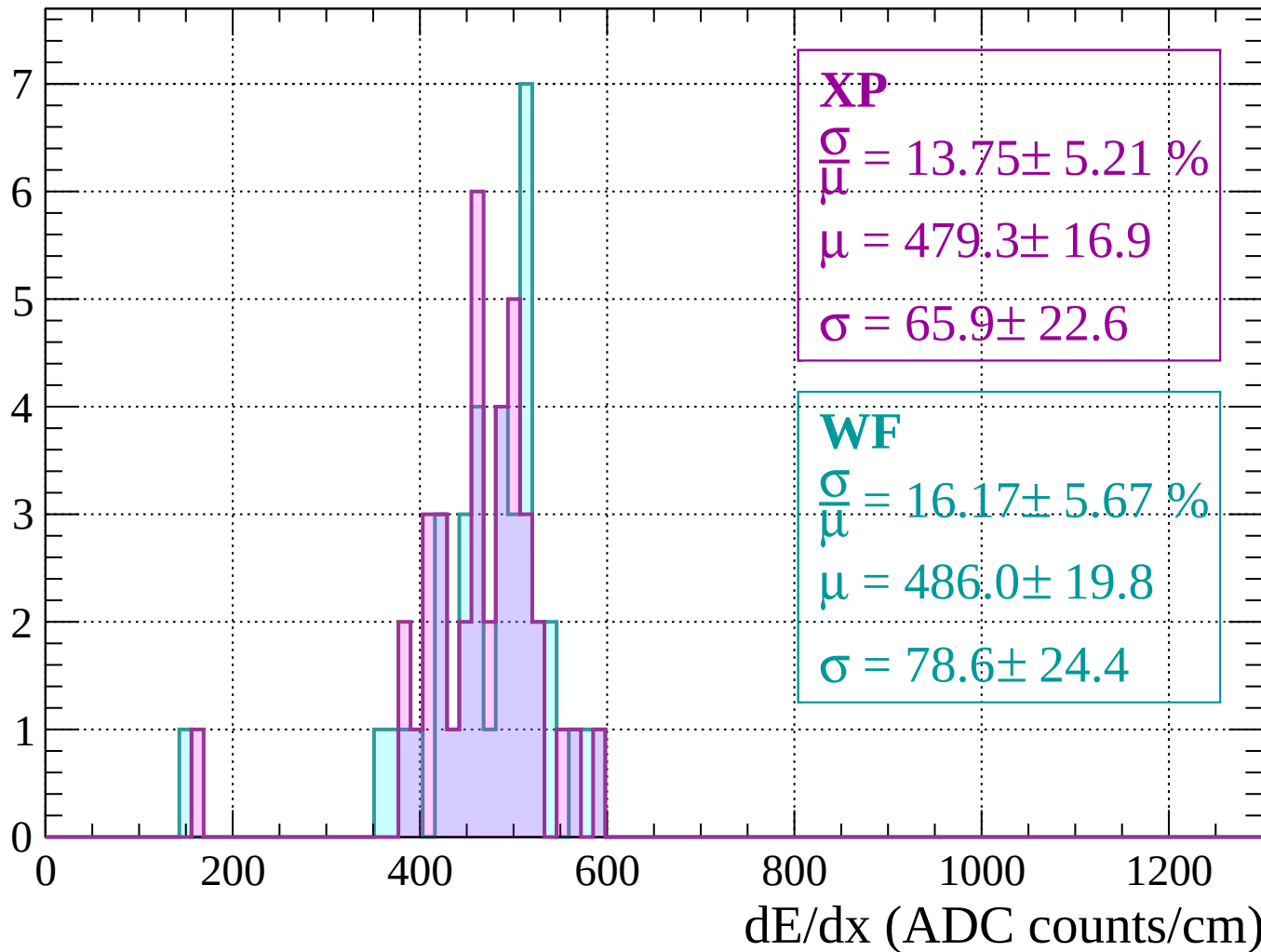






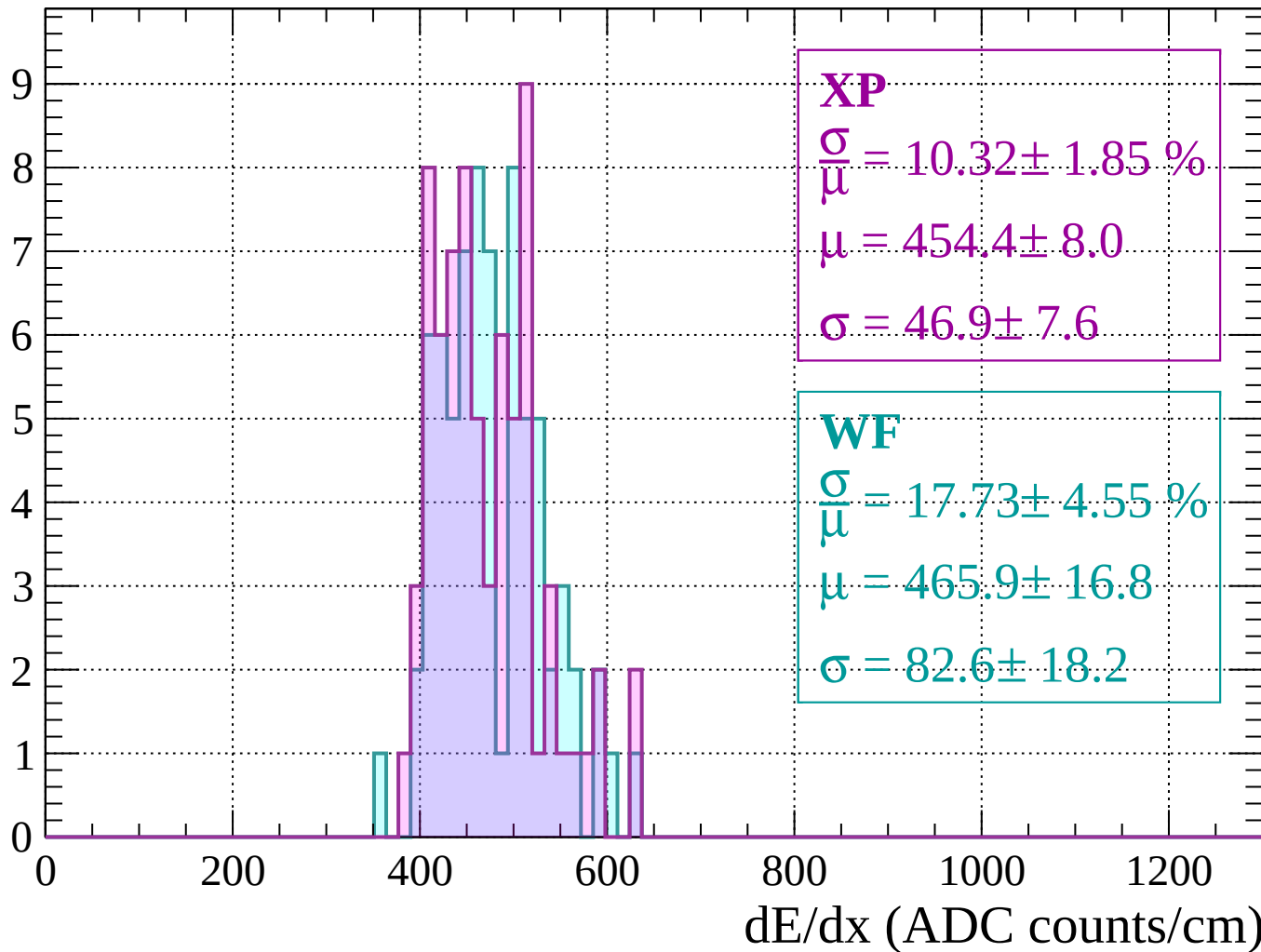
Energy loss | -2000 < p < -1960

Count



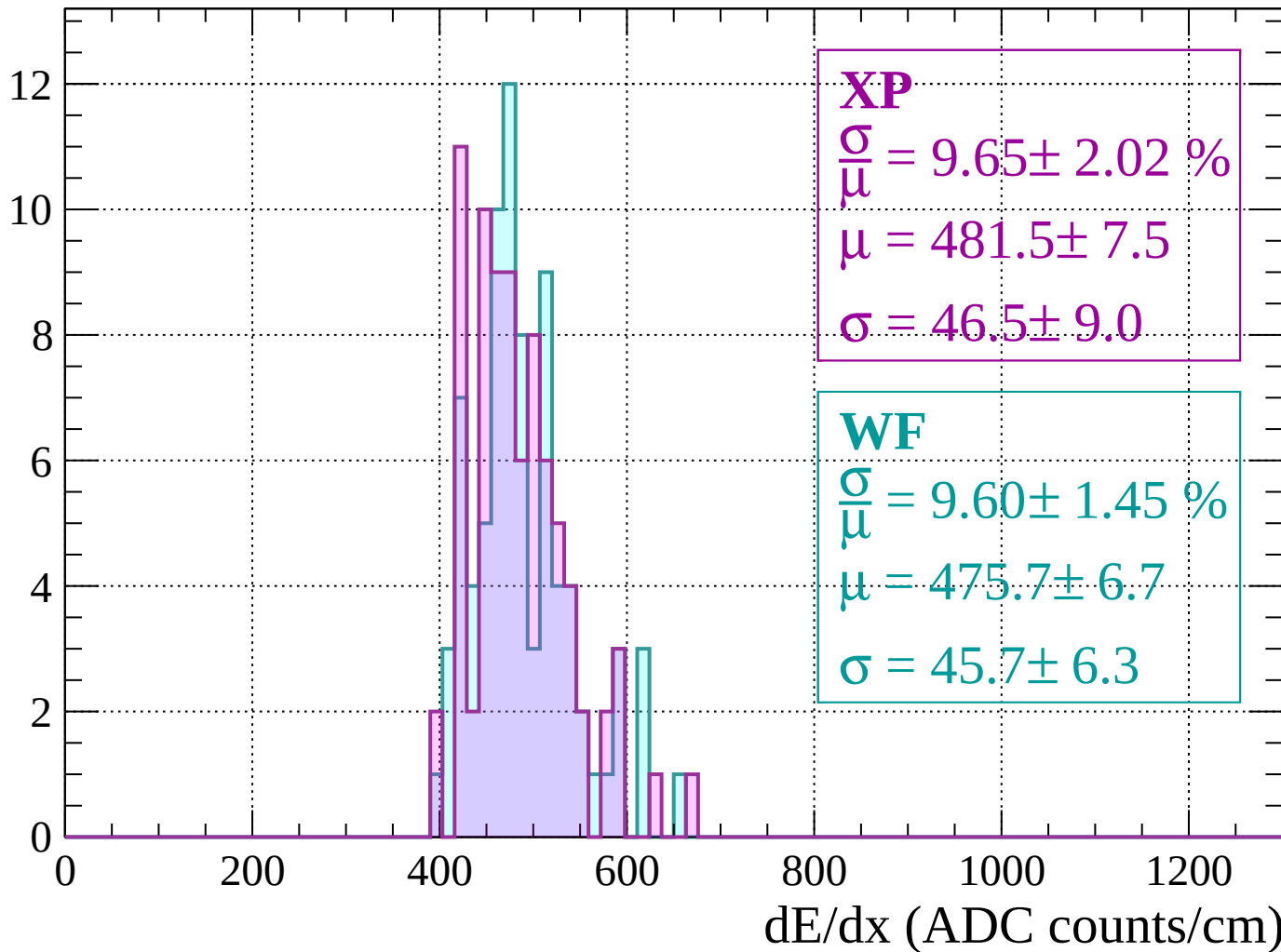
Energy loss | -1960 < p < -1920

Count



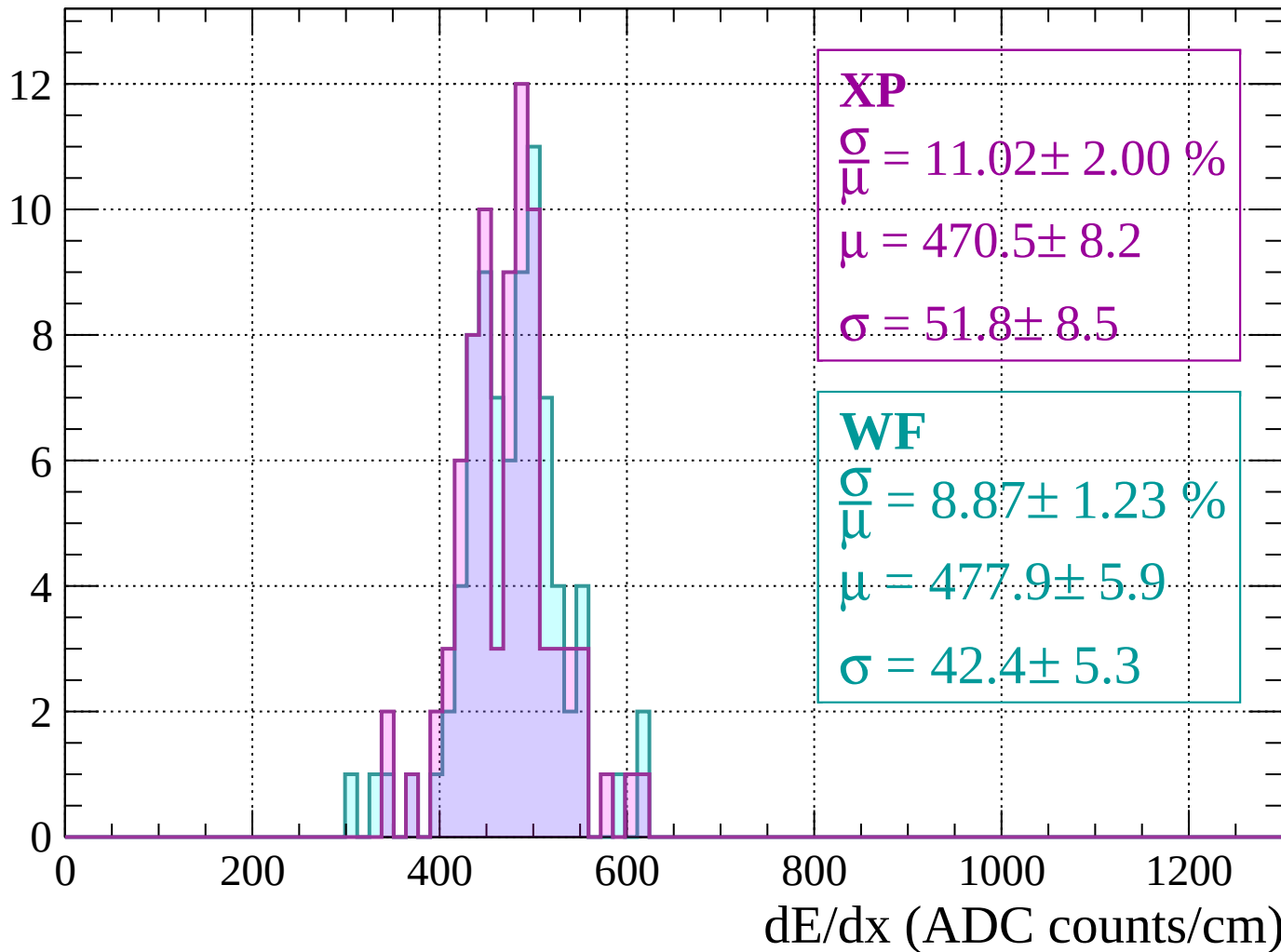
Energy loss | -1920 < p < -1880

Count



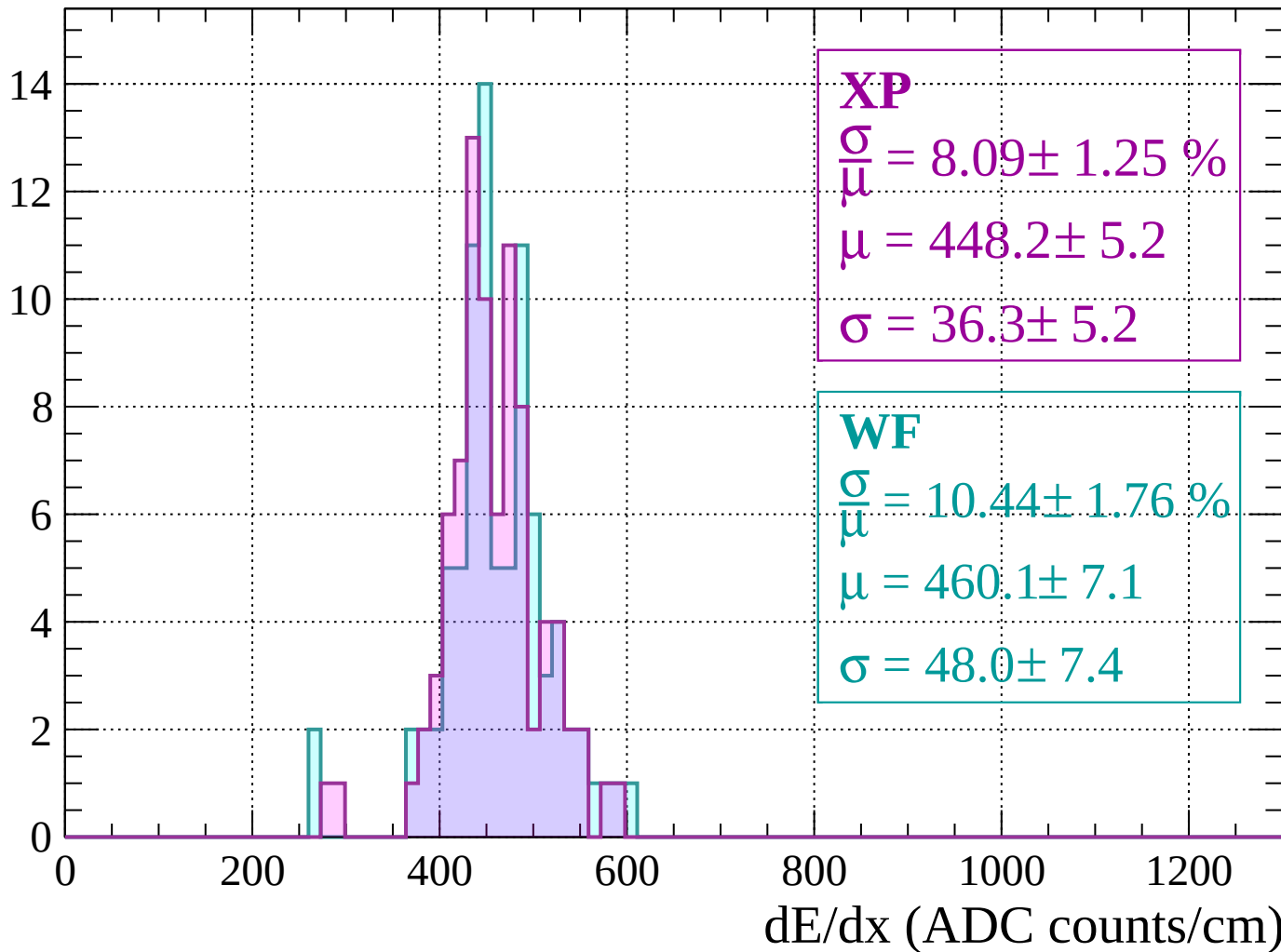
Energy loss | $-1880 < p < -1840$

Count



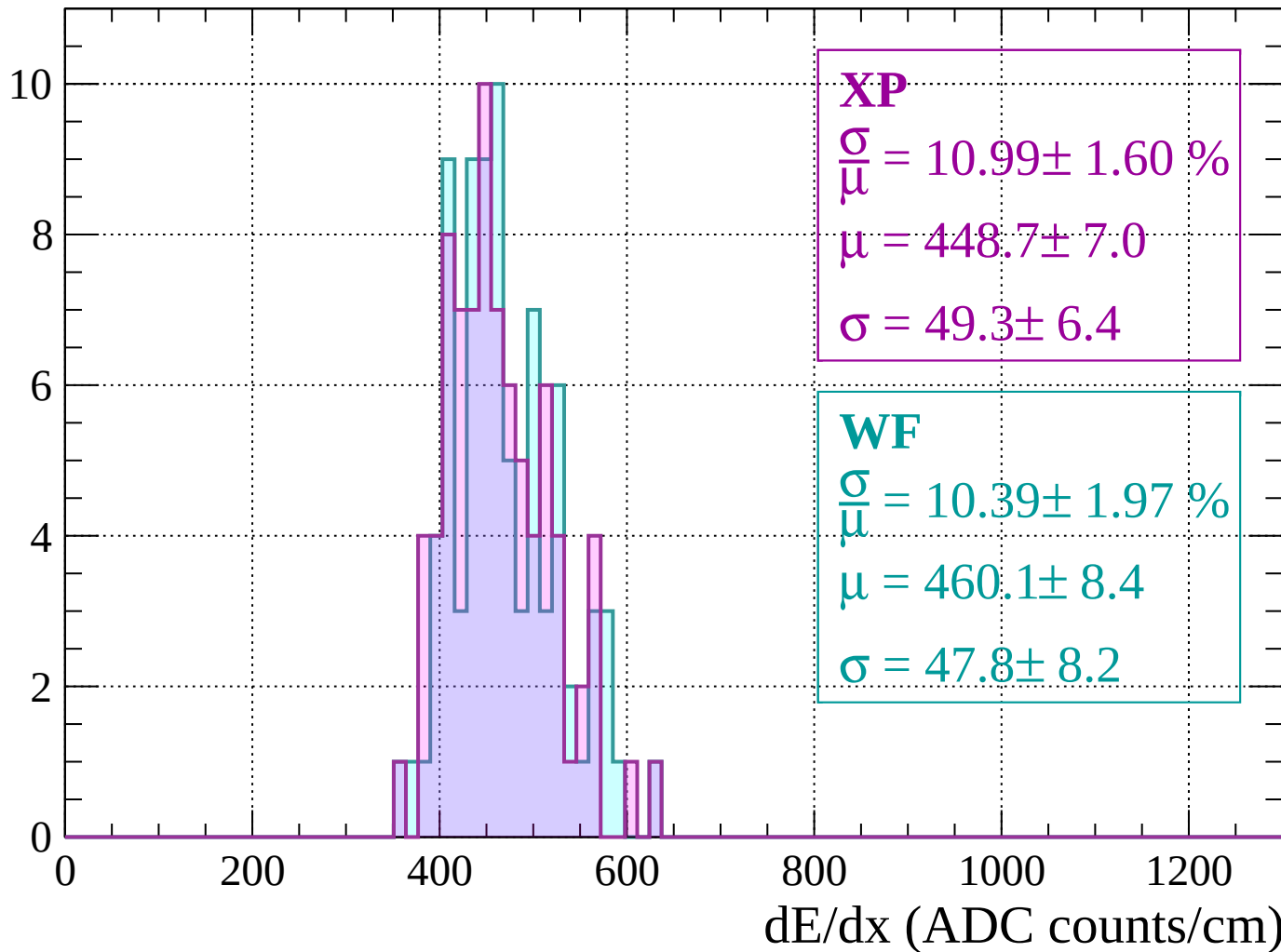
Energy loss | -1840 < p < -1800

Count



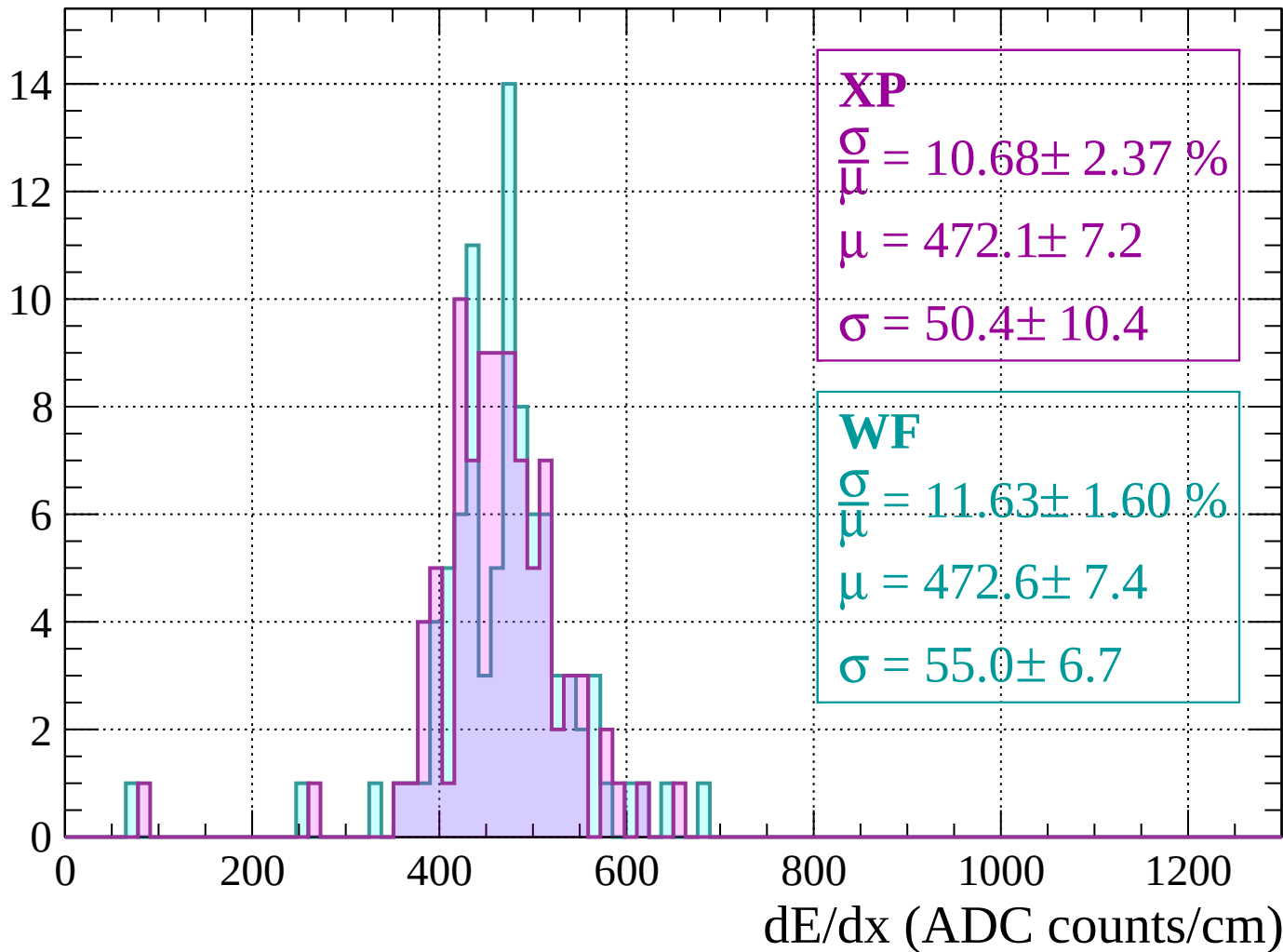
Energy loss | -1800 < p < -1760

Count



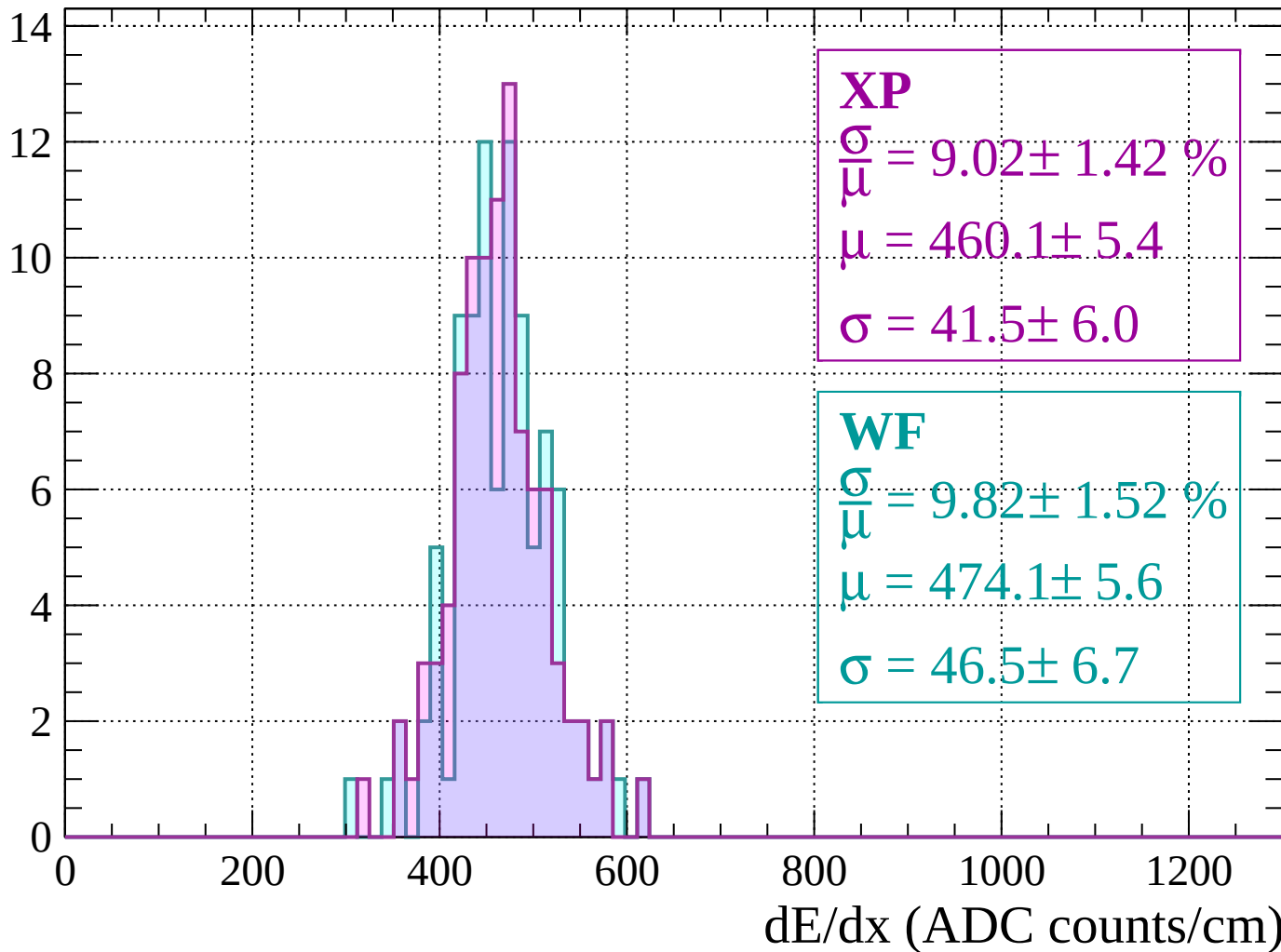
Energy loss | -1760 < p < -1720

Count



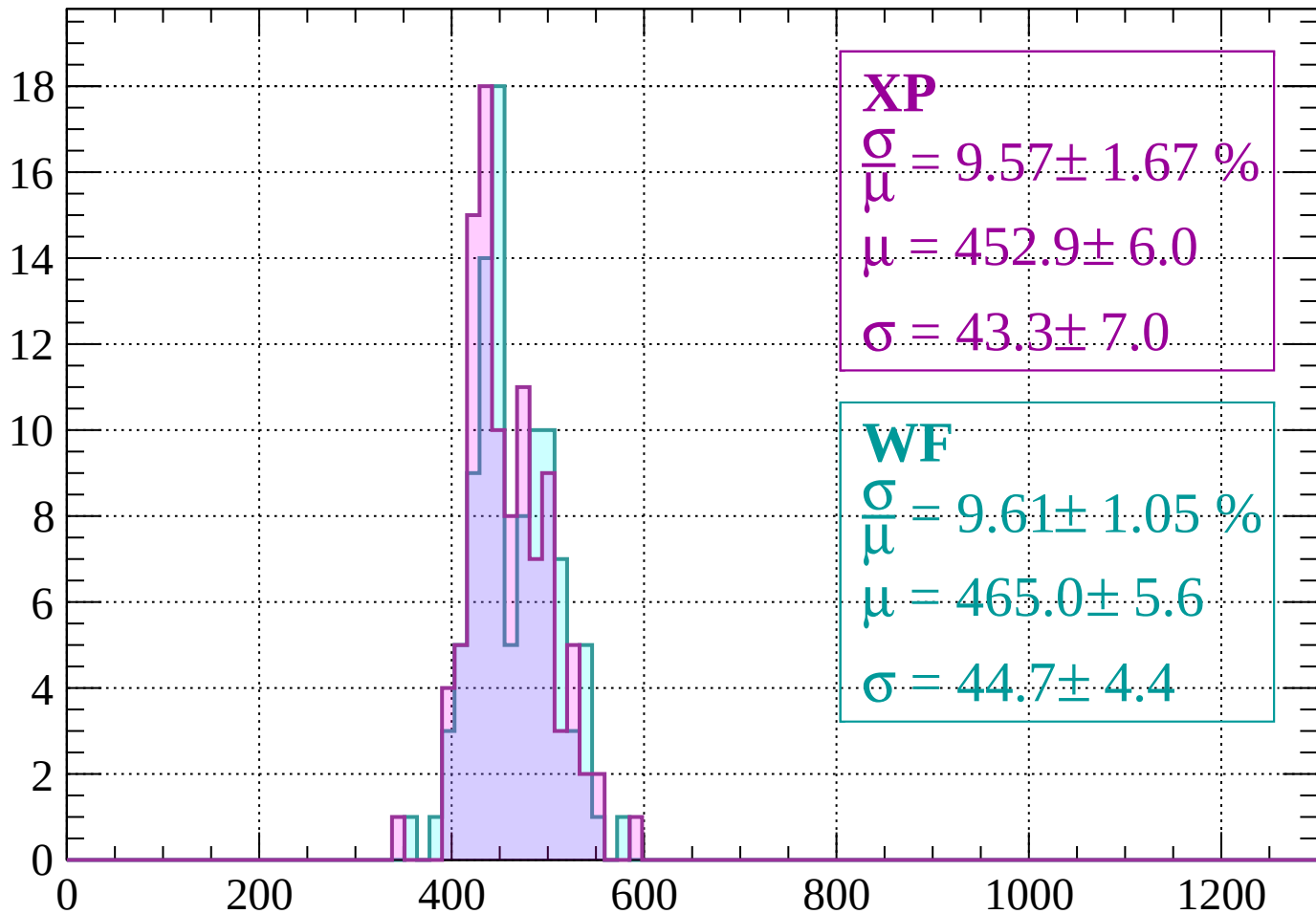
Energy loss | $-1720 < p < -1680$

Count



Energy loss | $-1680 < p < -1640$

Count



XP

$$\frac{\sigma}{\mu} = 9.57 \pm 1.67 \%$$

$$\mu = 452.9 \pm 6.0$$

$$\sigma = 43.3 \pm 7.0$$

WF

$$\frac{\sigma}{\mu} = 9.61 \pm 1.05 \%$$

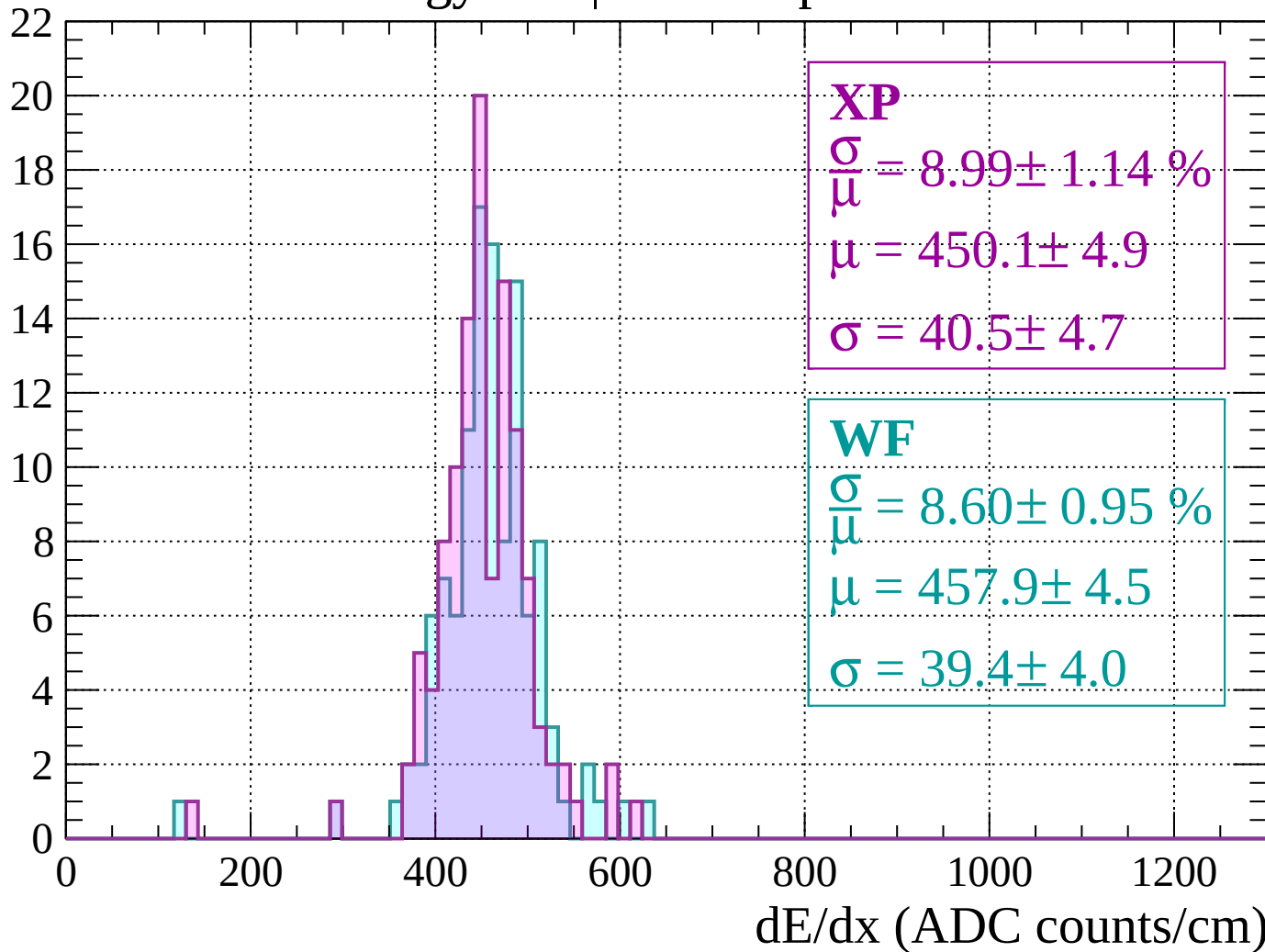
$$\mu = 465.0 \pm 5.6$$

$$\sigma = 44.7 \pm 4.4$$

dE/dx (ADC counts/cm)

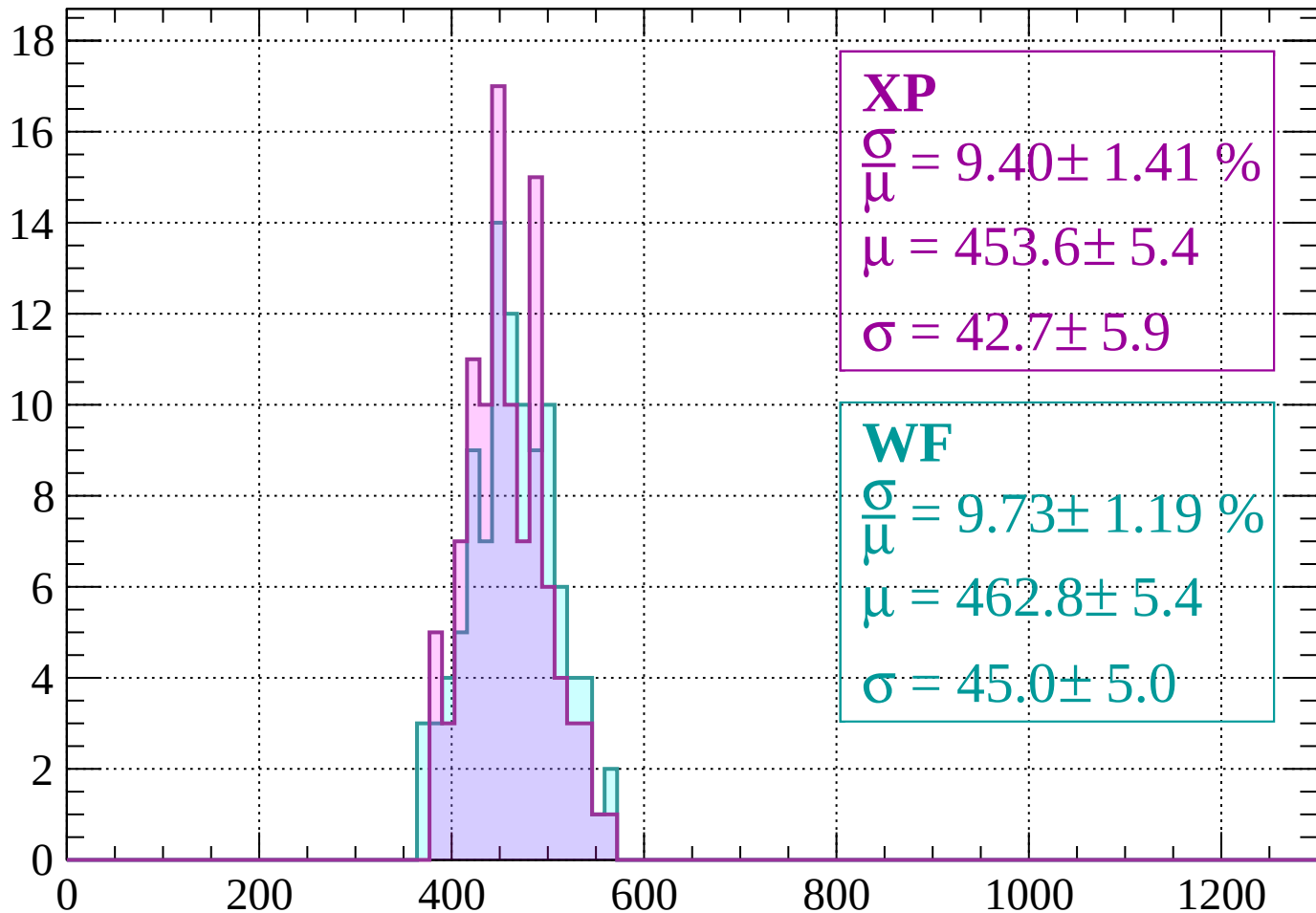
Energy loss | $-1640 < p < -1600$

Count



Energy loss | $-1600 < p < -1560$

Count



XP

$$\frac{\sigma}{\mu} = 9.40 \pm 1.41 \%$$

$$\mu = 453.6 \pm 5.4$$

$$\sigma = 42.7 \pm 5.9$$

WF

$$\frac{\sigma}{\mu} = 9.73 \pm 1.19 \%$$

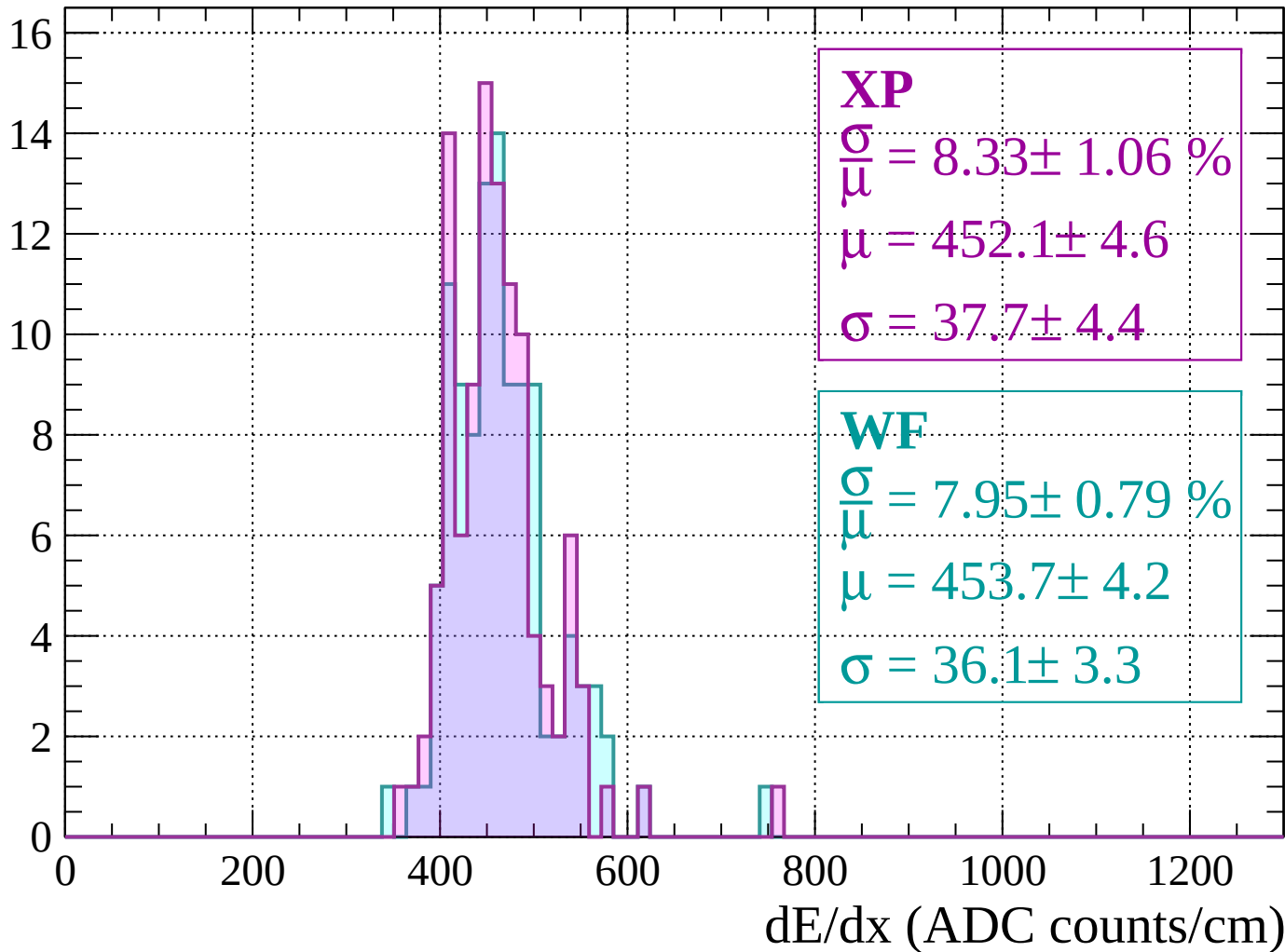
$$\mu = 462.8 \pm 5.4$$

$$\sigma = 45.0 \pm 5.0$$

dE/dx (ADC counts/cm)

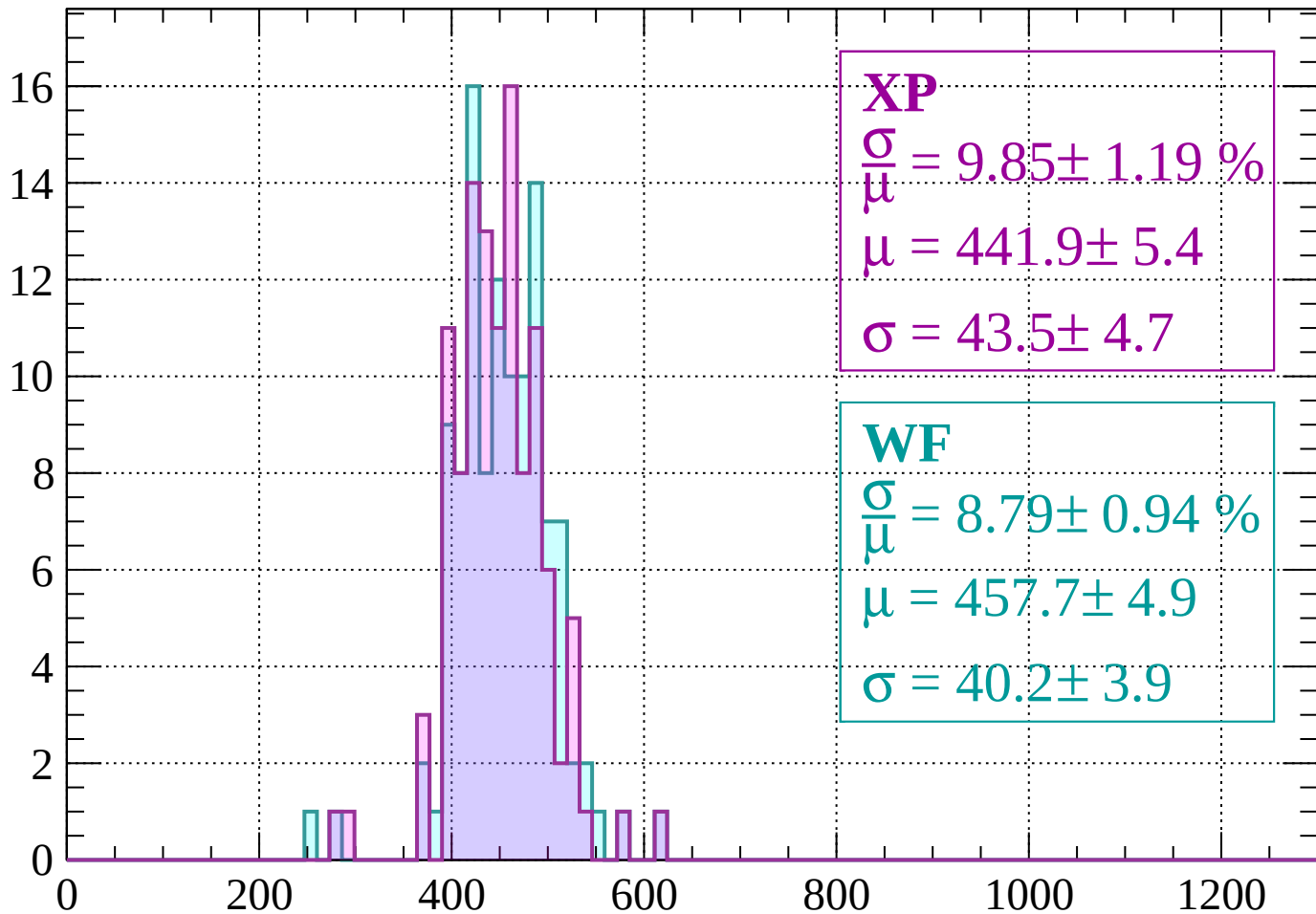
Energy loss | -1560 < p < -1520

Count



Energy loss | $-1520 < p < -1480$

Count



XP

$$\frac{\sigma}{\mu} = 9.85 \pm 1.19 \%$$

$$\mu = 441.9 \pm 5.4$$

$$\sigma = 43.5 \pm 4.7$$

WF

$$\frac{\sigma}{\mu} = 8.79 \pm 0.94 \%$$

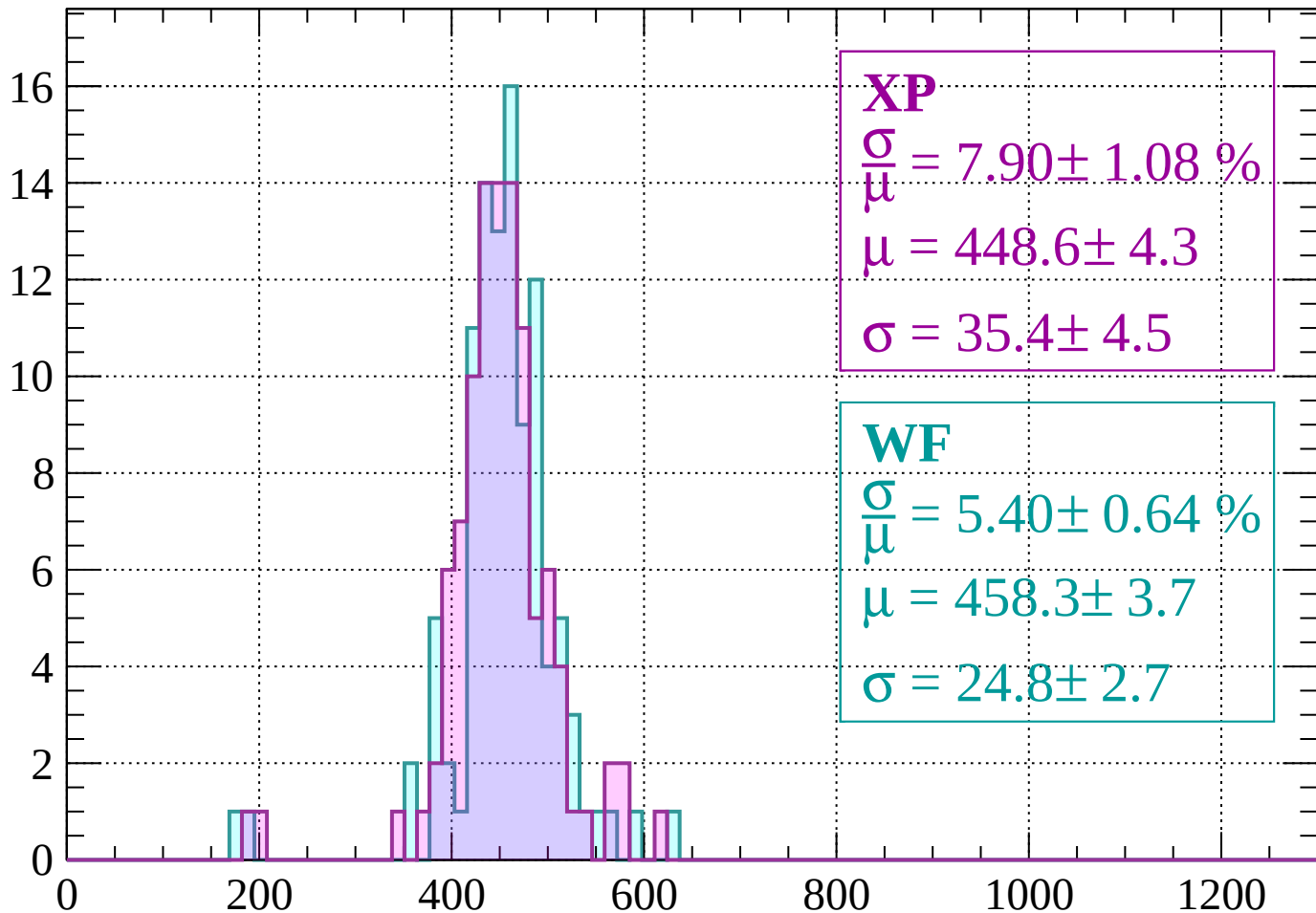
$$\mu = 457.7 \pm 4.9$$

$$\sigma = 40.2 \pm 3.9$$

dE/dx (ADC counts/cm)

Energy loss | $-1480 < p < -1440$

Count



XP

$$\frac{\sigma}{\mu} = 7.90 \pm 1.08 \%$$

$$\mu = 448.6 \pm 4.3$$

$$\sigma = 35.4 \pm 4.5$$

WF

$$\frac{\sigma}{\mu} = 5.40 \pm 0.64 \%$$

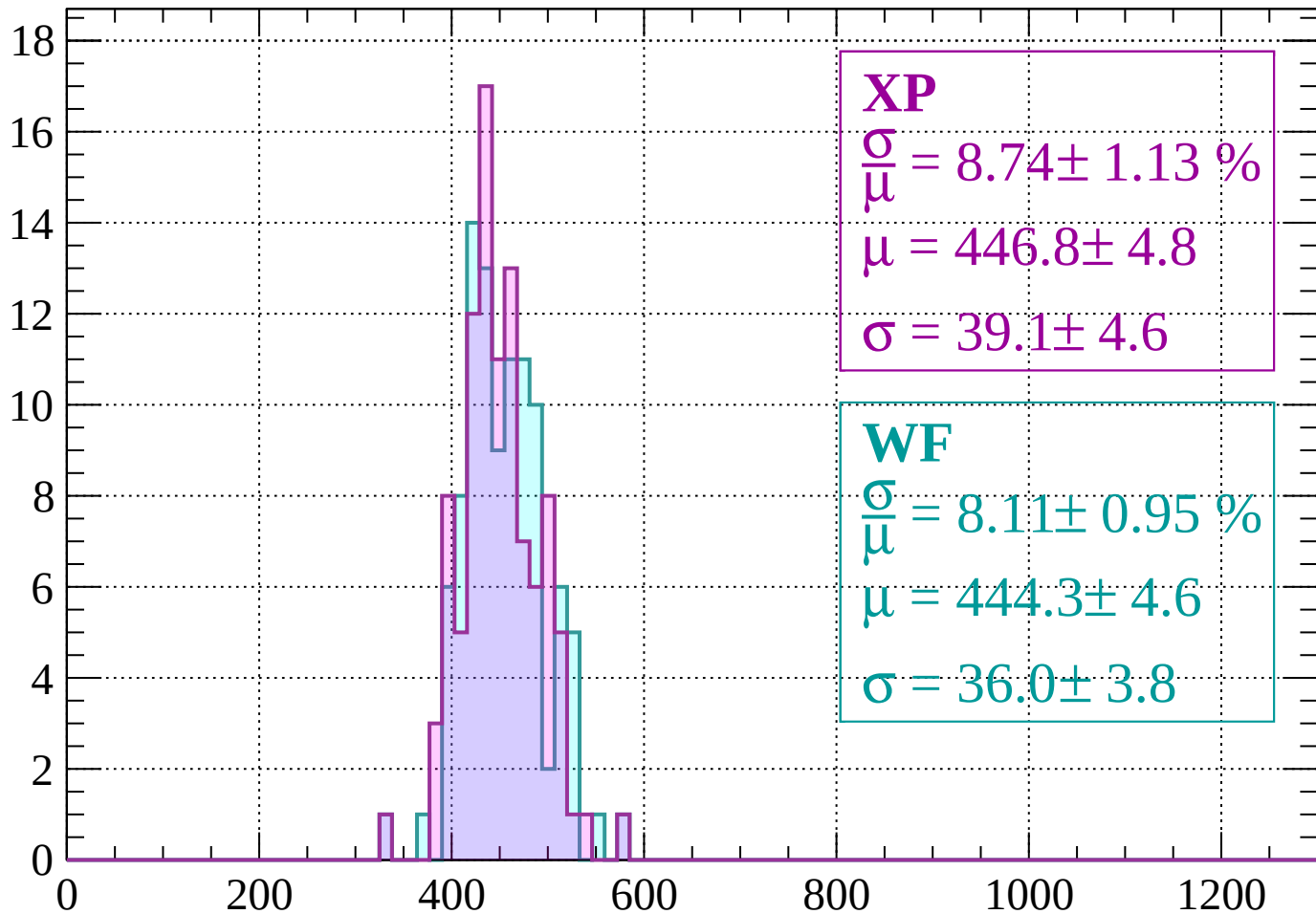
$$\mu = 458.3 \pm 3.7$$

$$\sigma = 24.8 \pm 2.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1440 < p < -1400$

Count



XP

$$\frac{\sigma}{\mu} = 8.74 \pm 1.13 \%$$

$$\mu = 446.8 \pm 4.8$$

$$\sigma = 39.1 \pm 4.6$$

WF

$$\frac{\sigma}{\mu} = 8.11 \pm 0.95 \%$$

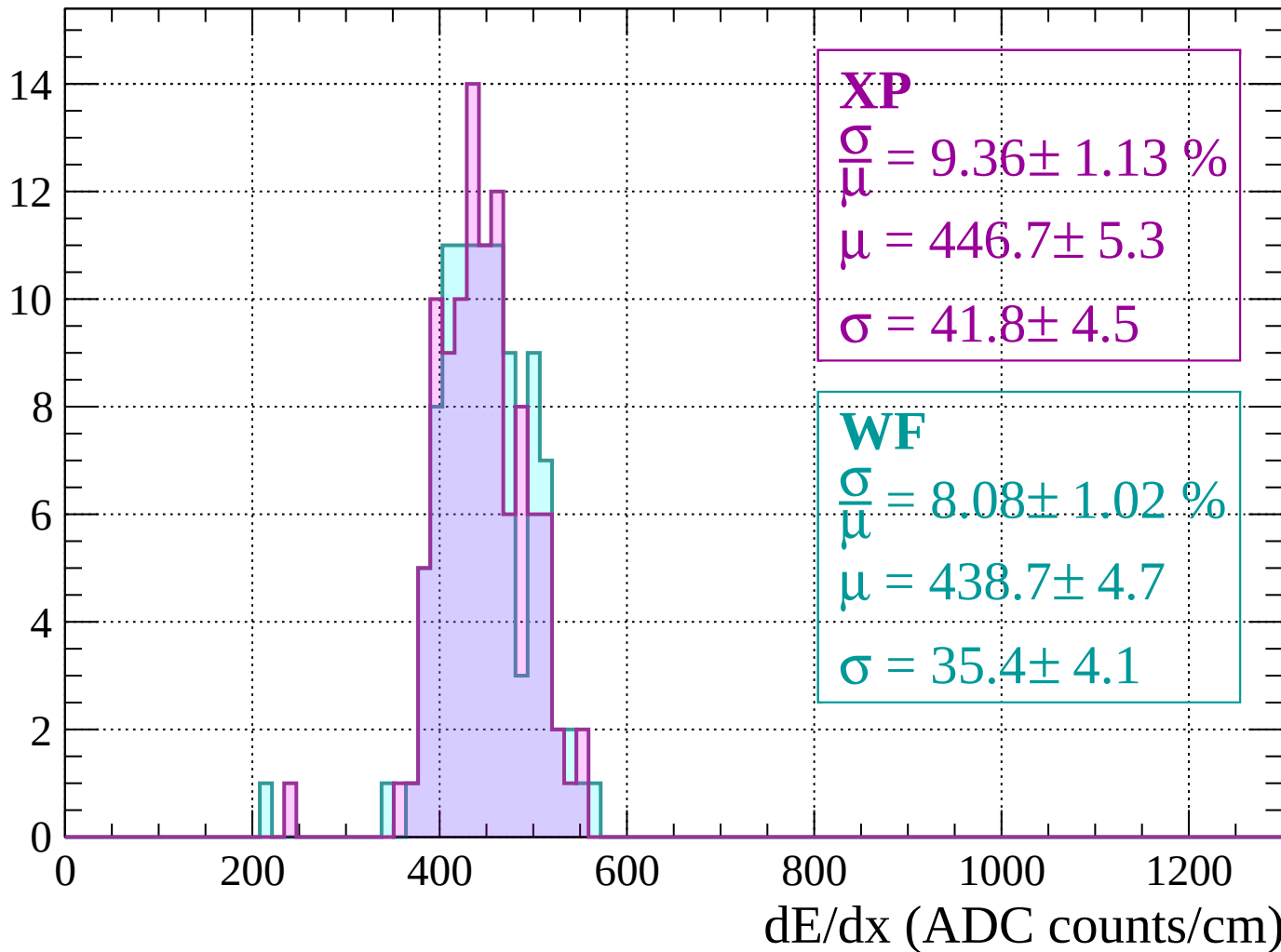
$$\mu = 444.3 \pm 4.6$$

$$\sigma = 36.0 \pm 3.8$$

dE/dx (ADC counts/cm)

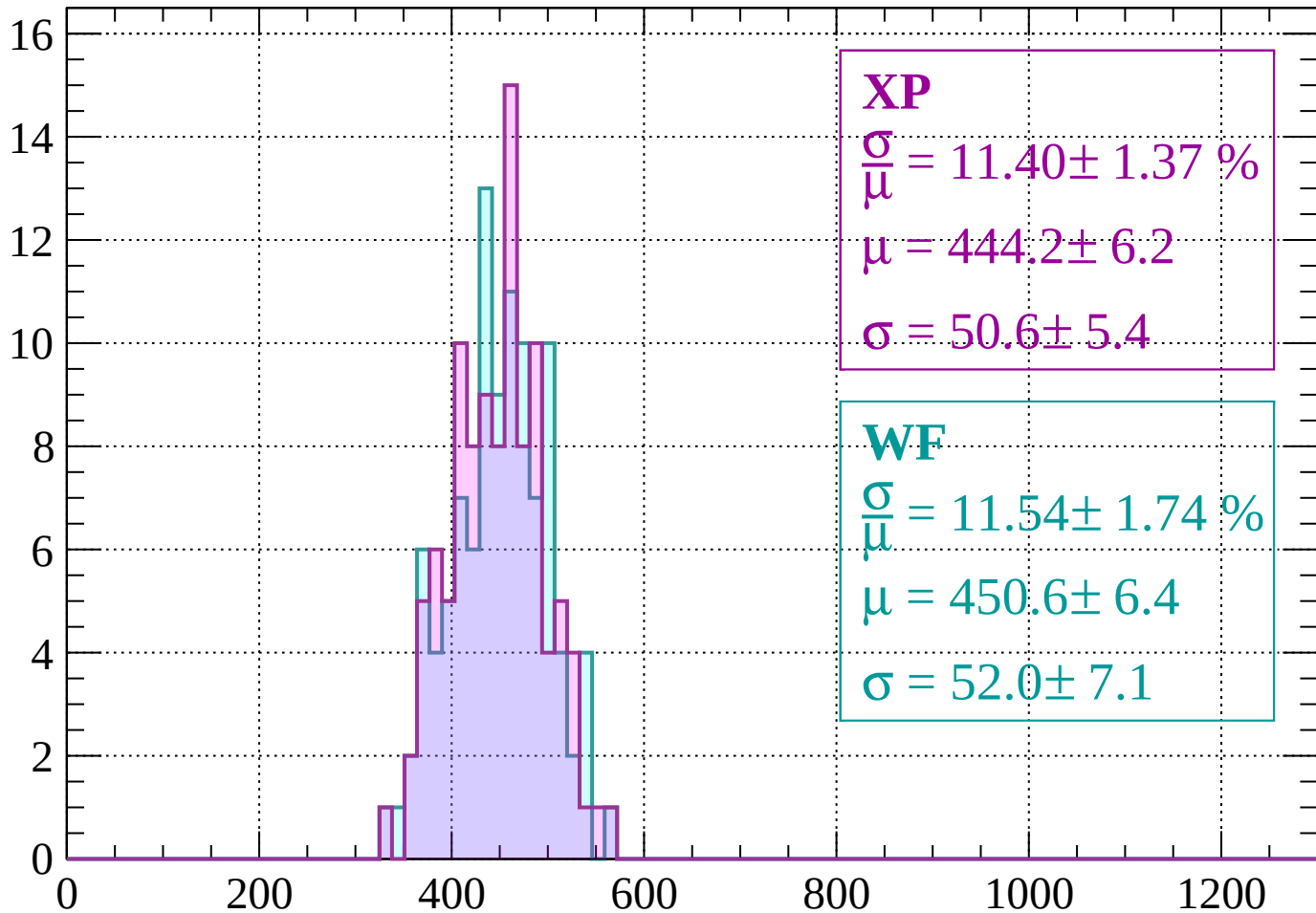
Energy loss | $-1400 < p < -1360$

Count



Energy loss | $-1360 < p < -1320$

Count



XP

$$\frac{\sigma}{\mu} = 11.40 \pm 1.37 \%$$

$$\mu = 444.2 \pm 6.2$$

$$\sigma = 50.6 \pm 5.4$$

WF

$$\frac{\sigma}{\mu} = 11.54 \pm 1.74 \%$$

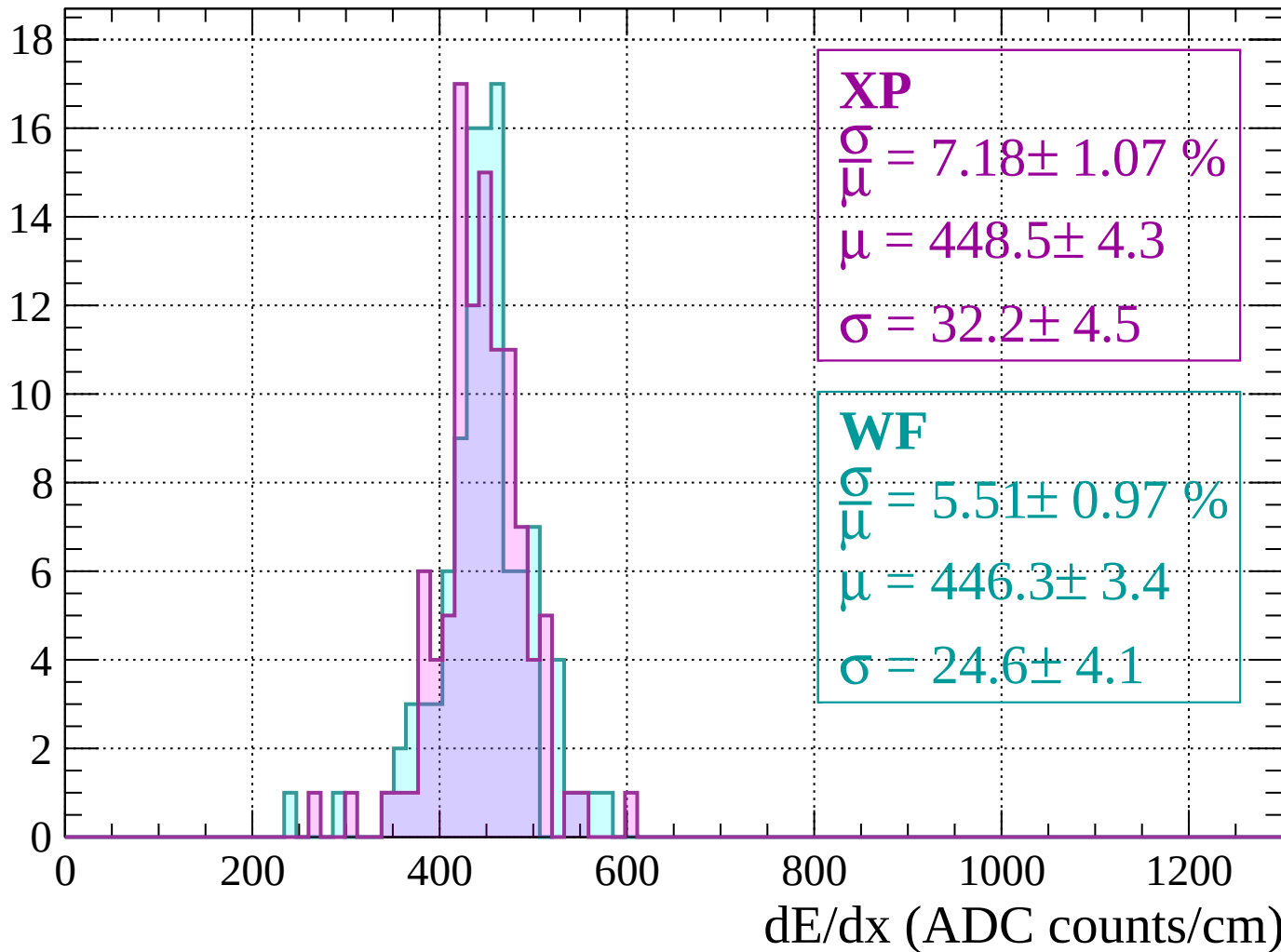
$$\mu = 450.6 \pm 6.4$$

$$\sigma = 52.0 \pm 7.1$$

dE/dx (ADC counts/cm)

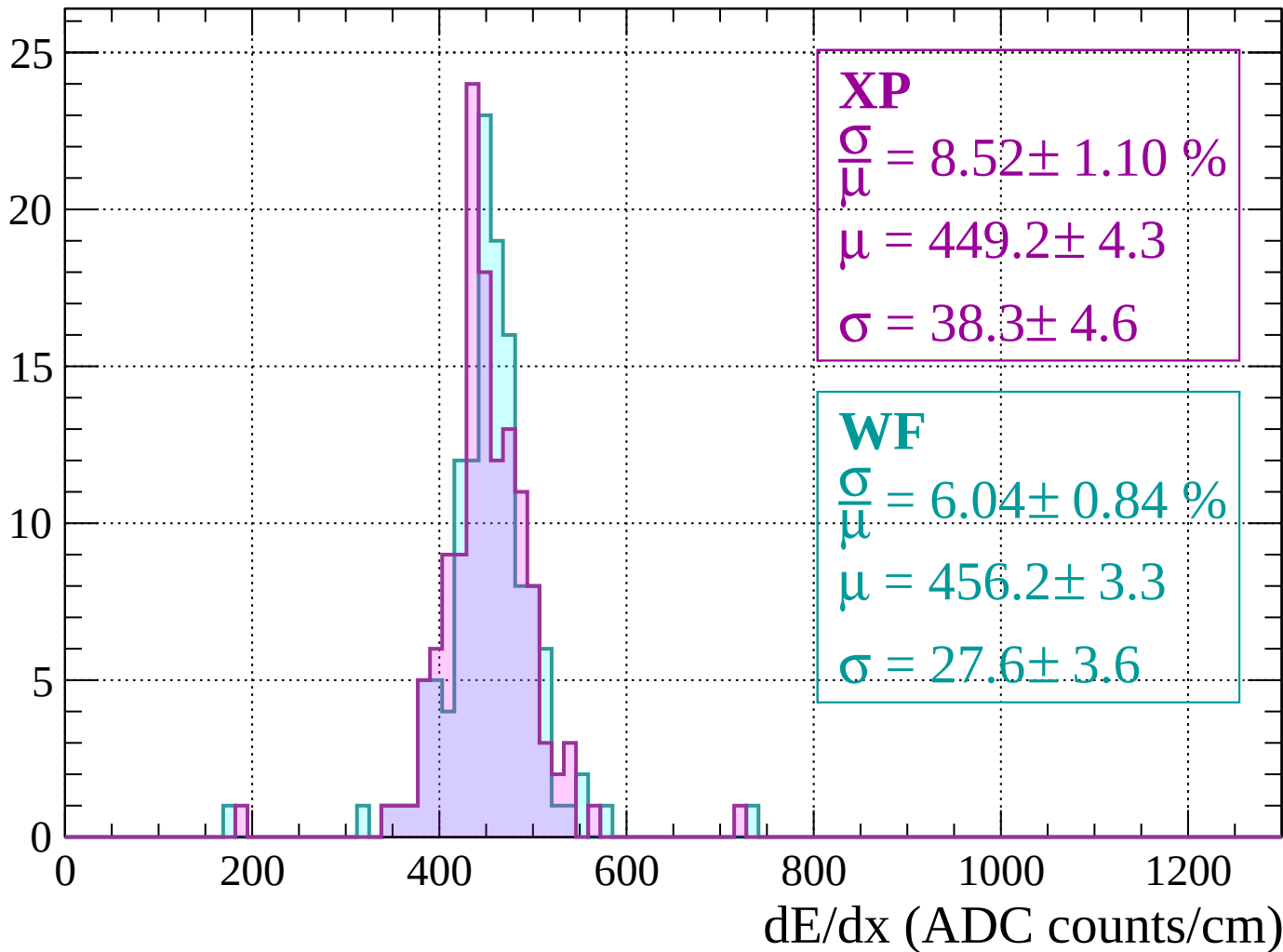
Energy loss | $-1320 < p < -1280$

Count



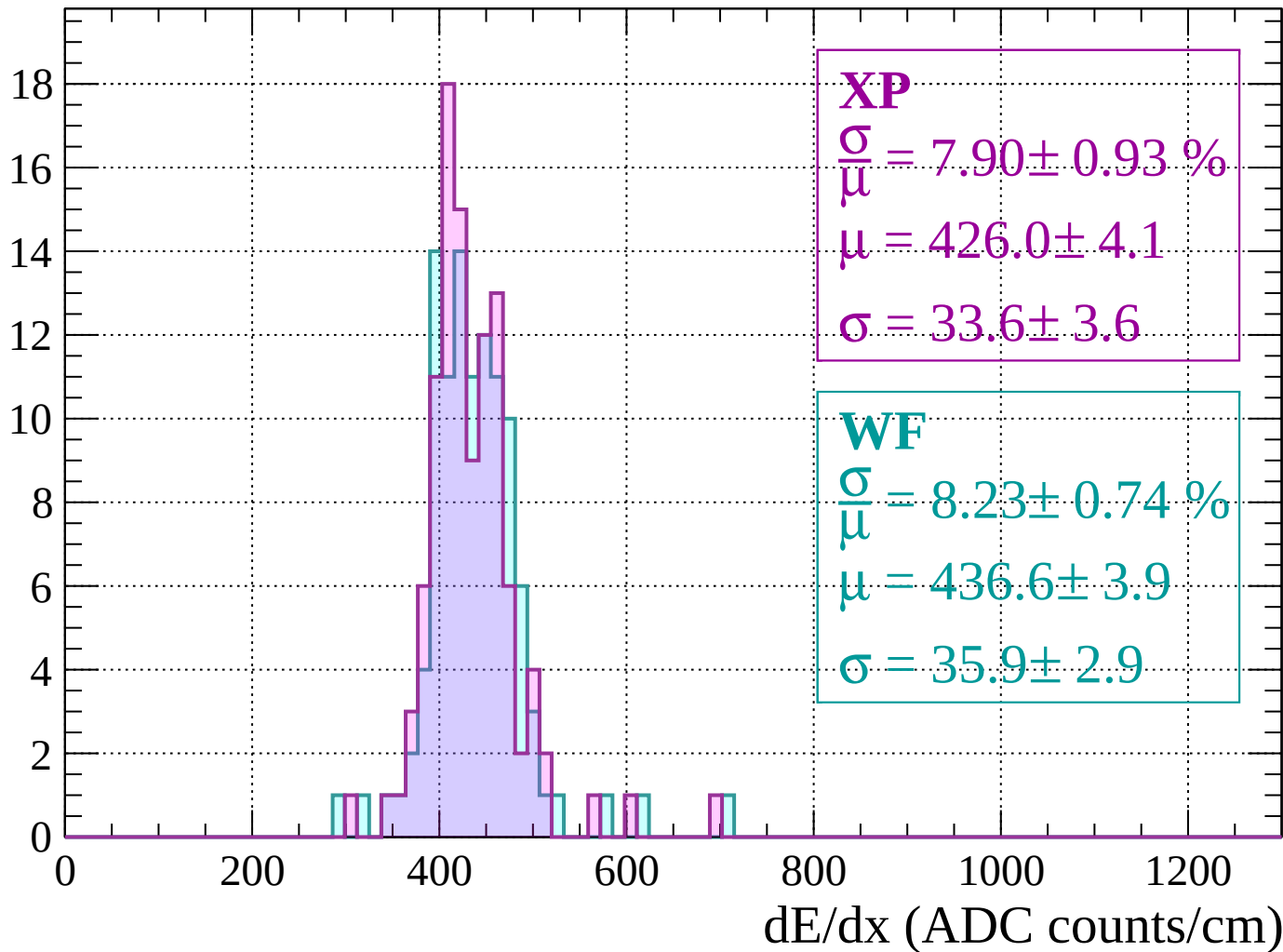
Energy loss | $-1280 < p < -1240$

Count



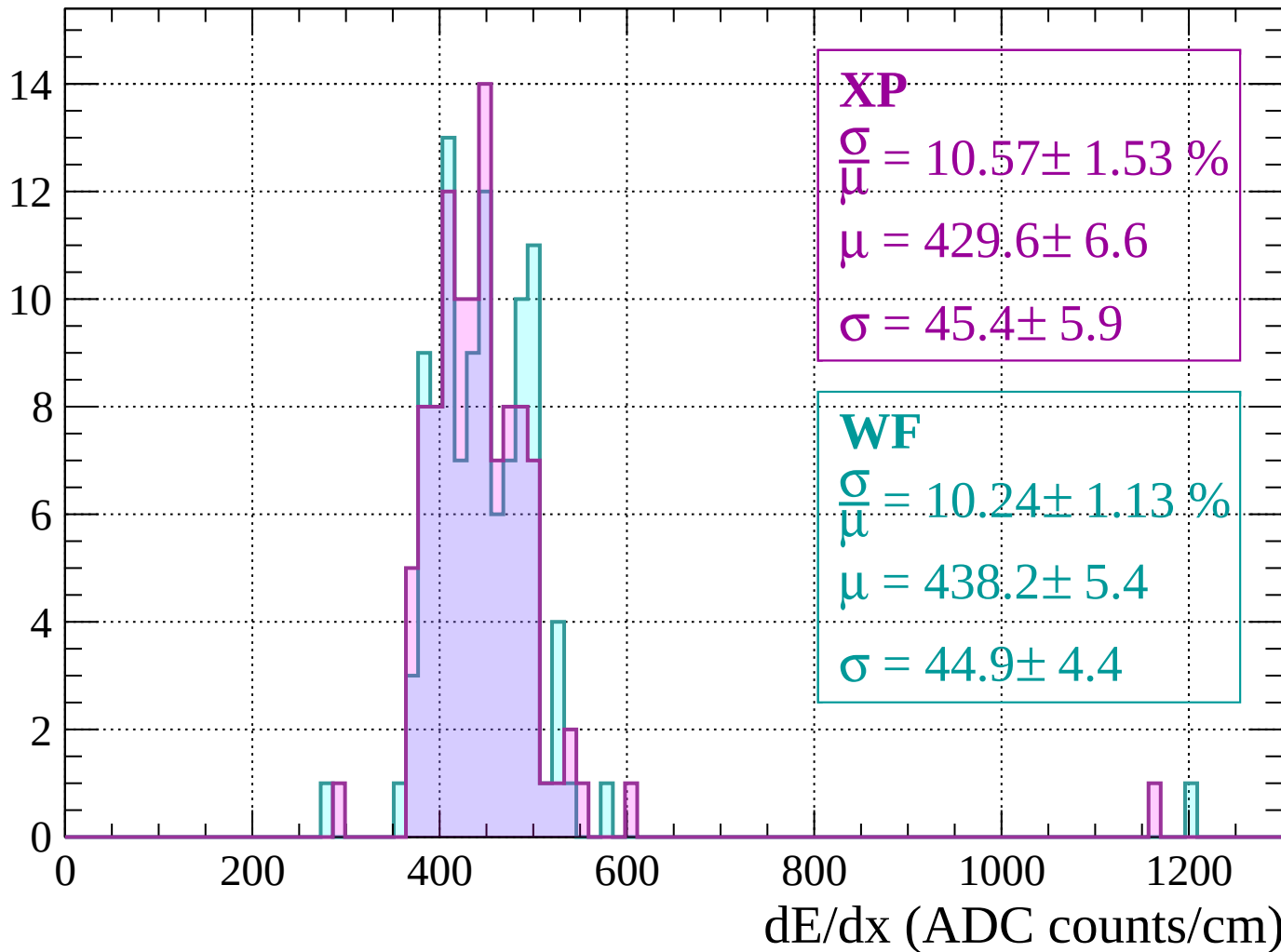
Energy loss | $-1240 < p < -1200$

Count



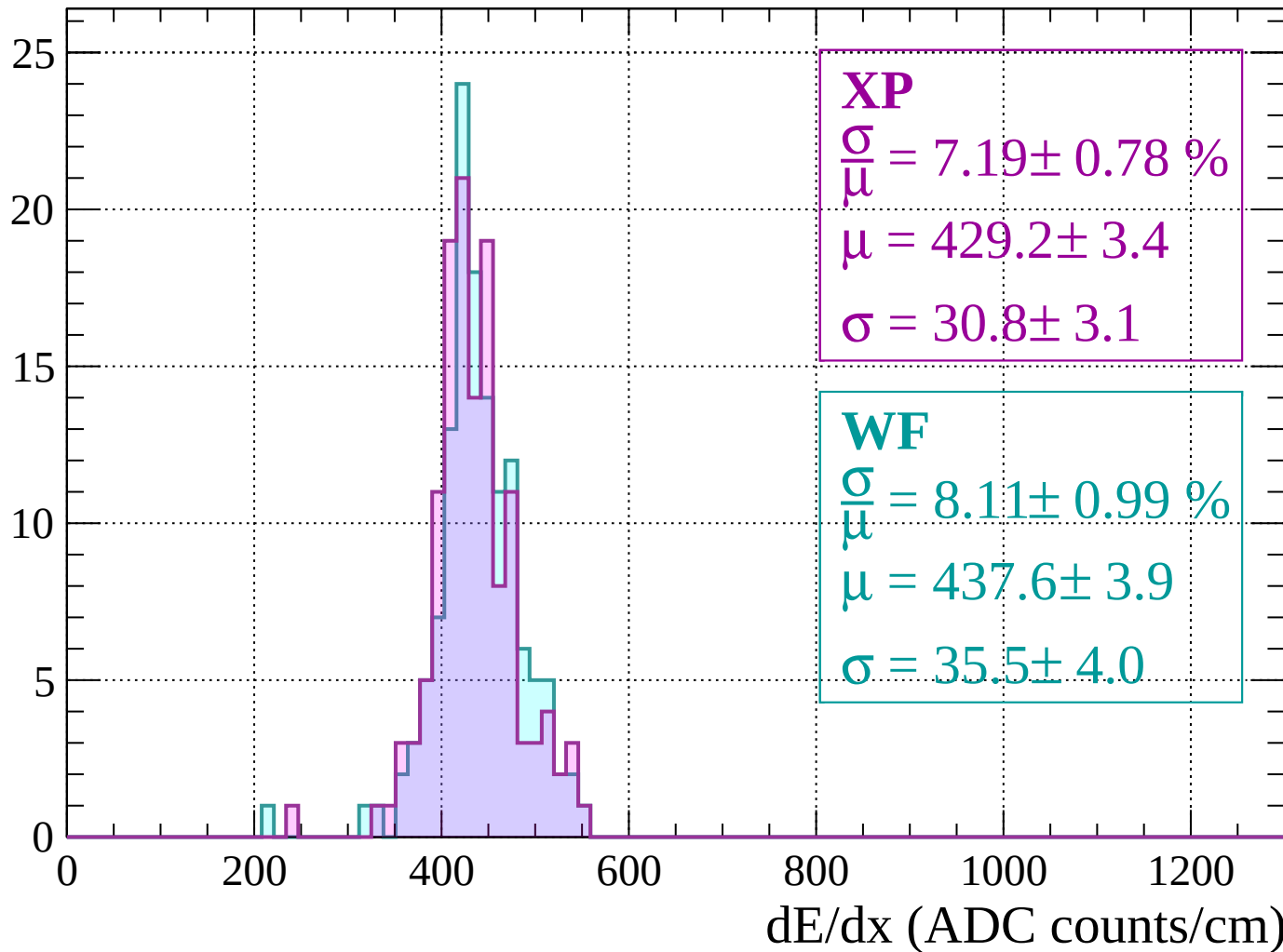
Energy loss | -1200 < p < -1160

Count



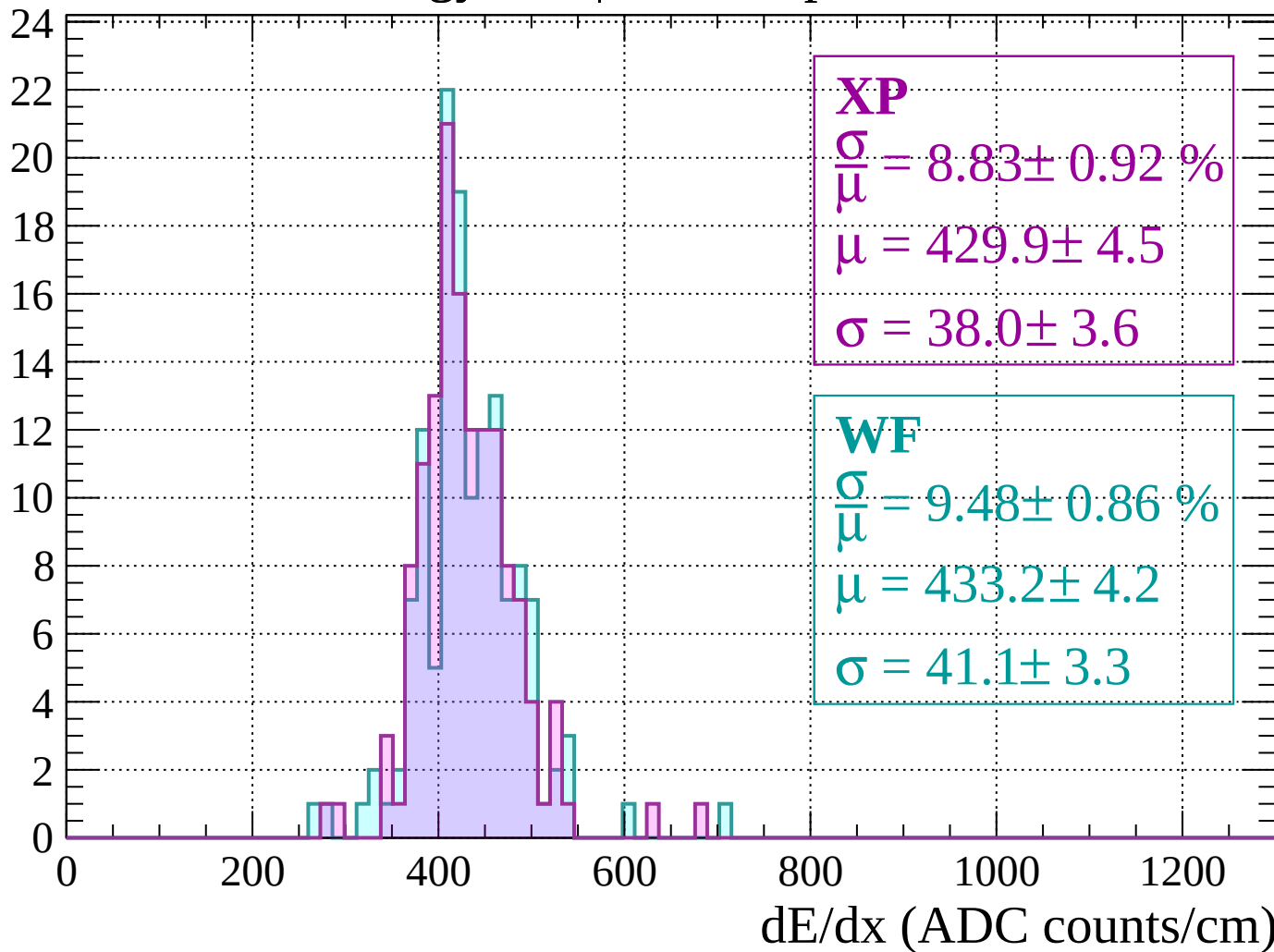
Energy loss | $-1160 < p < -1120$

Count



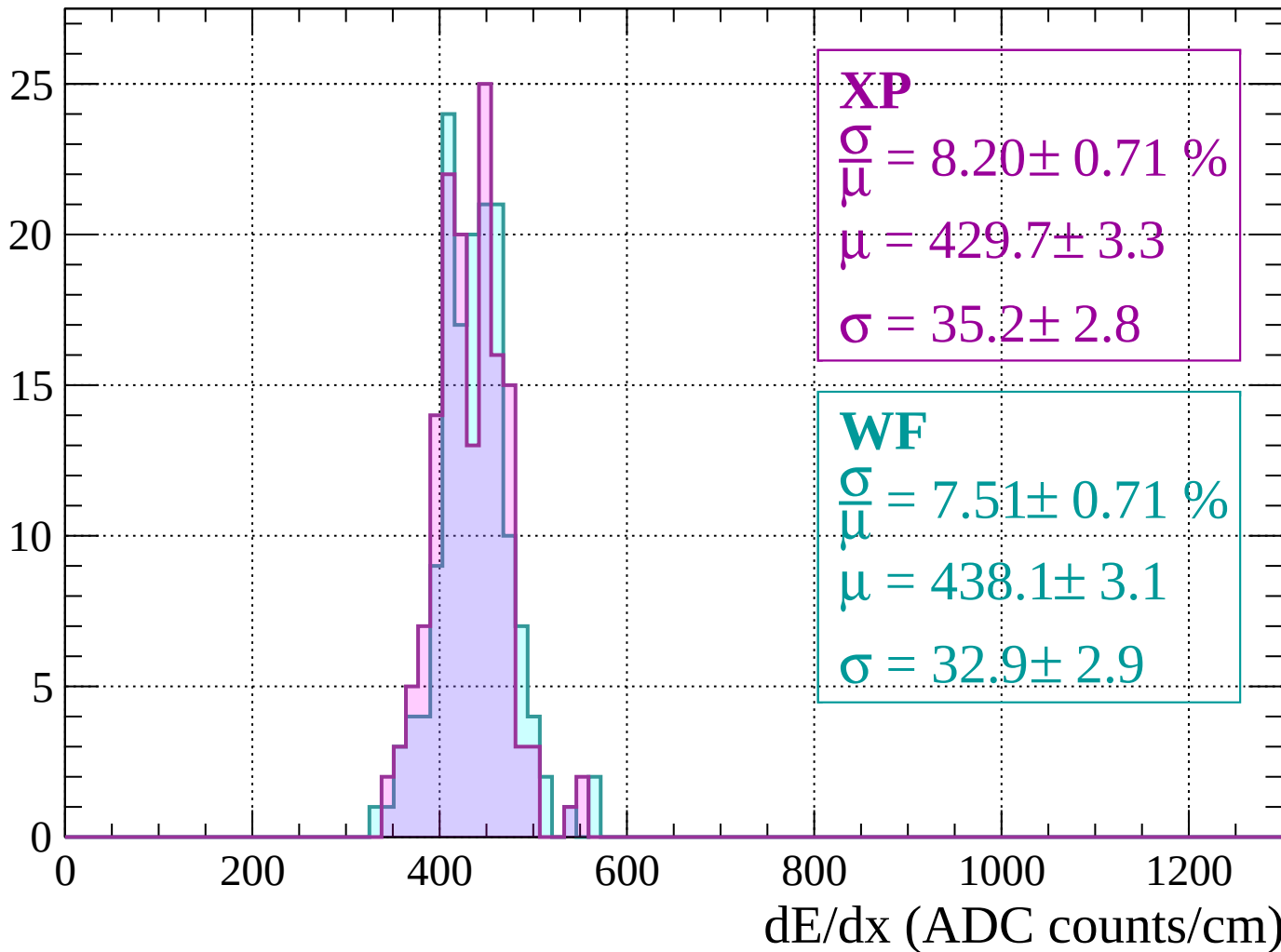
Energy loss | -1120 < p < -1080

Count



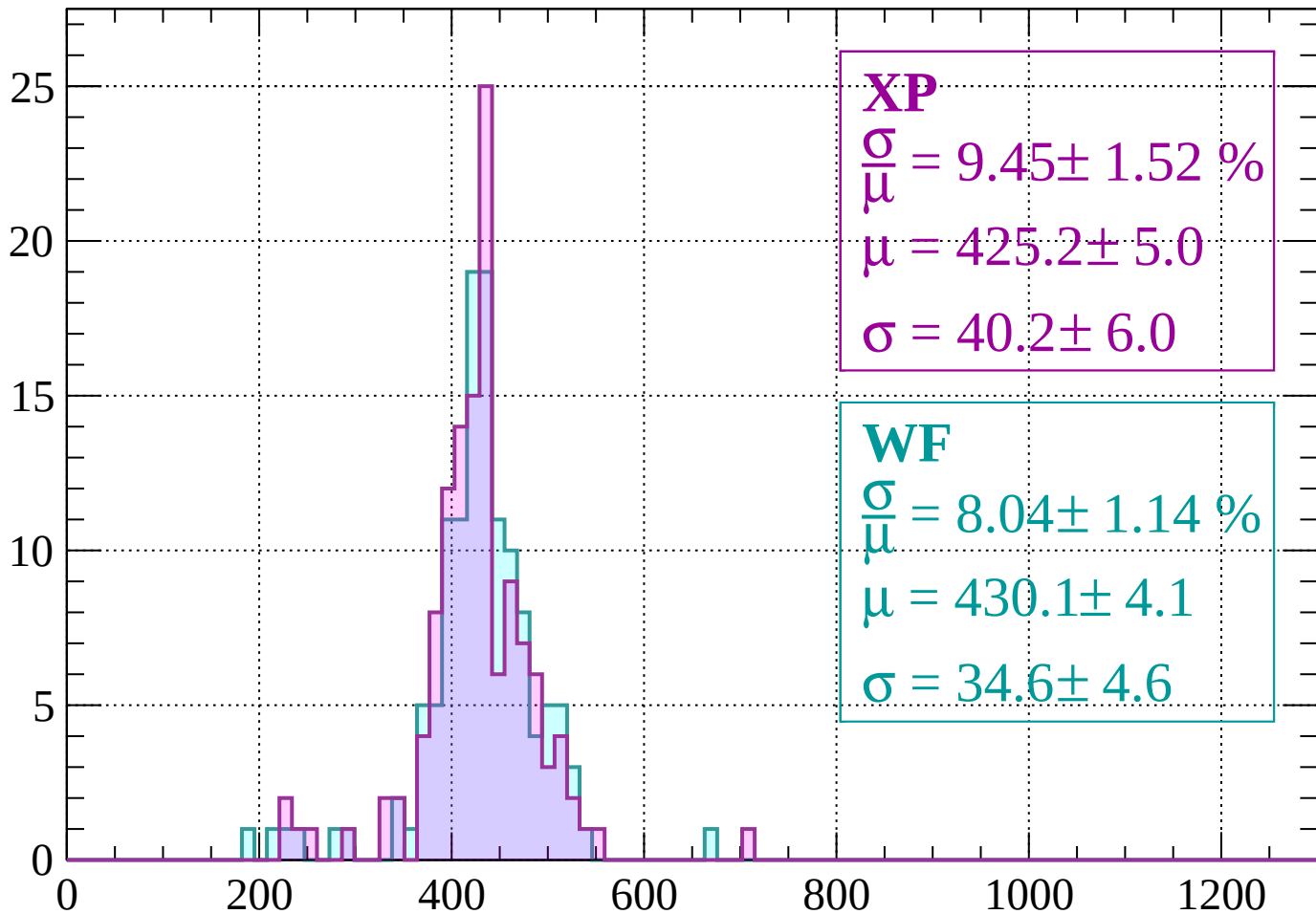
Energy loss | $-1080 < p < -1040$

Count



Energy loss | $-1040 < p < -1000$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 9.45 \pm 1.52 \%$$

$$\mu = 425.2 \pm 5.0$$

$$\sigma = 40.2 \pm 6.0$$

WF

$$\frac{\sigma}{\bar{\mu}} = 8.04 \pm 1.14 \%$$

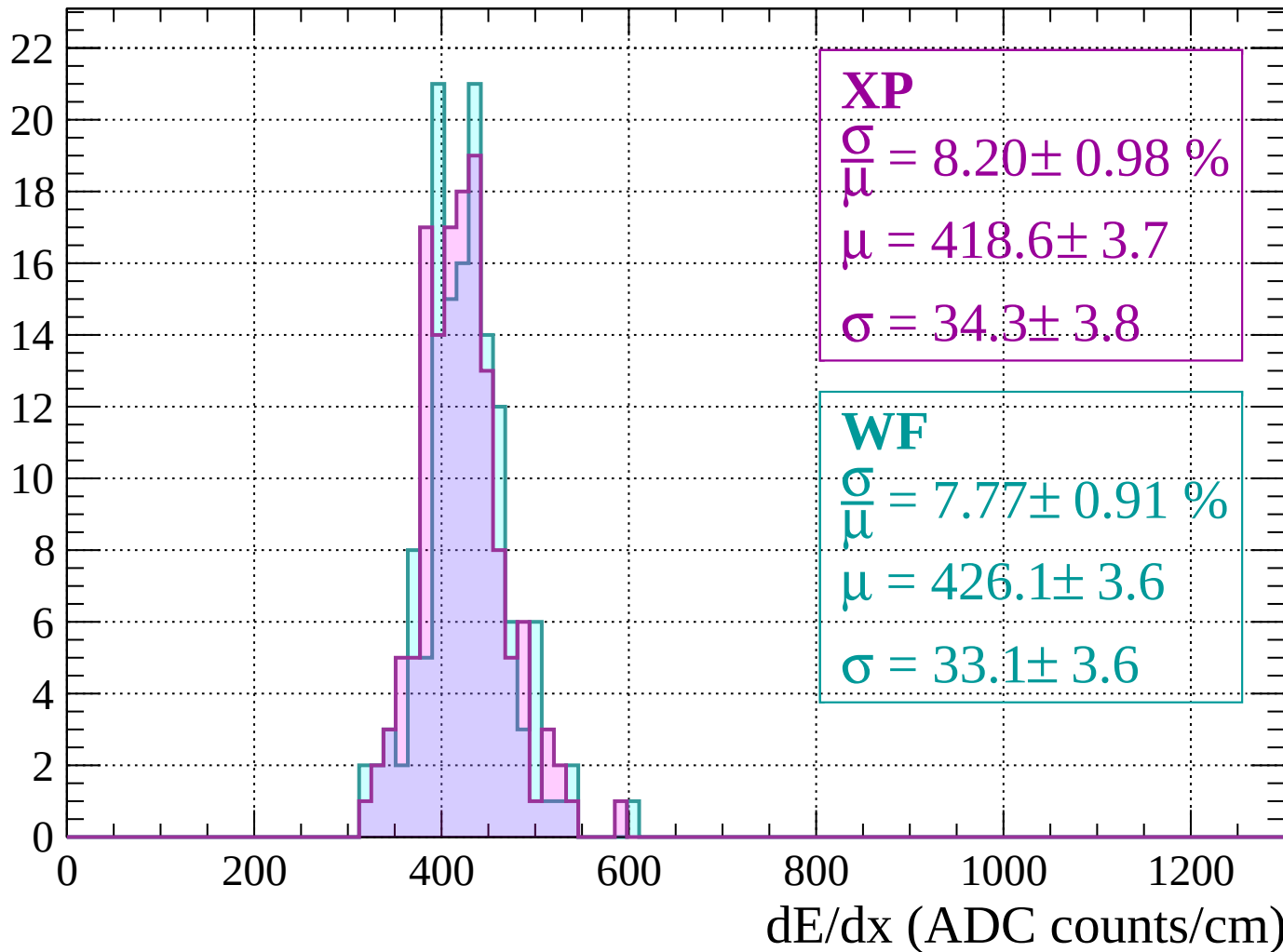
$$\mu = 430.1 \pm 4.1$$

$$\sigma = 34.6 \pm 4.6$$

dE/dx (ADC counts/cm)

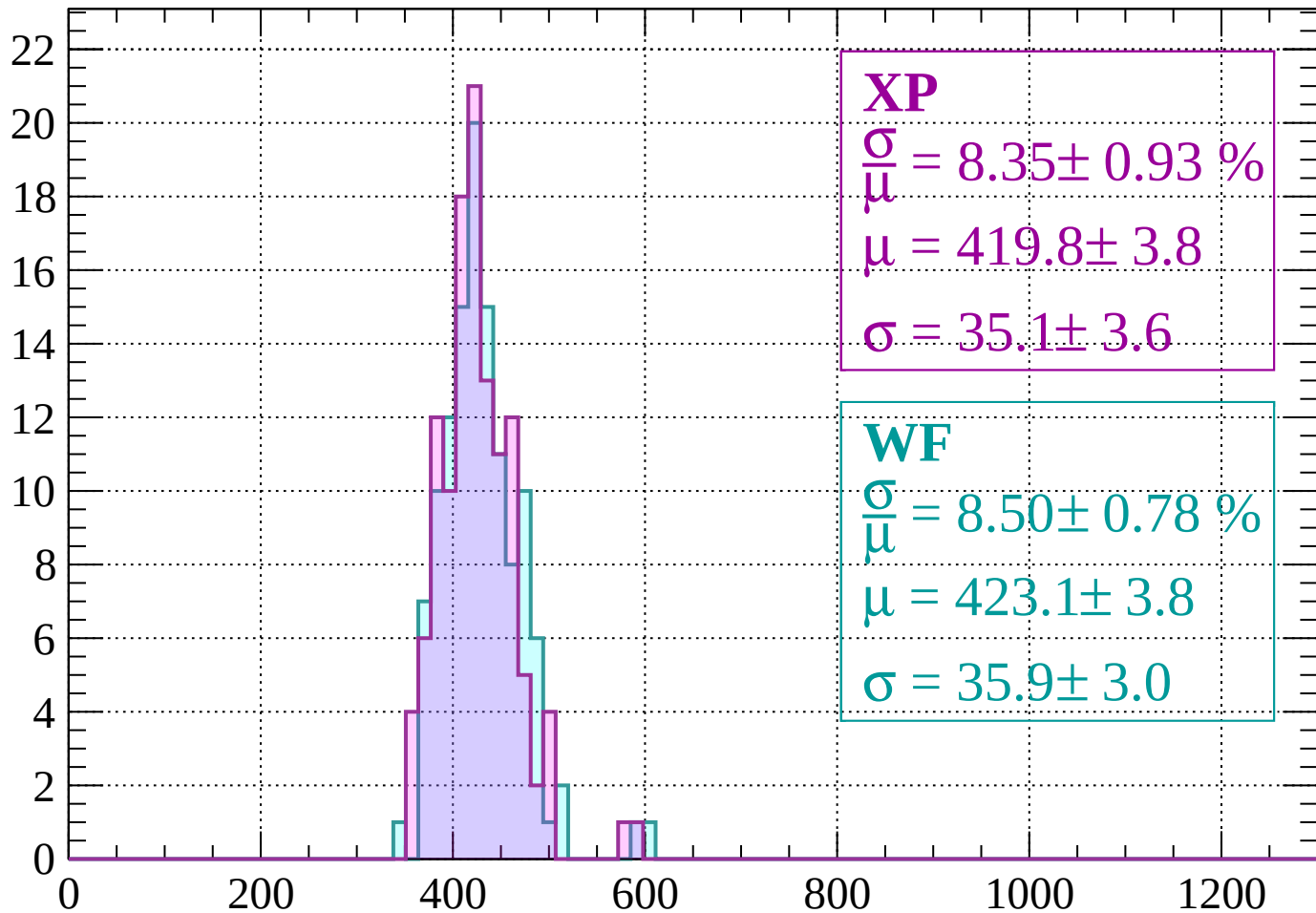
Energy loss | $-1000 < p < -960$

Count



Energy loss | $-960 < p < -920$

Count



XP

$$\frac{\sigma}{\mu} = 8.35 \pm 0.93 \%$$

$$\mu = 419.8 \pm 3.8$$

$$\sigma = 35.1 \pm 3.6$$

WF

$$\frac{\sigma}{\mu} = 8.50 \pm 0.78 \%$$

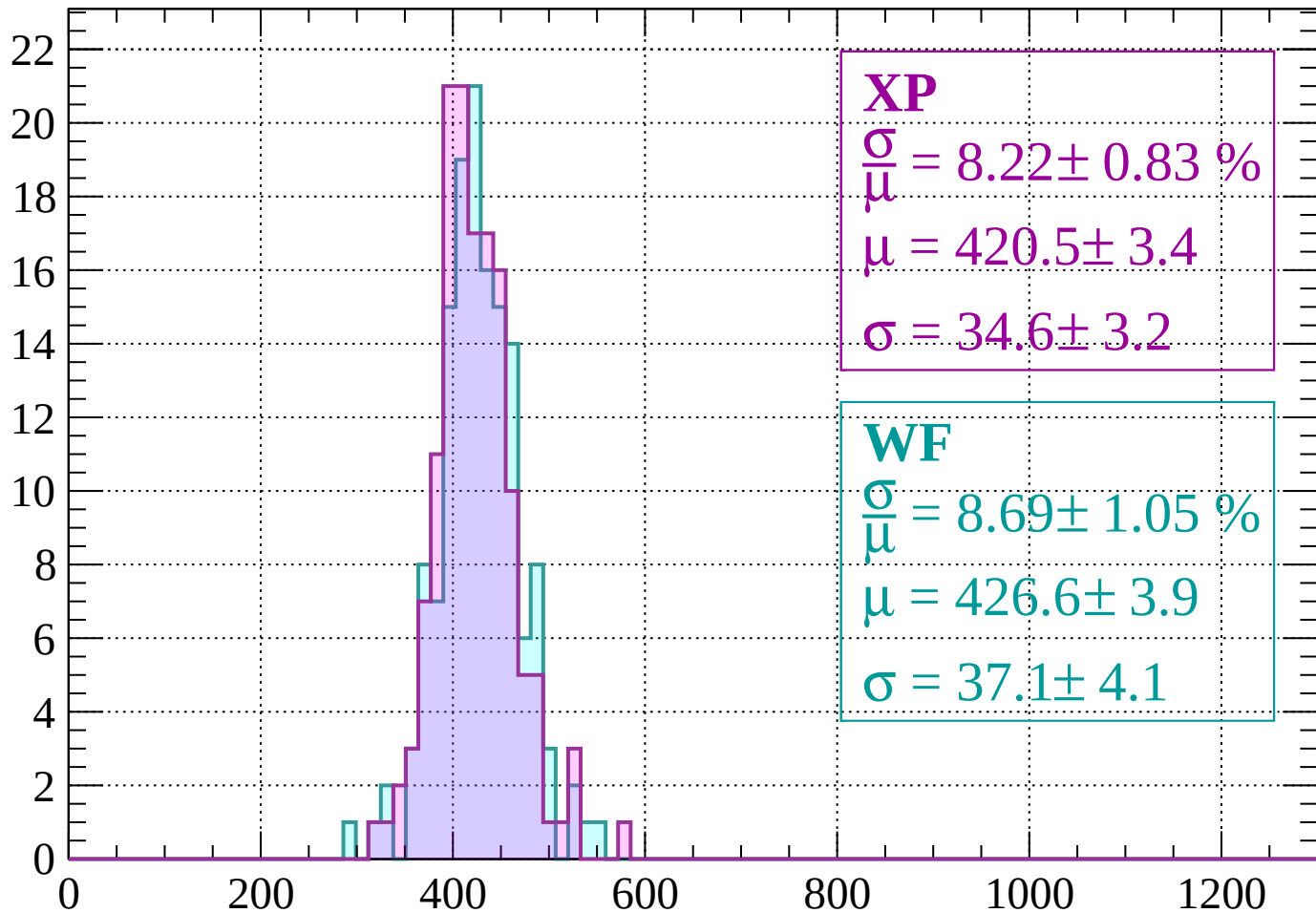
$$\mu = 423.1 \pm 3.8$$

$$\sigma = 35.9 \pm 3.0$$

dE/dx (ADC counts/cm)

Energy loss | $-920 < p < -880$

Count



XP

$$\frac{\sigma}{\mu} = 8.22 \pm 0.83 \%$$

$$\mu = 420.5 \pm 3.4$$

$$\sigma = 34.6 \pm 3.2$$

WF

$$\frac{\sigma}{\mu} = 8.69 \pm 1.05 \%$$

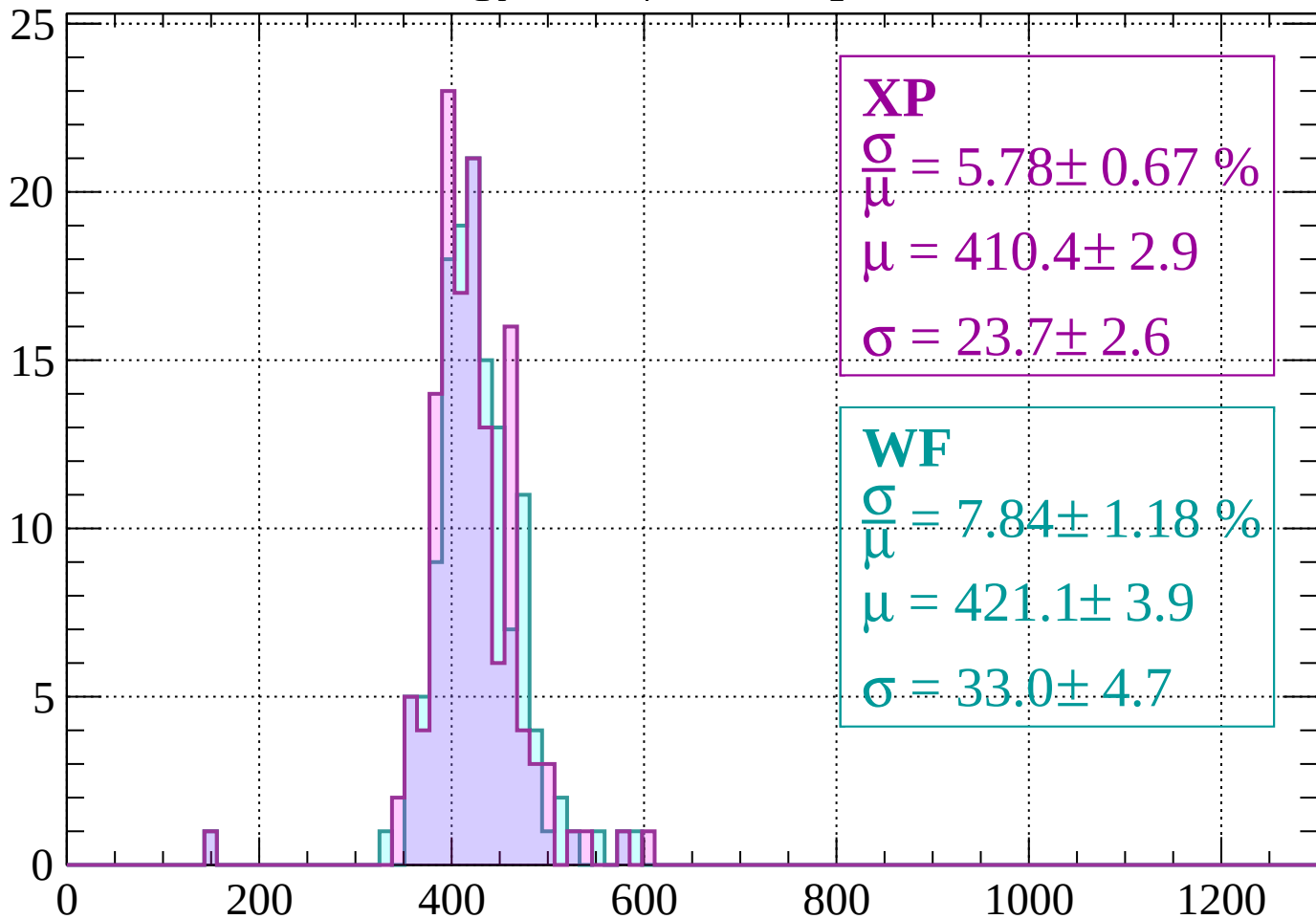
$$\mu = 426.6 \pm 3.9$$

$$\sigma = 37.1 \pm 4.1$$

dE/dx (ADC counts/cm)

Energy loss | $-880 < p < -840$

Count



XP

$$\frac{\sigma}{\mu} = 5.78 \pm 0.67 \%$$

$$\mu = 410.4 \pm 2.9$$

$$\sigma = 23.7 \pm 2.6$$

WF

$$\frac{\sigma}{\mu} = 7.84 \pm 1.18 \%$$

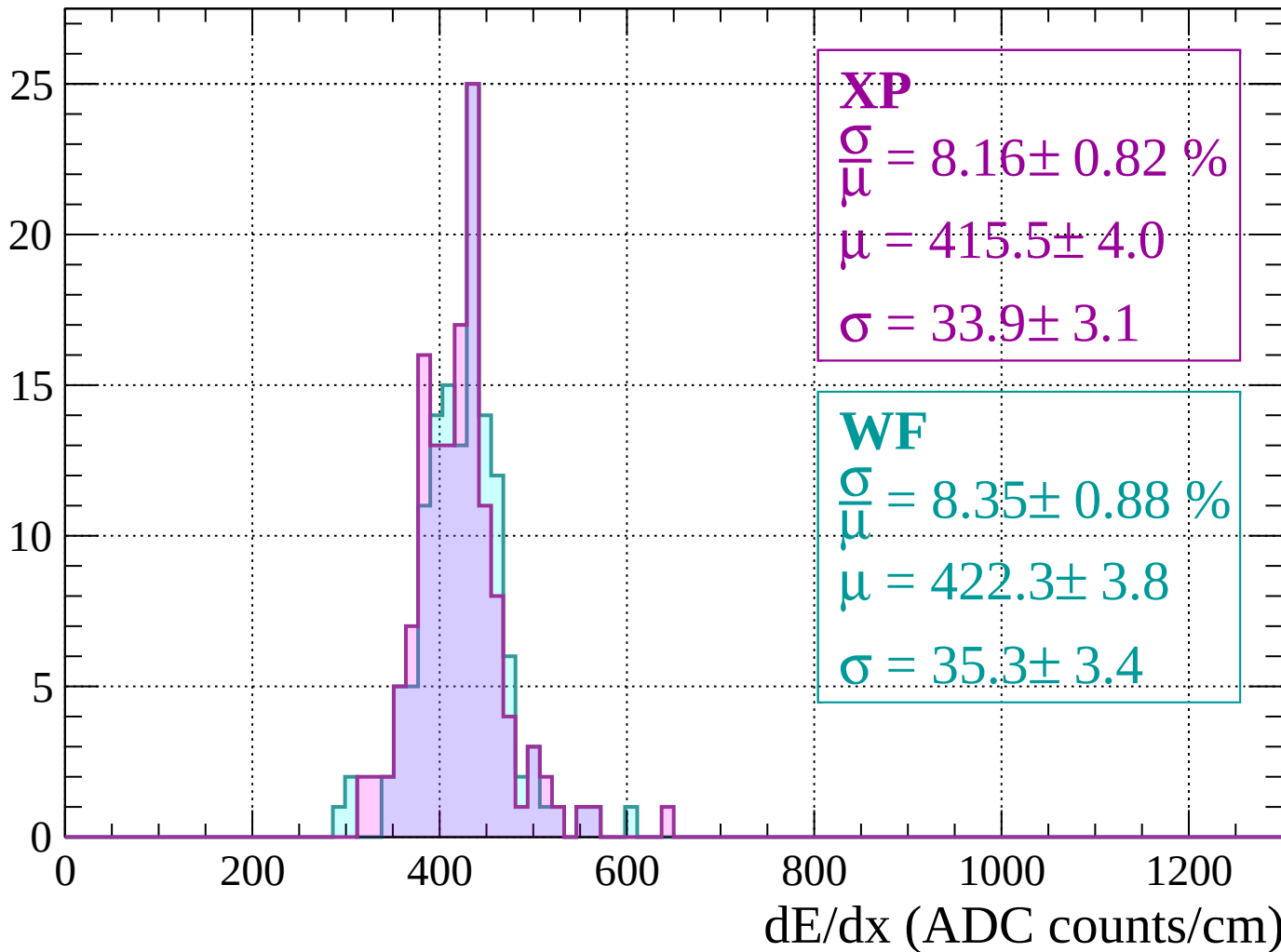
$$\mu = 421.1 \pm 3.9$$

$$\sigma = 33.0 \pm 4.7$$

dE/dx (ADC counts/cm)

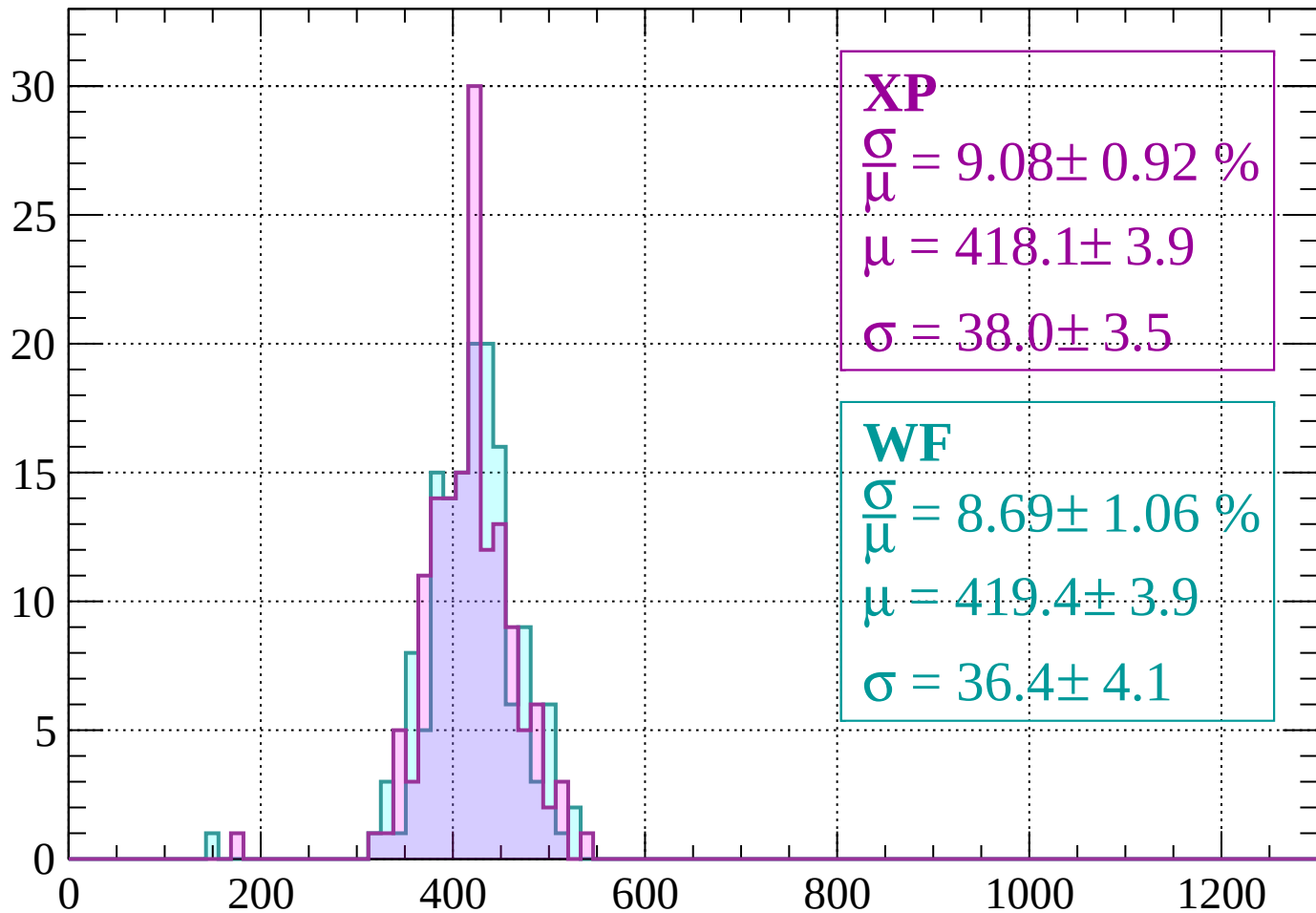
Energy loss | $-840 < p < -800$

Count



Energy loss | $-800 < p < -760$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 9.08 \pm 0.92 \%$$

$$\mu = 418.1 \pm 3.9$$

$$\sigma = 38.0 \pm 3.5$$

WF

$$\frac{\sigma}{\bar{\mu}} = 8.69 \pm 1.06 \%$$

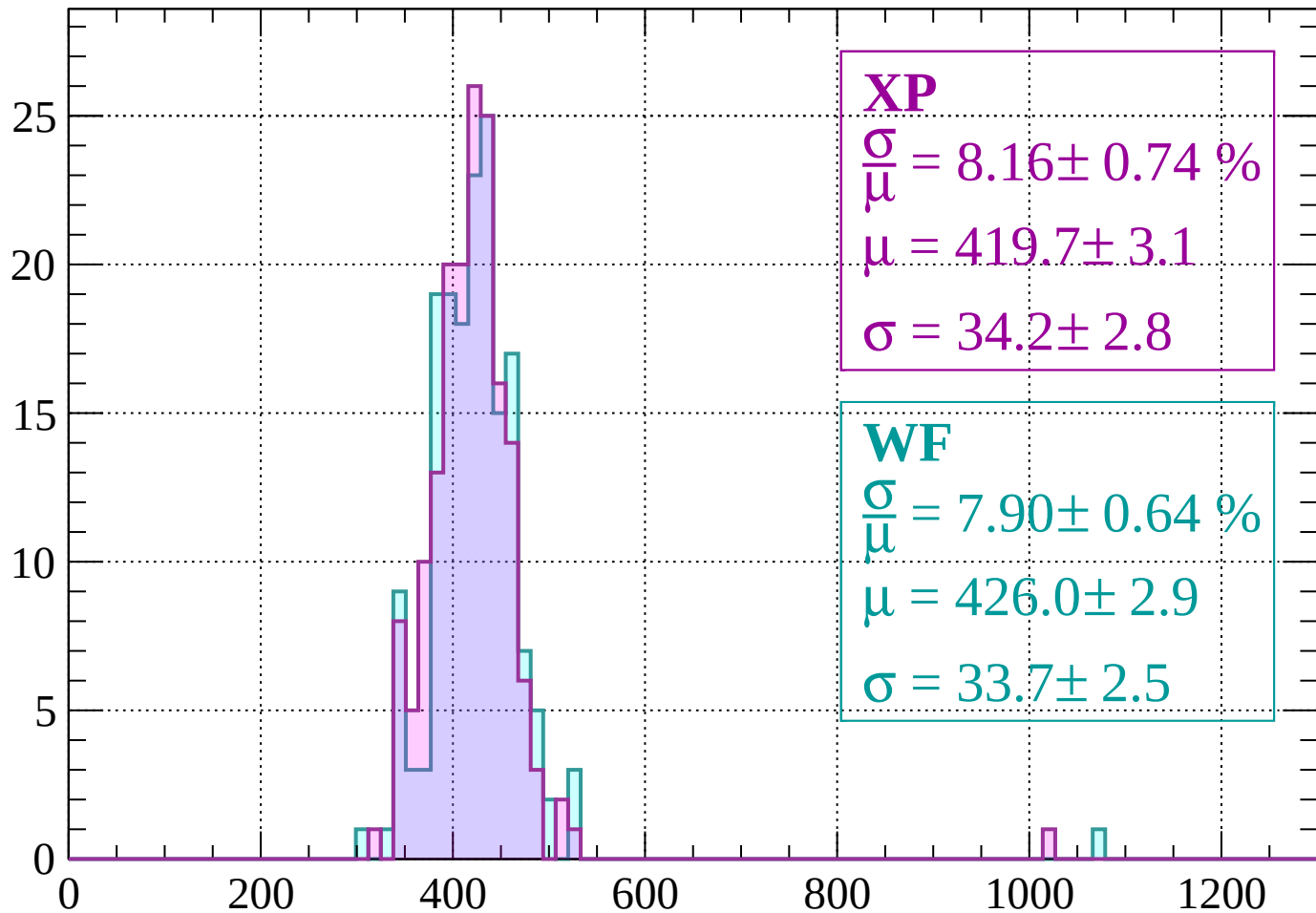
$$\mu = 419.4 \pm 3.9$$

$$\sigma = 36.4 \pm 4.1$$

dE/dx (ADC counts/cm)

Energy loss | $-760 < p < -720$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 8.16 \pm 0.74 \%$$

$$\mu = 419.7 \pm 3.1$$

$$\sigma = 34.2 \pm 2.8$$

WF

$$\frac{\sigma}{\bar{\mu}} = 7.90 \pm 0.64 \%$$

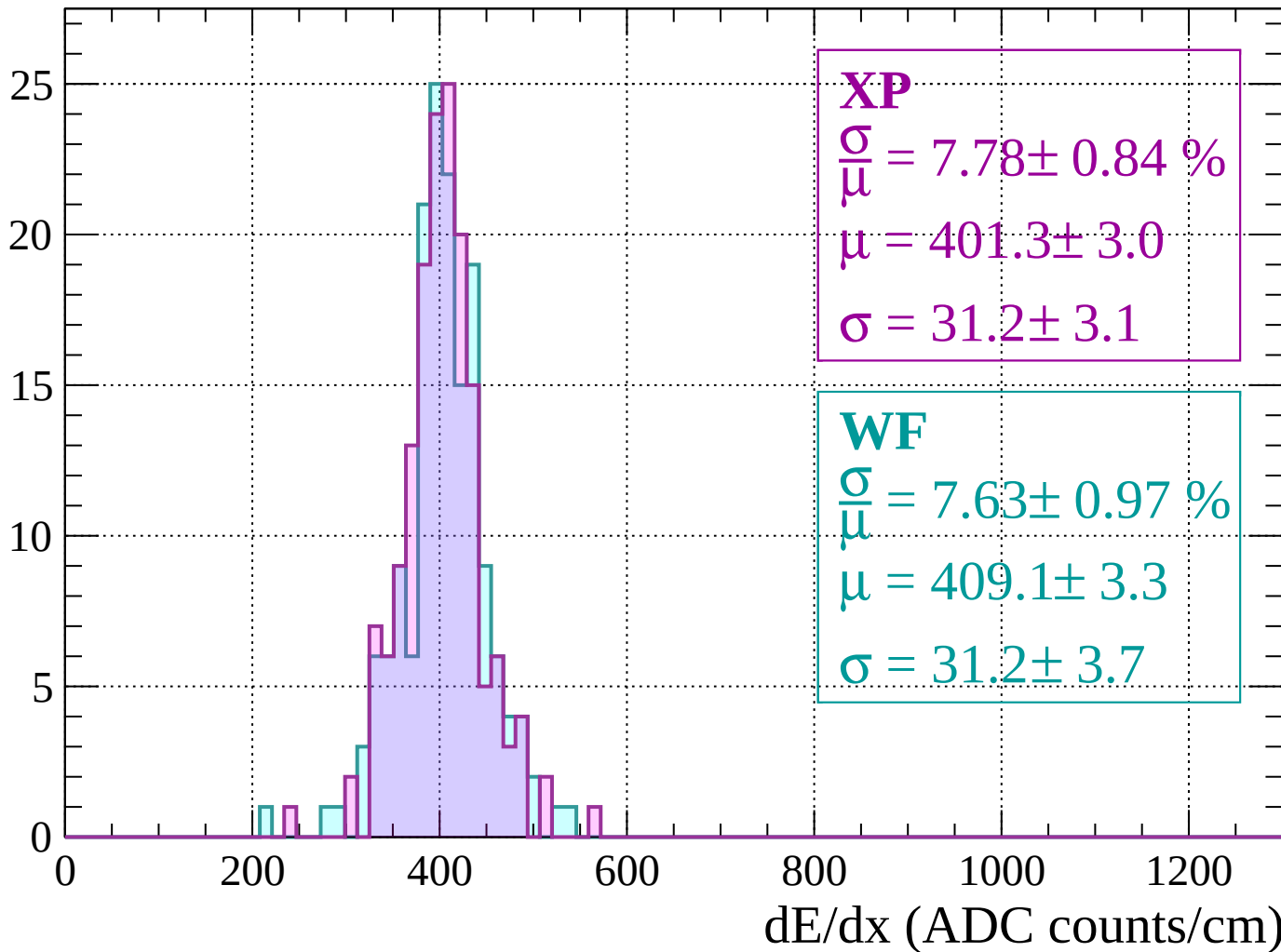
$$\mu = 426.0 \pm 2.9$$

$$\sigma = 33.7 \pm 2.5$$

dE/dx (ADC counts/cm)

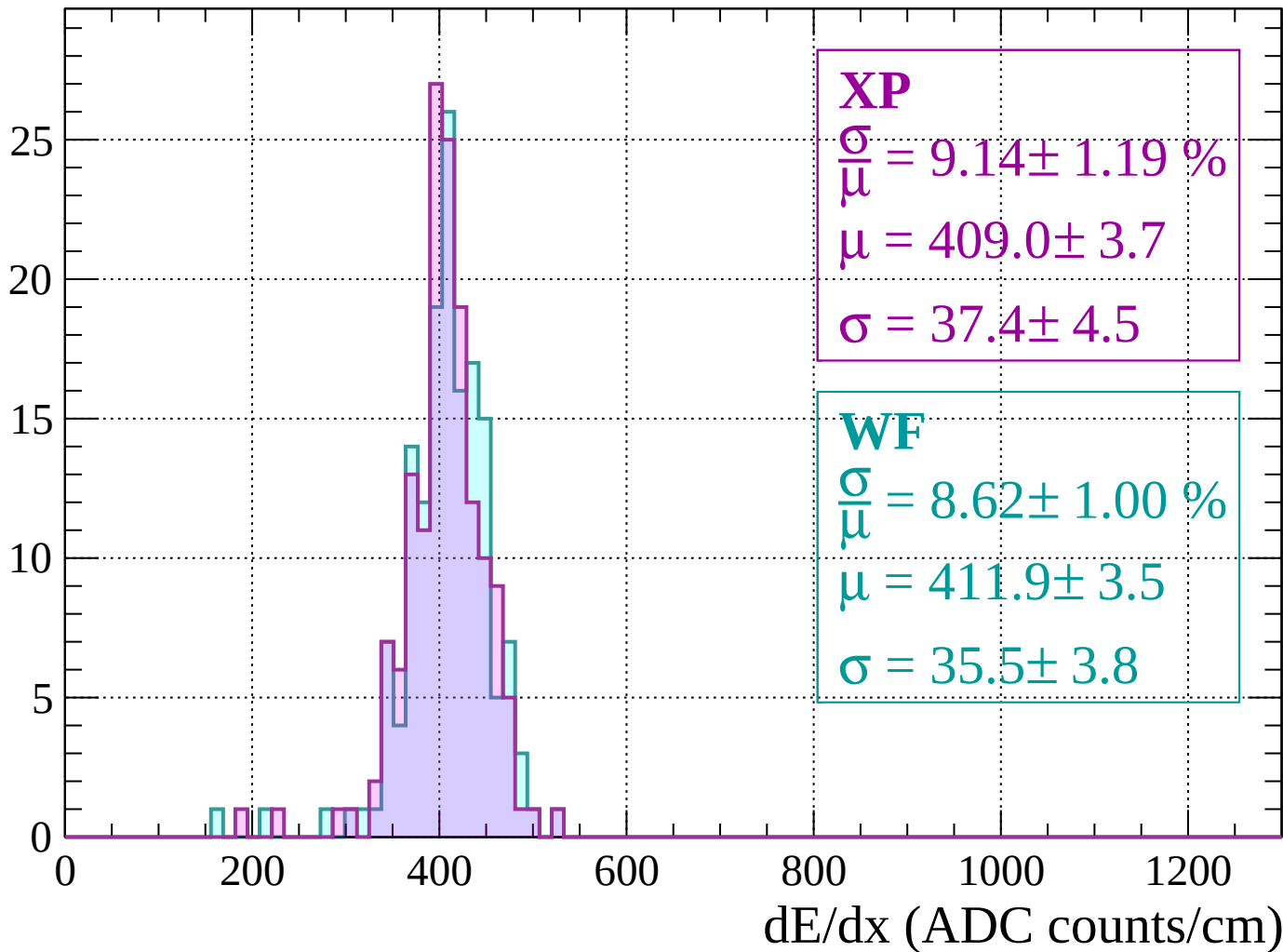
Energy loss | $-720 < p < -680$

Count



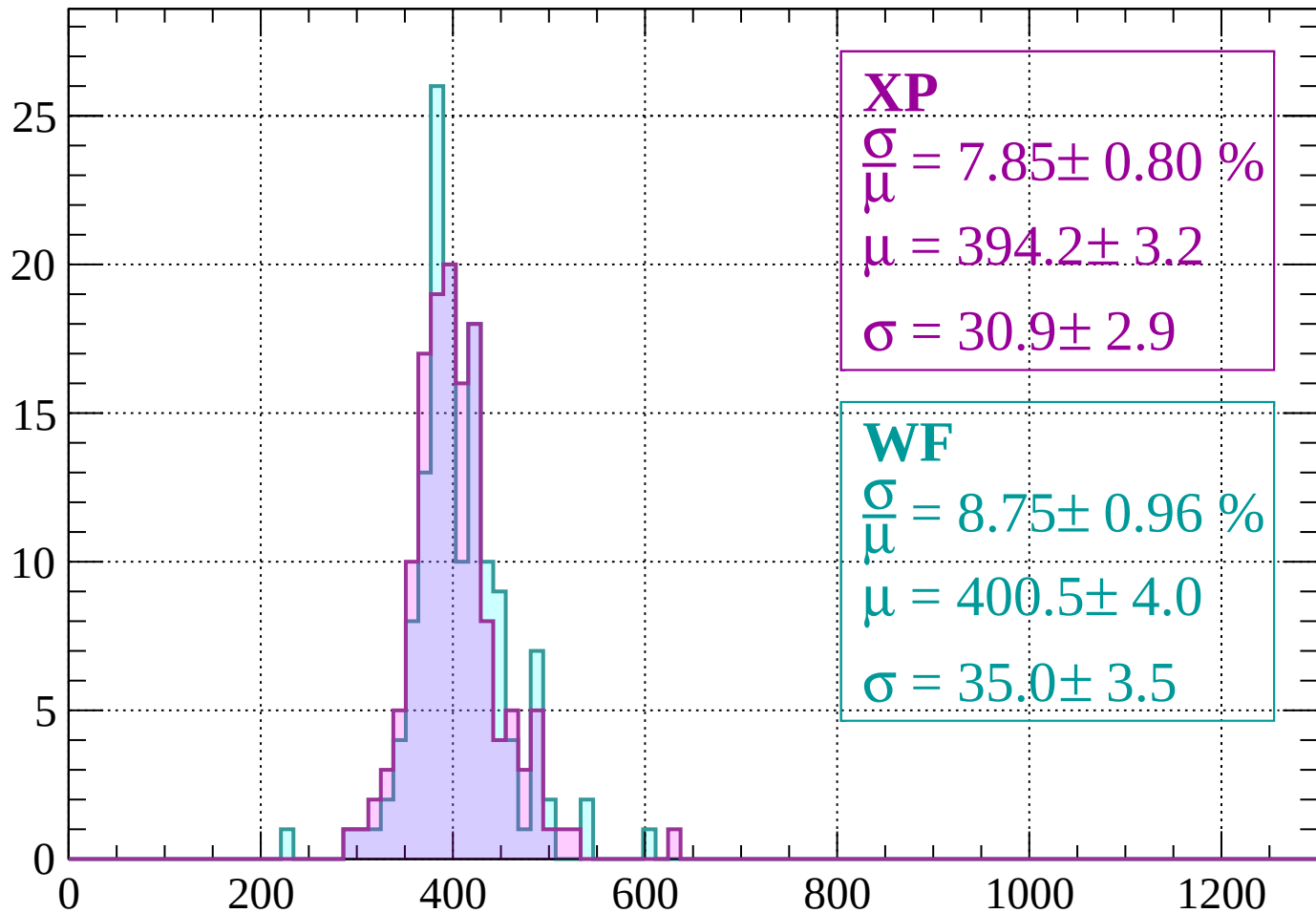
Energy loss | $-680 < p < -640$

Count



Energy loss | $-640 < p < -600$

Count



XP

$$\frac{\sigma}{\mu} = 7.85 \pm 0.80 \%$$

$$\mu = 394.2 \pm 3.2$$

$$\sigma = 30.9 \pm 2.9$$

WF

$$\frac{\sigma}{\mu} = 8.75 \pm 0.96 \%$$

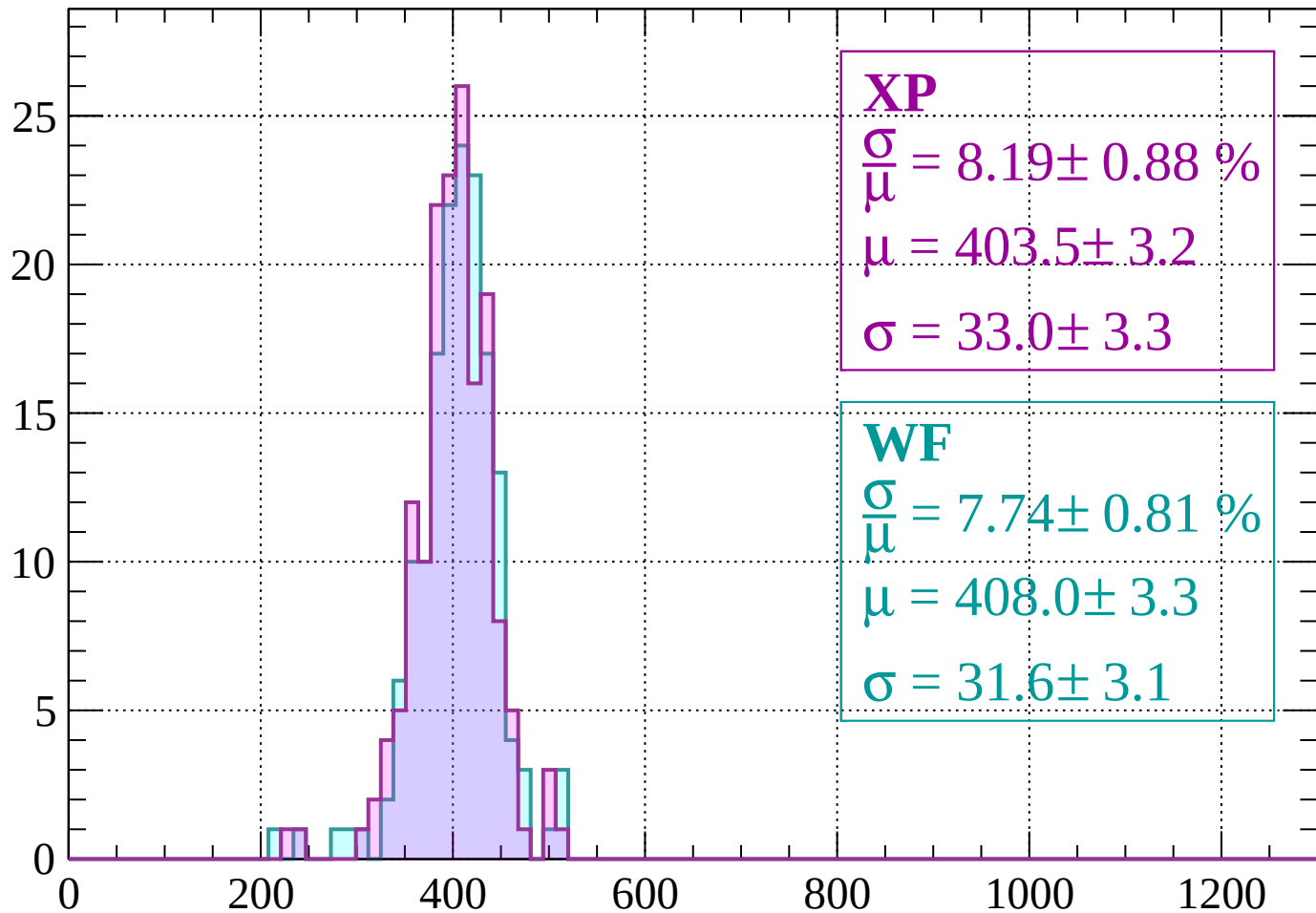
$$\mu = 400.5 \pm 4.0$$

$$\sigma = 35.0 \pm 3.5$$

dE/dx (ADC counts/cm)

Energy loss | $-600 < p < -560$

Count



XP

$$\frac{\sigma}{\mu} = 8.19 \pm 0.88 \%$$

$$\mu = 403.5 \pm 3.2$$

$$\sigma = 33.0 \pm 3.3$$

WF

$$\frac{\sigma}{\mu} = 7.74 \pm 0.81 \%$$

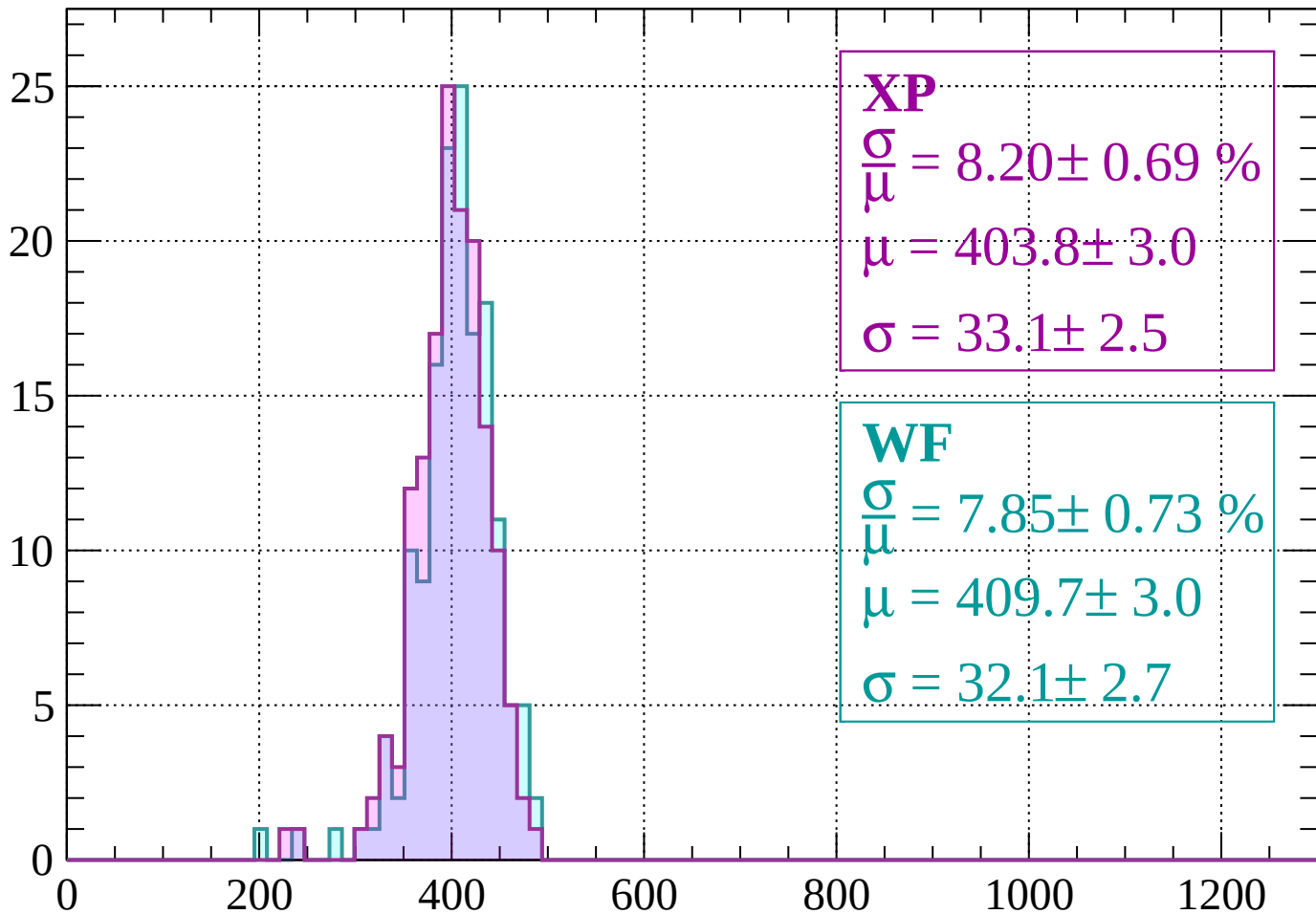
$$\mu = 408.0 \pm 3.3$$

$$\sigma = 31.6 \pm 3.1$$

dE/dx (ADC counts/cm)

Energy loss | $-560 < p < -520$

Count



XP

$$\frac{\sigma}{\mu} = 8.20 \pm 0.69 \%$$

$$\mu = 403.8 \pm 3.0$$

$$\sigma = 33.1 \pm 2.5$$

WF

$$\frac{\sigma}{\mu} = 7.85 \pm 0.73 \%$$

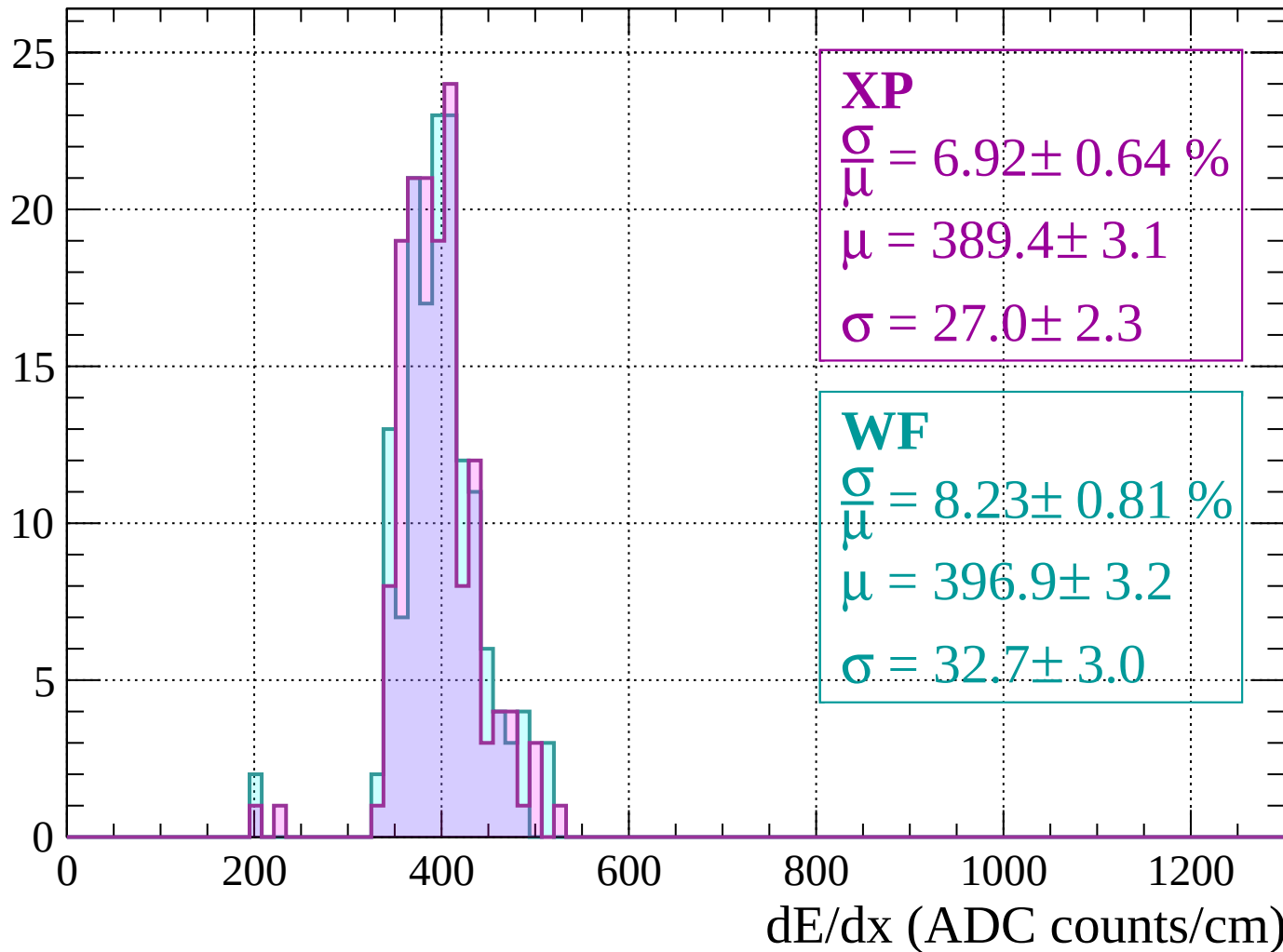
$$\mu = 409.7 \pm 3.0$$

$$\sigma = 32.1 \pm 2.7$$

dE/dx (ADC counts/cm)

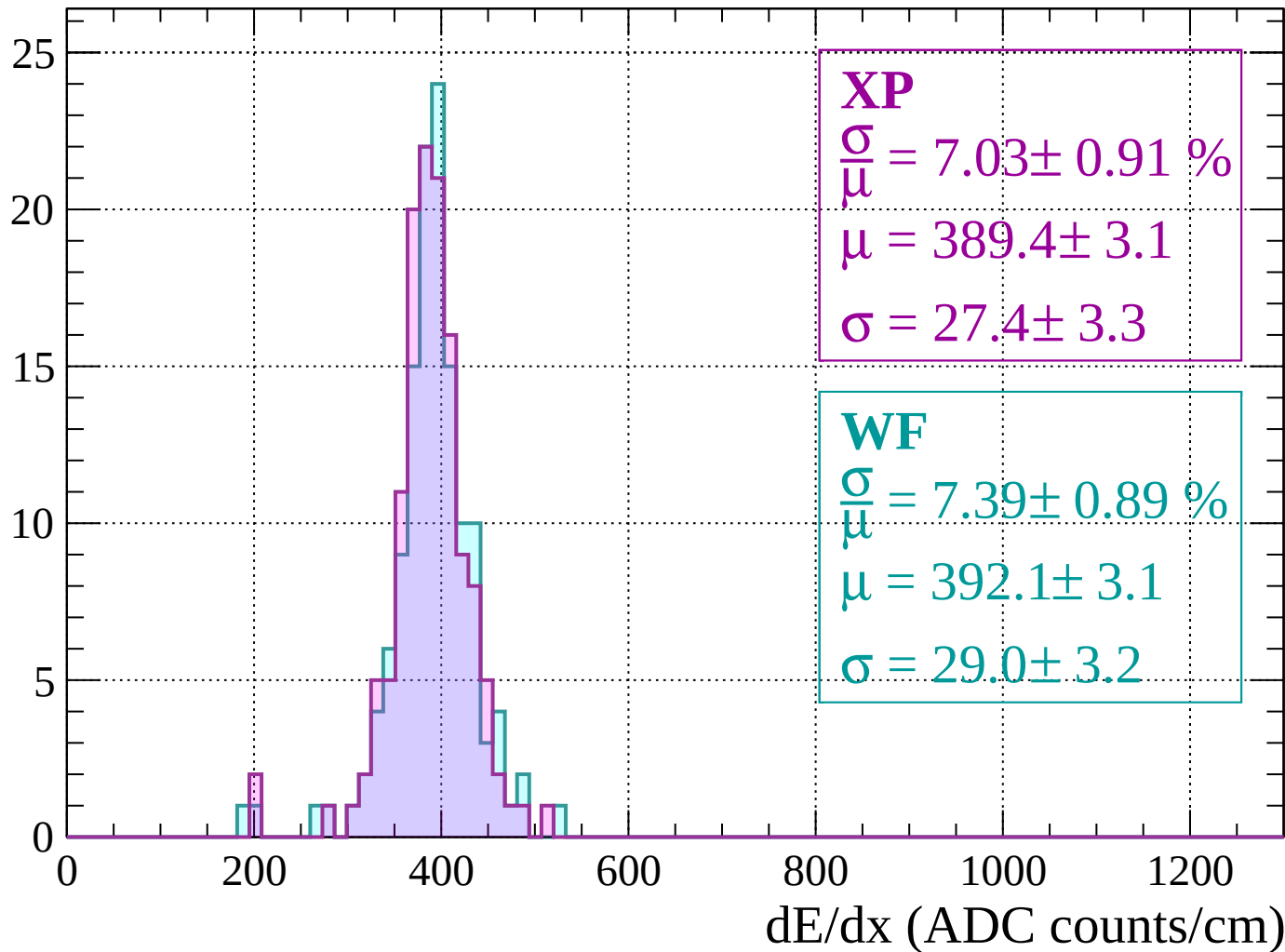
Energy loss | $-520 < p < -480$

Count



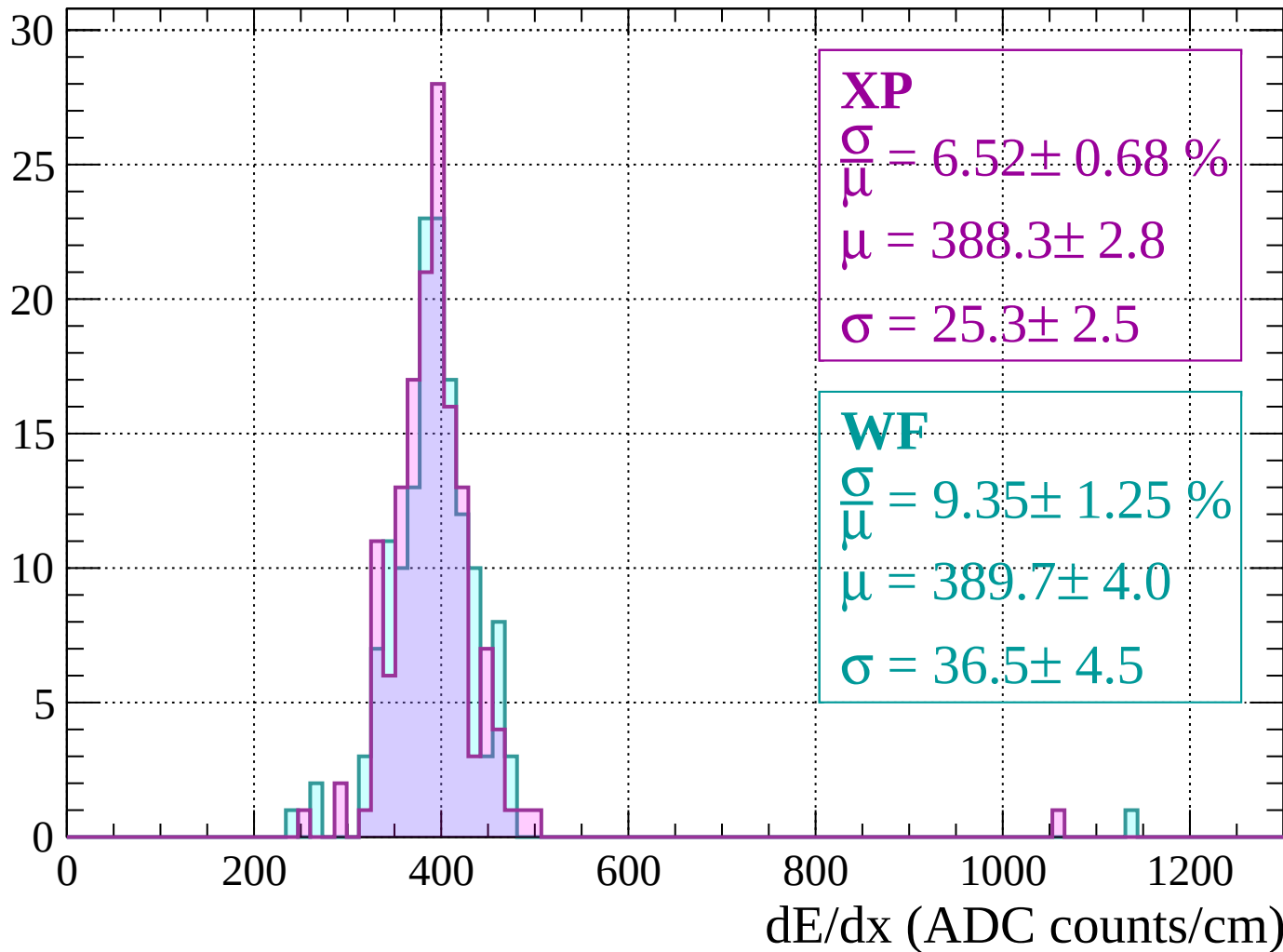
Energy loss | $-480 < p < -440$

Count



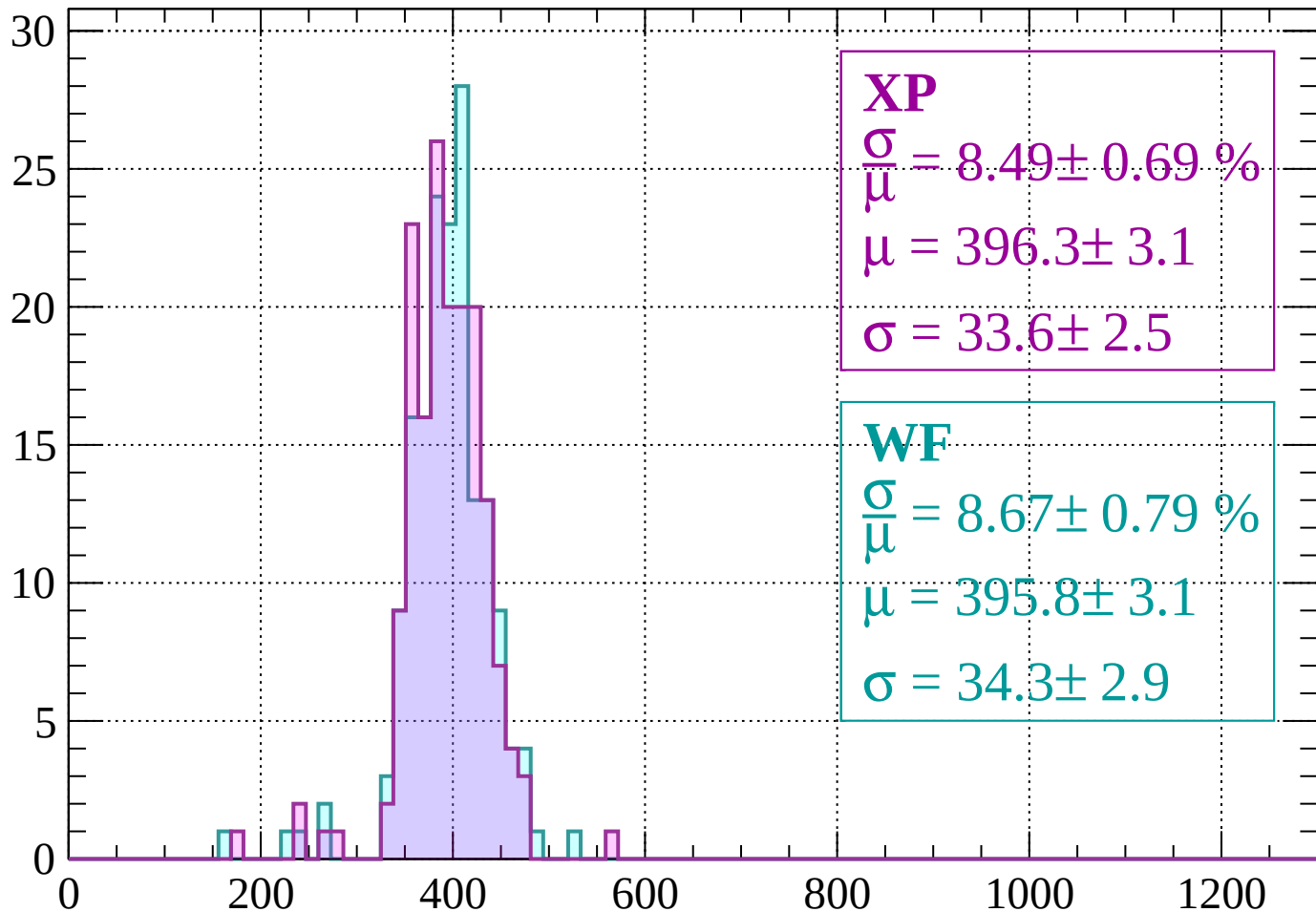
Energy loss | $-440 < p < -400$

Count



Energy loss | $-400 < p < -360$

Count



XP

$$\frac{\sigma}{\mu} = 8.49 \pm 0.69 \%$$

$$\mu = 396.3 \pm 3.1$$

$$\sigma = 33.6 \pm 2.5$$

WF

$$\frac{\sigma}{\mu} = 8.67 \pm 0.79 \%$$

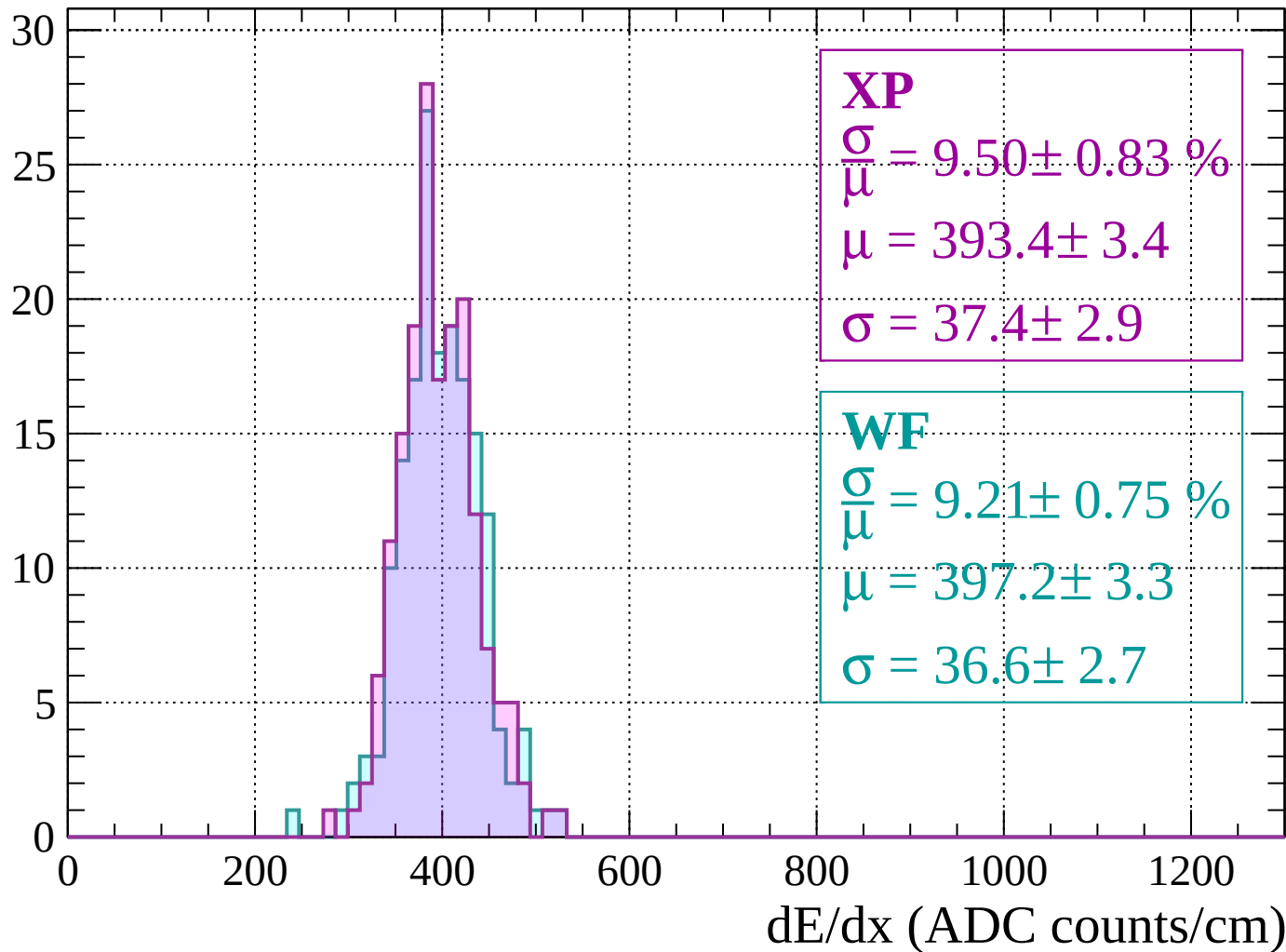
$$\mu = 395.8 \pm 3.1$$

$$\sigma = 34.3 \pm 2.9$$

dE/dx (ADC counts/cm)

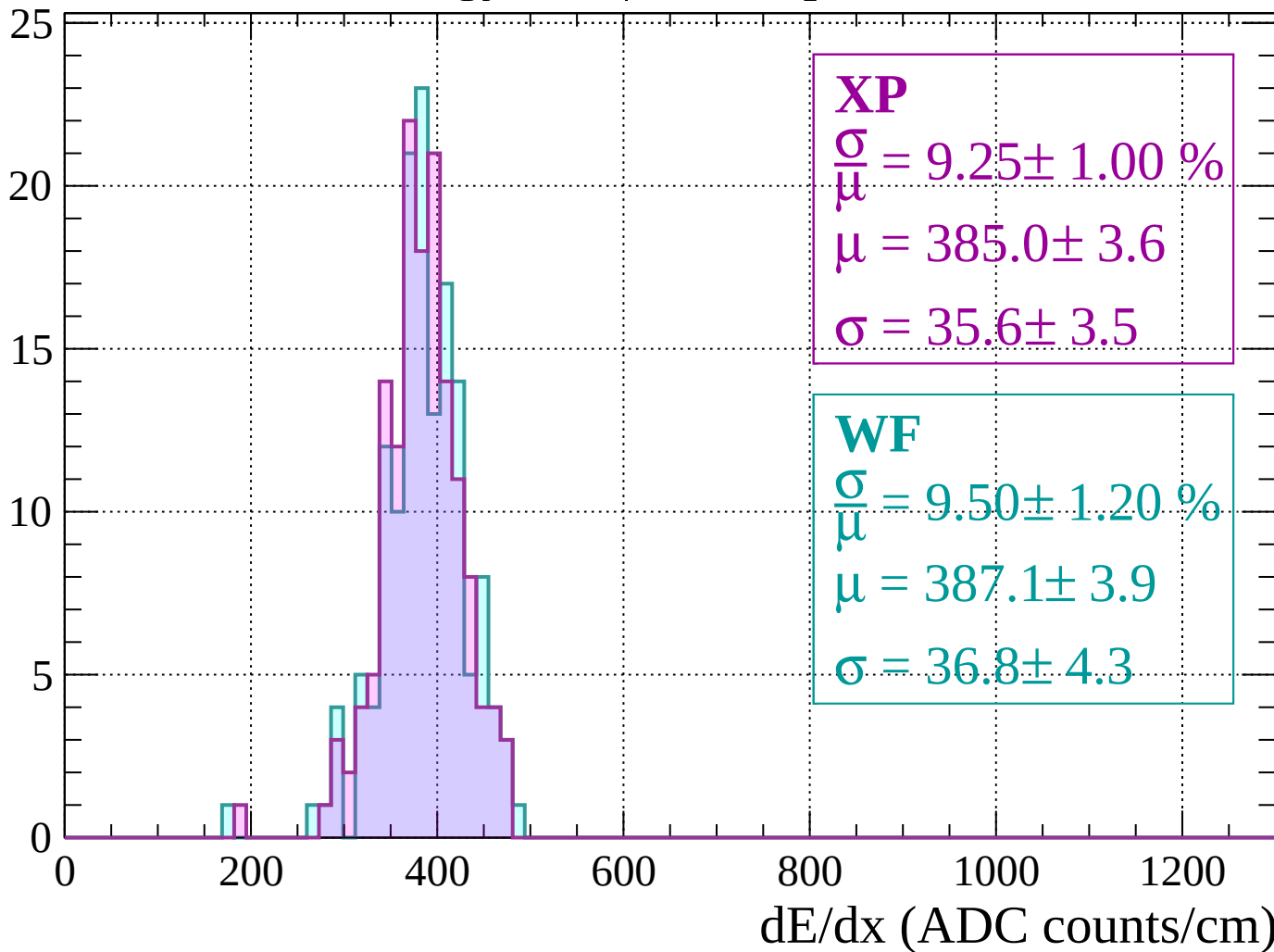
Energy loss | $-360 < p < -320$

Count



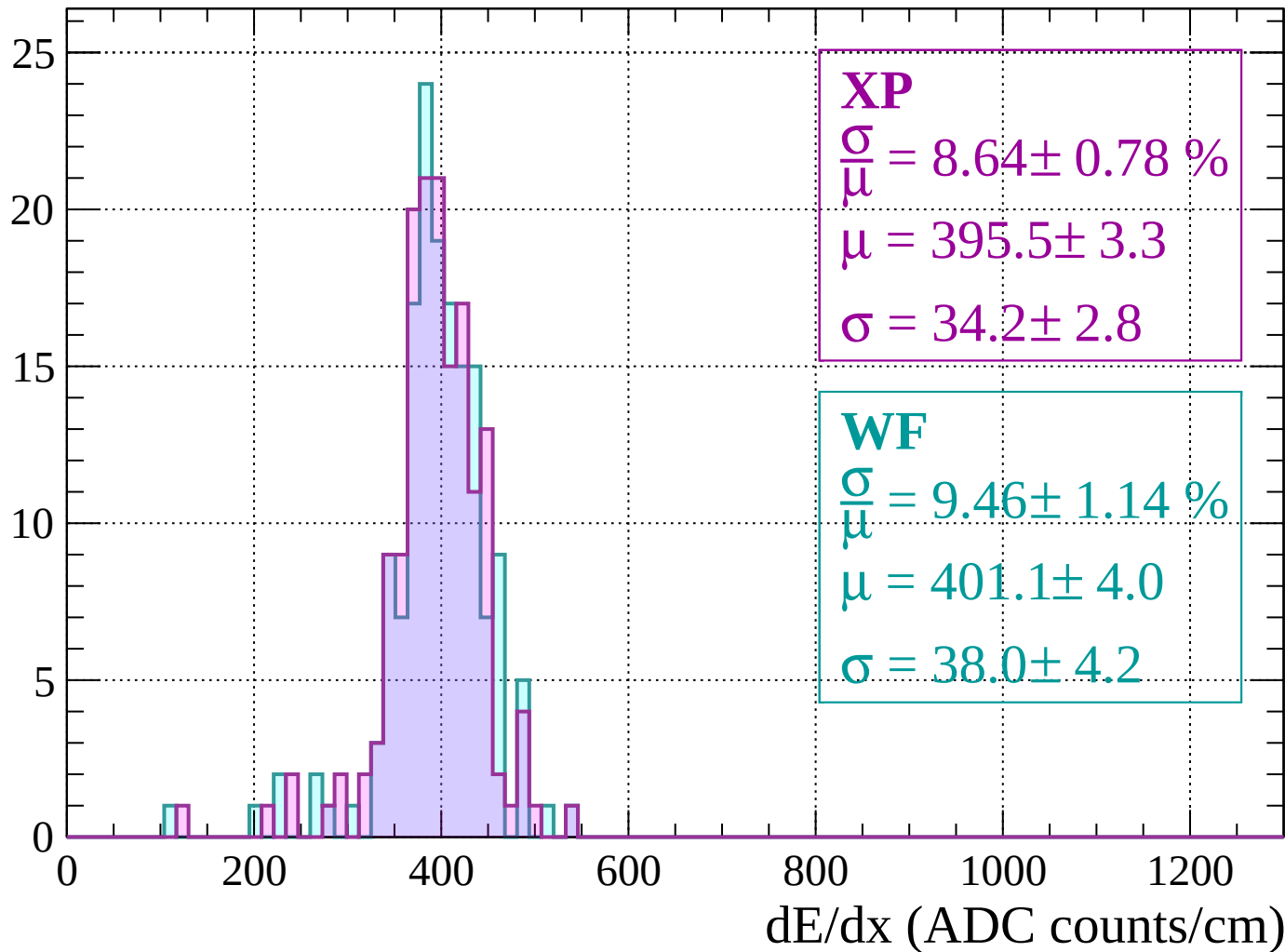
Energy loss | $-320 < p < -280$

Count



Energy loss | $-280 < p < -240$

Count



Energy loss | $-240 < p < -200$

Count

XP

$$\frac{\sigma}{\mu} = 7.95 \pm 0.86 \%$$

$$\mu = 411.1 \pm 3.6$$

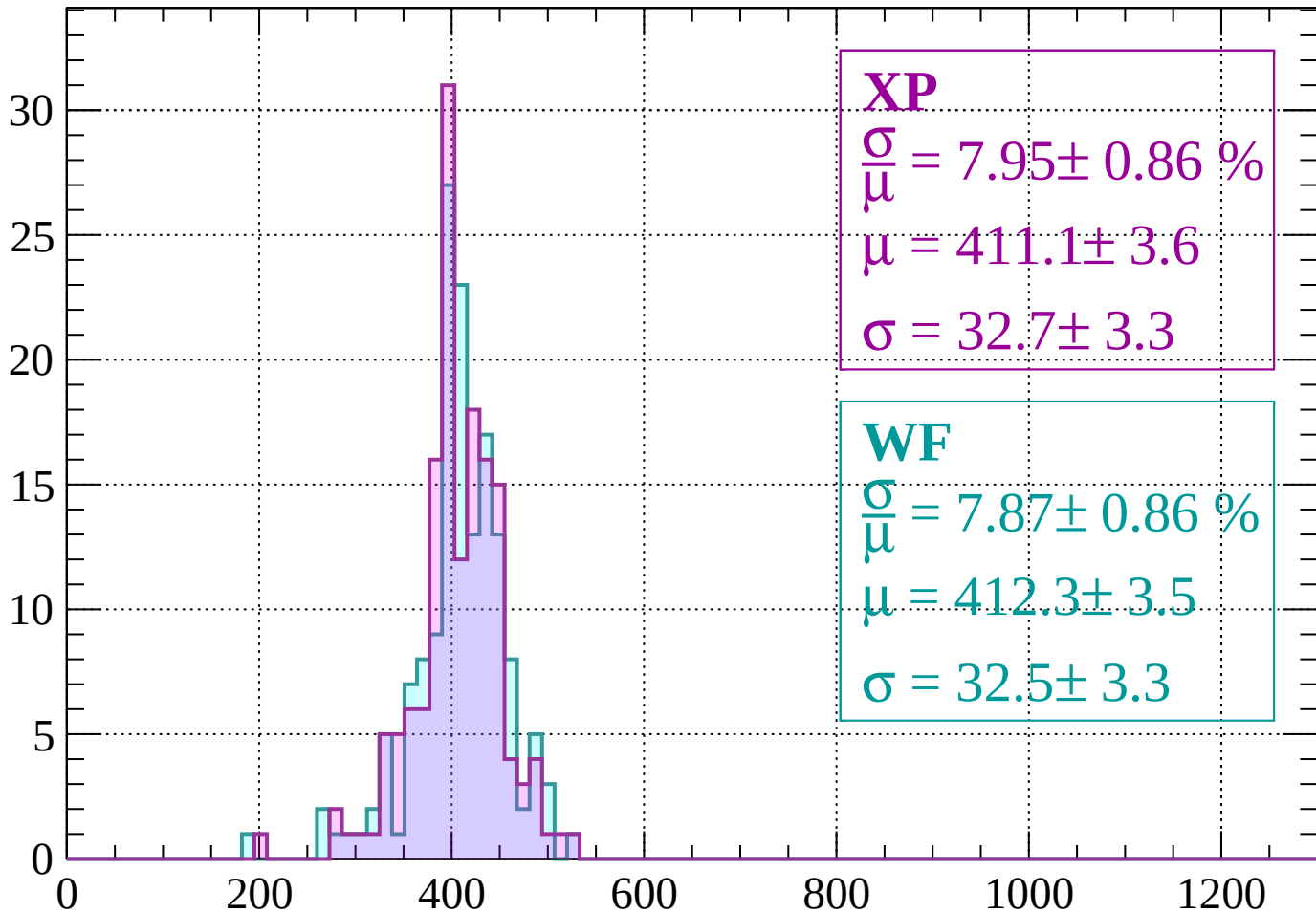
$$\sigma = 32.7 \pm 3.3$$

WF

$$\frac{\sigma}{\mu} = 7.87 \pm 0.86 \%$$

$$\mu = 412.3 \pm 3.5$$

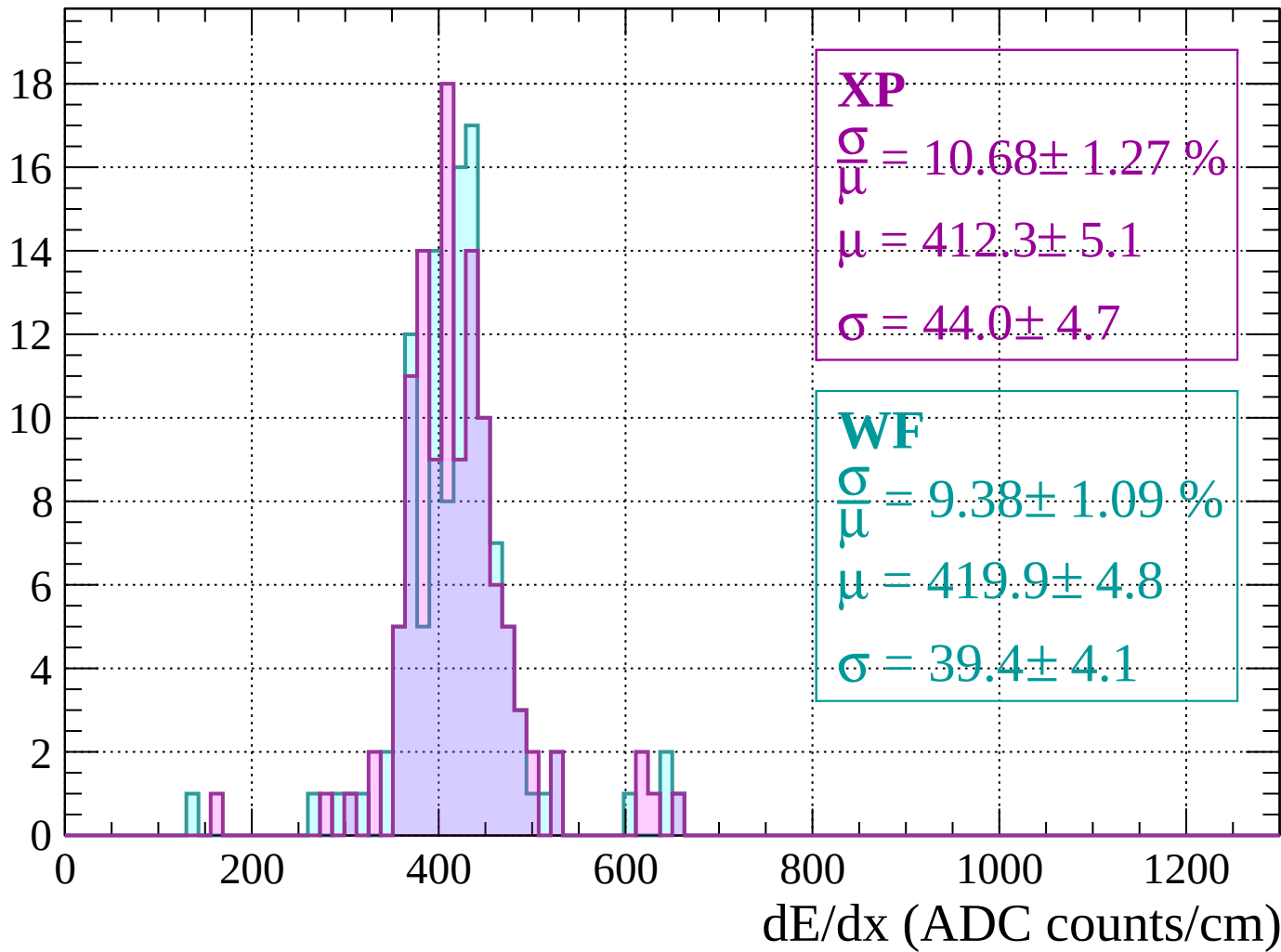
$$\sigma = 32.5 \pm 3.3$$



dE/dx (ADC counts/cm)

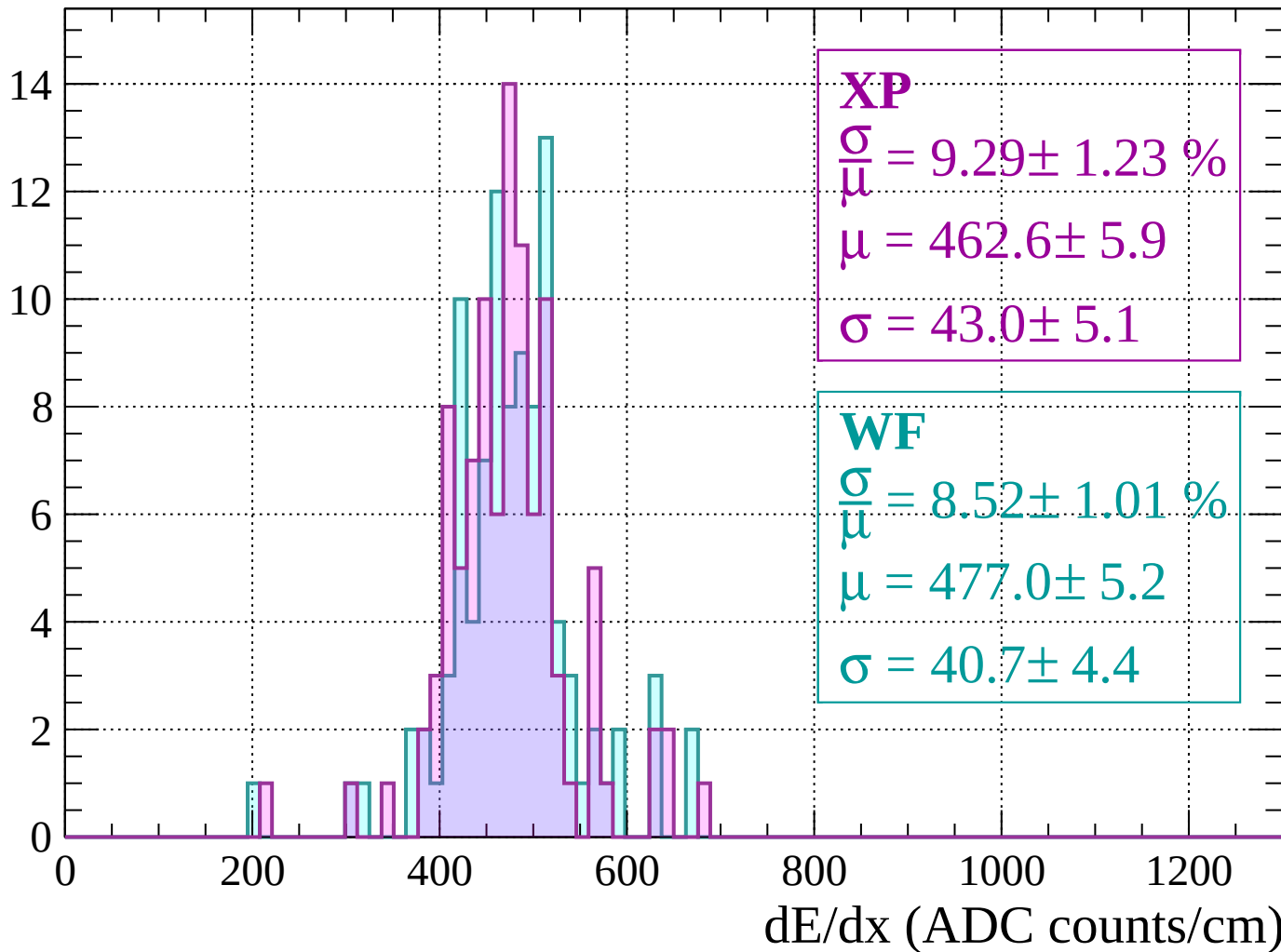
Energy loss | $-200 < p < -160$

Count



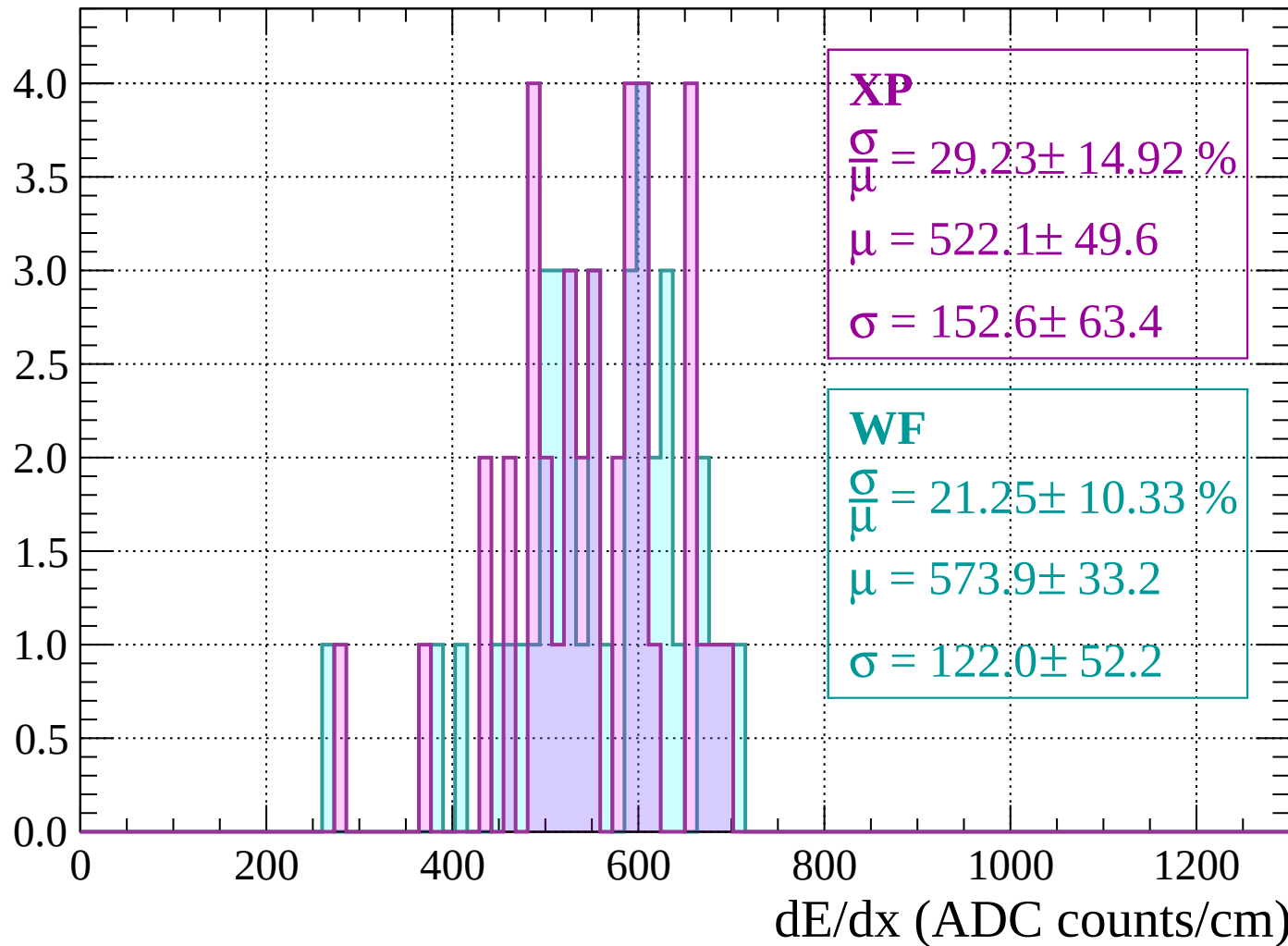
Energy loss | $-160 < p < -120$

Count



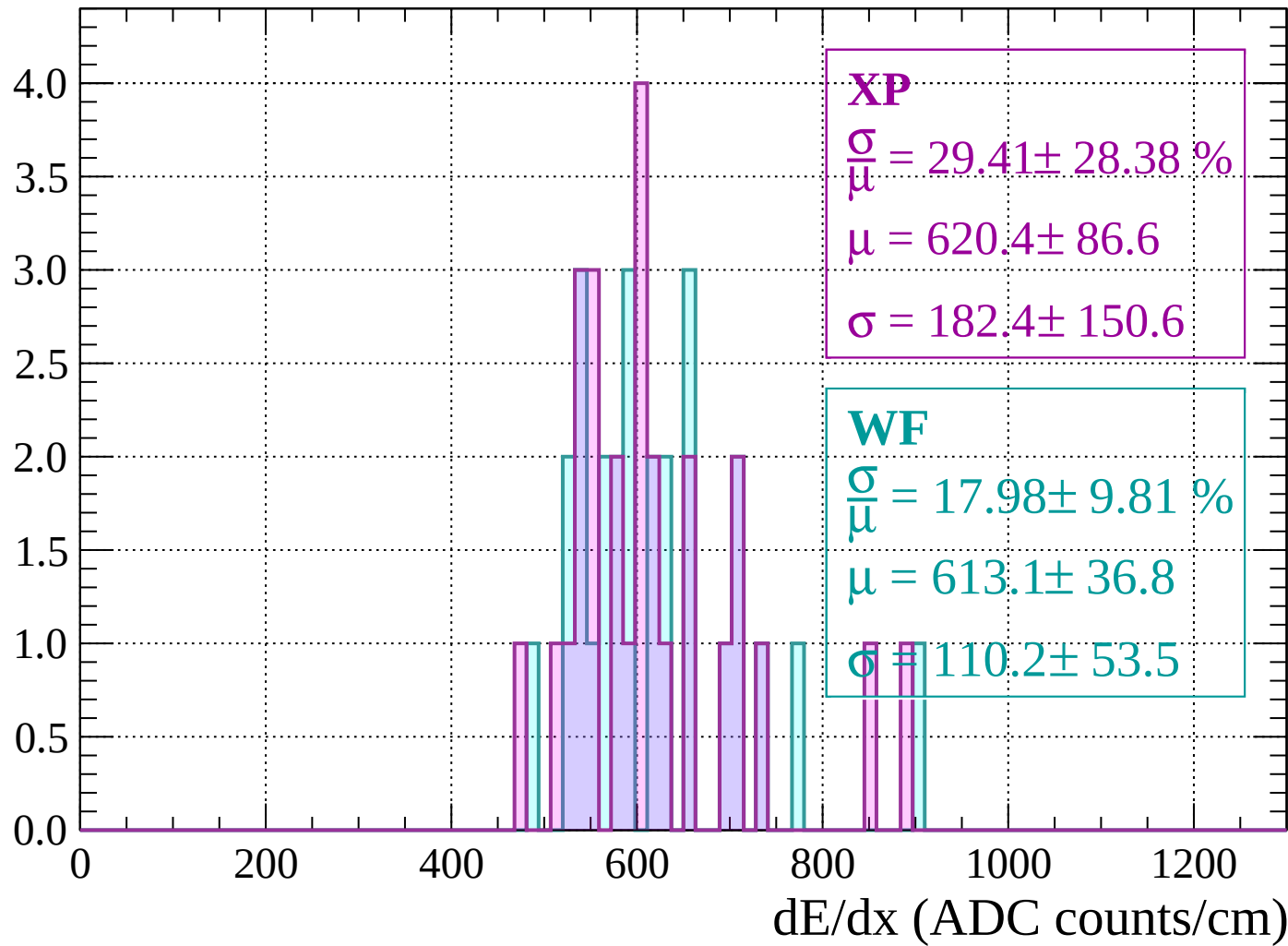
Energy loss | $-120 < p < -80$

Count



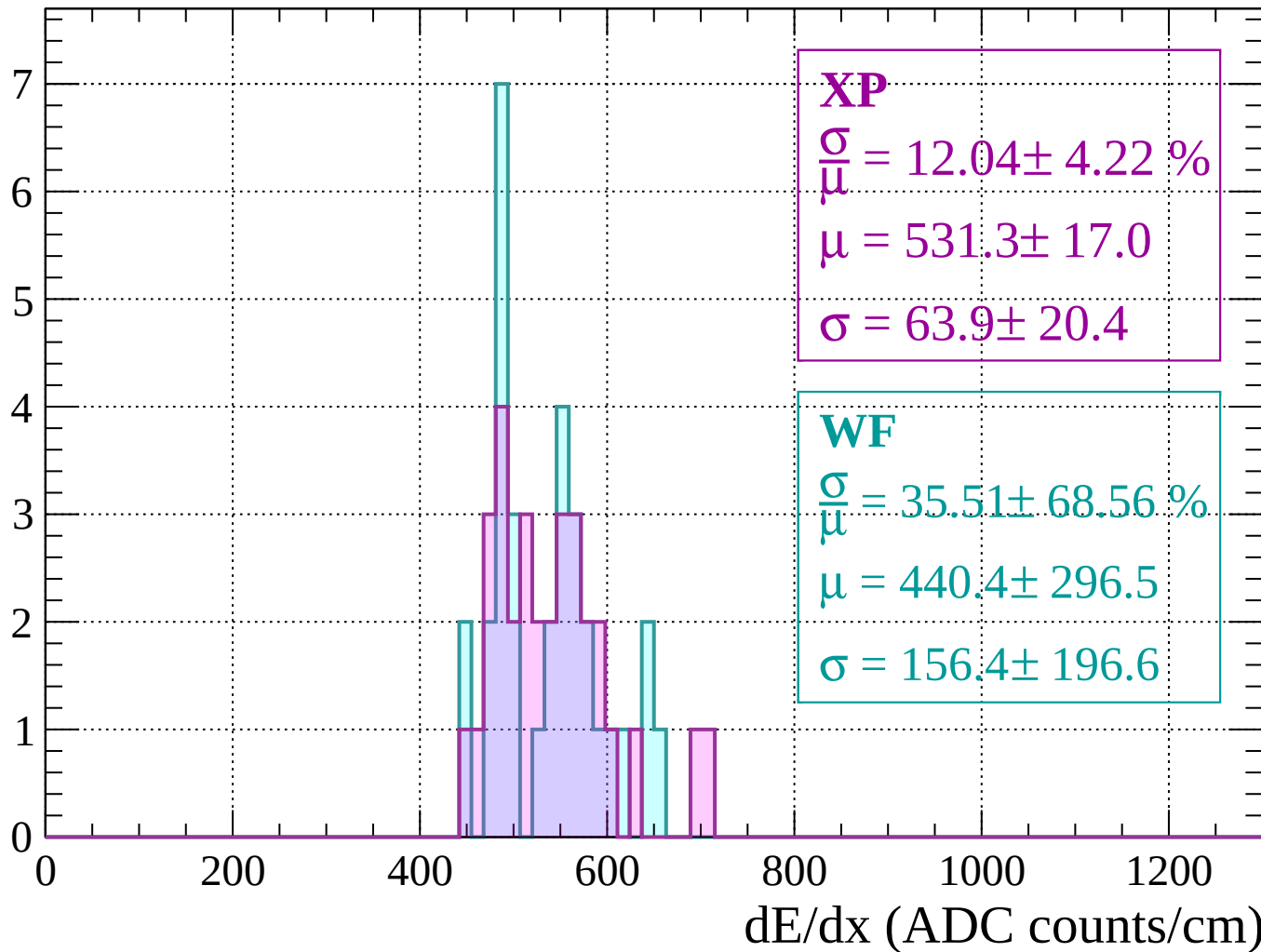
Energy loss | $-80 < p < -40$

Count



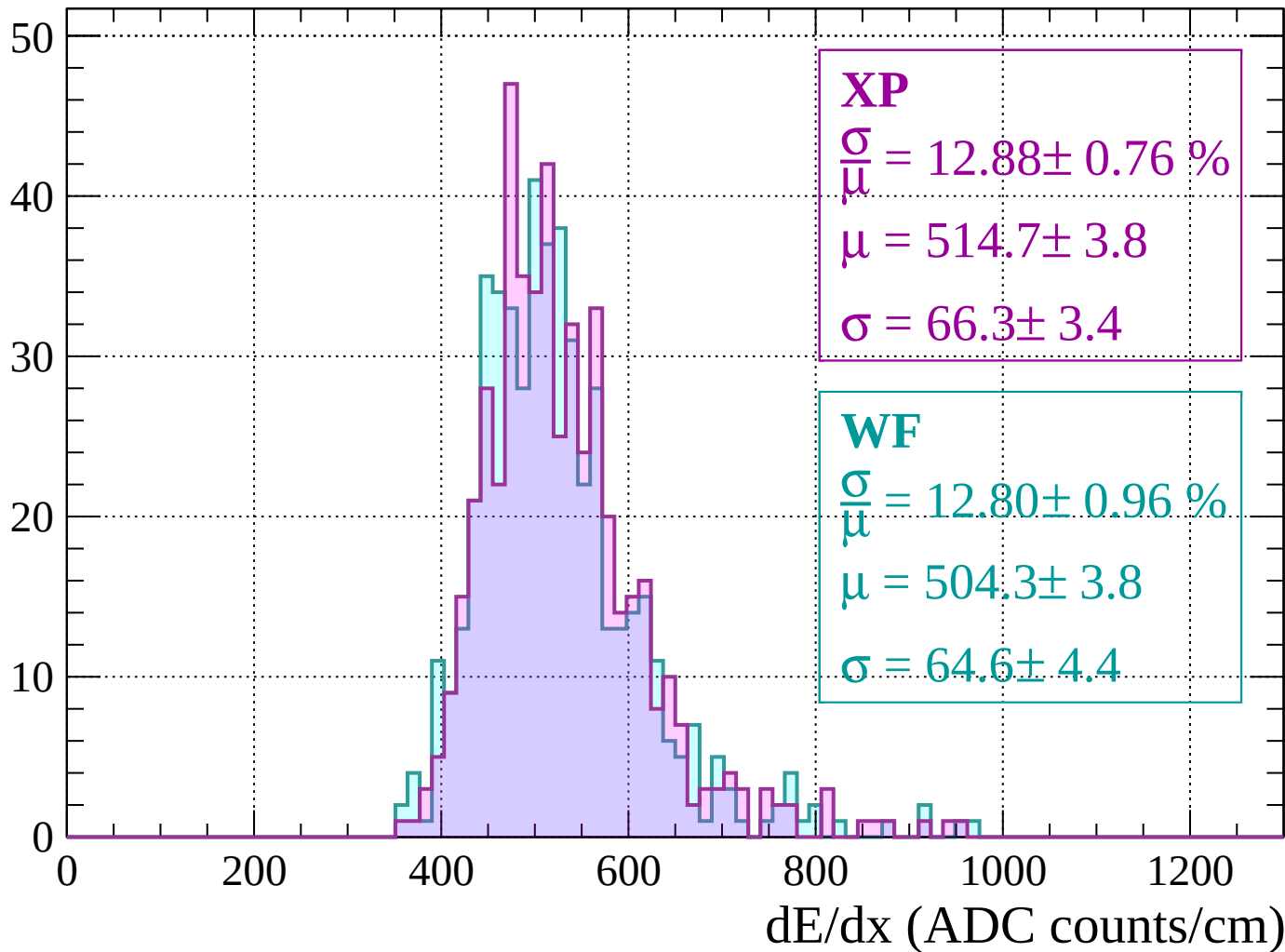
Energy loss | $-40 < p < 0$

Count



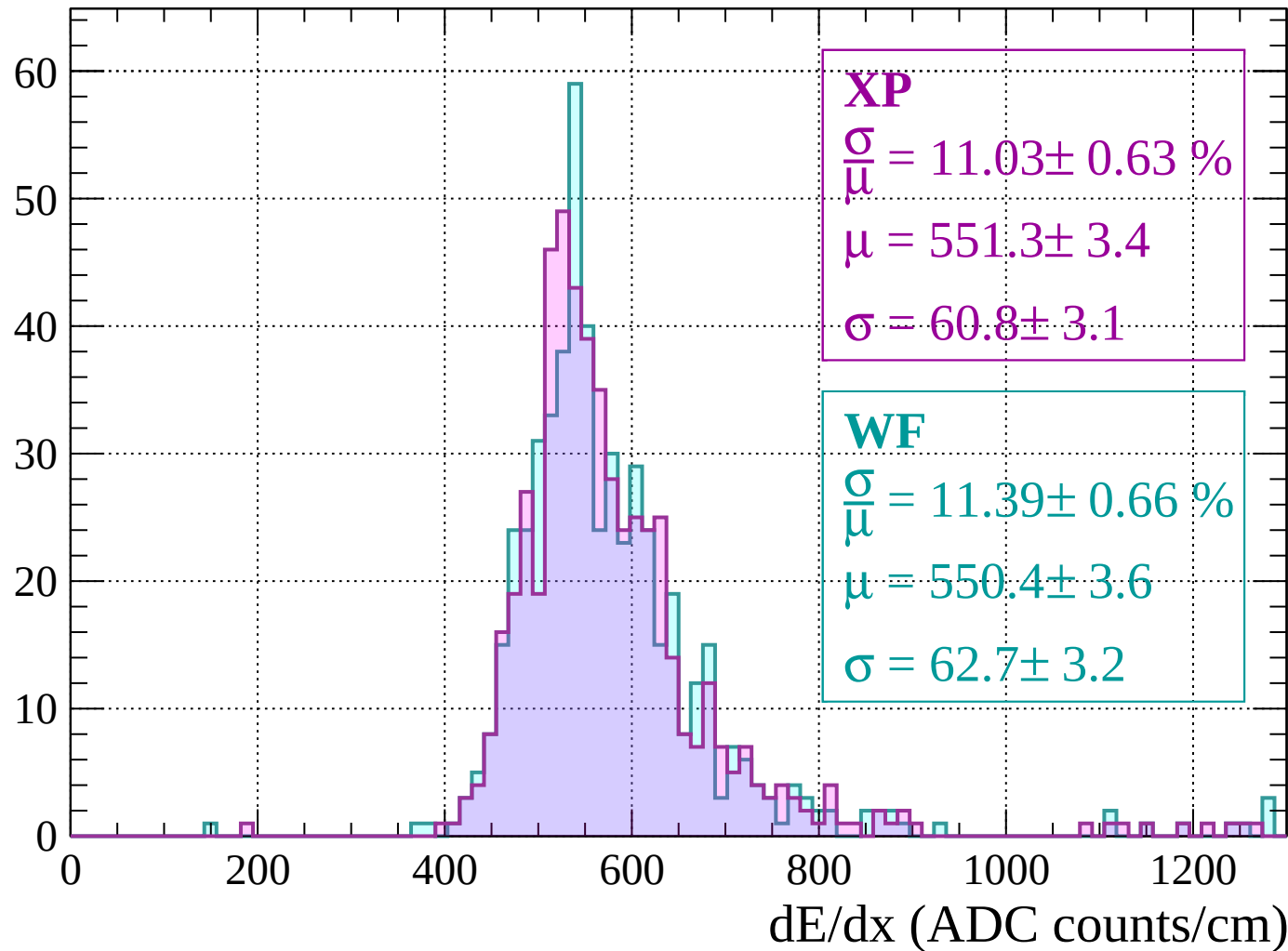
Energy loss | $0 < p < 40$

Count



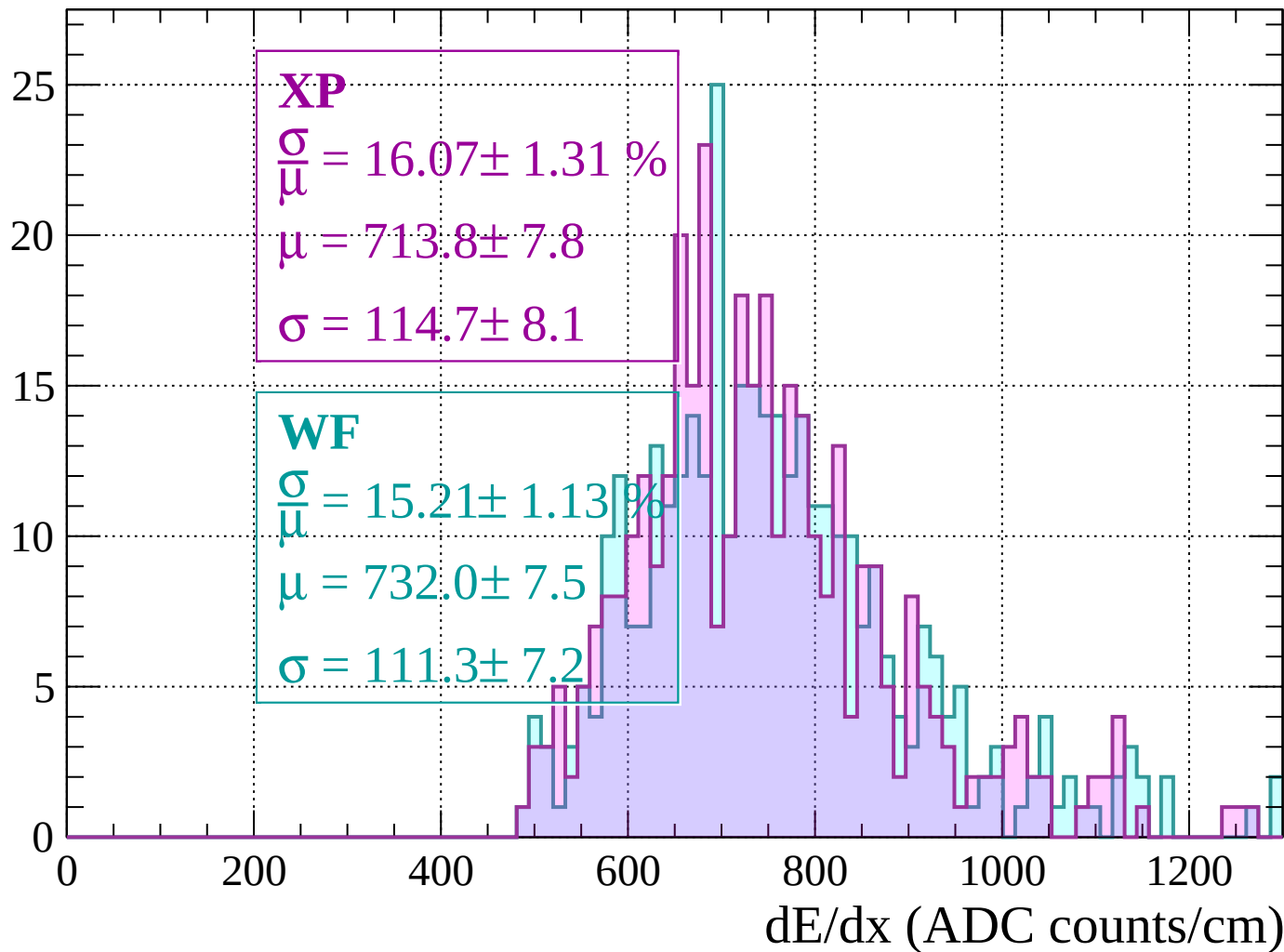
Energy loss | $40 < p < 80$

Count



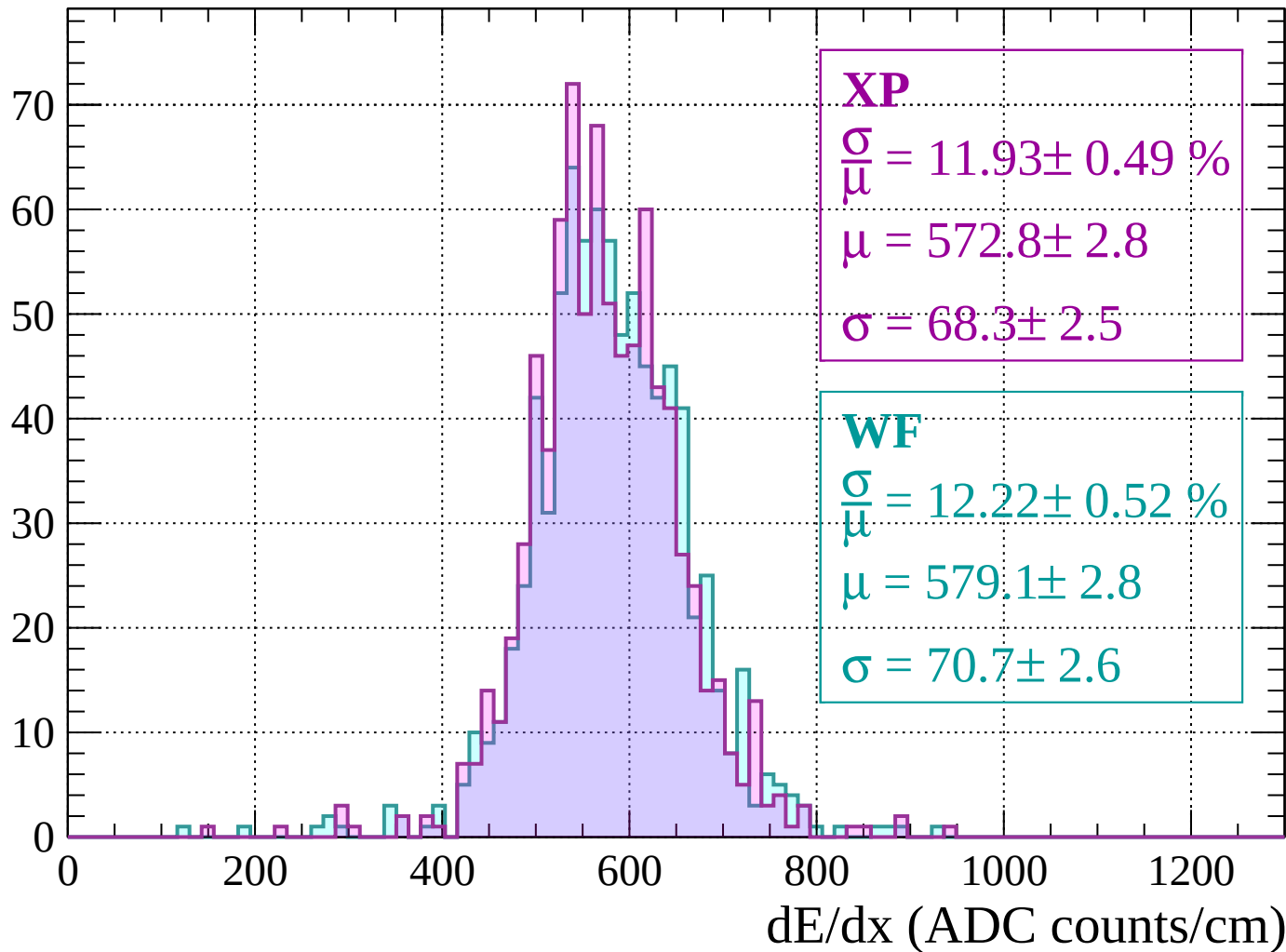
Energy loss | $80 < p < 120$

Count



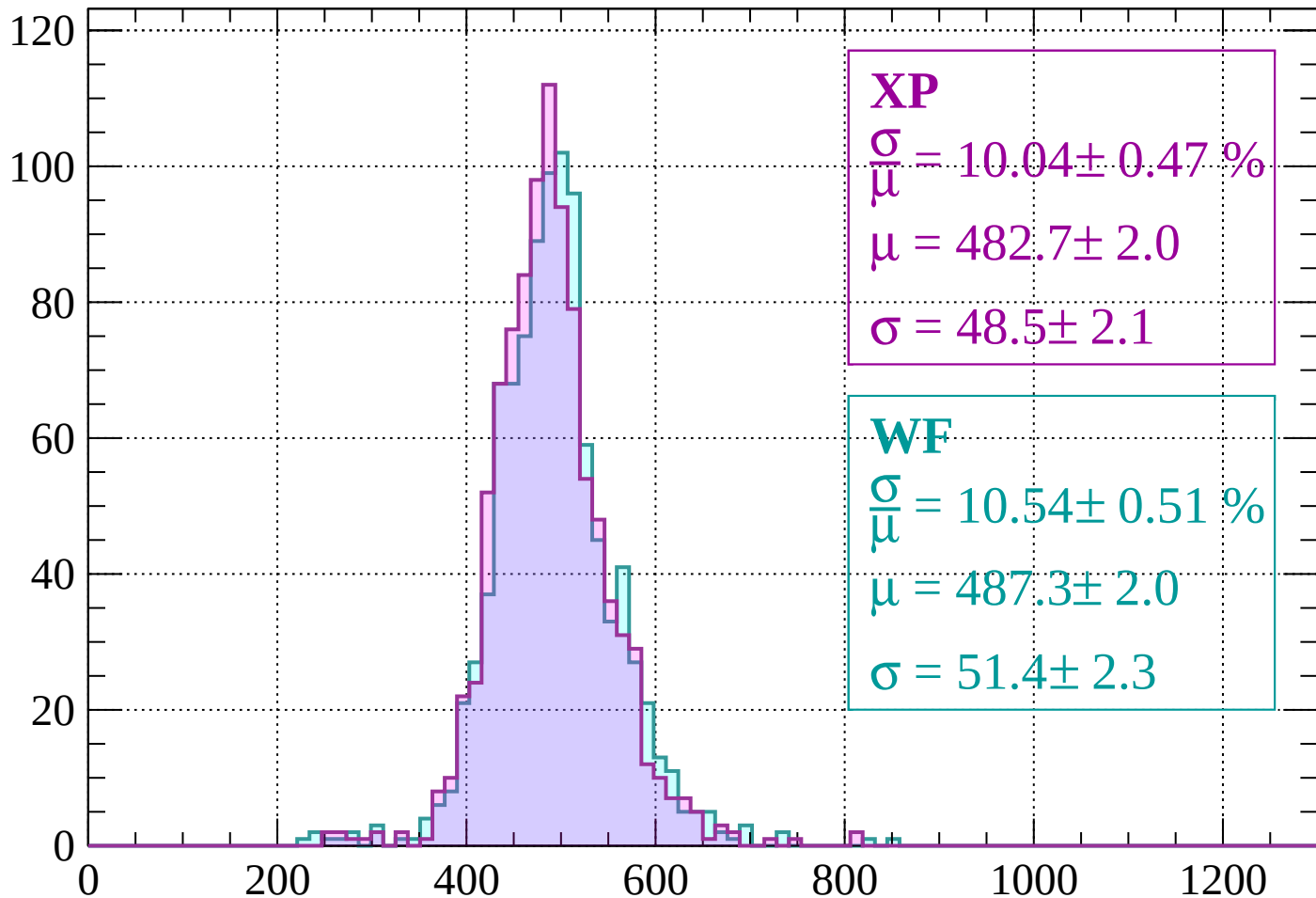
Energy loss | $120 < p < 160$

Count



Energy loss | $160 < p < 200$

Count



XP

$$\frac{\sigma}{\mu} = 10.04 \pm 0.47 \%$$

$$\mu = 482.7 \pm 2.0$$

$$\sigma = 48.5 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 10.54 \pm 0.51 \%$$

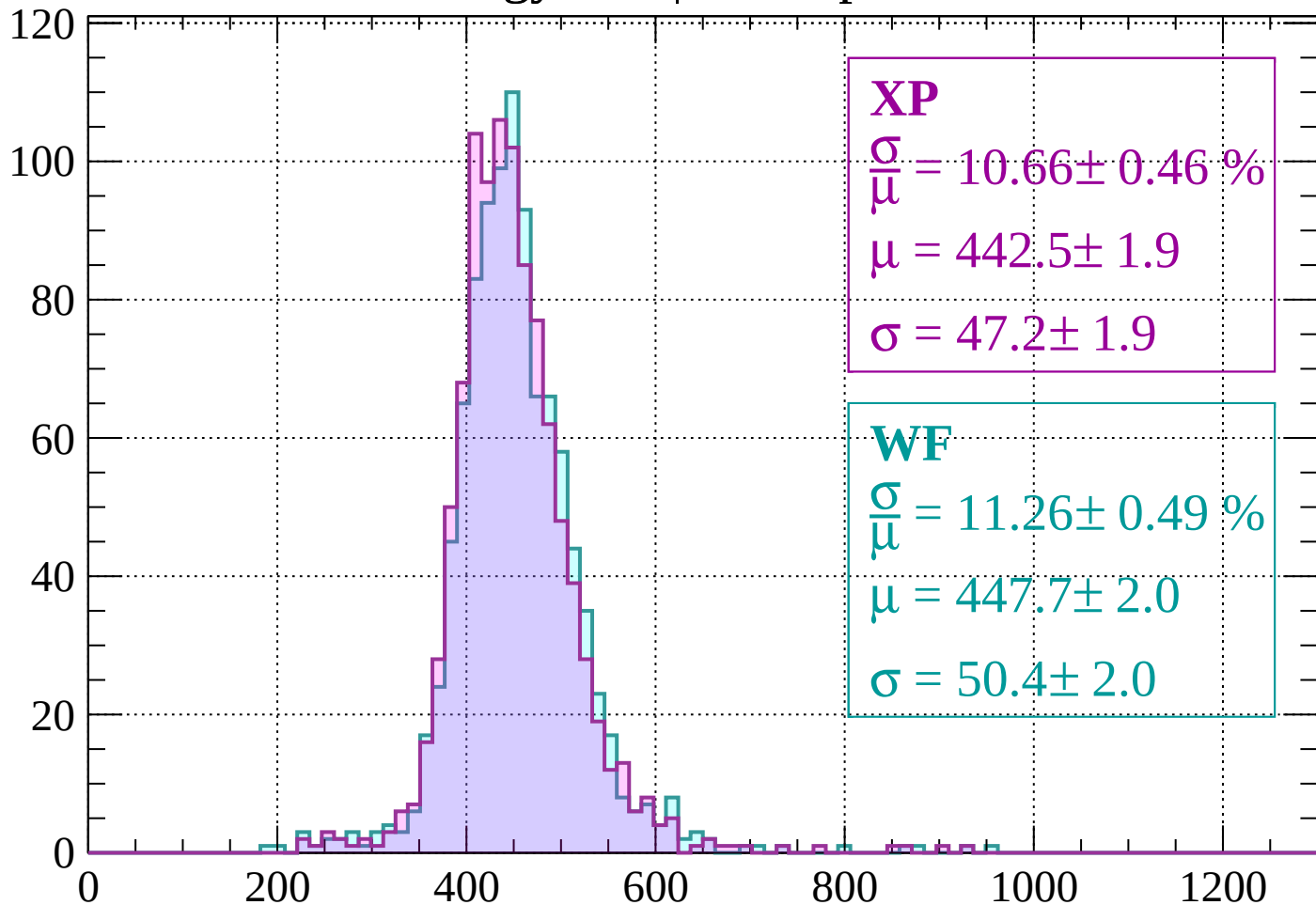
$$\mu = 487.3 \pm 2.0$$

$$\sigma = 51.4 \pm 2.3$$

dE/dx (ADC counts/cm)

Energy loss | $200 < p < 240$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 10.66 \pm 0.46 \%$$

$$\mu = 442.5 \pm 1.9$$

$$\sigma = 47.2 \pm 1.9$$

WF

$$\frac{\sigma}{\bar{\mu}} = 11.26 \pm 0.49 \%$$

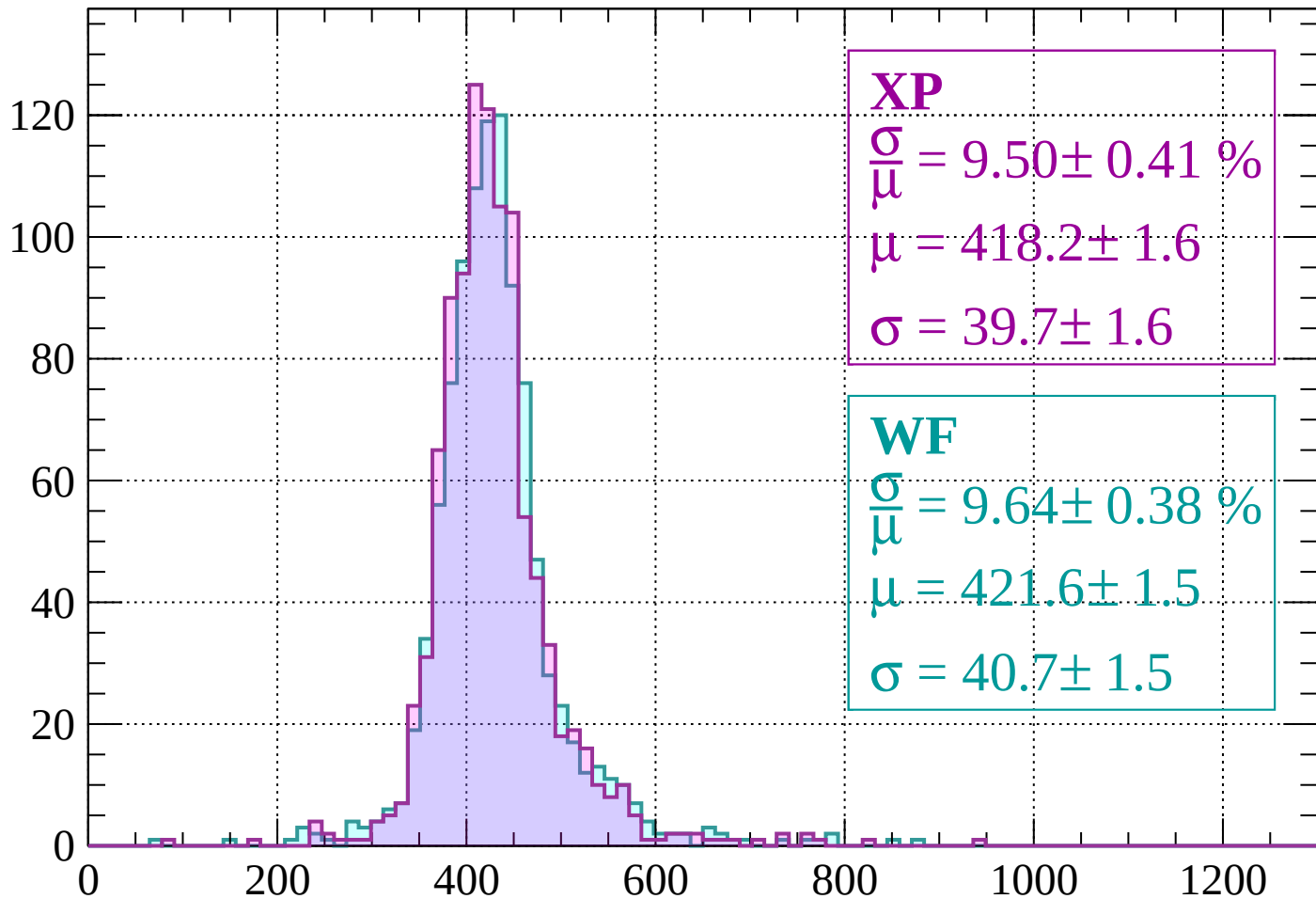
$$\mu = 447.7 \pm 2.0$$

$$\sigma = 50.4 \pm 2.0$$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 280$

Count



XP

$$\frac{\sigma}{\mu} = 9.50 \pm 0.41 \%$$

$$\mu = 418.2 \pm 1.6$$

$$\sigma = 39.7 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 9.64 \pm 0.38 \%$$

$$\mu = 421.6 \pm 1.5$$

$$\sigma = 40.7 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $280 < p < 320$

Count

120

100

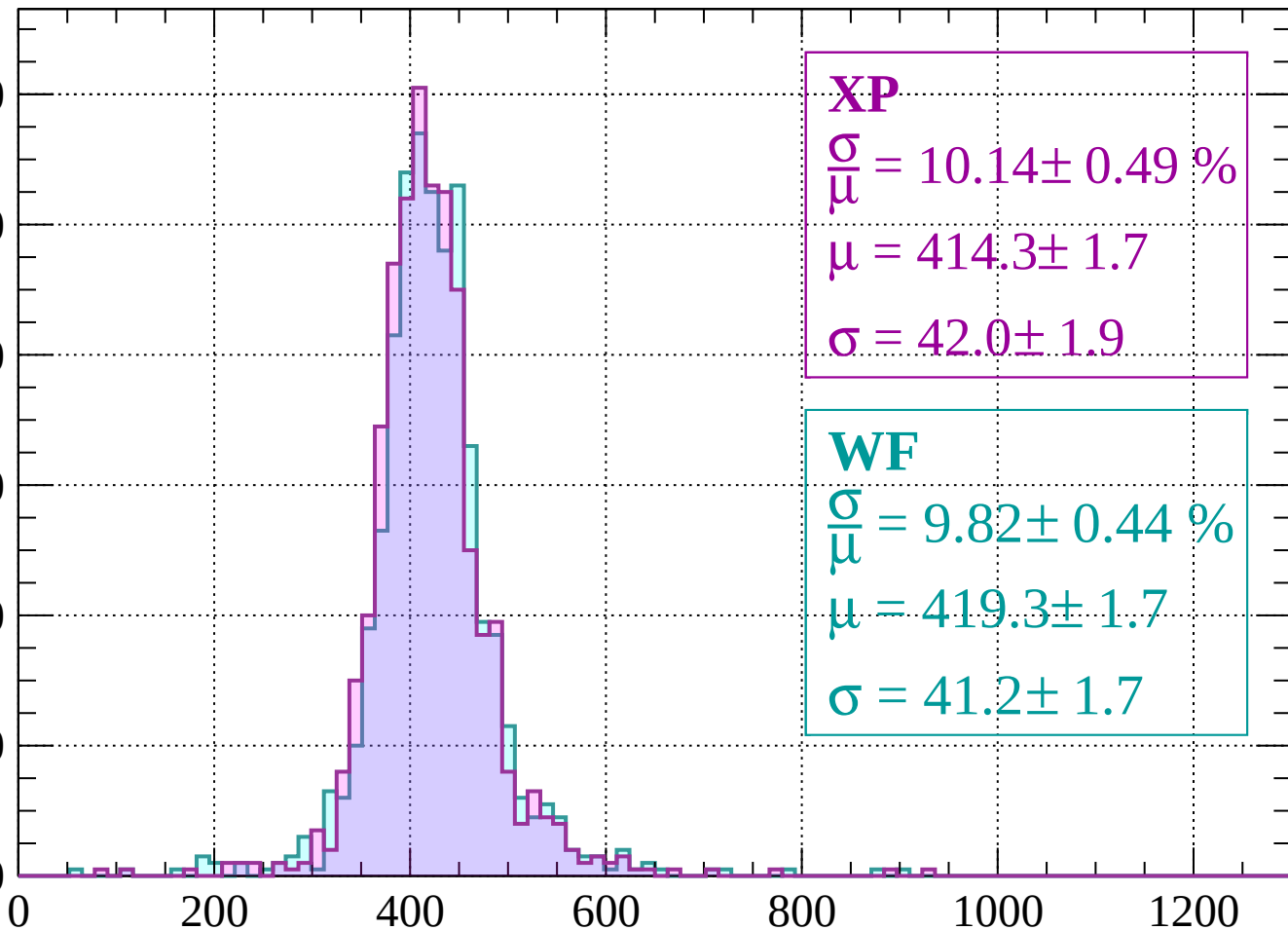
80

60

40

20

0



XP

$$\frac{\sigma}{\mu} = 10.14 \pm 0.49 \%$$

$$\mu = 414.3 \pm 1.7$$

$$\sigma = 42.0 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 9.82 \pm 0.44 \%$$

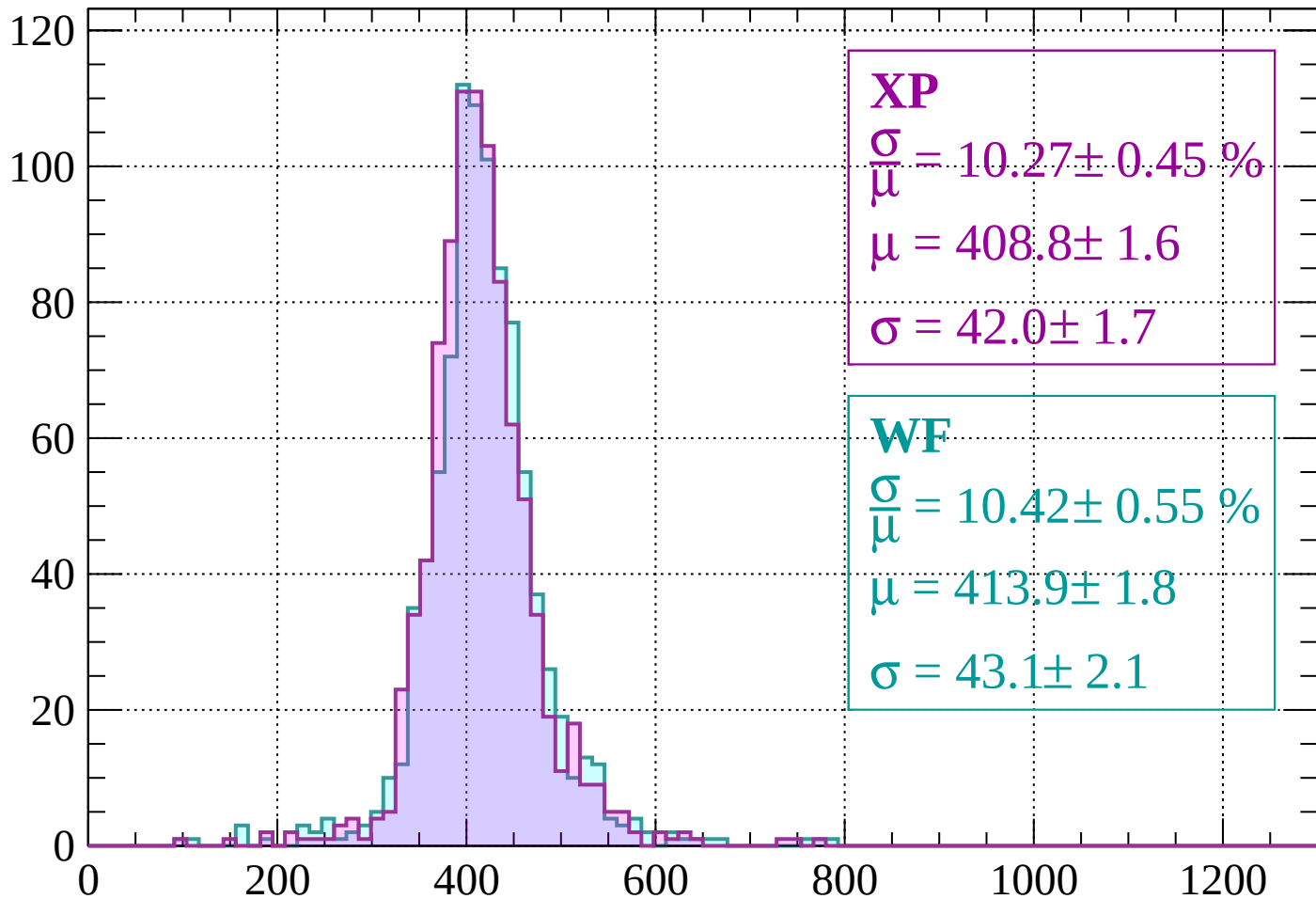
$$\mu = 419.3 \pm 1.7$$

$$\sigma = 41.2 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | $320 < p < 360$

Count



XP

$$\frac{\sigma}{\mu} = 10.27 \pm 0.45 \%$$

$$\mu = 408.8 \pm 1.6$$

$$\sigma = 42.0 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 10.42 \pm 0.55 \%$$

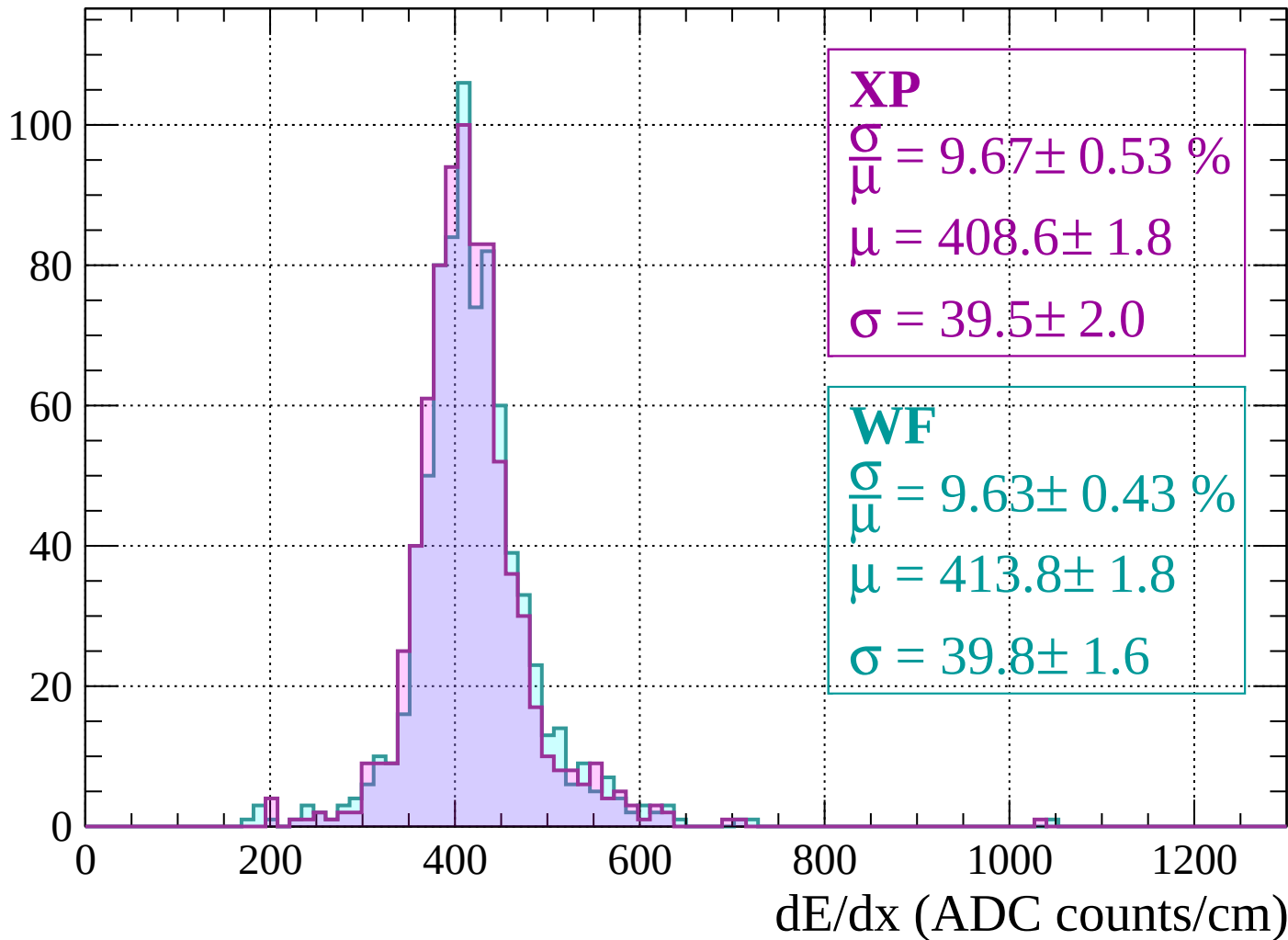
$$\mu = 413.9 \pm 1.8$$

$$\sigma = 43.1 \pm 2.1$$

dE/dx (ADC counts/cm)

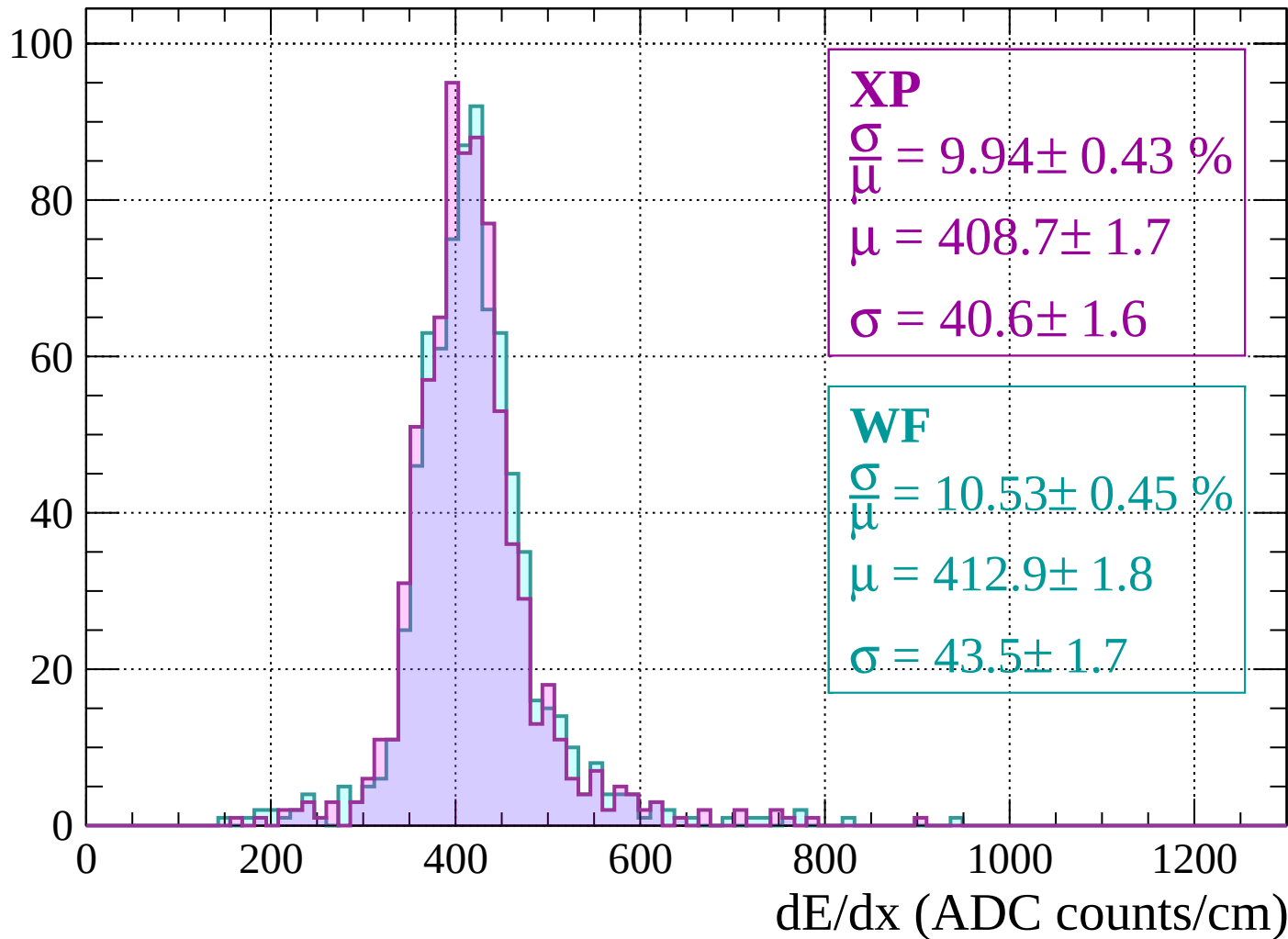
Energy loss | $360 < p < 400$

Count



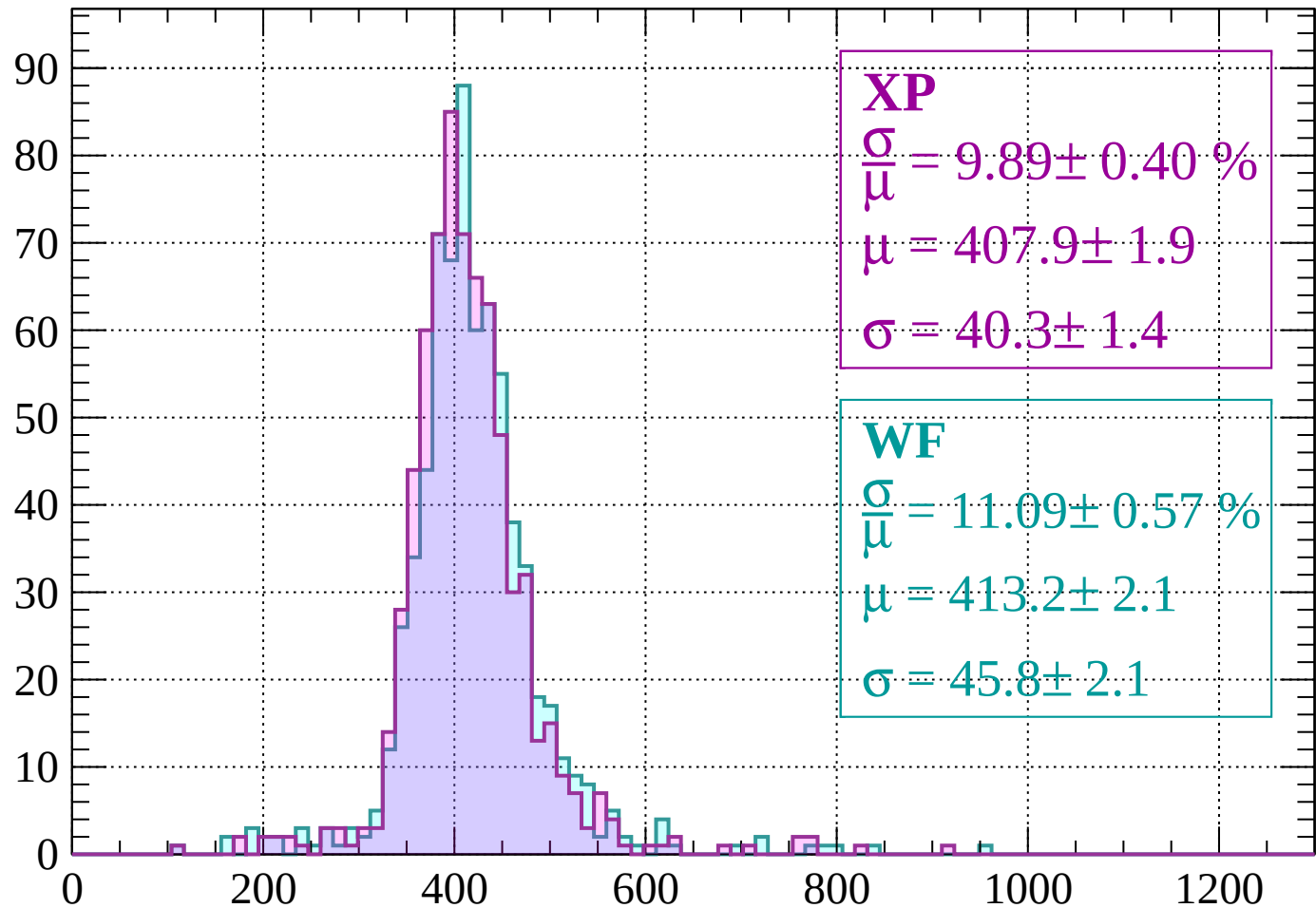
Energy loss | $400 < p < 440$

Count



Energy loss | $440 < p < 480$

Count



XP

$$\frac{\sigma}{\mu} = 9.89 \pm 0.40 \%$$

$$\mu = 407.9 \pm 1.9$$

$$\sigma = 40.3 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 11.09 \pm 0.57 \%$$

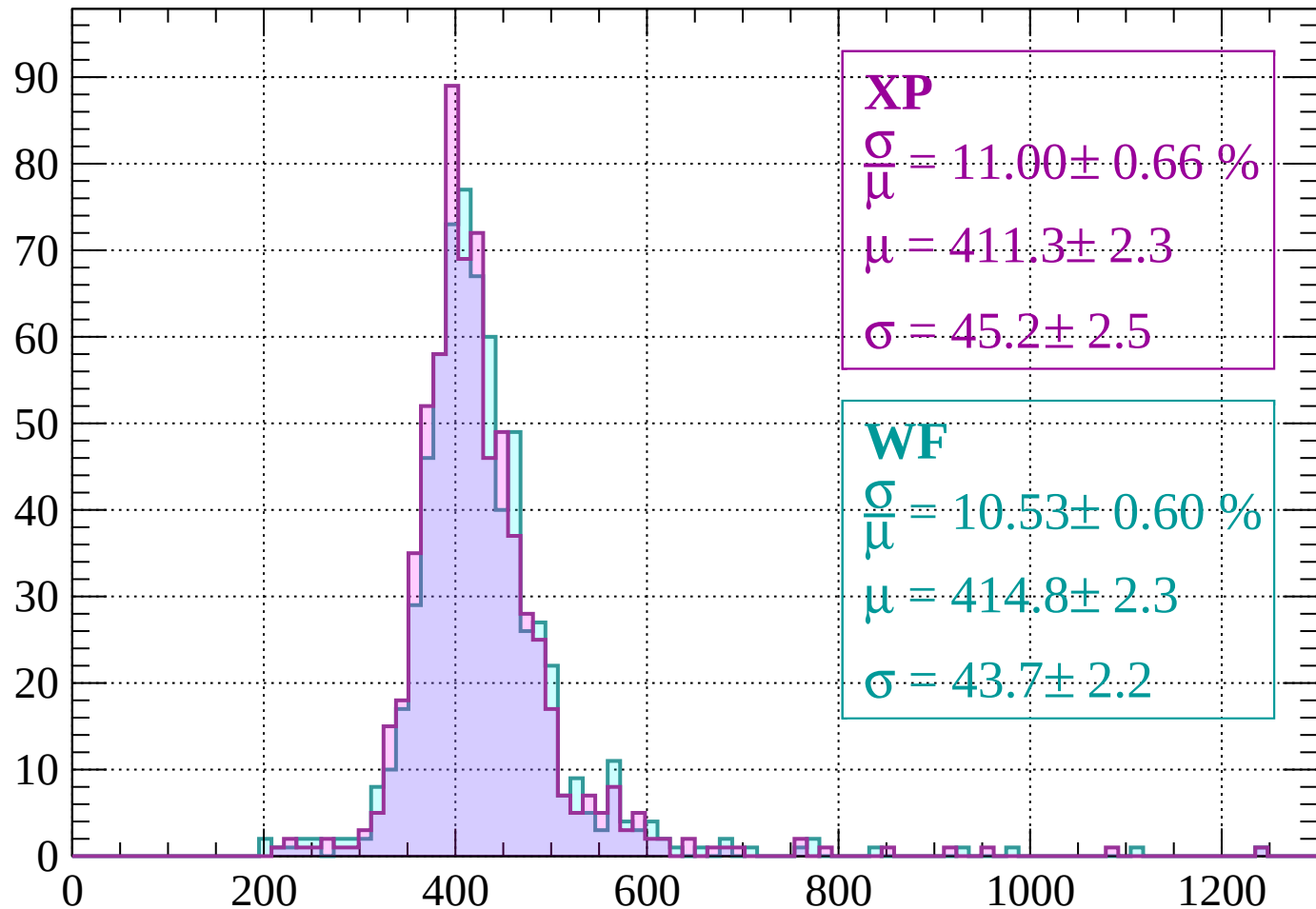
$$\mu = 413.2 \pm 2.1$$

$$\sigma = 45.8 \pm 2.1$$

dE/dx (ADC counts/cm)

Energy loss | $480 < p < 520$

Count



XP

$$\frac{\sigma}{\mu} = 11.00 \pm 0.66 \%$$

$$\mu = 411.3 \pm 2.3$$

$$\sigma = 45.2 \pm 2.5$$

WF

$$\frac{\sigma}{\mu} = 10.53 \pm 0.60 \%$$

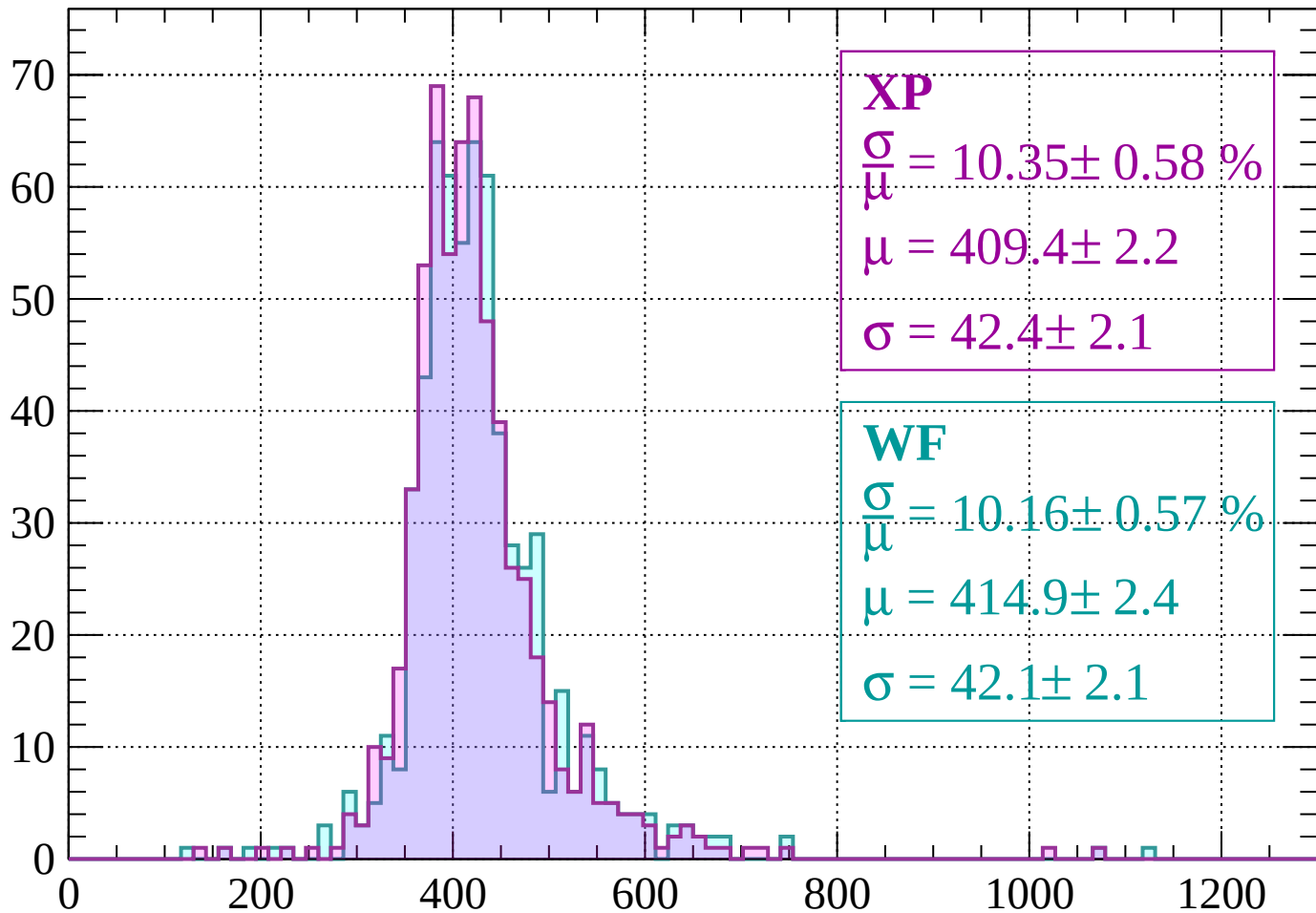
$$\mu = 414.8 \pm 2.3$$

$$\sigma = 43.7 \pm 2.2$$

dE/dx (ADC counts/cm)

Energy loss | $520 < p < 560$

Count



XP

$$\frac{\sigma}{\mu} = 10.35 \pm 0.58 \%$$

$$\mu = 409.4 \pm 2.2$$

$$\sigma = 42.4 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 10.16 \pm 0.57 \%$$

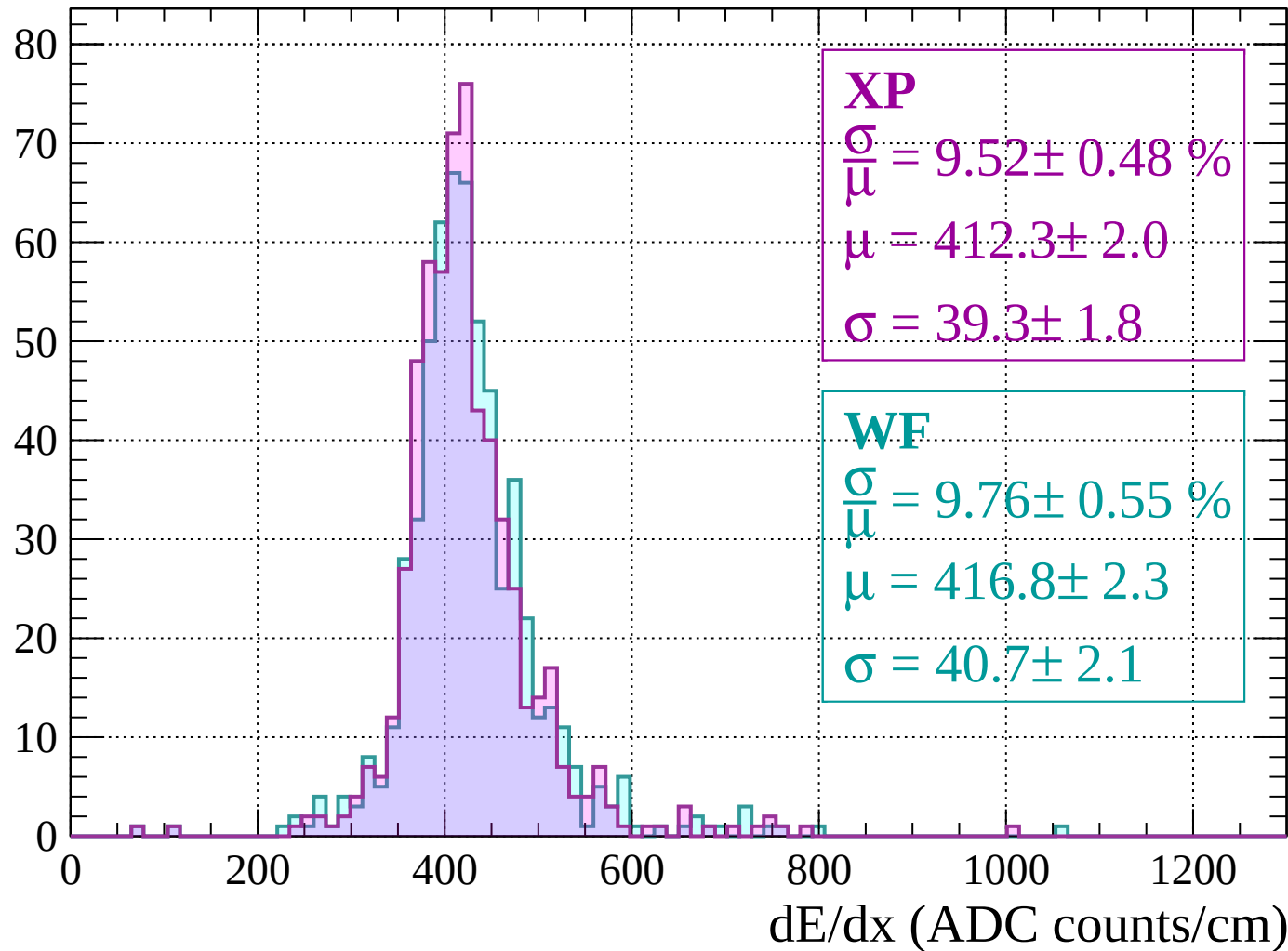
$$\mu = 414.9 \pm 2.4$$

$$\sigma = 42.1 \pm 2.1$$

dE/dx (ADC counts/cm)

Energy loss | $560 < p < 600$

Count



Energy loss | $600 < p < 640$

Count

XP

$$\frac{\sigma}{\mu} = 10.69 \pm 0.58 \%$$

$$\mu = 412.2 \pm 2.3$$

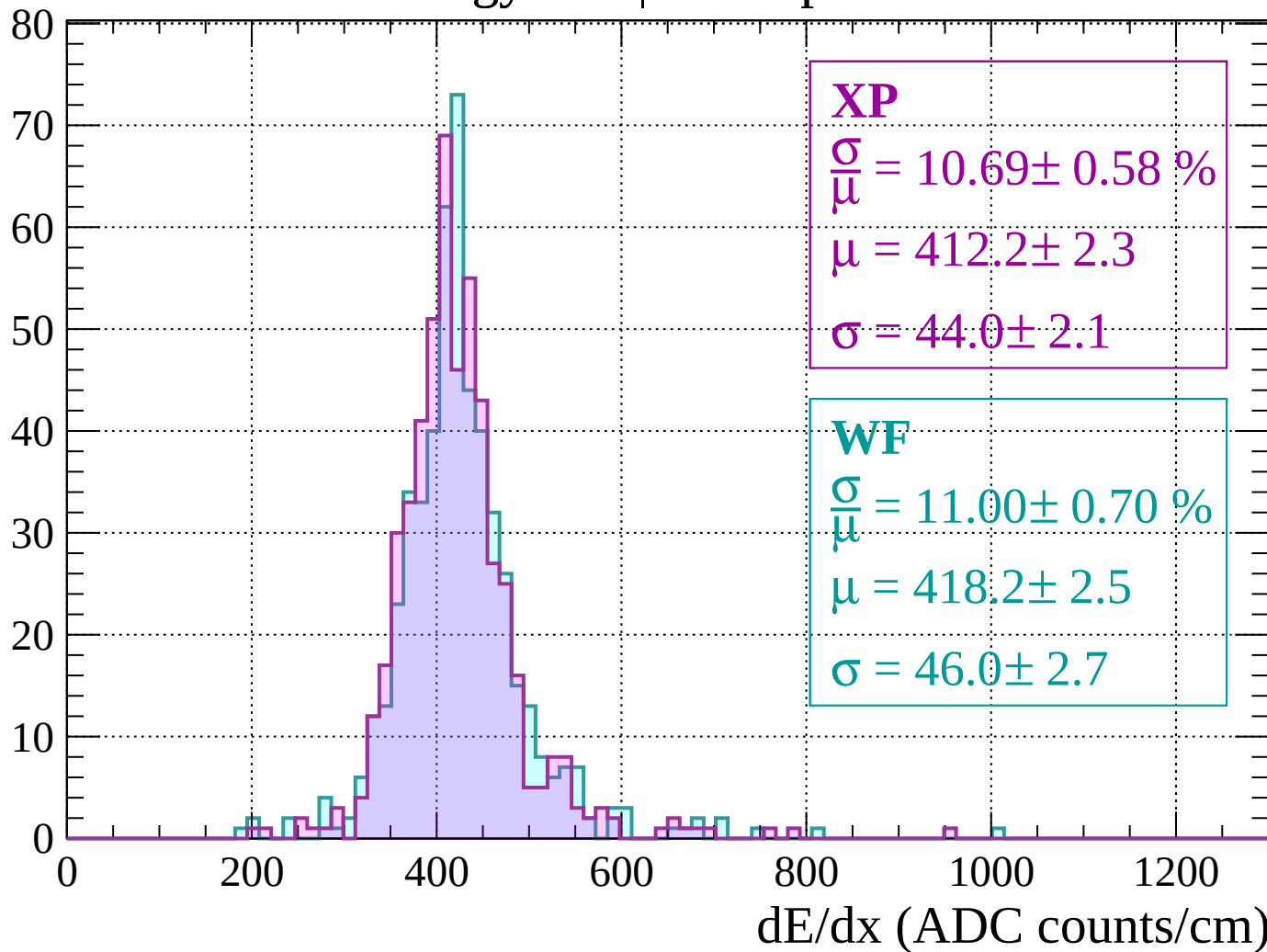
$$\sigma = 44.0 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 11.00 \pm 0.70 \%$$

$$\mu = 418.2 \pm 2.5$$

$$\sigma = 46.0 \pm 2.7$$



Energy loss | $640 < p < 680$

Count

XP

$$\frac{\sigma}{\mu} = 9.96 \pm 0.61 \%$$

$$\mu = 415.3 \pm 2.4$$

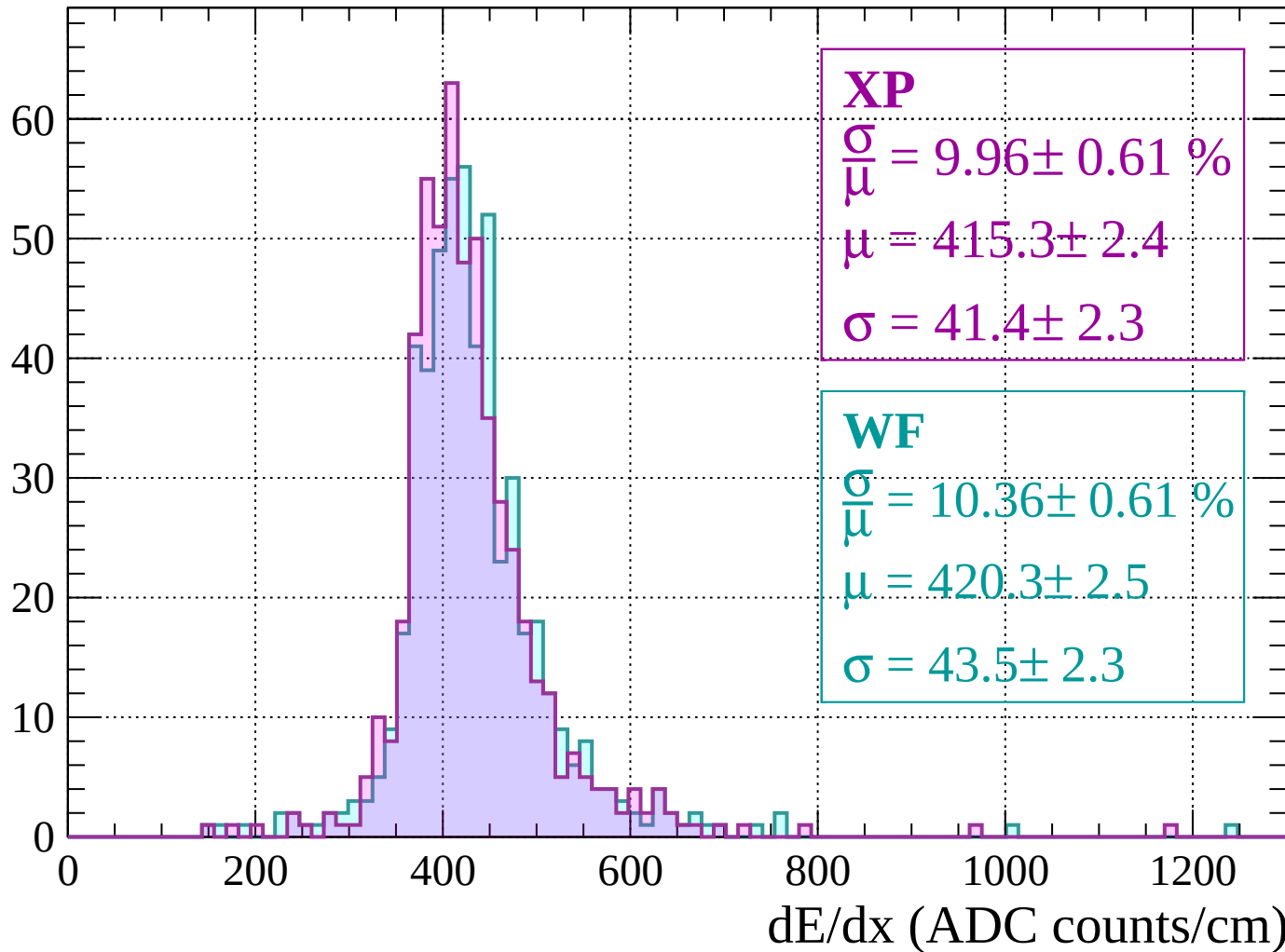
$$\sigma = 41.4 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 10.36 \pm 0.61 \%$$

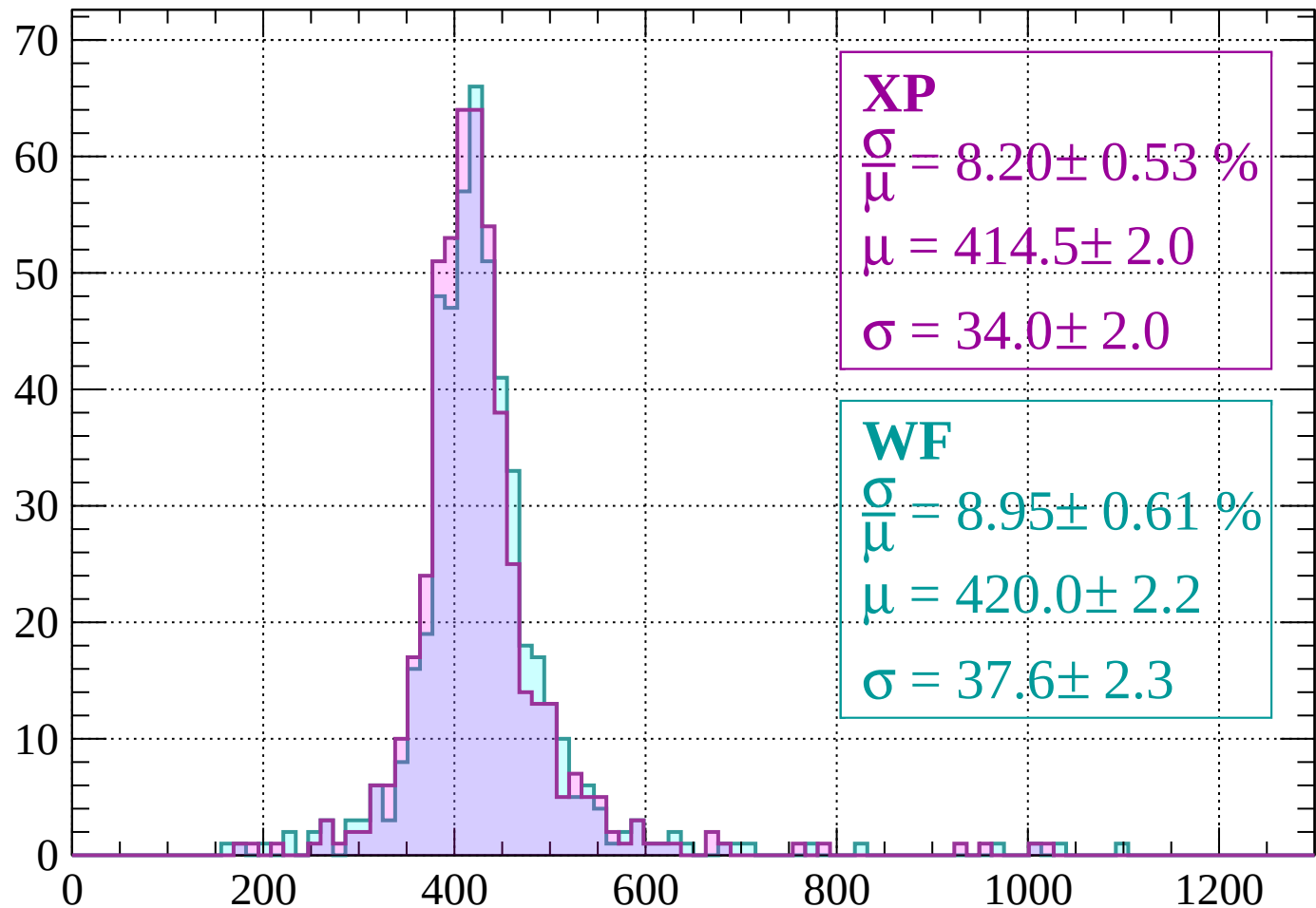
$$\mu = 420.3 \pm 2.5$$

$$\sigma = 43.5 \pm 2.3$$



Energy loss | $680 < p < 720$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 8.20 \pm 0.53 \%$$

$$\mu = 414.5 \pm 2.0$$

$$\sigma = 34.0 \pm 2.0$$

WF

$$\frac{\sigma}{\bar{\mu}} = 8.95 \pm 0.61 \%$$

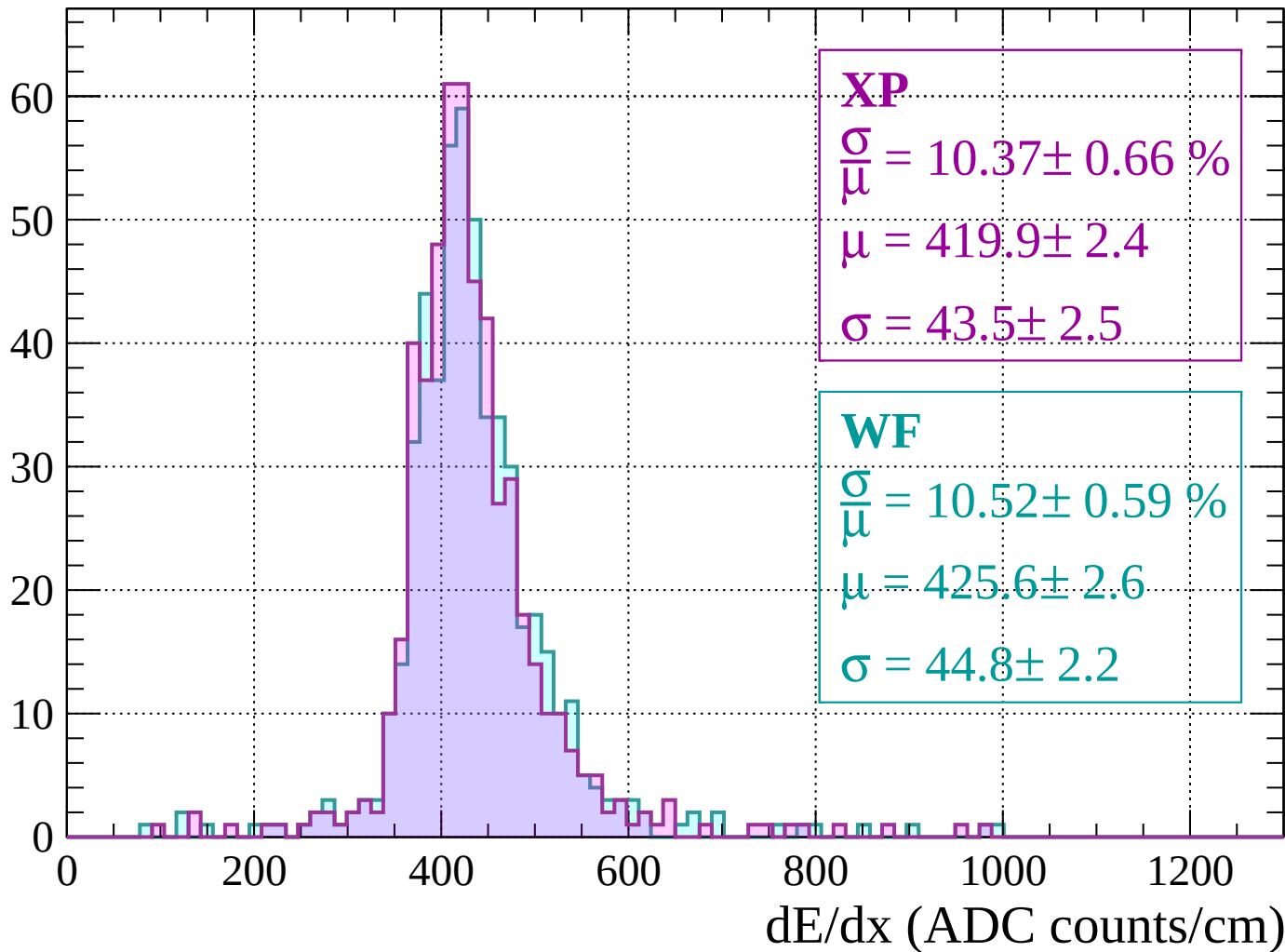
$$\mu = 420.0 \pm 2.2$$

$$\sigma = 37.6 \pm 2.3$$

dE/dx (ADC counts/cm)

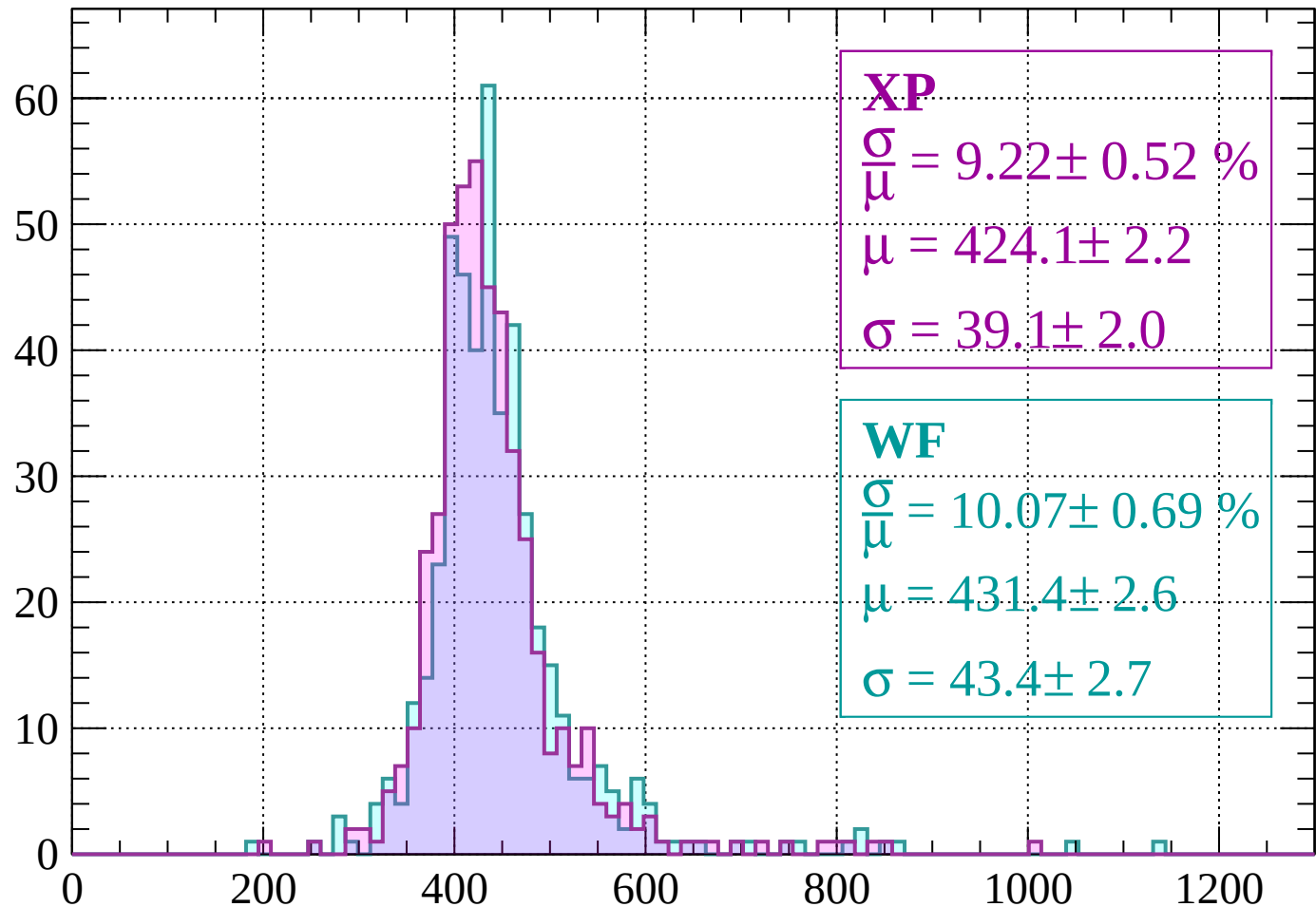
Energy loss | $720 < p < 760$

Count



Energy loss | $760 < p < 800$

Count



XP

$$\frac{\sigma}{\mu} = 9.22 \pm 0.52 \%$$

$$\mu = 424.1 \pm 2.2$$

$$\sigma = 39.1 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 10.07 \pm 0.69 \%$$

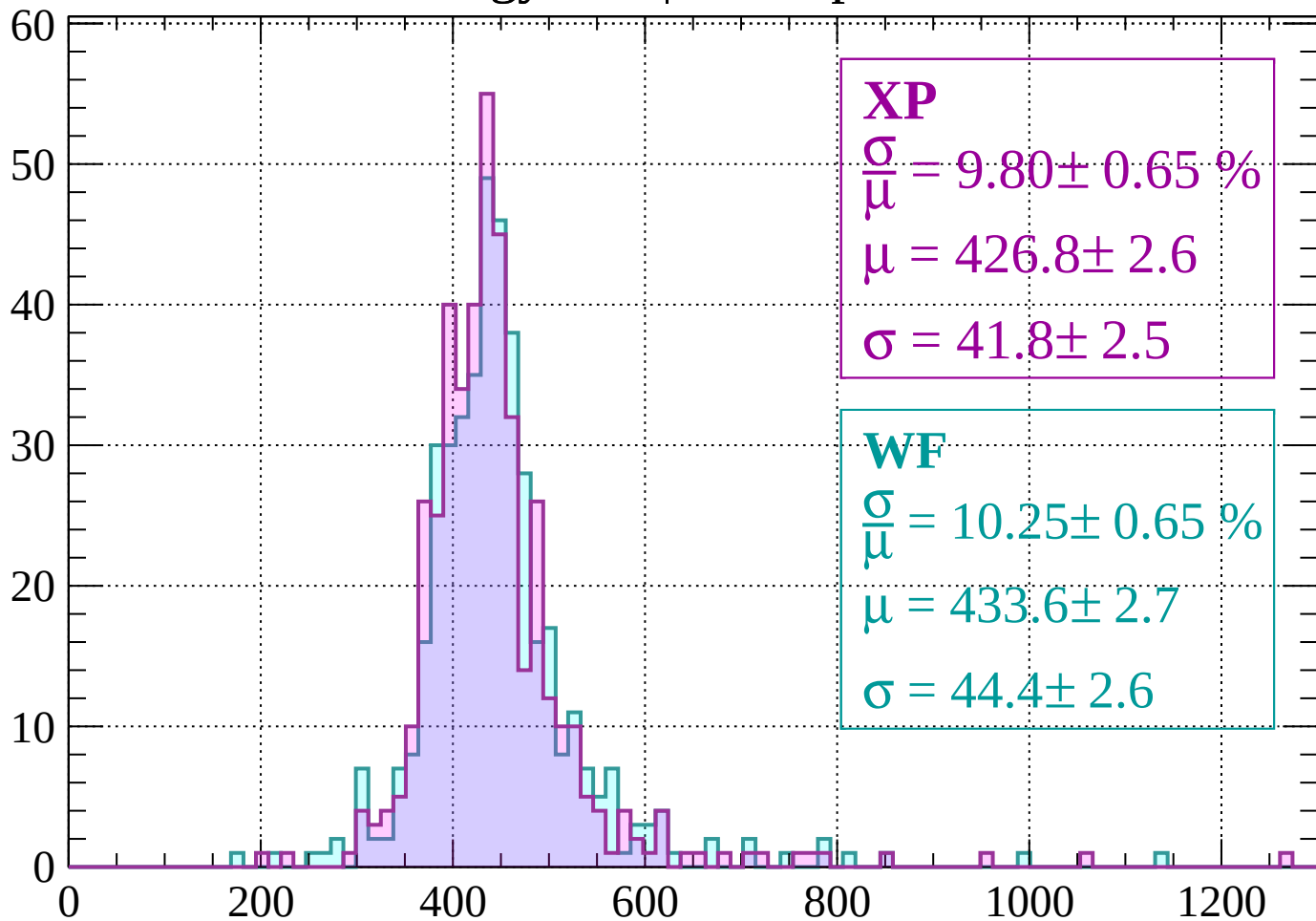
$$\mu = 431.4 \pm 2.6$$

$$\sigma = 43.4 \pm 2.7$$

dE/dx (ADC counts/cm)

Energy loss | $800 < p < 840$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 9.80 \pm 0.65 \%$$

$$\mu = 426.8 \pm 2.6$$

$$\sigma = 41.8 \pm 2.5$$

WF

$$\frac{\sigma}{\bar{\mu}} = 10.25 \pm 0.65 \%$$

$$\mu = 433.6 \pm 2.7$$

$$\sigma = 44.4 \pm 2.6$$

dE/dx (ADC counts/cm)

Energy loss | $840 < p < 880$

Count

XP

$$\frac{\sigma}{\mu} = 9.43 \pm 0.53 \%$$

$$\mu = 426.9 \pm 2.4$$

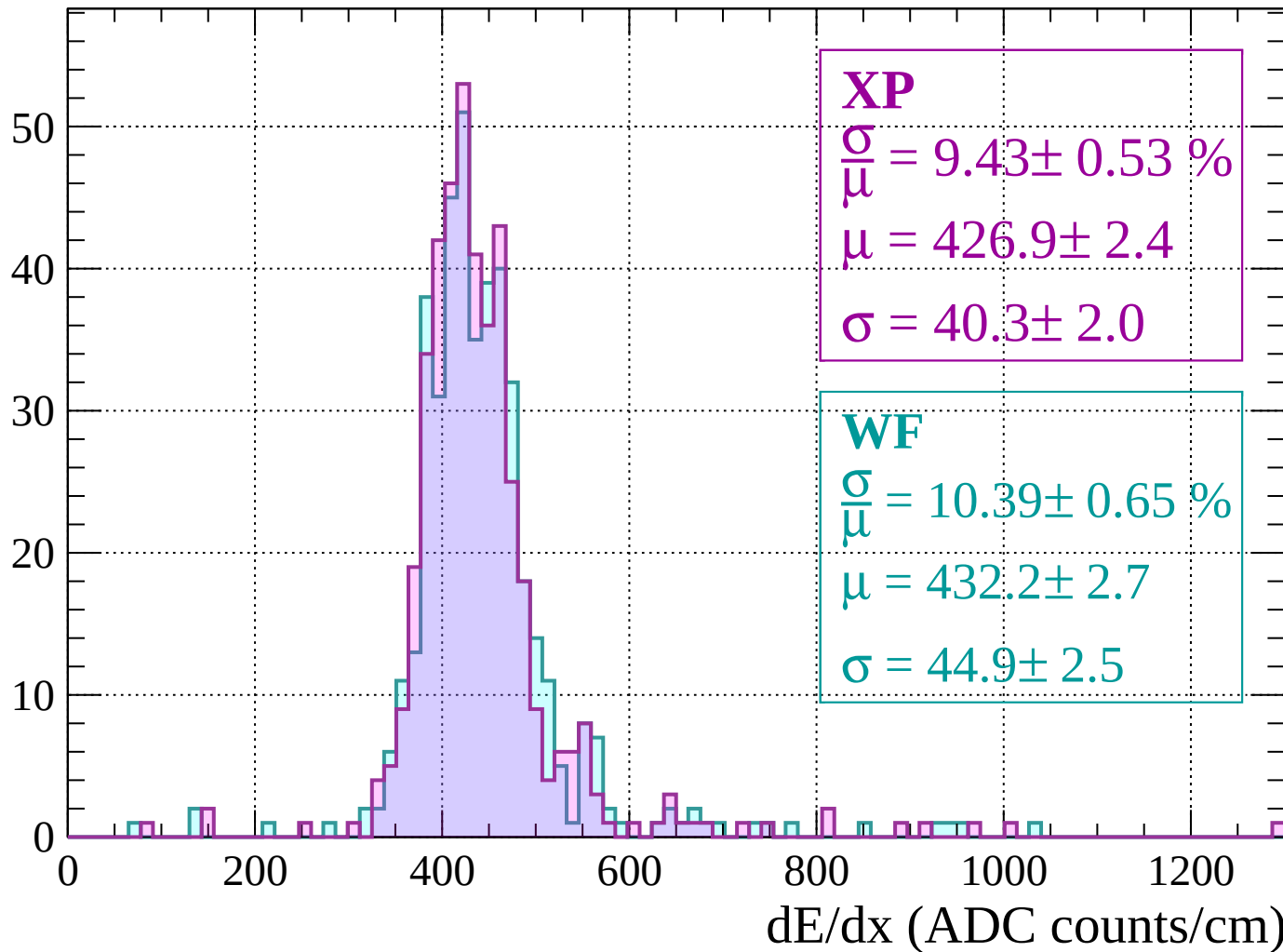
$$\sigma = 40.3 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 10.39 \pm 0.65 \%$$

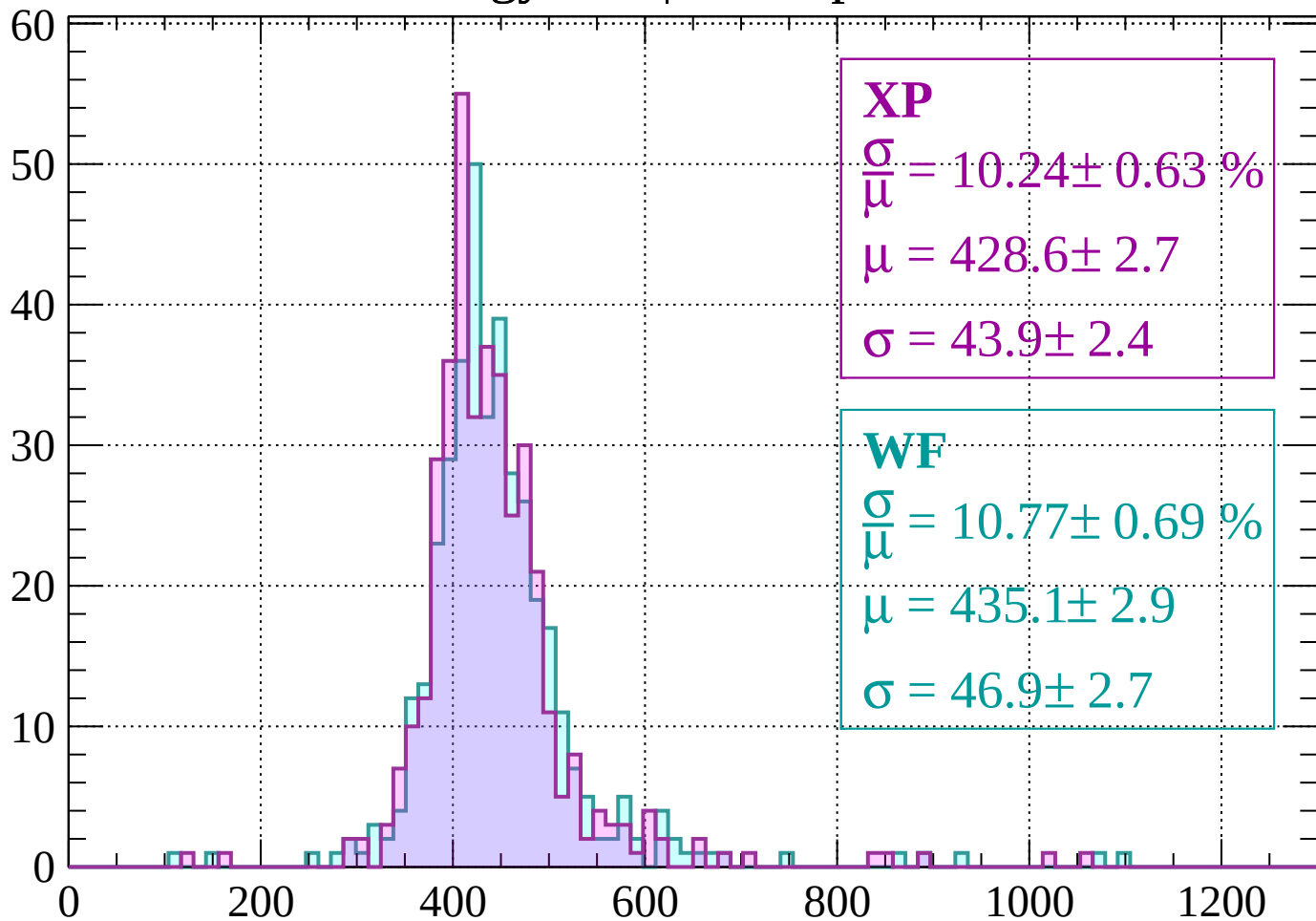
$$\mu = 432.2 \pm 2.7$$

$$\sigma = 44.9 \pm 2.5$$



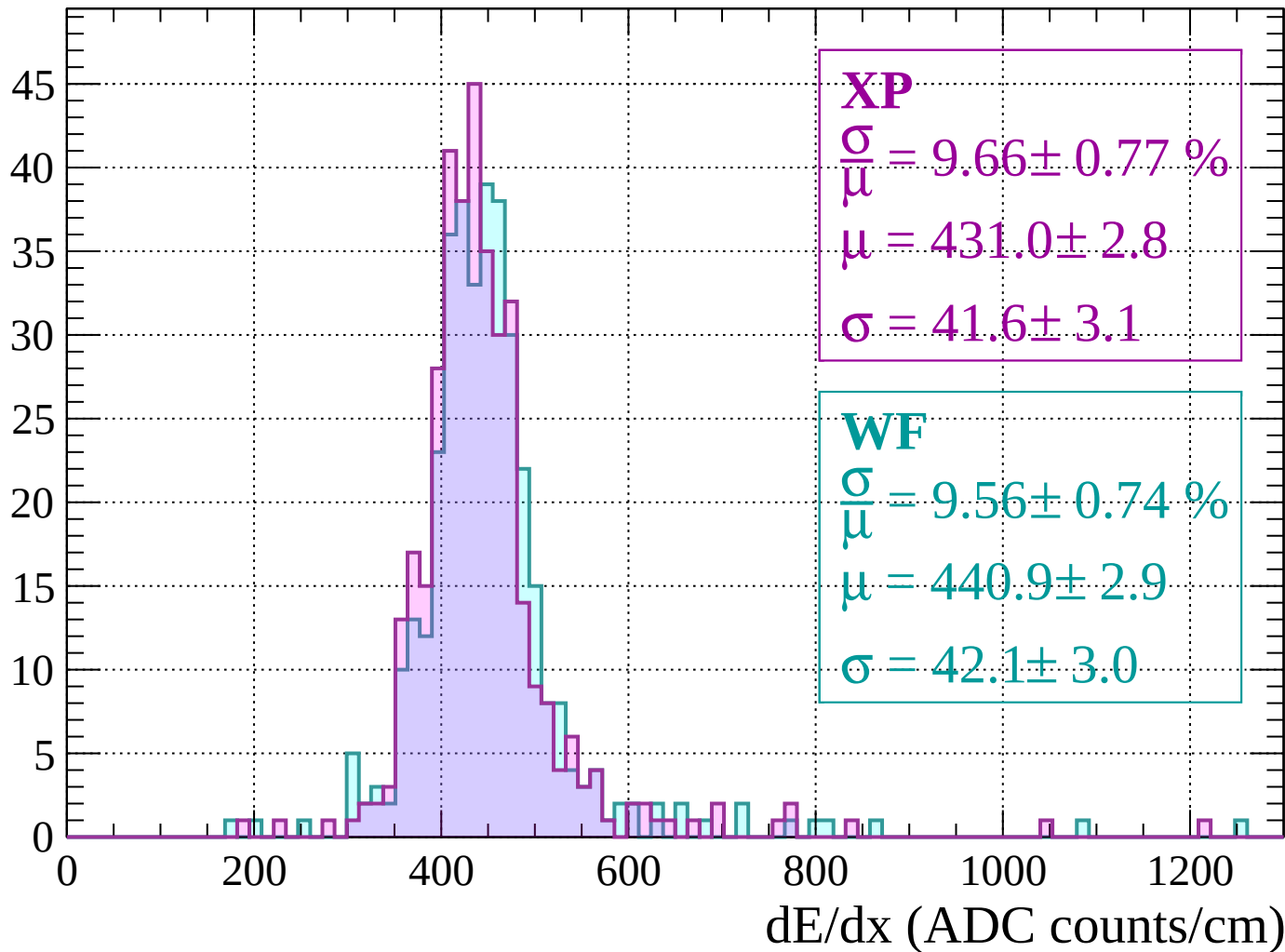
Energy loss | $880 < p < 920$

Count



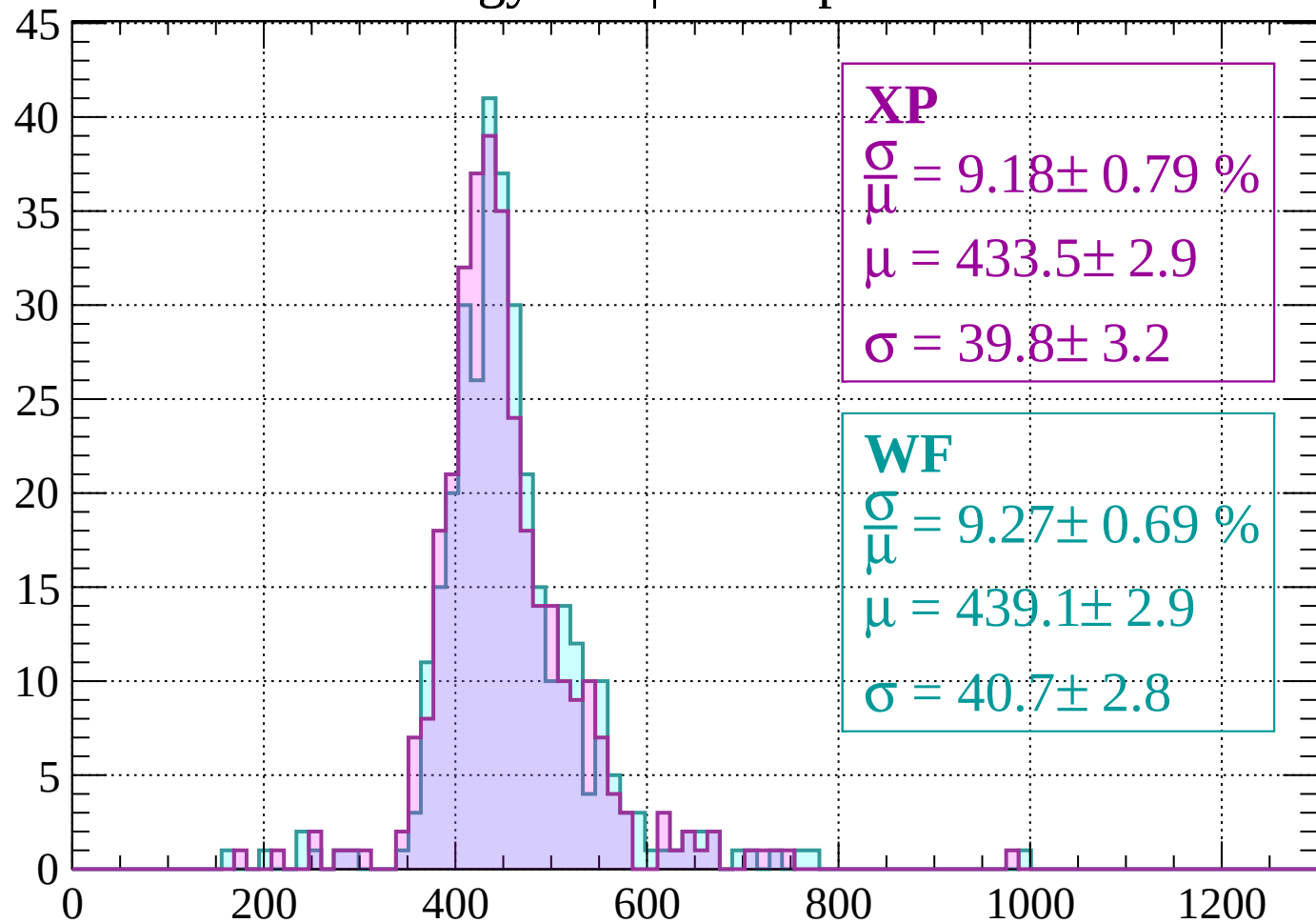
Energy loss | $920 < p < 960$

Count



Energy loss | $960 < p < 1000$

Count



XP

$$\frac{\sigma}{\mu} = 9.18 \pm 0.79 \%$$

$$\mu = 433.5 \pm 2.9$$

$$\sigma = 39.8 \pm 3.2$$

WF

$$\frac{\sigma}{\mu} = 9.27 \pm 0.69 \%$$

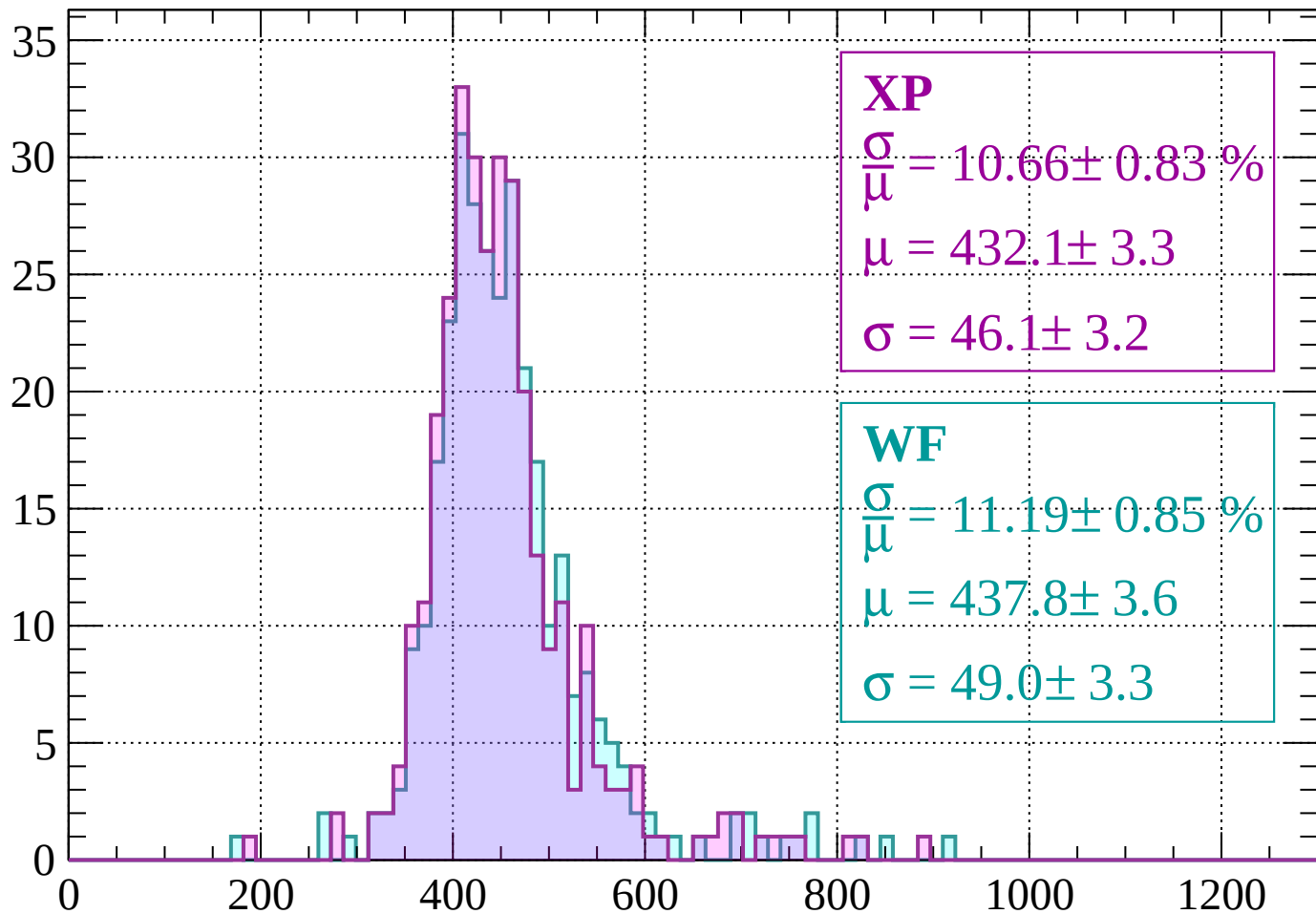
$$\mu = 439.1 \pm 2.9$$

$$\sigma = 40.7 \pm 2.8$$

dE/dx (ADC counts/cm)

Energy loss | $1000 < p < 1040$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 10.66 \pm 0.83 \%$$

$$\mu = 432.1 \pm 3.3$$

$$\sigma = 46.1 \pm 3.2$$

WF

$$\frac{\sigma}{\bar{\mu}} = 11.19 \pm 0.85 \%$$

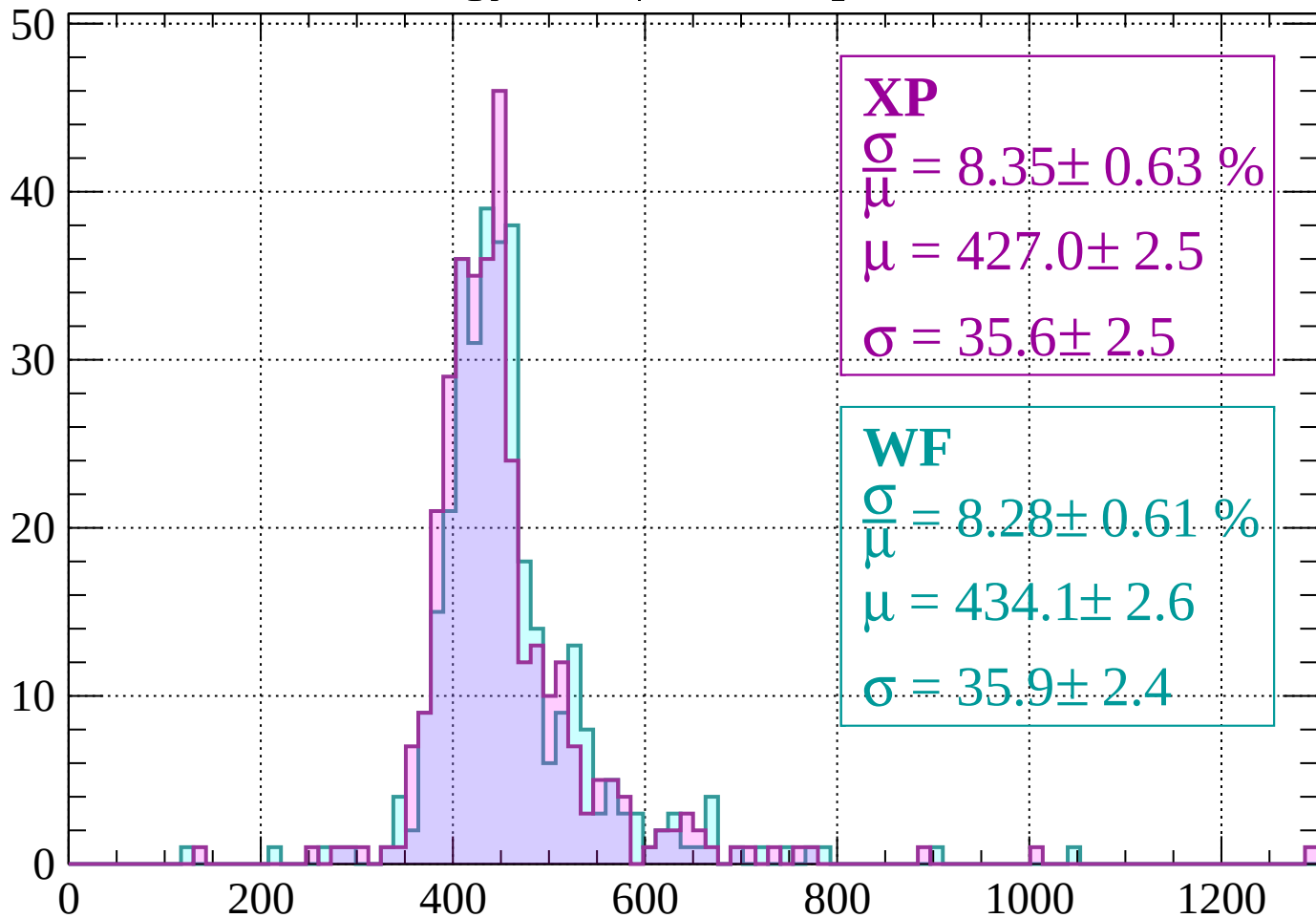
$$\mu = 437.8 \pm 3.6$$

$$\sigma = 49.0 \pm 3.3$$

dE/dx (ADC counts/cm)

Energy loss | $1040 < p < 1080$

Count



XP

$$\frac{\sigma}{\mu} = 8.35 \pm 0.63 \%$$

$$\mu = 427.0 \pm 2.5$$

$$\sigma = 35.6 \pm 2.5$$

WF

$$\frac{\sigma}{\mu} = 8.28 \pm 0.61 \%$$

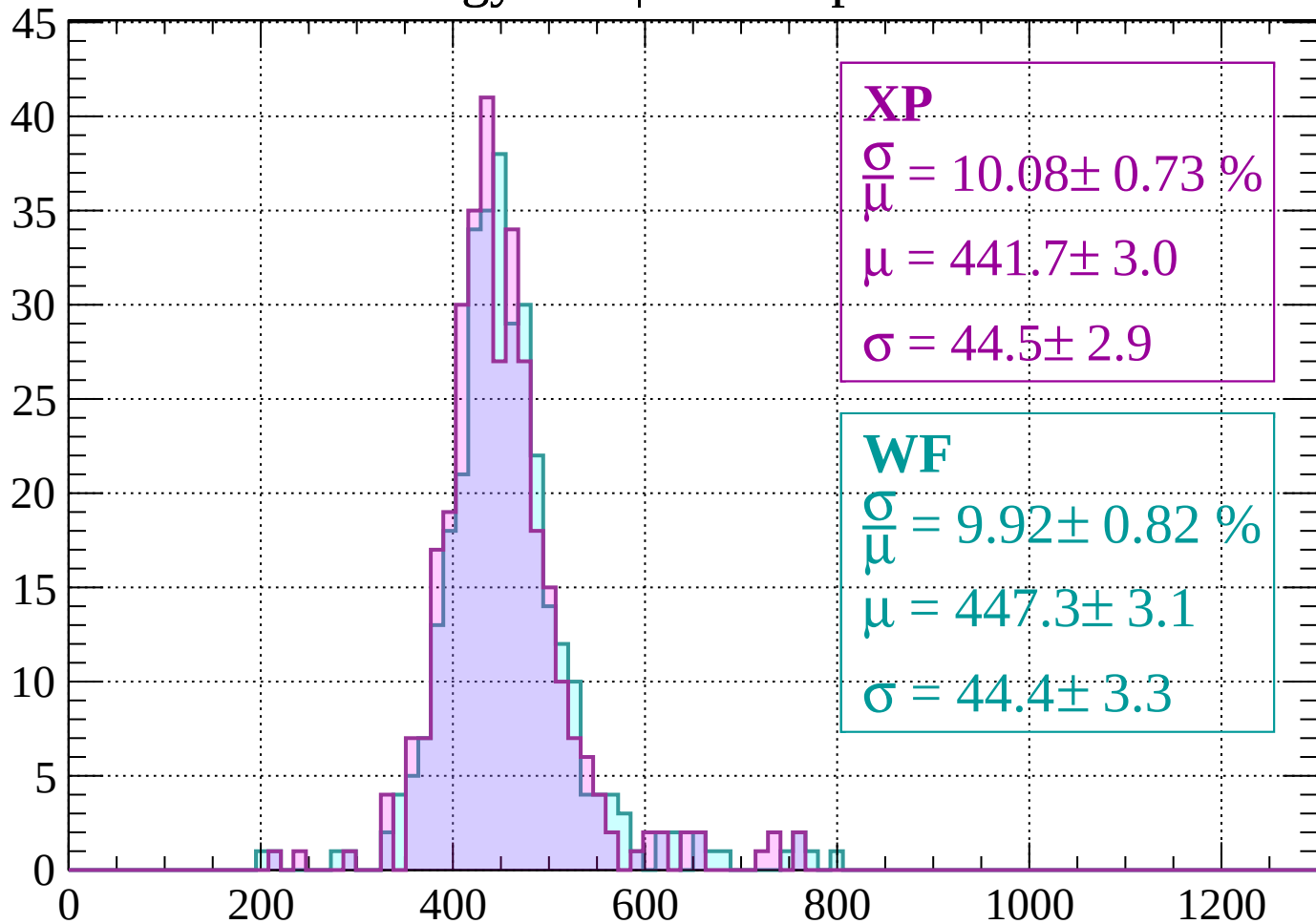
$$\mu = 434.1 \pm 2.6$$

$$\sigma = 35.9 \pm 2.4$$

dE/dx (ADC counts/cm)

Energy loss | $1080 < p < 1120$

Count



XP

$$\frac{\sigma}{\mu} = 10.08 \pm 0.73 \%$$

$$\mu = 441.7 \pm 3.0$$

$$\sigma = 44.5 \pm 2.9$$

WF

$$\frac{\sigma}{\mu} = 9.92 \pm 0.82 \%$$

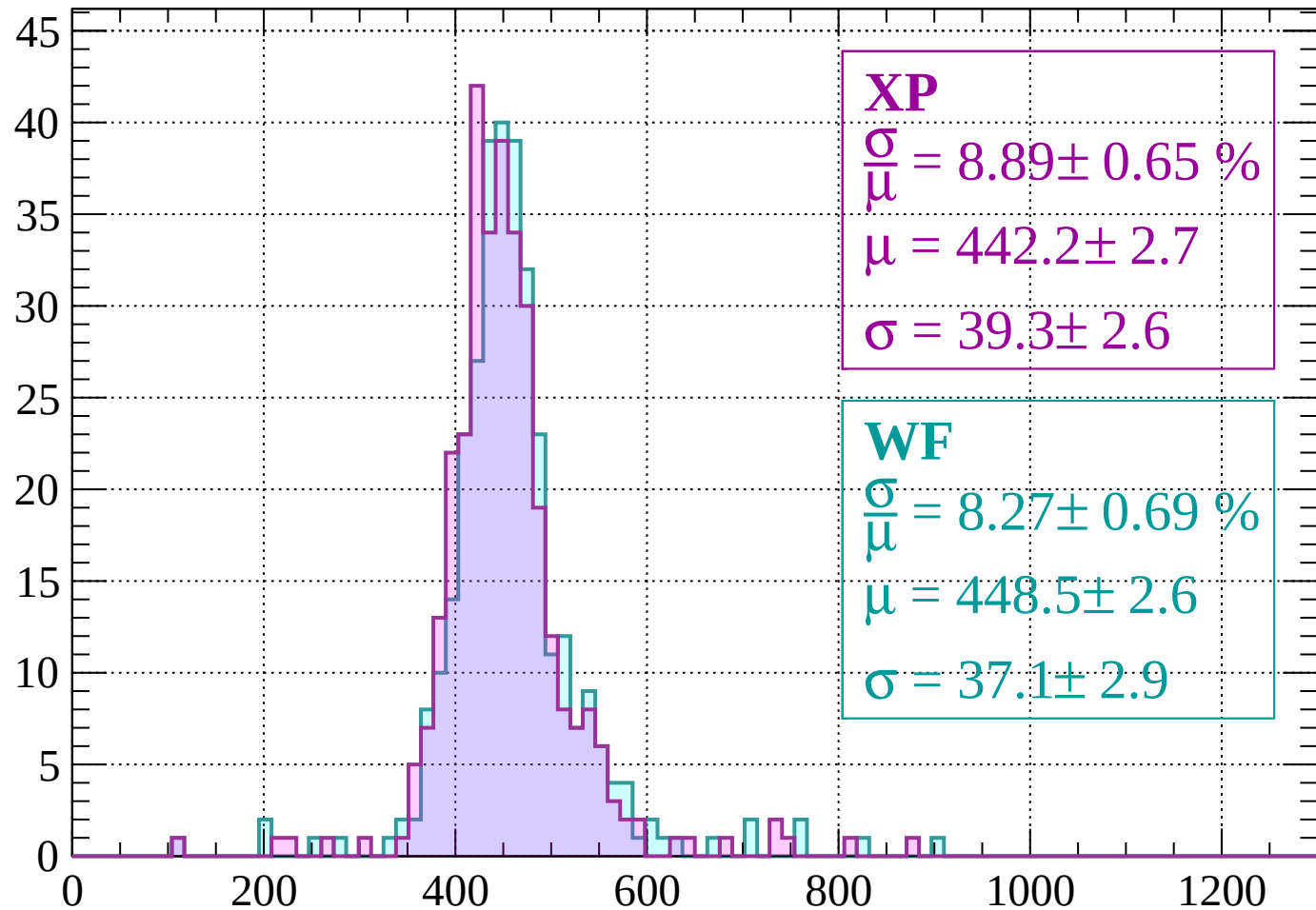
$$\mu = 447.3 \pm 3.1$$

$$\sigma = 44.4 \pm 3.3$$

dE/dx (ADC counts/cm)

Energy loss | $1120 < p < 1160$

Count



XP

$$\frac{\sigma}{\mu} = 8.89 \pm 0.65 \%$$

$$\mu = 442.2 \pm 2.7$$

$$\sigma = 39.3 \pm 2.6$$

WF

$$\frac{\sigma}{\mu} = 8.27 \pm 0.69 \%$$

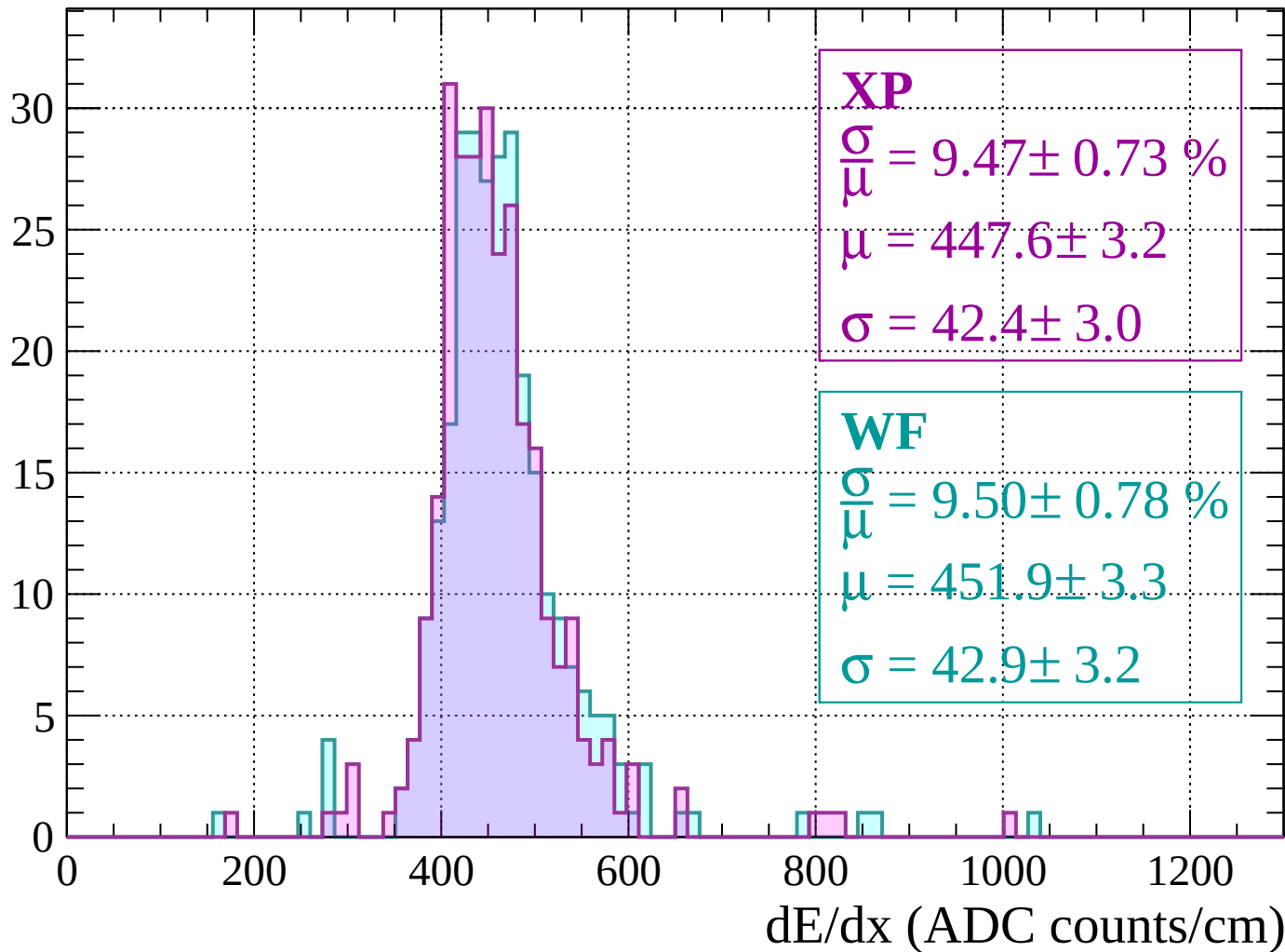
$$\mu = 448.5 \pm 2.6$$

$$\sigma = 37.1 \pm 2.9$$

dE/dx (ADC counts/cm)

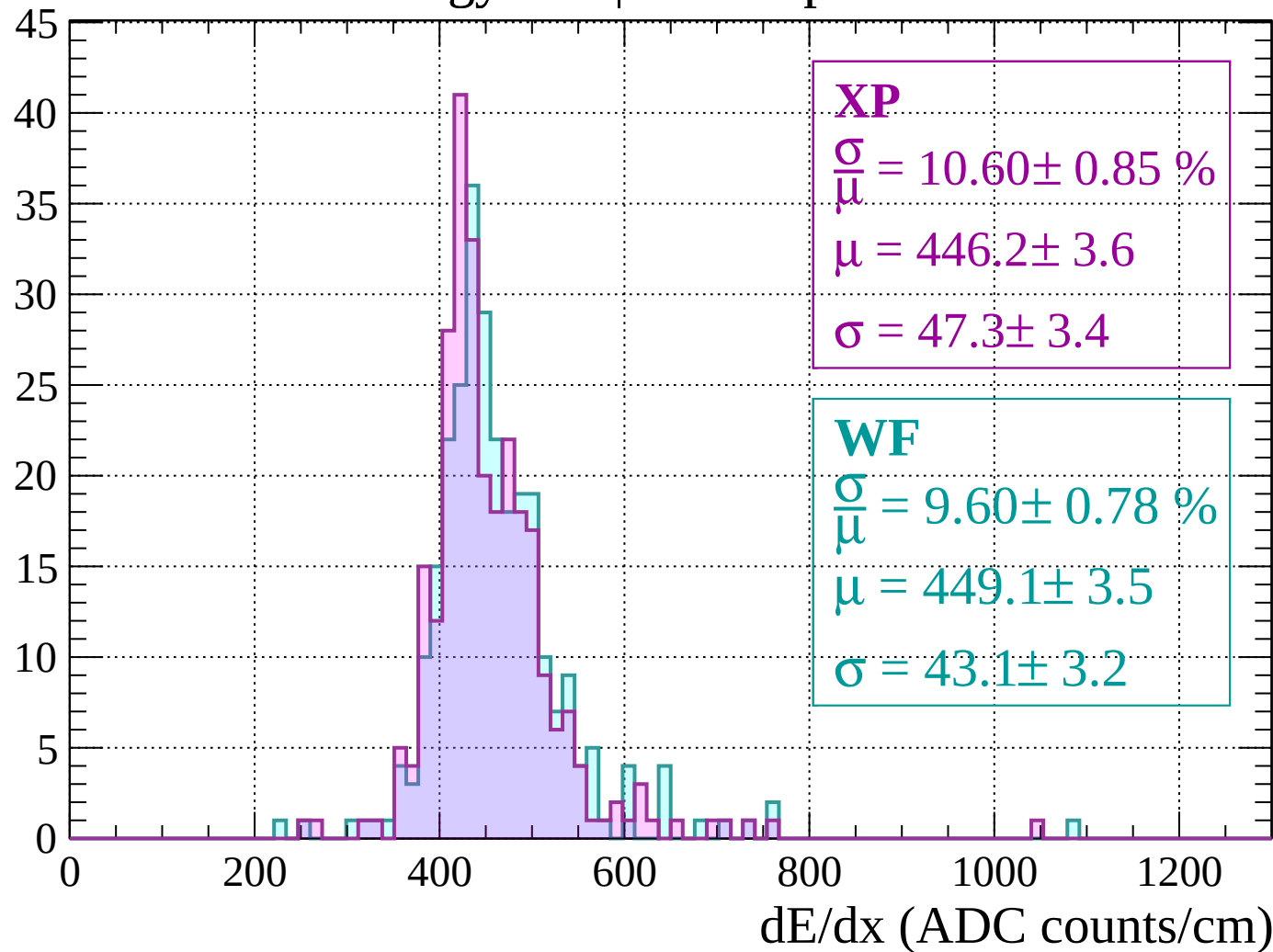
Energy loss | $1160 < p < 1200$

Count



Energy loss | $1200 < p < 1240$

Count



Energy loss | $1240 < p < 1280$

Count

XP

$$\frac{\sigma}{\bar{\mu}} = 9.54 \pm 0.79 \%$$

$$\mu = 451.4 \pm 3.6$$

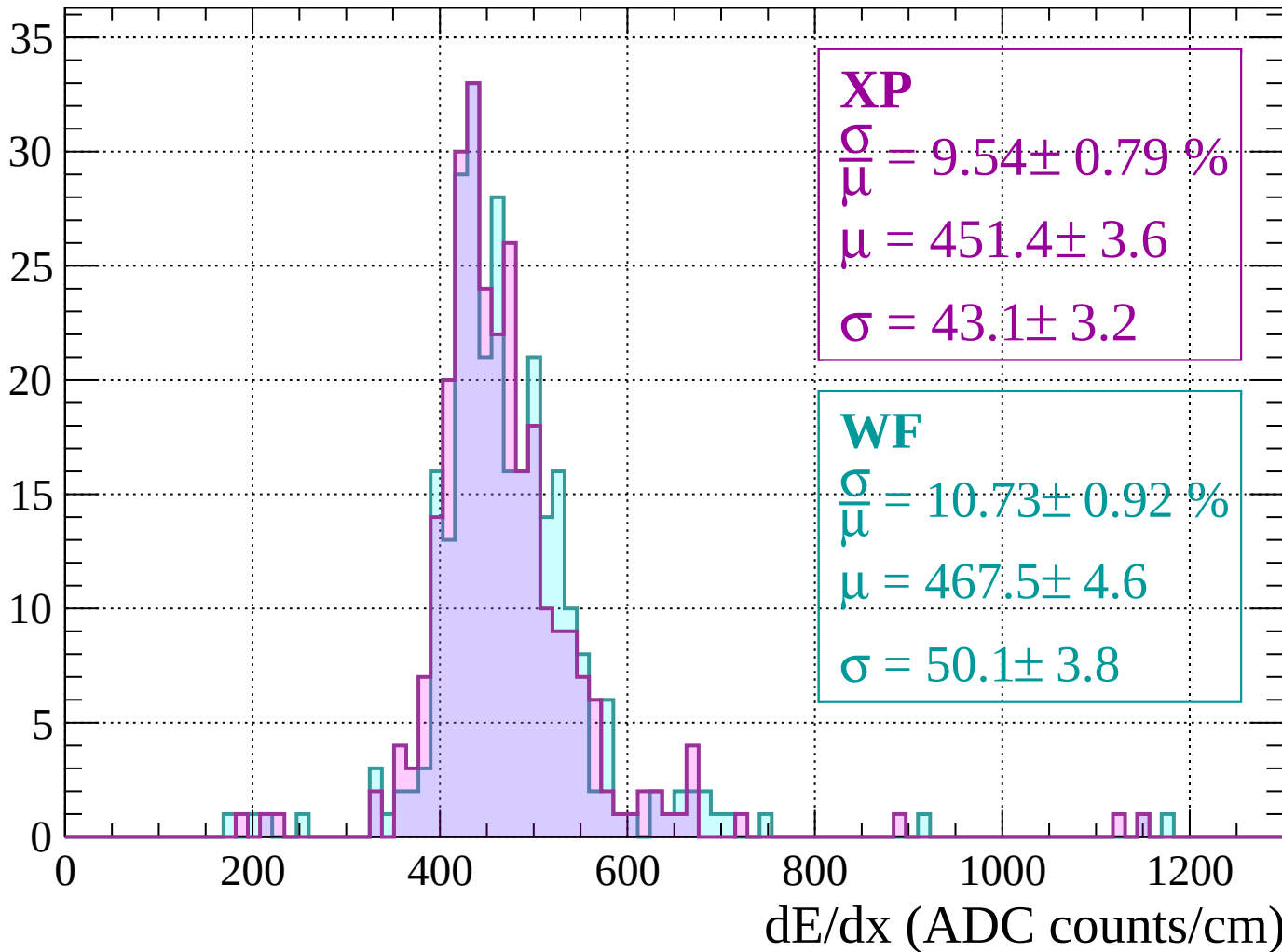
$$\sigma = 43.1 \pm 3.2$$

WF

$$\frac{\sigma}{\bar{\mu}} = 10.73 \pm 0.92 \%$$

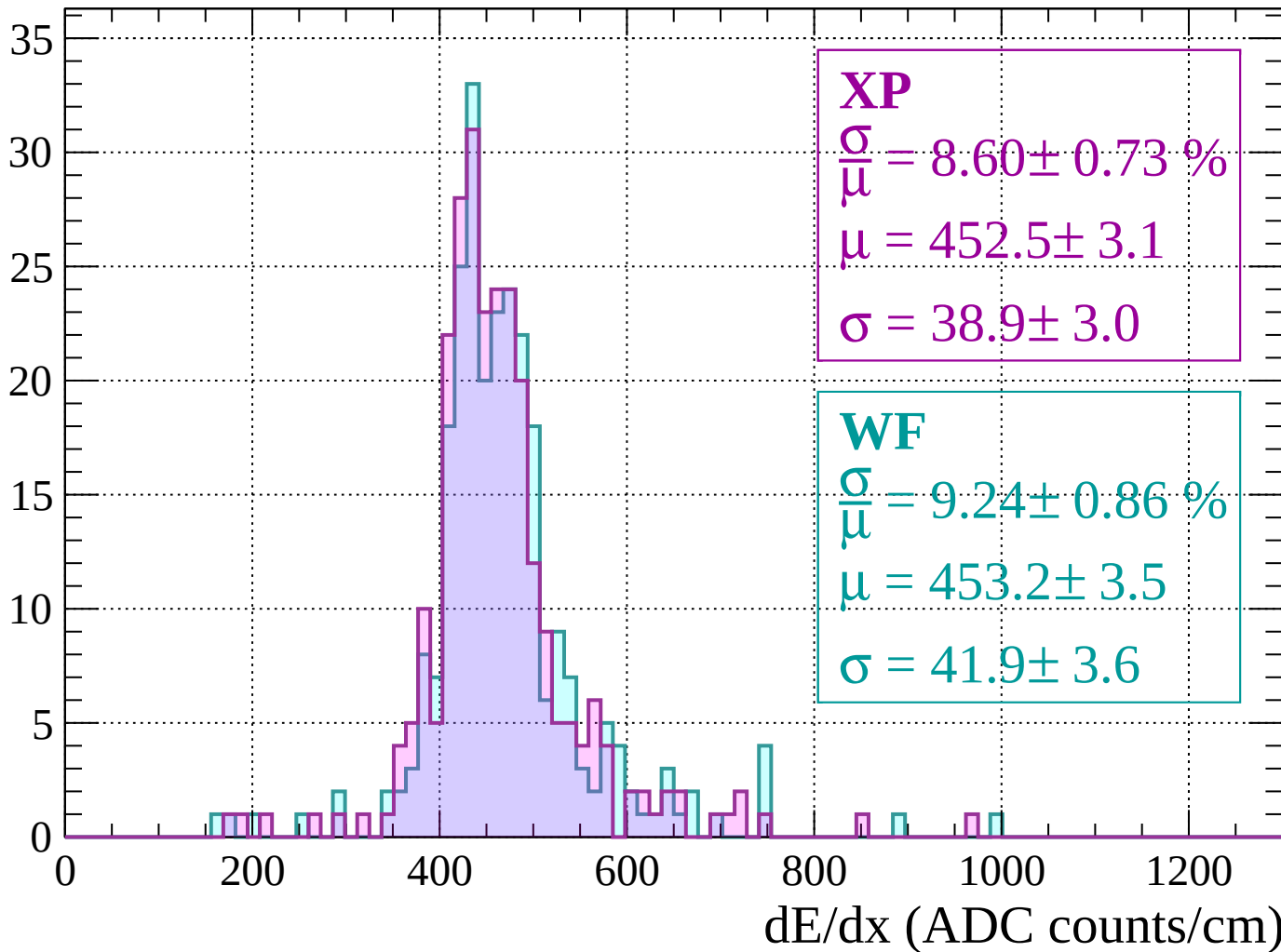
$$\mu = 467.5 \pm 4.6$$

$$\sigma = 50.1 \pm 3.8$$



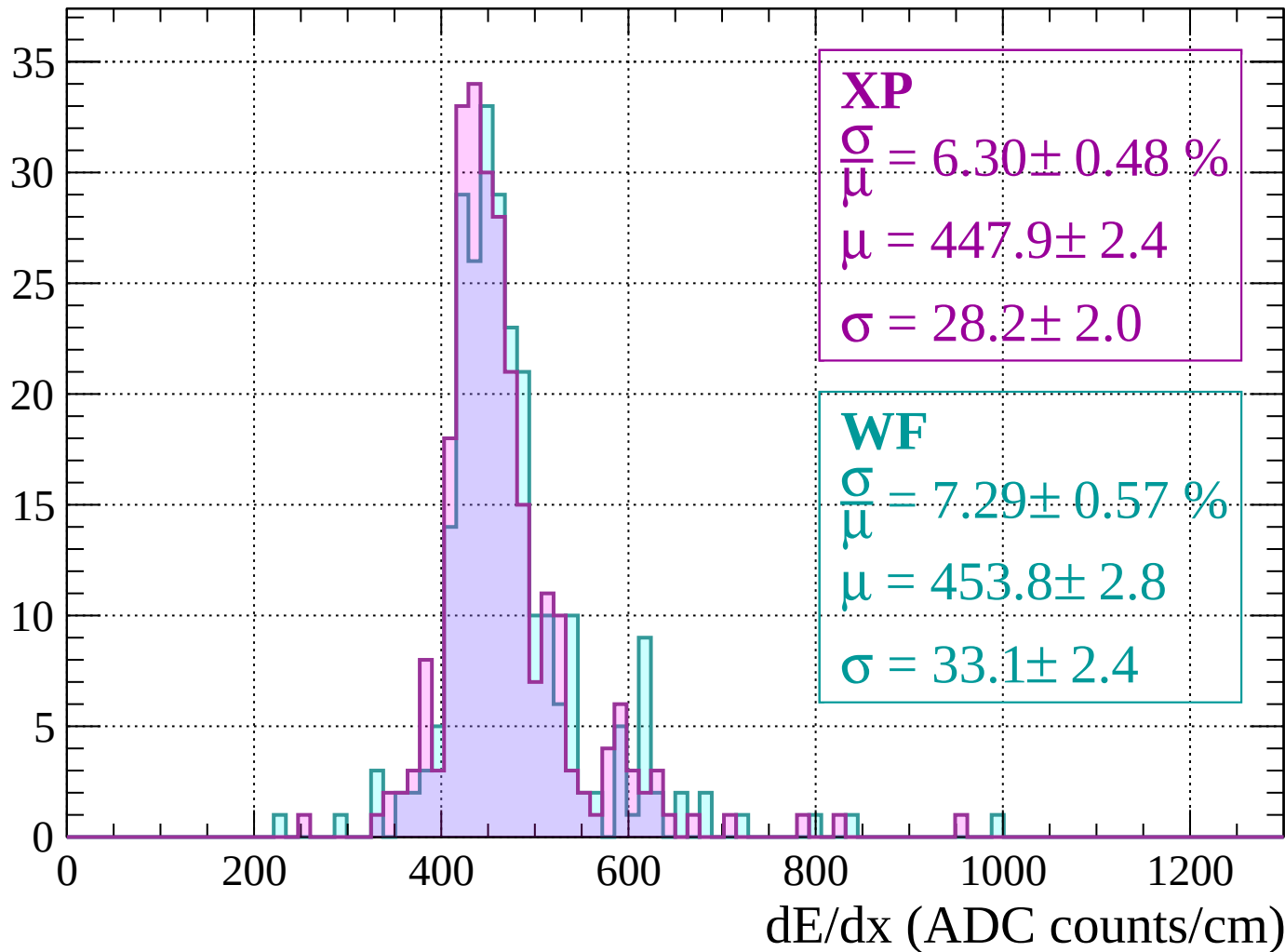
Energy loss | $1280 < p < 1320$

Count



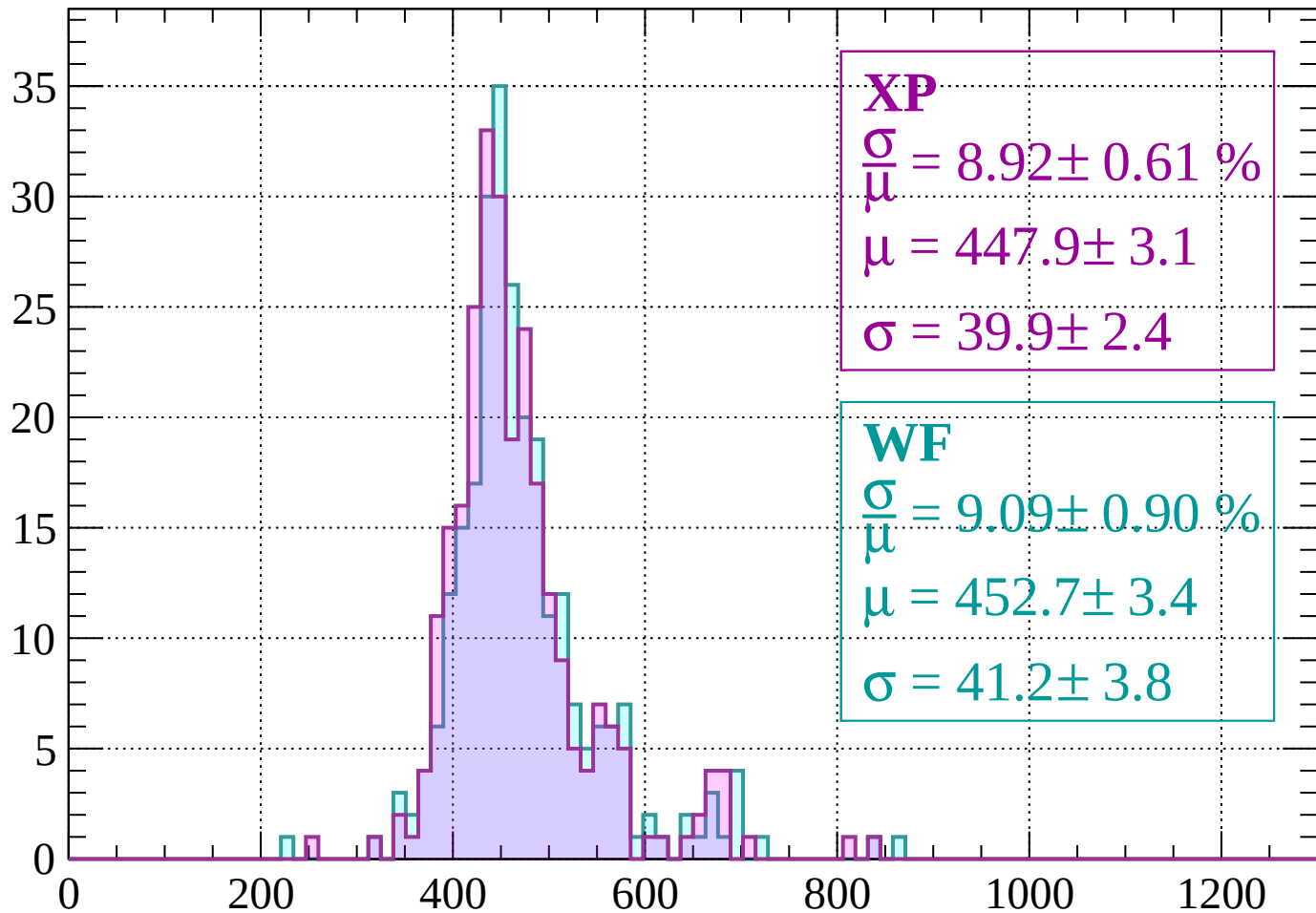
Energy loss | $1320 < p < 1360$

Count



Energy loss | $1360 < p < 1400$

Count



XP

$$\frac{\sigma}{\mu} = 8.92 \pm 0.61 \%$$

$$\mu = 447.9 \pm 3.1$$

$$\sigma = 39.9 \pm 2.4$$

WF

$$\frac{\sigma}{\mu} = 9.09 \pm 0.90 \%$$

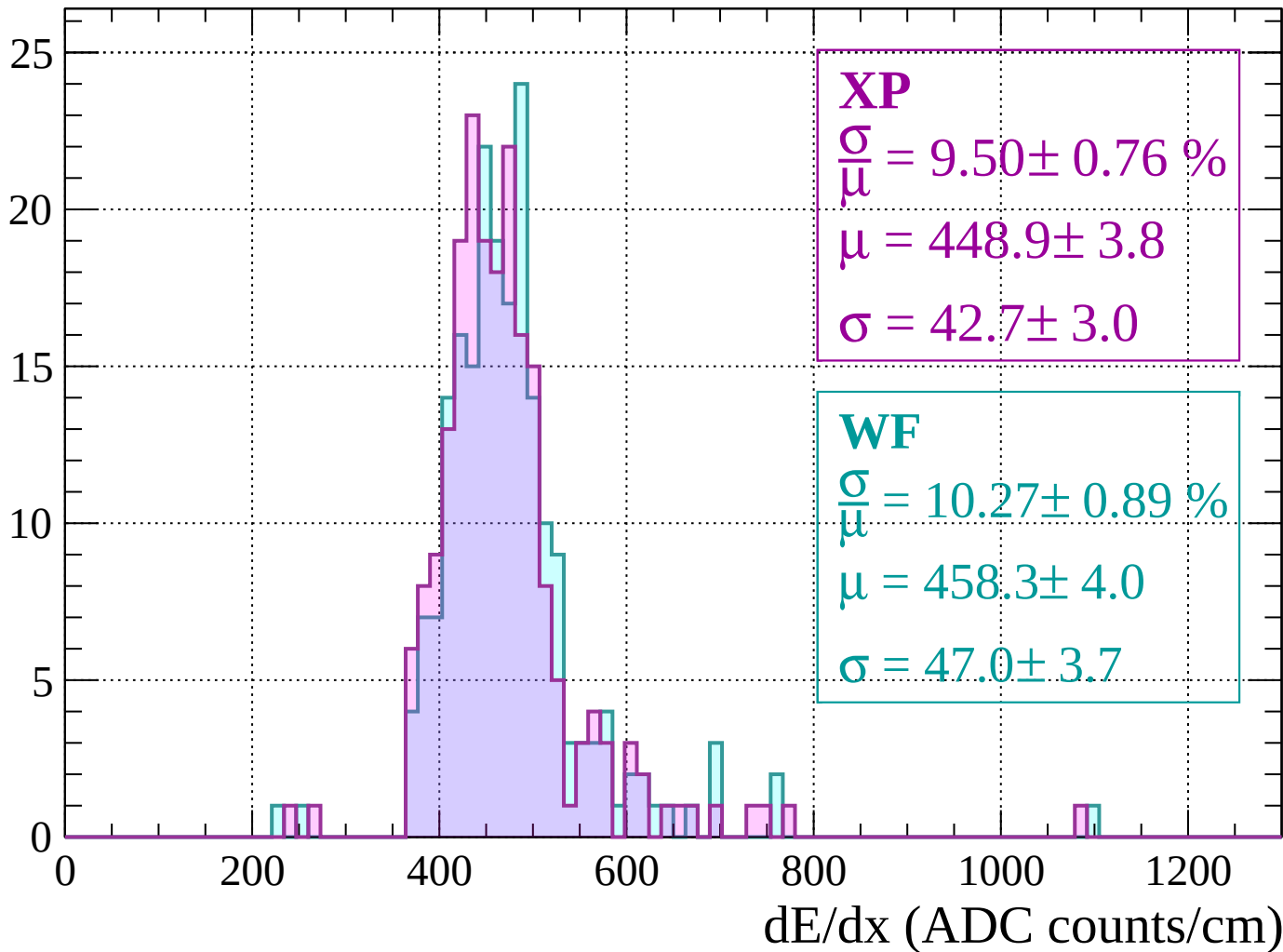
$$\mu = 452.7 \pm 3.4$$

$$\sigma = 41.2 \pm 3.8$$

dE/dx (ADC counts/cm)

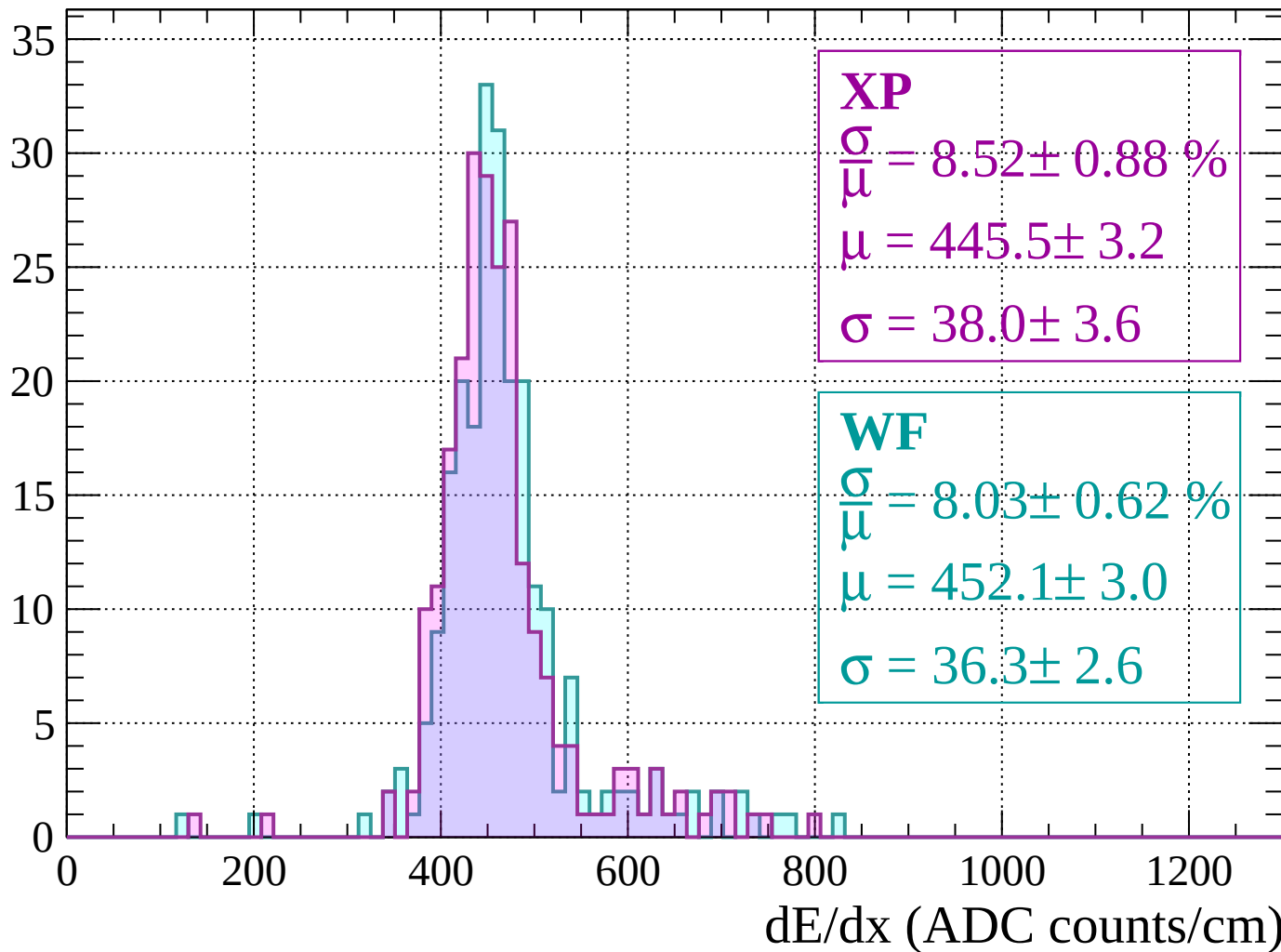
Energy loss | $1400 < p < 1440$

Count



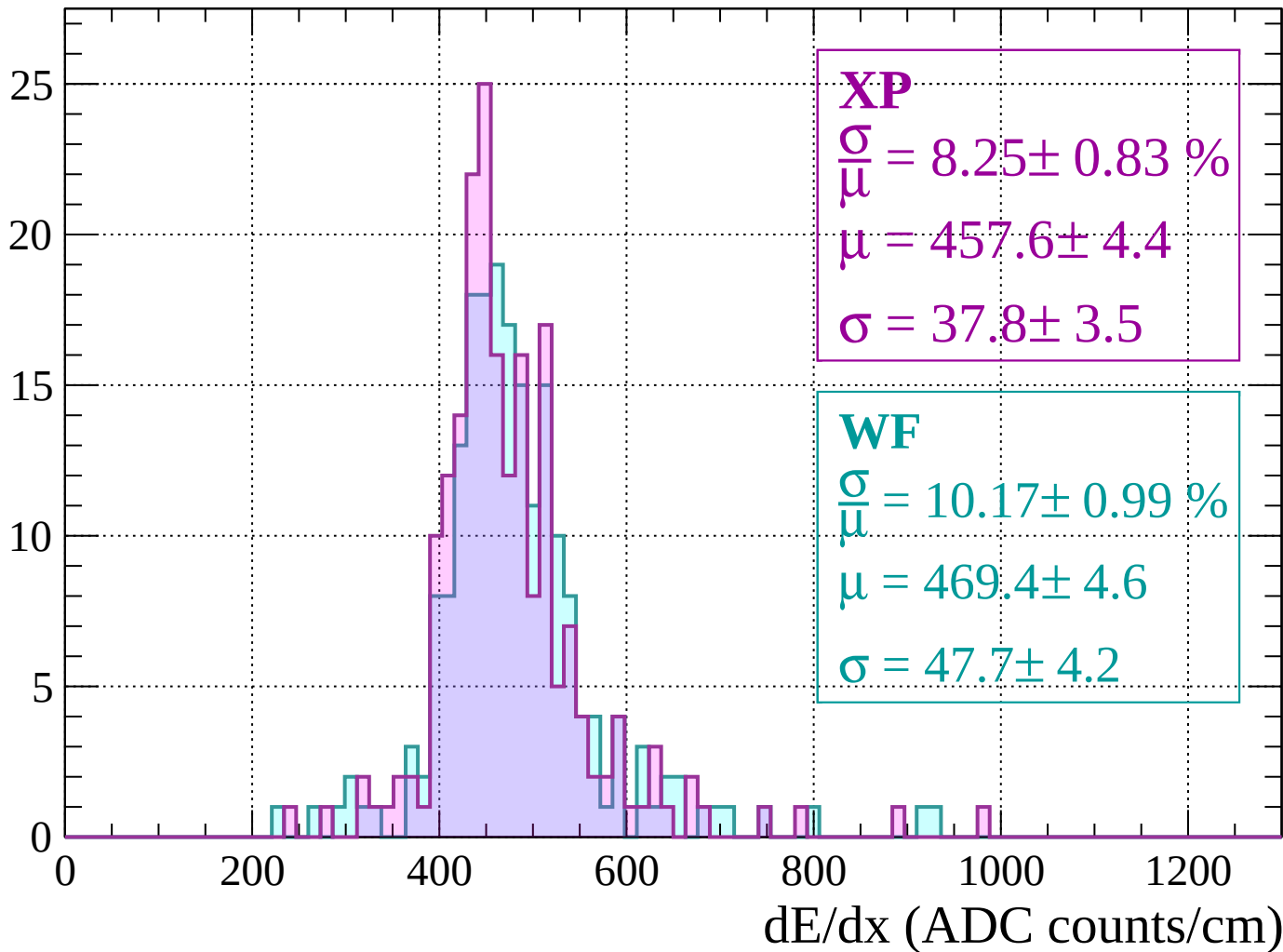
Energy loss | $1440 < p < 1480$

Count



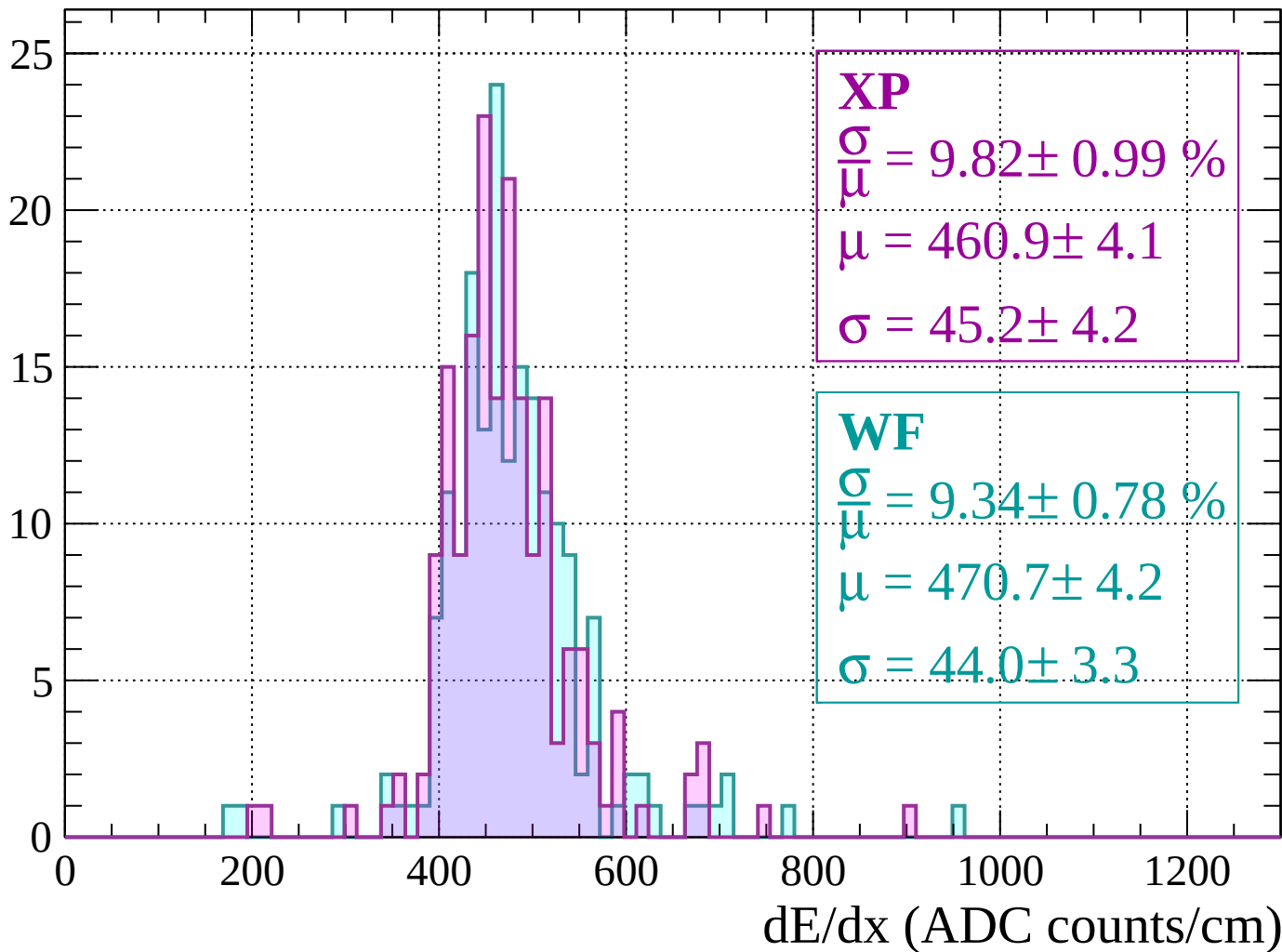
Energy loss | $1480 < p < 1520$

Count



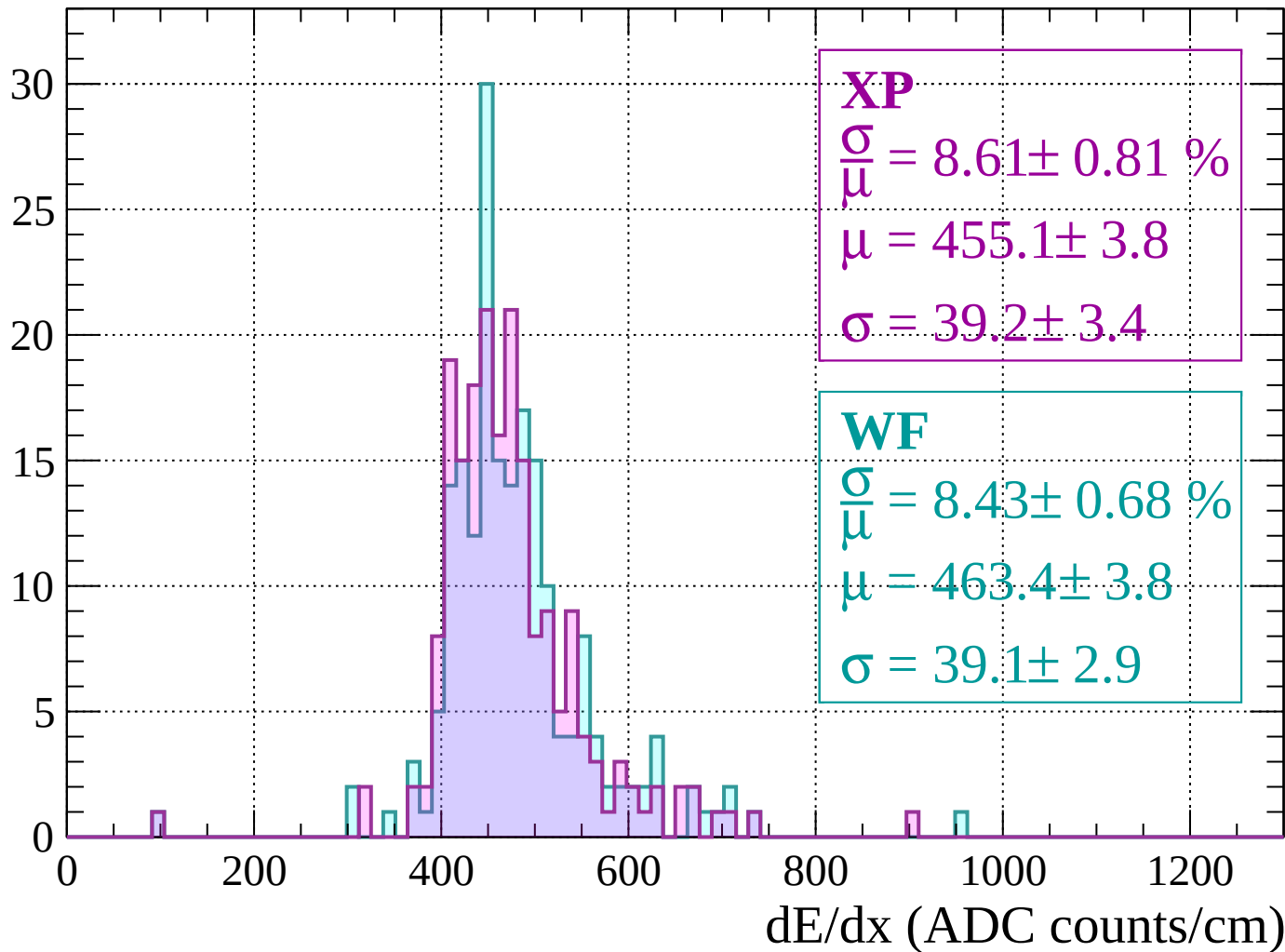
Energy loss | $1520 < p < 1560$

Count



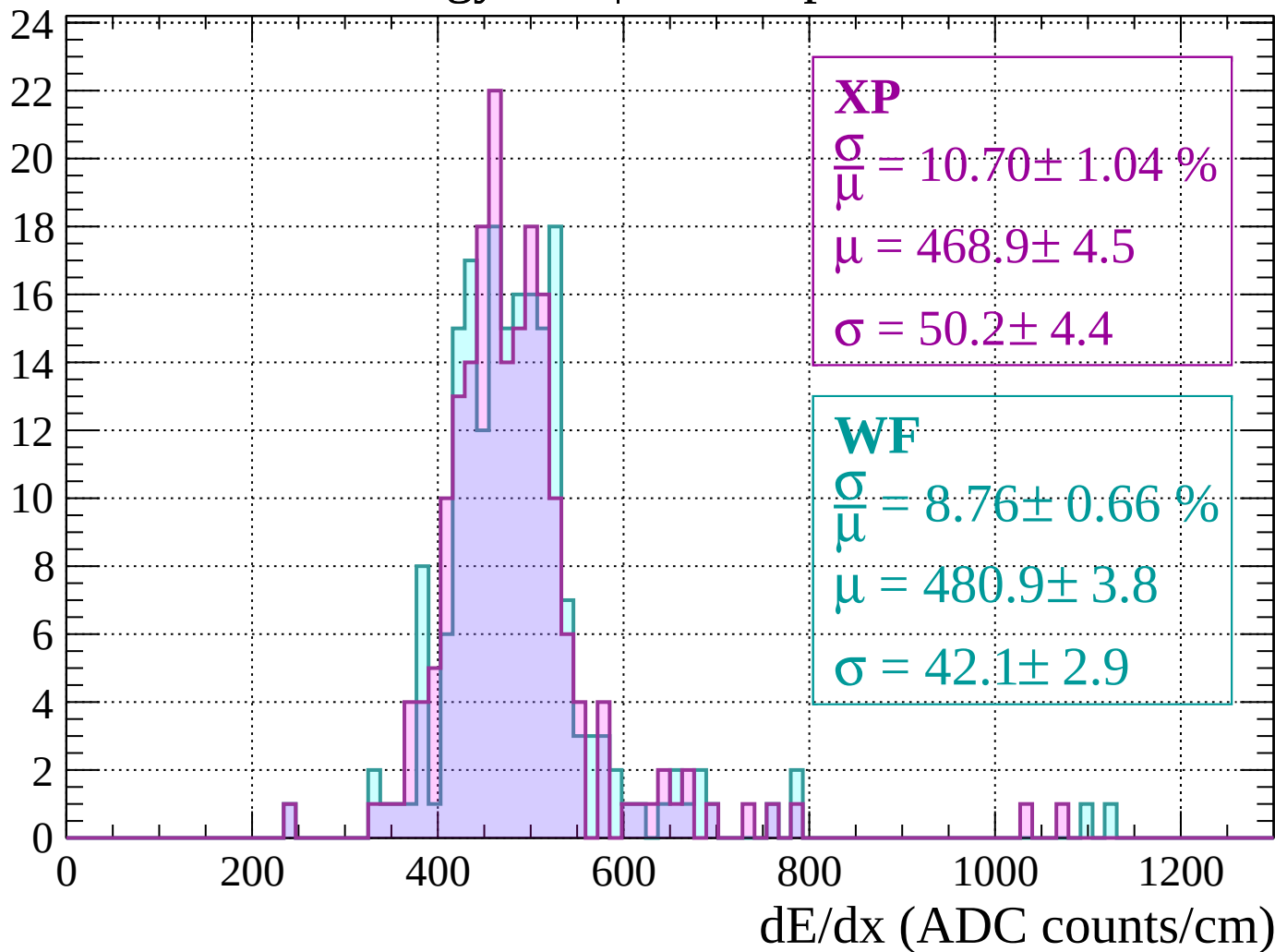
Energy loss | $1560 < p < 1600$

Count



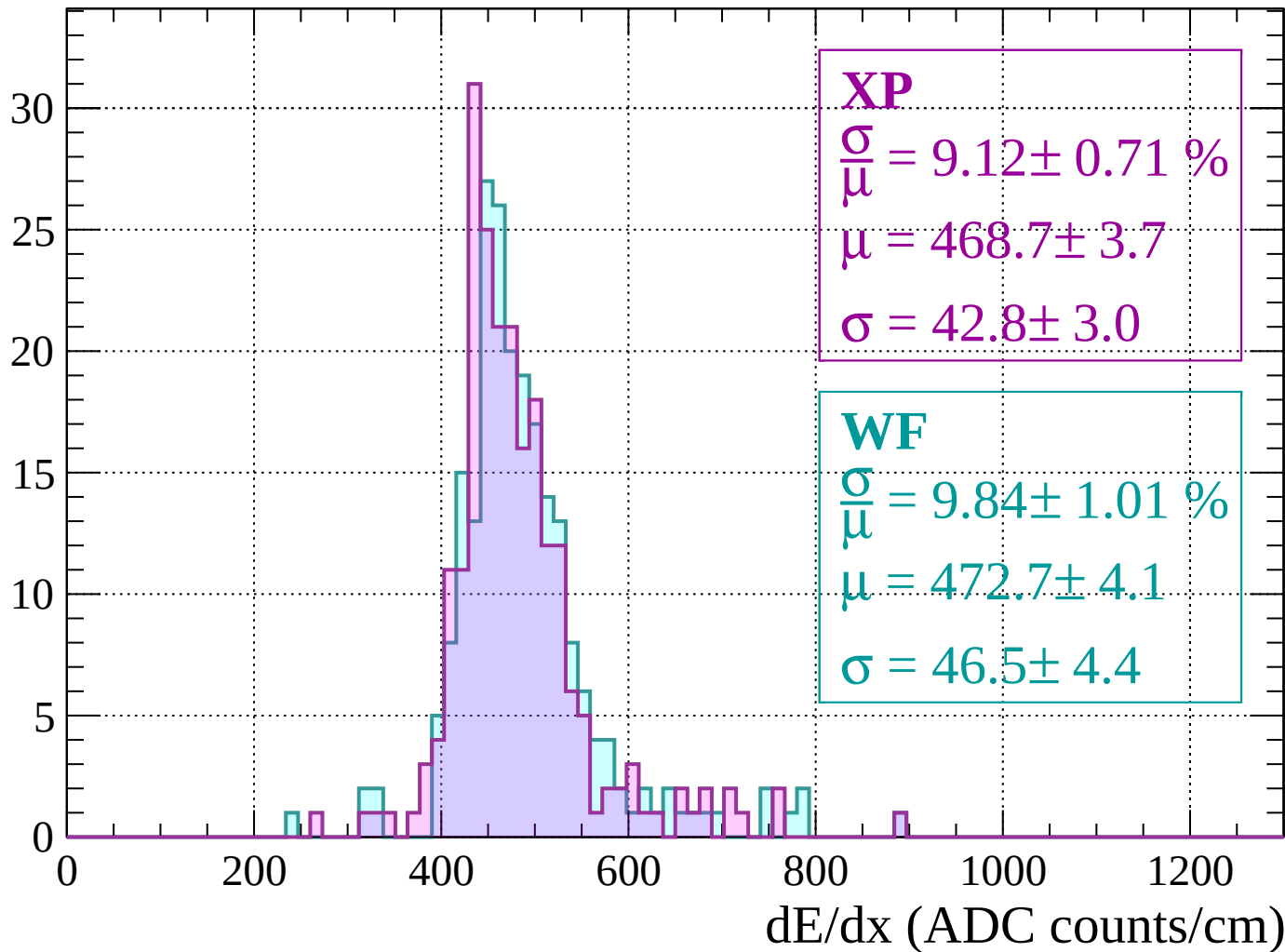
Energy loss | $1600 < p < 1640$

Count



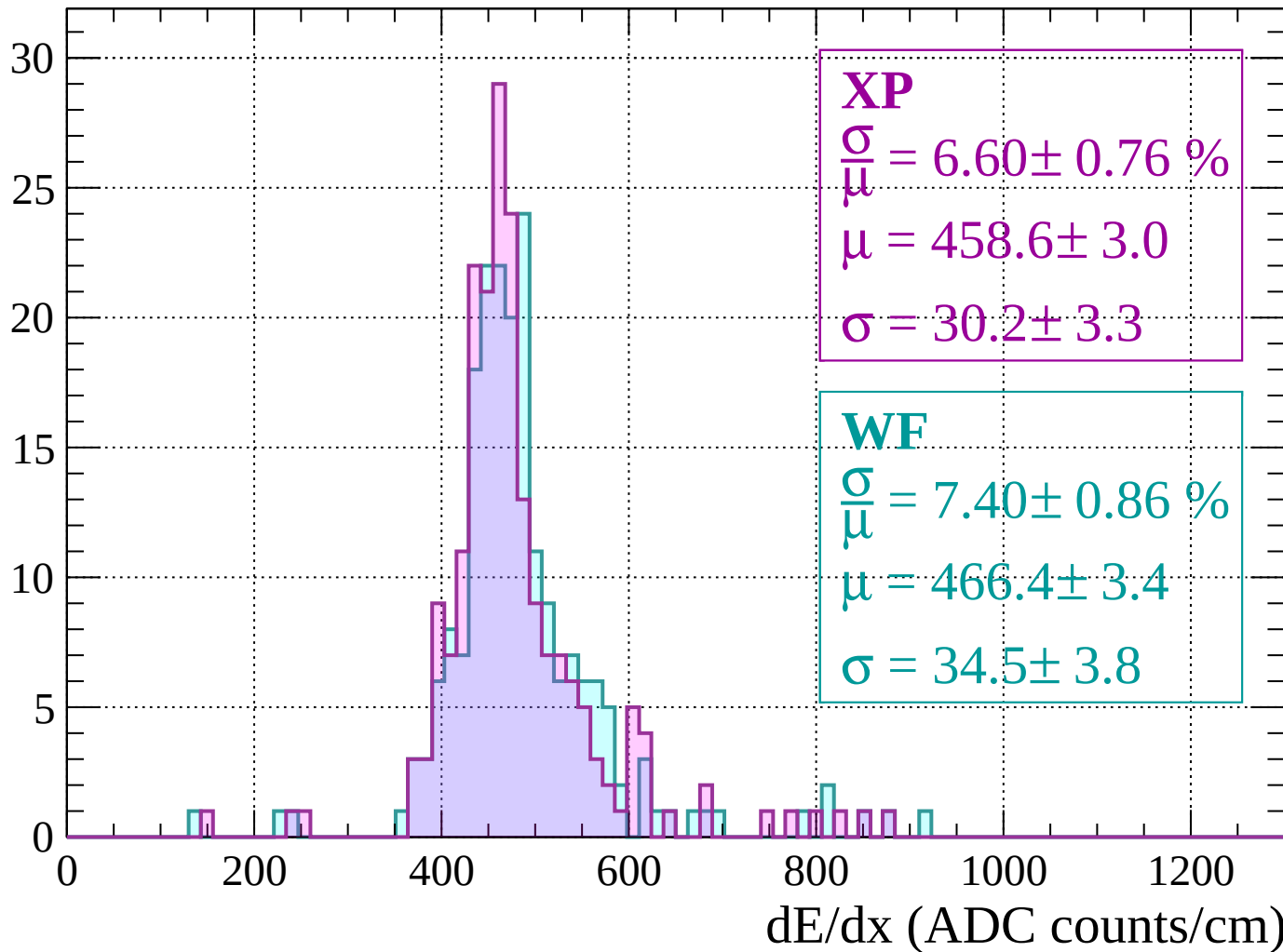
Energy loss | $1640 < p < 1680$

Count



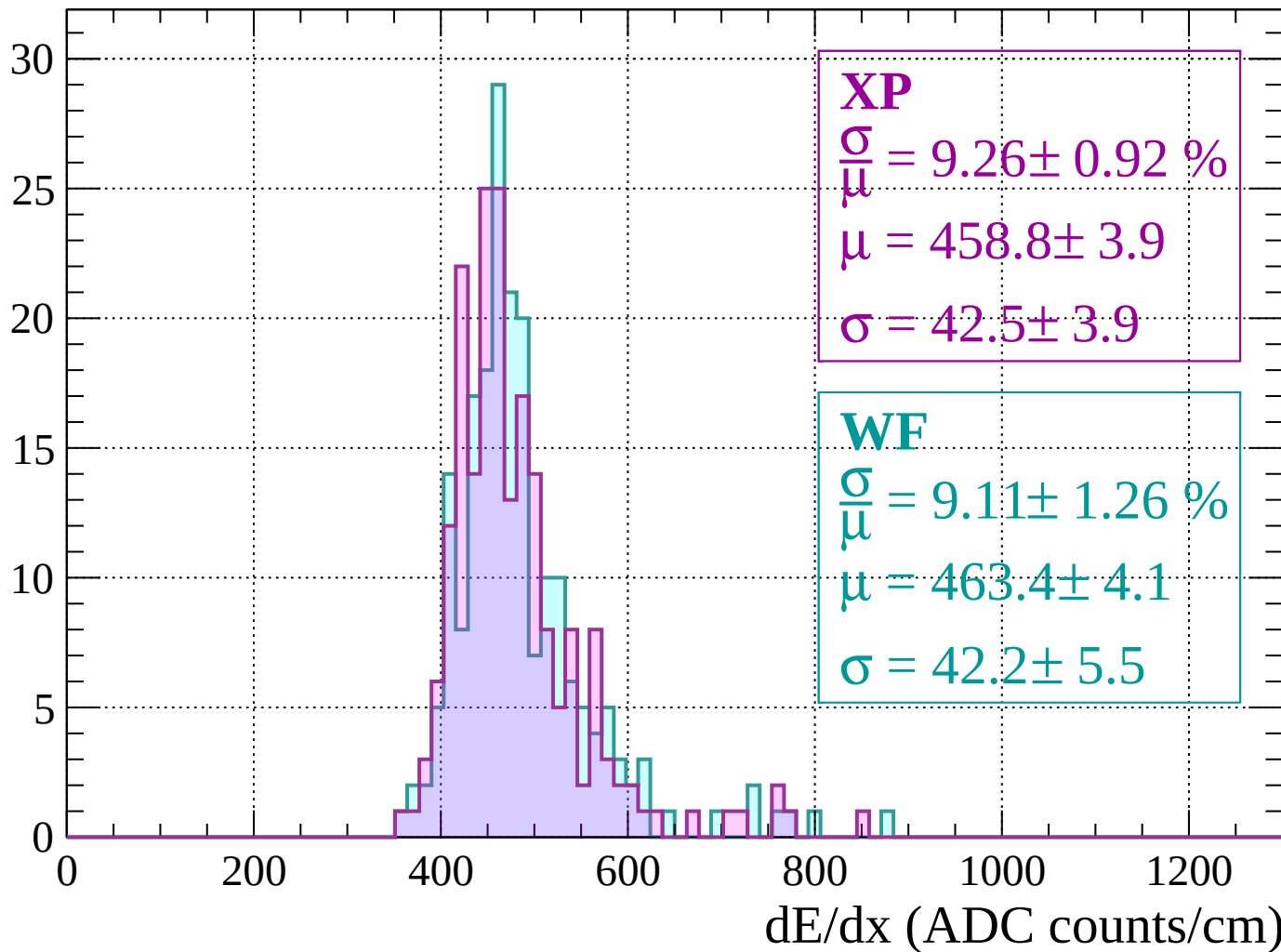
Energy loss | $1680 < p < 1720$

Count



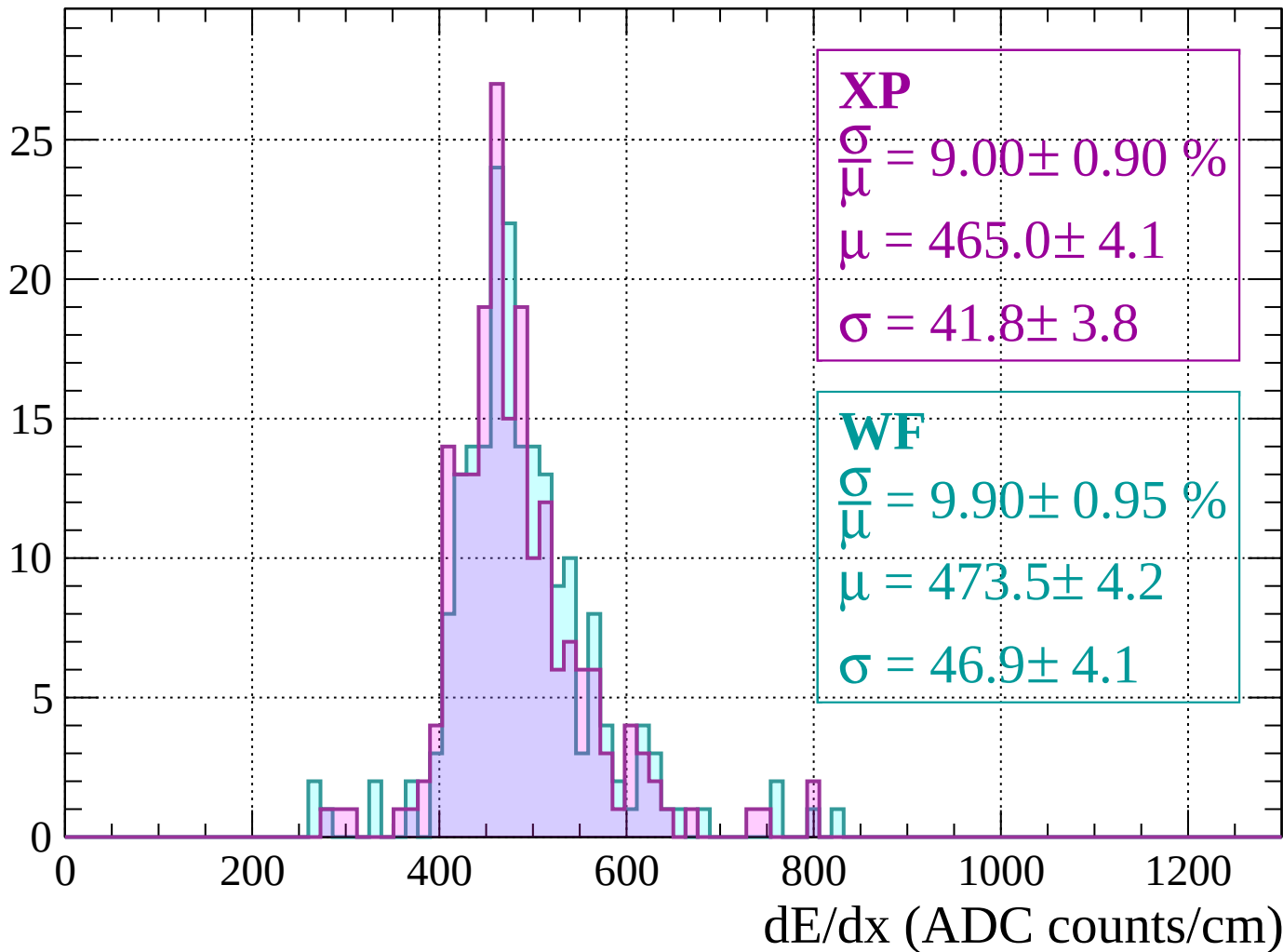
Energy loss | $1720 < p < 1760$

Count



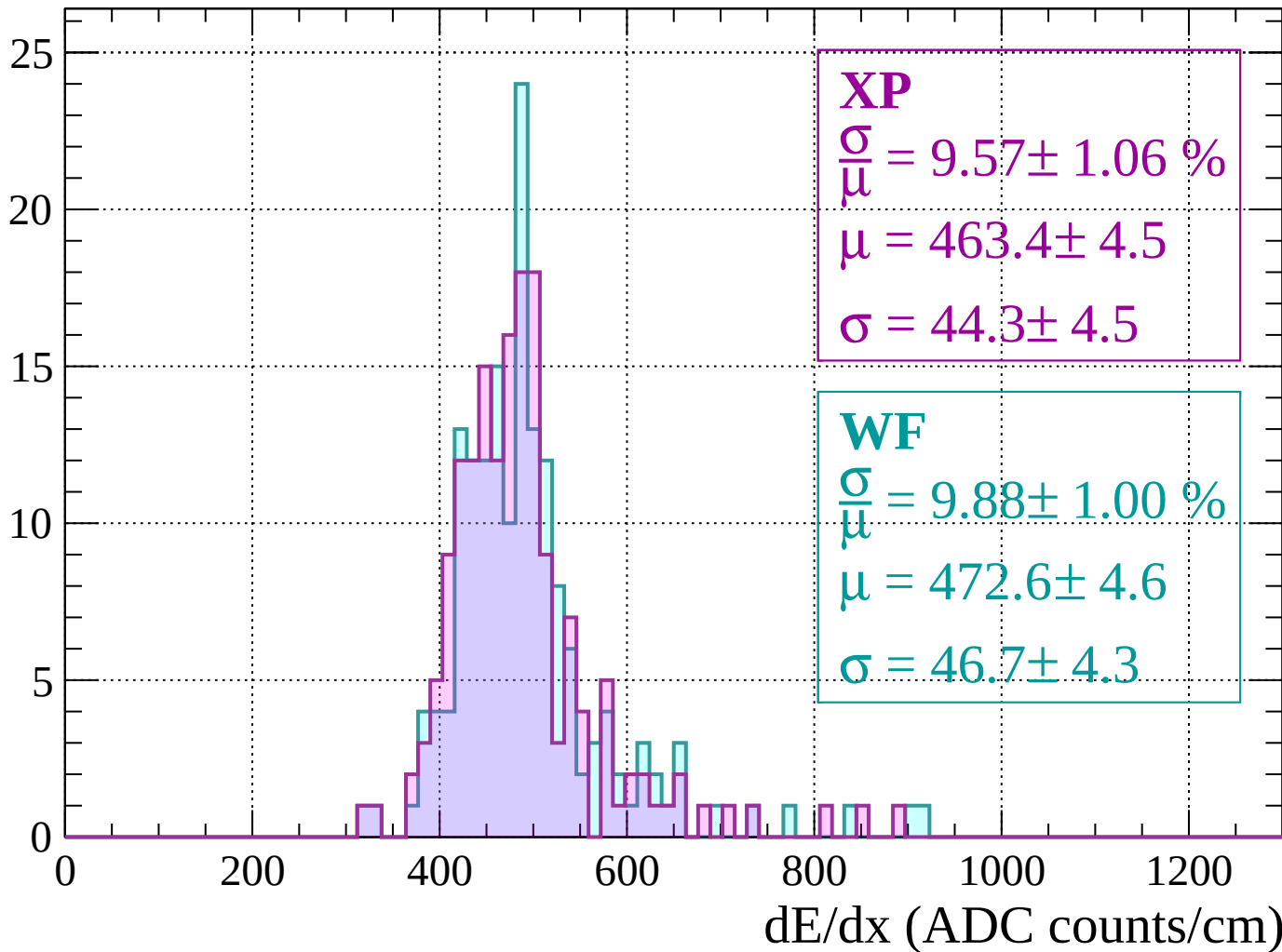
Energy loss | $1760 < p < 1800$

Count



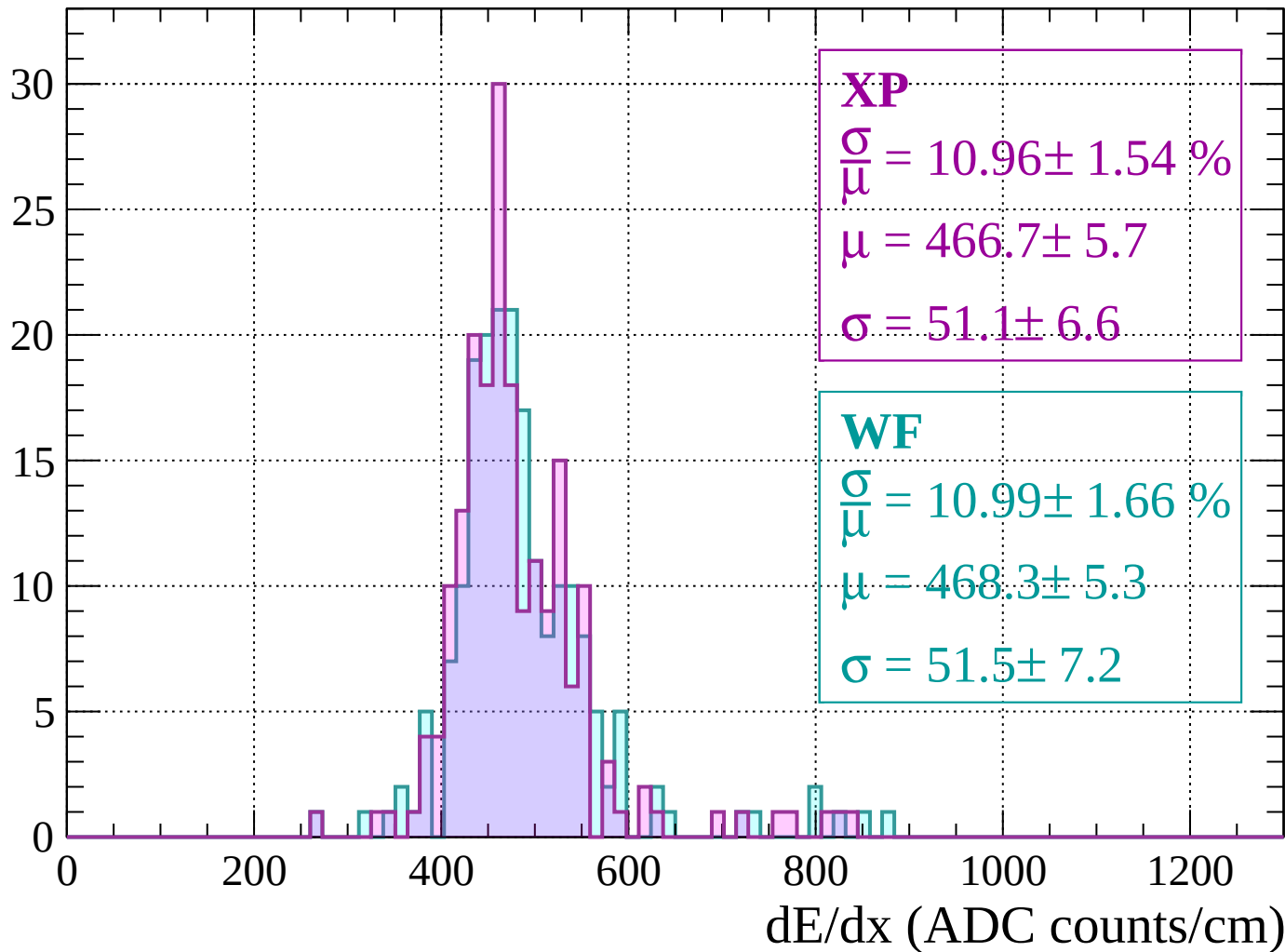
Energy loss | $1800 < p < 1840$

Count



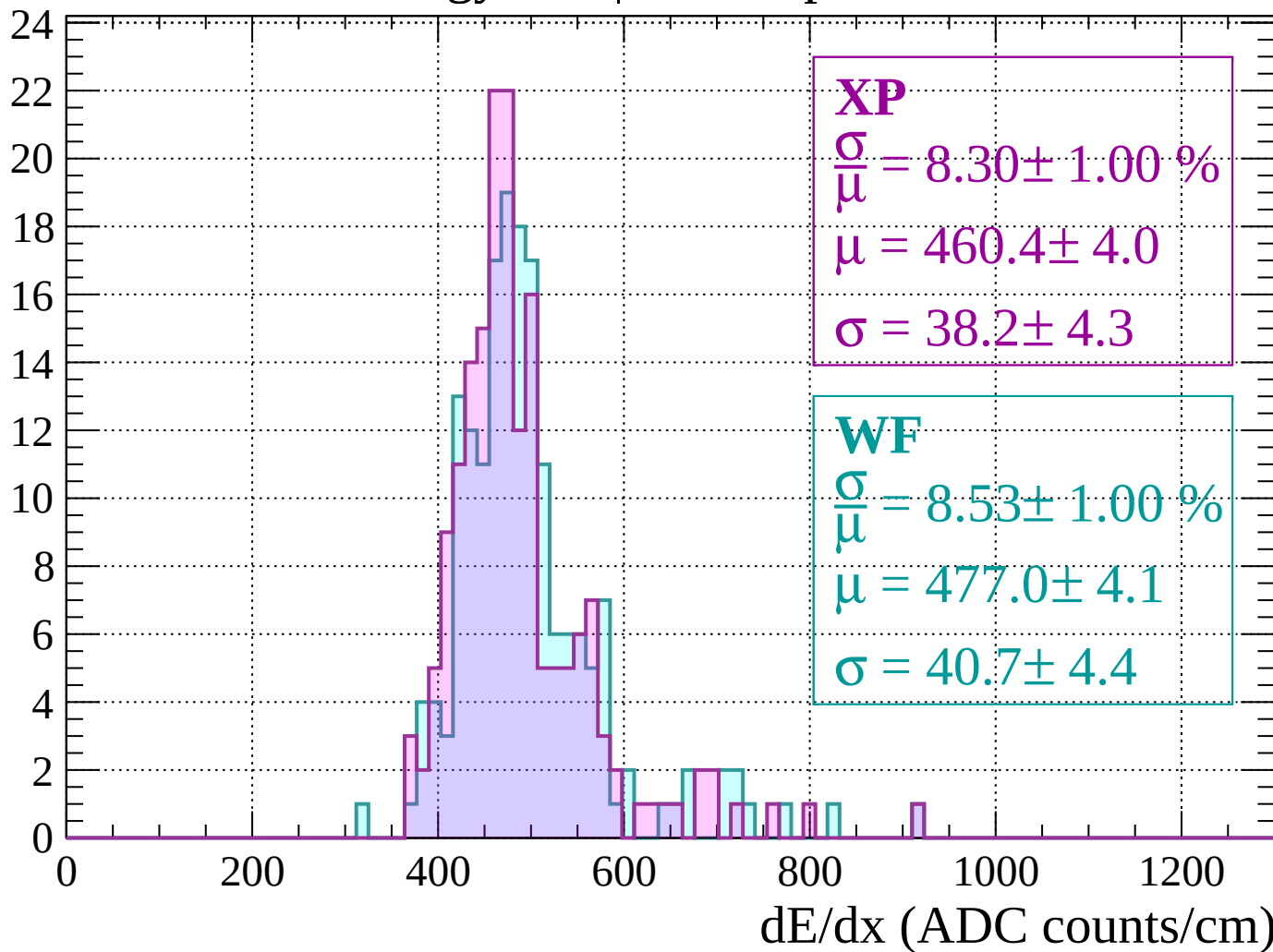
Energy loss | $1840 < p < 1880$

Count



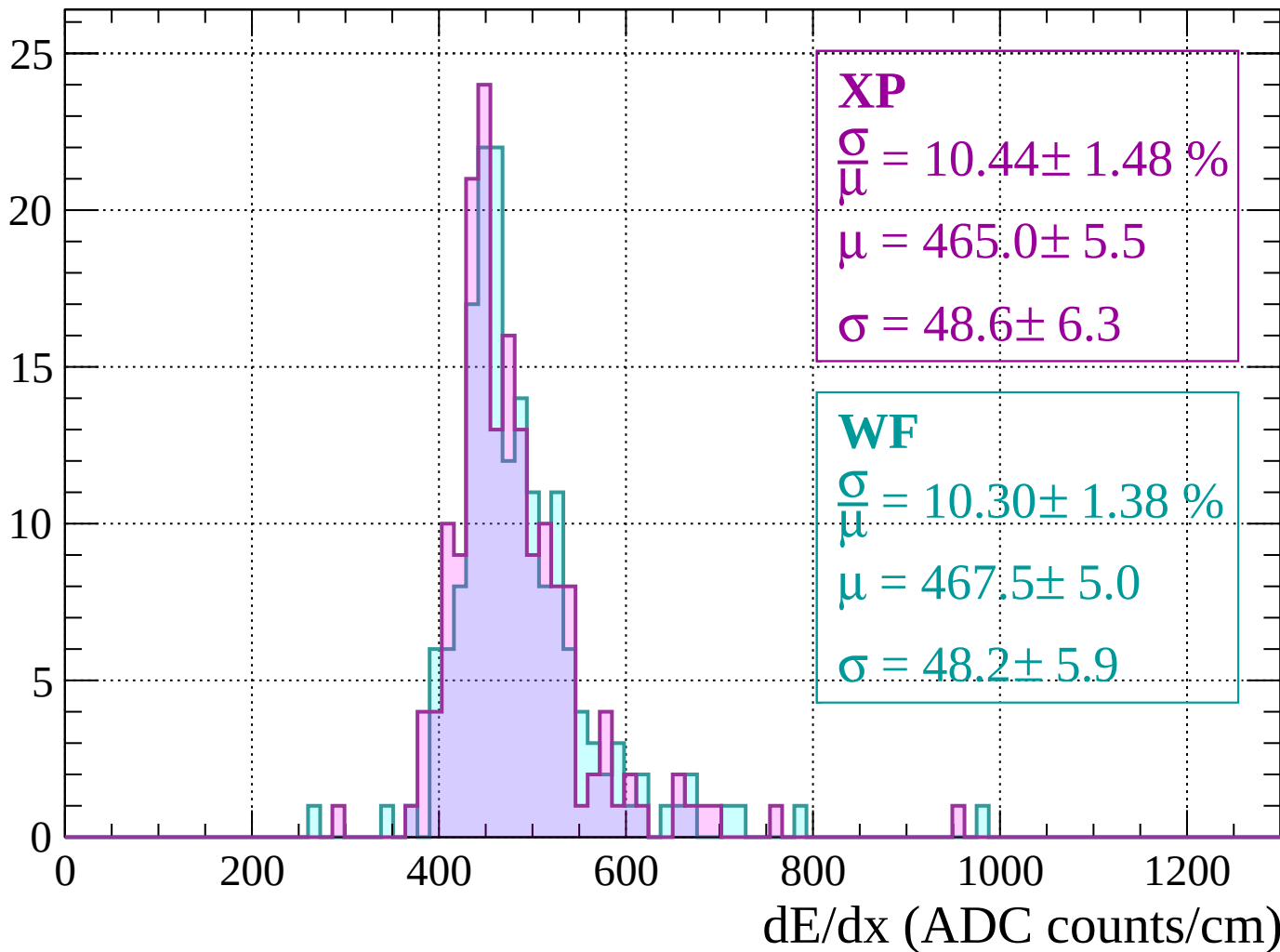
Energy loss | $1880 < p < 1920$

Count



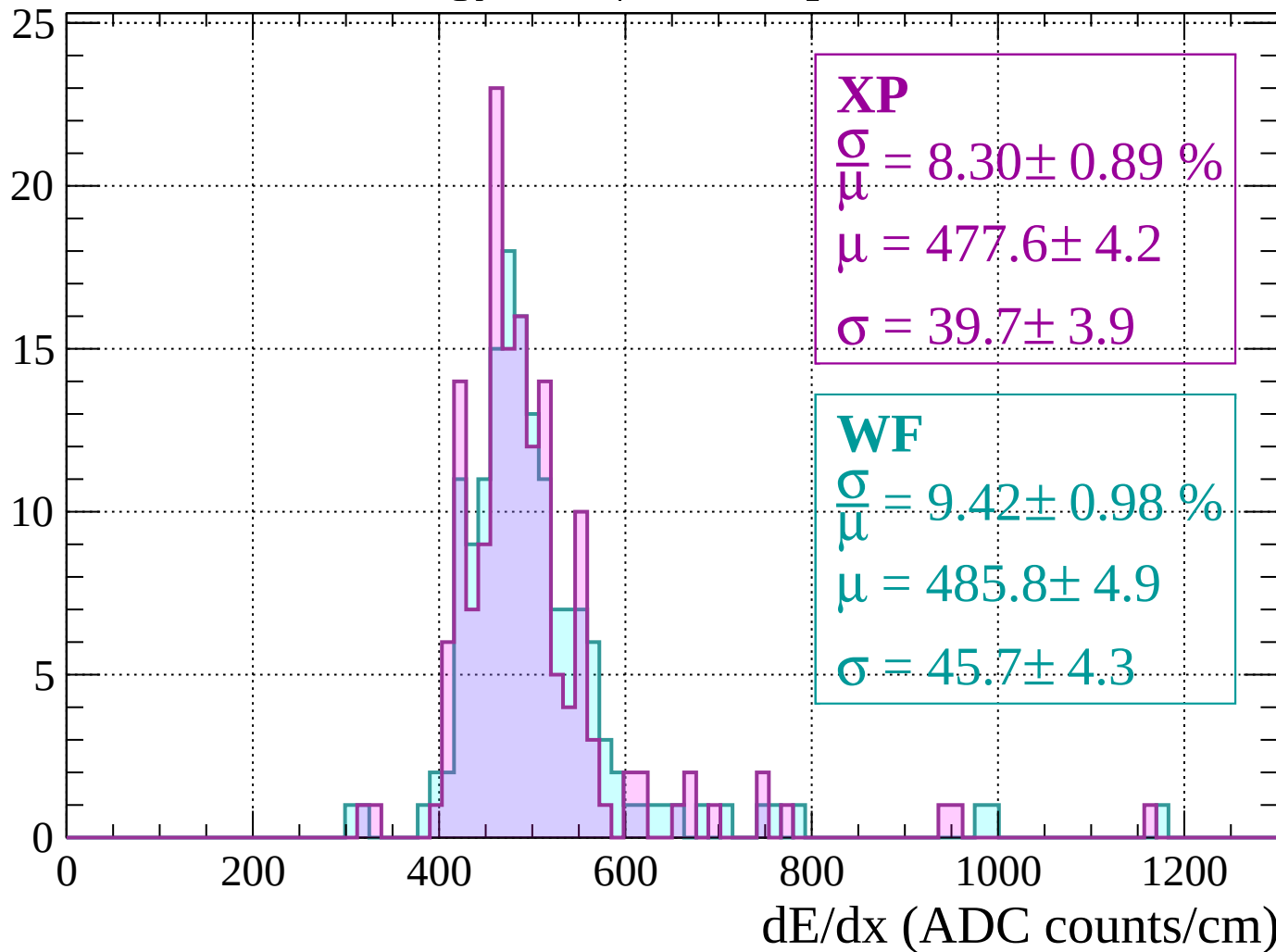
Energy loss | $1920 < p < 1960$

Count



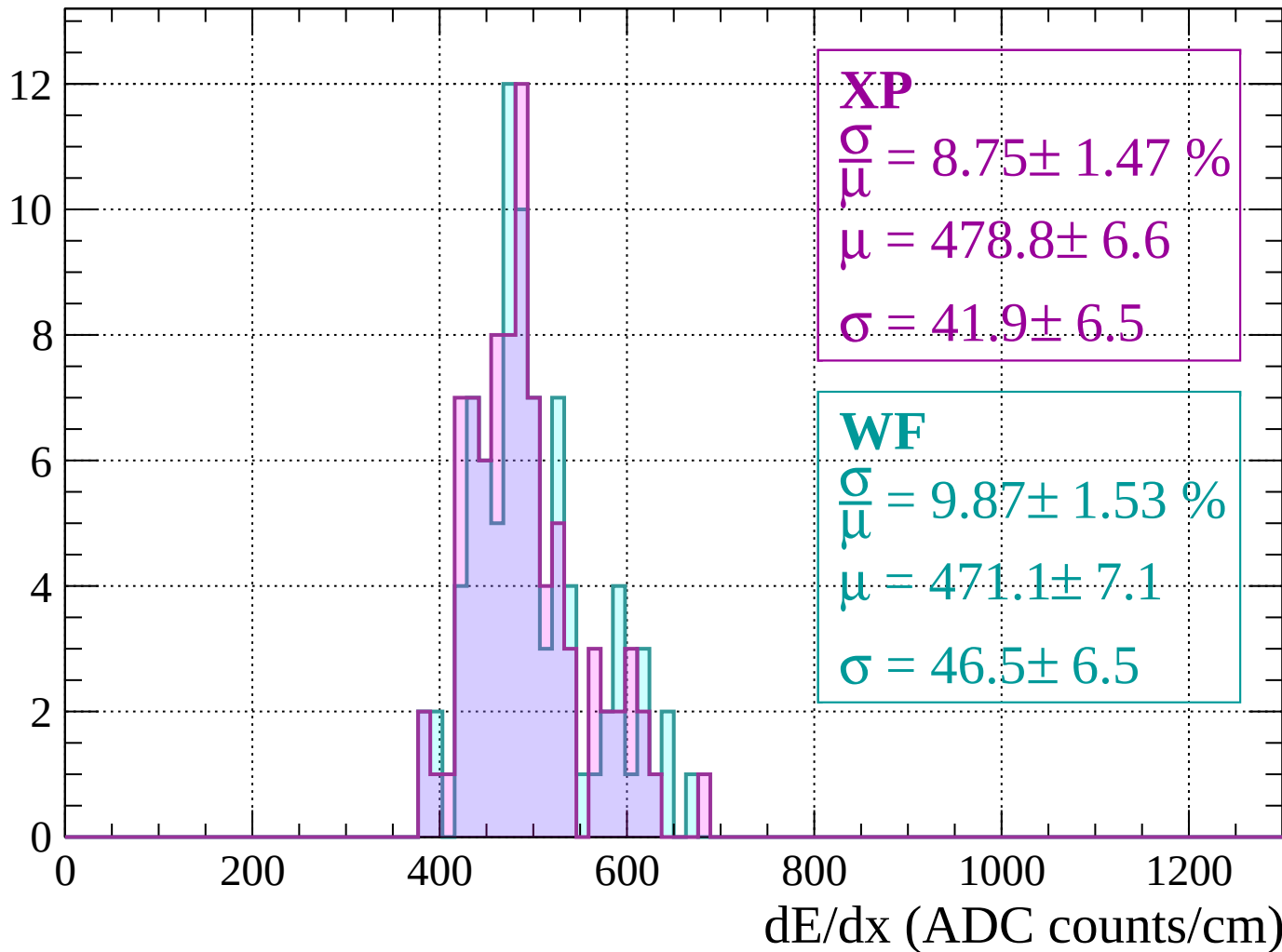
Energy loss | $1960 < p < 2000$

Count



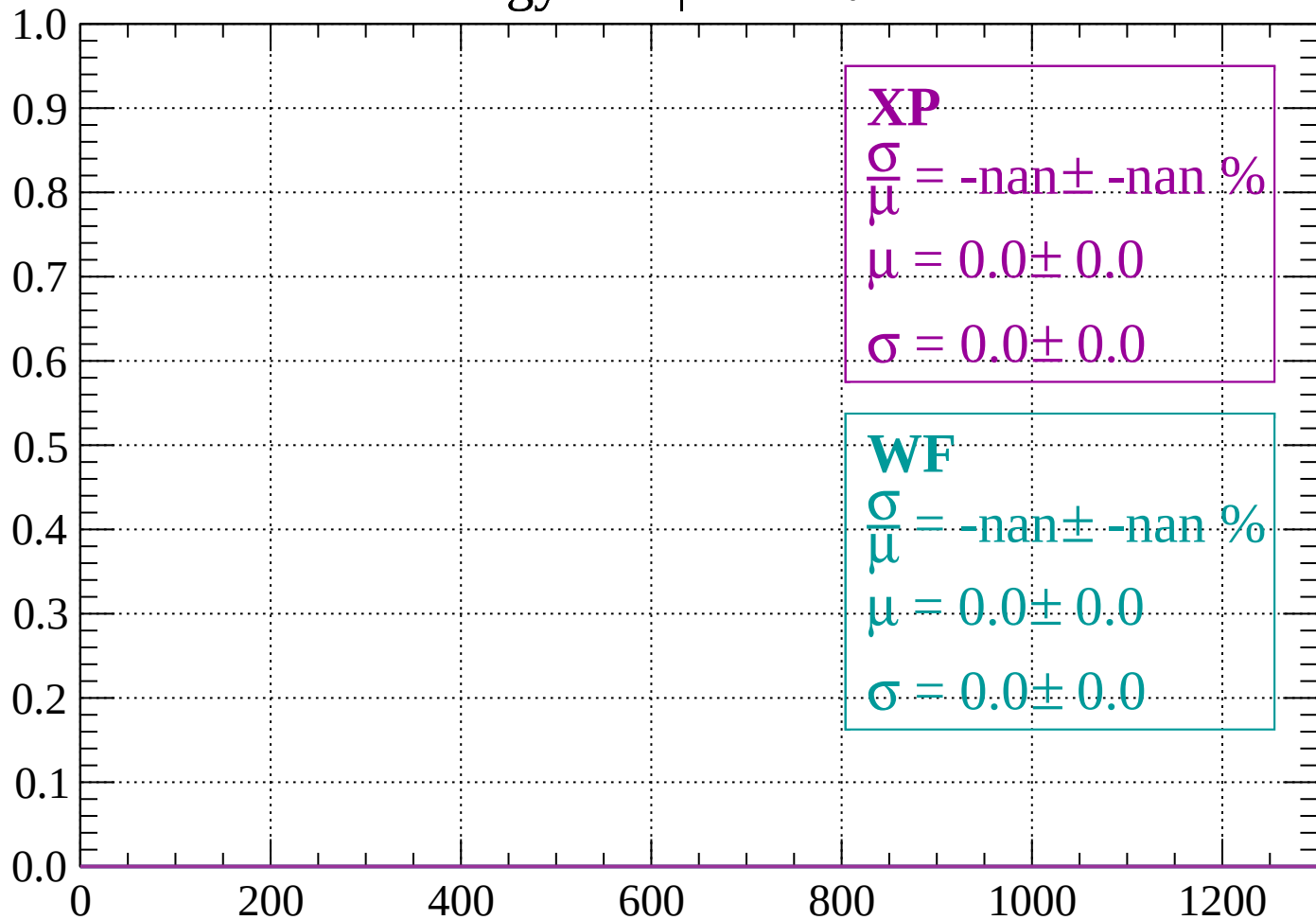
Energy loss | $2000 < p < 2040$

Count



Energy loss | $-90 < \theta < -88$

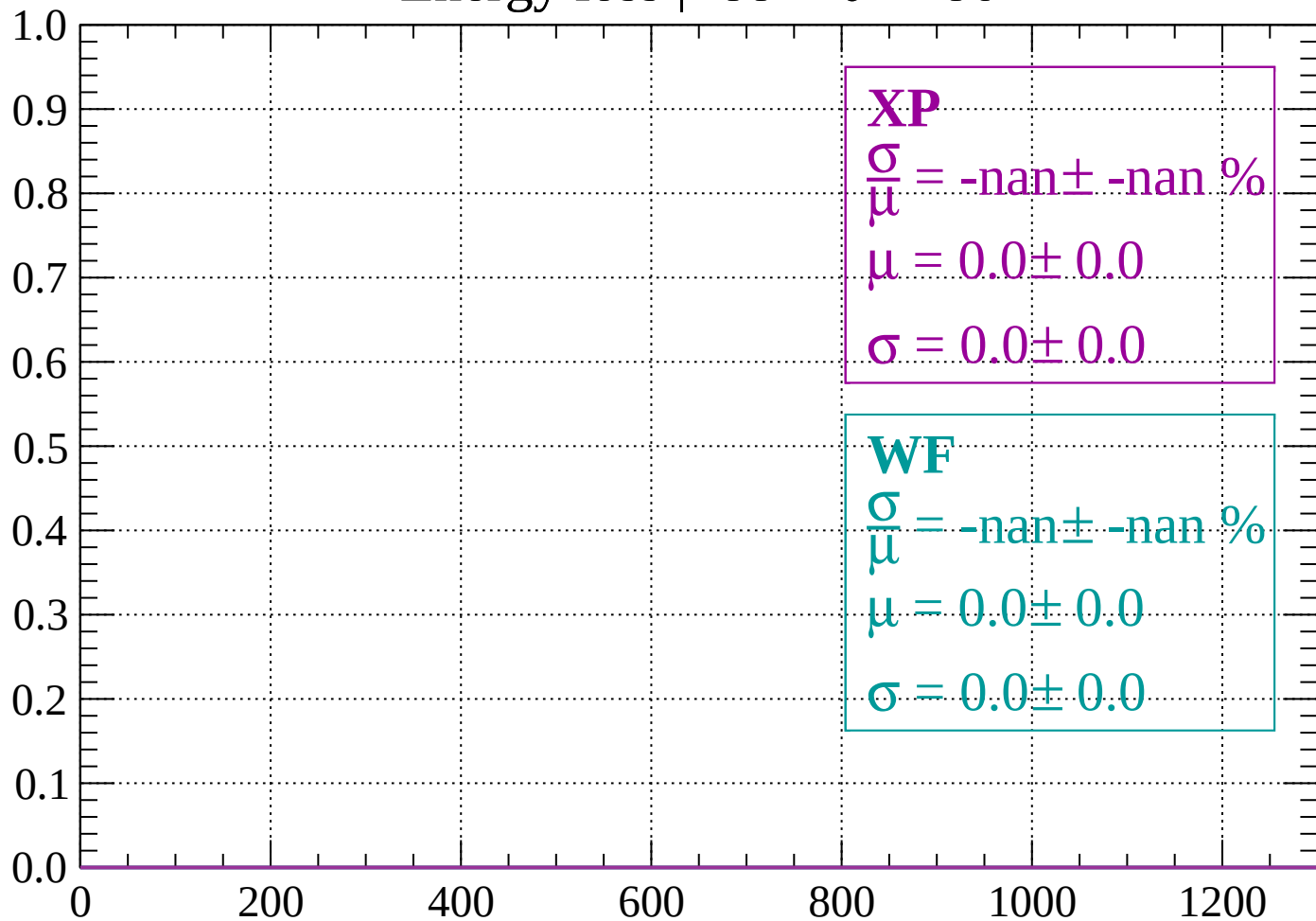
Count



dE/dx (ADC counts/cm)

Energy loss | $-88 < \theta < -86$

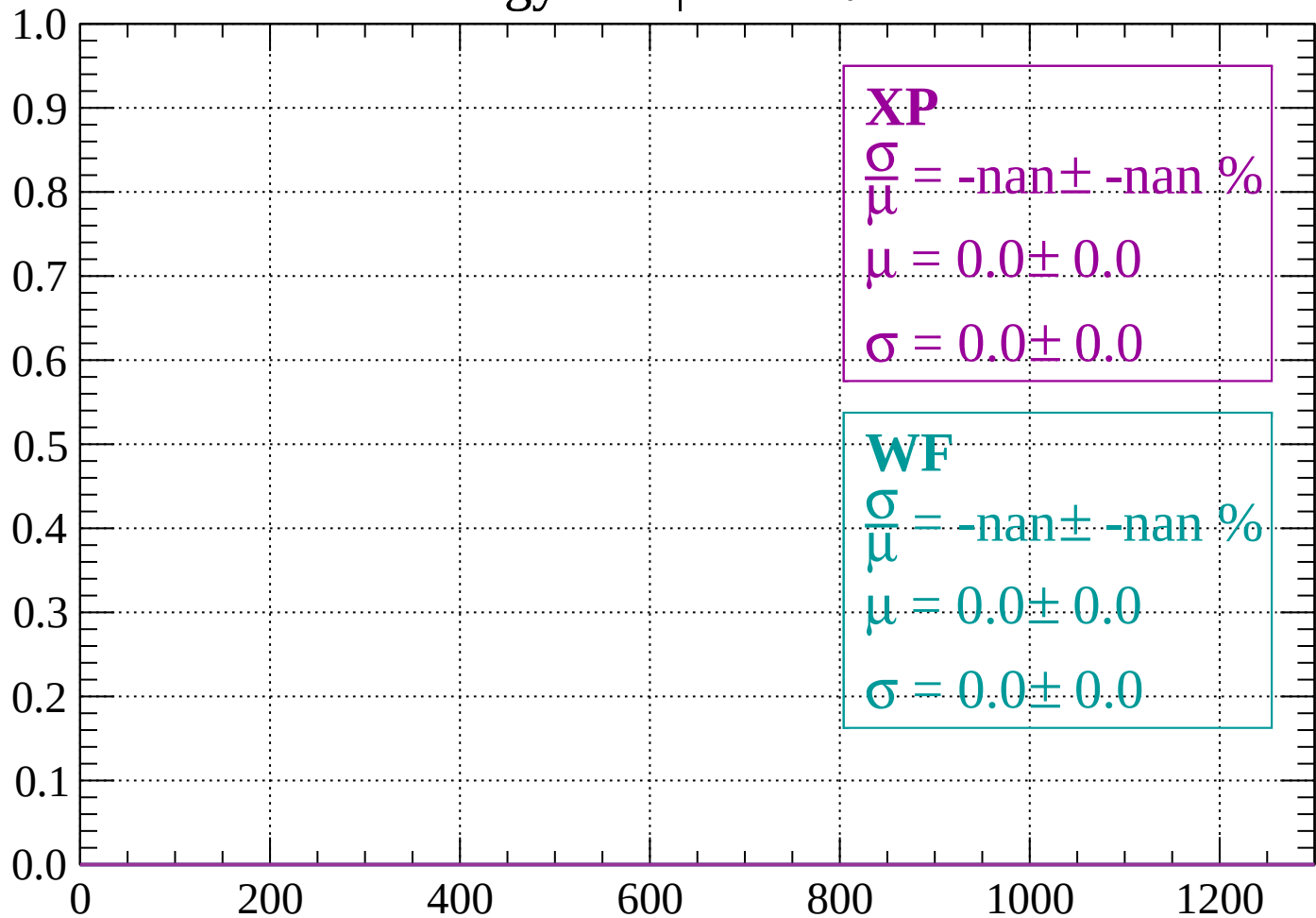
Count



dE/dx (ADC counts/cm)

Energy loss | $-86 < \theta < -84$

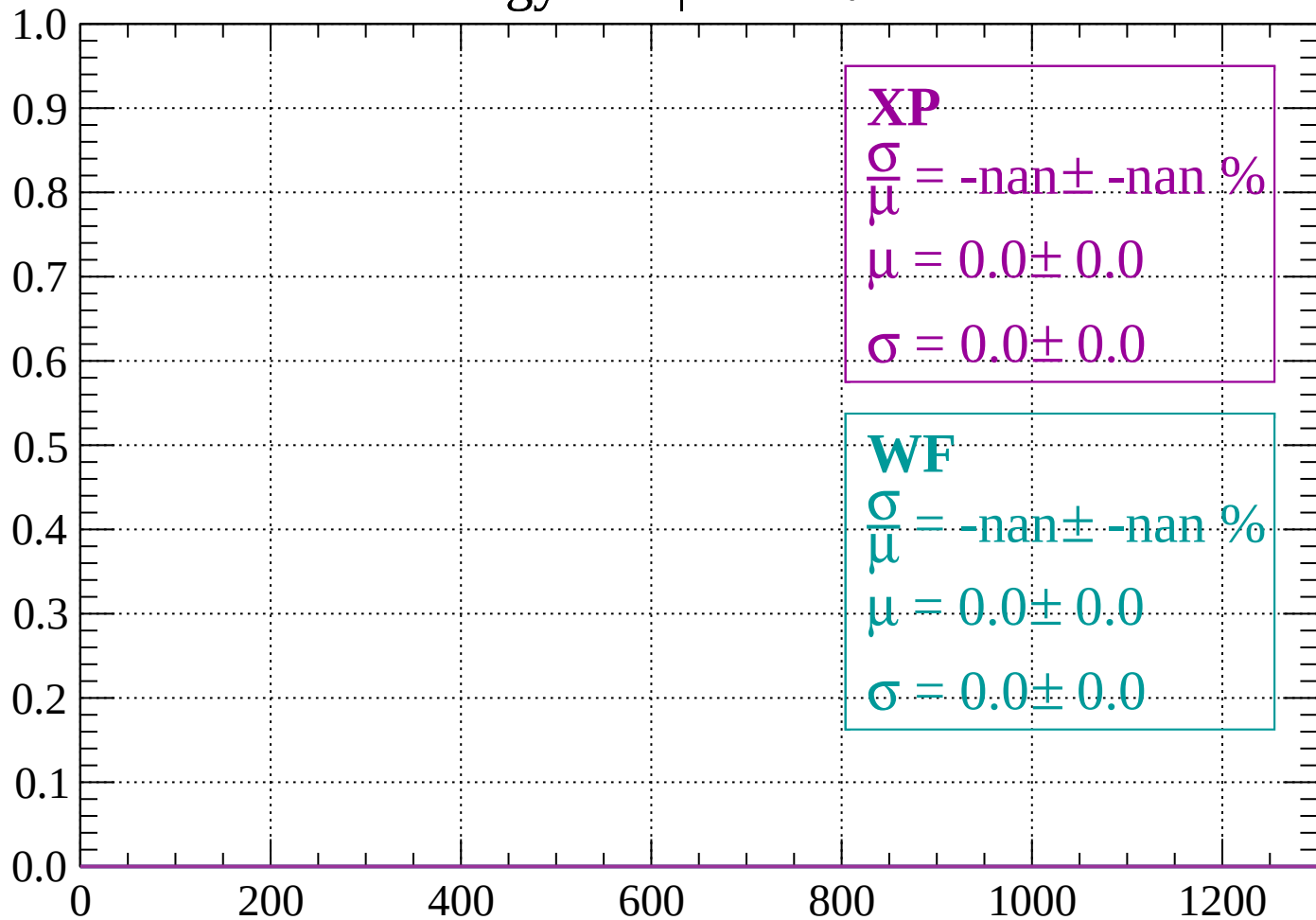
Count



dE/dx (ADC counts/cm)

Energy loss | $-84 < \theta < -82$

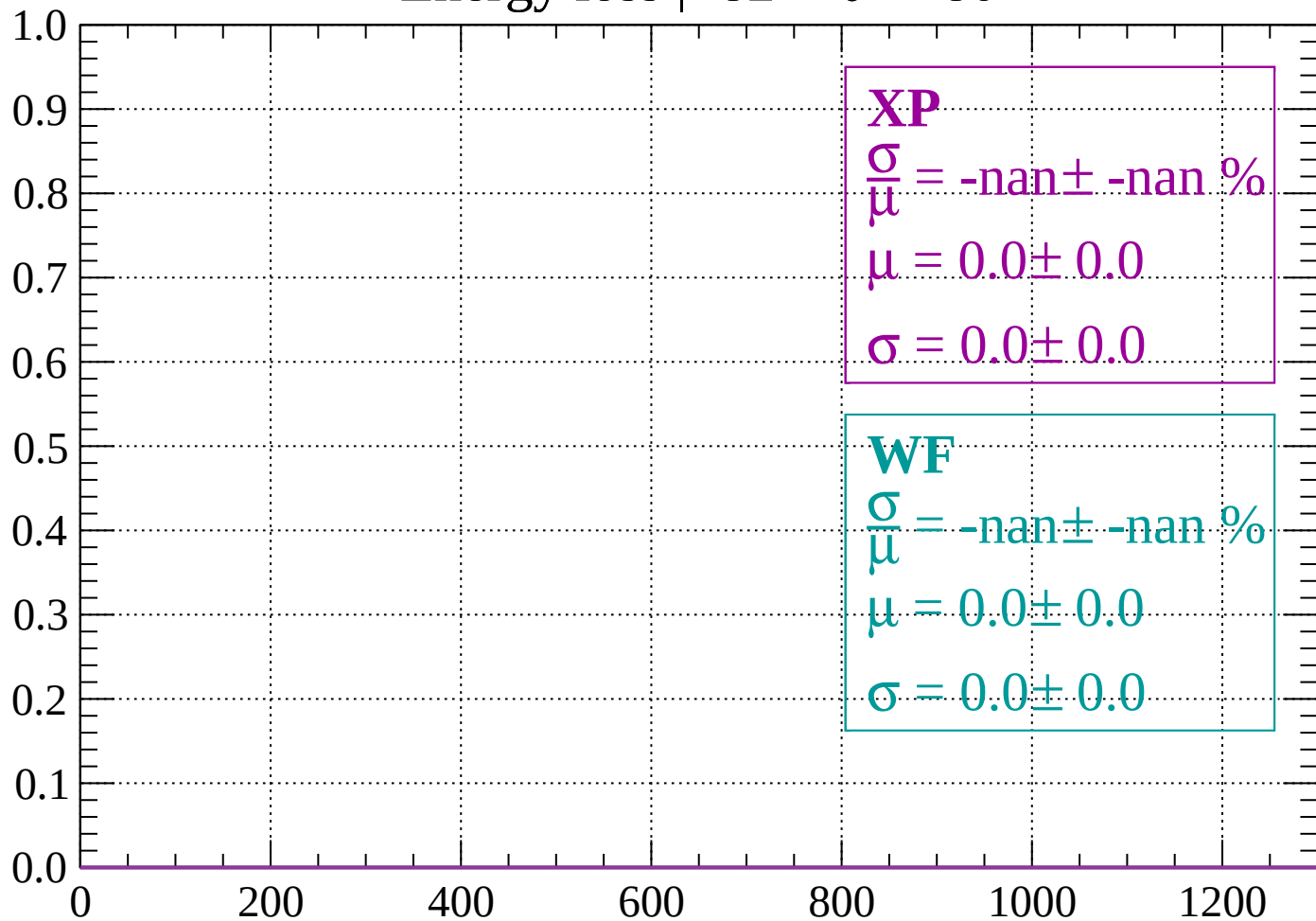
Count



dE/dx (ADC counts/cm)

Energy loss | $-82 < \theta < -80$

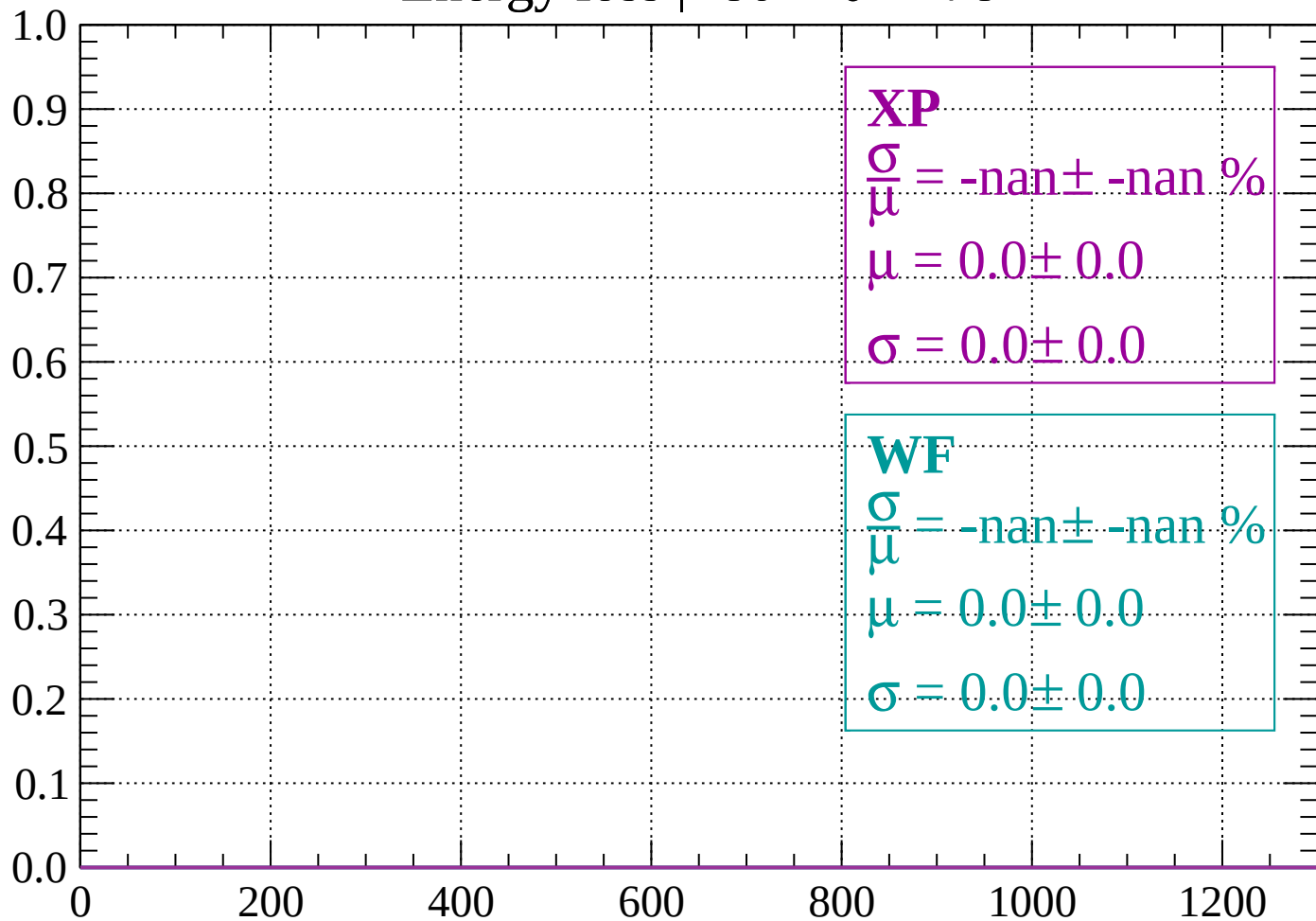
Count



dE/dx (ADC counts/cm)

Energy loss | $-80 < \theta < -78$

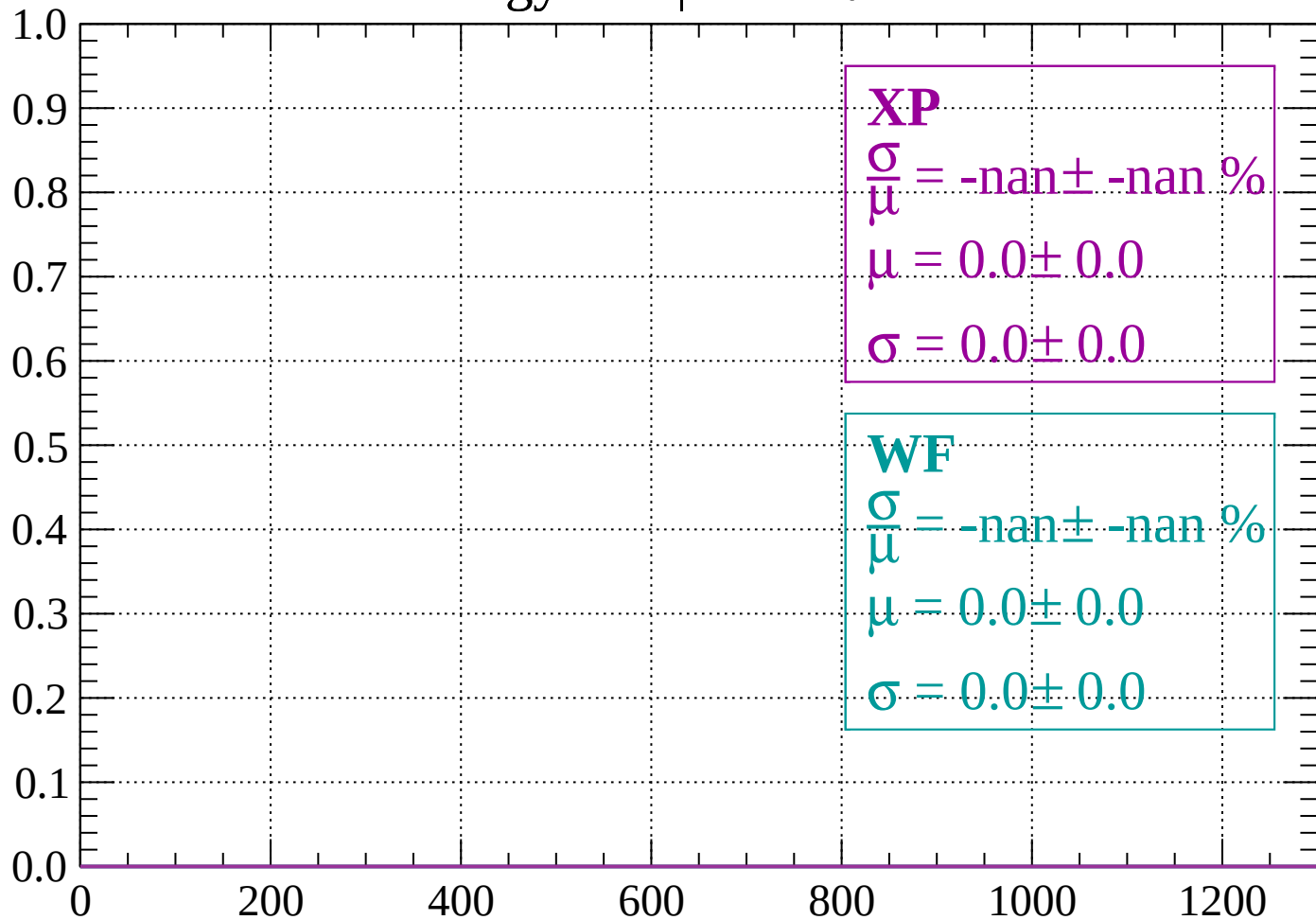
Count



dE/dx (ADC counts/cm)

Energy loss | $-78 < \theta < -76$

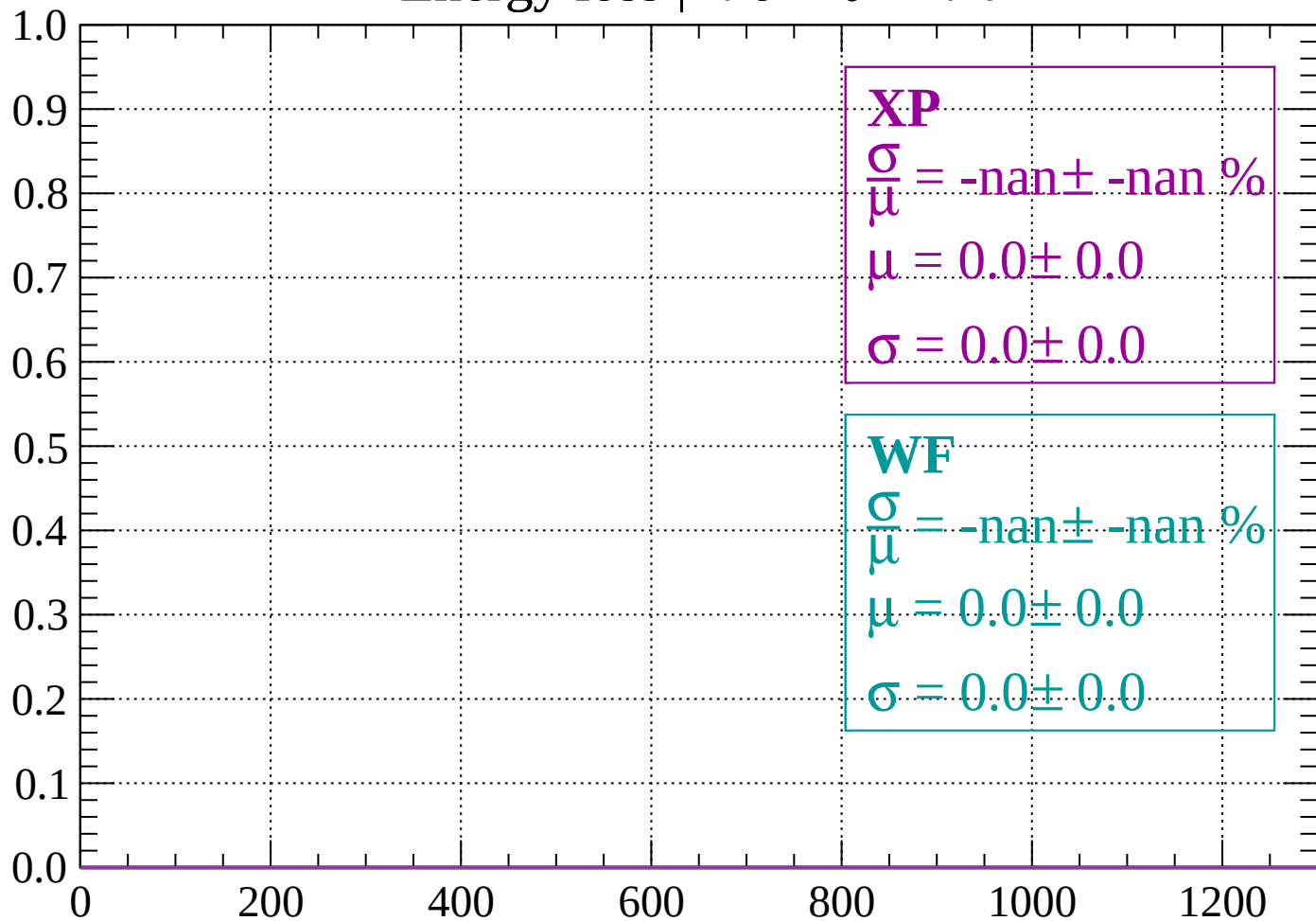
Count



dE/dx (ADC counts/cm)

Energy loss | $-76 < \theta < -74$

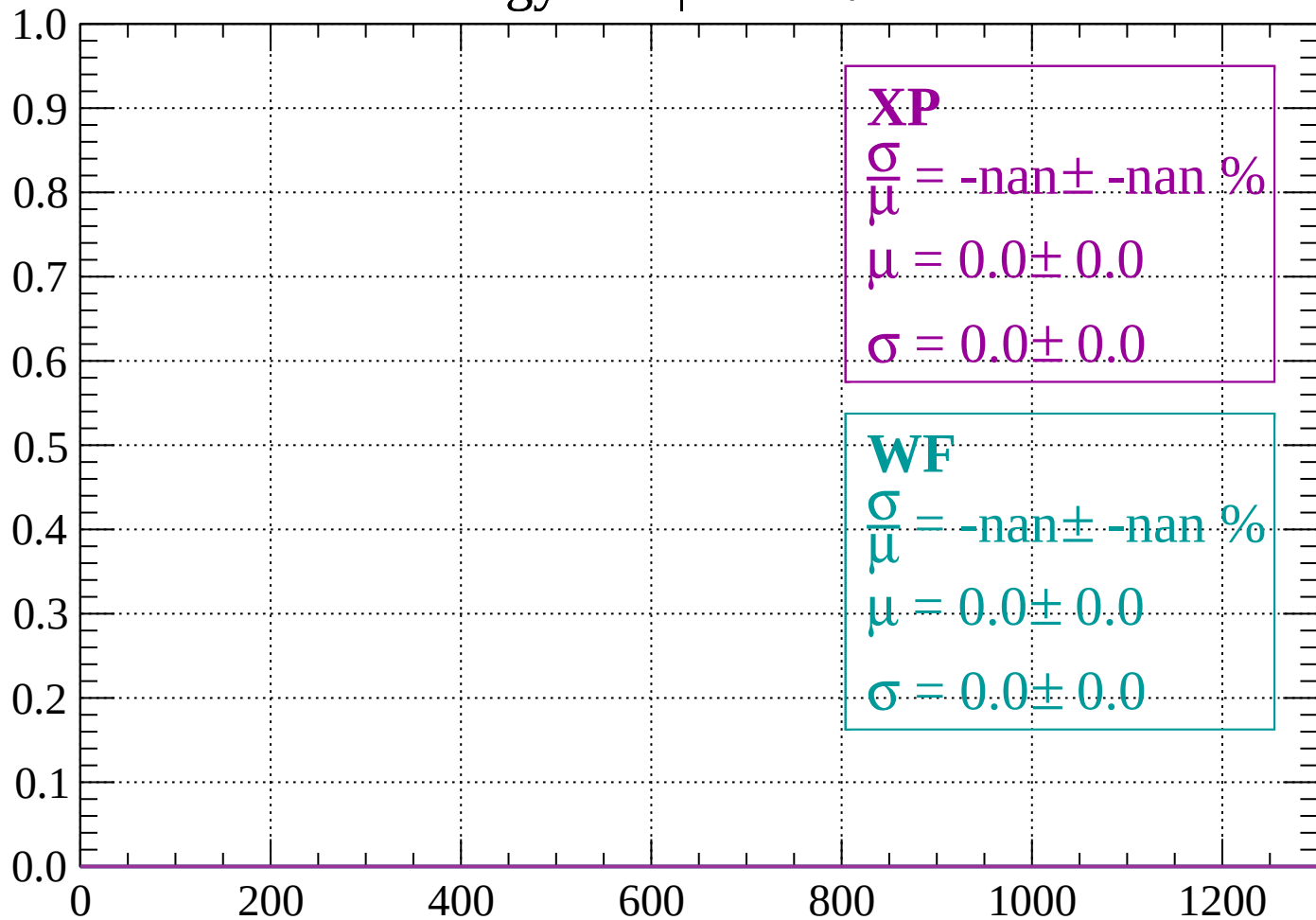
Count



dE/dx (ADC counts/cm)

Energy loss | $-74 < \theta < -72$

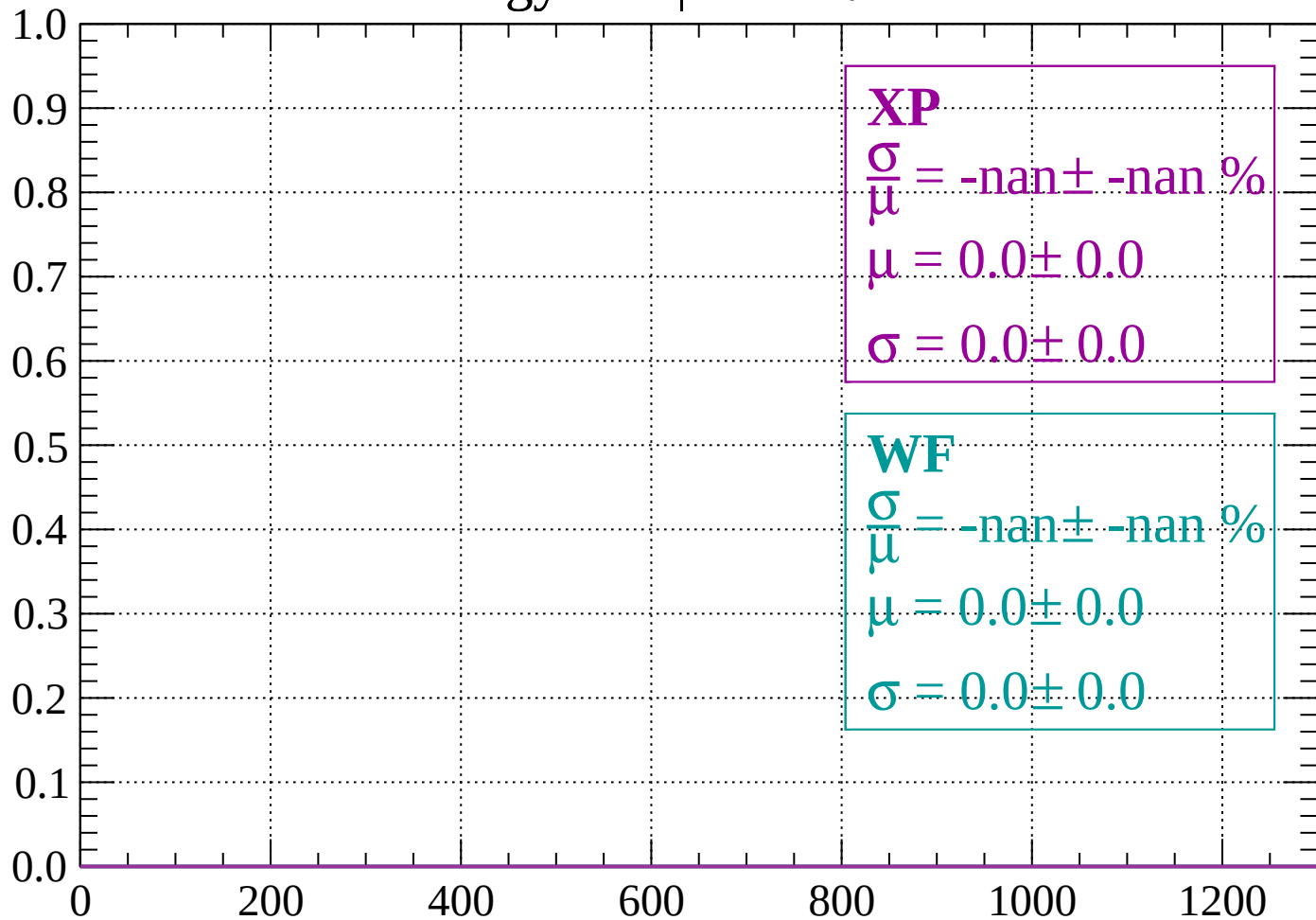
Count



dE/dx (ADC counts/cm)

Energy loss | $-72 < \theta < -70$

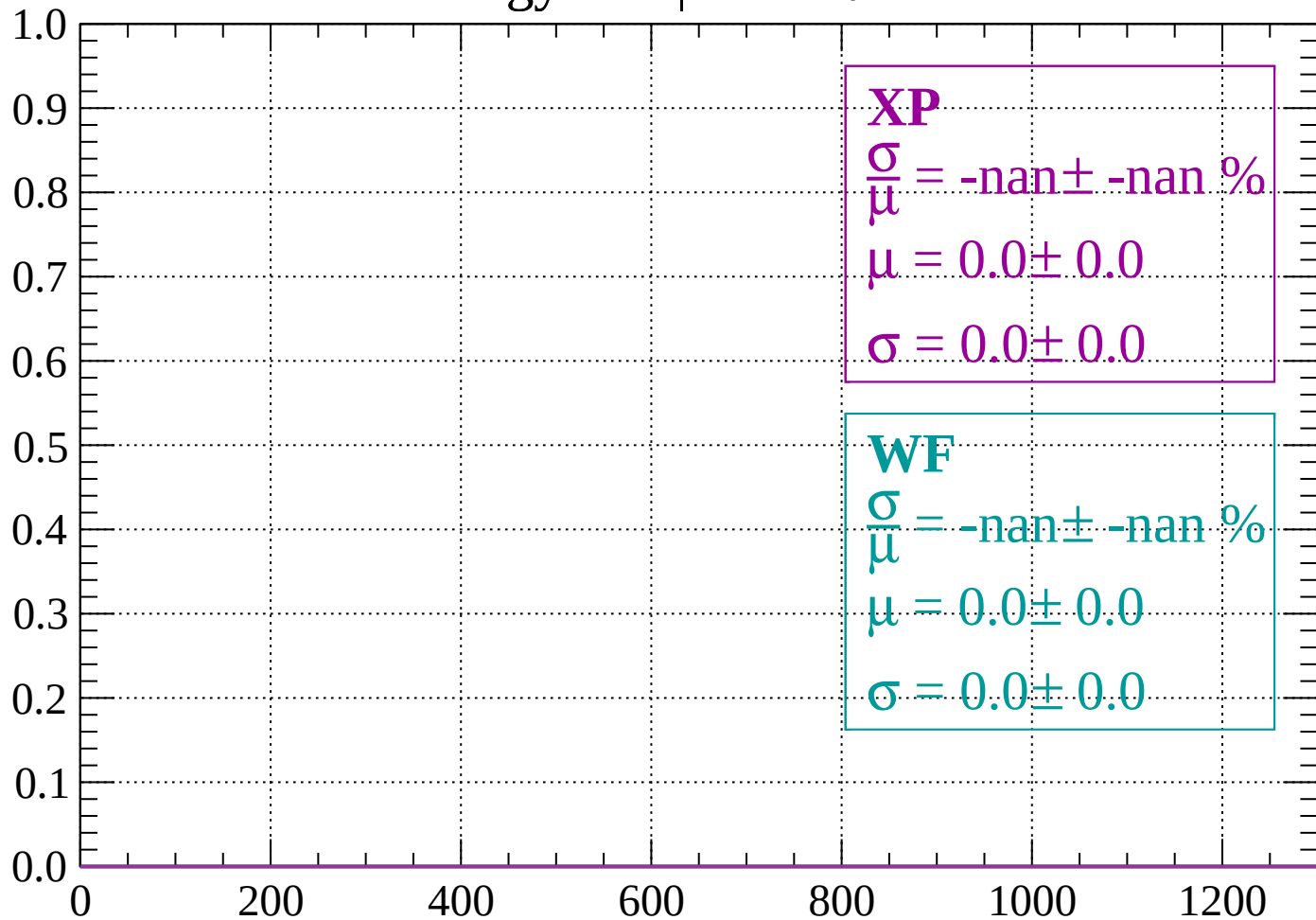
Count



dE/dx (ADC counts/cm)

Energy loss | $-70 < \theta < -68$

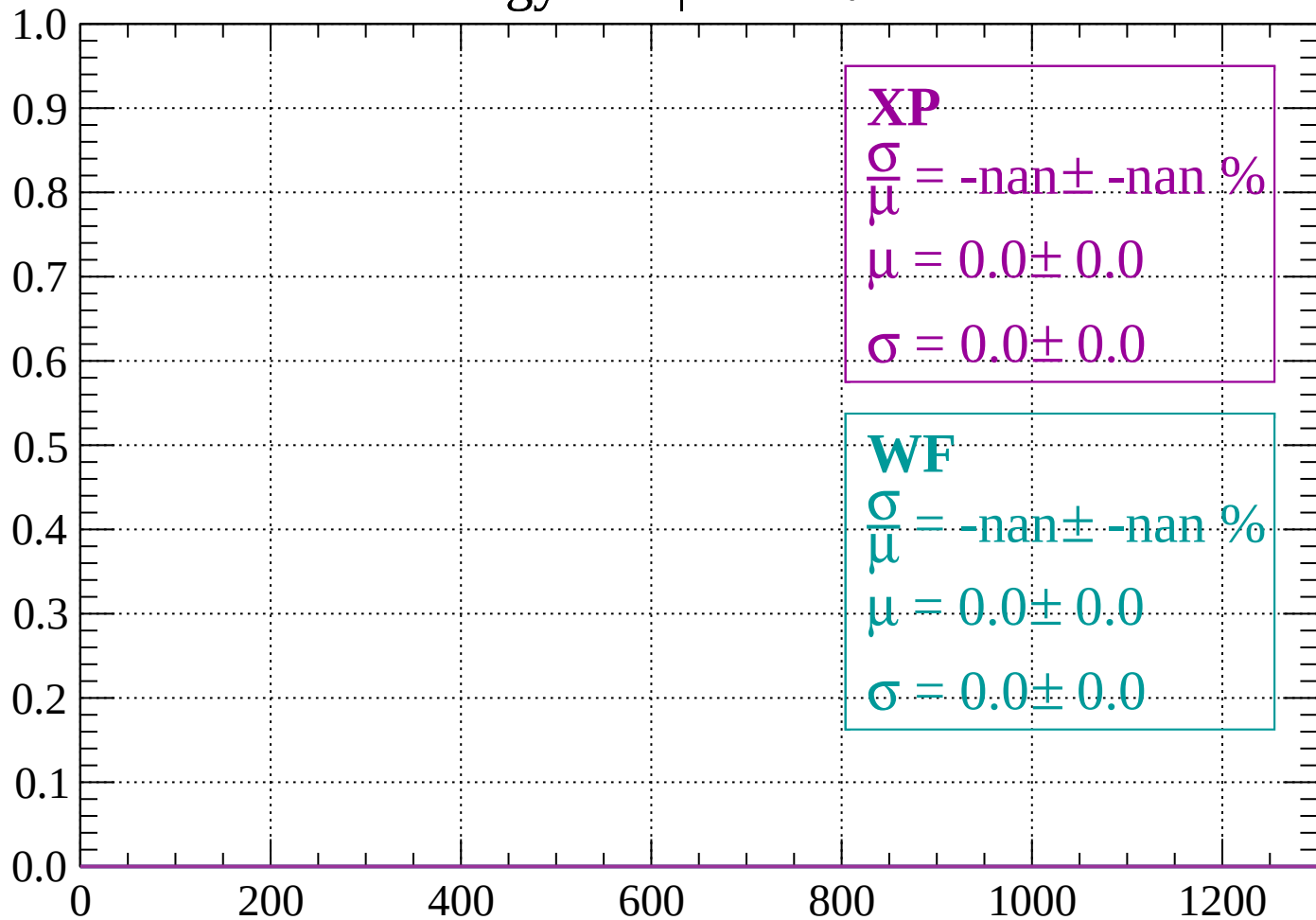
Count



dE/dx (ADC counts/cm)

Energy loss | $-68 < \theta < -66$

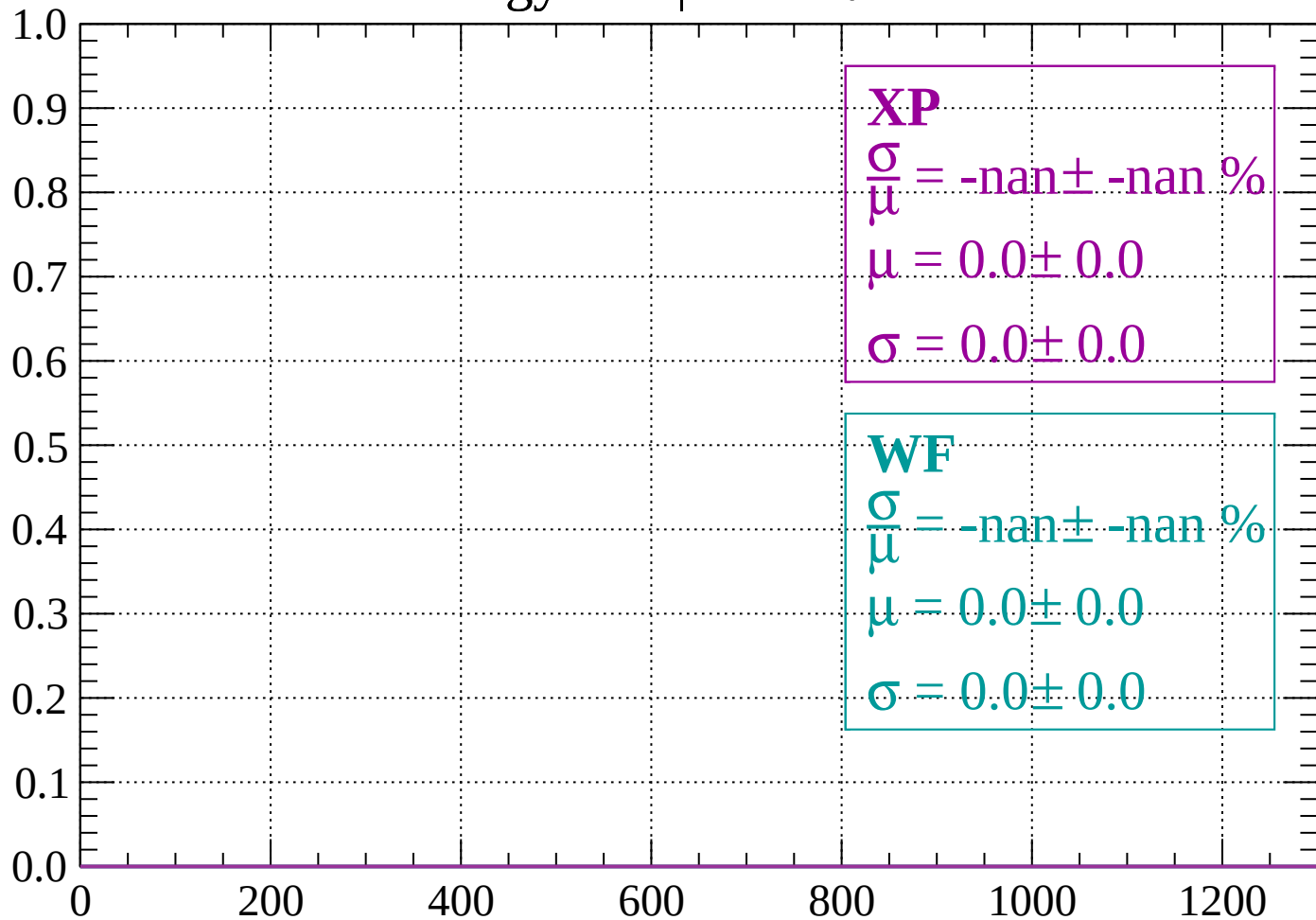
Count



dE/dx (ADC counts/cm)

Energy loss | $-66 < \theta < -64$

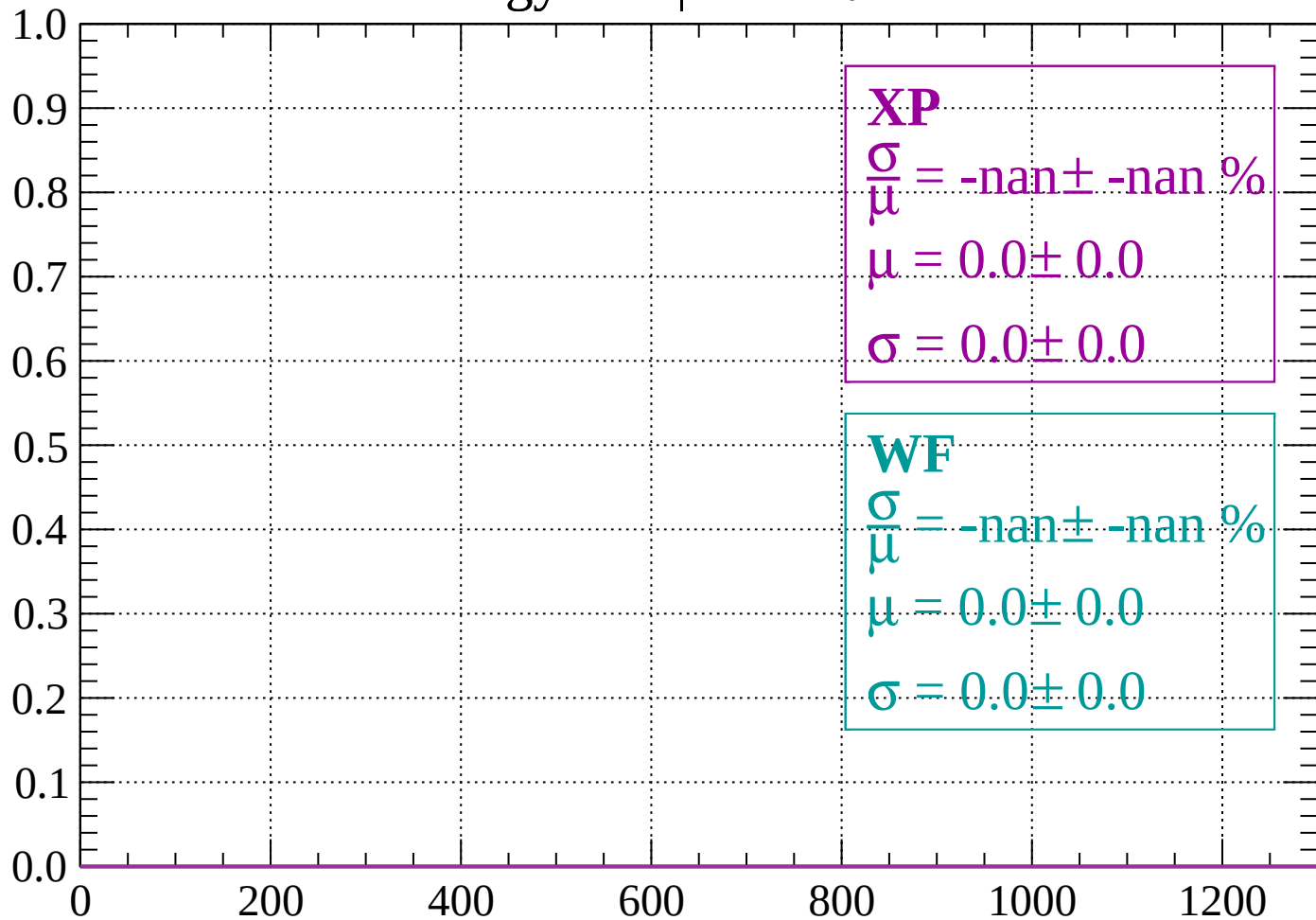
Count



dE/dx (ADC counts/cm)

Energy loss | $-64 < \theta < -62$

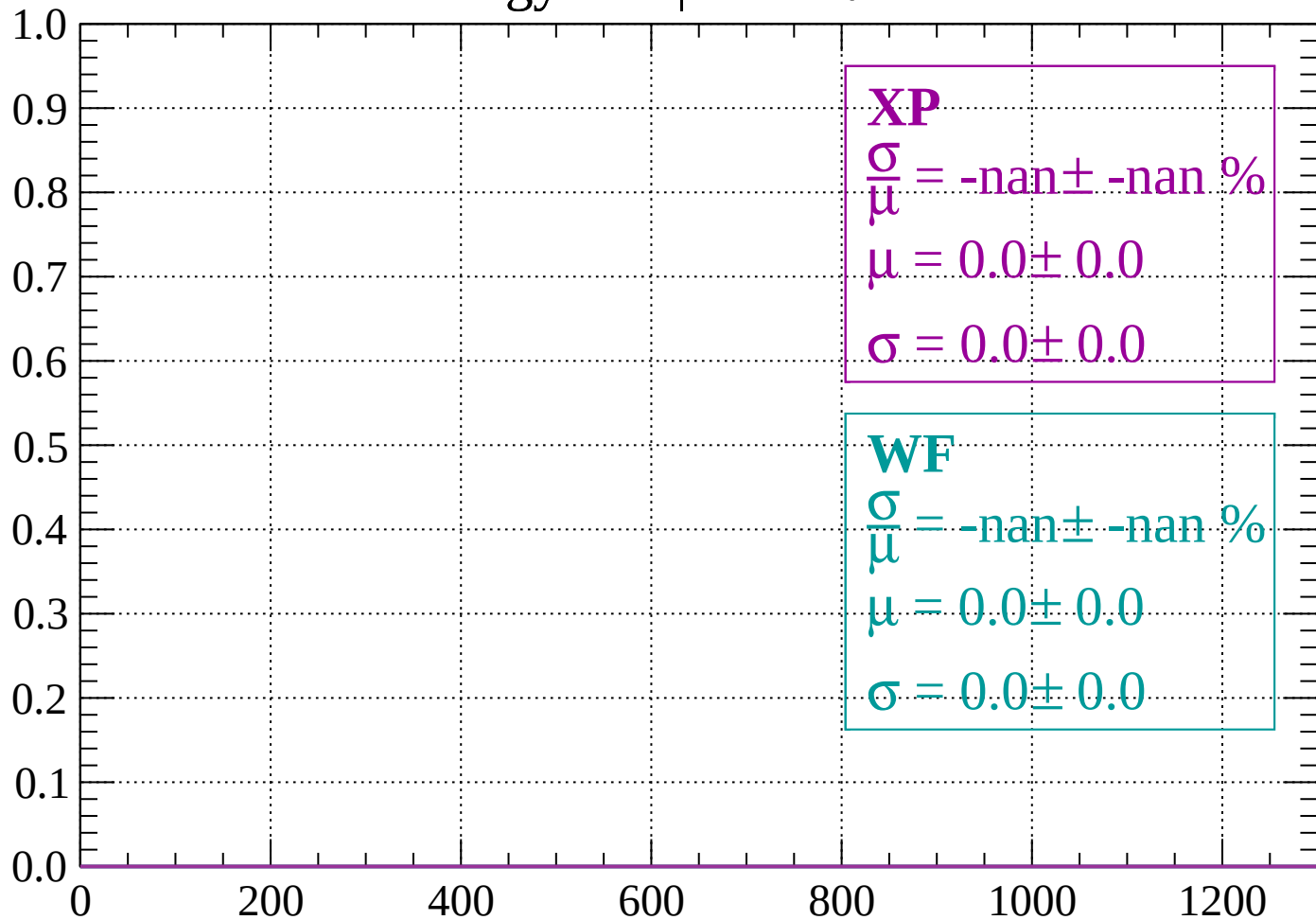
Count



dE/dx (ADC counts/cm)

Energy loss | $-62 < \theta < -60$

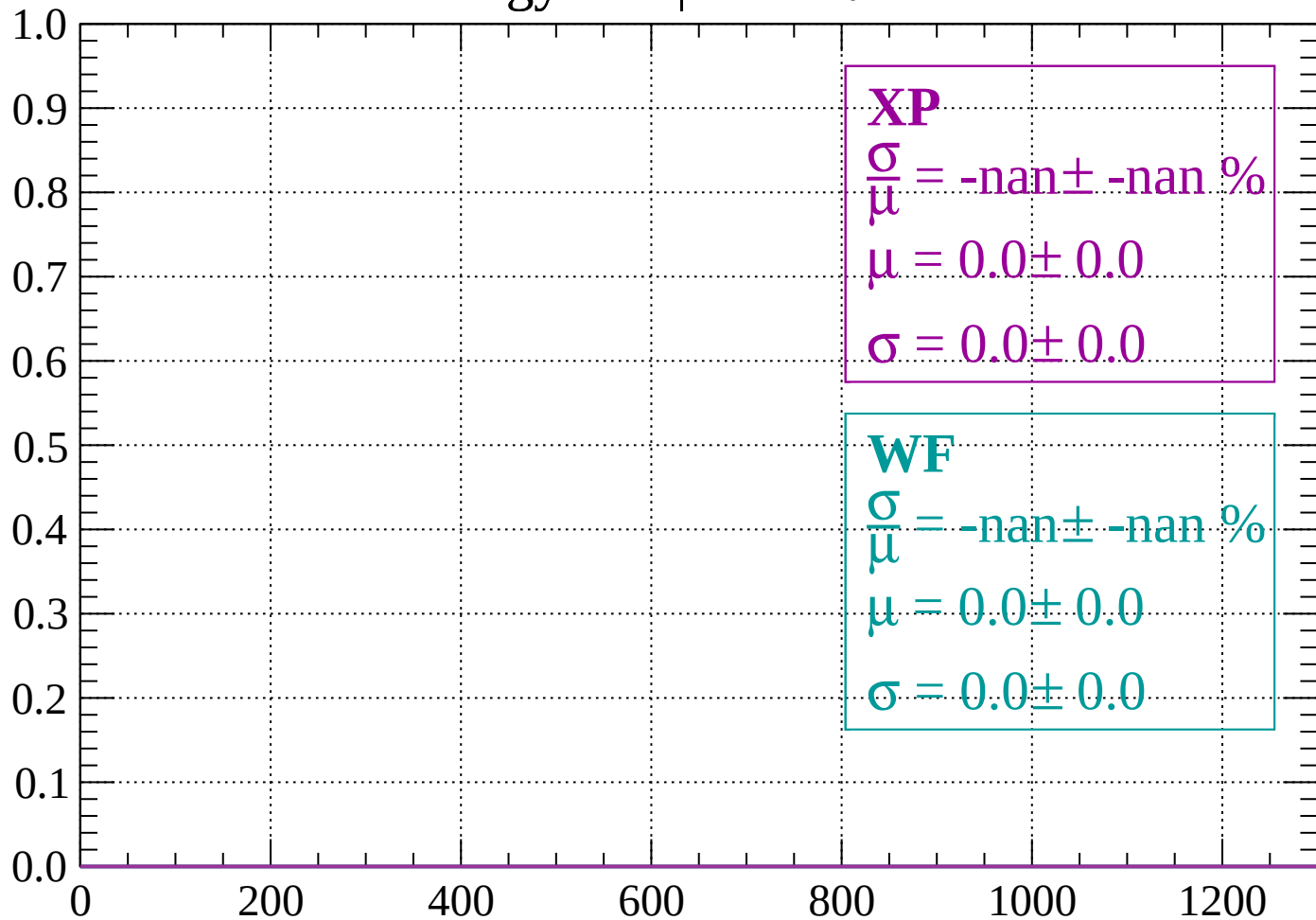
Count



dE/dx (ADC counts/cm)

Energy loss | $-60 < \theta < -58$

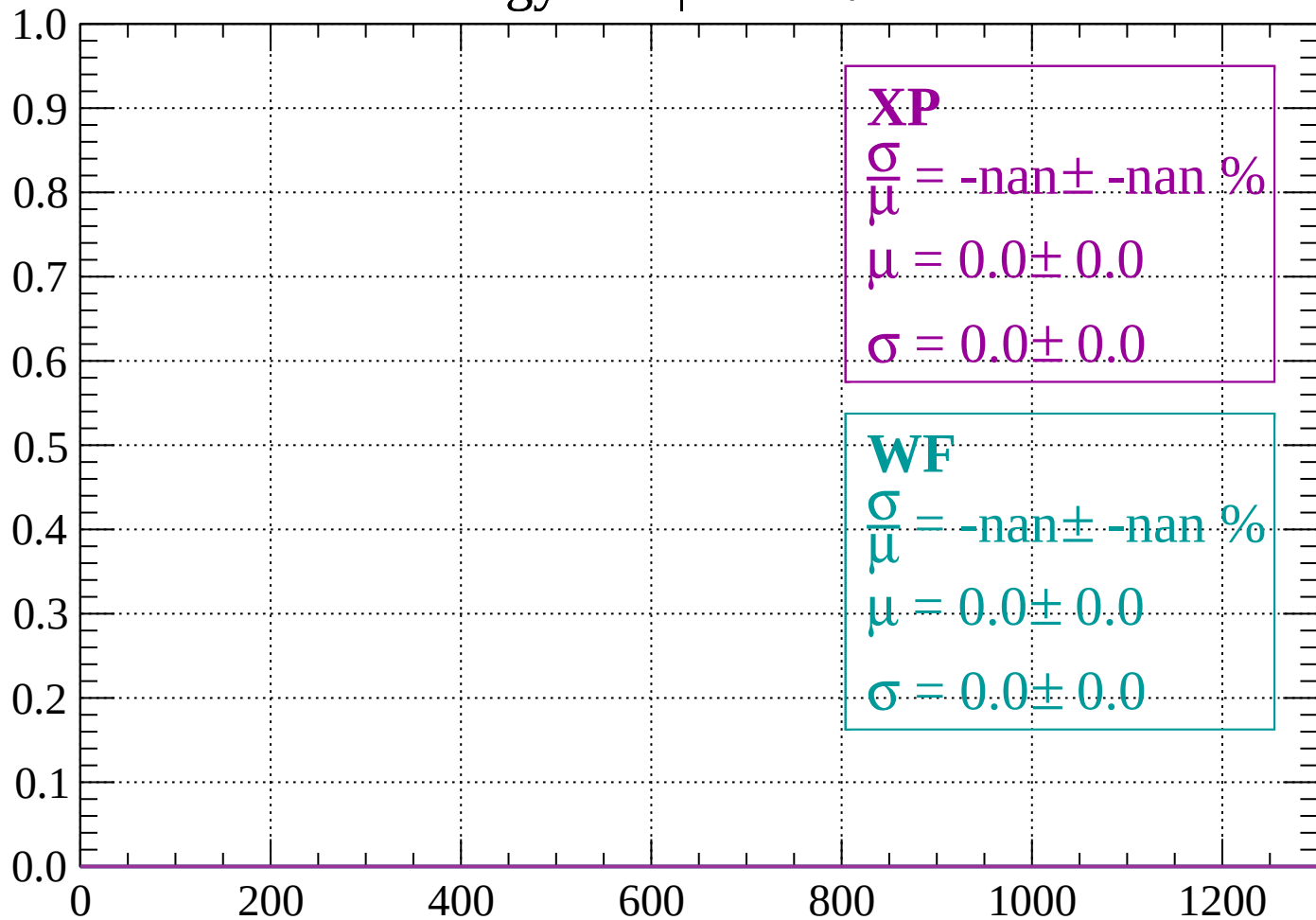
Count



dE/dx (ADC counts/cm)

Energy loss | $-58 < \theta < -56$

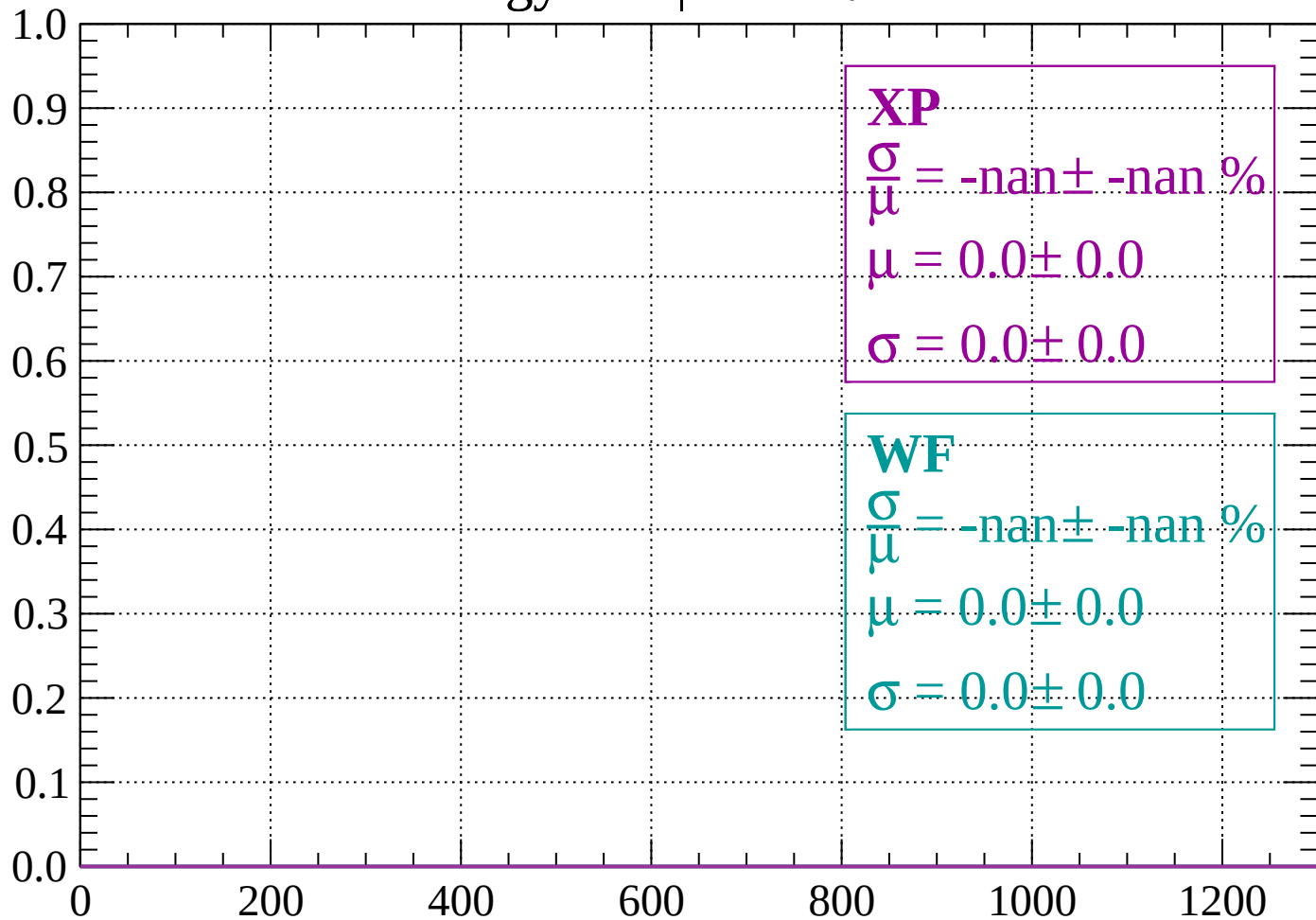
Count



dE/dx (ADC counts/cm)

Energy loss | $-56 < \theta < -54$

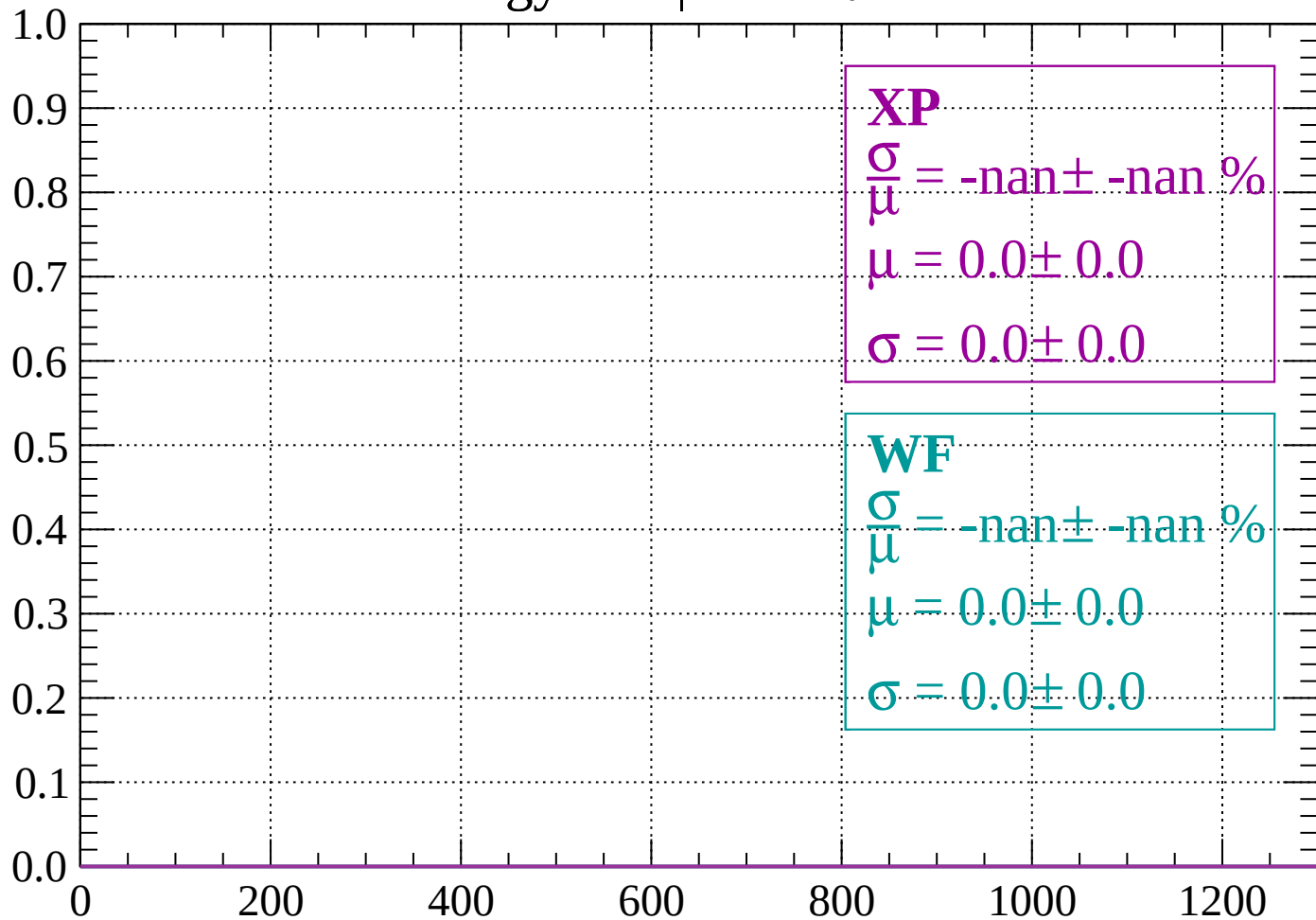
Count



dE/dx (ADC counts/cm)

Energy loss | $-54 < \theta < -52$

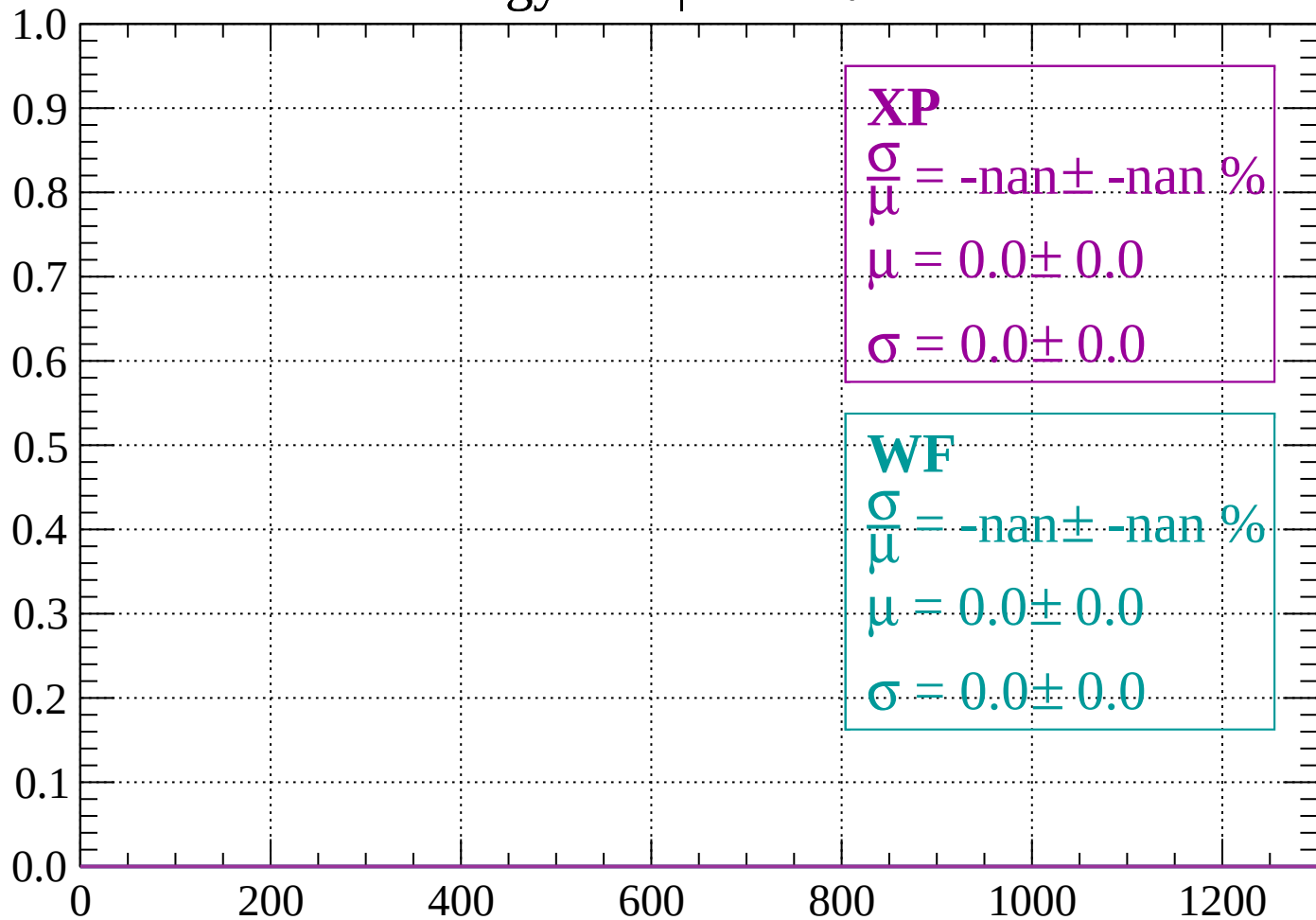
Count



dE/dx (ADC counts/cm)

Energy loss | $-52 < \theta < -50$

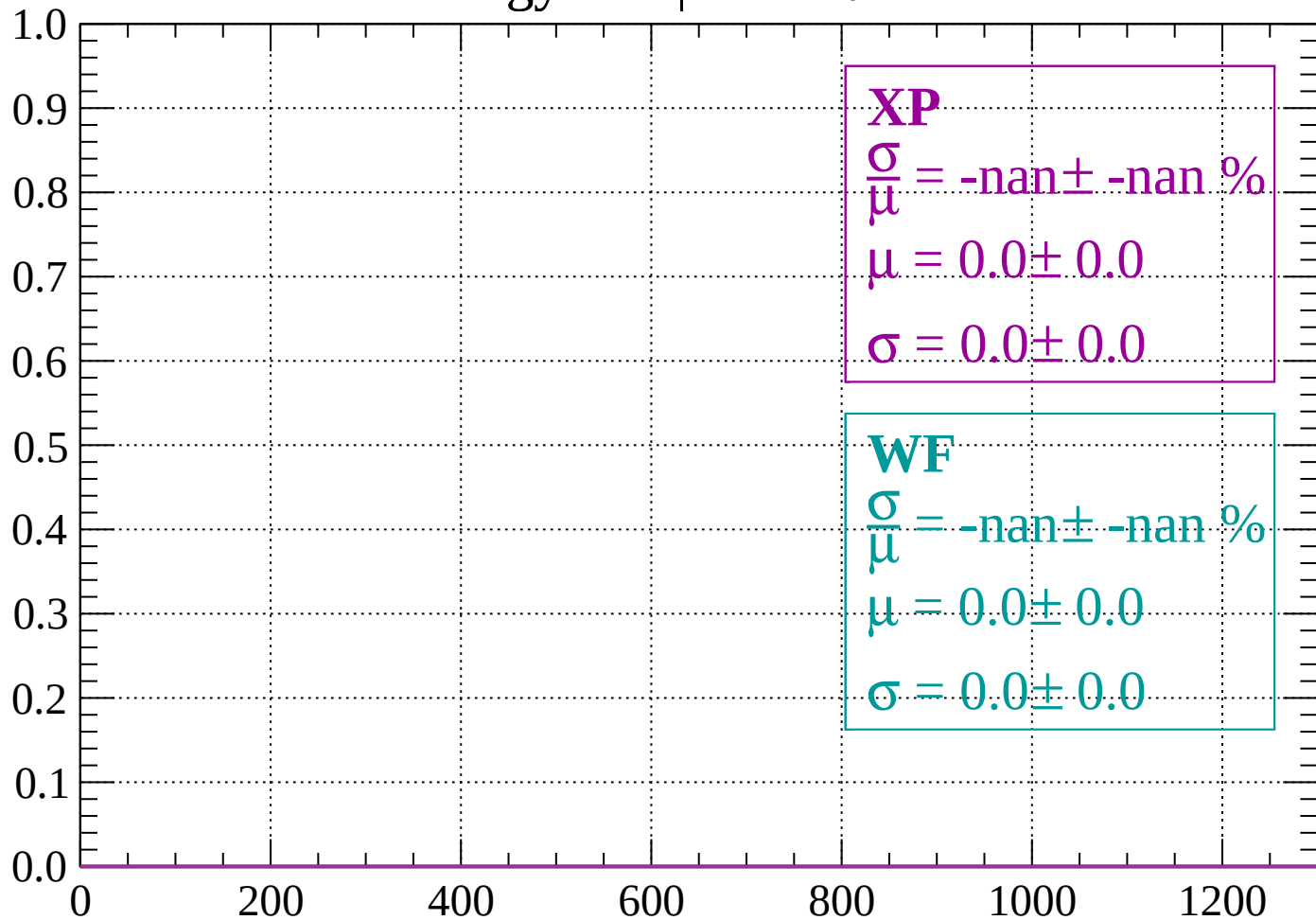
Count



dE/dx (ADC counts/cm)

Energy loss | $-50 < \theta < -48$

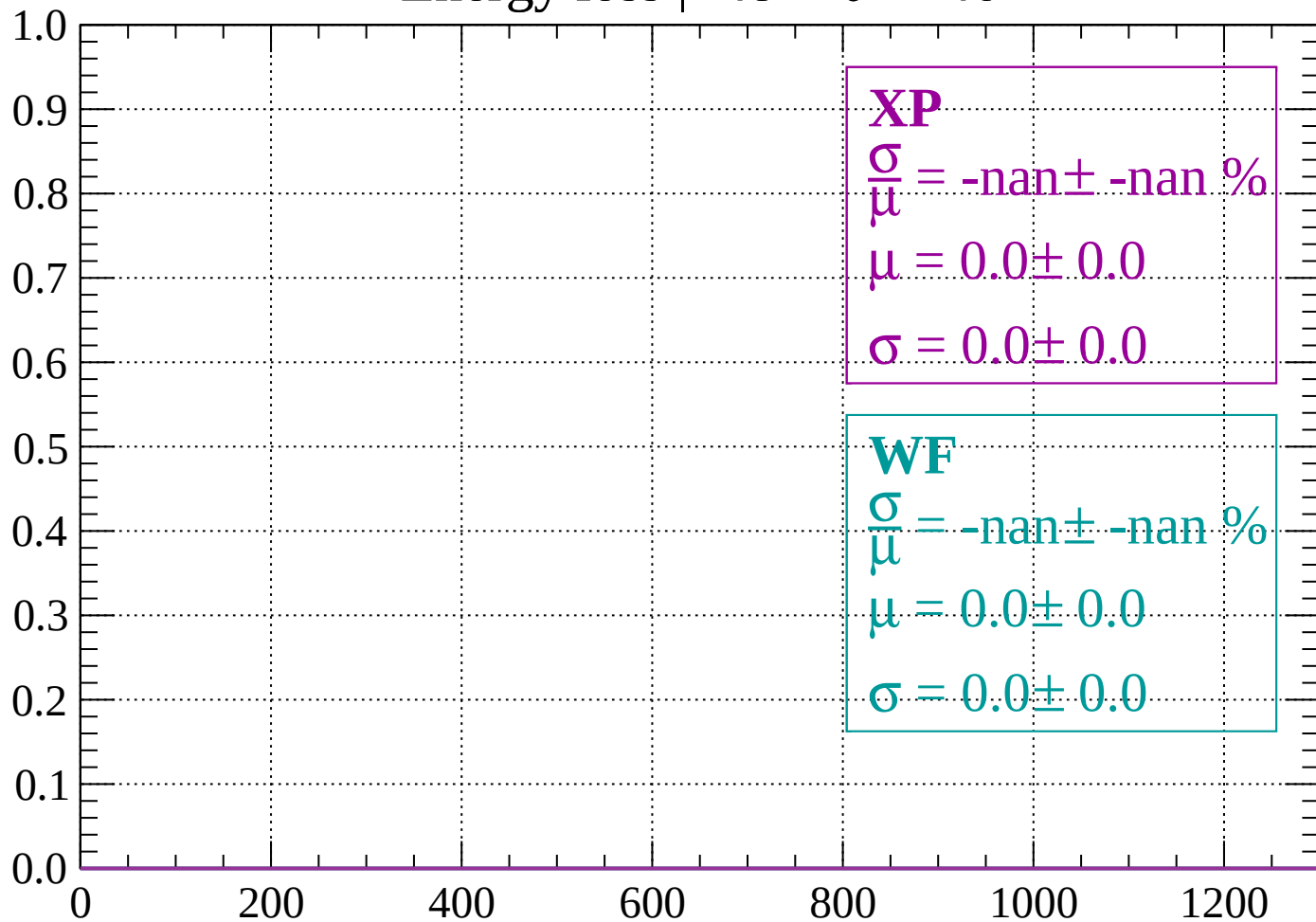
Count



dE/dx (ADC counts/cm)

Energy loss | $-48 < \theta < -46$

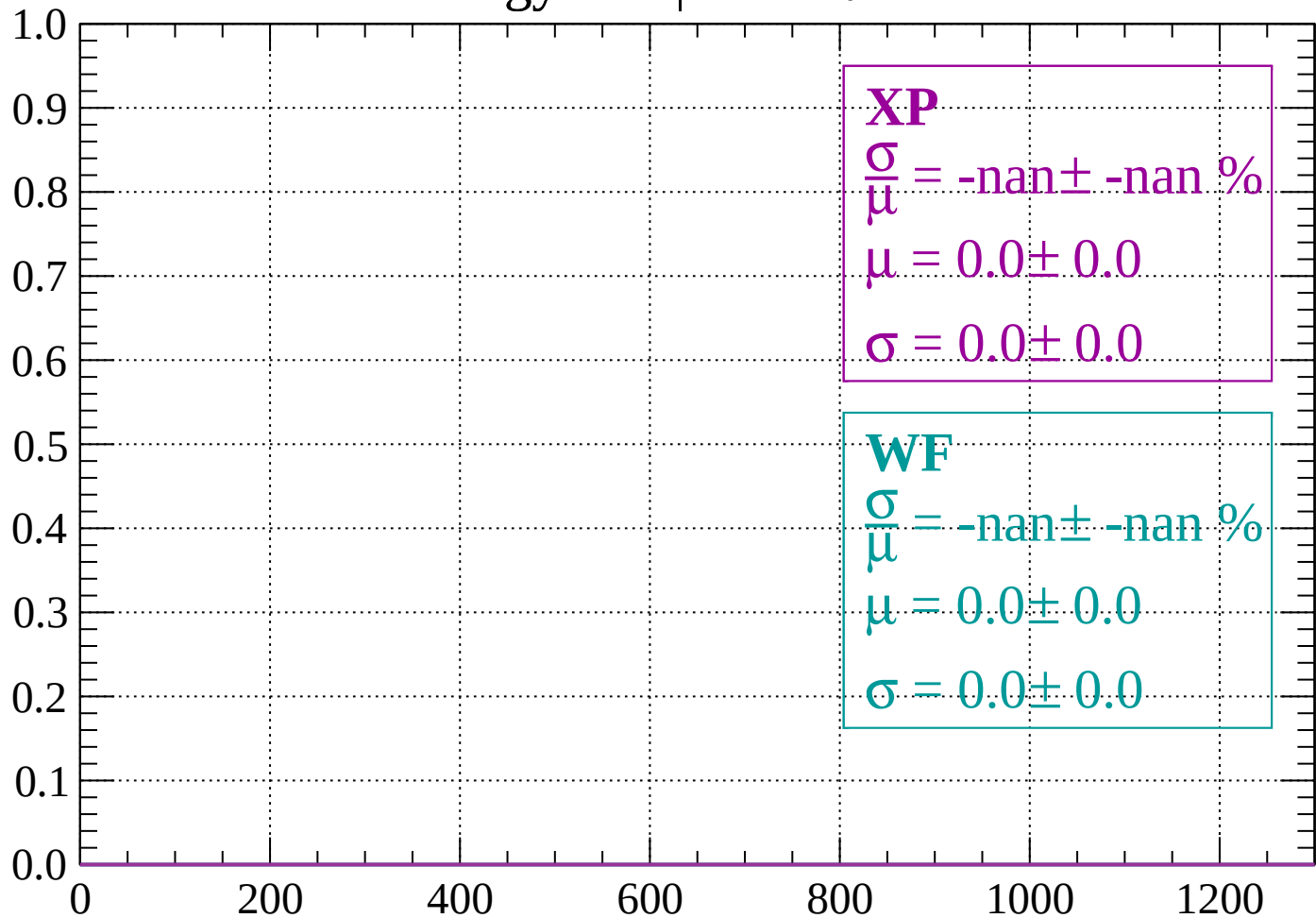
Count



dE/dx (ADC counts/cm)

Energy loss | $-46 < \theta < -44$

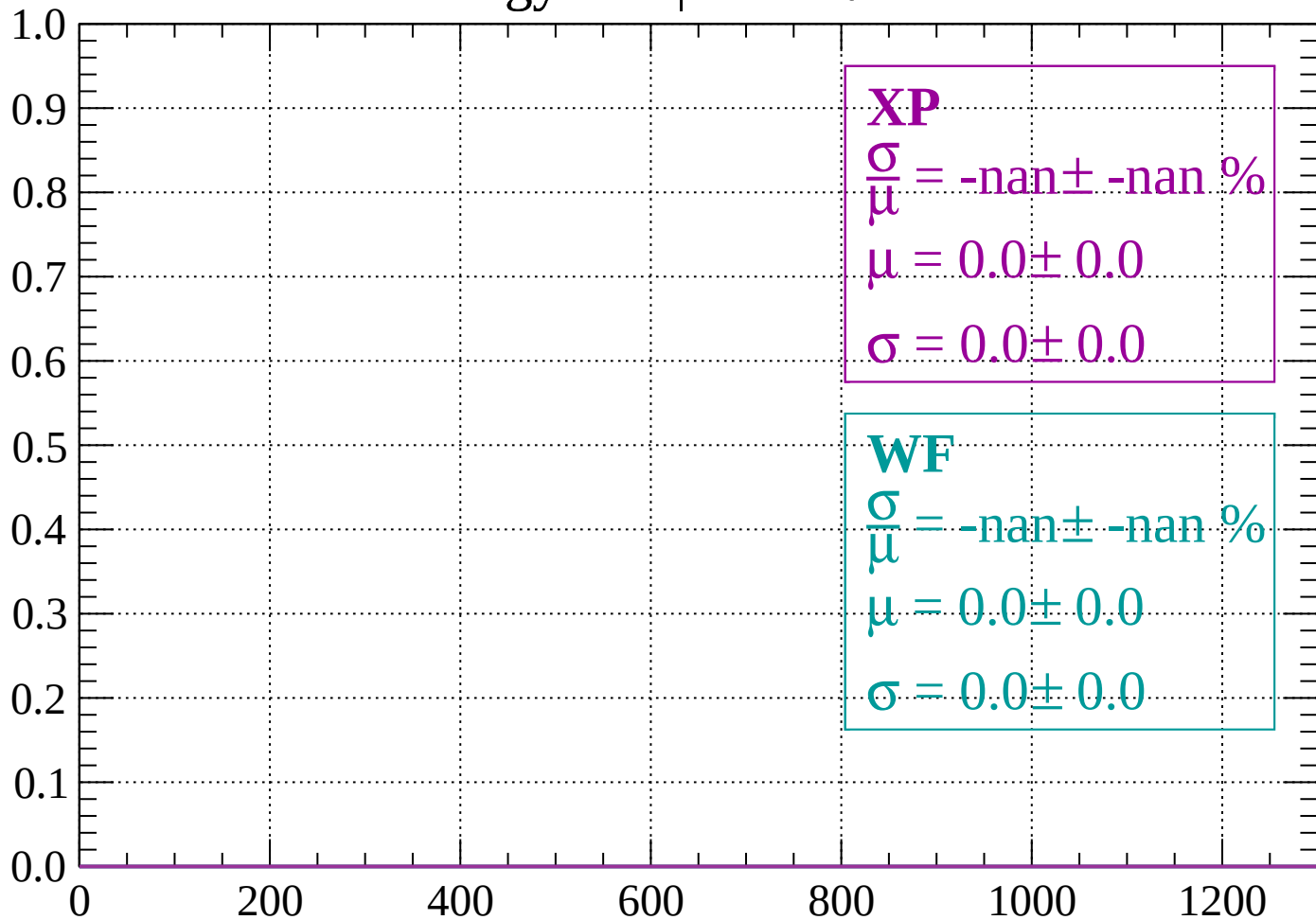
Count



dE/dx (ADC counts/cm)

Energy loss | $-44 < \theta < -42$

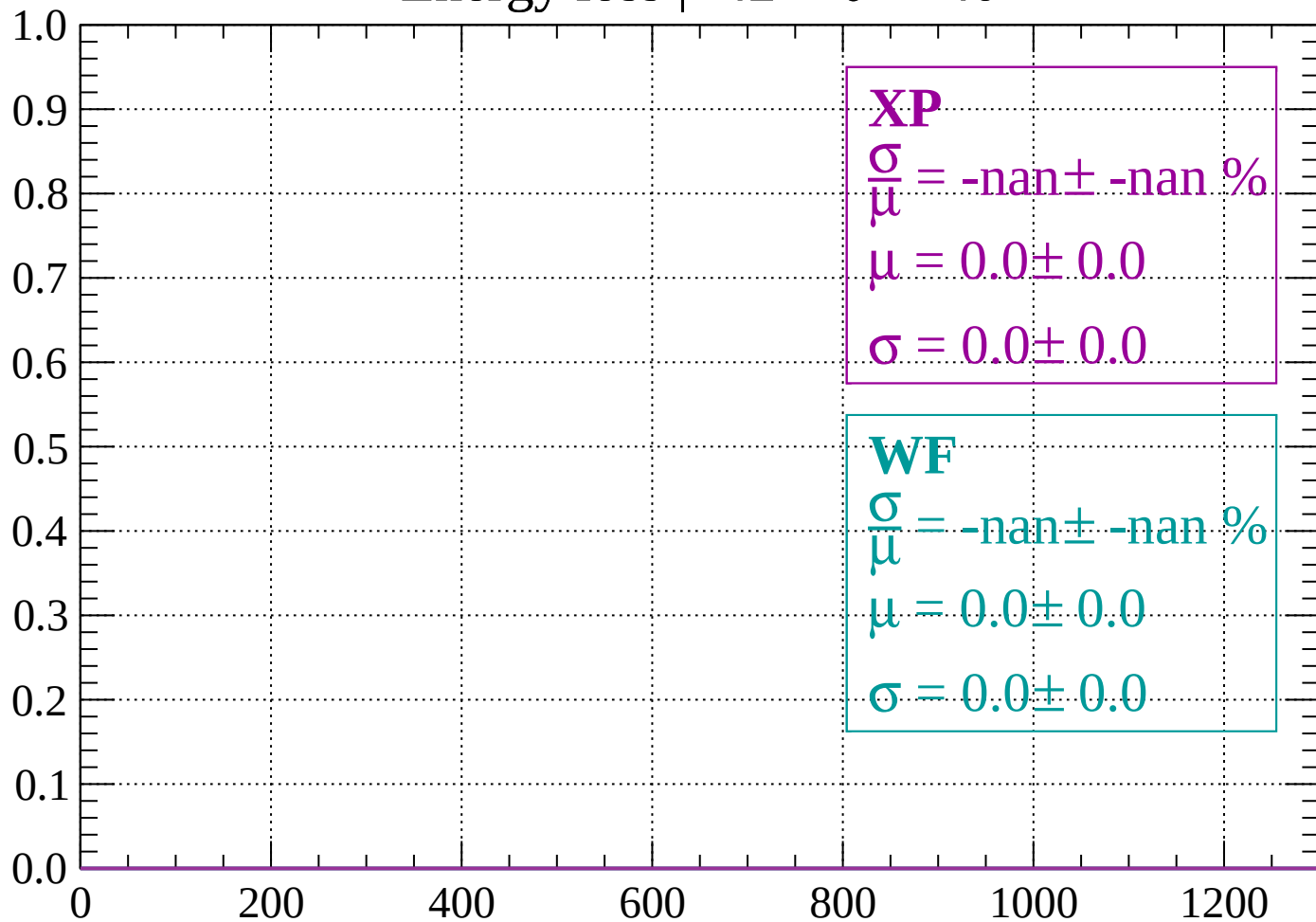
Count



dE/dx (ADC counts/cm)

Energy loss | $-42 < \theta < -40$

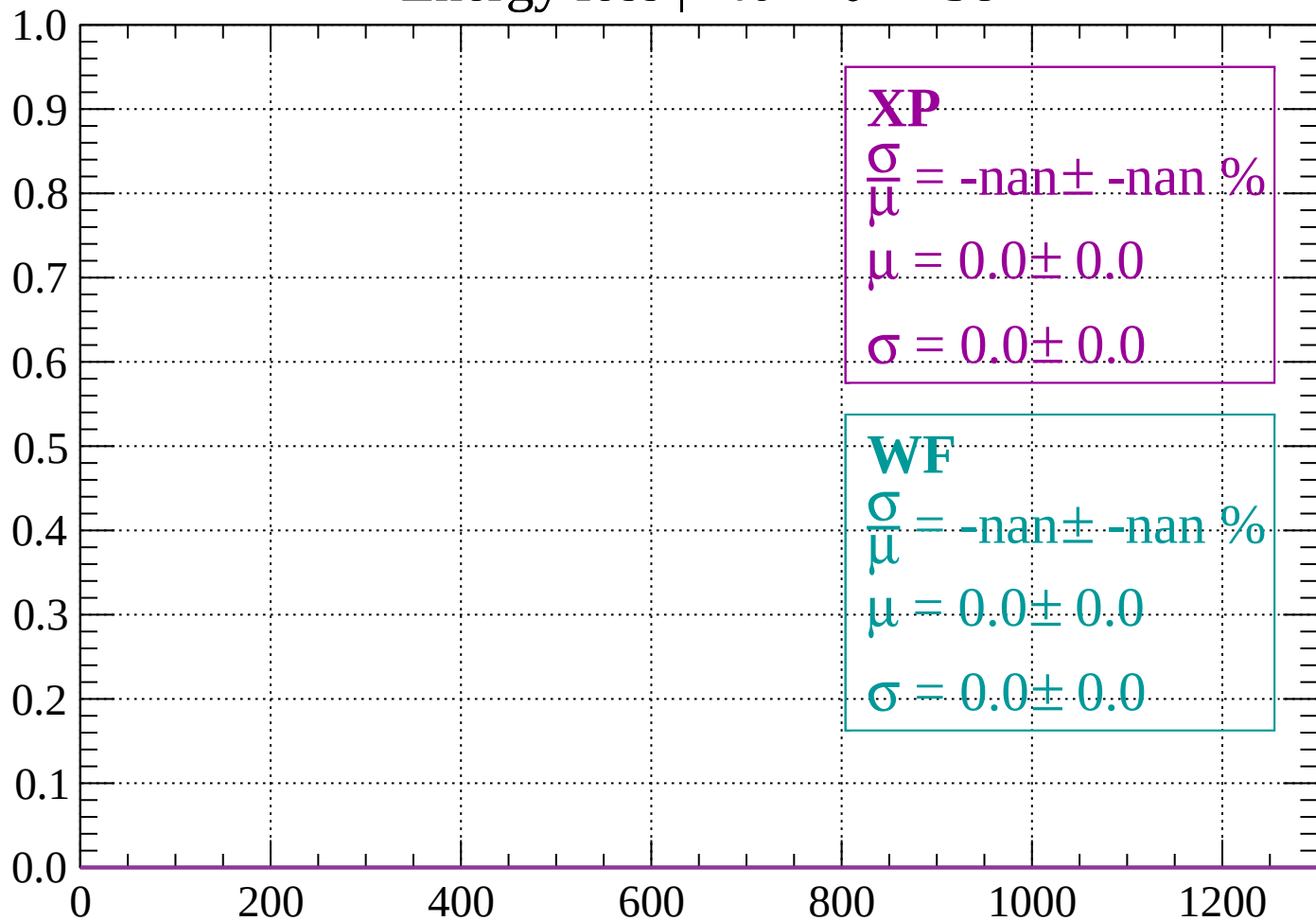
Count



dE/dx (ADC counts/cm)

Energy loss | $-40 < \theta < -38$

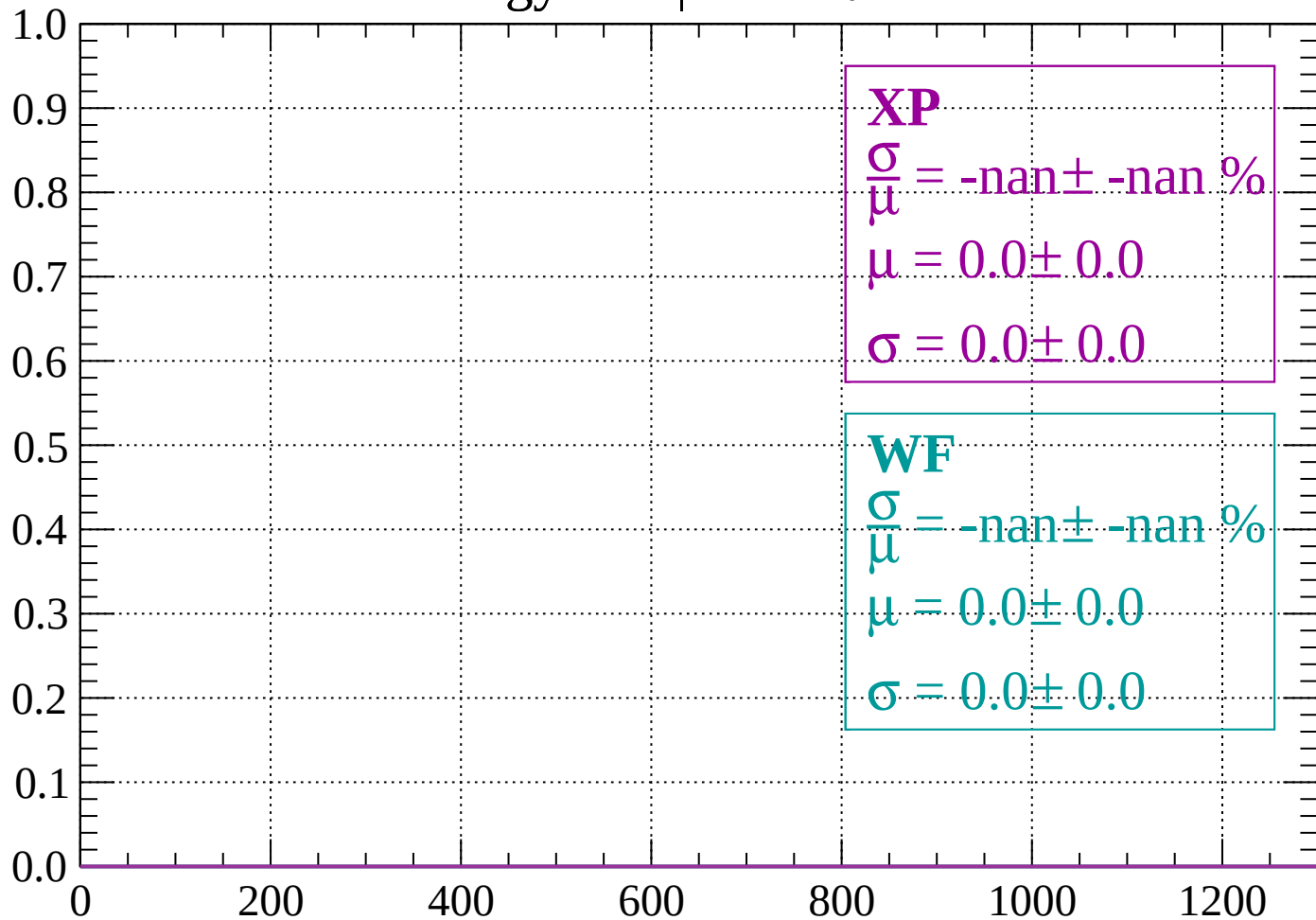
Count



dE/dx (ADC counts/cm)

Energy loss | $-38 < \theta < -36$

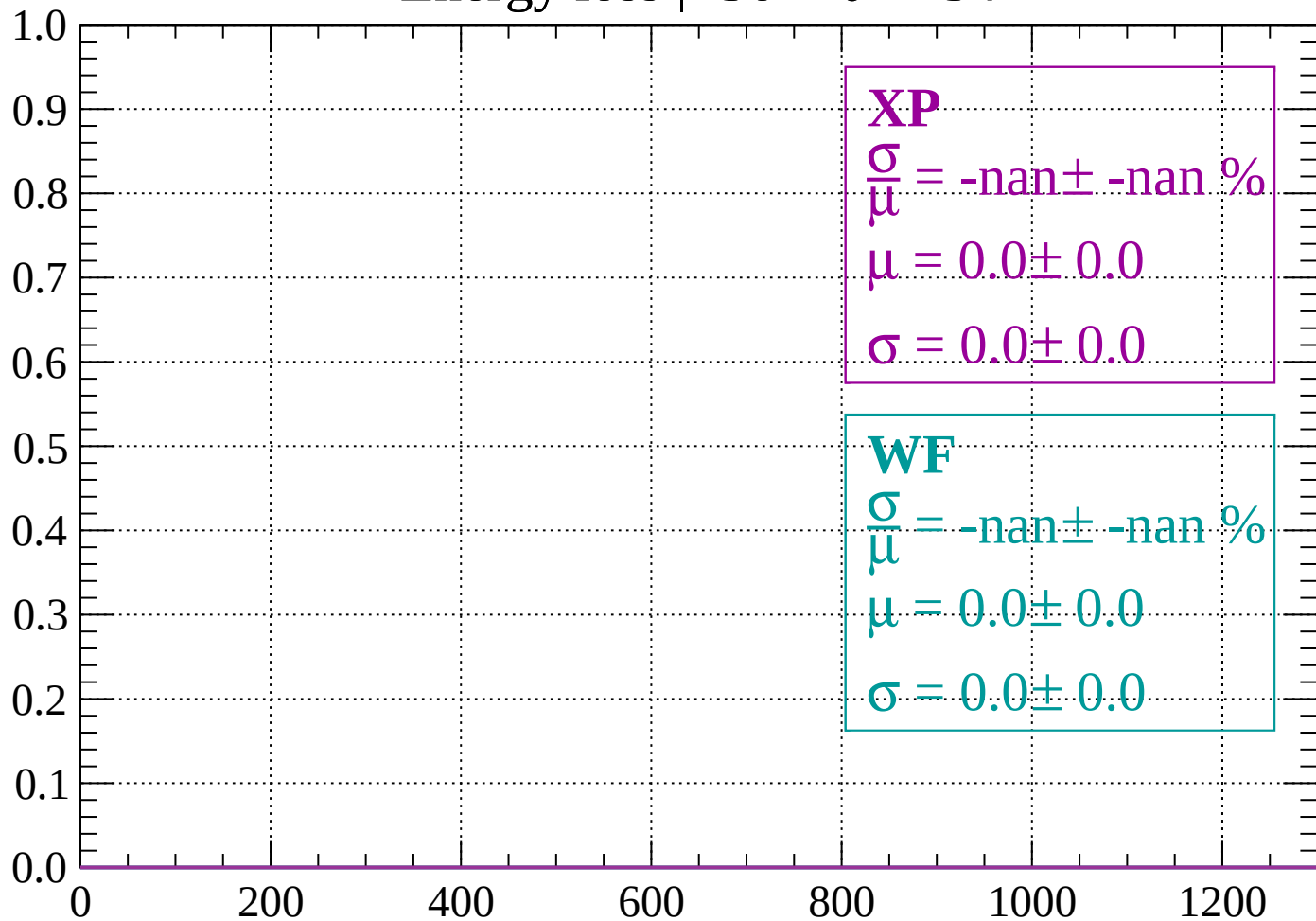
Count



dE/dx (ADC counts/cm)

Energy loss | $-36 < \theta < -34$

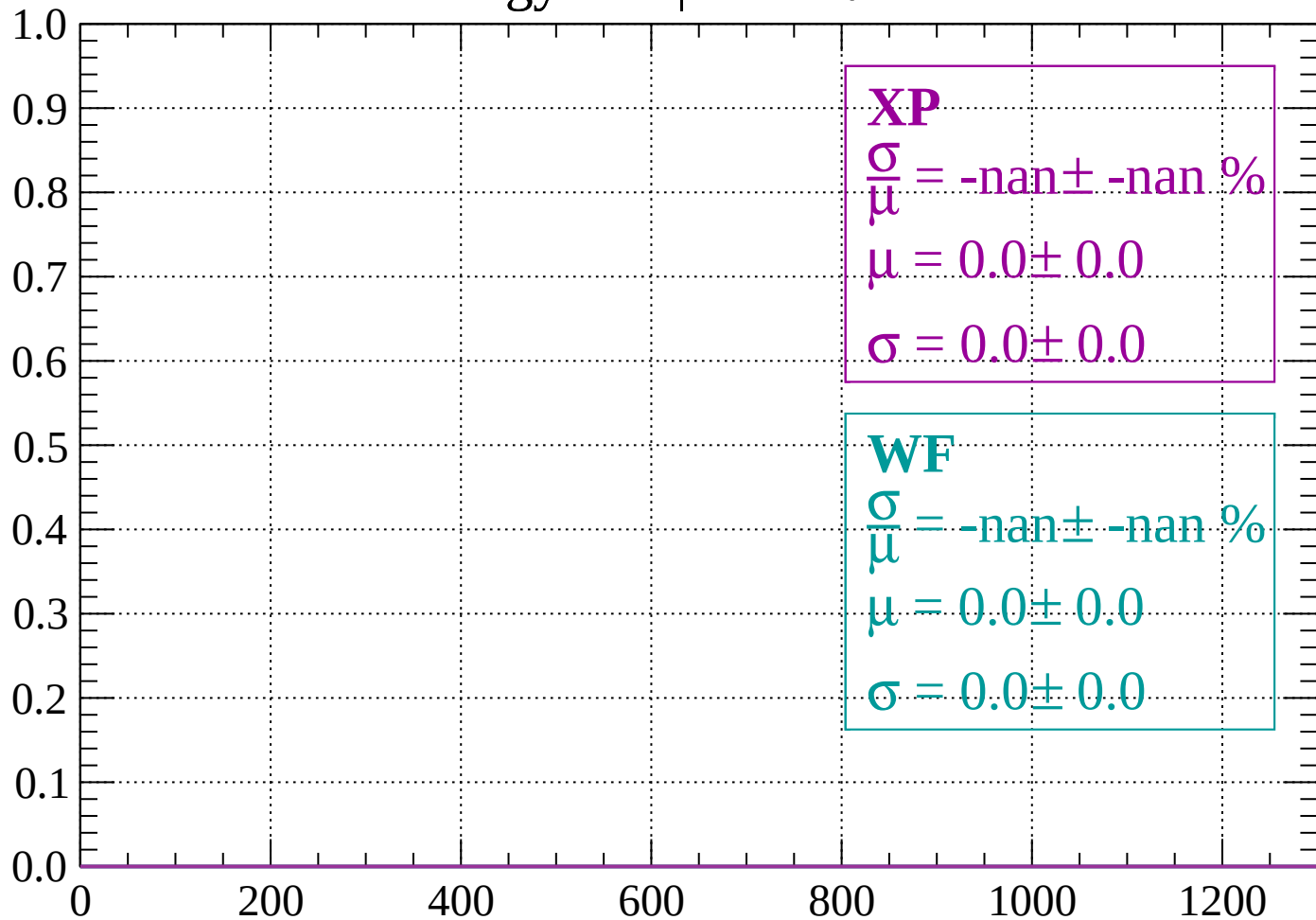
Count



dE/dx (ADC counts/cm)

Energy loss | $-34 < \theta < -32$

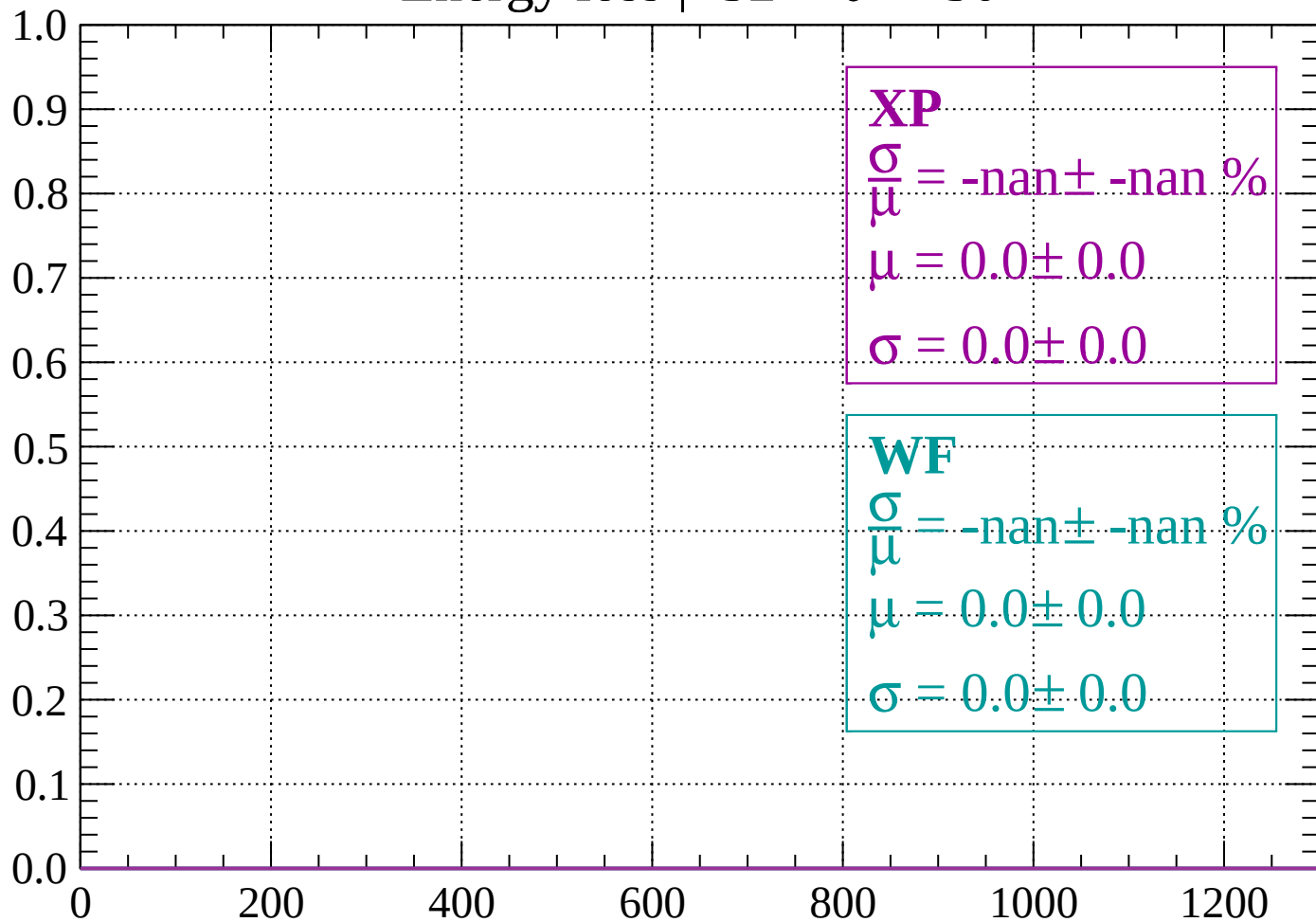
Count



dE/dx (ADC counts/cm)

Energy loss | $-32 < \theta < -30$

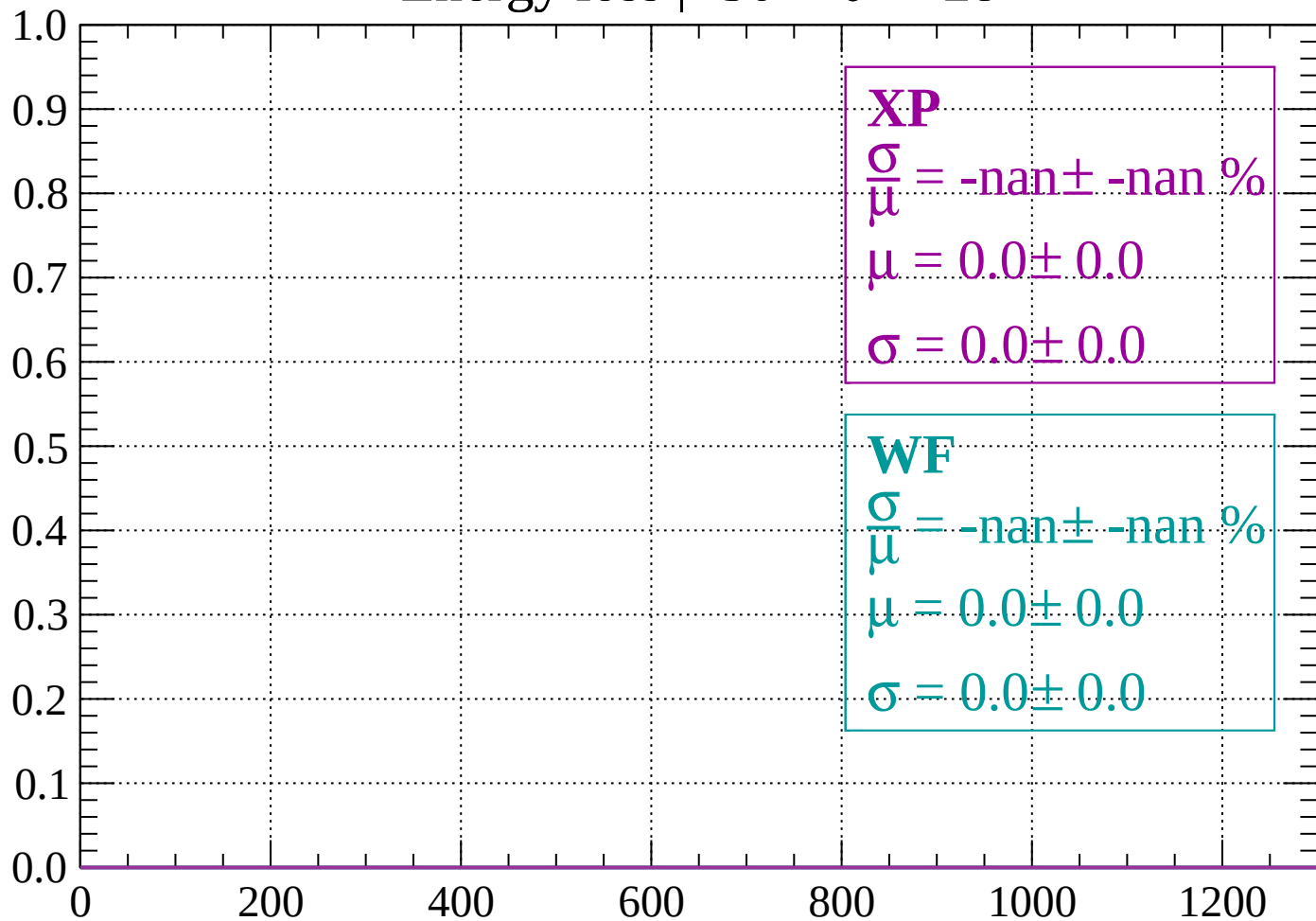
Count



dE/dx (ADC counts/cm)

Energy loss | $-30 < \theta < -28$

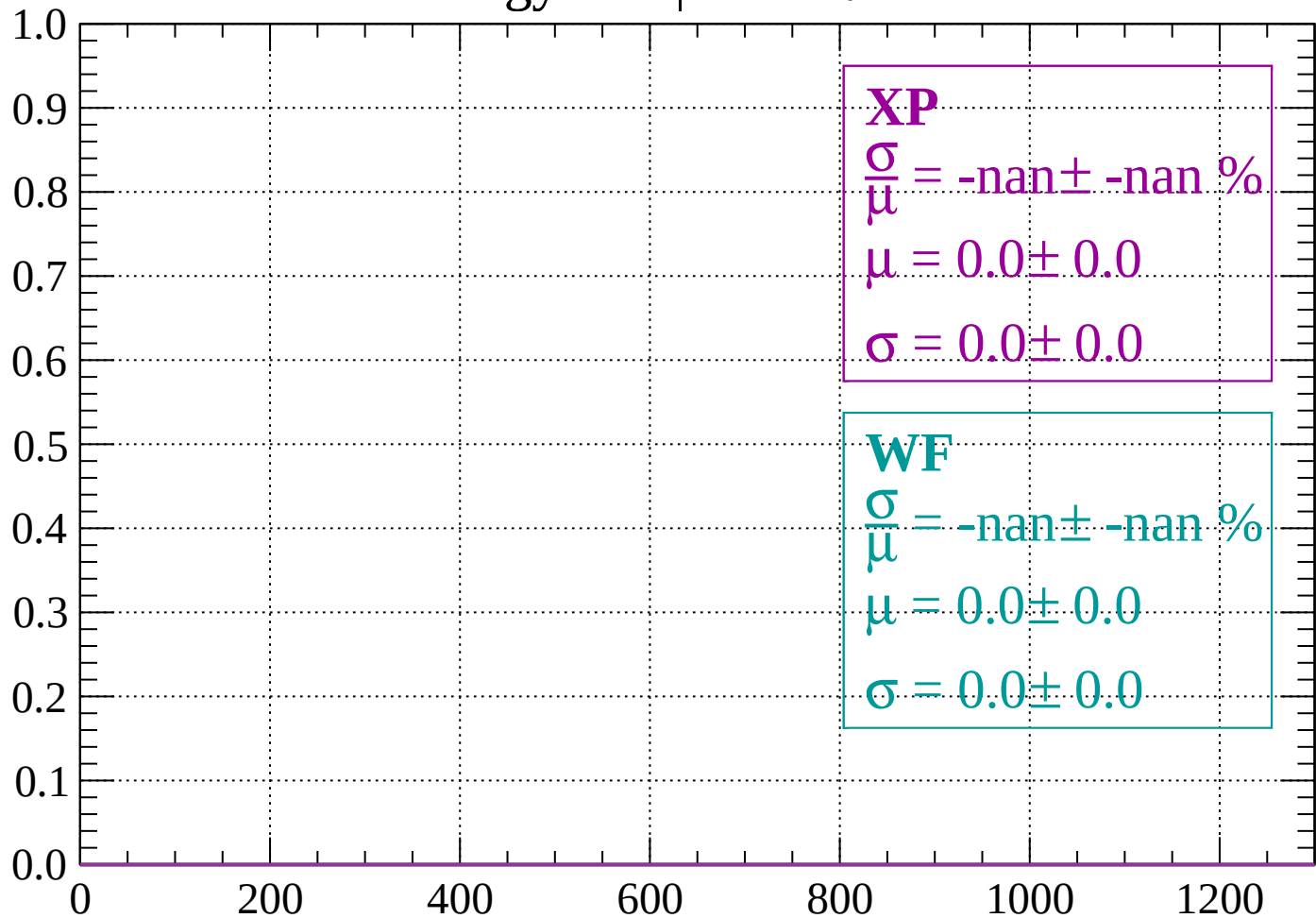
Count



dE/dx (ADC counts/cm)

Energy loss | $-28 < \theta < -26$

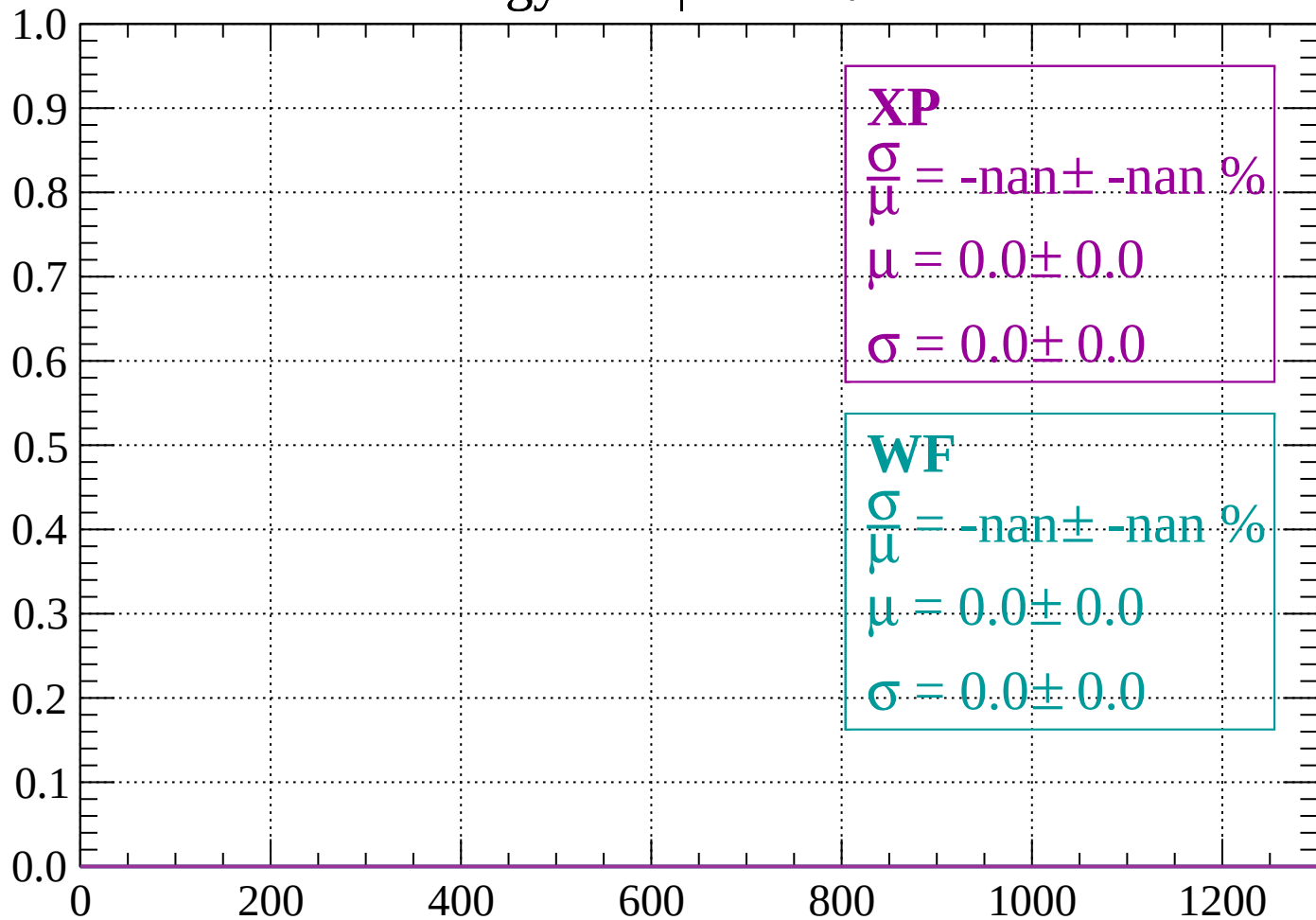
Count



dE/dx (ADC counts/cm)

Energy loss | $-26 < \theta < -24$

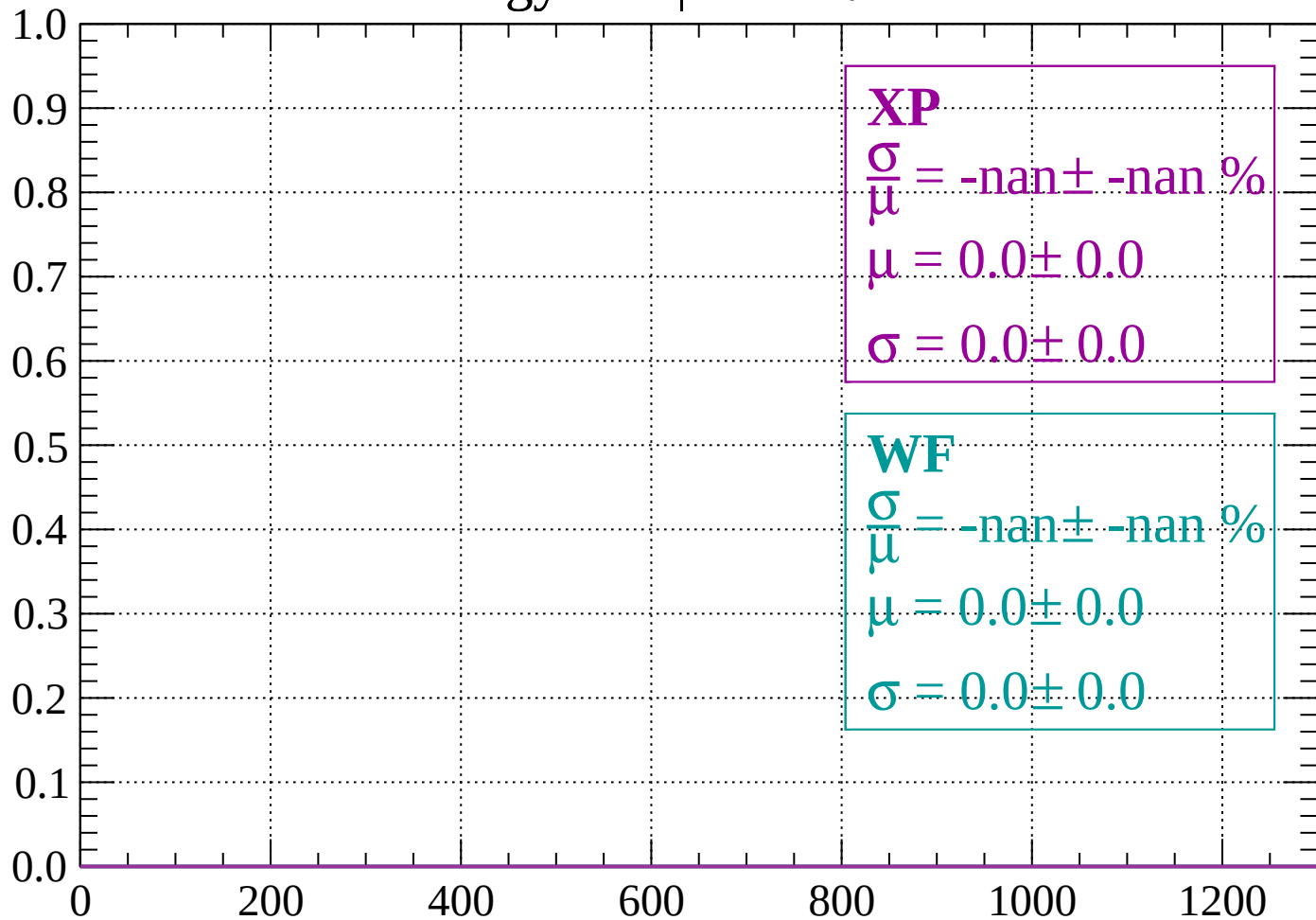
Count



dE/dx (ADC counts/cm)

Energy loss | $-24 < \theta < -22$

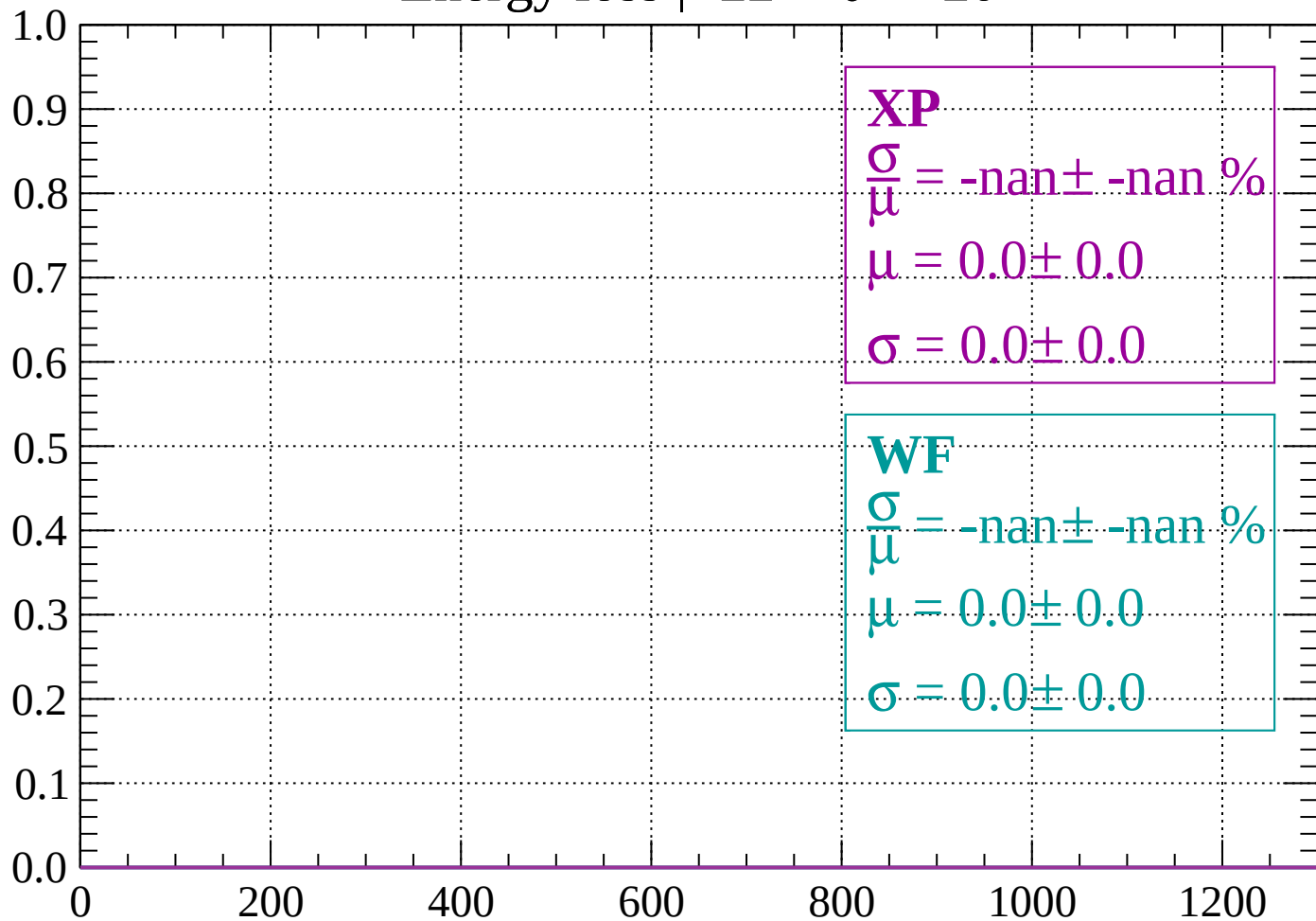
Count



dE/dx (ADC counts/cm)

Energy loss | $-22 < \theta < -20$

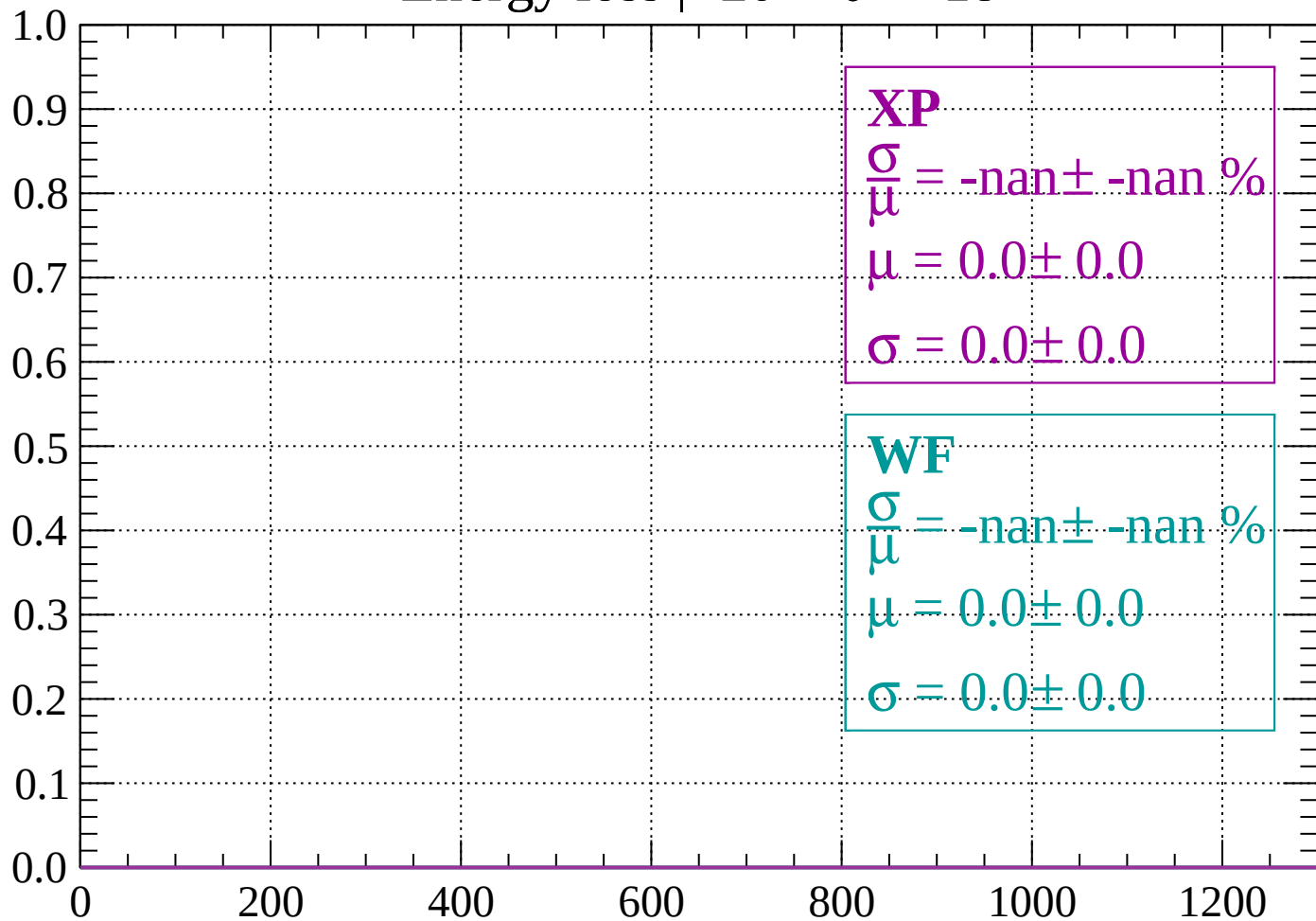
Count



dE/dx (ADC counts/cm)

Energy loss | $-20 < \theta < -18$

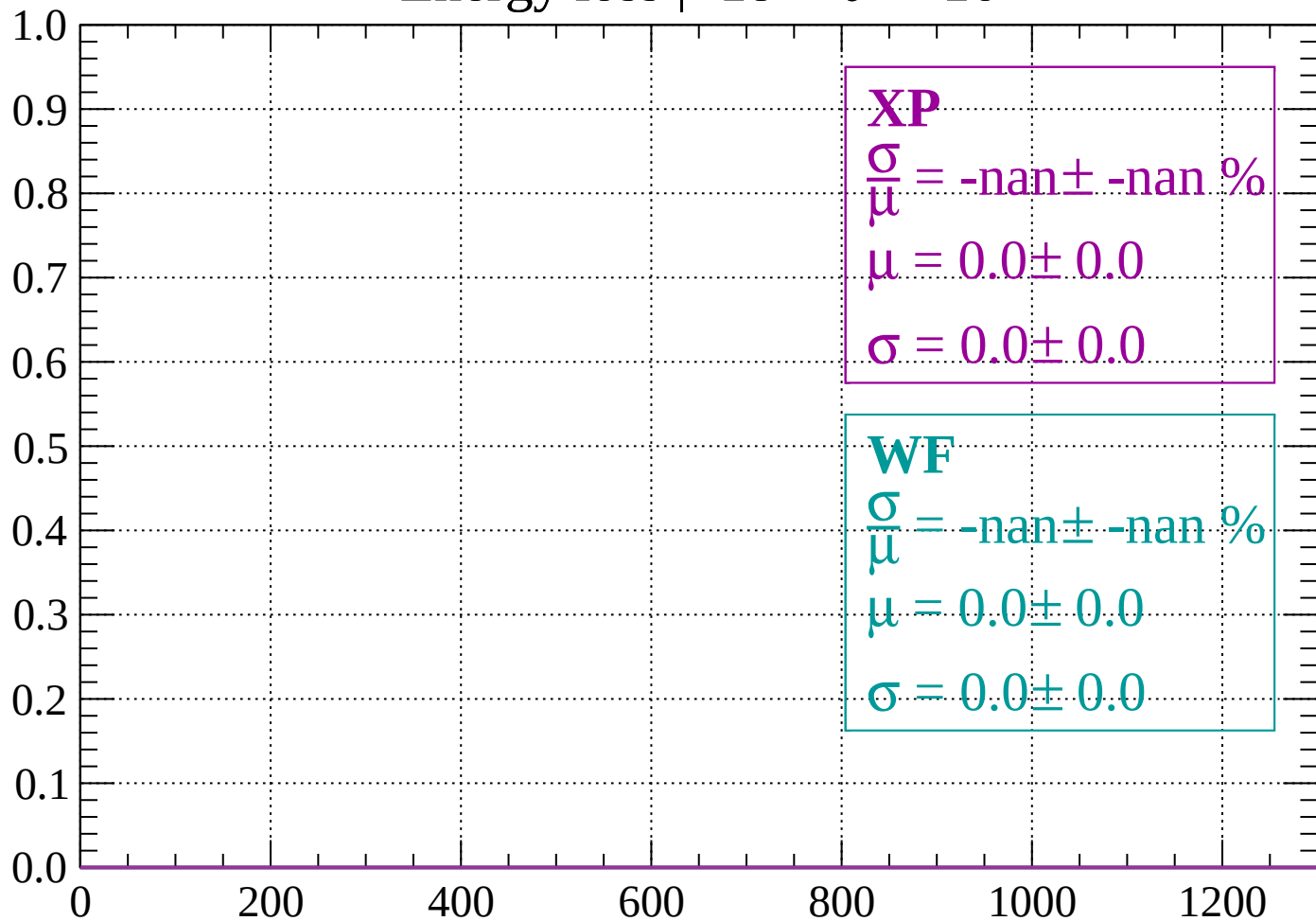
Count



dE/dx (ADC counts/cm)

Energy loss | $-18 < \theta < -16$

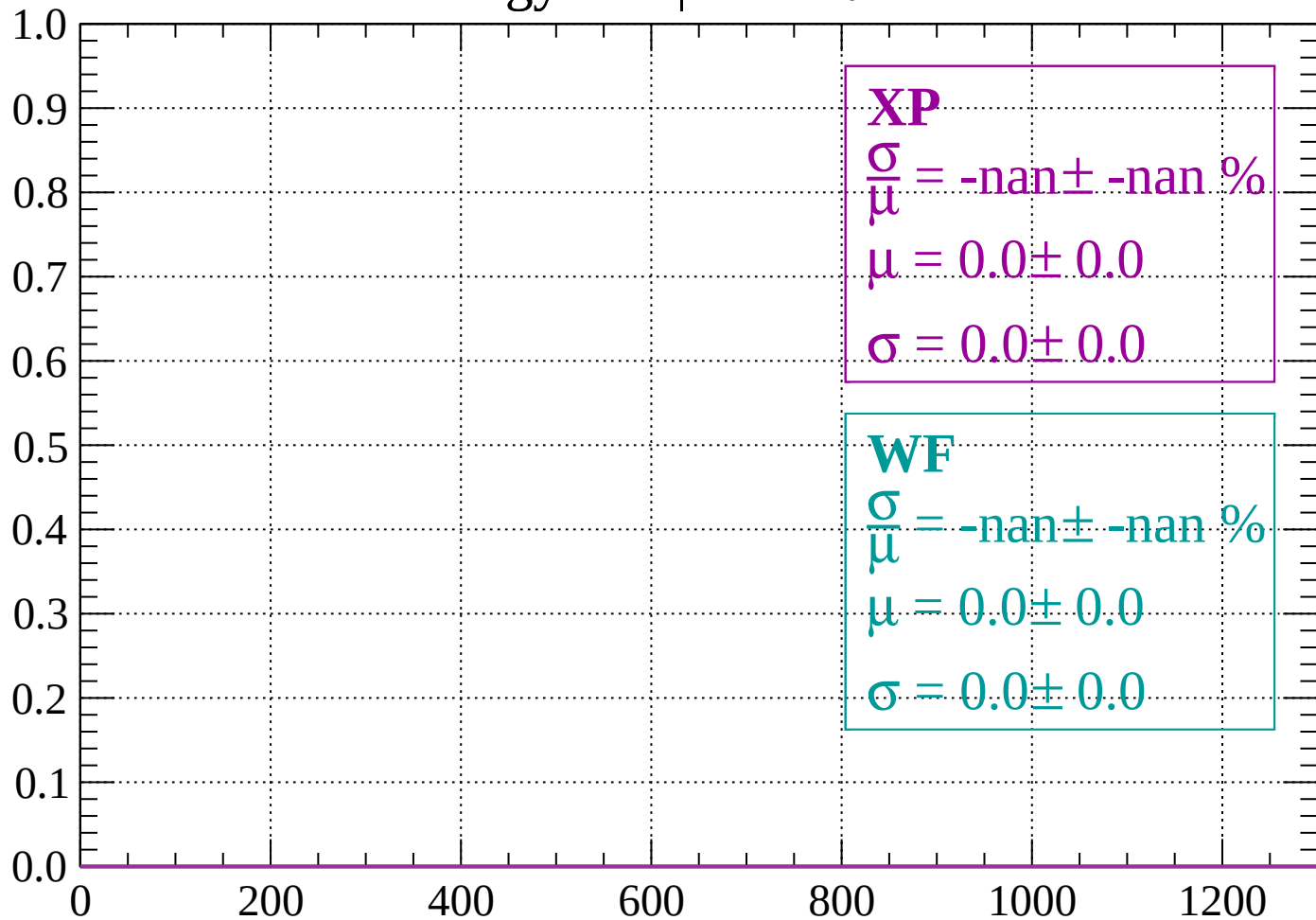
Count



dE/dx (ADC counts/cm)

Energy loss | $-16 < \theta < -14$

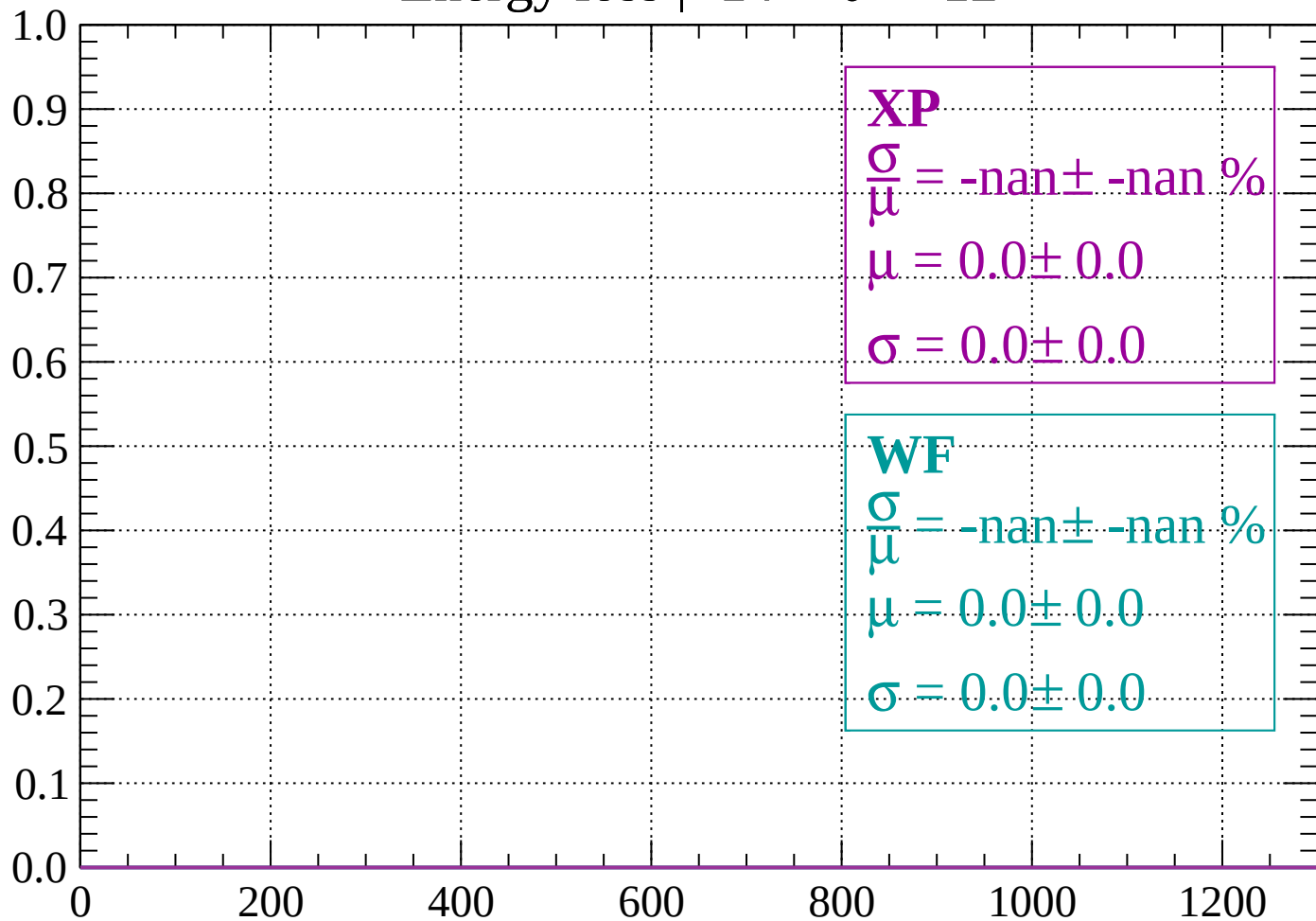
Count



dE/dx (ADC counts/cm)

Energy loss | $-14 < \theta < -12$

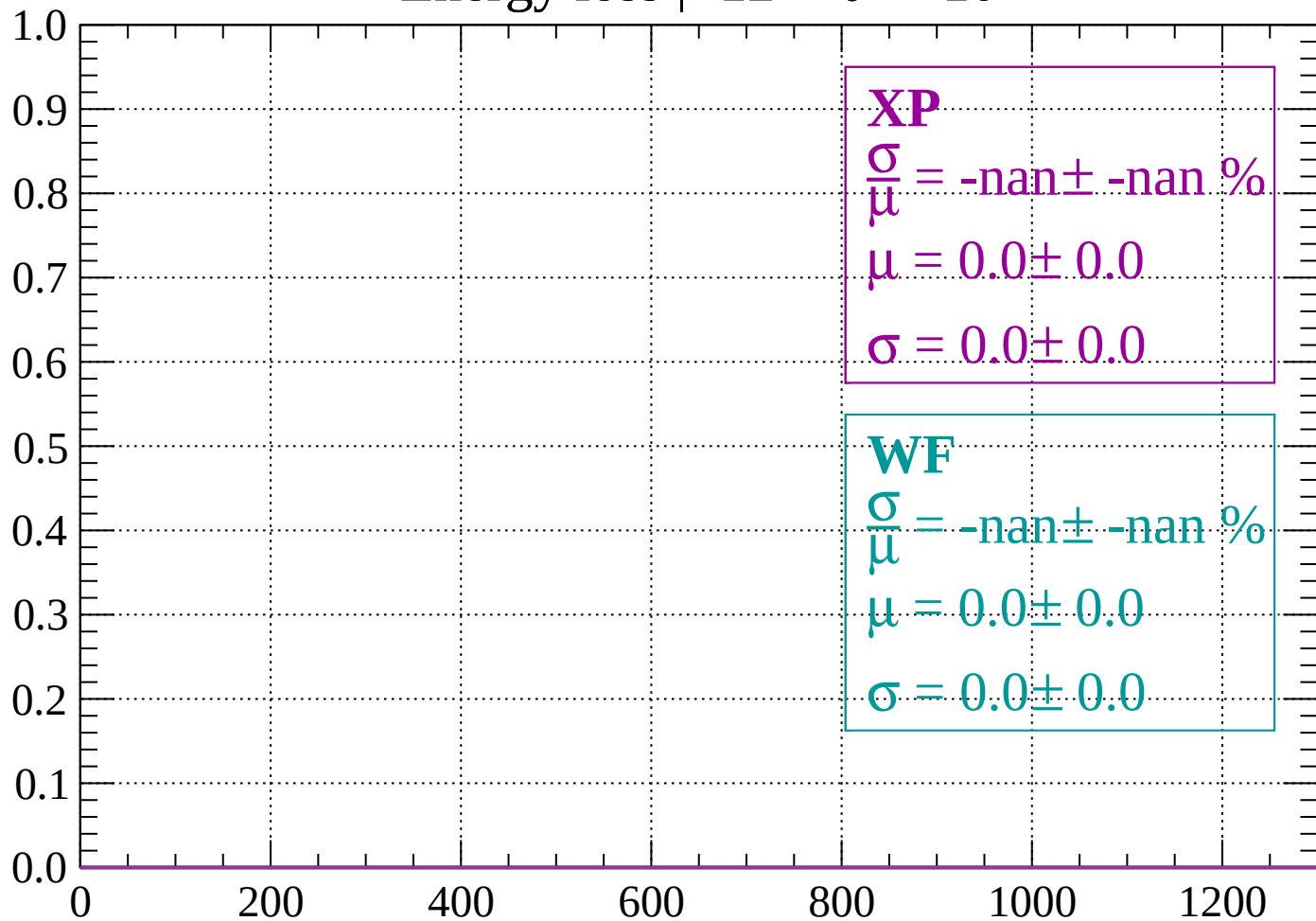
Count



dE/dx (ADC counts/cm)

Energy loss | $-12 < \theta < -10$

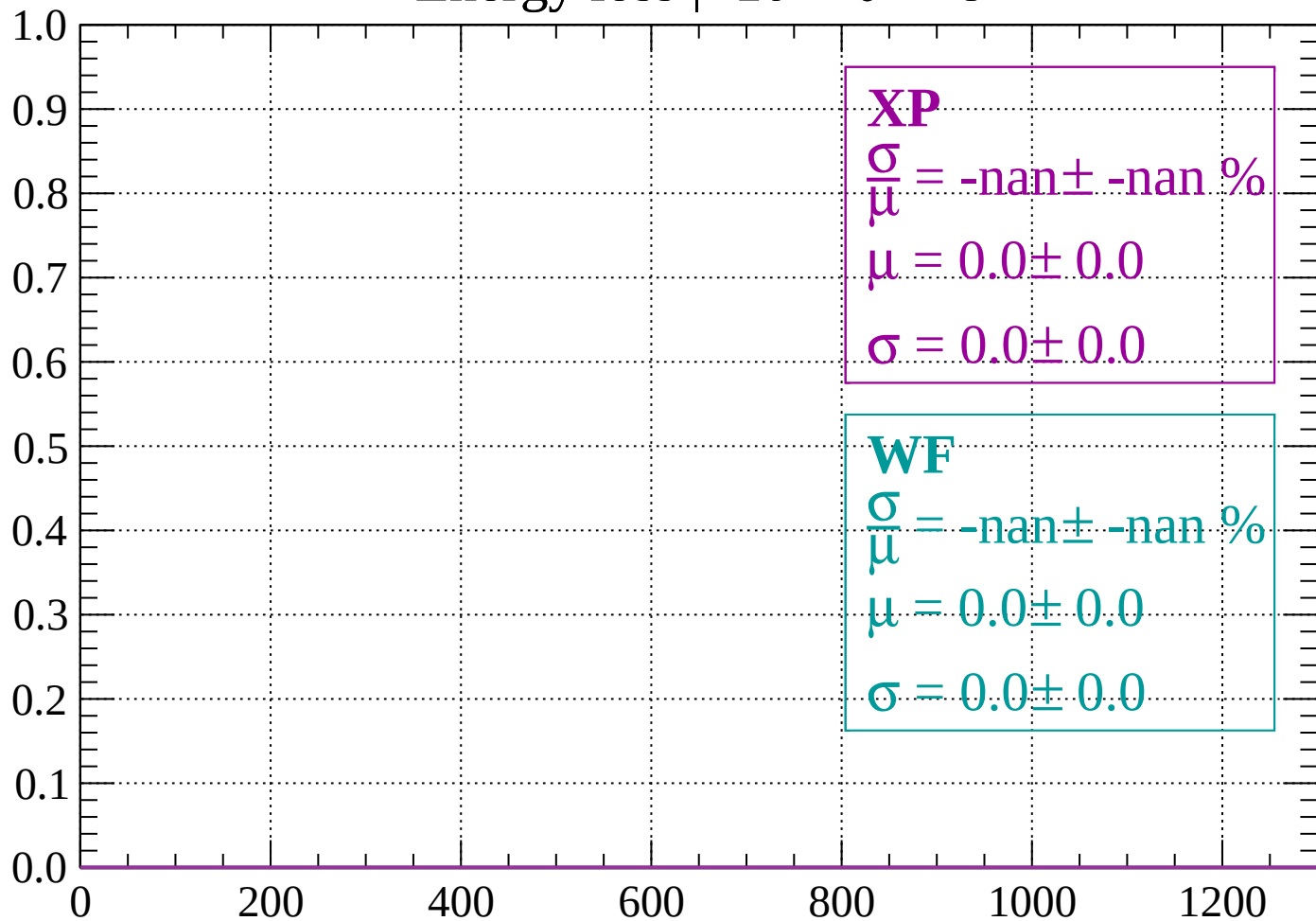
Count



dE/dx (ADC counts/cm)

Energy loss | $-10 < \theta < -8$

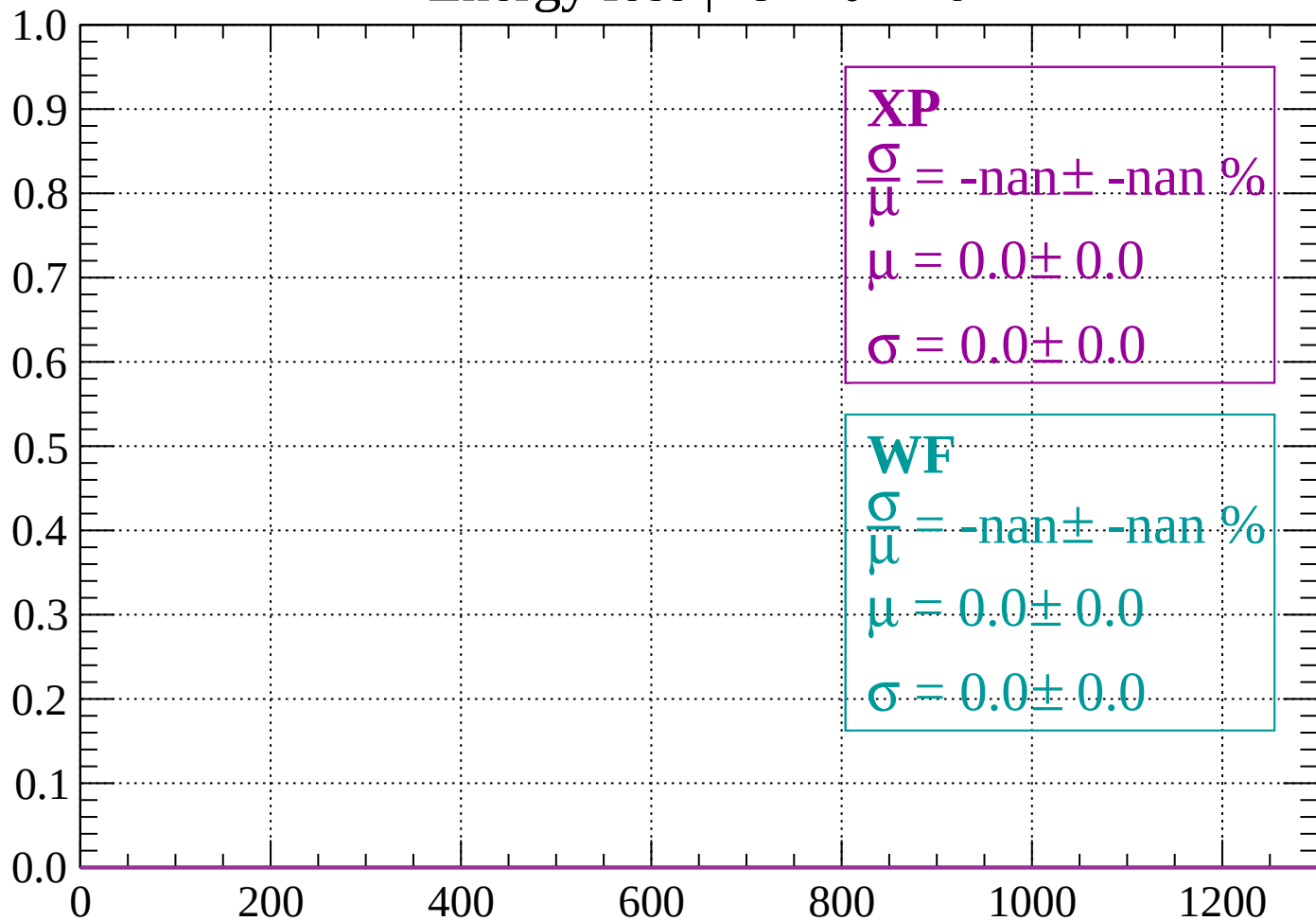
Count



dE/dx (ADC counts/cm)

Energy loss | $-8 < \theta < -6$

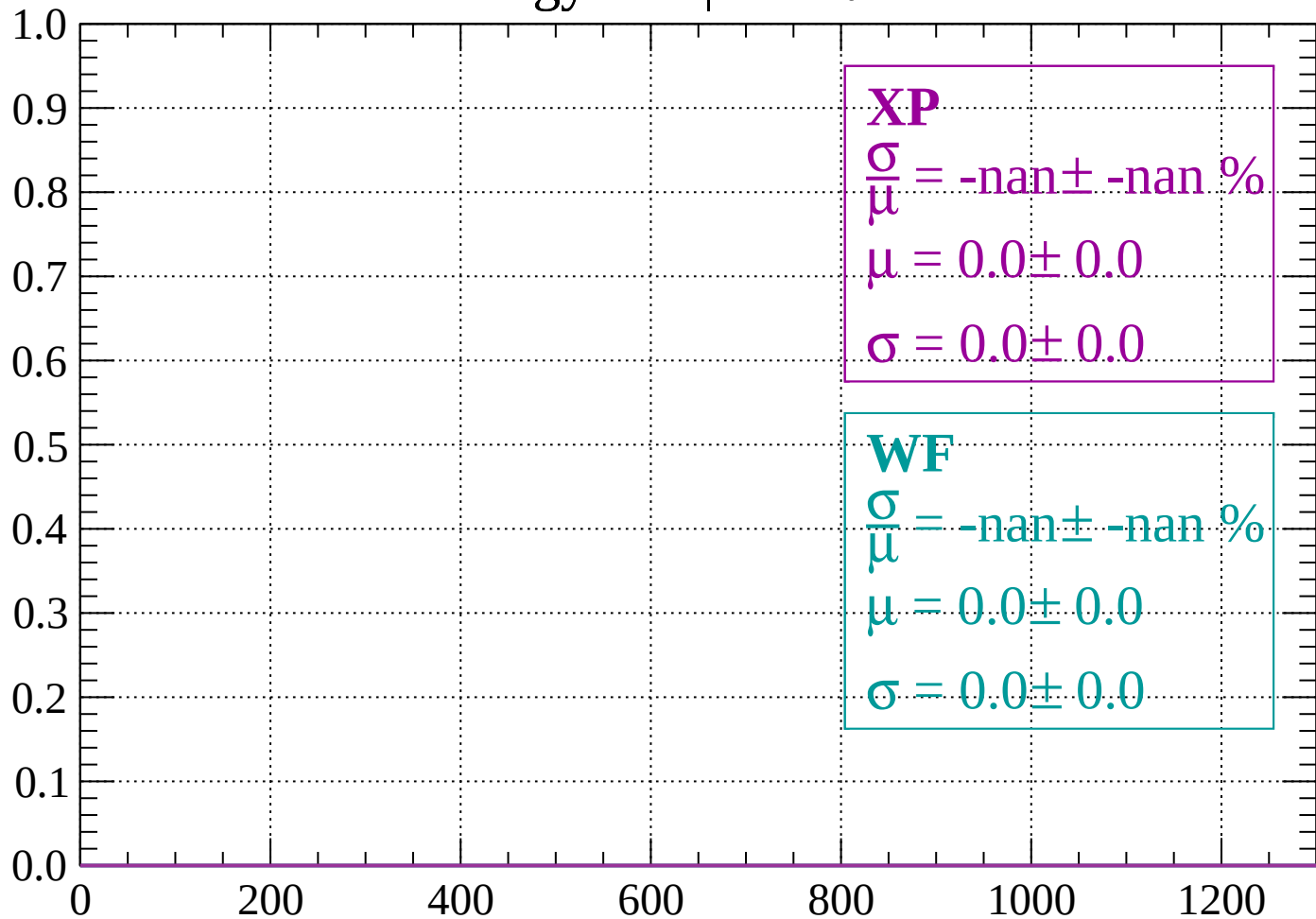
Count



dE/dx (ADC counts/cm)

Energy loss | $-6 < \theta < -4$

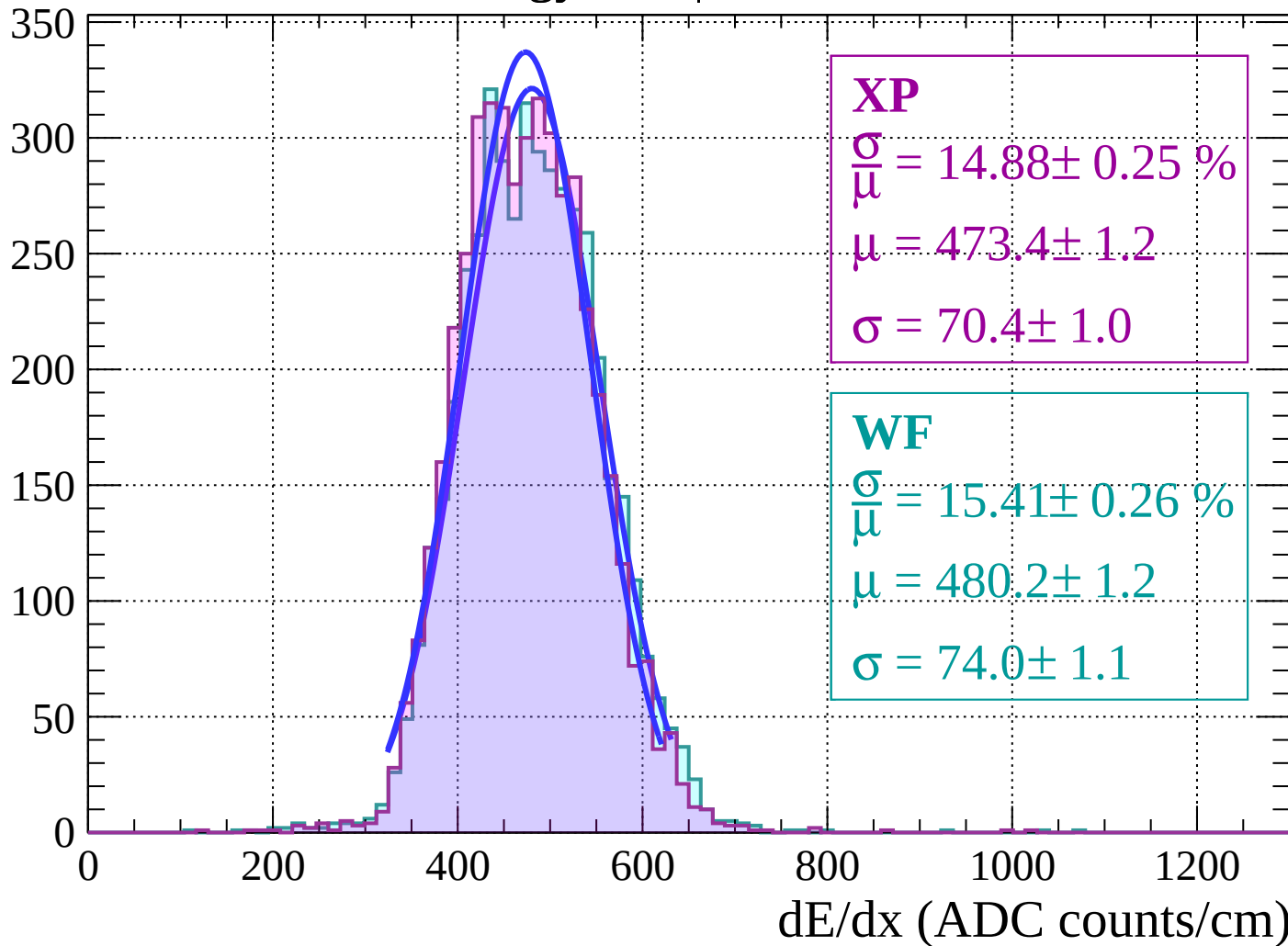
Count



dE/dx (ADC counts/cm)

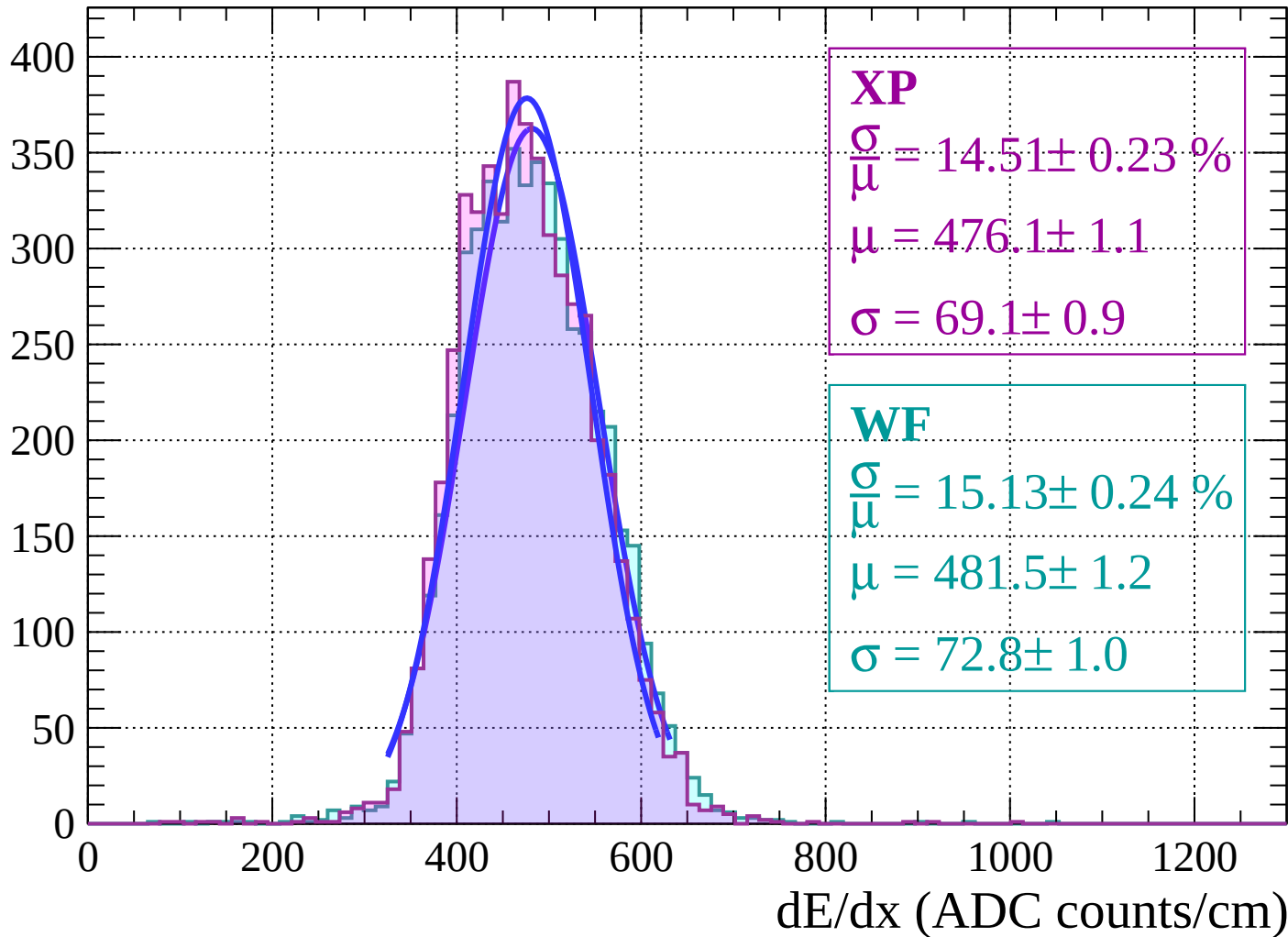
Energy loss | $-4 < \theta < -2$

Count

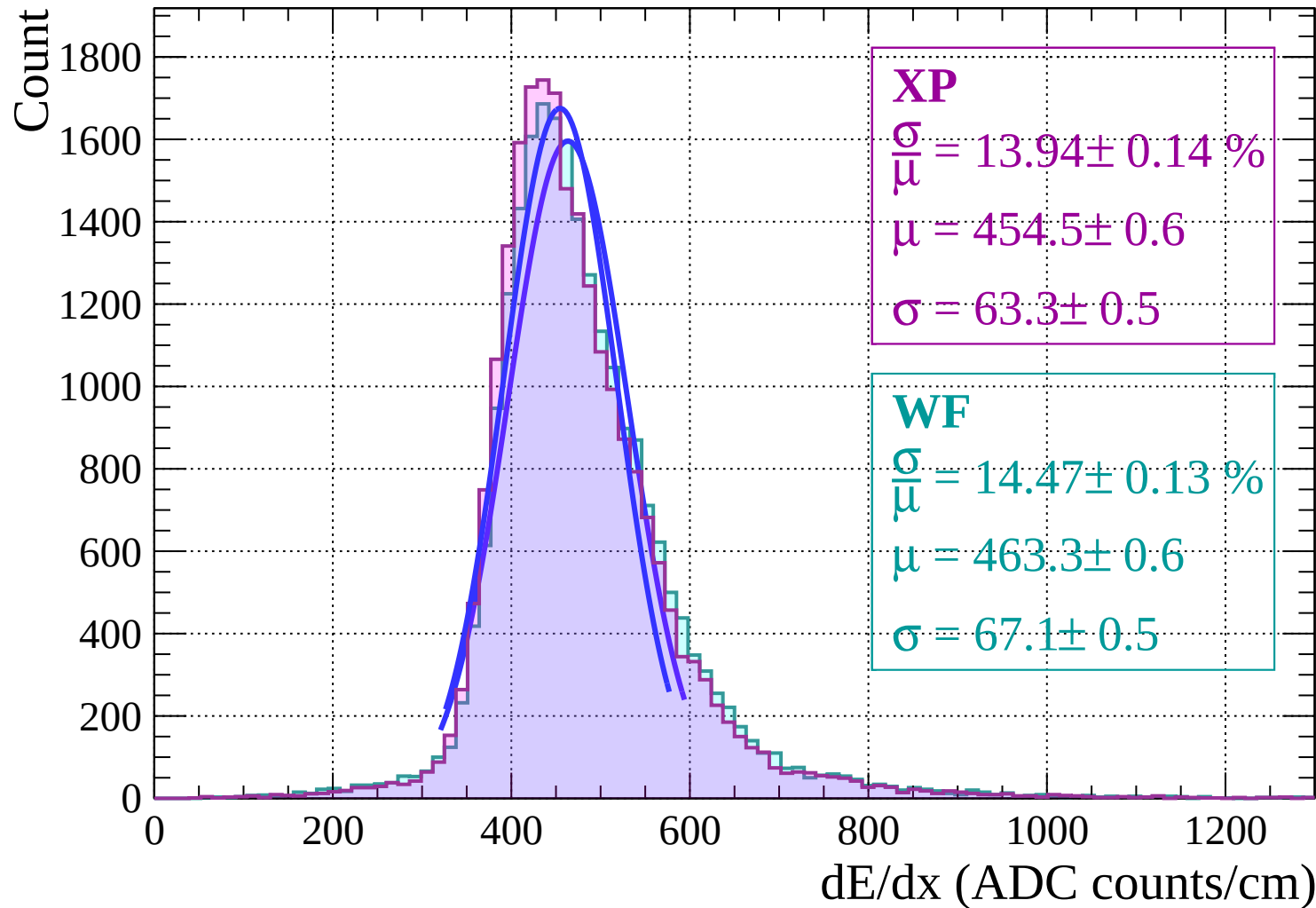


Energy loss | $-2 < \theta < 0$

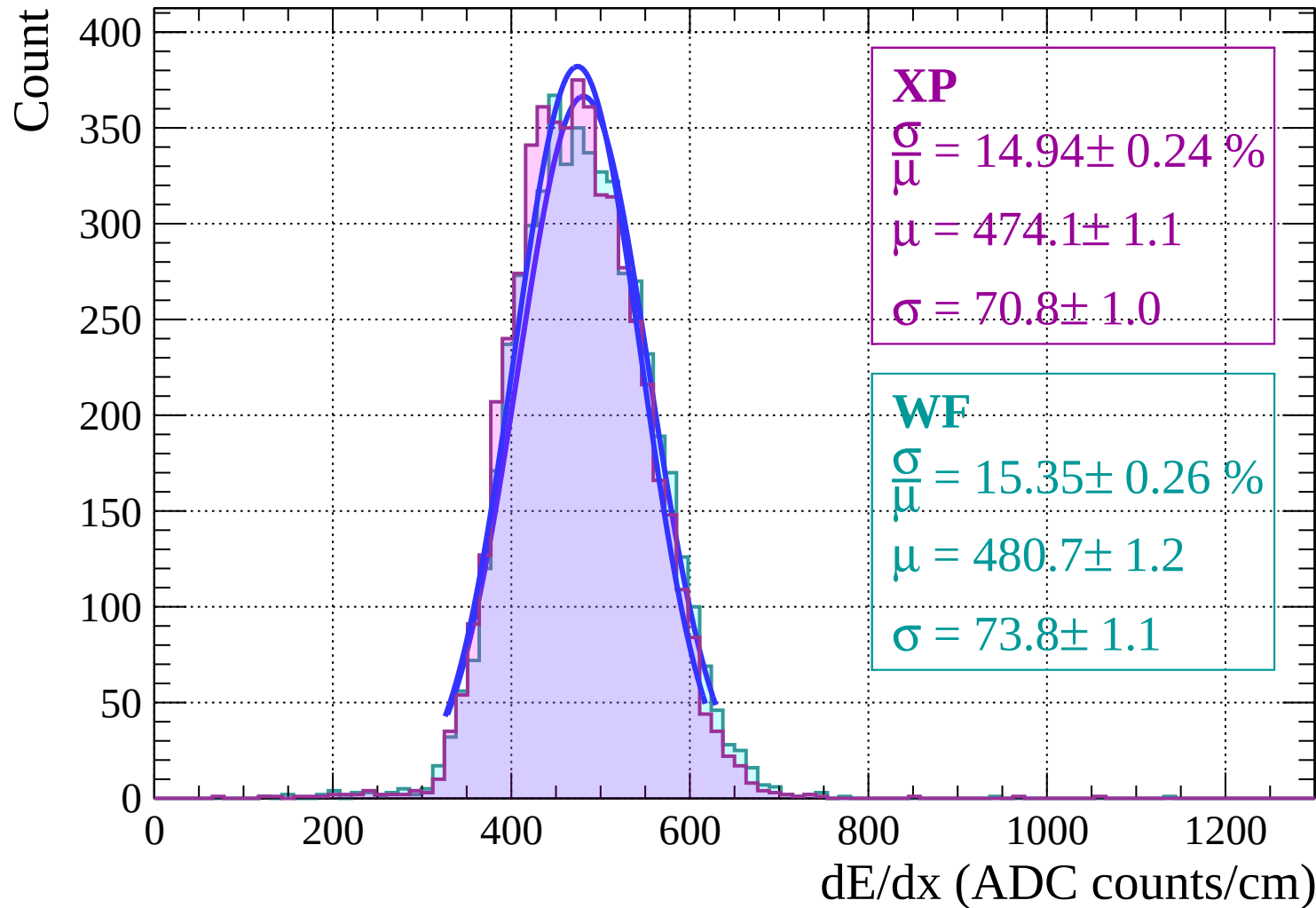
Count



Energy loss | $0 < \theta < 2$

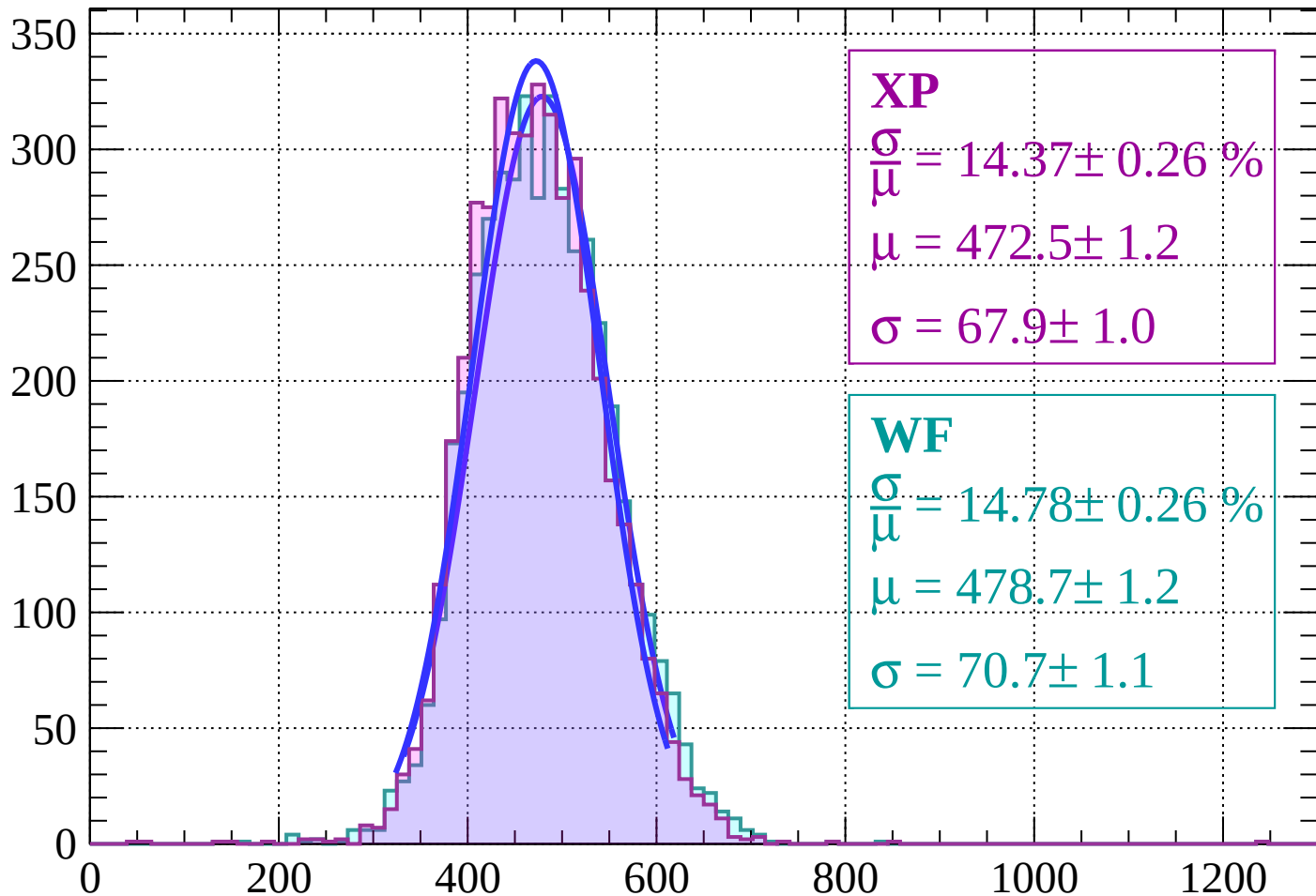


Energy loss | $2 < \theta < 4$



Energy loss | $4 < \theta < 6$

Count



XP

$$\frac{\sigma}{\mu} = 14.37 \pm 0.26 \%$$

$$\mu = 472.5 \pm 1.2$$

$$\sigma = 67.9 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 14.78 \pm 0.26 \%$$

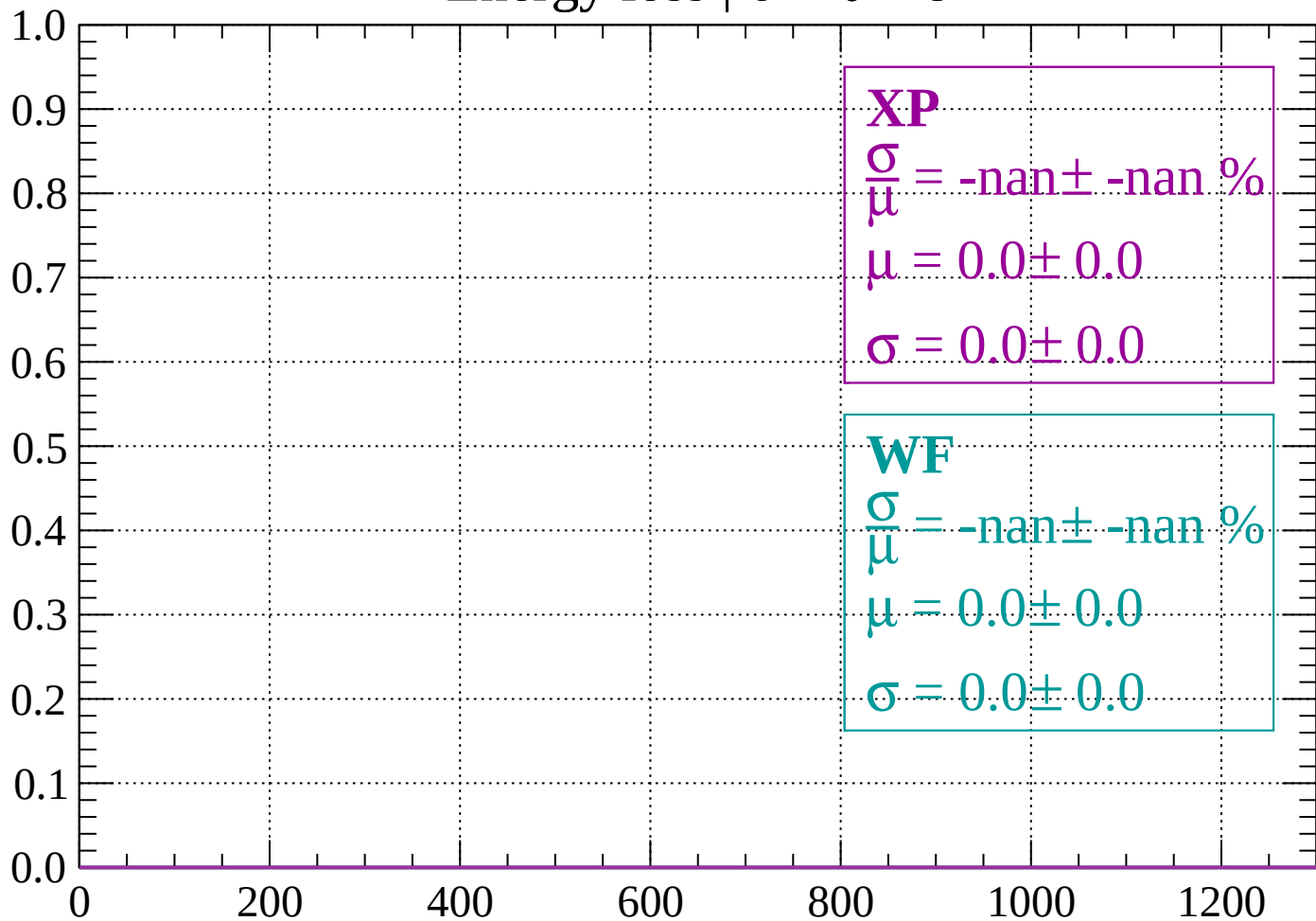
$$\mu = 478.7 \pm 1.2$$

$$\sigma = 70.7 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $6 < \theta < 8$

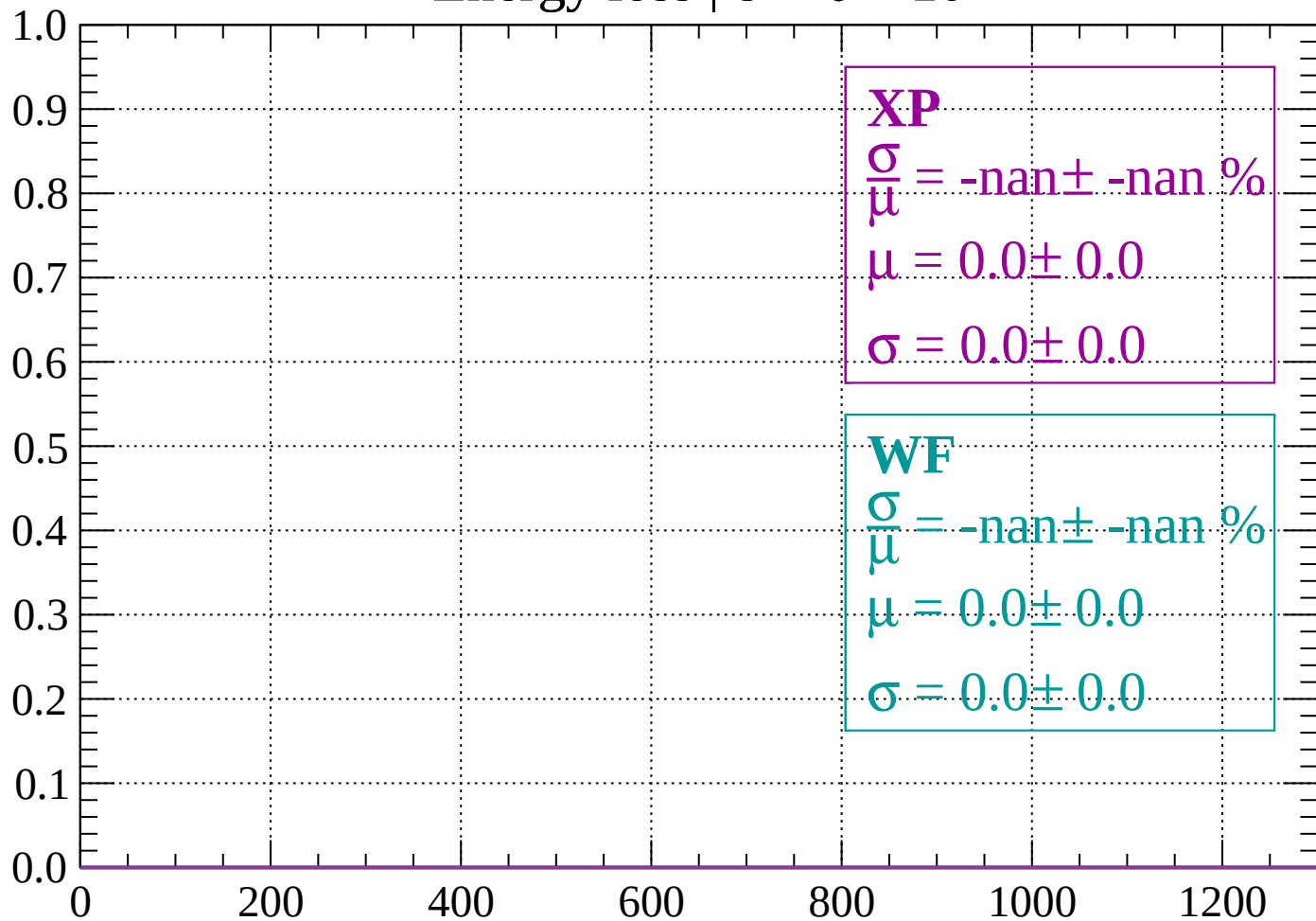
Count



dE/dx (ADC counts/cm)

Energy loss | $8 < \theta < 10$

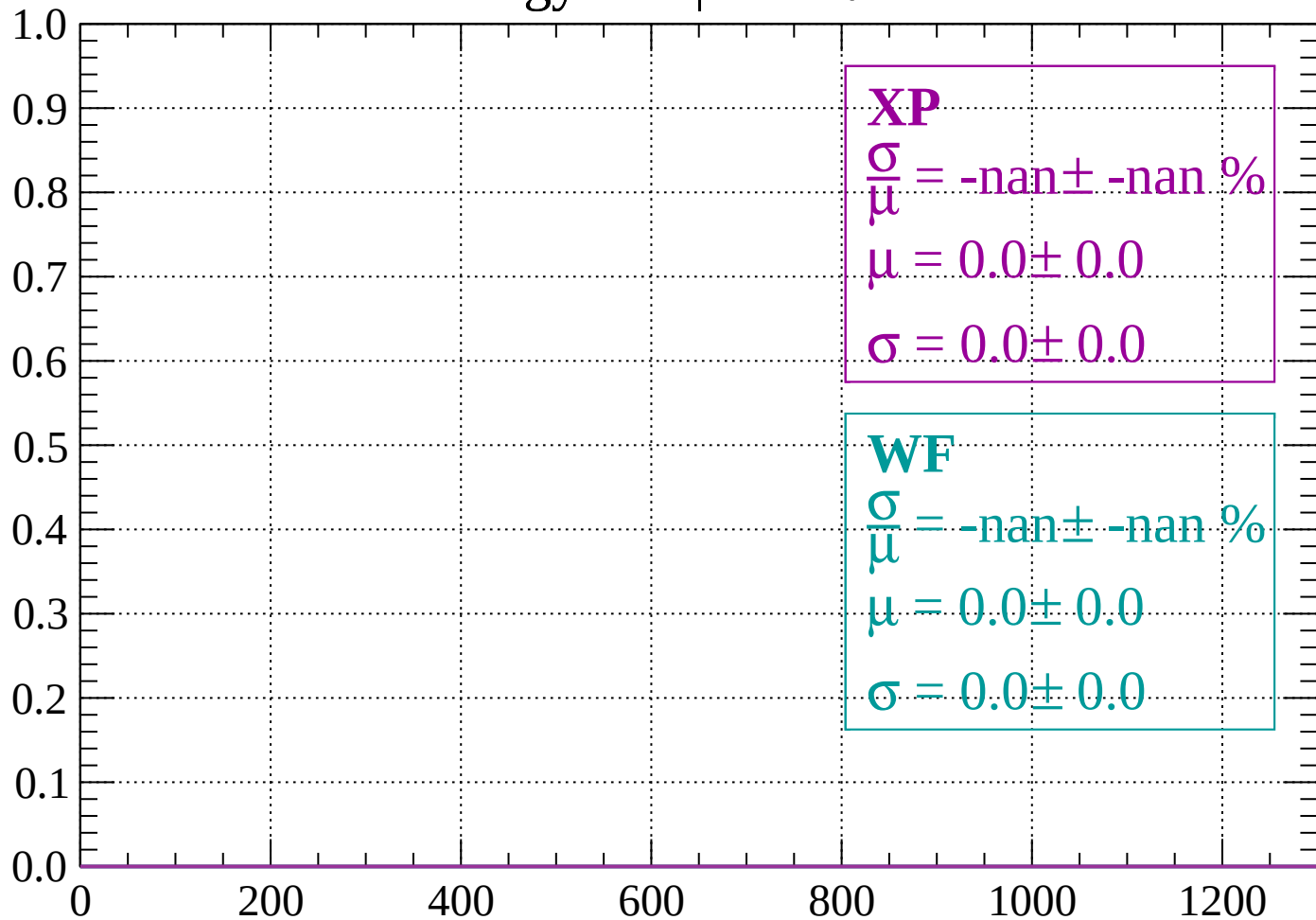
Count



dE/dx (ADC counts/cm)

Energy loss | $10 < \theta < 12$

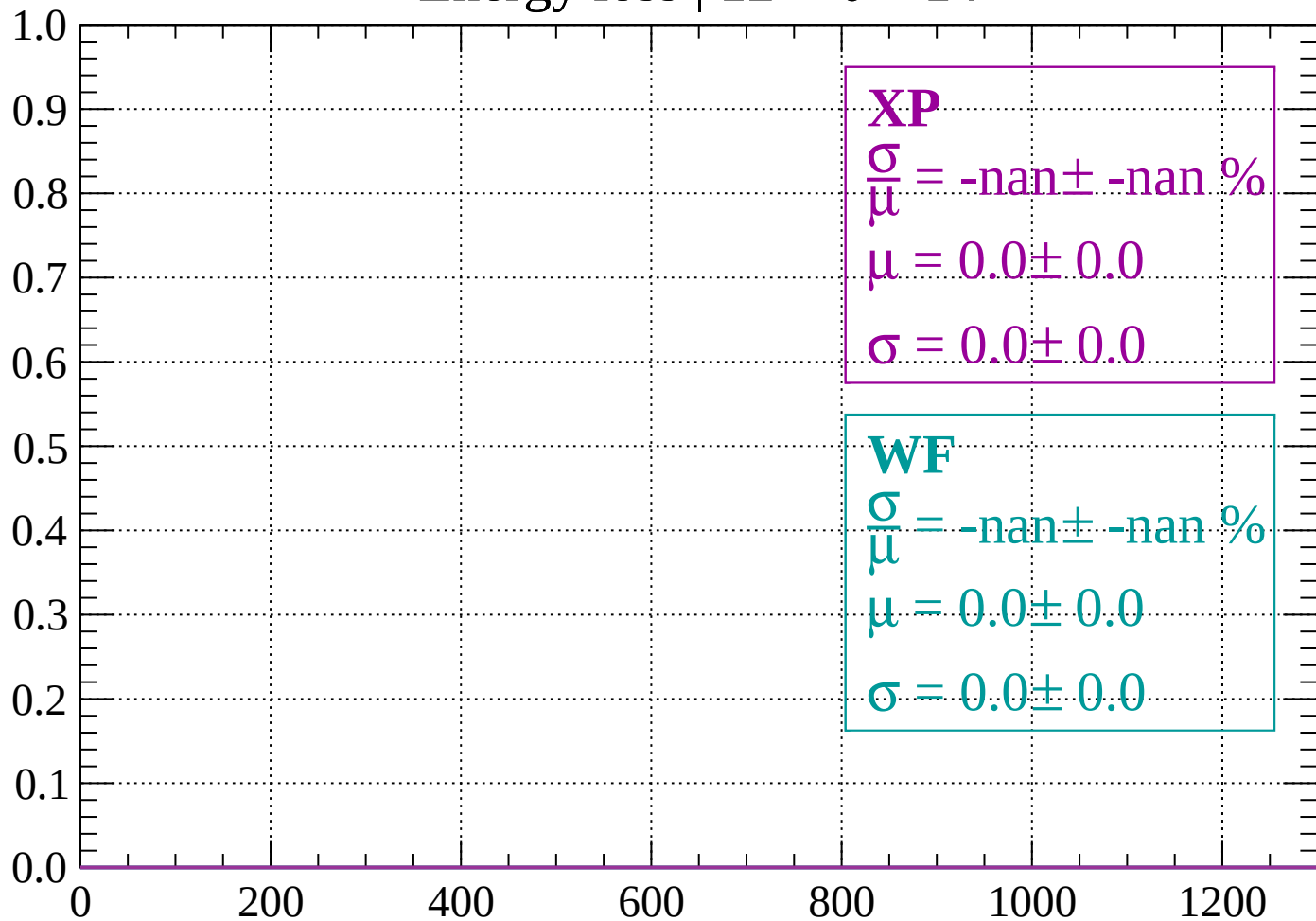
Count



dE/dx (ADC counts/cm)

Energy loss | $12 < \theta < 14$

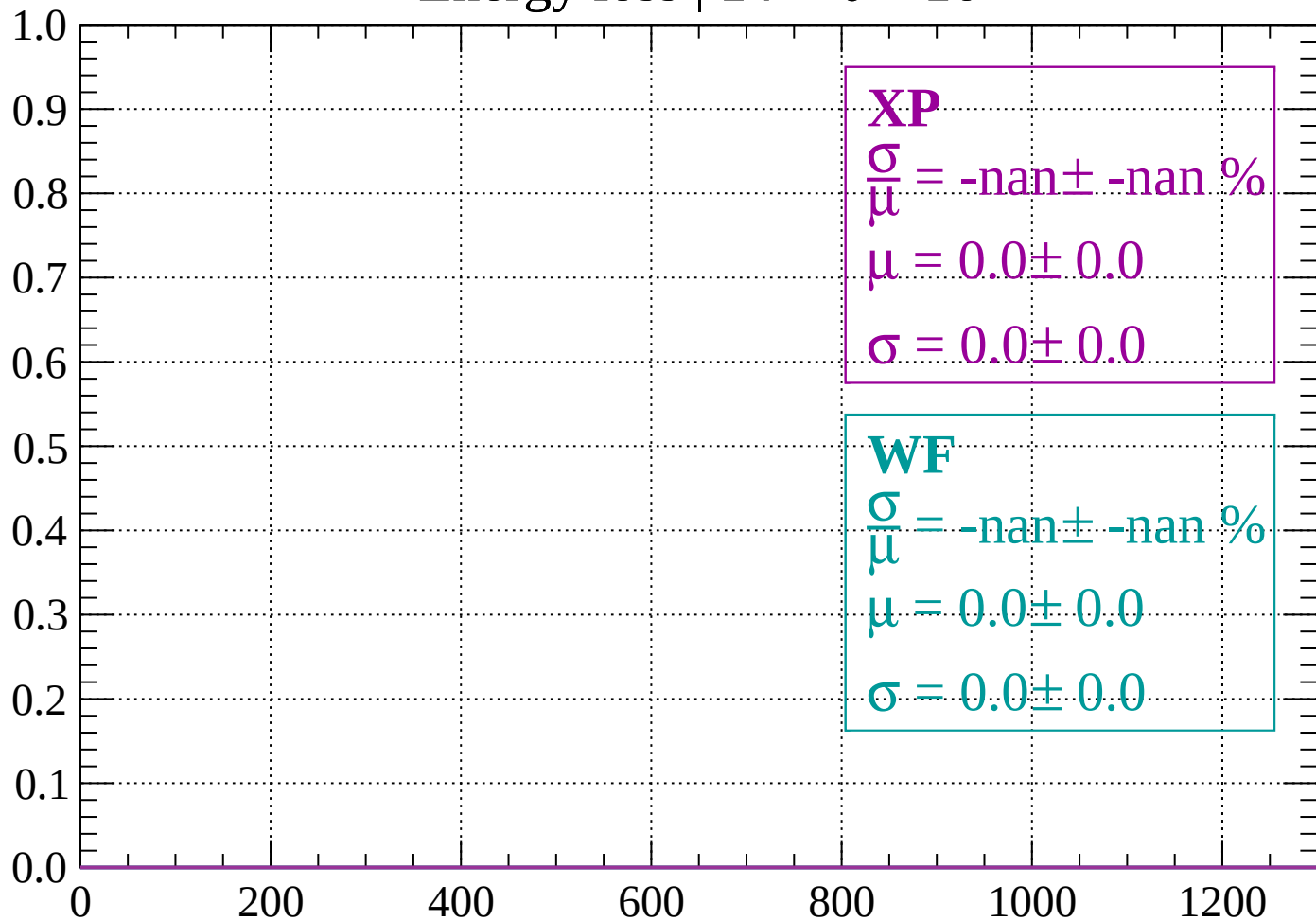
Count



dE/dx (ADC counts/cm)

Energy loss | $14 < \theta < 16$

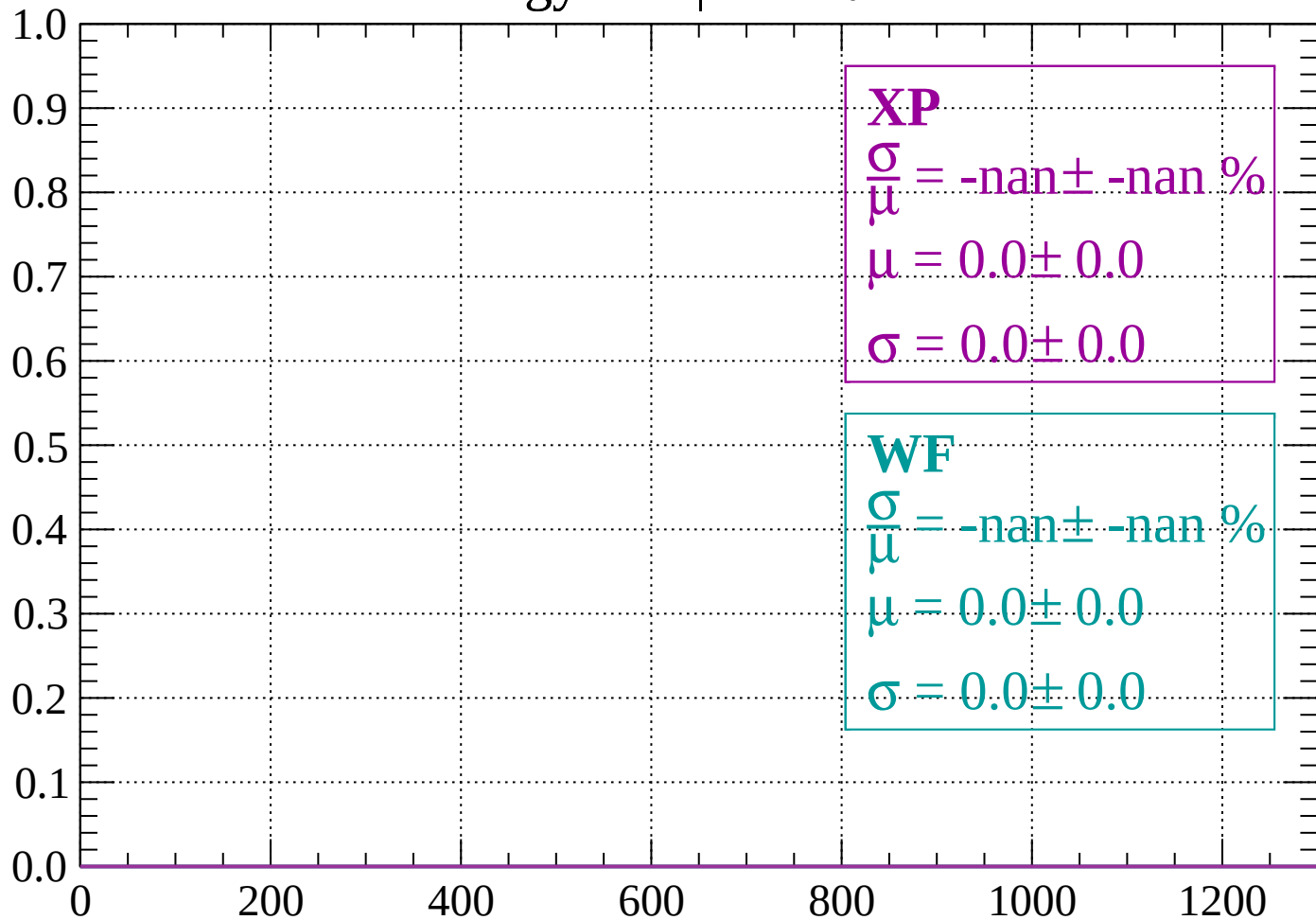
Count



dE/dx (ADC counts/cm)

Energy loss | $16 < \theta < 18$

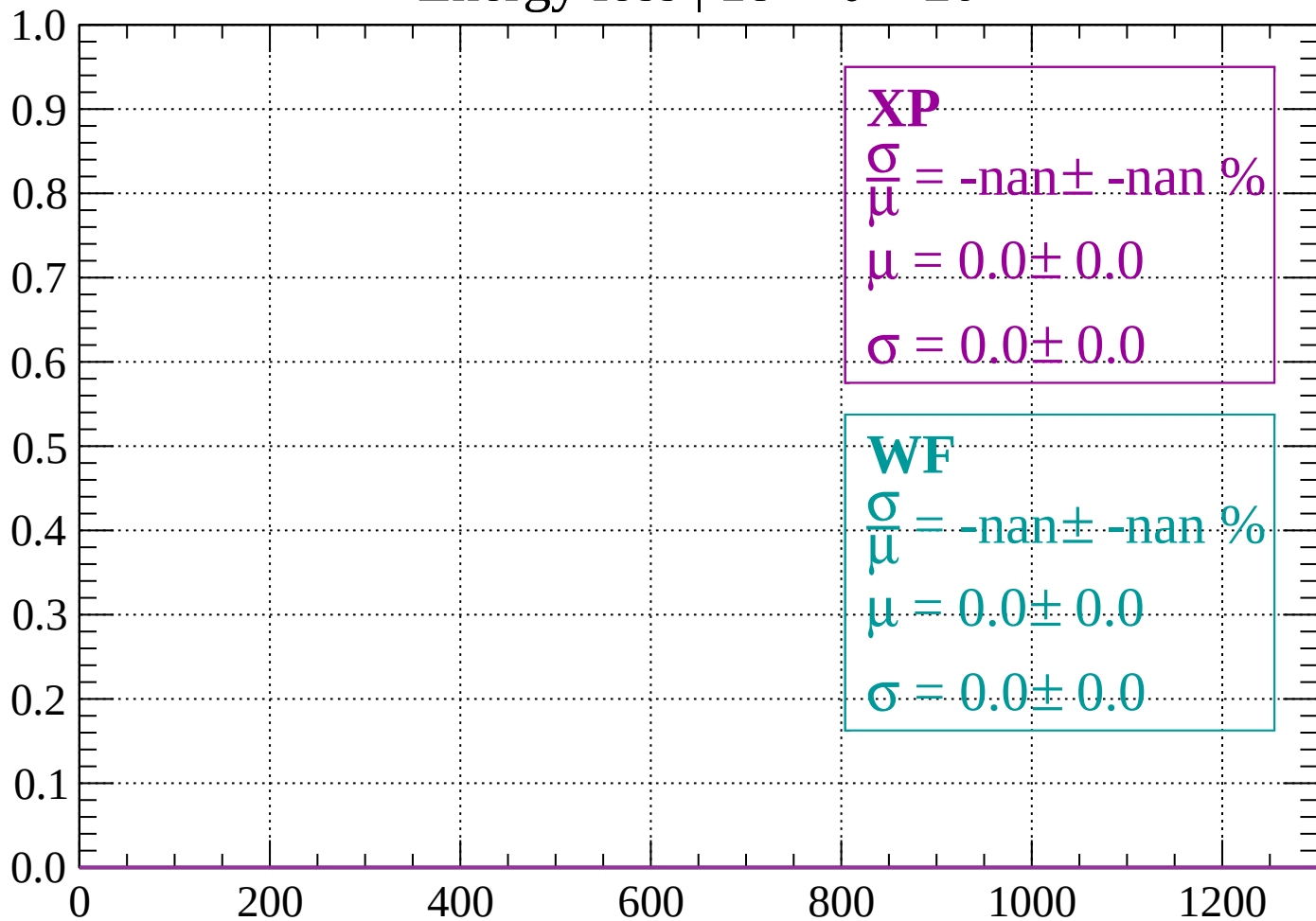
Count



dE/dx (ADC counts/cm)

Energy loss | $18 < \theta < 20$

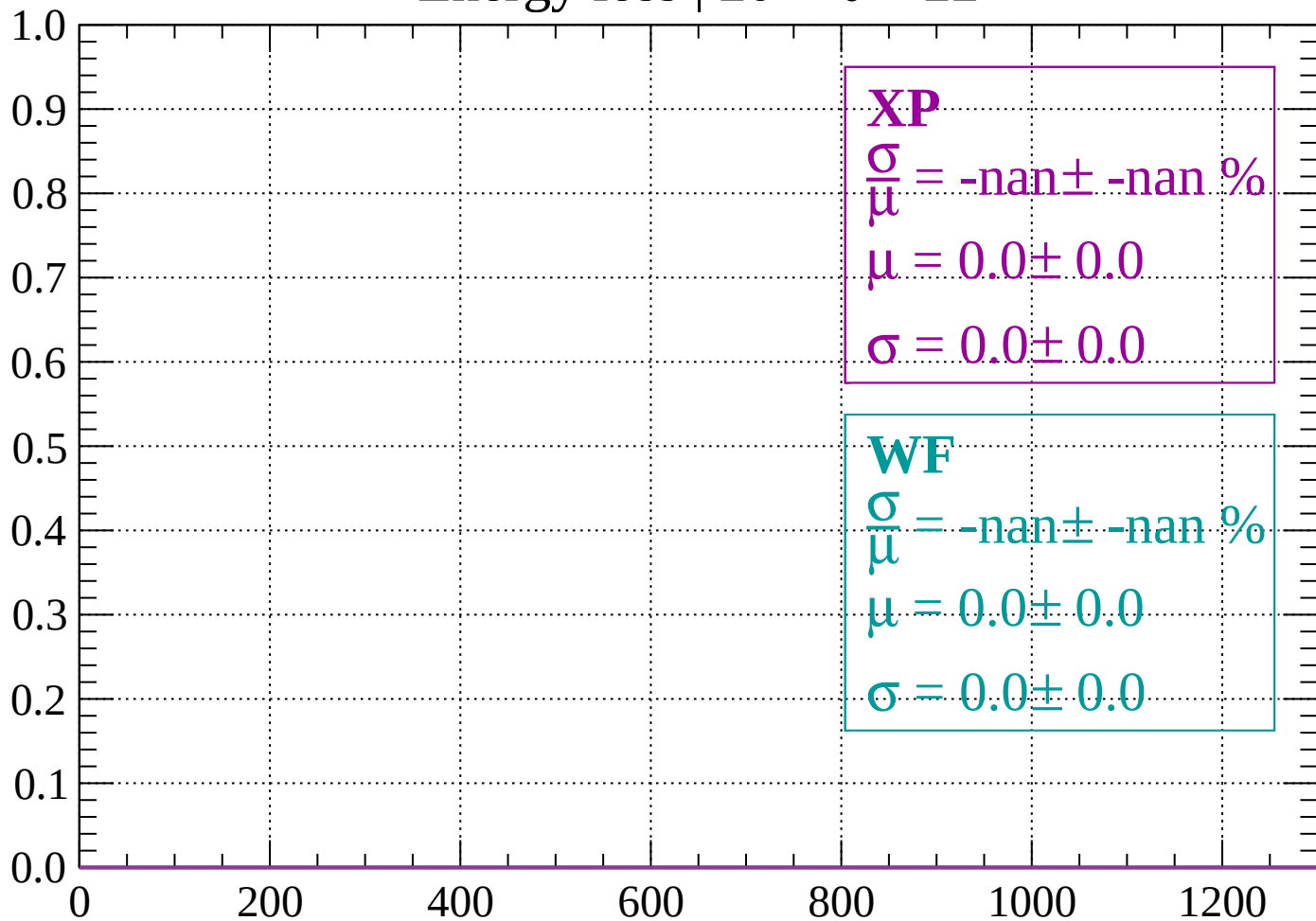
Count



dE/dx (ADC counts/cm)

Energy loss | $20 < \theta < 22$

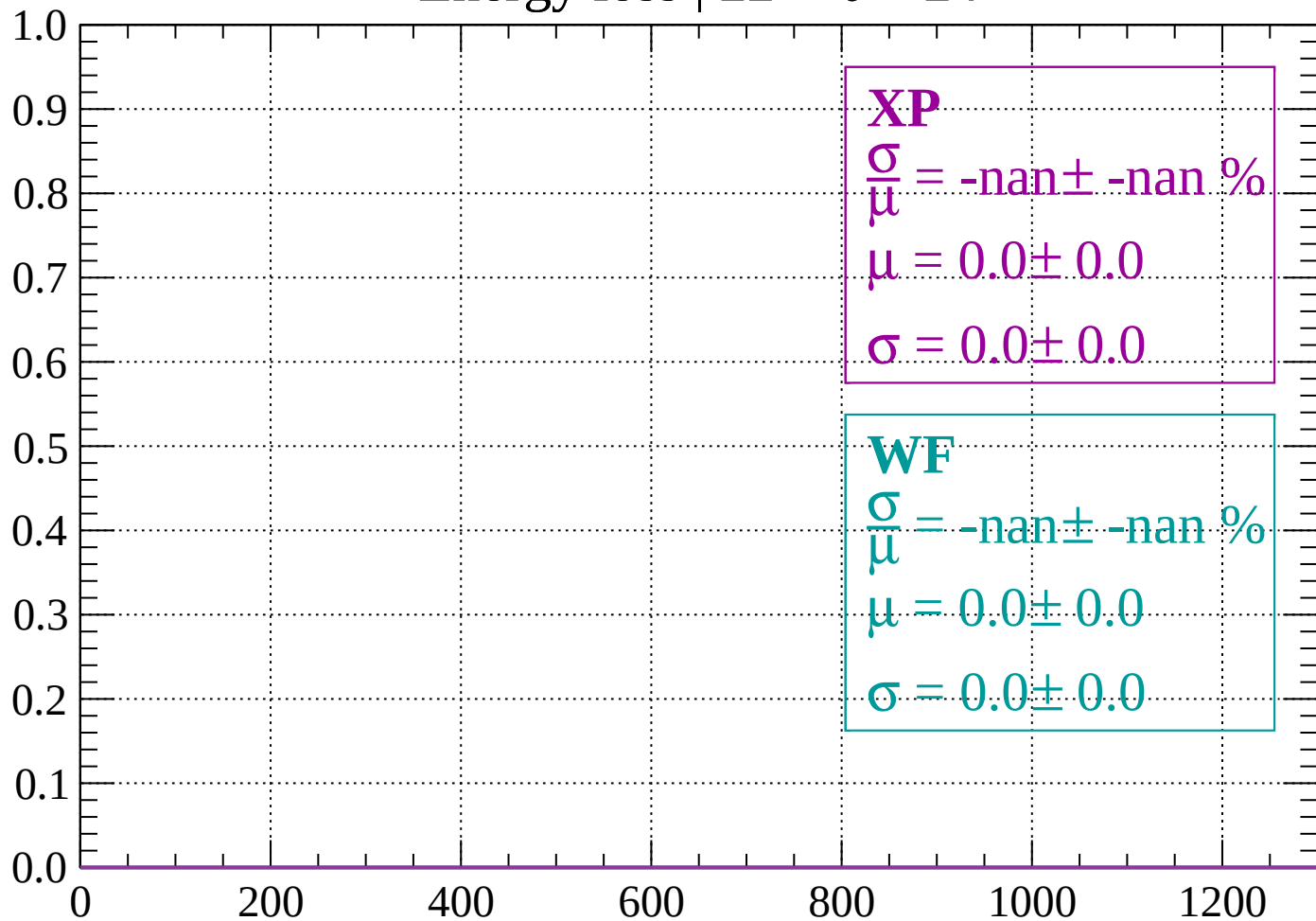
Count



dE/dx (ADC counts/cm)

Energy loss | $22 < \theta < 24$

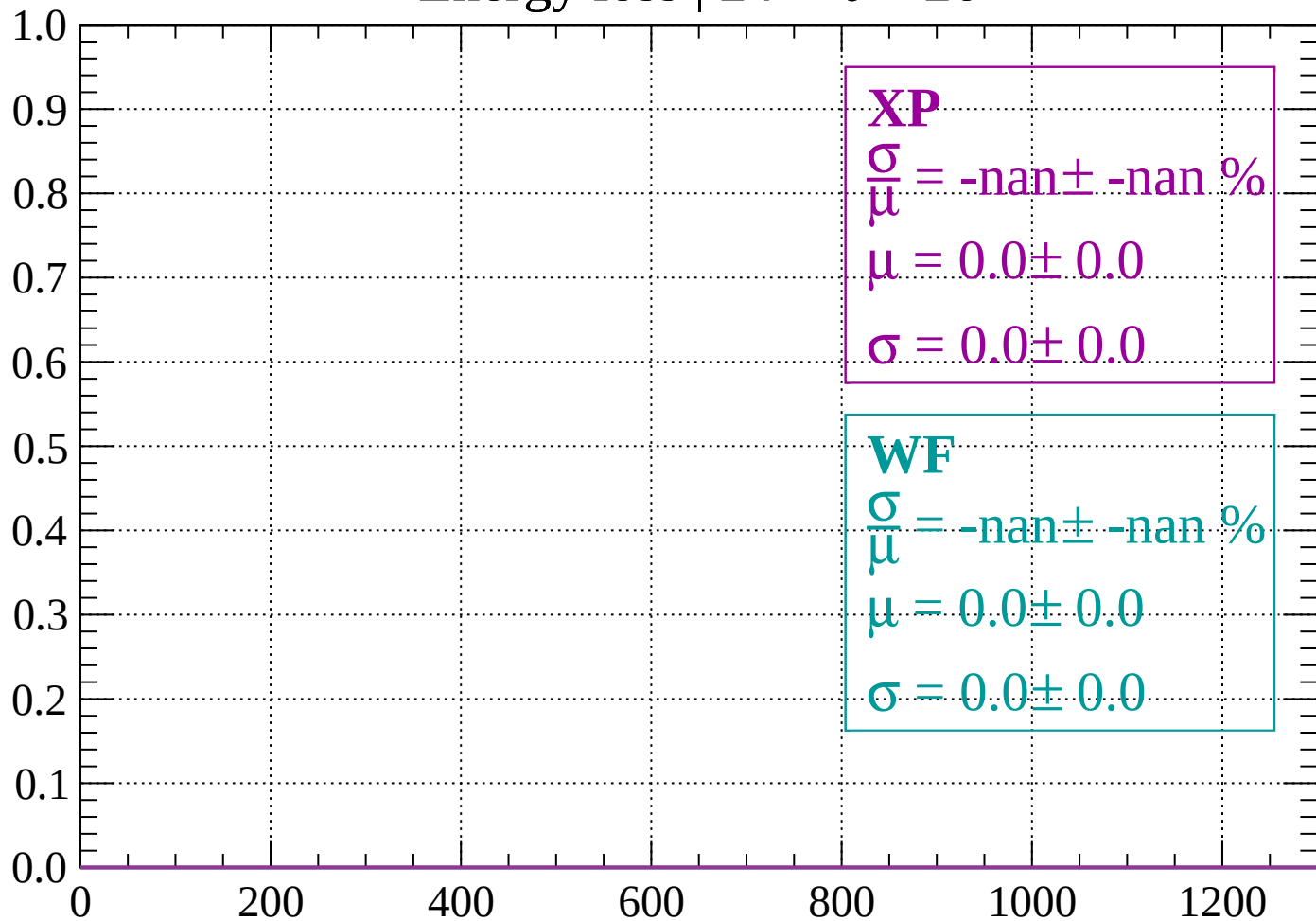
Count



dE/dx (ADC counts/cm)

Energy loss | $24 < \theta < 26$

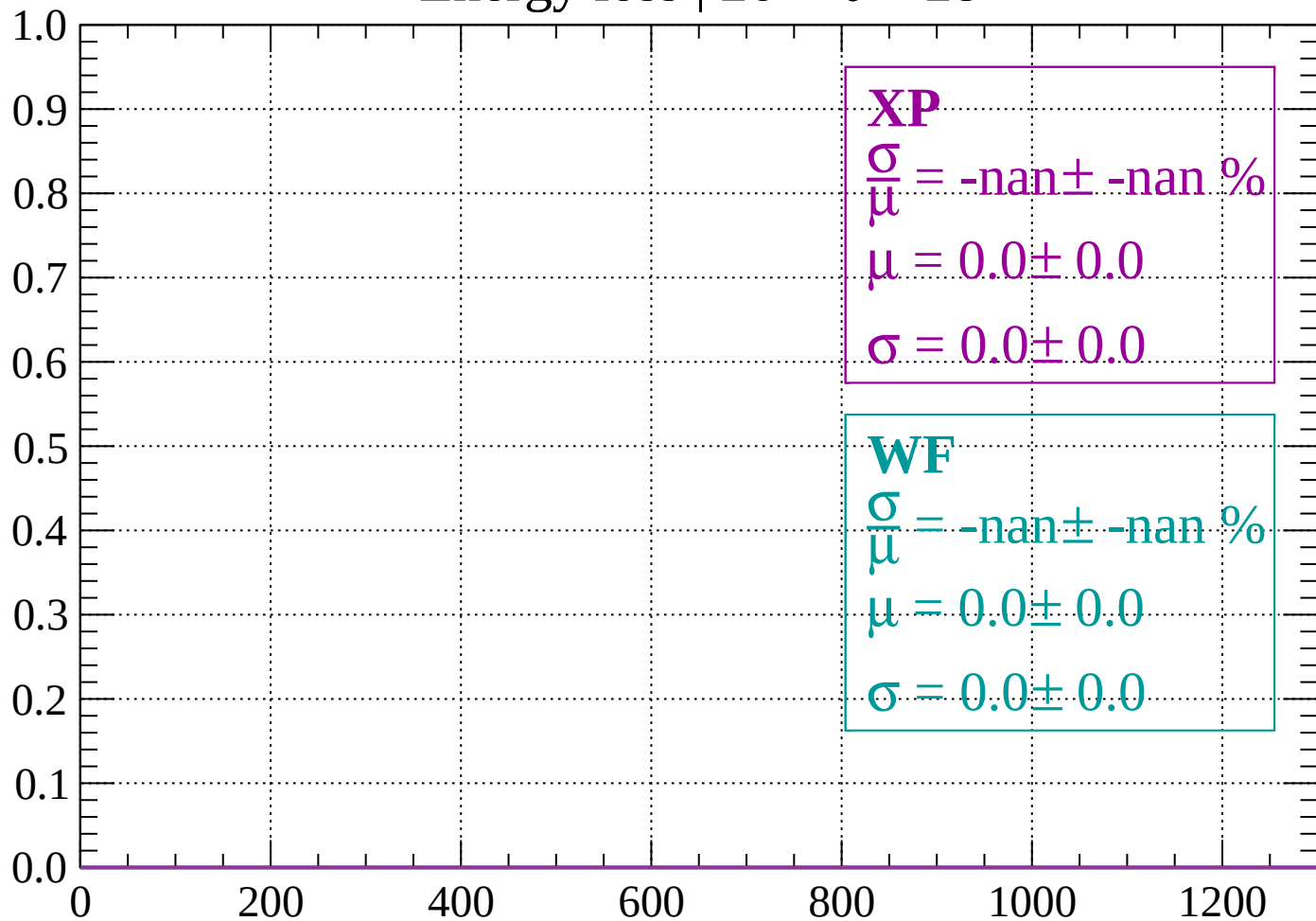
Count



dE/dx (ADC counts/cm)

Energy loss | $26 < \theta < 28$

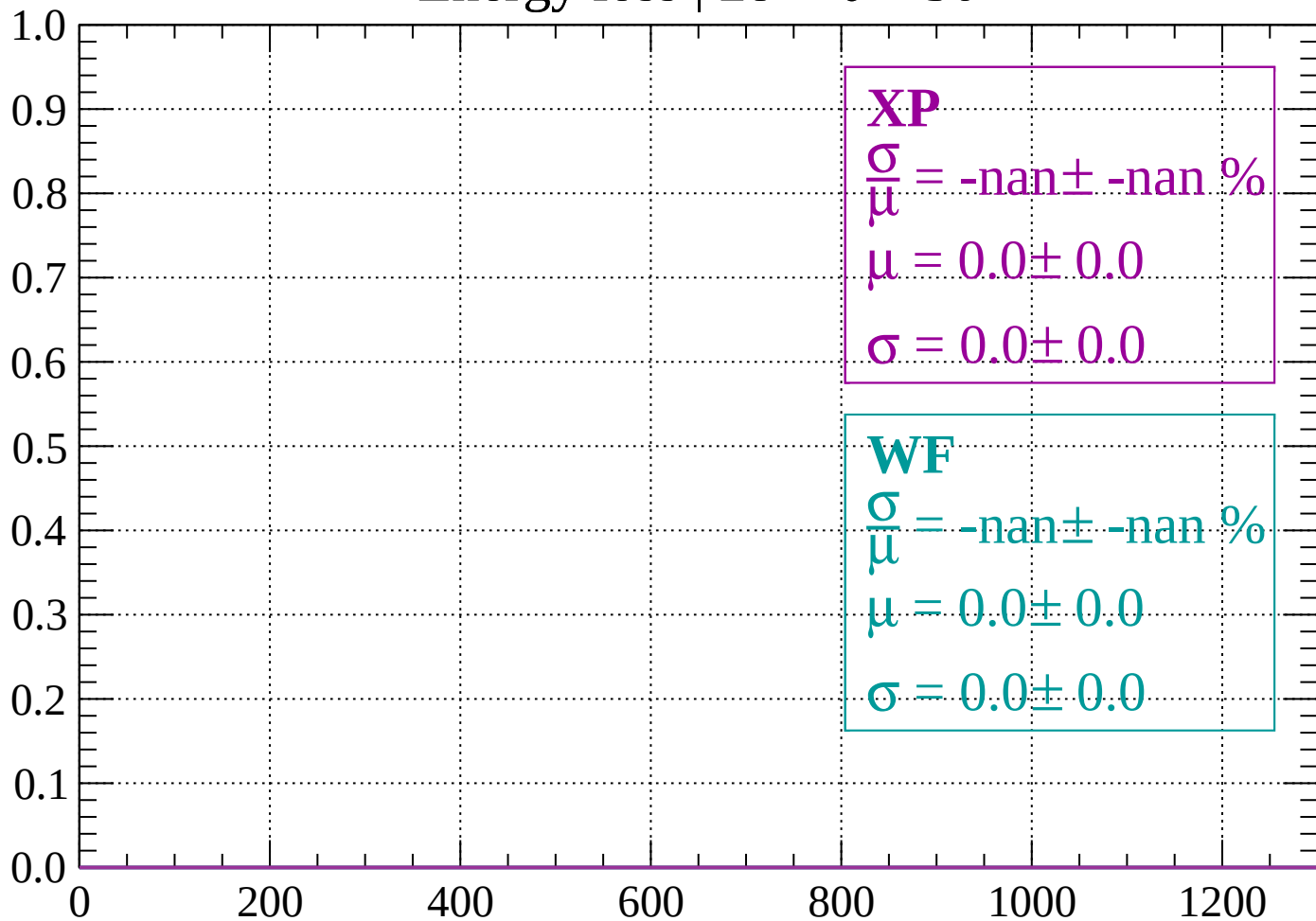
Count



dE/dx (ADC counts/cm)

Energy loss | $28 < \theta < 30$

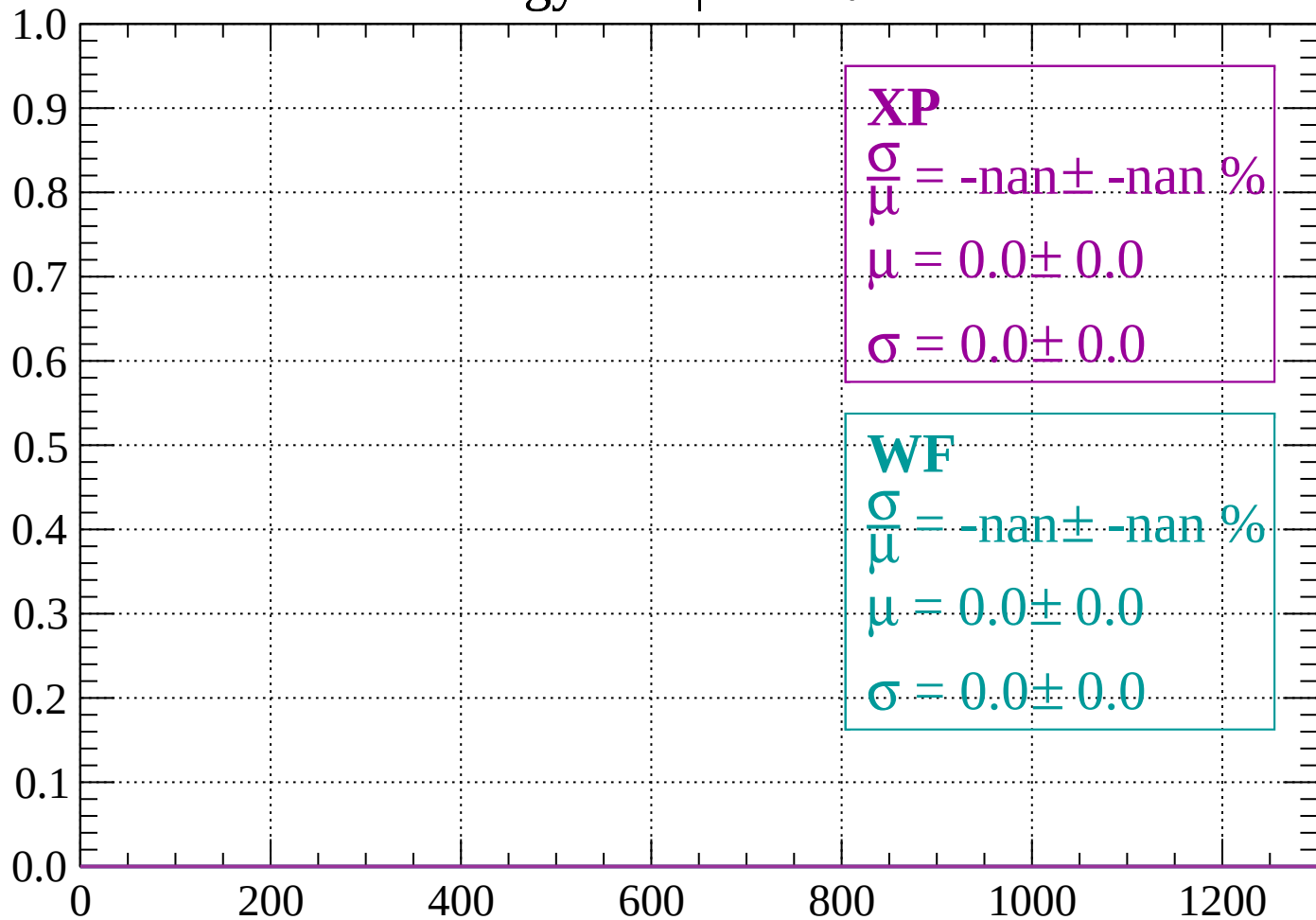
Count



dE/dx (ADC counts/cm)

Energy loss | $30 < \theta < 32$

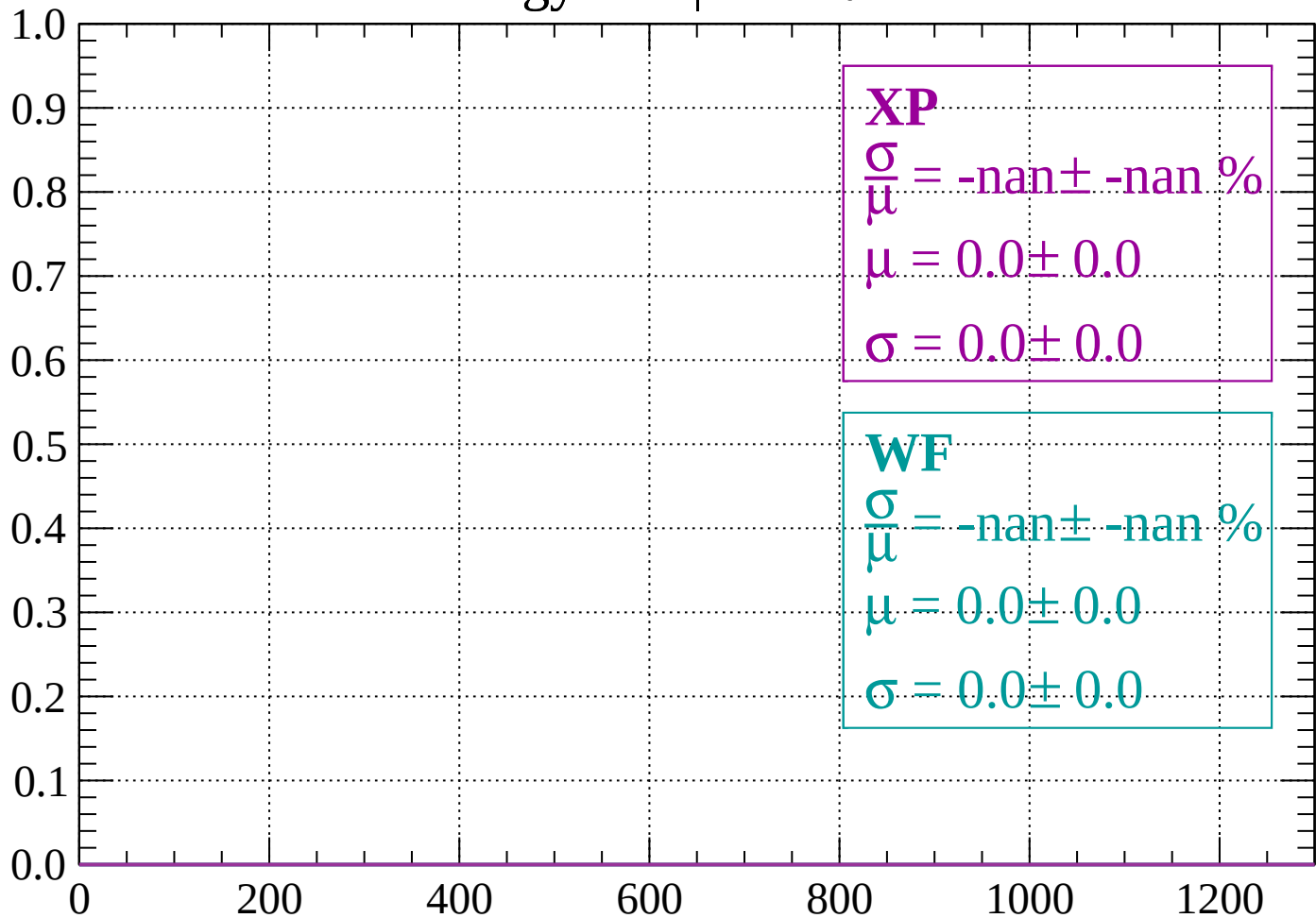
Count



dE/dx (ADC counts/cm)

Energy loss | $32 < \theta < 34$

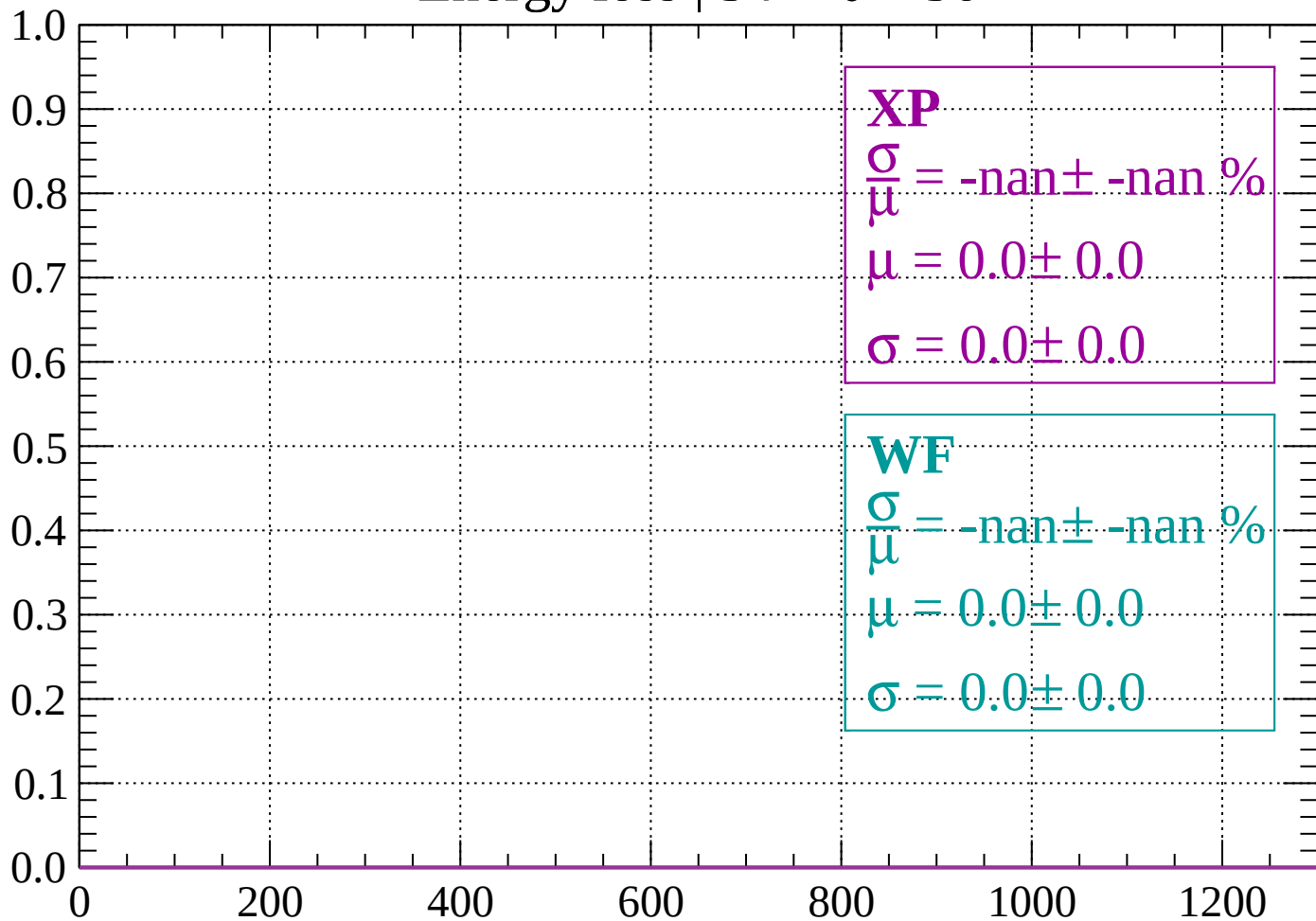
Count



dE/dx (ADC counts/cm)

Energy loss | $34 < \theta < 36$

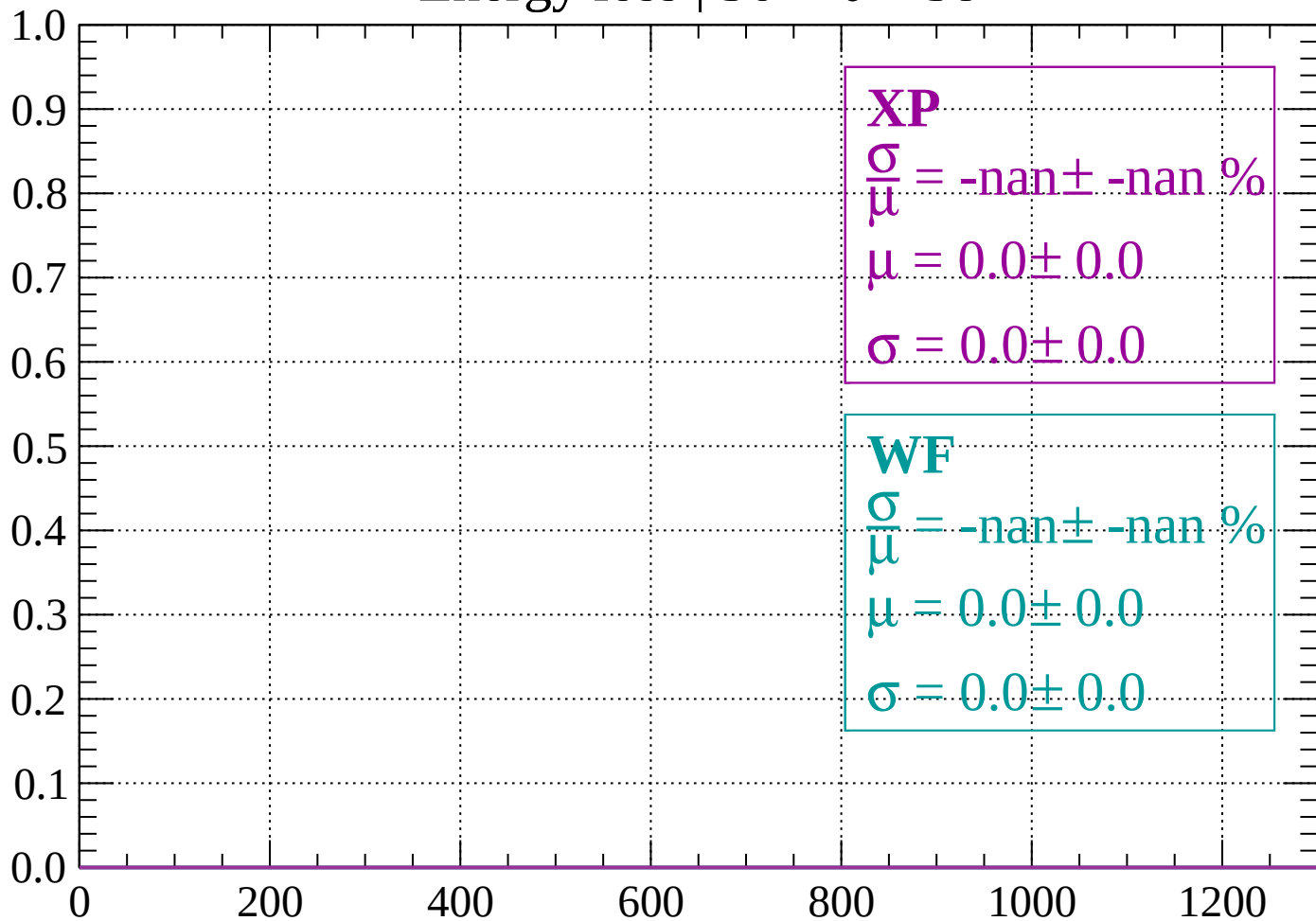
Count



dE/dx (ADC counts/cm)

Energy loss | $36 < \theta < 38$

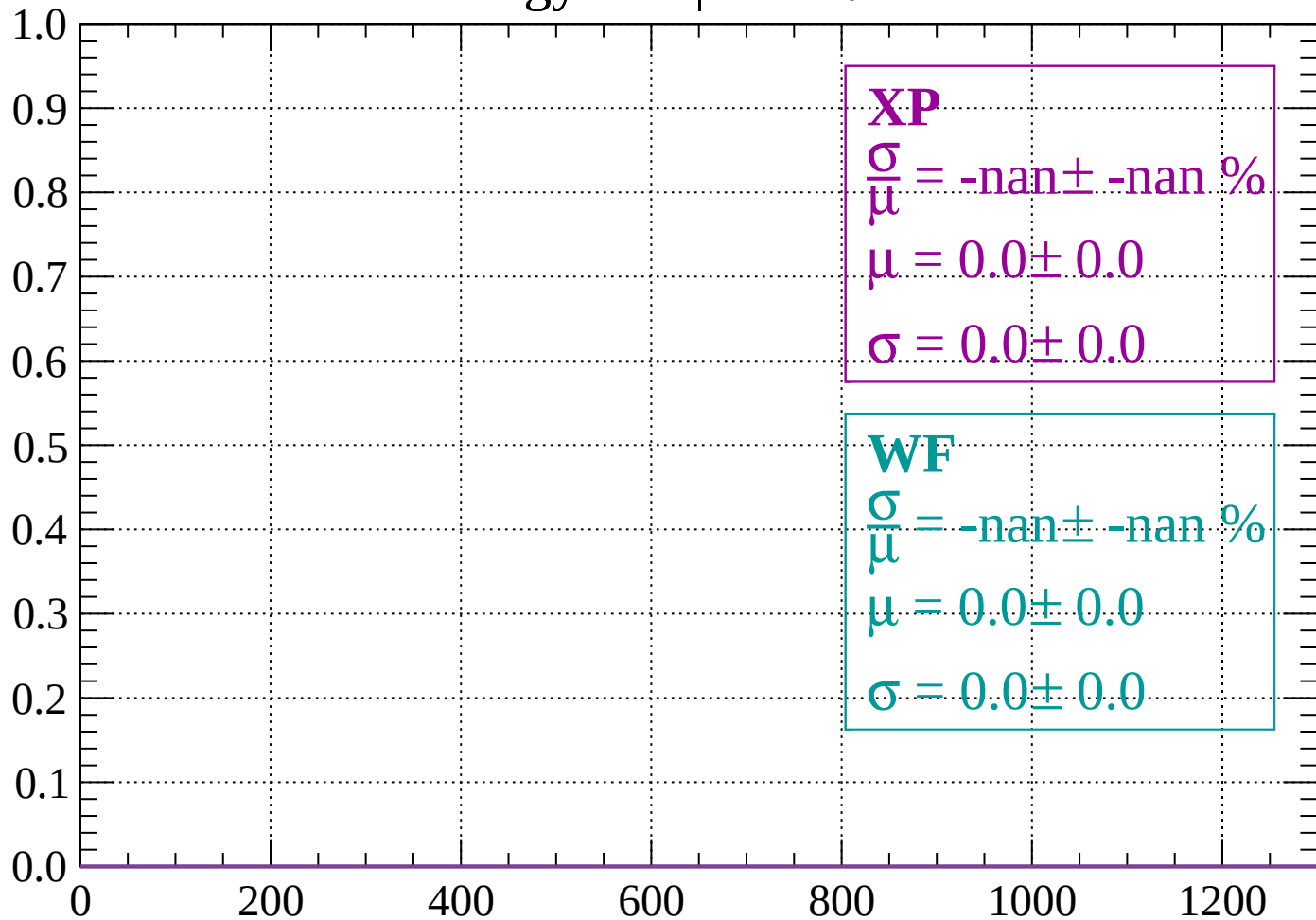
Count



dE/dx (ADC counts/cm)

Energy loss | $38 < \theta < 40$

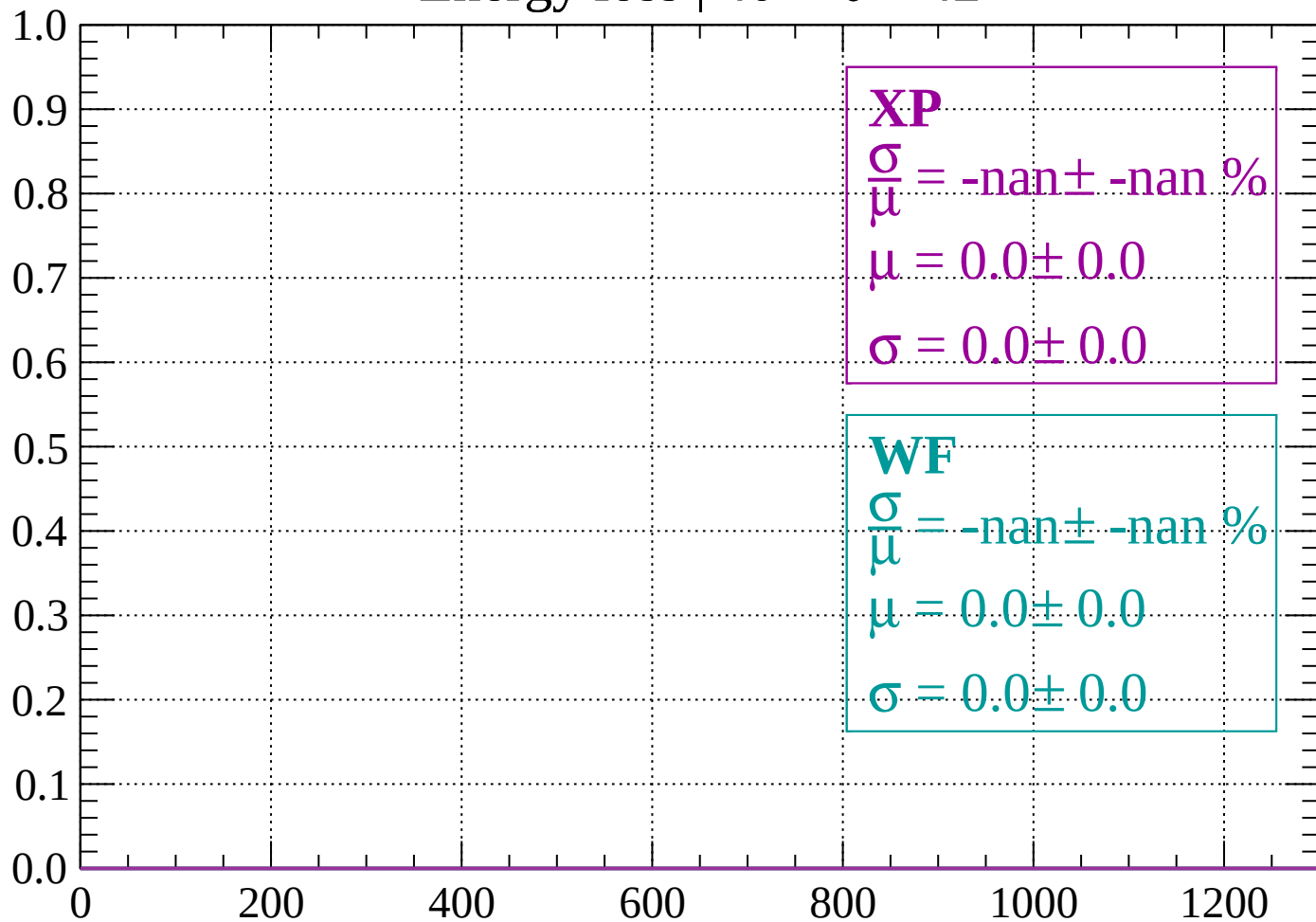
Count



dE/dx (ADC counts/cm)

Energy loss | $40 < \theta < 42$

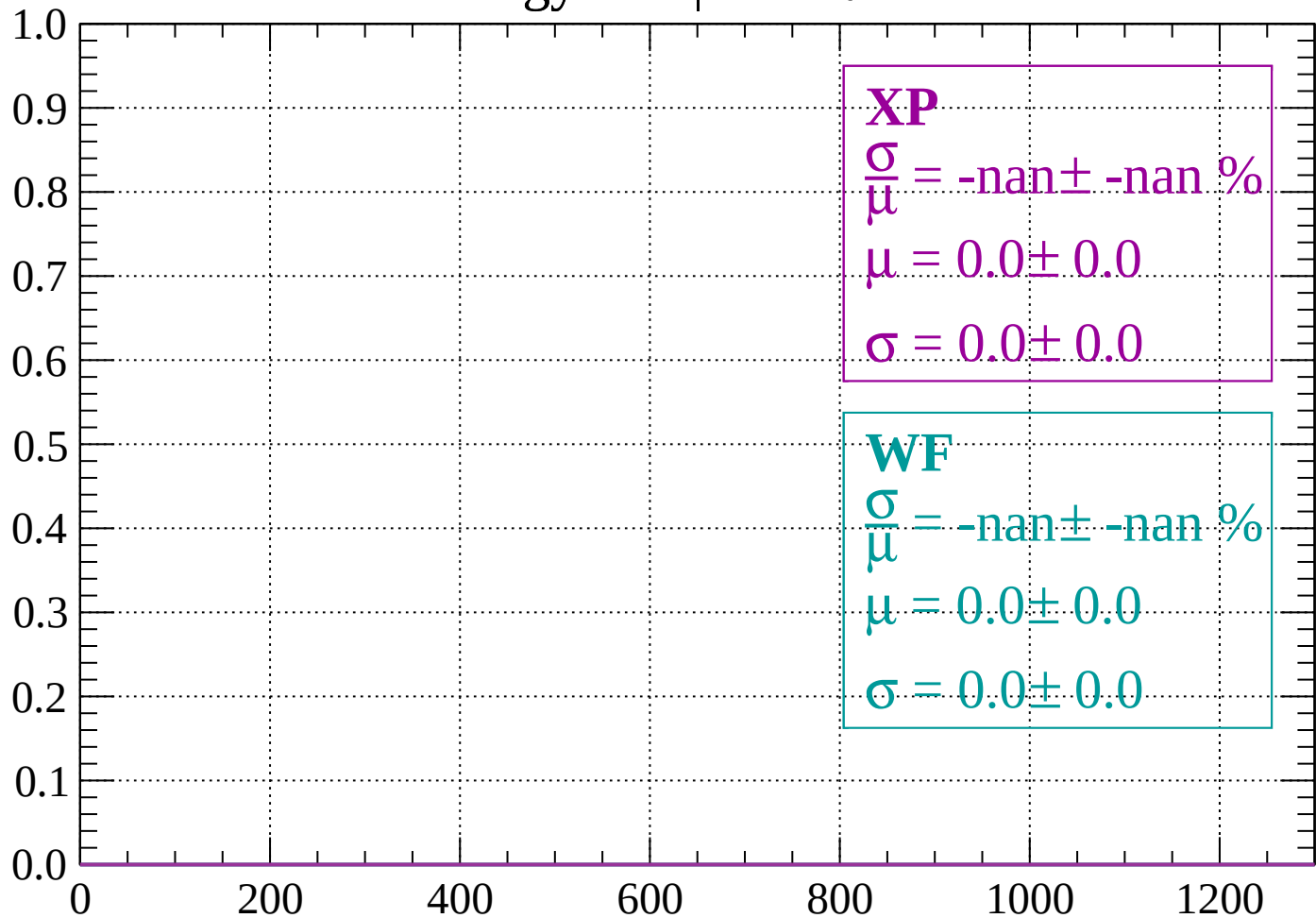
Count



dE/dx (ADC counts/cm)

Energy loss | $42 < \theta < 44$

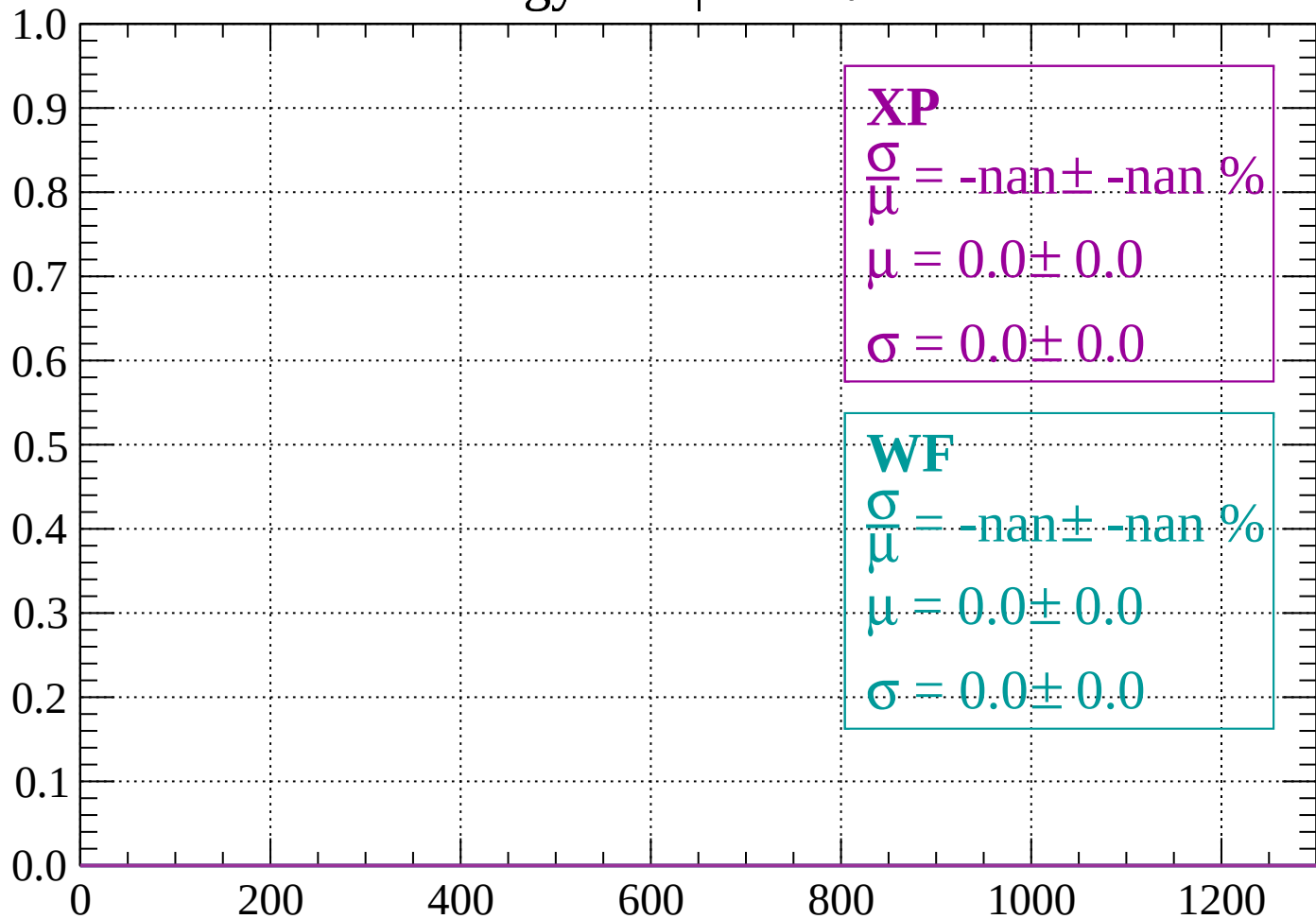
Count



dE/dx (ADC counts/cm)

Energy loss | $44 < \theta < 46$

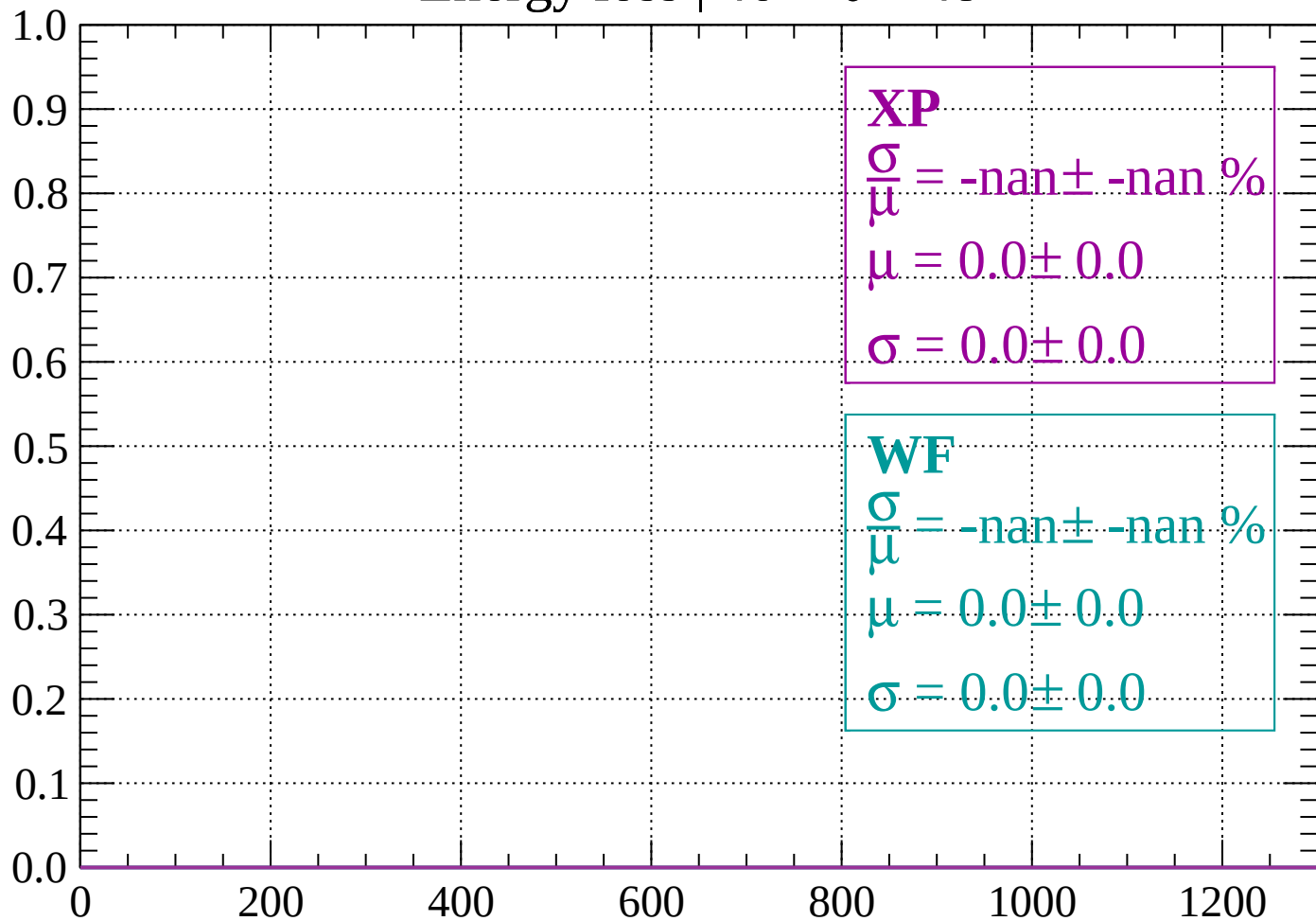
Count



dE/dx (ADC counts/cm)

Energy loss | $46 < \theta < 48$

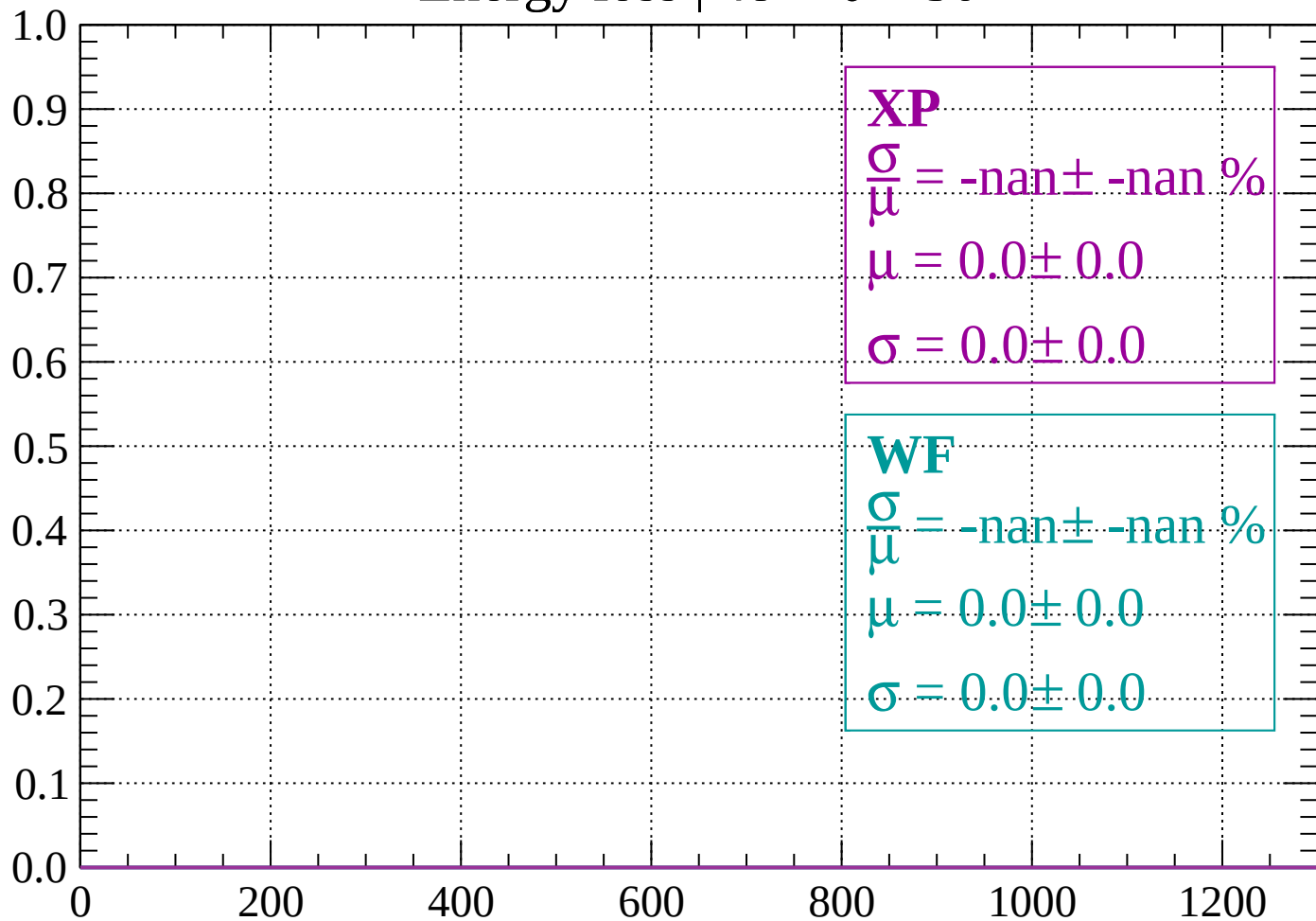
Count



dE/dx (ADC counts/cm)

Energy loss | $48 < \theta < 50$

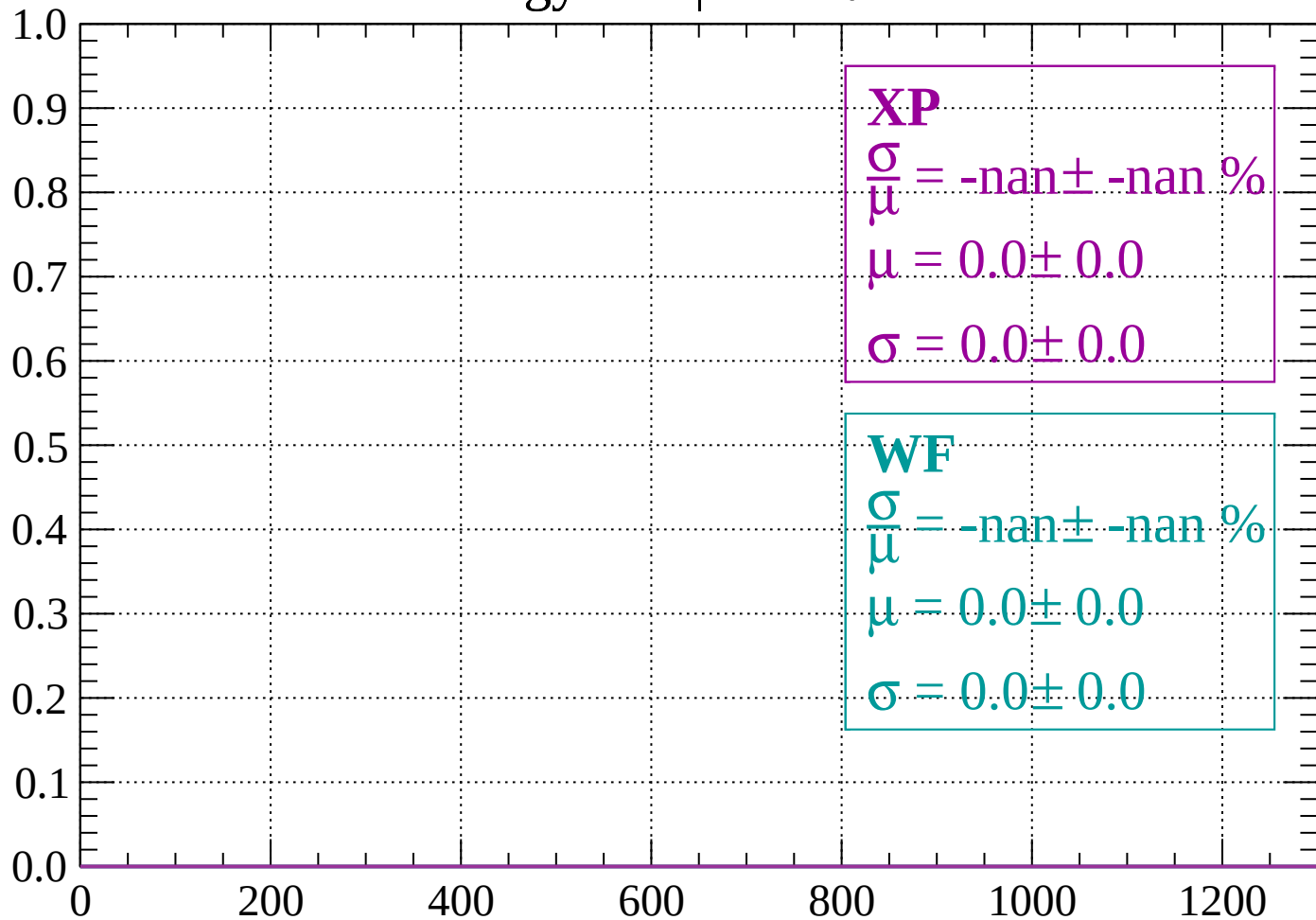
Count



dE/dx (ADC counts/cm)

Energy loss | $50 < \theta < 52$

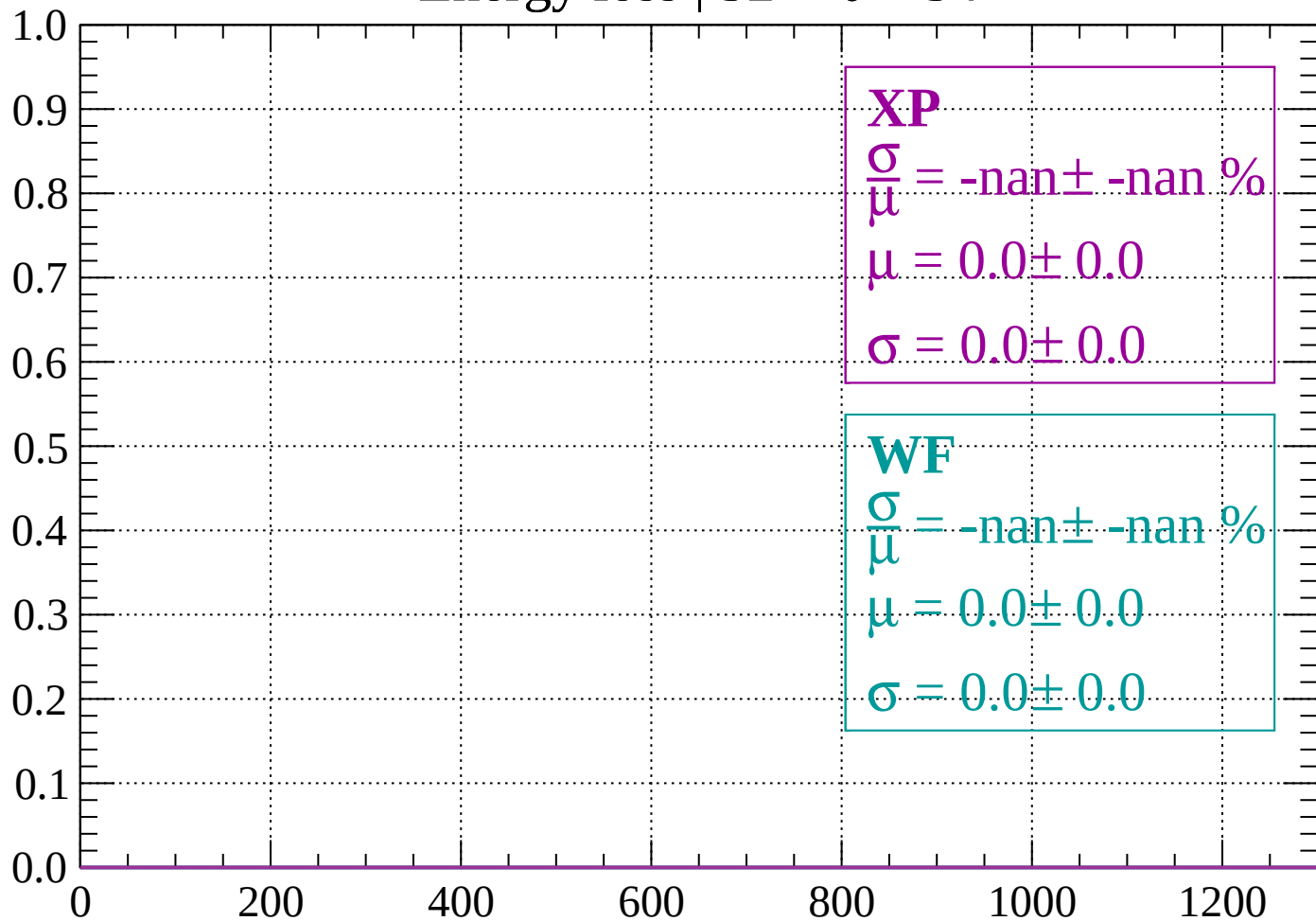
Count



dE/dx (ADC counts/cm)

Energy loss | $52 < \theta < 54$

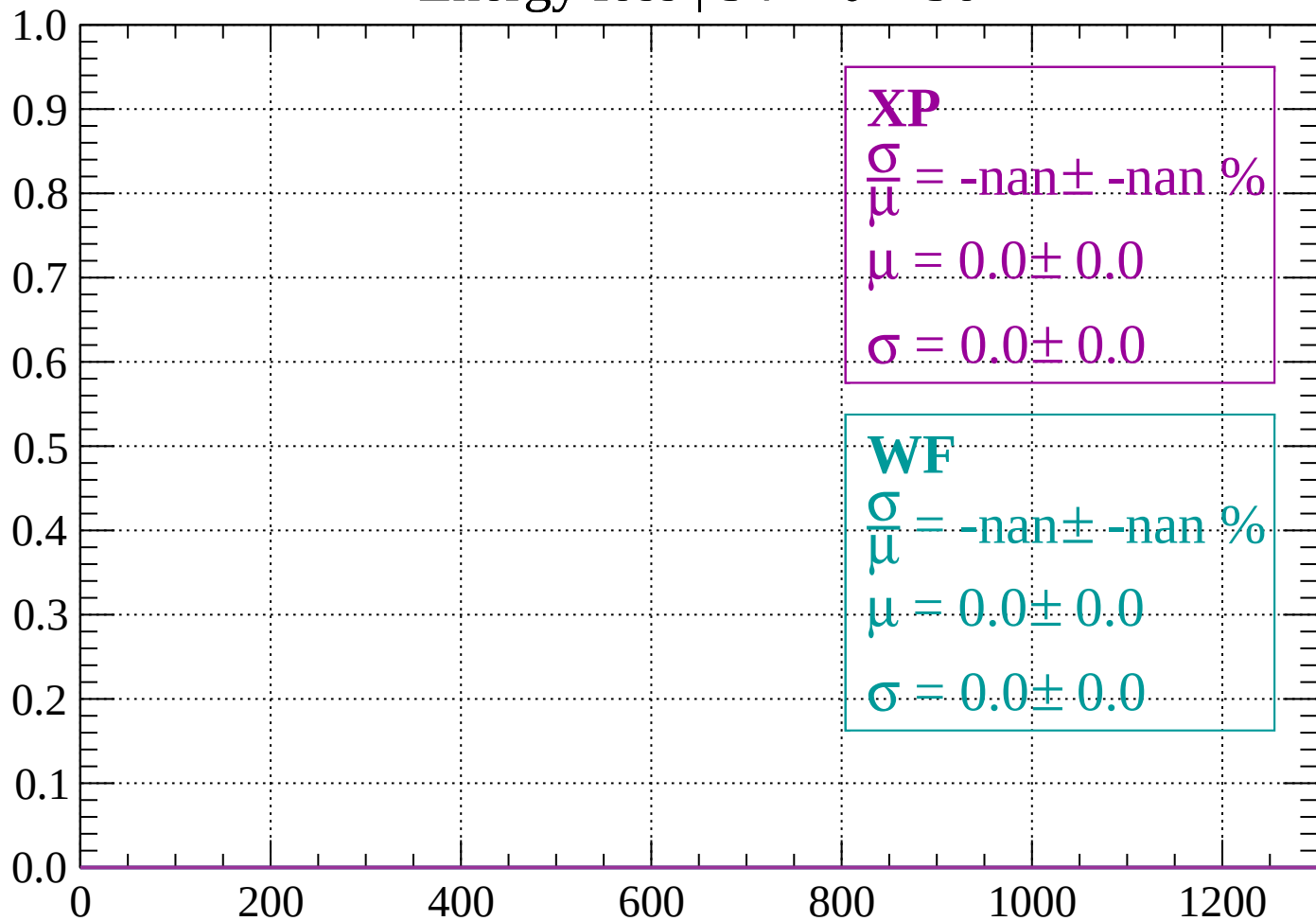
Count



dE/dx (ADC counts/cm)

Energy loss | $54 < \theta < 56$

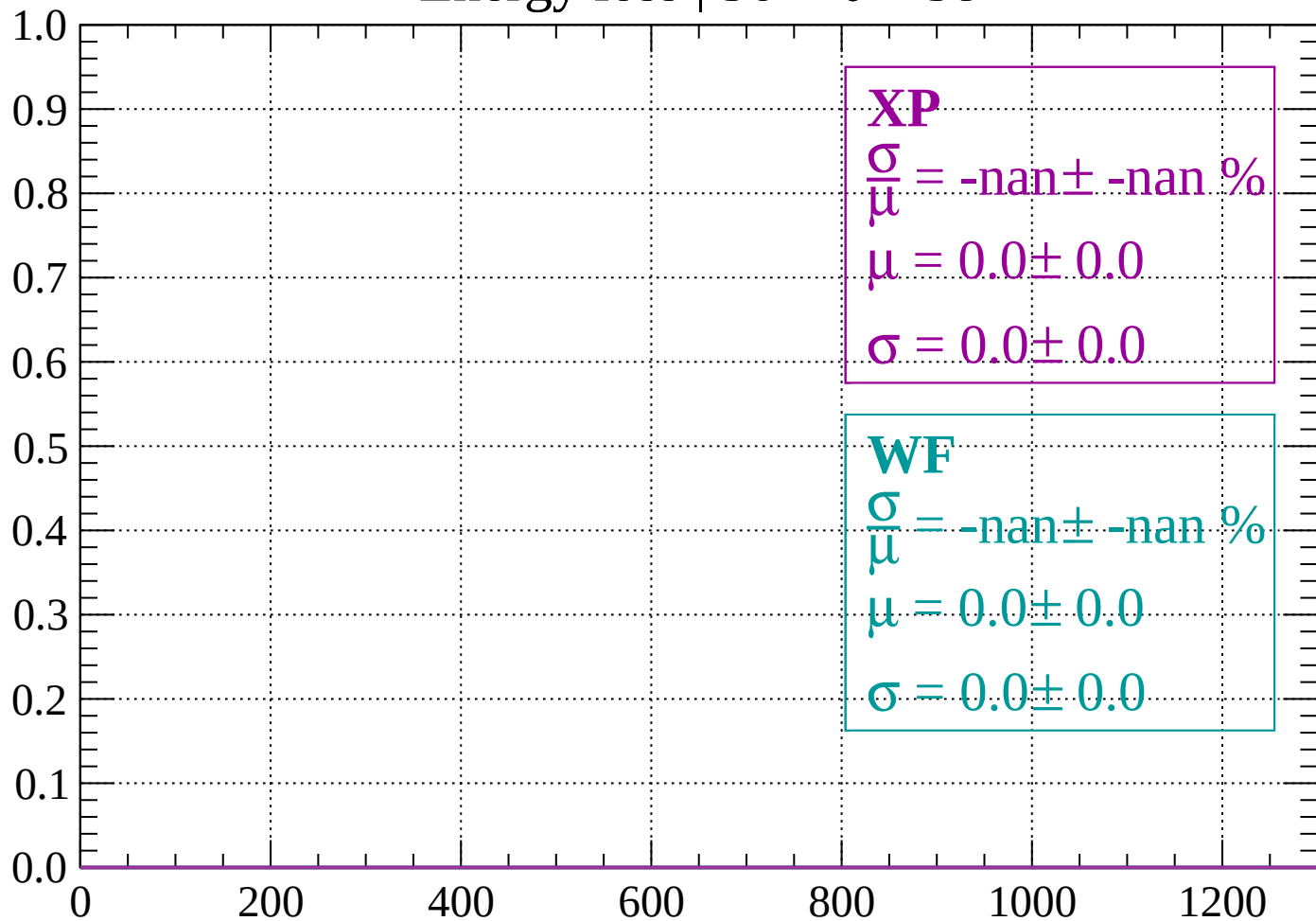
Count



dE/dx (ADC counts/cm)

Energy loss | $56 < \theta < 58$

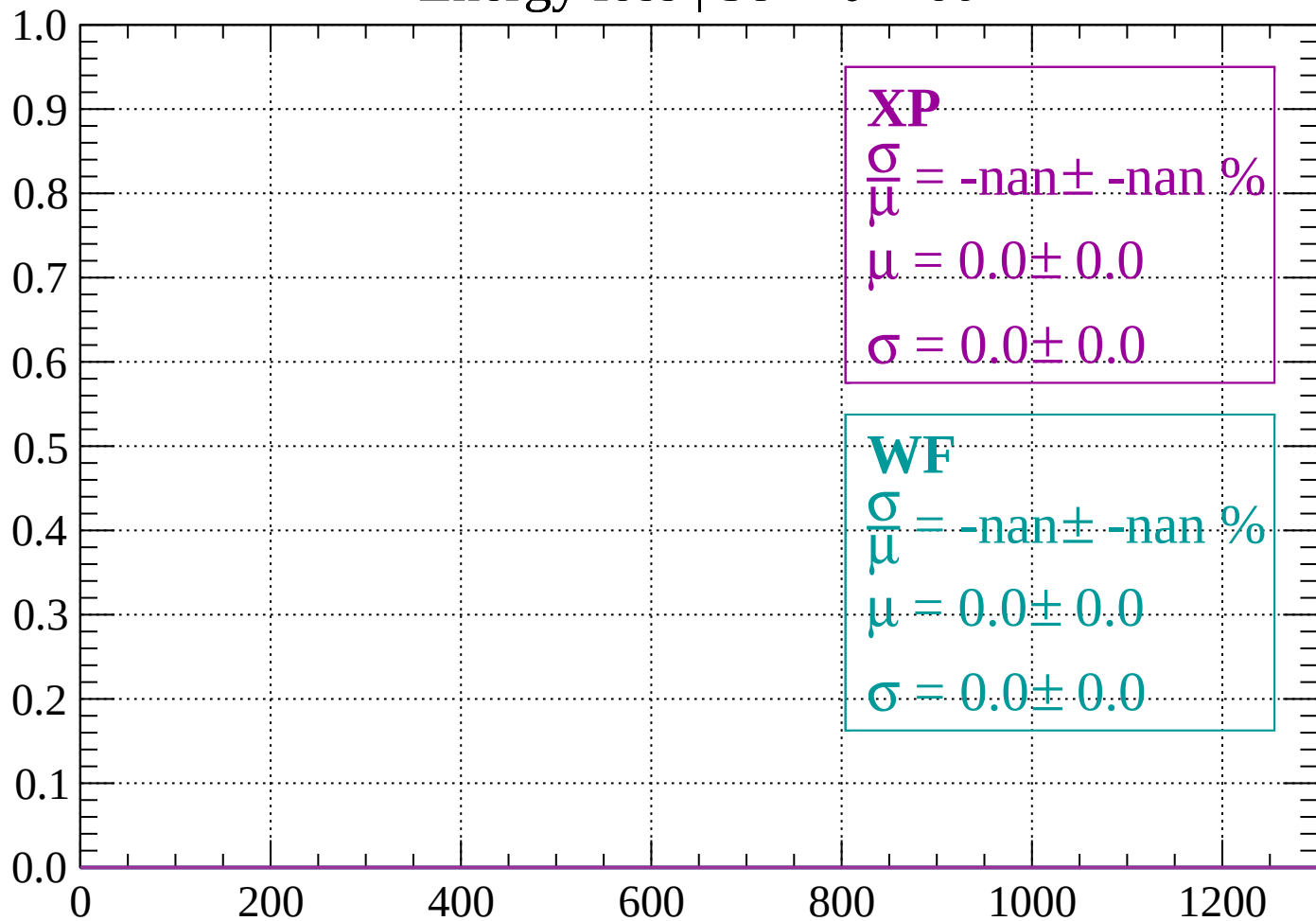
Count



dE/dx (ADC counts/cm)

Energy loss | $58 < \theta < 60$

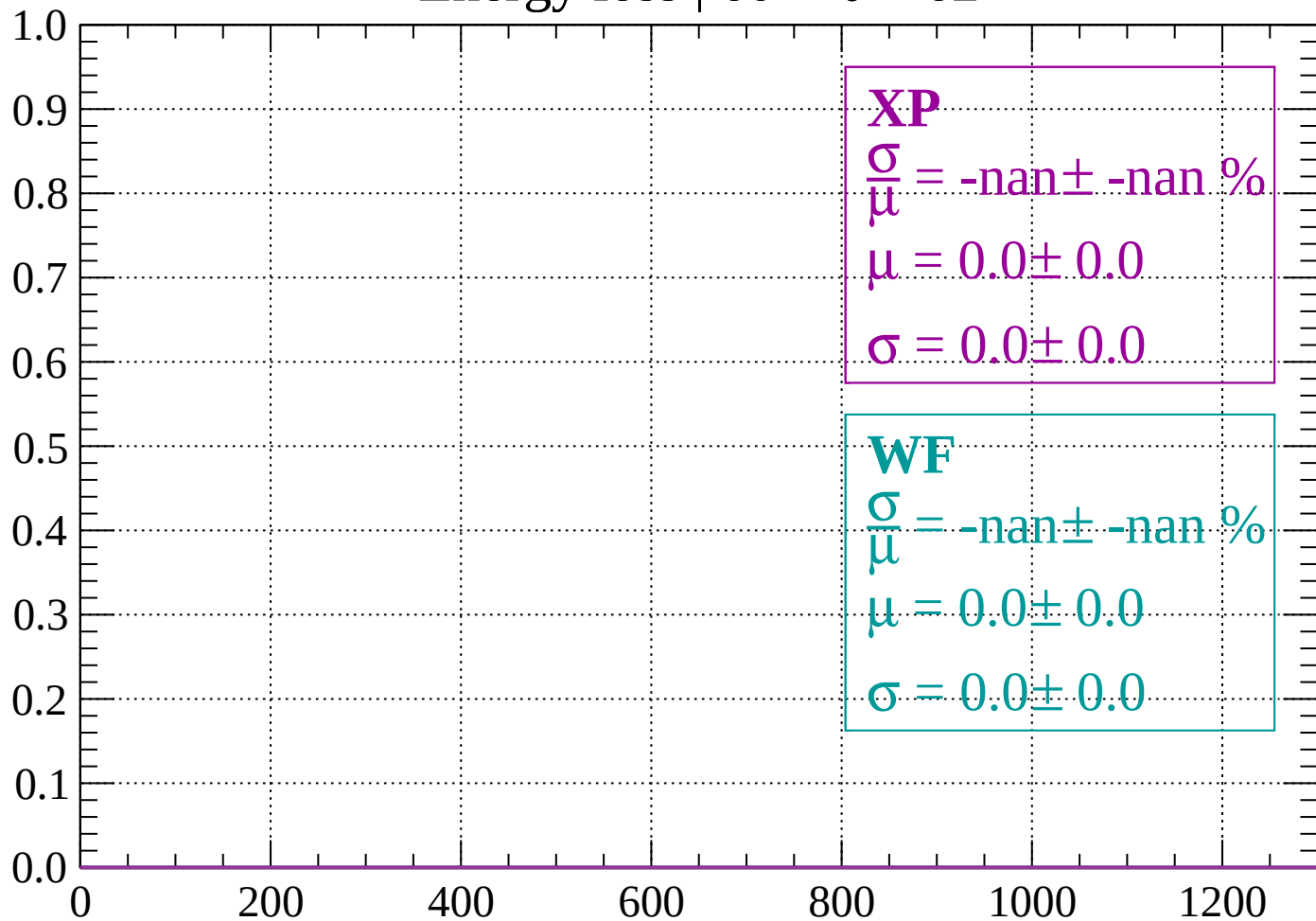
Count



dE/dx (ADC counts/cm)

Energy loss | $60 < \theta < 62$

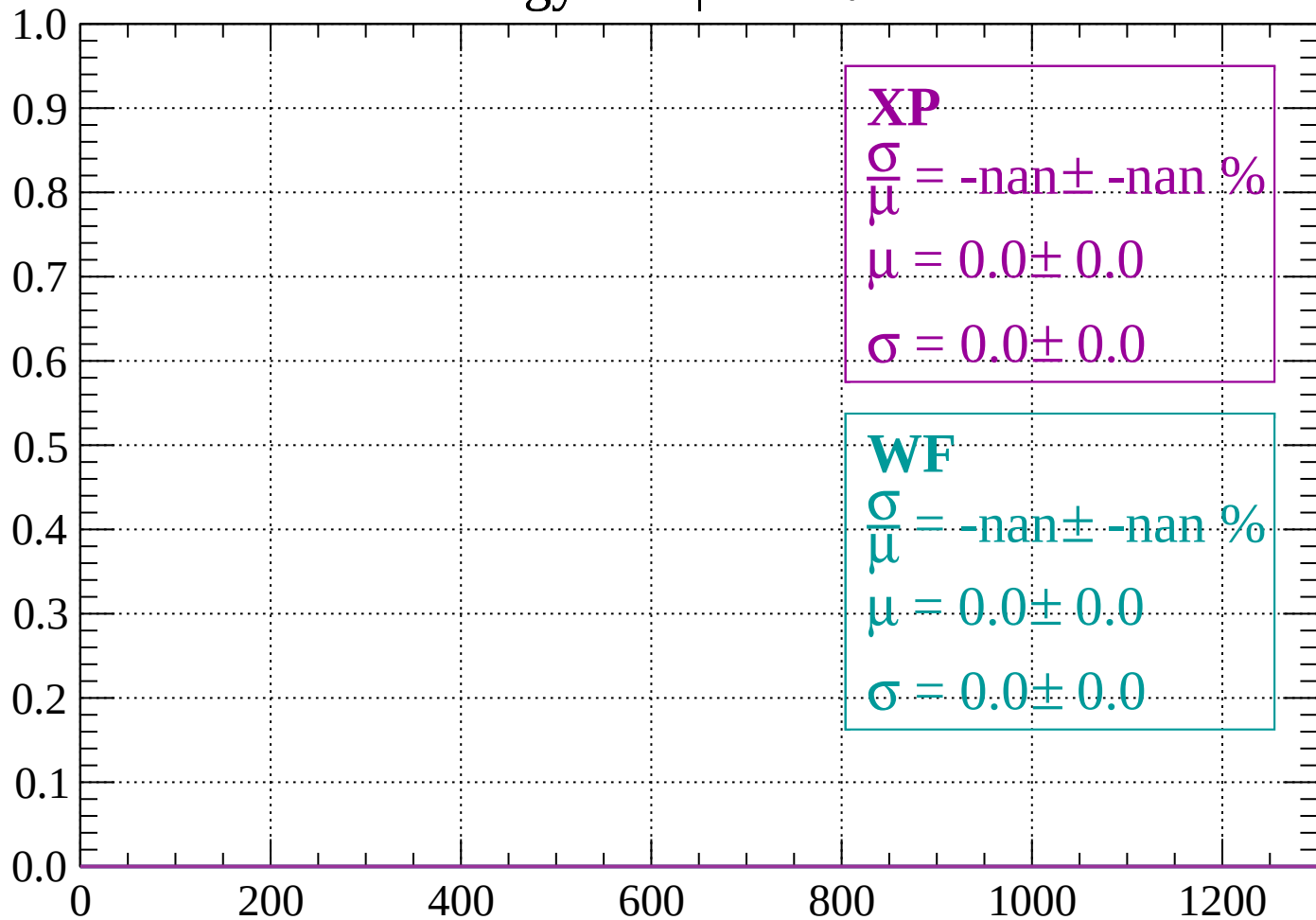
Count



dE/dx (ADC counts/cm)

Energy loss | $62 < \theta < 64$

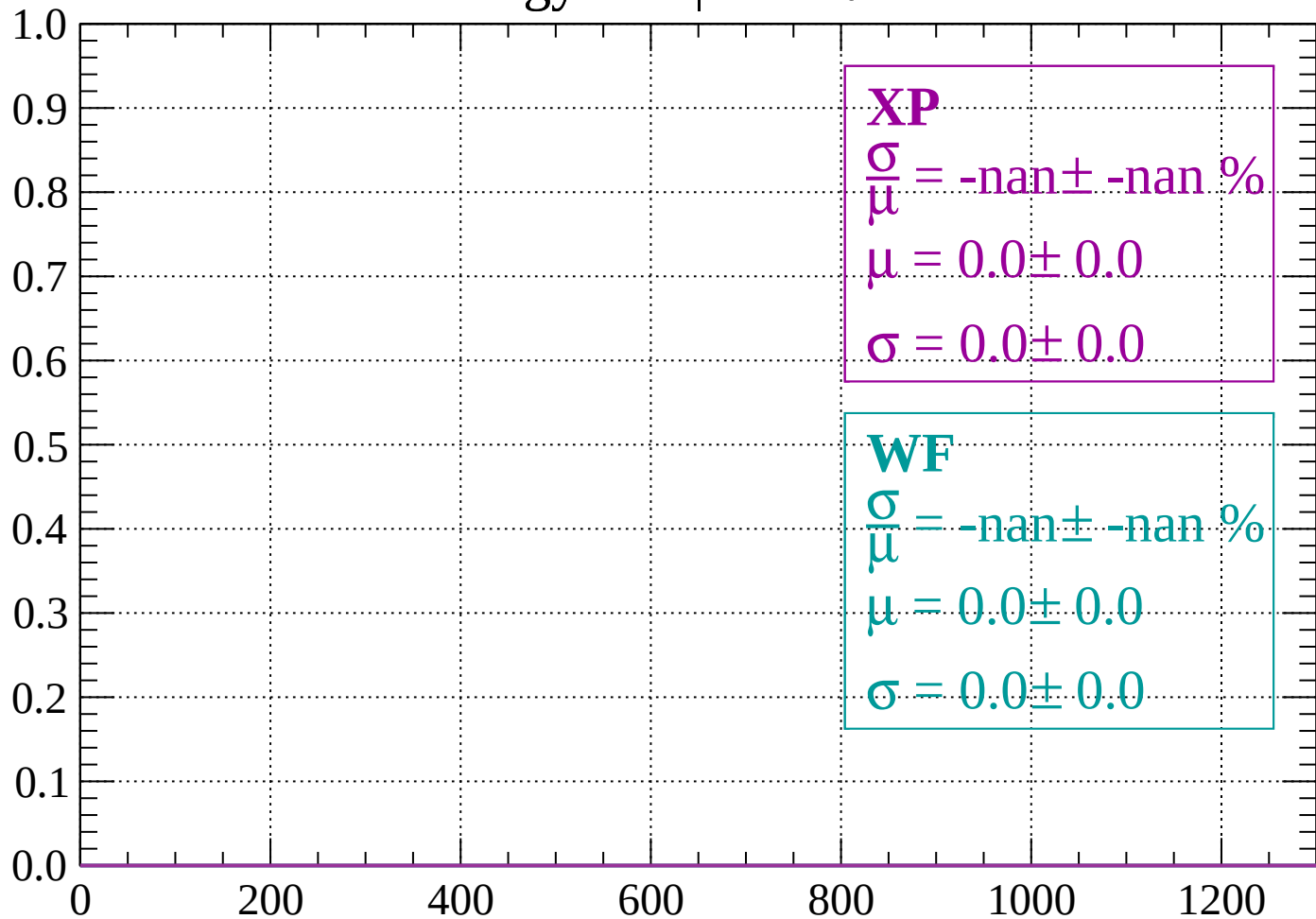
Count



dE/dx (ADC counts/cm)

Energy loss | $64 < \theta < 66$

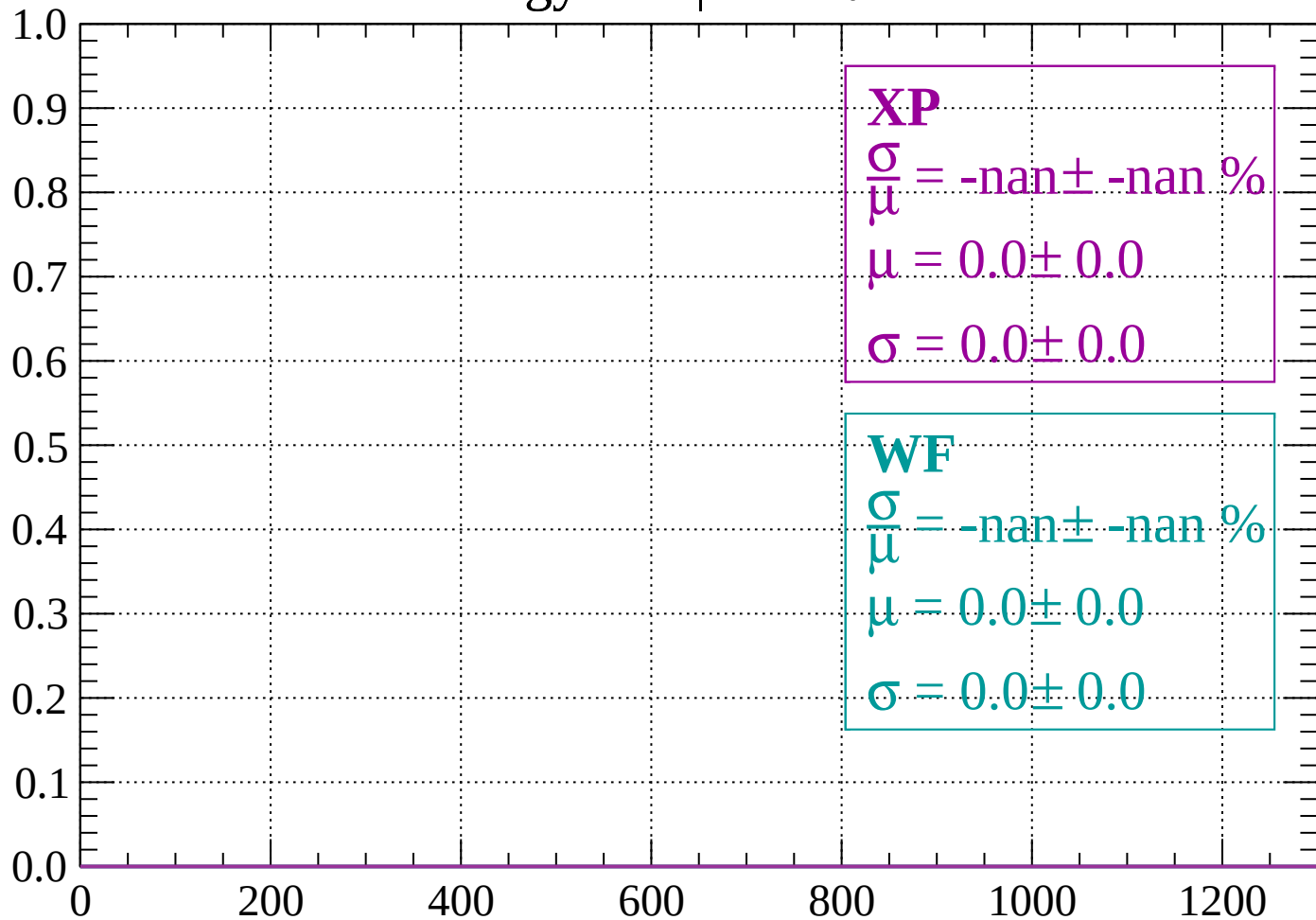
Count



dE/dx (ADC counts/cm)

Energy loss | $66 < \theta < 68$

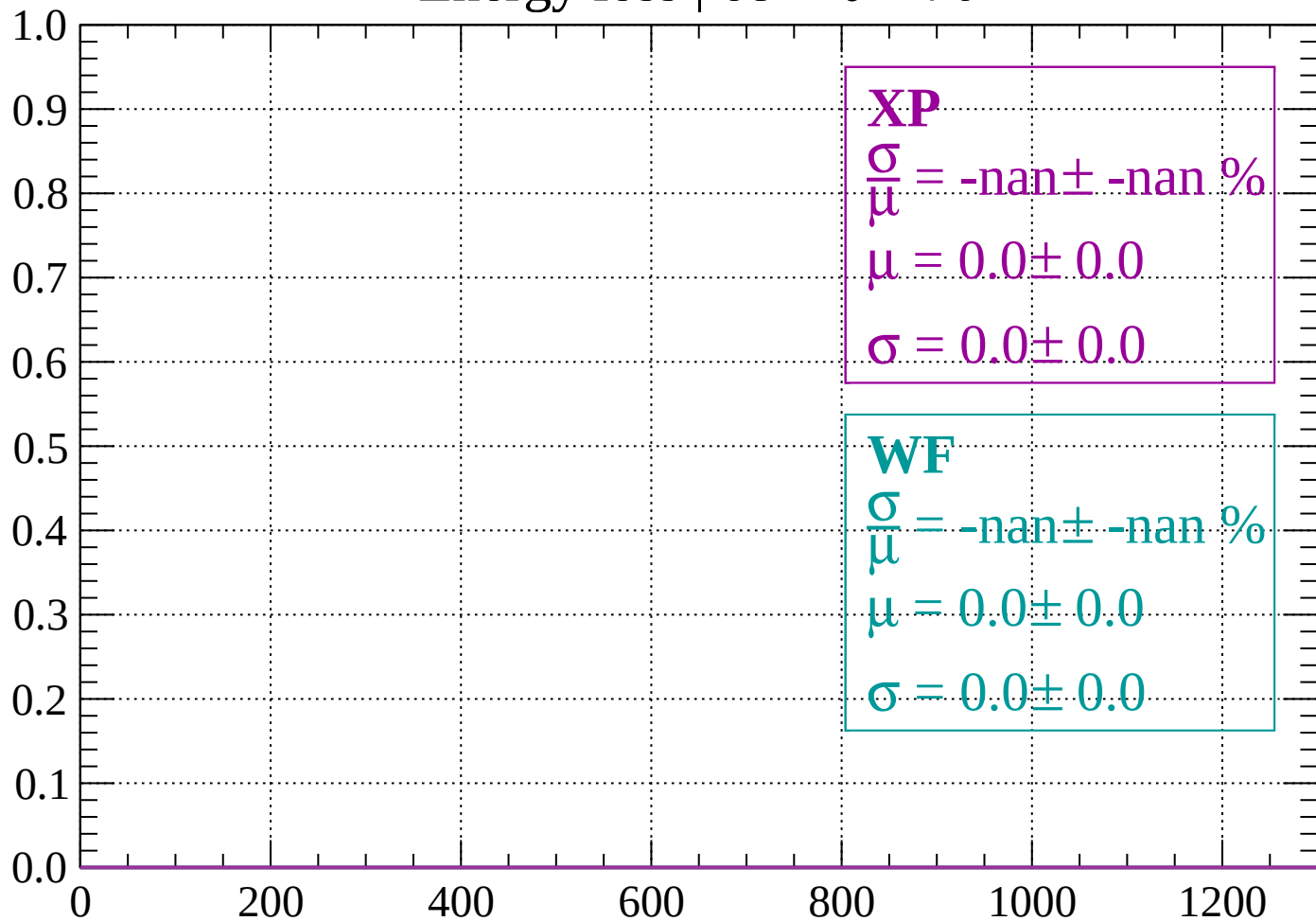
Count



dE/dx (ADC counts/cm)

Energy loss | $68 < \theta < 70$

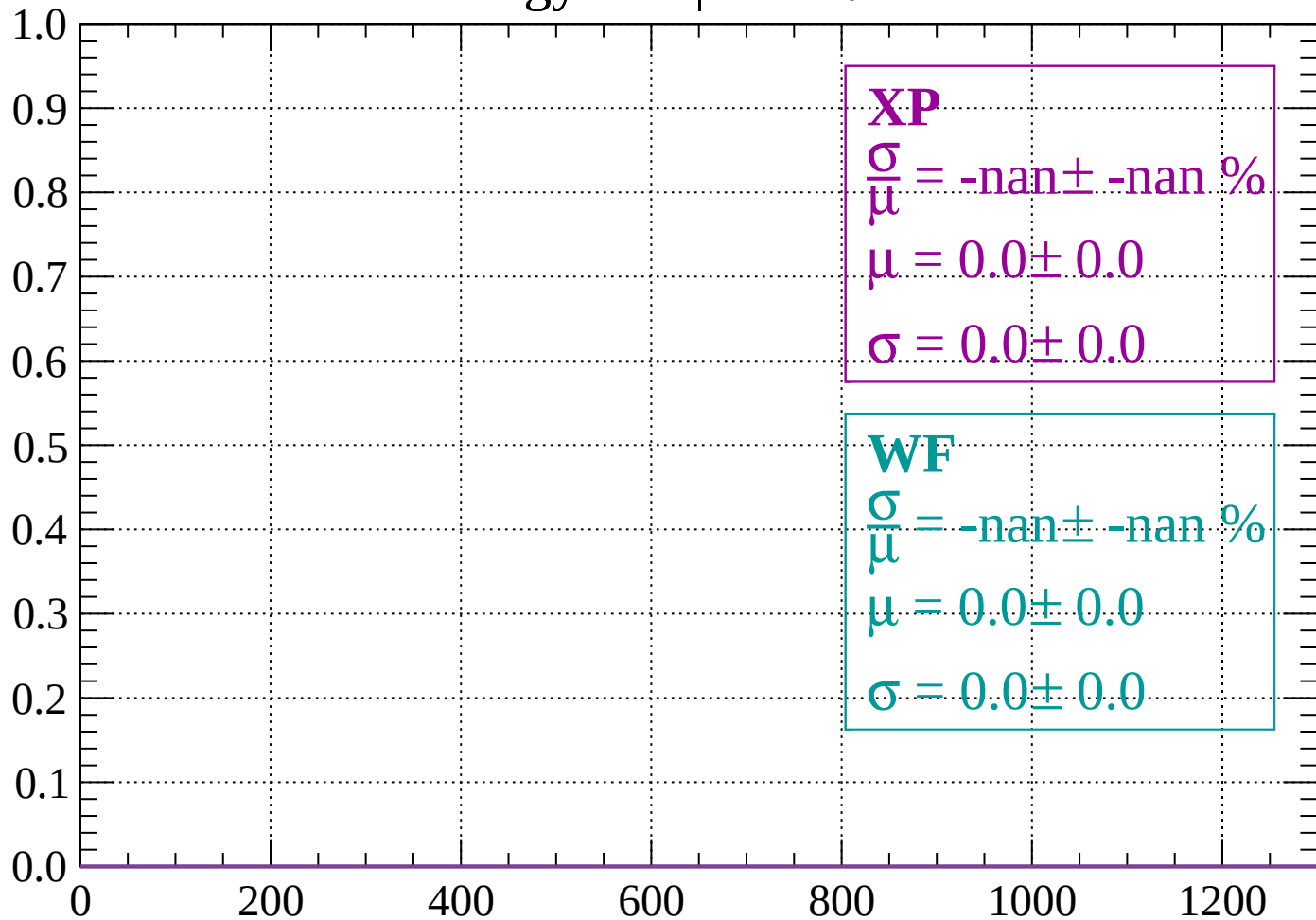
Count



dE/dx (ADC counts/cm)

Energy loss | $70 < \theta < 72$

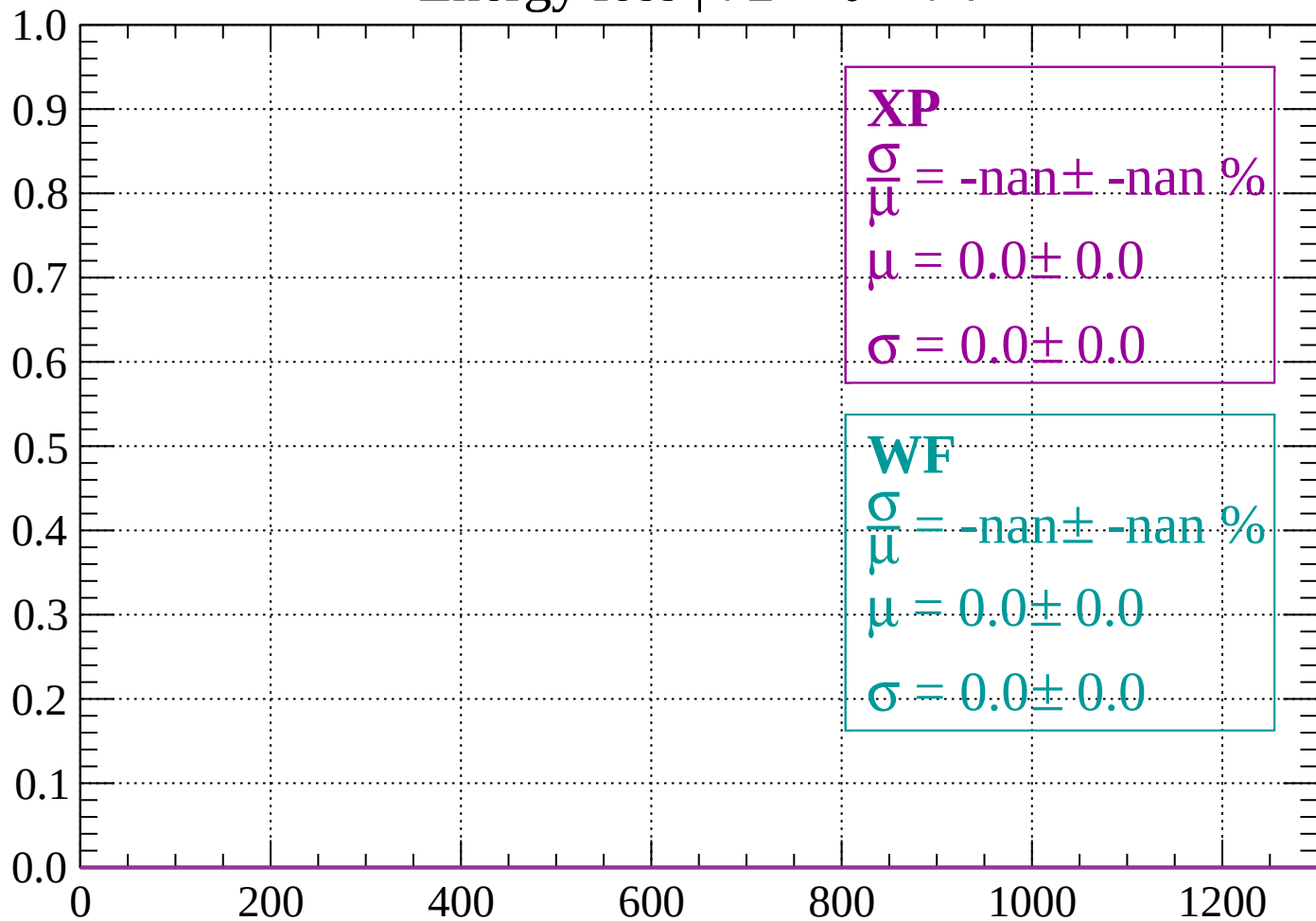
Count



dE/dx (ADC counts/cm)

Energy loss | $72 < \theta < 74$

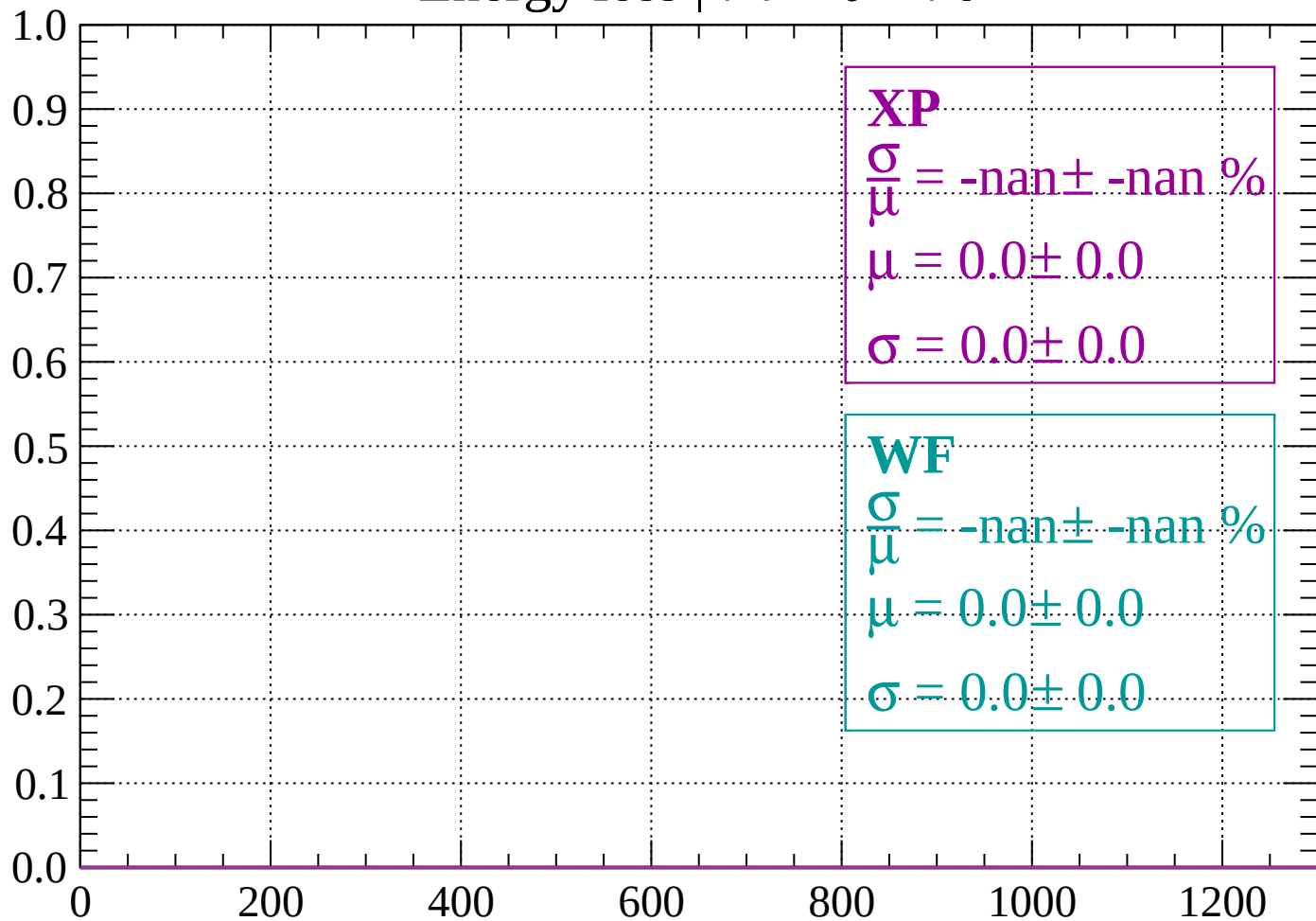
Count



dE/dx (ADC counts/cm)

Energy loss | $74 < \theta < 76$

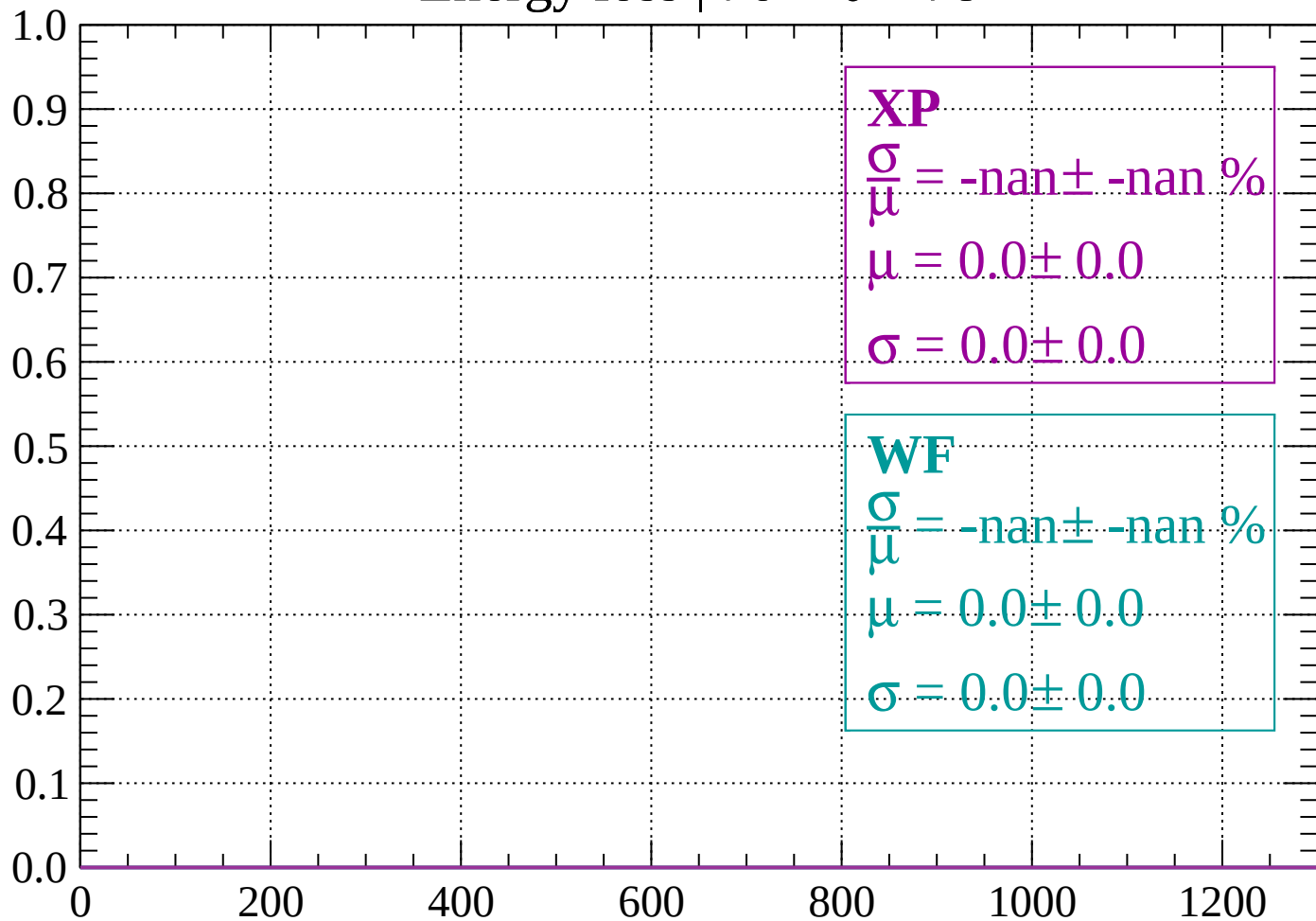
Count



dE/dx (ADC counts/cm)

Energy loss | $76 < \theta < 78$

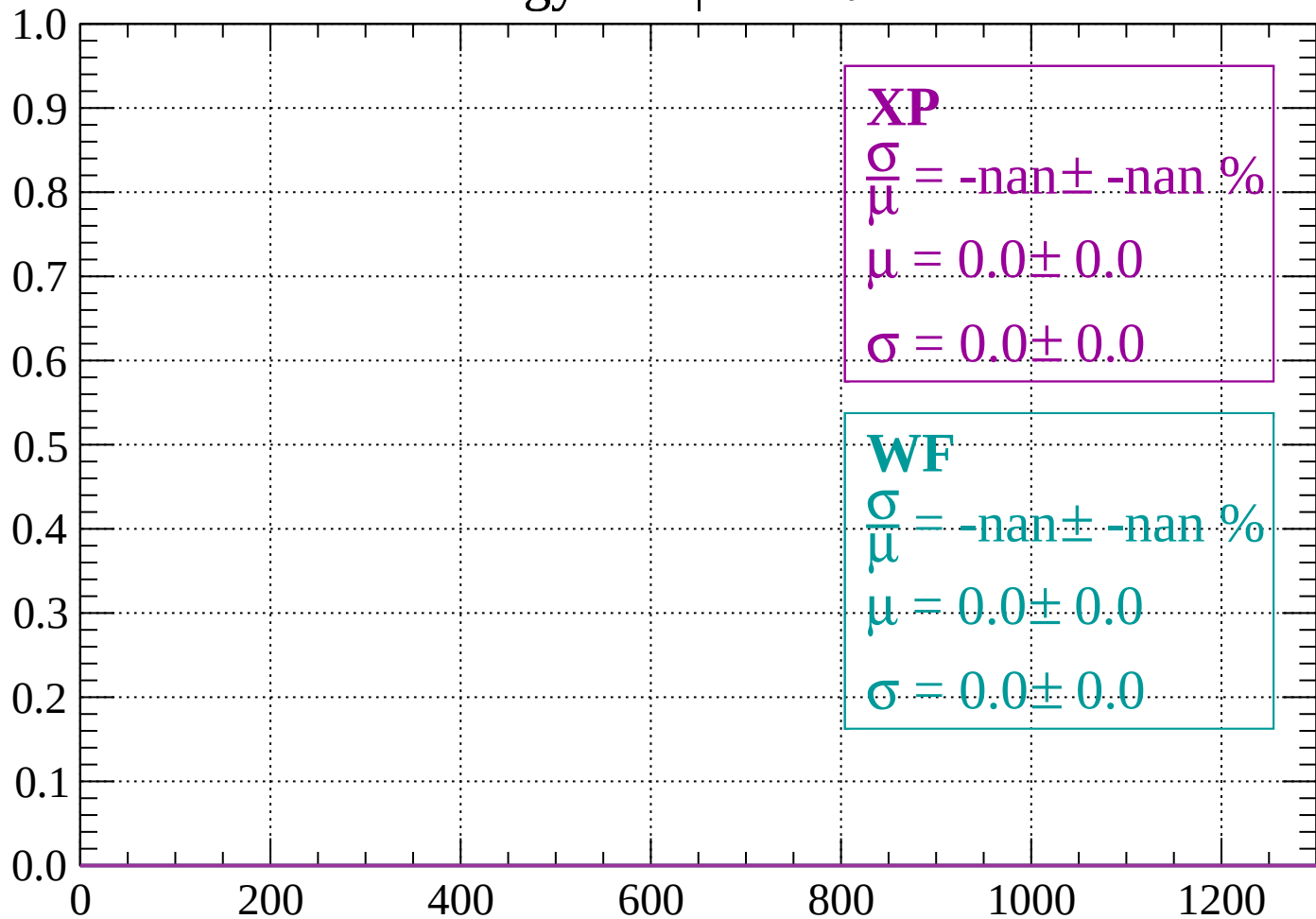
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 80$

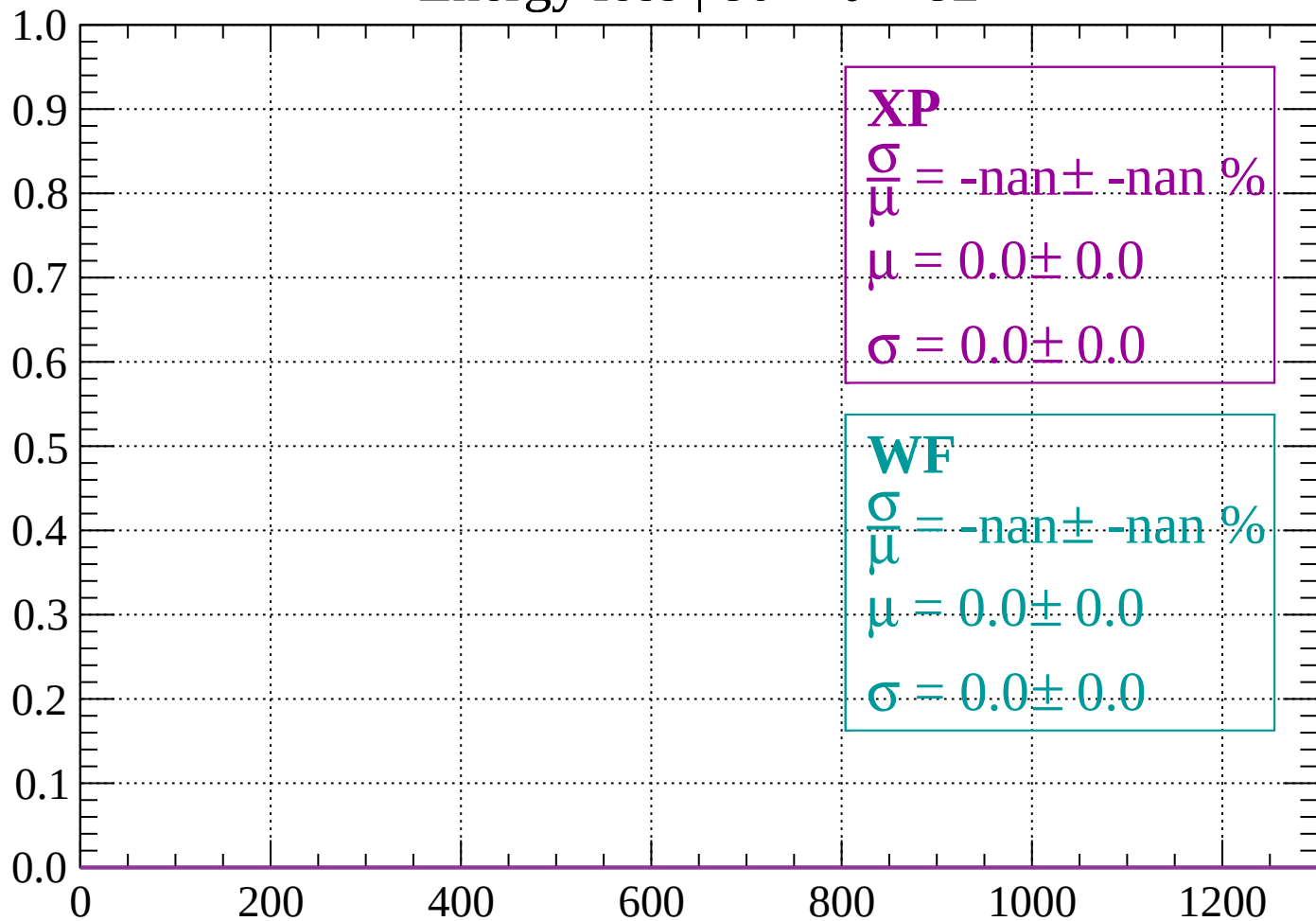
Count



dE/dx (ADC counts/cm)

Energy loss | $80 < \theta < 82$

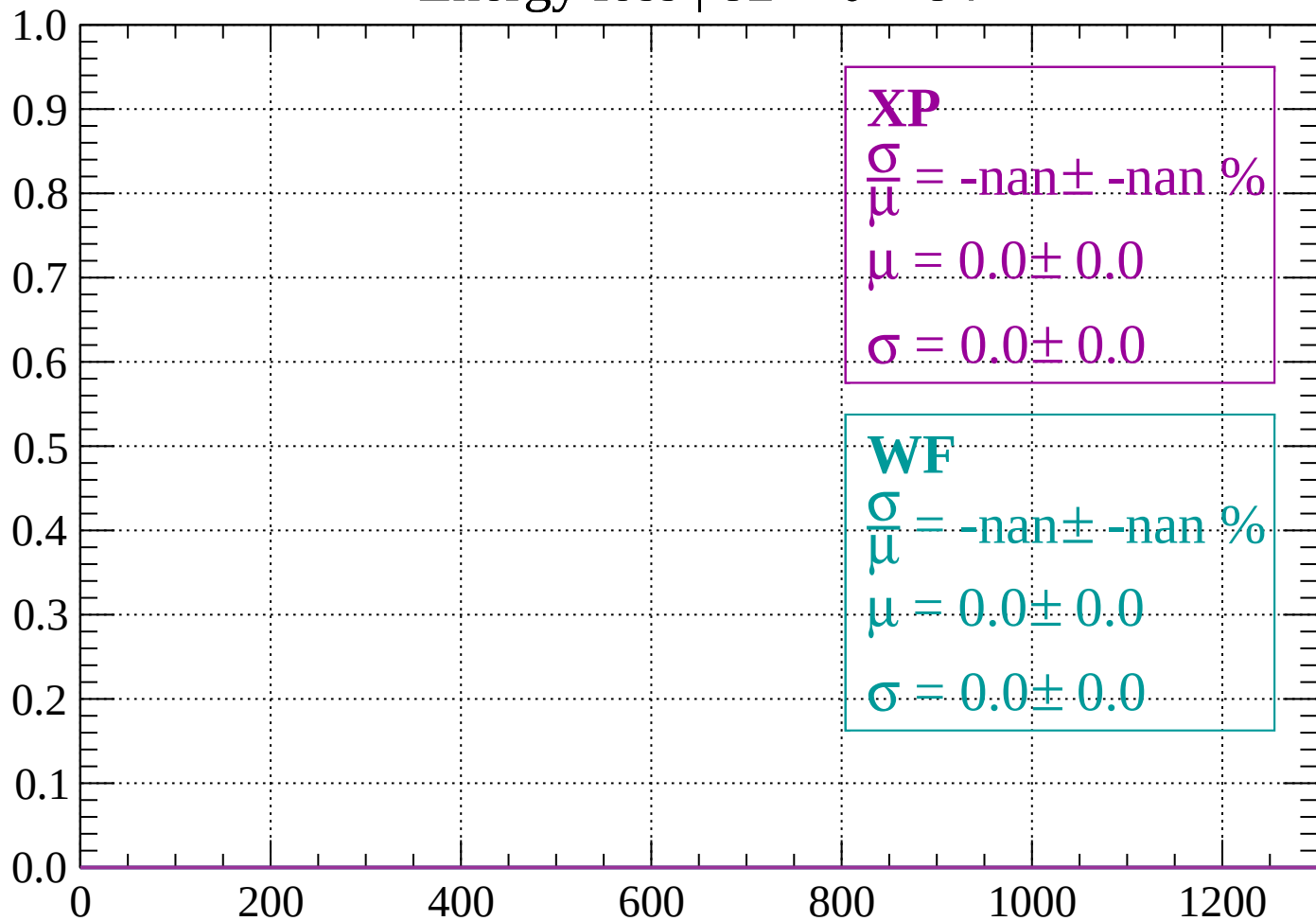
Count



dE/dx (ADC counts/cm)

Energy loss | $82 < \theta < 84$

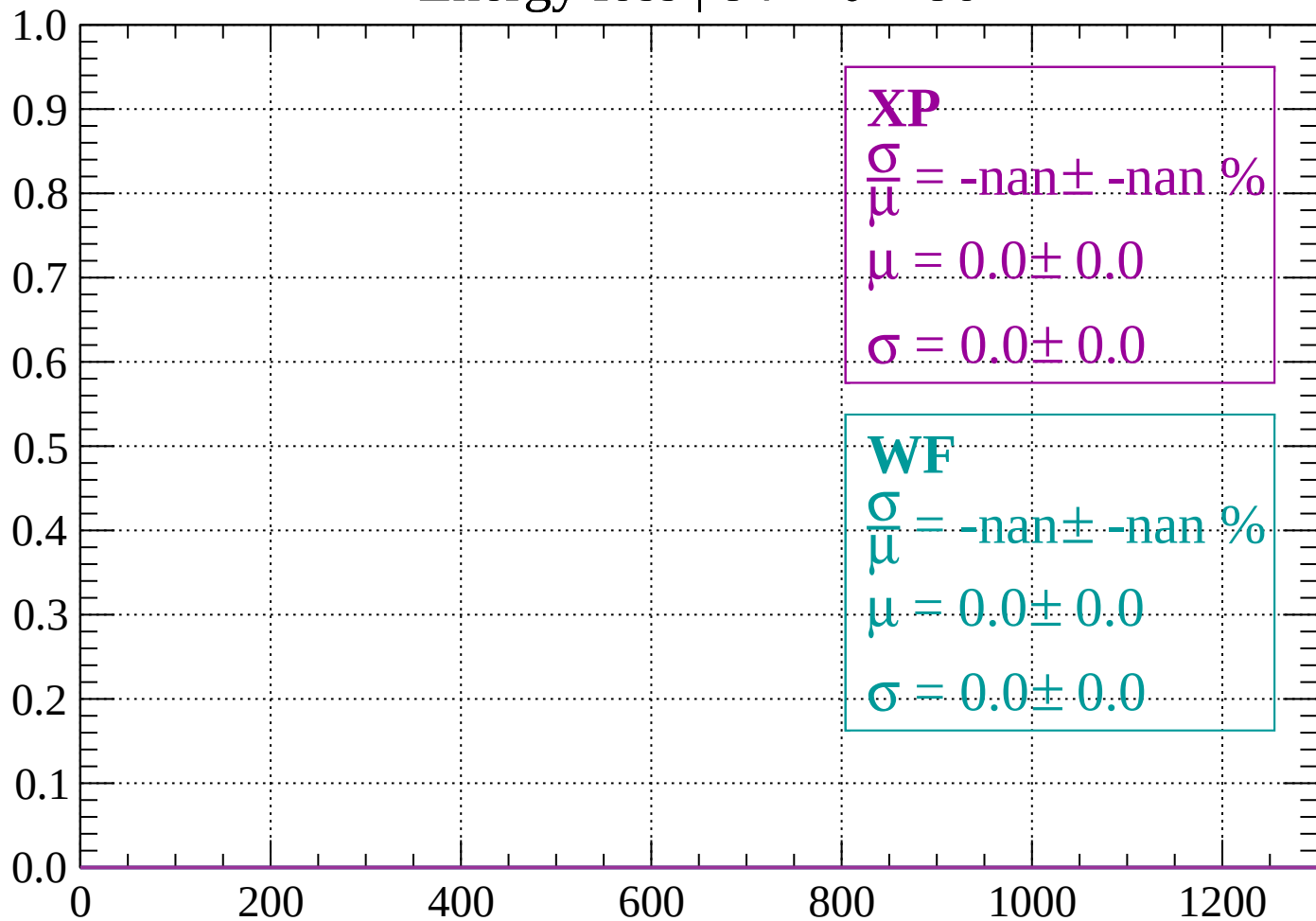
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 86$

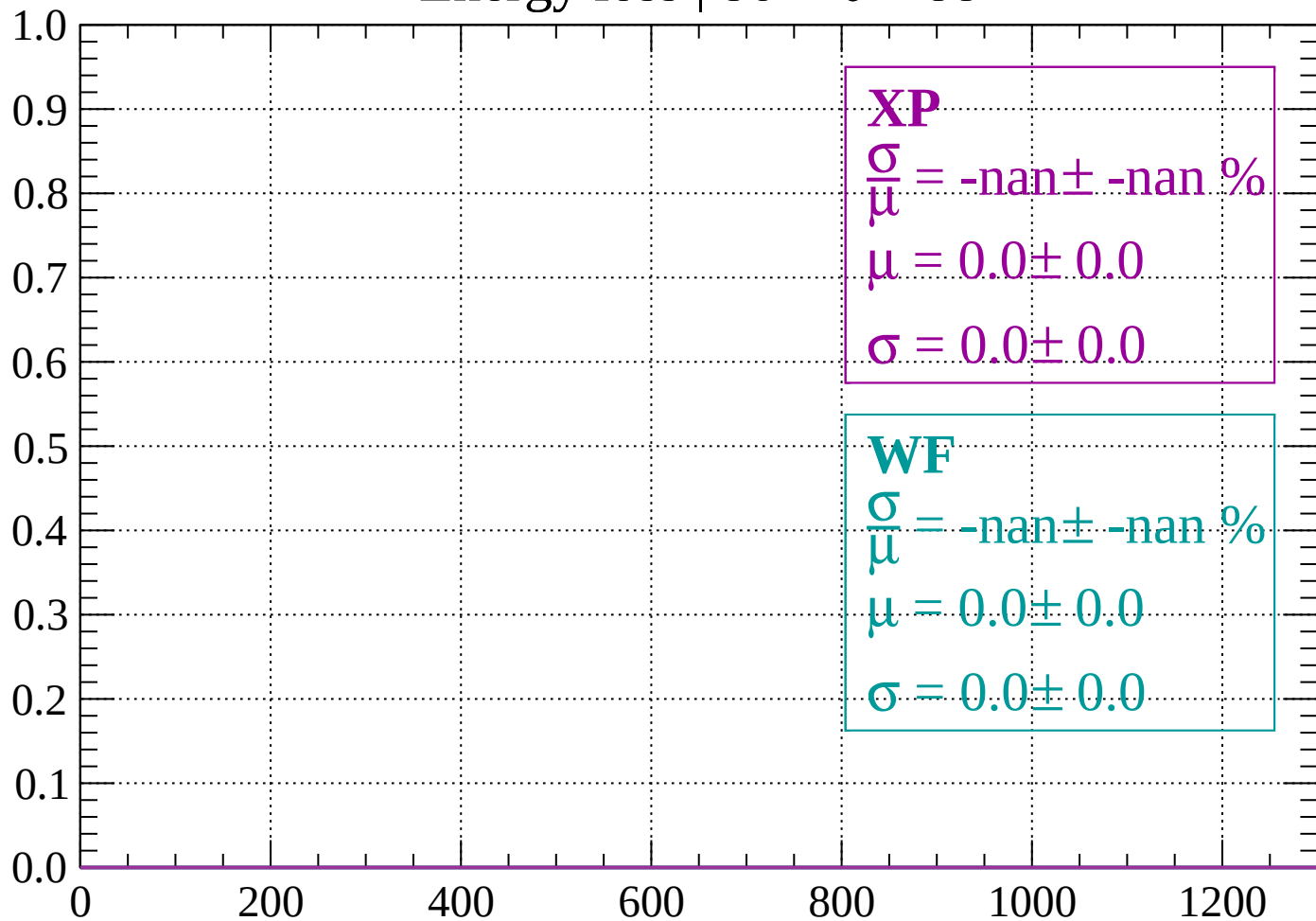
Count



dE/dx (ADC counts/cm)

Energy loss | $86 < \theta < 88$

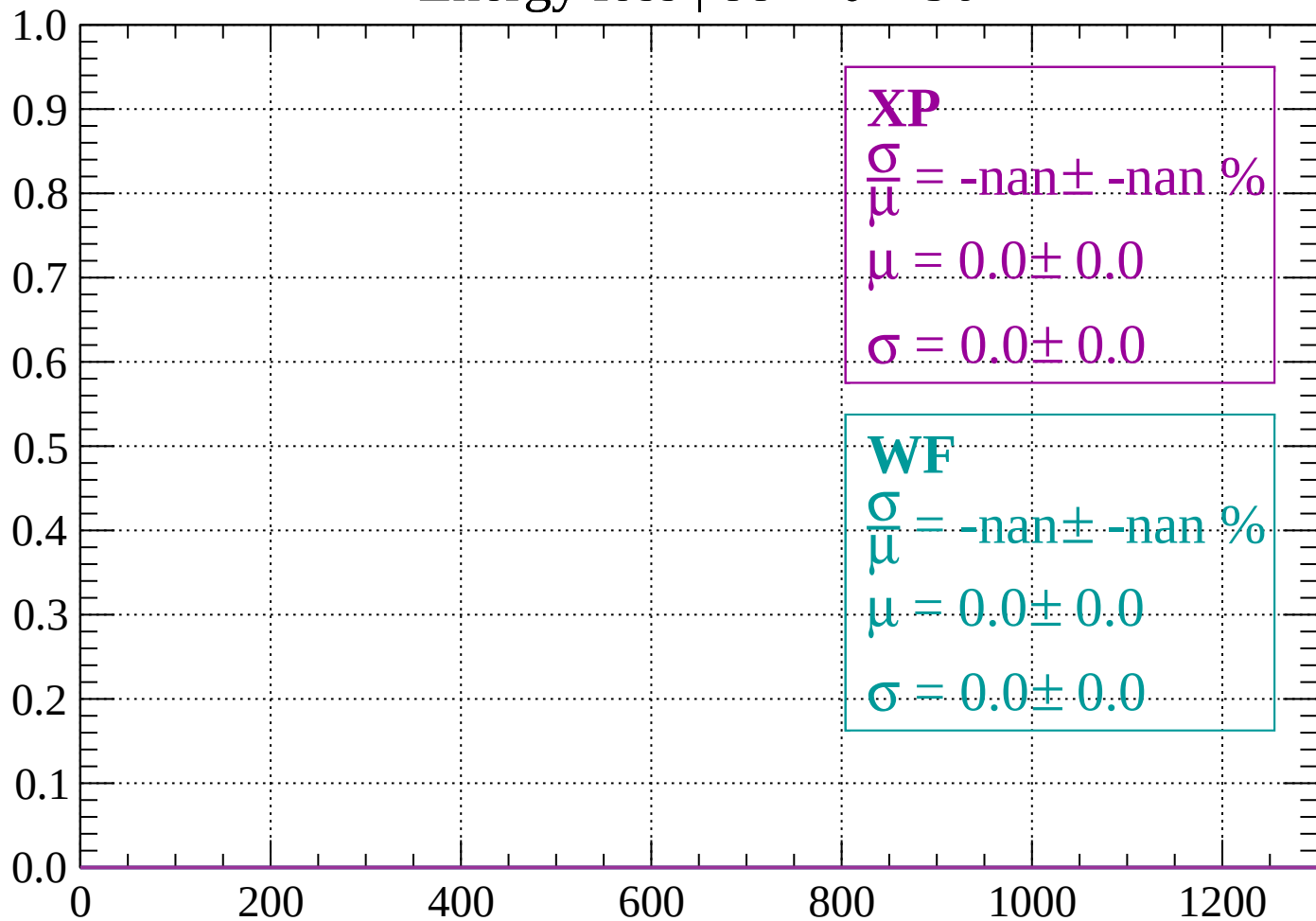
Count



dE/dx (ADC counts/cm)

Energy loss | $88 < \theta < 90$

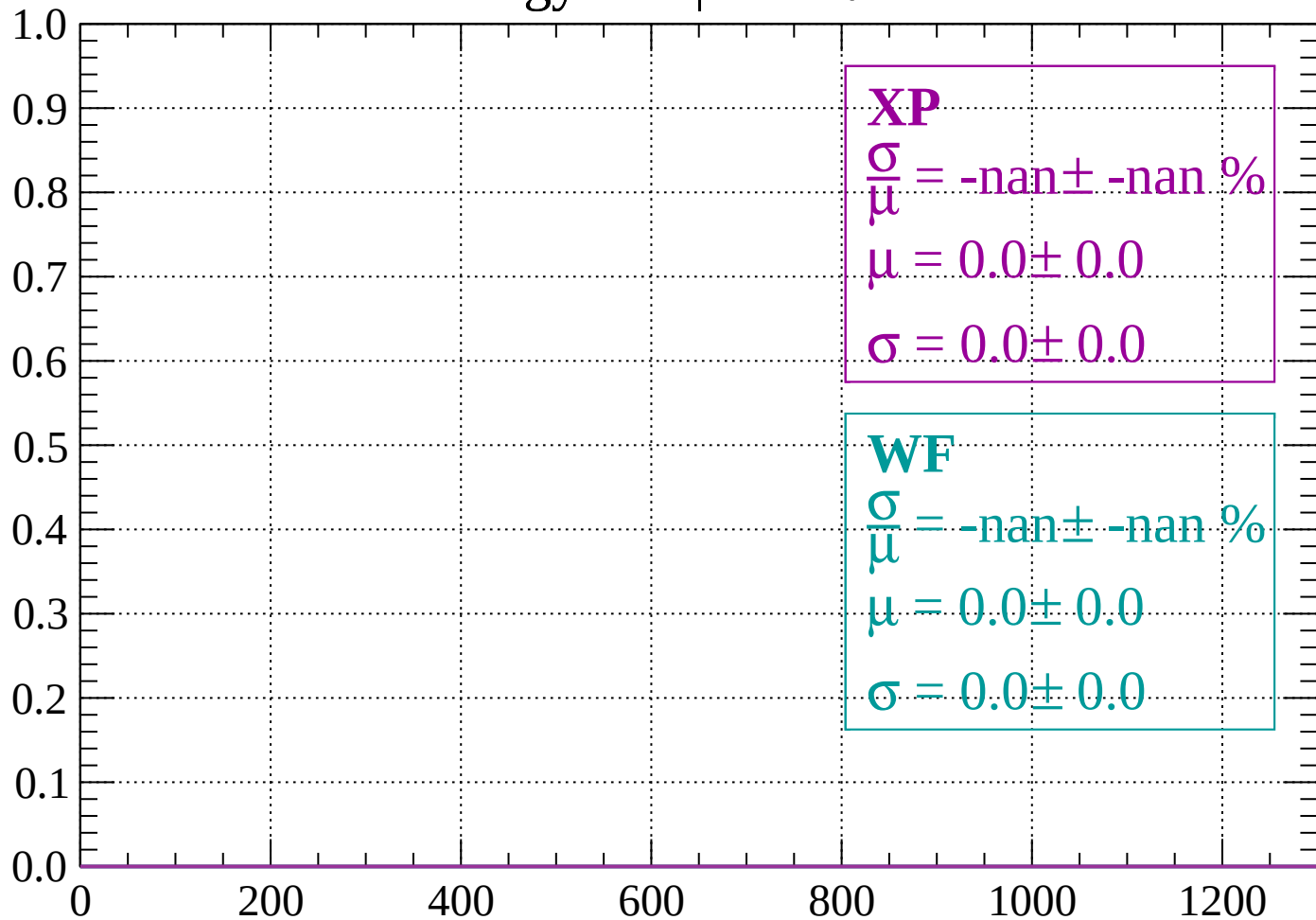
Count



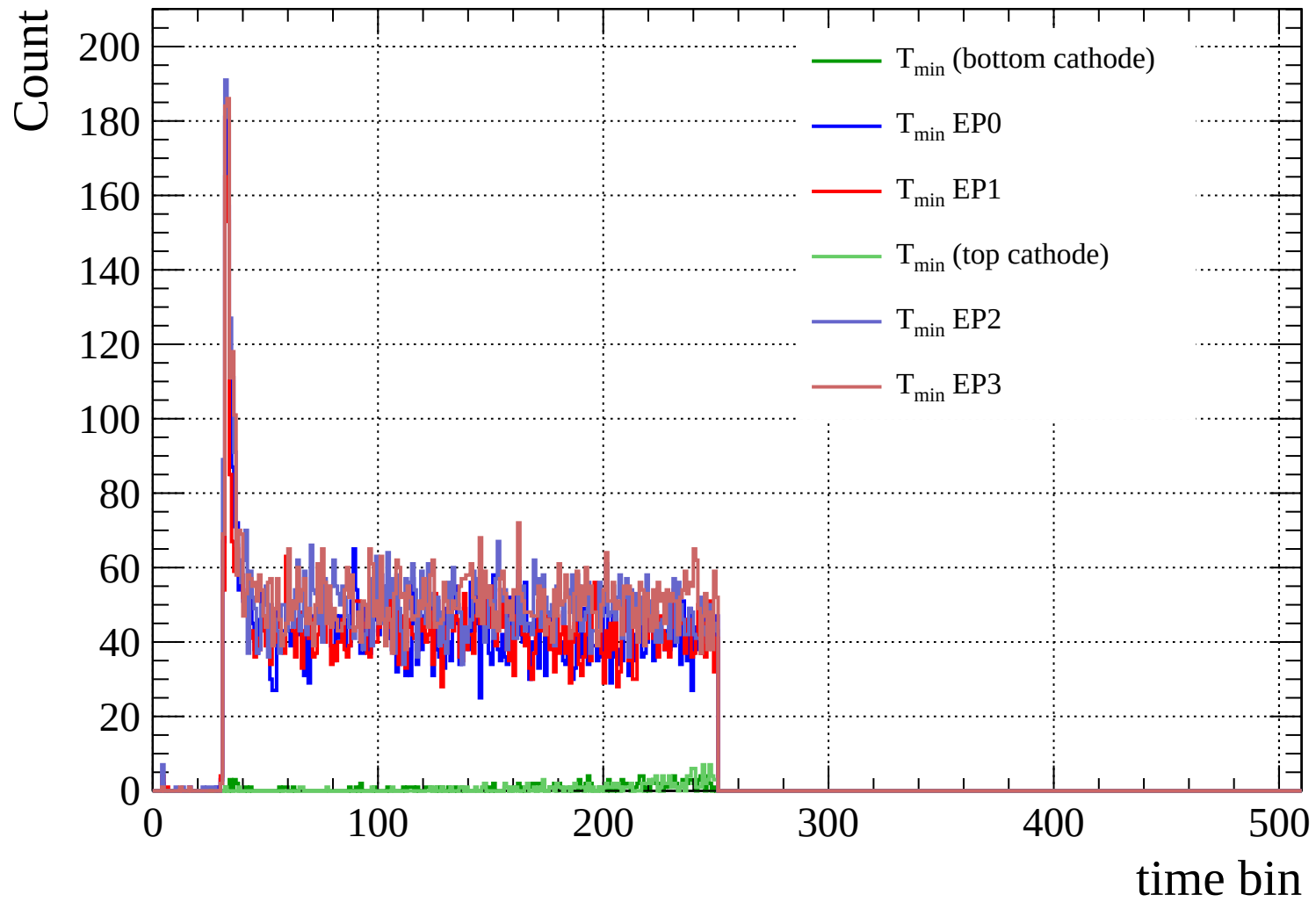
dE/dx (ADC counts/cm)

Energy loss | $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)



Count

